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# Foreword

This Technical Specification (TS) has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
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  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# 1 Scope

This present document specifies the measurement procedure for the conformance test of the mobile station that contain transmitting characteristics, receiving characteristics and performance requirements in addition to requirements for support of RRM (Radio Resource Management) in both UTRATDD modes. The two options are the 3,84 Mcps TDD Option and 1,28 Mcps TDD Option, respectively.

The requirements are listed in different clauses only if the corresponding parameters deviate. More generally, tests are only applicable to those mobiles that are intended to support the appropriate functionality. To indicate the circumstances in which tests apply, this is noted in the “definition and applicability” part of the test.

For example only release 5 and later UE declared to support HSDPA shall be tested for this functionality. In the event that for some tests different conditions apply for different releases, this is indicated within the text of the test itself.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TS 25.102 (V4.0.0): "UTRA (UE) TDD; Radio Transmission and Reception (TDD)".

[2] 3GPP TS 25.123: "Requirements for Support of Radio Resource Management (TDD)".

[3] 3GPP TS 34.108 "Common Test Environments for User Equipment (UE) Conformance Testing"

[4] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[5] 3GPP TS 25.224: "Physical Layer Procedures (TDD)".

[6] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[7] 3GPP TR 25.990: "Vocabulary".

[8] ITU-R Recommendation SM.328-9: "Spectra and bandwidth of emissions".

[9] 3GPP TS 25.331: "Radio Resource Control (RRC) Protocol Specification".

[10] 3GPP TS 25.433 "UTRAN Iub Interface NBAP Signalling".

[11] ITU-R Recommendation SM.329: "Spurious emissions".

[12] 3GPP TS 25.304: "UE Procedures in Idle Mode and Procedures for Cell Reselection in Connected Mode".

[13] 3GPP TS 25.303: "Interlayer Procedures in Connected Mode".

[14] 3GPP TS 25.321: "Medium Access Control (MAC) protocol specification".

[15] 3GPP TS 25.223: "Spreading and modulation (TDD)".

[16] ETSI ETR 273-1-2: "Improvement of radiated methods of measurement (using test sites) and evaluation of the corresponding measurement uncertainties; Part 1: Uncertainties in the measuremement of mobile radio equipment characteristics; Sub-part 2: Examples and annexes".

[17] 3GPP TR 25.926: "UE Radio Access Capabilities".

[18] 3GPP TR 21.904: "UE capability requirements".

[19] 3GPP TS 25.221: "Physical channels and mapping of transport channels onto physical channels (TDD)".

[20] 3GPP TS 45.008: "Digital cellular telecommunications system (Phase 2+); Radio subsystem link control".

[21] 3GPP TS 34.123-1: "User Equipment (UE) Conformance Specification; Part 1: Protocol Conformance Specification".

[22] 3GPP TS 25.225: "Physical Layer – Measurements (TDD)".

[23] 3GPP TS 51.010-1: " Mobile Station (MS) conformance specification; Part 1: Conformance specification".

[24] 3GPP TS 25.306: "UE Radio Access capabilities definition".

[25] ITU-T recommendation O.153: "Basic parameters for the measurement of error performance at bit rates below the primary rate".

# 3 Definitions, abbreviations and equations

For the purposes of the present document, the definitions, symbols, abbreviations and equations used in the present document are listed in TR 21.905 [6] and TR 25.990 [7].

## 3.1 Definitions

For the purpose of the present document, the following definition applies:

**Power Spectral Density:** The units of Power Spectral Density (PSD) are extensively used in this document. PSD is a function of power versus frequency and when integrated across a given bandwidth, the function represents the mean power in such a bandwidth. When the mean power is normalised to (divided by) the chip-rate it represents the mean energy per chip. Some signals are directly defined in terms of energy per chip, (DPCH_Ec, Ec, and P-CCPCH_Ec) and others defined in terms of PSD (Io, Ioc, Ior and $\hat{I}$or). There also exist quantities that are a ratio of energy per chip to PSD (DPCH_Ec/Ior, Ec/Ior etc.). This is the common practice of relating energy magnitudes in communication systems.

It can be seen that if both energy magnitudes in the ratio are divided by time, the ratio is converted from an energy ratio to a power ratio, which is more useful from a measurement point of view. It follows that an energy per chip of X dBm/3,84 MHz (3,84 Mcps TDD option) or X dBm/1,281,28 MHz (1,281,28 Mcps TDD option) can be expressed as a mean power per chip of X dBm. Similarly, a signal PSD of Y dBm/3,84 MHz (3,84 Mcps TDD option) or Y dBm/1,281,28 MHz (1,281,28 Mcps TDD option) can be expressed as a signal power of Y dBm.

**Maximum Output Power:** This is a measure of the maximum power the UE can transmit (i.e. the actual power as would be measured assuming no measurement error) in a bandwidth of at least (1+ $\alpha$) times the chip rate of the radio access mode. The period of measurement shall be a transmit timeslot excluding the guard period.

**Mean Power:** When applied to a CDMA modulated signal this is the power (transmitted or received) in a bandwidth of at least (1+ $\alpha$) times the chip rate of the radio access mode. The period of measurement shall be a transmit timeslot excluding the guard period unless otherwise stated.

**Output power:** The mean power of the UE delivered to a load with resistance equal to the nominal load impedance of the transmitter.

**RRC Filtered Mean Power:** The mean power as measured through a root raised cosine filter with roll-off factor $\alpha$ and a bandwidth equal to the chip rate of the radio access mode.

**Nominal Maximum Output Power:** This is the nominal power defined by the UE power class. The period of measurement shall be a transmit timeslot excluding the guard period.

**Received Signal Code Power (RSCP):** Given only signal power is received, the RRC filtered mean power of the received signal after despreading and combining.

**Interference Signal Code Power (ISCP):** Given only interference power is received, the RRC filtered mean power of the received signal after despreading to the code and combining. Equivalent to the RSCP value but now only interference is received instead of signal.

The following two definitions are applicable for LCR TDD multi-frequency operation.

**Inter-frequency cell:** In idle, CELL-FACH, CELL-PCH and URA-PCH state, inter-frequency cell is defined as the cell of which the primary frequency is different from the user's current frequency. In CELL-DCH state, inter-frequency cell is defined as the cell of which the frequency to be measured in that cell is different from the operating frequency for single carrier operation or is different from the associated DPCH operating frequency for multi-carrier operation.

**Intra-frequency cell:** In idle, CELL-FACH, CELL-PCH and URA-PCH state, intra-frequency cell is defined as the cell of which the primary frequency is the same as the user's current frequency. In CELL-DCH state, intra-frequency cell is defined as the cell of which the frequency to be measured in that cell is the same as the operating frequency for single carrier operation or is the same as the associated DPCH operating frequency for multi-carrier operation.

NOTE 1: The RRC filtered mean power of a perfectly modulated CDMA signal is 0.246 dB lower than the mean power of the same signal.

NOTE 2: The roll-off factor $\alpha$ is defined in section 6.8.1 of [1].

## 3.2 Abbreviations

For the purpose of the present document, the following abbreviations apply.

| | |
|---|---|
| ACLR | Adjacent Channel Leakage power Ratio |
| ACS | Adjacent Channel Selectivity |
| AFC | Automatic Frequency Control |
| ATT | Attenuator |
| CW | Continuous wave (unmodulated signal) |
| DPCH | Dedicated physical channel |
| DPCH_Ec | Average energy per PN chip for DPCH |
| EVM | Error Vector Magnitude |
| FFS | For Further Study |
| Fuw | Frequency of unwanted signal. This is specified in bracket in terms of an absolute frequency(s) or frequency offset from the assigned channel frequency. |
| HYB | Hybrid |
| $I_{BTS}$ | Interference signal power level at BTS in dBm, which is broadcasted on BCH |
| $I_{oac}$ | The power spectral density of the adjacent frequency channel as measured at the UE antenna connector. |
| Ioc | The power spectral density (integrated in a noise bandwidth equal to the chip rate and normalized to the chip rate) of a band limited white noise source (simulating interference from cells which are not defined in a test procedure) as measured at the UE antenna connector. |
| Ior | The total transmit power spectral density (integrated in a bandwidth of (1+α) times the chip rate and normalized to the chip rate) of the downlink signal at the BS antenna connector |
| $\hat{I}_{or}$ | The received power spectral density (integrated in a bandwidth of (1+α) times the chip rate and normalized to the chip rate) of the downlink signal as measured at the UE antenna connector |
| Iowc | Unwanted signal power level |
| OBW | Occupied Bandwidth |
| OCNS | Orthogonal Channel Noise Simulator, a mechanism used to simulate the users or control signals on the other orthogonal channels of a downlink. |
| PCDE | Peak Code Domain Error |
| PPM | Parts Per Million |
| PRBS | Pseudo Random Bit Sequence |
| RRC | Root-Raised Cosine |

| | |
|---|---|
| SCTD | Space Code Transmit Diversity |
| SIR | Signal to Interference ratio |
| SS | System Simulator |
| TBD | To Be Defined |
| TPC | Transmit Power Control |
| TS | Time Slot |

## 3.3 Equations

For the purpose of the present document, the following additional equations apply:

| | |
|---|---|
| $\frac{DPCH_Ec}{I_{or}}$ | The ratio of the average energy per PN chip of the DPCH to the total transmit power spectral density of the downlink at the BS antenna connector |
| $\frac{\Sigma\ DPCH_Ec}{I_{or}}$ | The ratio of the sum of DPCH_Ec for one service in case of multicode to the total transmit power spectral density of the downlink at the BS antenna connector |

# 4 Frequency bands and channel arrangement

## 4.1 General

The information presented in this clause is based on the chip rates of 3,84 Mcps TDD Option,1,28 Mcps TDD Option and 7.68Mcps Option.

## 4.2 Frequency bands

UTRA/TDD is designed to operate in the following bands;

a) 1 900 – 1 920 MHz: Uplink and downlink transmission
   2 010 – 2 025 MHz Uplink and downlink transmission

b) 1 850 – 1 910 MHz: Uplink and downlink transmission (note 1)
   1 930 – 1 990 MHz: Uplink and downlink transmission

c) 1 910 – 1 930 MHz: Uplink and downlink transmission (note 1)

d) 2 570 - 2 620 MHz: Uplink and downlink transmission (note 2)

NOTE 1: Used in ITU Region 2.

NOTE 2: Used in ITU Region 1.

Additional allocations in ITU region 2 are for further study.

Deployment in existing or other frequency bands is not precluded.

## 4.3 TX–RX frequency separation

### 4.3.1 3,84 Mcps TDD Option

No TX-RX frequency separation is required as Time Division Duplex (TDD) is employed. Each TDMA frame consists of 15 timeslots where each timeslot can be allocated to either transmit or receive.

### 4.3.2 1,28 Mcps TDD Option

No TX-RX frequency separation is required as Time Division Duplex (TDD) is employed. Each subframe consists of 7 main timeslots where all main timeslots (at least the first one) before the single switching point are allocated DL and all main timeslots (at least the last one) after the single switching point are allocated UL.

### 4.3.3 7,68 Mcps TDD Option

No TX-RX frequency separation is required as Time Division Duplex (TDD) is employed. Each TDMA frame consists of 15 timeslots where each timeslot can be allocated to either transmit or receive.

## 4.4 Channel arrangement

### 4.4.1 Channel spacing

#### 4.4.1.1 3,84 Mcps TDD Option

The nominal channel spacing is 5 MHz, but this can be adjusted to optimise performance in a particular deployment scenario.

#### 4.4.1.2 1,28 Mcps TDD Option

The nominal channel spacing is 1,6 MHz, but this can be adjusted to optimise performance in a particular deployment scenario.

#### 4.4.1.3 7,68 Mcps TDD Option

The nominal channel spacing is 10 MHz, but this can be adjusted to optimise performance in a particular deployment scenario.

### 4.4.2 Channel raster

The channel raster is 200 kHz, which means that the carrier frequency must be a multiple of 200 kHz.

### 4.4.3 Channel number

The carrier frequency is designated by the UTRA absolute radio frequency channel number (UARFCN). The value of the UARFCN in the IMT2000 band is defined as follows:

$N_t = 5 * F$ $0,0 \text{ MHz} \leq F \leq 3276,6 \text{ MHz}$ where F is the carrier frequency in MHz

### 4.4.4 UARFCN (3,84 Mcps TDD Option)

The following UARFCN range shall be supported for each band.

**Table 4.4.1: UTRA Absolute Radio Frequency Channel Number**

| Frequency Band | Frequency Range | UARFCN Uplink and Downlink transmission |
|---|---|---|
| For operation in frequency band as defined in subclause 5.2 (a) | 1900-1920 MHz<br>2010-2025 MHz | 9512 to 9588<br>10062 to 10113 |
| For operation in frequency band as defined in subclause 5.2 (b) | 1850-1910 MHz<br>1930-1990 MHz | 9262 to 9538<br>9662 to 9938 |
| For operation in frequency band as defined in subclause 5.2 (c) | 1910-1930 MHz | 9562 to 9638 |

### 4.4.5 UARFCN (7,68 Mcps TDD Option)

The following UARFCN range shall be supported for each band.

**Table 4.4.2: UTRA Absolute Radio Frequency Channel Number**

| Frequency Band | Frequency Range | UARFCN Uplink and Downlink transmission | Additional UARFCN Uplink and Downlink transmission |
|---|---|---|---|
| For operation in frequency band as defined in subclause 4.2 (a) | 1900-1920 MHz<br>2010-2025 MHz | 9512 to 9588<br>10062 to 10113 | - |
| For operation in frequency band as defined in subclause 4.2 (b) | 1850-1910 MHz<br>1930-1990 MHz | 9262 to 9538<br>9662 to 9938 | - |
| For operation in frequency band as defined in subclause 4.2 (c) | 1910-1930 MHz | 9562 to 9638 | - |
| For operation in frequency band as defined in subclause 4.2 (d) | 2570-2620 MHz | 12874 to 13076 | - |

# 5 Transmitter Characteristics

## 5.1 General

Transmitting performance test of the UE is implemented during communicating with the SS via air interface. The procedure is uses normal call protocol until the UE is communicating on traffic channel basically. (Refer to TS 34.108 [3] Common Test Environments for User Equipment (UE) Conformance Testing) On the traffic channel, the UE provides special function for testing that is called Logical Test Interface and the UE is tested using this function. (Refer to TS 34.109 [4] Logical Test Interface; Special conformance testing functions).

Unless detailed the transmitter characteristic are specified at the antenna connector of the UE. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed. Transmitter characteristics for UE(s) with multiple antennas/antenna connectors are for further study.

The UE antenna performance has a significant impact on system performance, and minimum requirements on the antenna efficiency are therefore intended to be included in future versions of the present document. It is recognized that different requirements and test methods are likely to be required for the different types of UE.

The common RF test conditions are defined in annex E, and each test conditions in this Clause should refer annex E. Individual test conditions are defined in the paragraph of each test.

# 5.2 User Equipment maximum output power

## 5.2.1 Definition and applicability

The nominal maximum output power and its tolerance are defined according to the Power Class of the UE.

The requirements in this test apply to all UTRA – TDD- UEs

Notes copied from TS 25.102 clause 6.2.1:

NOTE 1: For multi-code operation the nominal maximum output power will be reduced by the difference of peak to average ratio between single and multi-code transmission.

NOTE 2: The tolerance allowed for the nominal maximum power applies even at the multi-code transmission mode

NOTE 3: For UE using directive antennas for transmission, a class dependent limit will be placed on the maximum EIRP (Equivalent Isotropic Radiated Power).

## 5.2.2 Minimum Requirements

### 5.2.2.1 3.84 Mcps TDD option

The error of the UE maximum output power shall not exceed the tolerance shown in tables 5.2.2 1.a and b for single and multi-code for 3.84Mcps TDD option.

**Table 5.2.2.1.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +30 dBm | +1dB/-3dB |
| 2 | +24 dBm | +1dB/-3dB |
| 3 | +21 dBm | +2dB/-2dB |
| 4 | +10 dBm | +4dB/-4dB |

**Table 5.2.2.1.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +27 dBm (note) | +1dB/-3dB |
| 2 | 21 dBm (note) | +1dB/-3dB |
| 3 | 18 dBm (note) | +2dB/-2dB |
| 4 | +7 dBm (note) | +4dB/-4dB |
| NOTE: These figures are not mentioned in 25.102. Instead there is a note, saying: "For multi-code operation the maximum output power will be reduced by the difference of peak to average ratio between single and multi-code transmission." The figures are calculated from maximum output power single code (table 5.2.2.1.a) and UL multicode reference measurement channel (12,2 kbit/s) (annex C.2.2.1 for the 3,84 TDD Option) containing two code signals with equal level. | | |

The normative reference for this requirement is TS 25.102 [1] clause 6.2.1.

### 5.2.2.2 1.28 Mcps TDD option

The error of the UE maximum output power shall not exceed the tolerance shown in tables 5.2.2 2.a and b for single and multi-code for 1.28Mcps TDD option.

**Table 5.2.2.2.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +33 dBm | +1dB/-3dB |
| 2 | +24 dBm | +1dB/-3dB |
| 3 | +21 dBm | +2dB/-2dB |
| 4 | +27 dBm | +1dB/-3dB |

**Table 5.2.2.2.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +30 dBm (note) | +1dB/-3dB |
| 2 | 21 dBm (note) | +1dB/-3dB |
| 3 | 18 dBm (note) | +2dB/-2dB |
| 4 | +24 dBm (note) | +1dB/-3dB |
| NOTE: These figures are not mentioned in 25.102. Instead there is a note, saying: "For multi-code operation the maximum output power will be reduced by the difference of peak to average ratio between single and multi-code transmission." The figures are calculated from maximum output power single code (table 5.2.2.a) and UL multicode reference measurement channel (12,2 kbit/s) (annex C.2.2.2 for the 1,28 Mcps TDD Option) containing two code signals with equal level. | | |

The normative reference for this requirement is TS 25.102 [1] clause 6.2.1.

### 5.2.2.3 7.68 Mcps TDD option

The error of the UE maximum output power shall not exceed the tolerance shown in tables 5.2.2 1.a and b for single and multi-code for 7.68Mcps TDD option.

**Table 5.2.2.1.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +30 dBm | +1dB/-3dB |
| 2 | +24 dBm | +1dB/-3dB |
| 3 | +21 dBm | +2dB/-2dB |
| 4 | +10 dBm | +4dB/-4dB |

**Table 5.2.2.1.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +27 dBm (note) | +1dB/-3dB |
| 2 | 21 dBm (note) | +1dB/-3dB |
| 3 | 18 dBm (note) | +2dB/-2dB |
| 4 | +7 dBm (note) | +4dB/-4dB |
| NOTE: These figures are not mentioned in 25.102. Instead there is a note, saying: "For multi-code operation the maximum output power will be reduced by the difference of peak to average ratio between single and multi-code transmission." The figures are calculated from maximum output power single code (table 5.2.2.1.a) and UL multicode reference measurement channel (12,2 kbit/s) (annex C.2.2.3 for the 7.68 TDD Option) containing two code signals with equal level. | | |

The normative reference for this requirement is TS 25.102 [1] clause 6.2.1.

## 5.2.3 Test purpose

For the following reasons:

Limit interference.

Verify that the maximum output power is achievable.

It is the purpose of the test to verify that the UE's maximum output power is within its tolerance limits under all environmental conditions.

## 5.2.4 Method of test

### 5.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) Calls are set up according to the Generic call setup procedure using parameters as specified in tables 5.2.4.1.1a and b for the 3,84 Mcps TDD Option and in tables 5.2.4.1.2a and b for the 1,28 Mcps TDD Option, respectively.
3) Enter the UE into loopback test mode and start the loopback test.

#### 5.2.4.1.1 3,84 Mcps TDD Option

**Table 5.2.4.1.1a: Test parameters for Maximum Output Power single code (3,84 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| UL Reference measurement channel | 12,2 kbps, according to annex C.2.1.1 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power. |
| Data content | real life (sufficient irregular) |

**Table 5.2.4.1.1b: Test parameters for Maximum Output Power multicode (3,84 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2 kbps, according to annex C.2.2.1 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

#### 5.2.4.1.2 1,28 Mcps TDD Option

**Table 5.2.4.1.2a: Test parameters for Maximum Output Power single code (1,28 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| UL Reference measurement channel | 12,2 kbps, according to annex C.2.1.2. |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power. |
| Data content | real life (sufficient irregular) |

**Table 5.2.4.1.2b: Test parameters for Maximum Output Power multicode (1,28 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2 kbps, according to annex C.2.2.2 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

#### 5.2.4.1.3 7,68 Mcps TDD Option

**Table 5.2.4.1.3a: Test parameters for Maximum Output Power single code (7,68 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| UL Reference measurement channel | 12,2 kbps, according to annex C.2.1.3 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power. |
| Data content | real life (sufficient irregular) |

**Table 5.2.4.1.3b: Test parameters for Maximum Output Power multicode (7,68 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2 kbps, according to annex C.2.2.3 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

### 5.2.4.2 Procedure

1) Measure the mean power of the UE output signal.

2) Run step 1) for RF channels Low / Mid / High.

## 5.2.5 Test Requirements

5.2.5.1 3,84 Mcps TDD OptionThe output power, measured in step 2) of clause 5.2.4.2, shall not exceed the prescribed tolerance in table 5.2.5 a and b.

**Table 5.2.5.1.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +30 dBm | +1,7 dB / -3,7 dB |
| 2 | +24 dBm | +1,7 dB / -3,7dB |
| 3 | +21 dBm | +2,7 dB / -2,7dB |
| 4 | +10 dBm | +4,7 dB /-4,7dB |

**Table 5.2.5.1.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | 27 dBm | +1,7 dB / -3,7 dB |
| 2 | 21 dBm | +1,7dB / -3,7 dB |
| 3 | 18 dBm | +2,7dB / -2,7 dB |
| 4 | 7 dBm | +4,7 dB / -4,7 dB |

NOTE 1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

NOTE 2: Concerning multicode transmission this test applies only for UE power classes 2 and 3. It is intended, that additional test requirements for UE power classes 1 and 4 in this case are part of a later version of the present document.

### 5.2.5.2 1.28 Mcps TDD Option

The output power, measured in step 2) of clause 5.2.4.2, shall not exceed the prescribed tolerance in table 5.2.5.2 a and b.

**Table 5.2.5.2.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +33dBm | +1,7 dB / -3,7 dB |
| 2 | +24 dBm | +1,7 dB / -3,7dB |
| 3 | +21 dBm | +2,7 dB / -2,7dB |
| 4 | +27 dBm | +1,7 dB / -3,7 dB |

**Table 5.2.5.2.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | 30 dBm | +1,7 dB / -3,7 dB |
| 2 | 21 dBm | +1,7dB / -3,7 dB |
| 3 | 18 dBm | +2,7dB / -2,7 dB |
| 4 | 24 dBm | +1,7 dB / -3,7 dB |

NOTE 1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

NOTE 2: Concerning multicode transmission this test applies only for UE power classes 2 and 3. It is intended, that additional test requirements for UE power classes 1 and 4 in this case are part of a later version of the present document.

### 5.2.5.3 7.68 Mcps TDD Option

The output power, measured in step 2) of clause 5.2.4.2, shall not exceed the prescribed tolerance in table 5.2.5.3 a and b.

**Table 5.2.5.3.a: Maximum Output Power single code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | +33dBm | +1,7 dB / -3,7 dB |
| 2 | +24 dBm | +1,7 dB / -3,7dB |
| 3 | +21 dBm | +2,7 dB / -2,7dB |
| 4 | +27 dBm | +1,7 dB / -3,7 dB |

**Table 5.2.5.3.b: Maximum Output Power multi code**

| Power Class | Nominal maximum output power | Tolerance |
|---|---|---|
| 1 | 30 dBm | +1,7 dB / -3,7 dB |
| 2 | 21 dBm | +1,7dB / -3,7 dB |
| 3 | 18 dBm | +2,7dB / -2,7 dB |
| 4 | 24 dBm | +1,7 dB / -3,7 dB |

NOTE 1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

NOTE 2: Concerning multicode transmission this test applies only for UE power classes 2 and 3. It is intended, that additional test requirements for UE power classes 1 and 4 in this case are part of a later version of the present document.

# 5.3 UE frequency stability

## 5.3.1 Definition and applicability

The frequency stability is the difference of the modulated carrier frequency between the RF transmission from the UE and the RF transmission from the BS. The UE shall use the same frequency source for both RF frequency generation and chip clocking.

The requirements of this test apply to all types of UTRA- UE.

## 5.3.2 Minimum Requirements

The UE frequency stability, observed over a period of one timeslot, shall be within ±0,1 ppm compared to signals received from the BS.

The normative reference for this requirement is TS 25.102 [1] clause 6.3.

## 5.3.3 Test purpose

Reliable frequency stability of the UE's transmitter in certain tolerance limits is prerequisite for connectivity.

This test stresses the ability of the UE's receiver to derive correct frequency information from the received signal for the transmitter.

## 5.3.4 Method of test

### 5.3.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH, vibration; see clauses G.2.1, G.2.2 and G.2.3.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

#### 5.3.4.1.1 3,84 Mcps TDD Option

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the Generic call setup procedure using parameters as specified in table 5.3.4.1.1.
3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.3.4.1.1: Test parameters for Frequency Stability (3,84 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| SS level (Ior) | –105 dBm / 3,84 MHz (reference sensitivity) |
| UL reference measurement channel | 12,2 kbps according to annex C.2.1.1. |
| Data content | real life (sufficient irregular) |

#### 5.3.4.1.2 1,28 Mcps TDD Option

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the Generic call setup procedure using parameters as specified in table 5.3.4.1.2.
3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.3.4.1.2: Test parameters for Frequency Stability (1,28 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| SS level (Ior) | –108 dBm / 1,28 MHz (reference sensitivity) |
| UL reference measurement channel | 12,2 kbps according to annex C.2.1.2 |
| Data content | real life (sufficient irregular) |

#### 5.3.4.1.3 7,68 Mcps TDD Option

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the Generic call setup procedure using parameters as specified in table 5.3.4.1.3.
3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.3.4.1.3: Test parameters for Frequency Stability (7,68 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| SS level (Ior) | –102 dBm / 7,68 MHz (reference sensitivity) |
| UL reference measurement channel | 12,2 kbps according to annex C.2.1.3. |
| Data content | real life (sufficient irregular) |

### 5.3.4.2 Procedure

1) Measure the frequency error delta f across the TS according to annex B.
2) Repeat step 1) for 200 bursts (time slots).
3) Run Step 1) and 2) for RF channels Low /Mid/ High.

### 5.3.5 Test Requirements

For all measured bursts (time slots), the frequency error, derived in clause 5.3.4.2, shall not exceed ±(0,1 ppm + 10 Hz).

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4

## 5.4 Output Power Dynamics

Power control is used to limit the interference level.

### 5.4.1 Uplink power control

Uplink power control is the ability of the UE transmitter to sets its output power in accordance with measured downlink path loss, values determined by higher layer signalling and path loss weighting parameter $\alpha$ as defined in TS 25.331 [9]. The output power is defined as the RRC filtered mean power of the transmit timeslot.

#### 5.4.1.1 Initial accuracy (3,84 Mcps TDD Option)

##### 5.4.1.1.1 Definition and applicability

Initial Uplink power control is the ability of the UE transmitter to sets its output power in accordance with measured downlink path loss, and signalling values: $I_{BTS}$ and Constant value, received from the BCH and applicable for the PRACH.

The requirements and this test apply to all types of UTRA - UEs.

##### 5.4.1.1.2 Minimum requirements

The UE power control, initial accuracy, is given in table 5.4.1.1.2.

**Table 5.4.1.1.2: Initial uplink power control tolerance (3,84 Mcps TDD Option)**

| | |
|---|---|
| Normal conditions | ±9 dB |
| Extreme conditions | ±12 dB |

The reference for this requirement is TS 25.102 [1] clause 6.4.1.1.1.

##### 5.4.1.1.3 Test purpose

The power of the received signal at the UE and the BCH information control the power of the transmitted UE signal with the target to transmit at lowest power, acceptable for proper communication.

The test stresses the ability of the receiver to measure the received power over the receiver dynamic range and to derive from this correct transmitter-power.

##### 5.4.1.1.4 Method of test

###### 5.4.1.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

Connect the SS to the MS antenna connector as shown in figure A.1.

A call is set up according to the generic call setup procedure [3] using parameters as specified in table 5.4.1.1.4. The RACH procedure within the call setup is used for the test.

**Table 5.4.1.1.4: Test parameters for uplink Power Control (3,84 Mcps TDD Option)**

| | RX-Upper dynamic end | RX-middle | RX-Sensitivity level |
|---|---|---|---|
| SS transmit power | -25 dBm/3,84 MHz | -65 dBm/3,84 MHz | -105 dBm/3,84 MHz |
| Broadcasted transmit-power PCCPCH | 35 dBm | 35 dBm | 24 dBm |
| Simulated path loss = Broadcasted TX – SS TX Power | 60 dB | 100 dB | 129 dB |
| $I_{BTS\ (UL\ interference)}$ | -75 dBm | -100 dBm | -110 dBm |
| Constant value | -10 dB | -10 dB | -10 dB |
| Nominal expected UE TX power | -25 dBm | -10 dBm | +9 dBm (note 2) |
| NOTE 1: While the SS transmit power shall cover the UE receiver input dynamic range, the logical parameters: broadcasted transmit power, $I_{BTS}$, and RACH constant value are chosen to achieve a UE TX power, located within the TX output power dynamic range of a class 3 UE. | | | |
| NOTE 2: Nominal TX output power 9 dBm allows to check the uplink power control algorithm within the entire tolerance range (9 dBm +-12 dB: 9 dBm +12 dB =21 dBm = max power class 3). | | | |

#### 5.4.1.1.4.2 Procedure

1) Set the SS transmit power according to table 5.4.1.1.4.

2) Measure the RACH output power of the UE according to annex B.

3) Repeat the test for all SS transmit powers and parameters in table 5.4.1.1.4.

### 5.4.1.1.5 Test requirements

The deviation with respect to the nominal expected UE TX power (table 5.4.1.1.2), derived in step 2, shall not exceed the prescribed tolerance in table 5.4.1.1.5.

**Table 5.4.1.1.5: Test parameters for uplink Power Control**

| | | | |
|---|---|---|---|
| Expected UE TX power, normal conditions | -25 dBm ±10 dB | -10 dBm±10 dB | +9 dBm ±10 dB |
| Expected UE TX power, extreme conditions | -25 dBm ±13 dB | -10 dBm±13 dB | +9 dBm ±13 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.4.1.2 Differential accuracy, controlled input (3,84 Mcps TDD Option)

### 5.4.1.2.1 Definition and applicability

Uplink power control, differential accuracy, is the ability of the UE transmitter to sets its output power in accordance with measured downlink path loss, and the signalling values: $I_{BTS}$, $SIR_{Target}$, Constant Value, received from higher layers and applicable for the DPCH.

Specifically, the uplink power control, differential accuracy, controlled input, is defined as the error in the UE transmitter power step as a result of a step in $SIR_{TARGET}$, $I_{BTS}$ or DPCH Constant Value when the path loss weighting parameter $\alpha$=0.

The requirements of this test apply to all types of UTRA -UE.

#### 5.4.1.2.2 Minimum requirements

The step in $SIR_{TARGET}$ shall be rounded to the closest integer dB value. The power control error resulting from a change in $SIR_{TARGET}$, $I_{BTS}$ or DPCH Constant Value shall not exceed the values in table 5.4.1.2.2.

**Table 5.4.1.2.2: Transmitter power step tolerance as a result of control power step (3,84 Mcps TDD Option)**

| $\Delta SIR_{TARGET}$ [dB] | Transmitter power step tolerance [dB] |
|---|---|
| $\Delta SIR_{TARGET} \leq 1$ | ± 0,5 |
| $1 < \Delta SIR_{TARGET} \leq 2$ | ± 1 |
| $2 < \Delta SIR_{TARGET} \leq 3$ | ± 1,5 |
| $3 < \Delta SIR_{TARGET} \leq 10$ | ± 2 |
| $10 < \Delta SIR_{TARGET} \leq 20$ | ± 4 |
| $20 < \Delta SIR_{TARGET} \leq 30$ | ± 6 |
| $30 < \Delta SIR_{TARGET}$ | ± 9 (note) |
| NOTE: Value is given for normal conditions. For extreme conditions value is ±12. | |

The reference for this requirement is TS 25.102 [1] clause 6.4.1.1.2.

#### 5.4.1.2.3 Test purpose

It is verified if the UE sets correct uplink power steps in response to steps in the signalling value $SIR_{Target}$ and DPCH Constant Value, signalled via the downlink to the UE,under the following conditions: keeping the other signalling parameters constant and deactivating any influence due to varying pathloss.

#### 5.4.1.2.4 Method of test

##### 5.4.1.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS to the MS antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table 5.4.1.2.4.

**Table 5.4.1.2.4 : Test parameters for Uplink Power Control, Differential Accuracy, Controlled Input (3,84 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| UL reference measurement channel | 12,2 kbps according to annex C clause C.2.1.1 |
| BS Transmit to UE Transmit delay | 7 TSs--> $\alpha$=0 |
| SSTransmit power | -65 dBm |
| Reference transmit power broadcast on BCH | 35 dBm |
| $I_{BTS}$ | -100 |
| Constant value | -10 |
| Data content | real life (sufficient irregular) |

##### 5.4.1.2.4.2 Procedure

Using a combination of $SIR_{Target}$ and DPCH constant value signaled in the downlink,cover the UE-transmitter dynamic range by commanding the UEs power with the signalling value $SIR_{Target}$ in a step resolution (positive and negative direction) of:

1 dB approx. 68 steps up and 68 steps down

2 dB approx. 34 steps up and 34 steps down

3 dB approx. 22 steps up and 22 steps down

10 dB approx. 7 steps up and 7 steps down

20 dB approx. 3 steps up and 3 steps down

30 dB approx. 2 step up and 2 step down

maximum stepsize 1 step up and 1 step down

Measure the power according to annex B.

#### 5.4.1.2.5 Test requirements

For the UE output power laying between

Max Power minus tolerance and Min Power

the step response shall not exceed the prescribed tolerance in table 5.4.1.2.5.

**Table 5.4.1.2.5: Transmitter power step tolerance as a result of control power step**

| $\Delta SIR_{TARGET}$ [dB] | Transmitter power step tolerance [dB] |
|---|---|
| $\Delta SIR_{TARGET} \leq 1$ | ± 0,6 |
| $1 < \Delta SIR_{TARGET} \leq 2$ | ± 1,15 |
| $2 < \Delta SIR_{TARGET} \leq 3$ | ± 1,7 |
| $3 < \Delta SIR_{TARGET} \leq 10$ | ± 2,5 |
| $10 < \Delta SIR_{TARGET} \leq 20$ | ± 4,7 |
| $20 < \Delta SIR_{TARGET} \leq 30$ | ± 6,7 |
| $30 < \Delta SIR_{TARGET}$ | ± 10 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 5.4.1.2A Differential accuracy, controlled input

This is not tested.

### 5.4.1.3 Open loop power control (1,28 Mcps TDD Option)

#### 5.4.1.3.1 Definition and applicability

Open loop power control is the ability of the UE transmitter to sets its output power to a specific value. The open loop power control tolerance is given in table 5.4.1.3.2.

#### 5.4.1.3.2 Minimum requirements

The UE open loop power is defined as the average power in a timeslot or ON power duration, whichever is available, and they are measured with a filter that has a Root-Raised Cosine (RRC) filter response with a roll off $\alpha = 0.22$ and a bandwidth equal to the chip rate of 1,28 Mcps. The open loop power control tolerance is given in table 5.4.1.3.2.

**Table 5.4.1.3.2: Open loop power control tolerance (1,28 Mcps TDD Option)**

| | |
|---|---|
| Normal conditions | ±9 dB |
| Extreme conditions | ±12 dB |

The reference for this requirement is TS 25.102 [1] clause 6.4.1.2.1.1.

### 5.4.1.3.3 Test purpose

The power of the received signal and the BCCH information control the power of the transmitted signal with the target to transmit at lowest power acceptable for proper communication.

The test stresses the ability of the receiver to measure the received power correctly over the receiver dynamic range.

The test purpose is to verify that the UE open loop power control tolerance does not exceed the described value shown in table 5.4.1.3.2.

An excess error of the open loop power control decreases the system capacity.

### 5.4.1.3.4 Method of test

#### 5.4.1.3.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the Generic call setup procedure, and RF parameters are set up according to table 5.4.1.3.4a. The transmit power level by a UE on the UpPTS is measured. The network signals on BCH a power increment that is applied only for the access procedure. At each new transmission of a UpPTS burst during the access procedure, the transmit power level can be increased by this power increment. According to the test purpose this power increment is set to zero.

See TS 34.108 [3] for details regarding generic call setup procedure.

**Table 5.4.1.3.4a: Test parameters for Open Loop Power Control (UE) (1,28 Mcps TDD Option)**

| Parameter | Level / Status | Unit |
|---|---|---|
| $\hat{I}_{or}$ | See table 5.4.1.3.4b | dBm / 1,28 MHz |

**Table 5.4.1.3.4b: Test parameters for Open Loop Power Control (SS) (1,28 Mcps TDD Option) [1)]**

| Parameter | RX Upper dynamic end | RX-middle | RX-Sensitivity level |
|---|---|---|---|
| SS transmit power $\hat{I}_{or}$ | –25 dBm / 1,28 MHz | – 66 dBm / 1,28 MHz | – 108 dBm / 1,28 MHz [3)] |
| broadcasted Primary CCPCH transmit power on BCH | +35 dBm | +24 dBm | +11 dBm |
| Simulated path loss = broadcasted TX – SS TX power | +60 dB | +90 dB | +119 dB |
| $PRX_{UpPCHdes}$ | –85 dBm | –100 dBm | –110 dBm |
| $Pwr_{ramp}$ (Power Ramping Step) | 0 dB | 0 dB | 0 dB |
| I (Max SYNC_UL Transmissions) | 1 | 1 | 1 |
| Expected nominal UE TX power [5)] | -25 dBm | -10 dBm | +9 dBm [2)] |

NOTE 1: While the SS transmit power shall cover the receiver input dynamic range, the logical parameters: broadcasted TX power, Desired UpPCH RX power at the BS receiver and Power Ramp step are chosen to achieve a UE TX power, located within the TX output power dynamic range of a class 3 UE.

NOTE 2: Nominal TX output power 9 dBm allows to check the open loop power algorithm within the entire tolerance range (9 dBm ± 12 dB; 9 dBm + 12 dB = 21 dBm = max power class 3)

NOTE 3: The reference for this requirement is TS 25.102 [1] clause 7.3.1.2.

NOTE 4: This test applies only for max UE power classes 3. It is intended, that additional test requirements for UE power class 4 are part of a later release.

NOTE 5: The Expected nominal UE TX power is calculated by using the equation in the clause 8.5.7 Open Loop Power Control of [9]

#### 5.4.1.3.4.2 Procedure

1) Set the TX output level of the SS to obtain $\hat{I}_{or}$ at the UE antenna connector. $\hat{I}_{or}$ shall be according to table 5.4.1.3.4b (–25 dBm / 1,28 MHz).

2) Measure the UpPCH TX mean power of UE.

3) Repeat the above measurement for all SS levels in table 5.4.1.3.4b.

### 5.4.1.3.5 Test requirements

The measured UE TX power in step 2), shall not exceed the prescribed tolerance given in table 5.4.1.3.5.

**Table 5.4.1.3.5: Test parameters for open loop power control (1,28 Mcps TDD Option)**

| | | | |
|---|---|---|---|
| Expected UE TX power, normal conditions | -25 dBm ±10 dB | -10 dBm±10 dB | +9 dBm ±10 dB |
| Expected UE TX power, extreme conditions | -25 dBm ±13 dB | -10 dBm±13 dB | +9 dBm ±13 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.4.1.4 Closed loop power control (1,28 Mcps TDD Option)

### 5.4.1.4.1 Definition and applicability

Closed loop power control in the Uplink is the ability of the UE transmitter to adjust its output power in accordance with one or more TPC commands received in the downlink.

The power control step is the change in the UE transmitter output power in response to a single TPC command, TPC_cmd, arrived at the UE.

### 5.4.1.4.2 Minimum requirements

The UE transmitter shall have the capability of changing the output power with a step size of 1, 2 and 3 dB according to the value of $\Delta_{TPC}$ in the slot immediately after the TPC_cmd can be arrived.

a) The transmitter output power step due to closed loop power control shall be within the range shown in table 5.4.1.4.2a.

b) The transmitter average output power step due to closed loop power control shall be within the range shown in table 5.4.1.4.2b. Here a TPC_cmd group is a set of TPC_cmd values derived from a corresponding sequence of TPC commands of the same duration.

The closed loop power is defined as the relative power differences between RRC filtered mean power of original (reference) timeslot and RRC filtered mean power of the target timeslot without transient duration.

**Table 5.4.1.4.2a: Transmitter power control range**

| TPC_cmd | Transmitter power control range | | | | | |
|---|---|---|---|---|---|---|
| | 1 dB step size | | 2 dB step size | | 3 dB step size | |
| | Lower | Upper | Lower | Upper | Lower | Upper |
| Up | +0,5 dB | +1,5 dB | +1 dB | +3 dB | +1,5 dB | +4,5 dB |
| Down | -0,5 dB | -1,5 dB | -1 dB | -3 dB | -1,5 dB | -4,5 dB |

**Table 5.4.1.4.2b: Transmitter average power control range**

| TPC_cmd group | Transmitter power control range after 10 equal TPC_ cmd groups | | | | | |
|---|---|---|---|---|---|---|
| | 1 dB step size | | 2 dB step size | | 3 dB step size | |
| | Lower | Upper | Lower | Upper | Lower | Upper |
| Up | +8 dB | +12 dB | +16 dB | +24 dB | +24 dB | +36 dB |
| Down | -8 dB | -12 dB | -16 dB | -24 dB | -24 dB | -36 dB |

### 5.4.1.4.3 Test purpose

The purpose of this test is

- to verify that the UE inner loop power control size and response is meet to the described value shown in clause 5.4.1.4.2; and
- to verify that the TPC_cmd is correctly derived from received TPC commands.

### 5.4.1.4.4 Method of test

#### 5.4.1.4.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the Generic call setup procedure.
3) Enter the UE into loopback test mode and start the loopback test.

See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure and loopback test.

#### 5.4.1.4.4.2 Procedure

1) Before proceeding with paragraph (2) (Step A) below, set the output power of the UE, measured at the UE antenna connector, to be in the range –10± 9dBm. This may be achieved by setting the downlink signal ($\hat{I}_{or}$) to yield an appropriate open loop output power and/or by generating suitable downlink TPC commands from the SS.
2) Step A: Configure the uplink channel to set the TPC step size to 1 dB. When the Configuration is complete, transmit a sequence of TPC commands with the value 1 until the UE output power is above the maximum power threshold.
3) Step B: Transmit a sequence of 68 (note) TPC commands with the value 0.
4) Step C: Transmit a sequence of 68 (note) TPC commands with the value 1.

5) Step D: Reconfigure the uplink channel to set the TPC step size to 2dB. When the reconfiguration is complete, transmit a sequence of TPC commands with the value 1 until the UE output power is above the maximum power threshold. Transmit a sequence of 34 (note) TPC commands with the value 0.

6) Step E: Transmit a sequence of 34 (note) TPC commands with the value 1.

7) Step F: Reconfigure the uplink channel to set the TPC step size to 3 dB. When the reconfiguration is complete, transmit a sequence of TPC commands with the value 1 until the UE output power is above the maximum power threshold. Transmit a sequence of 22 (note) TPC commands with the value 0.

8) Step G: Transmit a sequence of 22 (note) TPC commands with the value 1.

NOTE: These numbers of TPC commands are given as examples. The actual number of TPC commands transmitted in these steps shall be sufficient to ensure that the UE reaches the relevant maximum or minimum power threshold.

#### 5.4.1.4.5 Test requirements

a) During Step B, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of −1 and step size of 1 dB, until the output power reaches (Minimum power threshold +0,5 dB).

b) During Step B, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of −1, and step size of 1 dB as given in table 5.4.1.4.2b, until the output power reaches (Minimum power threshold +0,5 dB).

c) During Step C, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of +1 and step size of 1 dB, until the output power reaches (Maximum power threshold −0,5 dB).

d) During Step C, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of +1, and step size of 1 dB as given in table 5.4.1.4.2b, until the output power reaches (Maximum power threshold −0,5 dB).

e) During Step D, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of −1 and step size of 2 dB, until the output power reaches (Minimum power threshold +1 dB).

f) During Step D, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of −1, and step size of 2 dB as given in table 5.4.1.4.2b, until the output power reaches (Minimum power threshold +1 dB).

g) During Step E, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of +1 and step size of 2 dB, until the output power reaches (Maximum power threshold −1 dB).

h) During Step E, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of +1, and step size of 2 dB as given in table 5.4.1.4.2b, until the output power reaches (Maximum power threshold −1 dB).

i) During Step F, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of −1 and step size of 3 dB, until the output power reaches (Minimum power threshold +1 dB).

j) During Step F, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of −1, and step size of 3 dB as given in table 5.4.1.4.2b, until the output power reaches (Minimum power threshold +1 dB).

k) During Step G, the difference in mean output power between adjacent slots shall be within the prescribed range given in table 5.4.1.4.2a for a TPC_cmd of +1 and step size of 3 dB, until the output power reaches (Maximum power threshold −1 dB).

l) During Step G, the change in mean output power over 10 consecutive slots shall be within the prescribed range for a TPC_cmd group of +1, and step size of 3 dB as given in table 5.4.1.4.2b, until the output power reaches (Maximum power threshold −1 dB).

## 5.4.1.5 Initial accuracy (7,68 Mcps TDD Option)

### 5.4.1.5.1 Definition and applicability

Initial Uplink power control is the ability of the UE transmitter to set its output power in accordance with measured downlink path loss, and signalling values: $I_{BTS}$ and Constant value, received from the BCH and applicable for the PRACH.

The requirements and this test apply to all types of UTRA - UEs.

### 5.4.1.5.2 Minimum requirements

The UE power control, initial accuracy, is given in table 5.4.1.5.2.

**Table 5.4.1.5.2: Initial uplink power control tolerance (7,68 Mcps TDD Option)**

| | |
|---|---|
| Normal conditions | ±9 dB |
| Extreme conditions | ±12 dB |

The reference for this requirement is TS 25.102 [1] clause 6.4.1.3.1.

### 5.4.1.5.3 Test purpose

The power of the received signal at the UE and the BCH information control the power of the transmitted UE signal with the target to transmit at lowest power, acceptable for proper communication.

The test stresses the ability of the receiver to measure the received power over the receiver dynamic range and to derive from this correct transmitter-power.

### 5.4.1.5.4 Method of test

#### 5.4.1.5.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

Connect the SS to the MS antenna connector as shown in figure A.1.

A call is set up according to the generic call setup procedure [3] using parameters as specified in table 5.4.1.5.4. The RACH procedure within the call setup is used for the test.

**Table 5.4.1.5.4: Test parameters for uplink Power Control (7,68 Mcps TDD Option)**

| | RX-Upper dynamic end | RX-middle | RX-Sensitivity level |
|---|---|---|---|
| SS transmit power | -25 dBm/7,68 MHz | -65 dBm/7,68 MHz | -102 dBm/7,68 MHz |
| Broadcasted transmit-power PCCPCH | 35 dBm | 35 dBm | 24 dBm |
| Simulated path loss = Broadcasted TX – SS TX Power | 60 dB | 100 dB | 126 dB |
| $I_{BTS}$ (UL interference) | -75 dBm | -100 dBm | -107 dBm |
| Constant value | -10 dB | -10 dB | -10 dB |
| Nominal expected UE TX power | -25 dBm | -10 dBm | +9 dBm (note 2) |
| NOTE 1: While the SS transmit power shall cover the UE receiver input dynamic range, the logical parameters: broadcasted transmit power, $I_{BTS}$, and RACH constant value are chosen to achieve a UE TX power, located within the TX output power dynamic range of a class 3 UE.<br>NOTE 2: Nominal TX output power 9 dBm allows to check the uplink power control algorithm within the entire tolerance range (9 dBm +-12 dB: 9 dBm +12 dB =21 dBm = max power class 3). | | | |

#### 5.4.1.5.4.2 Procedure

1) Set the SS transmit power according to table 5.4.1.1.4.

2) Measure the RACH output power of the UE according to annex B.

3) Repeat the test for all SS transmit powers and parameters in table 5.4.1.5.4.

### 5.4.1.5.5 Test requirements

The deviation with respect to the nominal expected UE TX power (table 5.4.1.5.2), derived in step 2, shall not exceed the prescribed tolerance in table 5.4.1.5.5.

**Table 5.4.1.5.5: Test parameters for uplink Power Control**

| | | | |
|---|---|---|---|
| Expected UE TX power, normal conditions | -25 dBm ±10 dB | -10 dBm±10 dB | +9 dBm ±10 dB |
| Expected UE TX power, extreme conditions | -25 dBm ±13 dB | -10 dBm±13 dB | +9 dBm ±13 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.4.2 Minimum output power

### 5.4.2.1 Definition and applicability

The minimum controlled output power of the UE is when the power is set to a minimum value. The minimum output power is defined as the mean power in one time slot excluding the guard period.

The normative requirements of this test apply to all types of UTRA- UE.

### 5.4.2.2 Minimum Requirements

#### 5.4.2.2.1 3,84Mcps TDD Option

The minimum output power shall be lower than or equal to –44 dBm.

The normative reference for this requirement is TS 25.102 [1] clause 6.4.2.1.1.

#### 5.4.2.2.2 1,28Mcps TDD Option

The minimum output power shall be better than–49 dBm.

The normative reference for this requirement is TS 25.102 [1] clause 6.4.2.1.2.

#### 5.4.2.2.3 7,68Mcps TDD Option

The minimum output power shall be lower than or equal to –41 dBm.

The normative reference for this requirement is TS 25.102 [1] clause 6.4.2.1.3.

### 5.4.2.3 Test purpose

The test purpose is to verify the ability of the UE to reduce its output power to a specified value.

### 5.4.2.4 Method of test

#### 5.4.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the Generic call setup procedure using parameters as specified in table E.3.1.2.
3) Enter the UE into loopback test mode and start the loopback test.

#### 5.4.2.4.2 Procedure

1) Configure the UE transmitter to enable power control steps of size 1 dB.
2) Set and send Down power control commands to the UE. The sequence shall be sufficiently long so that the UE output signal reached its minimum power.
2) Measure the mean power of the UE output signal according to annex B.

NOTE: Annex B returns the power in the decision points (displayed as reference power and power offset). This is equivalent to thermal power at the air-interface. Insofar 5.4.2.2.1 minimum output power for 3,84 Mcps TDD Option and 5.4.2.2.2 minimum output power for 1,28 Mcps TDD Option is consistent with 5.2 maximum output power.

3) Configure the UE transmitter to enable power control steps of 2 dB and of 3 dB, respectively, and repeat step 2).
4) Run step 2) for RF channels Low Mid and High.

### 5.4.2.5 Test requirements

#### 5.4.2.5.1 3,84 Mcps TDD Option

For all measurements, the minimum output power derived in step 3) and 4) of 5.4.2.4.2 shall be below –43 dBm.

#### 5.4.2.5.2 1,28 Mcps TDD Option

For all measurements, the minimum output power derived in step 3) and 4) of 5.4.2.4.2 shall be below -48 dBm.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

#### 5.4.2.5.3 7,68 Mcps TDD Option

For all measurements, the minimum output power derived in step 3) and 4) of 5.4.2.4.2 shall be below -40 dBm.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.4.3 Transmit OFF power

### 5.4.3.1 Definition and applicability

Transmit OFF power is defined as the RRC filtered mean power measured over one chip when the transmitter is off. The transmit OFF power state is when the UE does not transmit.

The requirements of this test apply to all types of UTRA-UE.

### 5.4.3.2 Minimum Requirements

The transmit OFF power shall be below −65 dBm.

The normative reference for this requirement is TS 25.102 clause 6.5.1.

### 5.4.3.3 Test purpose

Refer clause 5.4.4.3.

### 5.4.3.4 Method of test

Refer clause 5.4.4.4.

### 5.4.3.5 Test requirements

The transmit OFF power shall be below −63.5 dBm.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4

## 5.4.4 Transmit ON/OFF Time mask

### 5.4.4.1 Definition and applicability

The transmit ON/OFF time mask defines the ramping time allowed for the UE between transmit OFF power and transmit ON power.

This test applies for all UTRA TTD UEs.

### 5.4.4.2 Minimum requirements

#### 5.4.4.2.1 3,84Mcps TDD Option

The transmit power level versus time shall meet the mask specified in figure 5.4.4.2, where the transmission period refers to the burst without guard-period for a single transmission slot, and to the period from the beginning of the burst in the first transmission slot to the end of the burst without guard period in the last transmission timeslot for consecutive transmission slots.

The reference for this requirement is TS 25.102 [1] clause 6.5.2.1.1.

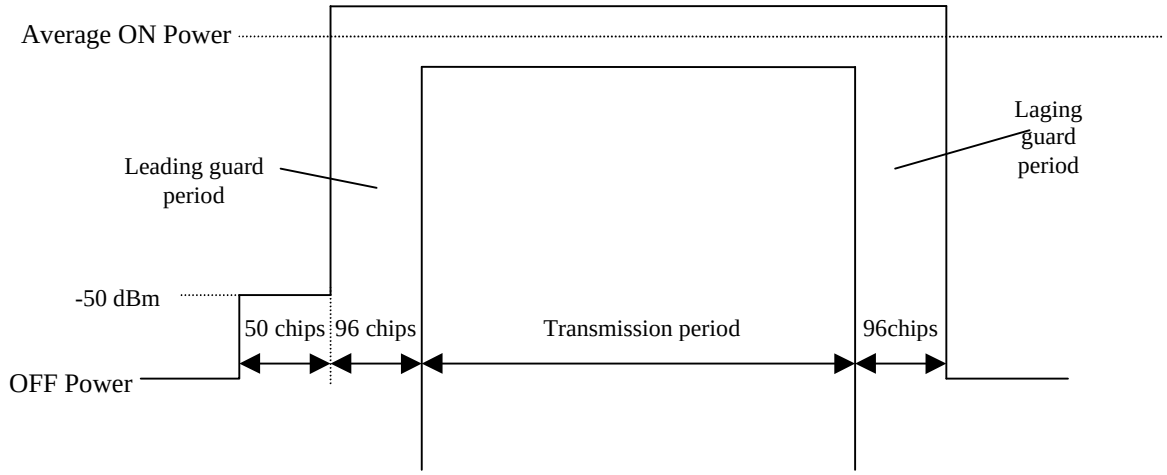



**Figure 5.4.4.2.1: Transmit ON/OFF template for 3,84 Mcps TDD Option**

### 5.4.4.2.2 1,28Mcps TDD Option

The transmit power level versus time shall meet the mask specified in figure 5.4.4.2.2, where the transmission period refers to the burst without guardperiod for a single transmission slot, and to the period from the beginning of the burst in the first transmission slot to the end of the burst without guard period in the last transmission timeslot for consecutive transmission slots.

The reference for this requirement is TS 25.102 [1] clause 6.5.2.1.2.

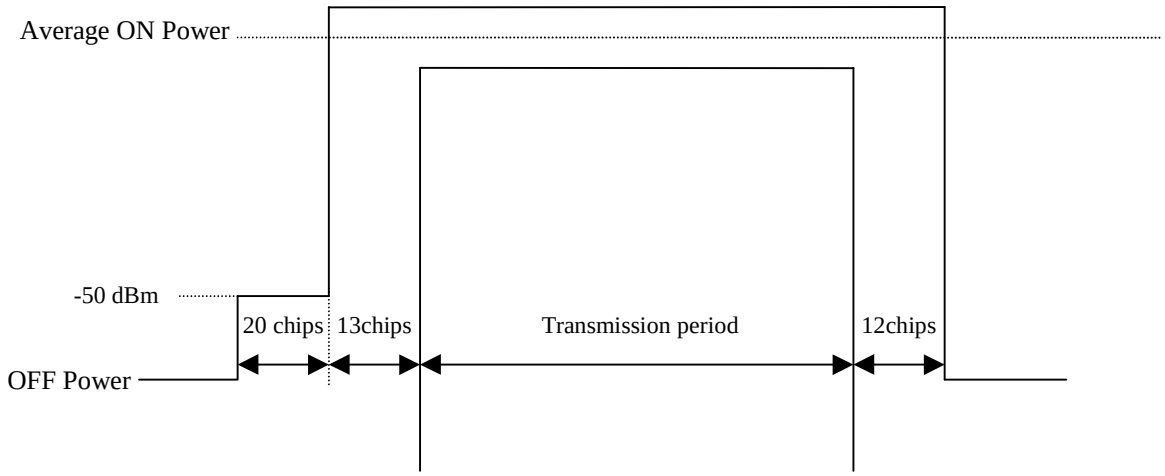



**Figure 5.4.4.2.2: Transmit ON/OFF template for 1,28Mcps TDD Option**

### 5.4.4.2.3 7,68Mcps TDD Option

The transmit power level versus time shall meet the mask specified in figure 5.4.4.2.3, where the transmission period refers to the burst without guard-period for a single transmission slot, and to the period from the beginning of the burst in the first transmission slot to the end of the burst without guard period in the last transmission timeslot for consecutive transmission slots.

The reference for this requirement is TS 25.102 [1] clause 6.5.2.1.3.

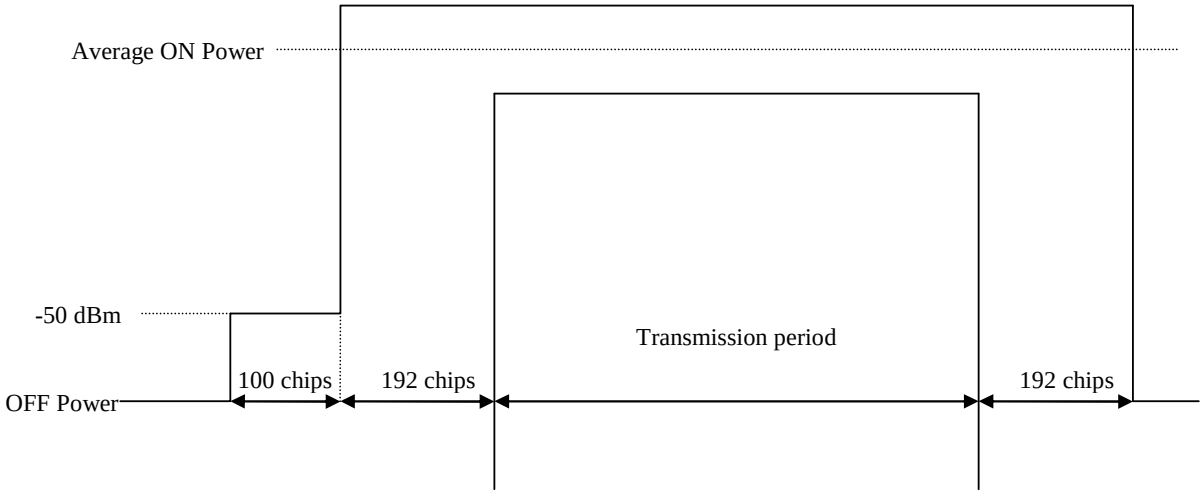



Figure 5.4.4.2.3: Transmit ON/OFF template for 7,68 Mcps TDD Option

### 5.4.4.3 Test Purpose

It is tested if the UE TX signal uses the guard period for on-to-off and off-to-on transitions, where the time position of guard period is derived from the burst under test itself.

It is furtheron tested, if the UE TX signal is below certain limits outside transmission period and guard periods where the position in time is derived from the burst under test itself.

With this test interference to other UTRA TDD users are limited

### 5.4.4.4 Method of test

#### 5.4.4.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

Connect the SS to the UE antenna connector as shown in figure A.1.

A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.

Enter the UE into loopback test mode and start the loopback test.

#### 5.4.4.4.2 Procedure

1) The time position of the midamble of the burst under test (TimeSlot s in Frame f) shall be the reference for the time position of the leading and laging guard-periods of the burst under test and, alternatively, for the equivalent guard periods of the next 2 bursts.

2) Record the following time periods with at least 2 samples /chip through a matched filter (RRC 0.22, BW equal to the chiprate) : TS s-1 and TS s+1 in frame f or f+1 or f+2

3) Calculate power samples by averaging the recorded samples of one chip duration.

### 5.4.4.5 Test requirements

Each power sample shall be below the limits (off Power (clause 5.4.3) and –50 dBm), indicated in figure 5.4.4.2.1 for 3,84 Mcps TDD Option,figure 5.4.4.2.2 for 1,28 Mcps TDD Option and figure 5.4.4.2.3 for 7,68 Mcps TDD Option, respectively.

NOTE: In this test no power limits apply during guard period.

## 5.4.5 Out-of-synchronisation handling of output power for continuous transmission

### 5.4.5.1 Definition and applicability

The UE shall monitor the DPCH quality in order to detect a loss of the signal on Layer 1, as specified in TS 25.224. [5] The thresholds $Q_{out}$ and $Q_{in}$ specify at what DPCH quality levels the UE shall shut its power off and when it shall turn its power on, respectively. The thresholds are not defined explicitly, but are defined by the conditions under which the UE shall shut its transmitter off and turn it on, as stated in this clause.

The requirement of this clause shall apply to all types of UTRA-UE.

### 5.4.5.2 Minimum Requirement

#### 5.4.5.2.1 3,84 Mcps TDD Option

When the UE estimates the DPCH quality over the last 160 ms period to be worse than a threshold $Q_{out}$, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the DPCH quality exceeds an acceptable level $Q_{in}$. When the UE estimates the DPCH quality over the last 160 ms period to be better than a threshold $Q_{in}$, the UE shall again turn its transmitter on within 40 ms.

The quality levels at the thresholds $Q_{out}$ and $Q_{in}$ correspond to different signal levels depending on the downlink conditions DCH parameters. For the conditions in table 5.4.5.2.1, a signal with the quality at the level $Q_{out}$ is generated by a ΣDPCH_Ec/Ior ratio of -13 dB, and a signal with $Q_{in}$ by a ΣDPCH_Ec/Ior ratio of -9 dB. In this test, the DL reference measurement channel (12,2) kbps specified in clause C.3.1, where the CRC bits are replaced by data bits, and with static propagation conditions is used.

**Table 5.4.5.2.1: DCH parameters the of Out-of-synch handling test case test case – 3,84 Mcps TDD option – continuous transmission**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 1.1 |
| $I_{oc}$ | dBm/3,84 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.5.2.1 |
| Information Data Rate | kbps | 13 |
| TFCI | - | On |

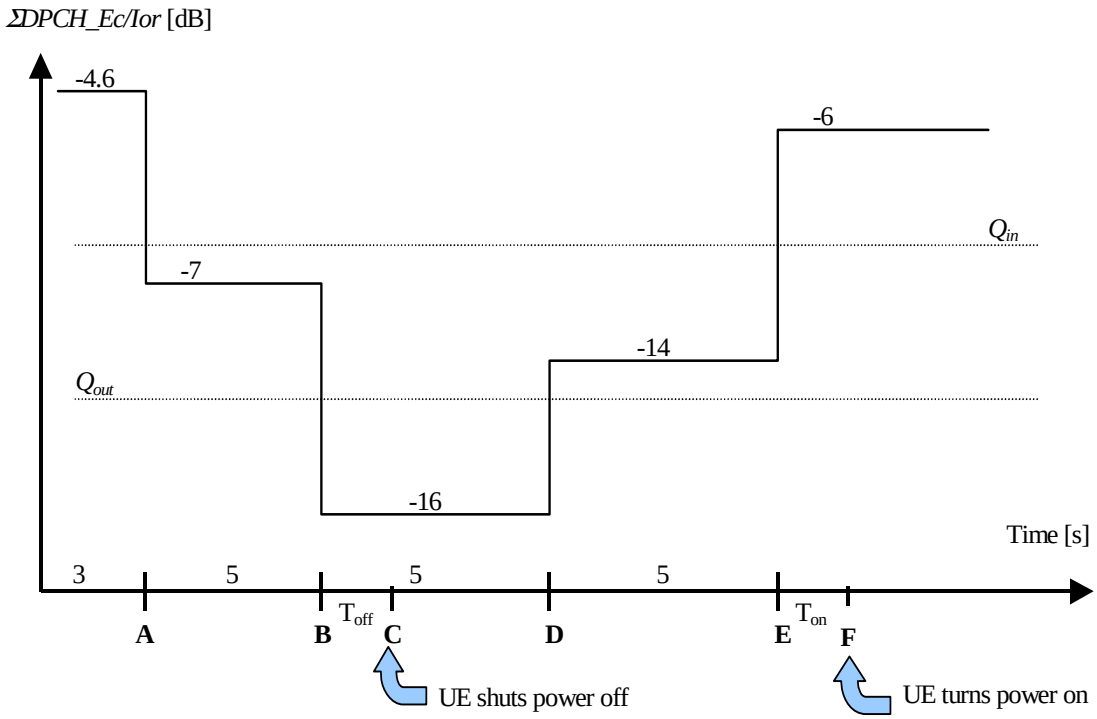



**Figure 5.4.5.2.1: Test case for out-of-synch handling in the UE. Conditions apply for 3,84 Mcps TDD Option – continuous transmission**

The requirements for the UE are that:

1) The UE shall not shut its transmitter off before point B.
2) The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3) The UE shall not turn its transmitter on between points C and E.
4) The UE shall turn its transmitter on before point F, which is Ton = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.1.

### 5.4.5.2.2 1,28 Mcps TDD Option

The parameters in table 5.4.5.2.2 are defined using the DL reference measurement channel (12,2) kbps specified in annex C where the CRC bits are replaced by data bits, and with static propagation conditions.

**Table 5.4.5.2.2: DCH parameters for test of Out-of-synch handling**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | -1 |
| $I_{oc}$ | dBm/1,28 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.5.2.2 |
| Information Data Rate | kbps | 12,2 |
| TFCI | - | On |

The conditions for when the UE shall shut its transmitter off and when it shall turn it on are defined by the parameters in table 5.4.5.1.2 together with the DPCH power level as defined in figure 5.4.5.1.

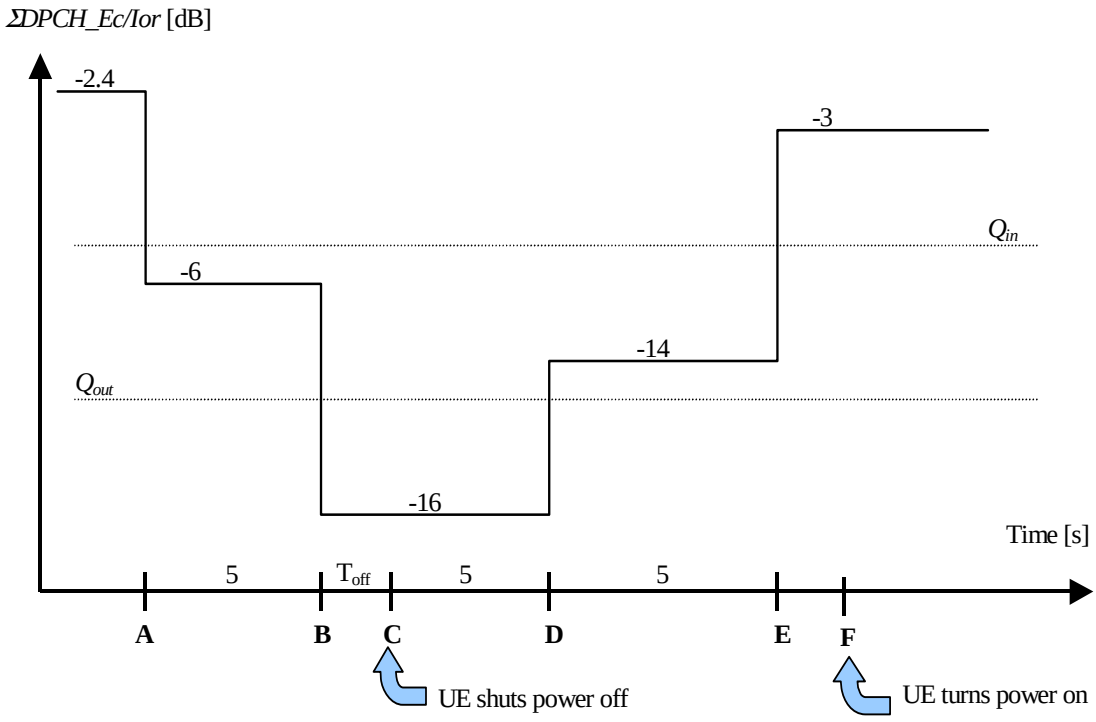



**Figure 5.4.5.2.2: Conditions for out-of-synch handling in the UE. The indicated thresholds $Q_{out}$ and$Q_{in}$ are only informative. Conditions apply for 1,28 Mcps TDD Option – continuous transmission**

The requirements for the UE are that:

1. The UE shall not shut its transmitter off before point B.
2. The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3. The UE shall not turn its transmitter on between points C and E.
4. The UE shall turn its transmitter on before point F, which is $T_{on}$ = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.2.

### 5.4.5.2.3 7,68 Mcps TDD Option

When the UE estimates the DPCH quality over the last 160 ms period to be worse than a threshold $Q_{out}$, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the DPCH quality exceeds an acceptable level $Q_{in}$. When the UE estimates the DPCH quality over the last 160 ms period to be better than a threshold $Q_{in}$, the UE shall again turn its transmitter on within 40 ms.

The quality levels at the thresholds $Q_{out}$ and $Q_{in}$ correspond to different signal levels depending on the downlink conditions DCH parameters. For the conditions in table 5.4.5.2.3, a signal with the quality at the level $Q_{out}$ is generated by a ΣDPCH_Ec/Ior ratio of -16 dB, and a signal with $Q_{in}$ by a ΣDPCH_Ec/Ior ratio of -12 dB. In this test, the DL reference measurement channel (12,2) kbps specified in clause C.3.1.3, where the CRC bits are replaced by data bits, and with static propagation conditions is used.

**Table 5.4.5.2.3: DCH parameters the of Out-of-synch handling test case test case – 7,68 Mcps TDD option – continuous transmission**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 1.1 |
| $I_{oc}$ | dBm/3,84 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.5.2.3 |
| Information Data Rate | kbps | 12,2 |
| TFCI | - | On |

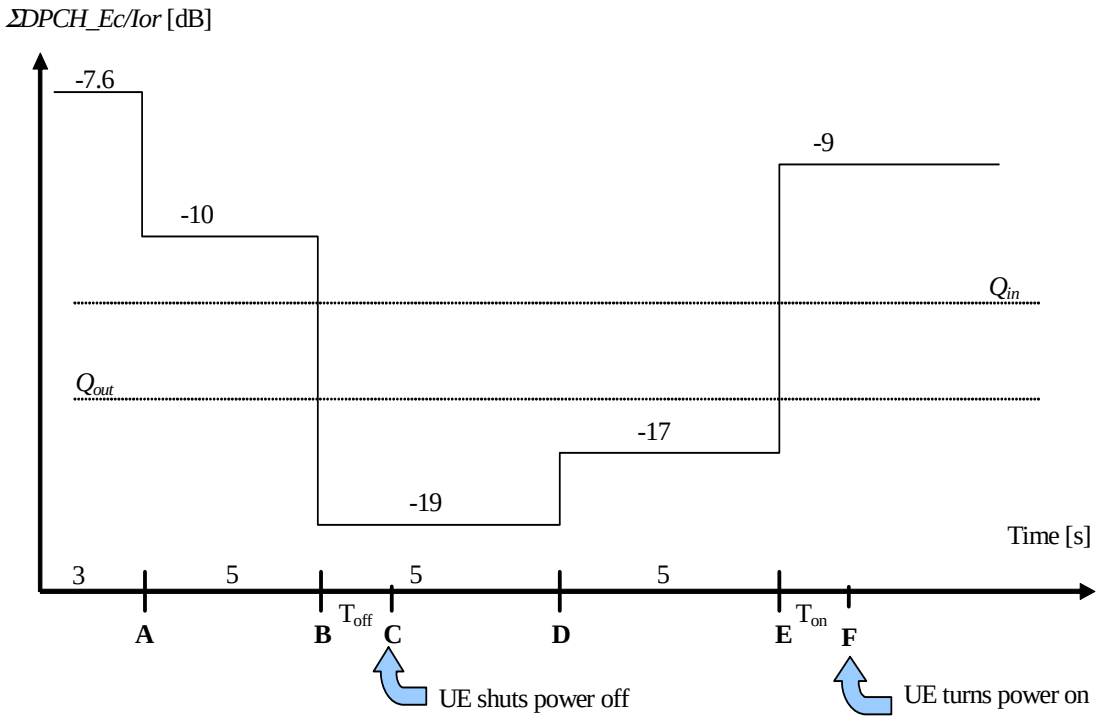



**Figure 5.4.5.2.3: Test case for out-of-synch handling in the UE. Conditions apply for 7,68 Mcps TDD Option – continuous transmission**

The requirements for the UE are that:

1) The UE shall not shut its transmitter off before point B.
2) The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3) The UE shall not turn its transmitter on between points C and E.
4) The UE shall turn its transmitter on before point F, which is Ton = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.3.

## 5.4.5.3 Test purpose

To verify that the UE monitors the DPCH quality and turns its transmitter on or off according to DPCH level diagram specified in figure5.4.5.1

## 5.4.5.4 Method of test

### 5.4.5.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) Calls are set up according to the Generic call setup procedure using parameters as specified in table 5.4.5.1
3) Enter the UE into loopback test mode and start the loopback test.
4) The handover triggering level shall be set very high [TBD] to ensure that the beacon channel power never exceeds the value of 10dB above it. Therefore the averaging time for signal quality will always be 160 milliseconds.

### 5.4.5.4.2 Procedure

#### 5.4.5.4.2.1 3,84 Mcps TDD Option

1) SS level and signalling values are set that the UE transmits maximum power (see annex E clause E.3.1)
2) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -4.6[+0.4 - 0] dB and verify that the UE TX signal is on.
3) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -7[+0.4 - 0]dB and verify that the UE TX signal remains on continuously for at least 5 seconds.
4) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -16[+0 -0.4] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.
5) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -14[+0 -0.4] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.
6) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -6[+0.4 - 0]dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

#### 5.4.5.4.2.2 1,28 Mcps TDD Option

1) The SS sends continuously Up power control commands to the UE until the UE transmitter power reaches maximum level
2) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -2.4 [+0.3 - 0] dB and verify that the UE TX signal is on.
3) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -6[+0.3 - 0] dB and verify that the UE TX signal remains on continuously for at least 5 seconds.
4) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -16[+0 - 0.3] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.

5) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -14[+0 - 0.3] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.

6) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -3[+0.3 - 0] dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

#### 5.4.5.4.2.3 7,68 Mcps TDD Option

1) SS level and signalling values are set that the UE transmits maximum power (see annex E clause E.3.1)

2) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -7.6[+0.4 - 0] dB and verify that the UE TX signal is on.

3) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -10[+0.4 - 0]dB and verify that the UE TX signal remains on continuously for at least 5 seconds.

4) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -19[+0 -0.4] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.

5) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -17[+0 -0.4] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.

6) Set the SS TX signal quality to $\frac{\Sigma DPCH_E_c}{I_{or}}$ = -9[+0.4 - 0]dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

### 5.4.5.5 Test Requirements

The UE TX on-criterion including tolerance window is derived from the initial conditions and is verified with the method of 5.4.2.4 minimum transmit power related to minimum requirements according to clause 5.4.2.2.1 for 3,84 Mcps TDD Option,5.4.2.2.2 for 1,28 Mcps TDD Option and 5.4.2.2.3 for 7.68Mcps TDD option, respectively. The UE transmitter is considered to be on if the UE transmitted power is higher than the minimum output power.

The UE TX off criterion including tolerance is verified according to clause 5.4.3 of the present document (Transmit off power). The UE transmitter is considered to be off if the UE transmitted power is lower than the transmit OFF power.

To pass the test, steps 1 through 6 of the procedure must be fulfilled.

## 5.4.6 Out-of-synchronisation handling of output power for discontinuous transmission

### 5.4.6.1 Definition and applicability

Normally the UE monitors the DPCH quality in order to detect a loss of the signal on Layer 1, as specified in TS 25.224. [5] The thresholds $Q_{out}$ and $Q_{in}$ specify at what DPCH quality levels the UE shall shut its power off and when it shall turn its power on, respectively. The thresholds are not defined explicitly, but are defined by the conditions under which the UE shall shut its transmitter off and turn it on, as stated in this clause.

However, during DTX, there are periods when the UE will receive no data from the UTRAN. As specified in TS 25.224, in order to keep synchronization, Special Bursts shall be transmitted by the UTRAN during these periods of no data.

During these periods, the conditions for when the UE shall shut its transmitter on or off are defined by the power level of the received Special Bursts.

When the UE does not detect at least one special burst with a quality above a threshold $Q_{sbout}$ over the last 160 ms period, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the special burst quality exceeds an acceptable level $Q_{sbin}$. When the UE estimates the special burst quality to be better than a threshold $Q_{sbin}$ over the last 160 ms, the UE shall again turn its transmitter on within 40 ms.

The requirement of this clause shall apply to all types of UTRA-UE.

### 5.4.6.2 Minimum Requirement

#### 5.4.6.2.1 3,84 Mcps TDD Option

When the UE estimates the DPCH quality over the last 160 ms period to be worse than a threshold $Q_{out}$, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the DPCH quality exceeds an acceptable level $Q_{in}$. When the UE estimates the DPCH quality over the last 160 ms period to be better than a threshold $Q_{in}$, the UE shall again turn its transmitter on within 40 ms.

The quality levels at the thresholds $Q_{out}$ and $Q_{in}$ correspond to different signal levels depending on the downlink conditions DCH parameters. For the conditions in table 5.4.6.2.1, a signal with the quality at the level $Q_{out}$ is generated by a DPCH_Ec/Ior ratio of -16 dB during special bursts, and a signal with $Q_{in}$ by a DPCH_Ec/Ior ratio of -12 dB.

**Table 5.4.6.2.1: DCH parameters the of Out-of-synch handling test case test case – 3,84 Mcps TDD option – discontinuous transmission**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 1.1 |
| $I_{oc}$ | dBm/3,84 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.6.2.1 |
| Bits/burst (including TFCI bits) | bits | 244 |
| TFCI | - | On |

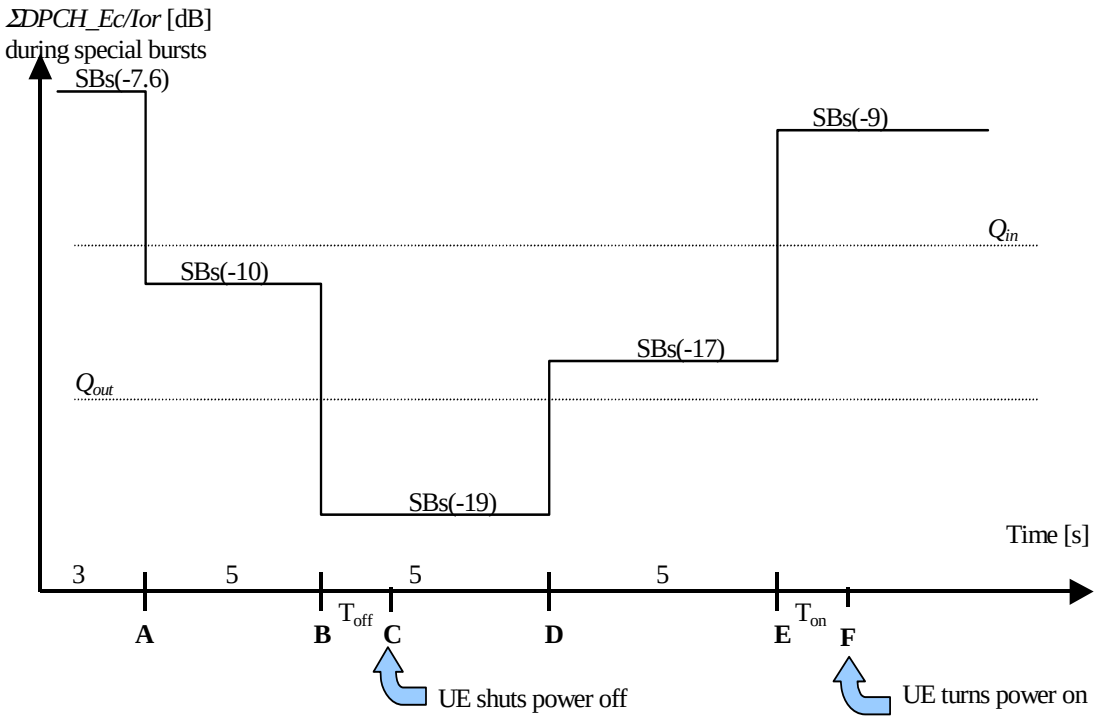



**Figure 5.4.6.2.1: Test case for out-of-synch handling in the UE. Conditions apply for 3,84 Mcps TDD Option – discontinuous transmission**

The requirements for the UE are that:

1) The UE shall not shut its transmitter off before point B.
2) The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3) The UE shall not turn its transmitter on between points C and E.
4) The UE shall turn its transmitter on before point F, which is Ton = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.1.

### 5.4.6.2.2 1,28 Mcps TDD Option

When the UE estimates the DPCH quality over the last 160 ms period to be worse than a threshold $Q_{out}$, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the DPCH quality exceeds an acceptable level $Q_{in}$. When the UE estimates the DPCH quality over the last 160 ms period to be better than a threshold $Q_{in}$, the UE shall again turn its transmitter on within 40 ms.

The quality levels at the thresholds $Q_{out}$ and $Q_{in}$ correspond to different signal levels depending on the downlink conditions DCH parameters. For the conditions in table 5.4.6.2.2, a signal with the quality at the level $Q_{out}$ is generated by a DPCH_Ec/Ior ratio of -16 dB during special bursts, and a signal with $Q_{in}$ by a DPCH_Ec/Ior ratio of -12 dB.

**Table 5.4.6.2.2: DCH parameters for test of Out-of-synch handling**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | -1 |
| $I_{oc}$ | dBm/1,28 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.6.2.2 |
| Bits/burst (including TFCI bits) | bits | 88 per subframe |
| TFCI | - | On |

The conditions for when the UE shall shut its transmitter off and when it shall turn it on are defined by the parameters in table 5.4.6.2.2 together with the DPCH power level as defined in figure 5.4.6.2.

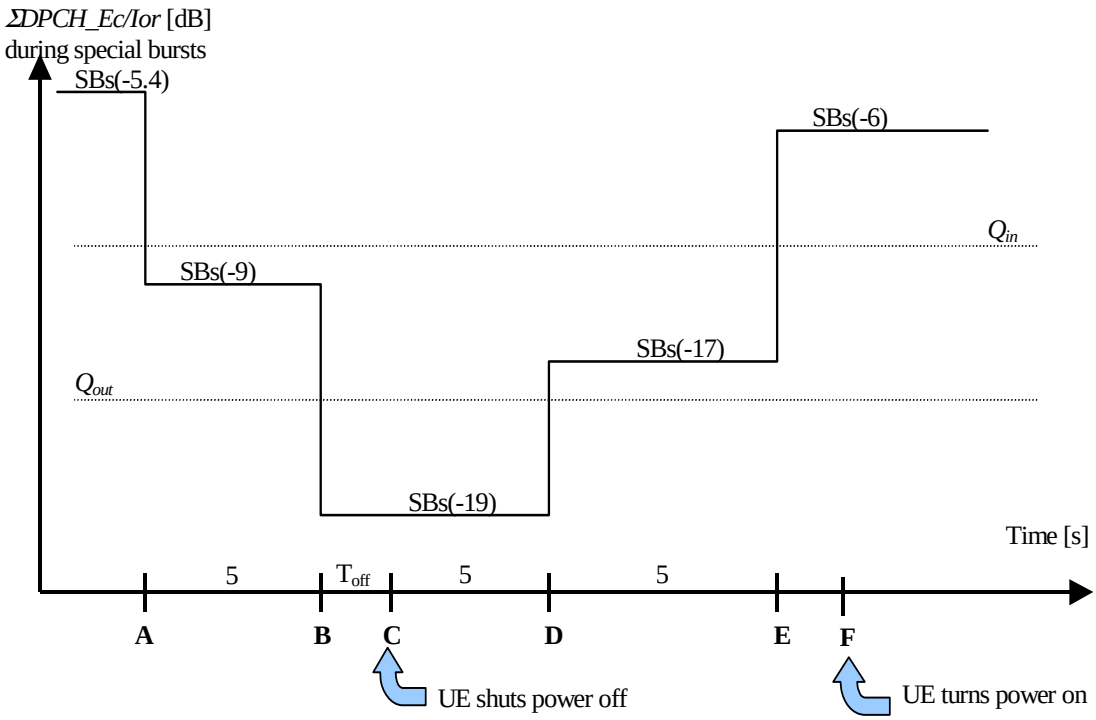



**Figure 5.4.6.2.2: Conditions for out-of-synch handling in the UE. The indicated thresholds $Q_{out}$ and$Q_{in}$ are only informative. Conditions apply for 1,28 Mcps TDD Option– discontinuous transmission**

The requirements for the UE are that:

1. The UE shall not shut its transmitter off before point B.
2. The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3. The UE shall not turn its transmitter on between points C and E.
4. The UE shall turn its transmitter on before point F, which is $T_{on}$ = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.2.

### 5.4.6.2.3 7,68 Mcps TDD Option

When the UE estimates the DPCH quality over the last 160 ms period to be worse than a threshold $Q_{out}$, the UE shall shut its transmitter off within 40 ms. The UE shall not turn its transmitter on again until the DPCH quality exceeds an acceptable level $Q_{in}$. When the UE estimates the DPCH quality over the last 160 ms period to be better than a threshold $Q_{in}$, the UE shall again turn its transmitter on within 40 ms.

The quality levels at the thresholds $Q_{out}$ and $Q_{in}$ correspond to different signal levels depending on the downlink conditions DCH parameters. For the conditions in table 5.4.6.2.3, a signal with the quality at the level $Q_{out}$ is generated by a DPCH_Ec/Ior ratio of -19 dB during special bursts, and a signal with $Q_{in}$ by a DPCH_Ec/Ior ratio of -15 dB.

**Table 5.4.6.2.3: DCH parameters the of Out-of-synch handling test case test case – 7,68 Mcps TDD option – discontinuous transmission**

| Parameter | Unit | Value |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 1.1 |
| $I_{oc}$ | dBm/3,84 MHz | -60 |
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | See figure 5.4.6.2.3 |
| Bits/burst (including TFCI bits) | bits | 244 |
| TFCI | - | On |

The conditions for when the UE shall shut its transmitter off and when it shall turn it on are defined by the parameters in table 5.4.6.2.3 together with the DPCH power level as defined in figure 5.4.6.2.3.

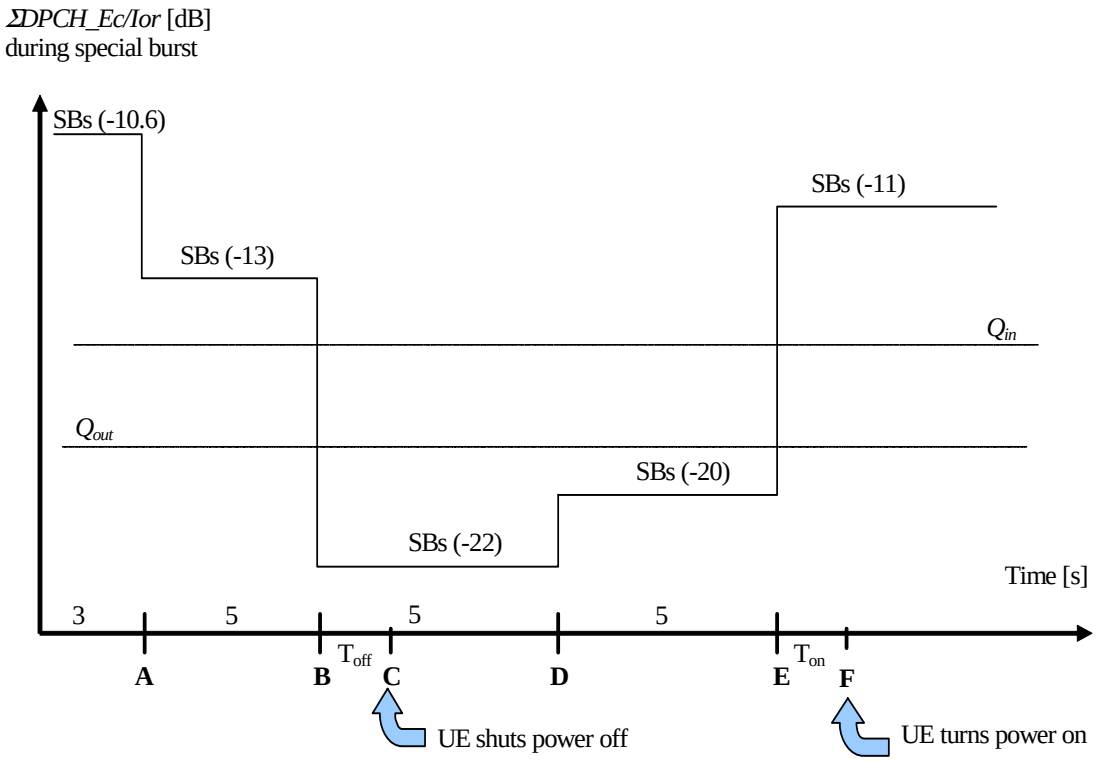



**Figure 5.4.6.2.3: Test case for out-of-synch handling in the UE. Conditions apply for 7,68 Mcps TDD Option – discontinuous transmission**

The requirements for the UE are that:

1) The UE shall not shut its transmitter off before point B.
2) The UE shall shut its transmitter off before point C, which is $T_{off}$ = 200 ms after point B
3) The UE shall not turn its transmitter on between points C and E.
4) The UE shall turn its transmitter on before point F, which is Ton = 200 ms after Point E.

The normative reference for this test is TS 25.102 [1] clause 6.4.3.1.3.

## 5.4.6.3 Test purpose

To verify that the UE monitors the DPCH quality and turns its transmitter on or off according to DPCH level diagram specified in figure5.4.6.1

### 5.4.6.4 Method of test

#### 5.4.6.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) Calls are set up according to the Generic call setup procedure using parameters as specified in table 5.4.6.1
3) Enter the UE into loopback test mode and start the loopback test.
4) The handover triggering level shall be set very high [TBD] to ensure that the beacon channel power never exceeds the value of 10dB above it. Therefore the averaging time for signal quality will always be 160 milliseconds.

#### 5.4.6.4.2 Procedure

##### 5.4.6.4.2.1 3,84 Mcps TDD Option

1) SS level and signalling values are set that the UE transmits maximum power (see annex E clause E.3.1)
2) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -7.6[+0.4 –0] dB and verify that the UE TX signal is on.
3) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -10[+0.4 –0] dB and verify that the UE TX signal remains on continuously for at least 5 seconds.
4) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -19[+0 –0.4] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.
5) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -17[+0 –0.4] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.
6) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -9[+0.4 –0] dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

##### 5.4.6.4.2.2 1,28 Mcps TDD Option

1) The SS sends continuously Up power control commands to the UE until the UE transmitter power reaches maximum level
2) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -5.4[+0.3 - 0] dB and verify that the UE TX signal is on.
3) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -9+[+0,3-0] dB and verify that the UE TX signal remains on continuously for at least 5 seconds.
4) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -19-[+0-0,3] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.

5) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -17-[=0-0,3] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.

6) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -6+[+0,3-0] dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

#### 5.4.6.4.2.3 7,68 Mcps TDD Option

1) SS level and signalling values are set that the UE transmits maximum power (see annex E clause E.3.1)

2) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -10.6[+0.4 –0] dB and verify that the UE TX signal is on.

3) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -13[+0.4 –0] dB and verify that the UE TX signal remains on continuously for at least 5 seconds.

4) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -22[+0 –0.4] dB and verify that the UE TX signal turns off 200 ms or earlier with respect to that instant.

5) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -20[+0 –0.4] dB and verify that the UE TX signal remains off continuously for at least 5 seconds.

6) Set the SS TX signal quality to $\frac{DPCH_E_c}{I_{or}}$ = -11[+0.4 –0] dB and verify that the UE TX signal is switched on 200 ms or earlier with respect to that instant.

### 5.4.6.5 Test Requirements

The UE TX on-criterion including tolerance window is derived from the initial conditions and is verified with the method of 5.4.2.4 minimum transmit power related to minimum requirements according to clause 5.4.2.2.1 for 3,84 Mcps TDD Option, 5.4.2.2.2 for 1,28 Mcps TDD Option and 5.4.2.2.3 for 7.68Mcps TDD option, respectively. The UE transmitter is considered to be on if the UE transmitted power is higher than the minimum output power.

The UE TX off criterion including tolerance is verified according to clause 5.4.3 of the present document (Transmit off power). The UE transmitter is considered to be off if the UE transmitted power is lower than the transmit OFF power.

To pass the test, steps 1 through 6 of the procedure must be fulfilled.

# 5.5 Output RF spectrum emissions

## 5.5.1 Occupied bandwidth

### 5.5.1.1 Definition and applicability

Occupied bandwidth is a measure of the bandwidth containing 99 % of the total integrated power for transmitted spectrum and is centered on the assigned channel frequency.

The requirements in this clause shall apply to all types of UTRA - UE.

### 5.5.1.2 Minimum Requirements

#### 5.5.1.2.1 3,84Mcps TDD Option

The occupied bandwidth shall be less than 5 MHz based on a chip rate of 3,84 Mcps.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.1.1.

#### 5.5.1.2.2 1,28Mcps TDD Option

The occupied channel bandwidth shall be less than 1.6 MHz based on a chip rate of 1,28 Mcps.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.1.2.

#### 5.5.1.2.3 7,68Mcps TDD Option

The occupied bandwidth shall be less than 10 MHz based on a chip rate of 7,68 Mcps.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.1.3.

### 5.5.1.3 Test purpose

The occupied bandwidth, defined in the Radio Regulations of the International Telecommunication Union ITU, is a useful concept for specifying the spectral properties of a given emission in the simplest possible manner; see also ITU-R Recommendation SM.328-9 [8].

The test purpose is to verify that the emission of the UE is sufficiently concentrated in the bandwidth for the service to be provided and is, therefore, not likely to create interference to other users of the spectrum beyond undue limits.

### 5.5.1.4 Method of test

#### 5.5.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.
3) Enter the UE into loopback test mode and start the loopback test.

#### 5.5.1.4.2 Procedure

1) Measure the power of the transmitted signal with a measurement filter of bandwidth [30 kHz]. The characteristic of the filter shall be approximately Gaussian (typical spectrum analyzer filter). The center frequency of the filter shall be stepped in contiguous 30 kHz steps from a minimum frequency, which shall be [7,5 – 0,015] MHz for the 3,84 Mcps TDD Option, [2,4 – 0,015] MHz for the 1,28 Mcps TDD Option and [15 - 0,015] MHz for 7.68 Mcps TDD option, respectively, below the assigned channel frequency of the transmitted signal, up to a maximum frequency, which shall be [7,5 – 0,015] MHz for the 3,84 Mcps TDD Option, [2,4 – 0,015] MHz for the 1,28 Mcps TDD Option and [15 - 0,015] MHz for 7.68 Mcps TDD option, respectively, above the assigned channel frequency of the transmitted signal. The step duration shall be sufficient slow to capture the active TS. The measured power shall be recorded for each step.
2) Determine the total transmitted power by accumulating the recorded power measurements results of all steps.
3) Sum up the power upward from the lower boundary of the measured frequency range in '(2)' and seek the limit frequency point by which this sum becomes 0.5 % of "Total Power" and save this point as "Lower Frequency".

4) Sum up the power downward from the upper boundary of the measured frequency range in '(2)' and seek the limit frequency point by which this sum becomes 0.5 % of "Total Power" and save this point as "Upper Frequency".

5) Calculate the difference ("Upper Frequency" – "Lower Frequency" = "Occupied Bandwidth") between two limit frequencies obtained in '(4)' and '(5)'.

#### 5.5.1.5 Test requirements

##### 5.5.1.5.1 3,84 Mcps TDD Option

The measured Occupied Bandwidth, derived in step 5), shall not exceed 5 MHz for the 3,84 Mcps TDD Option.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

##### 5.5.1.5.2 1,28 Mcps TDD Option

The measured Occupied Bandwidth, derived in step 5), shall not exceed 1.6 MHz for the 1,28 Mcps TDD Option.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

##### 5.5.1.5.3 7,68 Mcps TDD Option

The measured Occupied Bandwidth, derived in step 5), shall not exceed 10 MHz for the 7,68 Mcps TDD Option.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.5.2 Out of band emission

Out of band emissions are unwanted emissions immediately outside the nominal channel resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a spectrum emission mask and adjacent channel leakage power ratio (ACLR).

### 5.5.2.1 Spectrum emission mask

#### 5.5.2.1.1 Definition and applicability

##### 5.5.2.1.1.1 3,84 Mcps TDD Option

The spectrum emission mask of the UE is a requirement that applies to frequencies which are between 2,5 MHz and 12,5 MHz on both sides of the UE centre carrier frequency. The out of channel emission is specified relative to the RRC filtered mean power of the UE carrier.

The requirements of this test apply to all types of UTRA-UE.

##### 5.5.2.1.1.2 1,28 Mcps TDD Option

The spectrum emission mask of the UE applies to frequencies, which are between 0,8 MHz and 4,0 MHz on both sides of the centre carrier frequency. The out of channel emission is specified relative to the RRC filtered mean power of the UE carrier.

#### 5.5.2.1.1.3 7,68 Mcps TDD Option

The spectrum emission mask of the UE is a requirement that applies to frequencies which are between 5 MHz and 25 MHz on both sides of the UE centre carrier frequency. The out of channel emission is specified relative to the RRC filtered mean power of the UE carrier.

### 5.5.2.1.2 Minimum Requirements

#### 5.5.2.1.2.1 3,84 Mcps TDD Option

The power of any UE emission shall not exceed the levels specified in table 5.5.2.1.2.1.

The normative reference for this requirement is TS 25.102 clause 6.6.2.1.1.1.

**Table 5.5.2.1.2.1: Spectrum Emission Mask Requirement (3,84 Mcps TDD Option)**

| Δf in MHz (note 1) | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 2.5 - 3.5 | $\left\{-35-15\cdot\left(\frac{\Delta f}{MHz}-2.5\right)\right\}dBc$ | 30 kHz (note 2) |
| 3.5 - 7.5 | $\left\{-35-1\cdot\left(\frac{\Delta f}{MHz}-3.5\right)\right\}dBc$ | 1 MHz (note 3) |
| 7.5 - 8.5 | $\left\{-39-10\cdot\left(\frac{\Delta f}{MHz}-7.5\right)\right\}dBc$ | 1 MHz (note 3) |
| 8.5 - 12.5 | -49 dBc | 1 MHz |
| Note 1: Δf is the separation between the carrier frequency and the centre of the measuring filter.<br>Note 2: The first and last measurement position with a 30 kHz filter is at Δf equals to 2.515 MHz and 3.485 MHz.<br>Note 3: The first and last measurement position with a 1 MHz filter is at Δf equals to 4 MHz and 12 MHz. As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth. | | |
| The lower limit shall be –50dBm/3,84 MHz or the minimum requirement presented in this table which ever is the higher. | | |

#### 5.5.2.1.2.2 1,28 Mcps TDD Option

The power of any UE emission shall not exceed the levels specified in table 5.5.2.1.2.2.

The normative reference for this requirement is TS 25.102 clause 6.6.2.1.1.2.

**Table 5.5.2.1.2.2: Spectrum Emission Mask Requirement (1,28 Mcps TDD Option)**

| Δf (note 1) in MHz | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 0.8 | -35 dBc | 30 kHz (note 2) |
| 0.8-1.8 | $\left\{-35-14\cdot\left(\frac{\Delta f}{MHz}-0.8\right)\right\}dBc$ | 30 kHz (note 2) |
| 1.8-2.4 | $\left\{-49-17\cdot\left(\frac{\Delta f}{MHz}-1.8\right)\right\}dBc$ | 30 kHz (note 2) |
| 2.4 – 4.0 | -44 dBc | 1MHz (note 3) |
| NOTE 1: Δf is the separation between the carrier frequency and the centre of the measuring filter. | | |
| NOTE 2: The first and last measurement position with a 30 kHz filter is at Δf equals to 0.815 MHz and 2.385 MHz. | | |
| NOTE 3: The first and last measurement position with a 1 MHz filter is at Δf equals to 2.9MHz and 3.5MHz .As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth. | | |
| The lower limit shall be –55dBm/1,281,28 MHz or the minimum requirement presented in this table which ever is the higher. | | |

5.5.2.1.2.3 7,68 Mcps TDD Option

The power of any UE emission shall not exceed the levels specified in table 5.5.2.1.2.3.

The normative reference for this requirement is TS 25.102 clause 6.6.2.1.3.1.

**Table 5.5.2.1.2.3: Spectrum Emission Mask Requirement (7,68 Mcps TDD Option)**

| Δf* in MHz | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 5.0 - 5.75 | $\left\{-38-10.67\cdot\left(\frac{\Delta f}{MHz}-5.0\right)\right\}dBc$ | 30 kHz ** |
| 5.75 - 7.0 | $\left\{-46-5.6\cdot\left(\frac{\Delta f}{MHz}-5.75\right)\right\}dBc$ | 30 kHz** |
| 7.0 - 15 | $\left\{-38-0.5\cdot\left(\frac{\Delta f}{MHz}-7.0\right)\right\}dBc$ | 1 MHz *** |
| 15.0 - 17.0 | $\left\{-42-5.0\cdot\left(\frac{\Delta f}{MHz}-15.0\right)\right\}dBc$ | 1 MHz *** |
| 17.0 - 25.0 | -53 dBc | 1 MHz *** |
| * Δf is the separation between the carrier frequency and the centre of the measuring filter. | | |
| ** The first and last measurement position with a 30 kHz filter is at Δf equals to 5.015 MHz and 6.985 MHz | | |
| *** The first and last measurement position with a 1 MHz filter is at Δf equals to 7.5 MHz and 24.5 MHz. As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth. | | |
| Note: The lower limit shall be -47dBm/7.68 MHz or the minimum requirement presented in this table which ever is the higher. | | |

#### 5.5.2.1.3 Test purpose

This test supplements Occupied Bandwidth (verifying the spectral concentration of the UE's emissions) and Adjacent Channel Leakage Ratio (simulating the perception of other UTRA receivers) in a system independent way. It is the purpose of this test to limit interferences to other systems (wideband or narrowband).

#### 5.5.2.1.4 Method of test

##### 5.5.2.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.

3) Enter the UE into loopback test mode and start the loopback test.

##### 5.5.2.1.4.2 Procedure

1) Measure the power of the transmitted signal with a measurement filter of bandwidths according to table 5.5.2.1.2.1 for the 3,84 Mcps TDD Option, 5.5.2.1.2.2 for the 1,28 Mcps TDD Option and 5.5.2.1.2.3 for the 7,68 Mcps TDD Option, respectively. The characteristic of the filter shall be approximately Gaussian (typical spectrum analyzer filter). The center frequency of the filter shall be stepped in contiguous steps according to table 5.5.2.1.2.1 for the 3,84 Mcps TDD Option, 5.5.2.1.2.2 for the 1,28 Mcps TDD Option and 5.5.2.1.2.3 for the 7,68 Mcps TDD Option, respectively. The step duration shall be sufficient slow to capture the active TS. The measured power shall be recorded for each step.

2) Measure the RRC filtered mean power centered on the assigned channel frequency according to annex B.

3) Display the results of 1) in dBc with respect to 2).

### 5.5.2.1.5 Test requirements

#### 5.5.2.1.5.1 3,84 Mcps TDD Option

The result 5.5.2.1.4.2. step 3) shall fulfil the requirements of table 5.5.2.1.5.1.

**Table 5.5.2.1.5.1: Spectrum Emission Mask Requirement (3,84 Mcps TDD Option)**

| Δf in MHz (note 1) | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 2.5 - 3.5 | $\left\{-33.5-15\cdot\left(\frac{\Delta f}{MHz}-2.5\right)\right\}dBc$ | 30 kHz |
| 3.5 - 7.5 | $\left\{-33.5-1\cdot\left(\frac{\Delta f}{MHz}-3.5\right)\right\}dBc$ | 1 MHz |
| 7.5 - 8.5 | $\left\{-37.5-10\cdot\left(\frac{\Delta f}{MHz}-7.5\right)\right\}dBc$ | 1 MHz |
| 8.5 - 12.5 | -47.5 dBc | 1 MHz |
| Note 1: Δf is the separation between the carrier frequency and the centre of the measuring filter.<br>Note 2: The first and last measurement position with a 30 kHz filter is at Δf equals to 2.515 MHz and 3.485 MHz.<br>Note 3: The first and last measurement position with a 1 MHz filter is at Δf equals to 4 MHz and 12 MHz. As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.. | | |
| The lower limit shall be –48.5dBm/3,84 MHz or the minimum requirement presented in this table which ever is the higher. | | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

5.5.2.1.5.2 1,28 Mcps TDD Option

The result 5.5.2.1.4.2. step 3) shall fulfil the requirements of table 5.5.2.1.5.2.

**Table 5.5.2.1.5.2: Spectrum Emission Mask Requirement (1,28 Mcps TDD Option)**

| Δf (note 1) in MHz | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 0.8 | -33.5 dBc | 30 kHz (note 2) |
| 0.8-1.8 | $\left\{-33.5-14\cdot\left(\frac{\Delta f}{MHz}-0.8\right)\right\}dB$ | 30 kHz (note 2) |
| 1.8-2.4 | $\left\{-47.5-17\cdot\left(\frac{\Delta f}{MHz}-1.8\right)\right\}dB$ | 30 kHz (note 2) |
| 2.4 – 4.0 | -42.5 dBc | 1MHz (note 3) |
| NOTE 1: Δf is the separation between the carrier frequency and the centre of the measuring filter. | | |
| NOTE 2: The first and last measurement position with a 30 kHz filter is at Δf equals to 0.815 MHz and 2.385 MHz. | | |
| NOTE 3: The first and last measurement position with a 1 MHz filter is at Δf equals to 2.9MHz and 3.5MHz .As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth. | | |
| The lower limit shall be –53.5 dBm/1,281,28 MHz or the minimum requirement presented in this table which ever is the higher. | | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

#### 5.5.2.1.5.3 7,68 Mcps TDD Option

The result 5.5.2.1.4.2. step 3) shall fulfil the requirements of table 5.5.2.1.5.3.

**Table 5.5.2.1.5.3: Spectrum Emission Mask Requirement (7,68 Mcps TDD Option)**

| Δf* in MHz | Minimum requirement | Measurement bandwidth |
|---|---|---|
| 5.0 - 5.75 | $\left\{-36.5-10.67\cdot\left(\frac{\Delta f}{MHz}-5.0\right)\right\}dBc$ | 30 kHz ** |
| 5.75 - 7.0 | $\left\{-44.5-5.6\cdot\left(\frac{\Delta f}{MHz}-5.75\right)\right\}dBc$ | 30 kHz** |
| 7.0 - 15 | $\left\{-36.5-0.5\cdot\left(\frac{\Delta f}{MHz}-7.0\right)\right\}dBc$ | 1 MHz *** |
| 15.0 - 17.0 | $\left\{-40.5-5.0\cdot\left(\frac{\Delta f}{MHz}-15.0\right)\right\}dBc$ | 1 MHz *** |
| 17.0 - 25.0 | -51.5 dBc | 1 MHz *** |
| * | Δf is the separation between the carrier frequency and the centre of the measuring filter. | |
| ** | The first and last measurement position with a 30 kHz filter is at Δf equals to 5.015 MHz and 6.985 MHz | |
| *** | The first and last measurement position with a 1 MHz filter is at Δf equals to 7.5 MHz and 24.5 MHz. As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth can be different from the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth. | |
| Note: | The lower limit shall be -47dBm/7.68 MHz or the minimum requirement presented in this table which ever is the higher. | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

### 5.5.2.2 Adjacent Channel Leakage power Ratio (ACLR)

#### 5.5.2.2.1 Definition and applicability

Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the RRC filtered mean power centered on the assigned channel frequency to the RRC filtered mean power centered on an adjacent channel frequency.

The requirements in this clause shall apply to all types of UTRA-UE.

#### 5.5.2.2.2 Minimum Requirements

##### 5.5.2.2.2.1 3,84Mcps TDD Option

If the adjacent channel RRC filtered mean power is greater than –50 dBm then the ACLR shall be higher than the value specified in table 5.5.2.2.2.1.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.2.2.1.1.

**Table 5.5.2.2.2.1: UE ACLR (3,84 Mcps TDD Option)**

| Power Class | Adjacent channel | ACLR limit |
|---|---|---|
| 2, 3 | UE-channel ± 5 MHz | 33 dB |
| 2, 3 | UE-Channel ± 10 MHz | 43 dB |

#### 5.5.2.2.2.2 1,28Mcps TDD Option

If the adjacent channel RRC filtered mean power is greater than –55 dBm then the ACLR shall be better than the value specified in table 5.5.2.2.2.2.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.2.2.1.2.

**Table 5.5.2.2.2.2: UE ACLR (1,28Mcps TDD Option)**

| Power Class | adjacent channel | ACLR limit |
|---|---|---|
| 2, 3 | UE channel ± 1.6 MHz | 33 dB |
| 2, 3 | UE channel ± 3.2 MHz | 43 dB |

NOTE 1: The requirement shall still be met in the presence of switching transients.

NOTE 2: The ACLR requirements reflect what can be achieved with present state of the art technology.

NOTE 3: Requirement on the UE shall be reconsidered when the state of the art technology progresses.

#### 5.5.2.2.2.3 7,68Mcps TDD Option

If the adjacent channel RRC filtered mean power is greater than -50dBm measured with a 3.84 Mcps RRC filter then the ACLR shall be higher than the value specified in table 5.5.2.2.2.3.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.2.2.1.3.

**Table 5.5.2.2.2.3: UE ACLR (7,68 Mcps TDD Option)**

| Power Class | adjacent channel | Chip Rate for RRC Measurement Filter | ACLR limit |
|---|---|---|---|
| 2, 3 | UE channel ± 7.5 MHz | 3.84 MHz | 33 dB |
| 2, 3 | UE channel ± 12.5 MHz | 3.84 MHz | 43 dB |
| 2 ,3 | UE channel ± 10.0 MHz | 7.68 MHz | 33 dB |
| 2 ,3 | UE channel ± 20.0 MHz | 7.68 MHz | 43 dB |

NOTE:

1) The requirement shall still be met in the presence of switching transients.

2) The ACLR requirements reflect what can be achieved with present state of the art technology.

### 5.5.2.2.3 Test purpose

The test purpose is to verify the ability of the UE to limit the interference produced by the transmitted signal to other UTRA receivers operating at the first or second adjacent RF channel.

### 5.5.2.2.4 Method of test

#### 5.5.2.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.

3) Enter the UE into loopback test mode and start the loopback test.

#### 5.5.2.2.4.2 Procedure

1) Measure the RRC filtered mean power centered on the assigned channel frequency.

2) Measure RRC filtered mean power centered on the first lower adjacent channel frequency.

3) Calculate the ACLR by dividing the power measured in 1) by the power measured in 2).

4) Repeat steps 2) and 3) for the second lower adjacent RF channel (center frequency 10 MHz for the 3,84 Mcps TDD Option, 3,2 MHz for the 1,28 Mcps TDD Option and 20MHz for the 7,68 Mcps TDD option with the 7,68Mcps RRC filter, respectively, below the assigned channel frequency of the transmitted signal) and also for the first and second upper adjacent RF channel (center frequency 5 MHz for the 3,84 Mcps TDD Option, 1,6 MHz for the 1,28 Mcps TDD Option and 10MHz for the 7,68Mcps TDD option with the 7,68Mcps RRC filter, respectively, and 10 MHz, for the 3,84 Mcps TDD Option, 3,2 MHz for the 1,28 Mcps TDD Option and 20MHz for the 7,68Mcps TDD option with the 7,68Mcps RRC filter, respectively).

7) Run step 1) to 4) for RF channels Low/Mid/High.

8) For the 7,68Mcps TDD option repeat steps 1 to 7 except use the adjacent channel frequency of +/- 7.5MHz and 12.5MHz using the 3.84Mcps RRC filter..

### 5.5.2.2.5 Test requirements

#### 5.5.2.2.5.1 3,84 Mcps TDD Option

The ACLR calculated in steps 3) and 4) of clause 5.5.2.2.4.2 shall be equal or greater than the limits given in table 5.5.2.2.5.1 for the 3,84 Mcps TDD Option.

**Table 5.5.2.2.5.1: UE ACLR (3,84 Mcps TDD Option)**

| Power Class | Adjacent channel | ACLR limit |
|---|---|---|
| 2, 3 | UE-channel ± 5 MHz | 32.2 dB |
| 2, 3 | UE-Channel ± 10 MHz | 42.2 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F.

#### 5.5.2.2.5.2 1,28 Mcps TDD Option

The ACLR calculated in steps 3) and 4) of clause 5.5.2.2.4.2 shall be equal or greater than the limits given in table 5.5.2.2.5.2 for the 1,28 Mcps TDD Option.

**Table 5.5.2.2.5.2: UE ACLR (1,28 Mcps TDD Option)**

| Power Class | Adjacent channel | ACLR limit |
|---|---|---|
| 2, 3 | UE-channel ± 1.6 MHz | 32.2 dB |
| 2, 3 | UE-Channel ± 3.2 MHz | 42.2 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F.

#### 5.5.2.2.5.3 7,68 Mcps TDD Option

The ACLR calculated in steps 3) and 4) of clause 5.5.2.2.4.2 shall be equal or greater than the limits given in table 5.5.2.2.5.3 for the 7,68 Mcps TDD Option.

**Table 5.5.2.2.5.3: UE ACLR (7,68 Mcps TDD Option)**

| Power Class | adjacent channel | Chip Rate for RRC Measurement Filter | ACLR limit |
|---|---|---|---|
| 2, 3 | UE channel ± 7.5 MHz | 3.84 MHz | 32.8 dB |
| 2, 3 | UE channel ± 12.5 MHz | 3.84 MHz | 42.2 dB |
| 2 ,3 | UE channel ± 10.0 MHz | 7.68 MHz | 32.8 dB |
| 2 ,3 | UE channel ± 20.0 MHz | 7.68 MHz | 42.2 dB |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F.

## 5.5.3 Spurious emissions

### 5.5.3.1 Definition and applicability

Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emission, intermodulation products and frequency conversion products, but exclude out of band emissions.

The frequency boundary and the detailed transitions of the limits between the requirement for out band emissions and spectrum emissions are based on ITU-R Recommendations SM.329 [8].

### 5.5.3.2 Minimum Requirements

#### 5.5.3.2.1 3,84 Mcps TDD Option

These requirements are only applicable for frequencies which are greater than 12.5 MHz away from the UE center carrier frequency.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.3.1.1.

**Table 5.5.3.2.1a: General Spurious emissions requirements (3,84 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Minimum requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1 GHz ≤ f < 12.75 GHz | 1 MHz | -30 dBm |

**Table 5.5.3.2.1b: Additional Spurious emissions requirements (3,84 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Minimum requirement |
|---|---|---|
| 925 MHz ≤ f ≤ 935 MHz | 100 kHz | -67 dBm* |
| 935 MHz < f ≤ 960 MHz | 100 kHz | -79 dBm* |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 kHz | -71 dBm* |
| 1884.5 MHz ≤ f ≤ 1919.6 MHz | 300kHz | -41 dBm** |

NOTE: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.2.1a are permitted for each UARFCN used in the measurement.

** Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a).

#### 5.5.3.2.2 1,28Mcps TDD Option

These requirements are only applicable for frequencies which are greater than 4 MHz away from the UE center carrier frequency.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.3.1.2.

**Table 5.5.3.2.2a : General Spurious emissions requirements (1,28 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Minimum requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1 GHz ≤ f < 12.75 GHz | 1 MHz | -30 dBm |

**Table 5.5.3.2.2b : Additional Spurious emissions requirements (1,28 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Minimum requirement |
|---|---|---|
| 925 MHz ≤ f ≤ 935 MHz | 100 KHz | -67 dBm* |
| 935 MHz < f ≤ 960 MHz | 100 KHz | -79 dBm* |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 KHz | -71 dBm* |

NOTE: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.2.2a are permitted for each UARFCN used in the measurement.

#### 5.5.3.2.3 7,68 Mcps TDD Option

These requirements are only applicable for frequencies which are greater than 25 MHz away from the UE center carrier frequency.

The normative reference for this requirement is TS 25.102 [1] clause 6.6.3.1.3.

**Table 5.5.3.2.3a: General Spurious emissions requirements (7,68 Mcps TDD Option)**

| Frequency Bandwidth | Measurement Bandwidth | Minimum requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1 GHz ≤ f < 12.75 GHz | 1 MHz | -30 dBm |

**Table 5.5.3.2.3b: Additional Spurious emissions requirements (7,68 Mcps TDD Option)**

| Frequency Bandwidth | Measurement Bandwidth | Minimum requirement |
|---|---|---|
| 921 MHz ≤ f < 925 MHz | 100 kHz | -60 dBm (note 1) |
| 925 MHz ≤ f ≤ 935 MHz | 100 kHz | -67 dBm (note 1) |
| 935 MHz < f ≤ 960 MHz | 100 kHz | -79 dBm (note 1) |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 kHz | -71 dBm (note 1) |
| 2620 MHz ≤ f ≤ 2690 MHz | 3.84 MHz | -37 dBm (note 1) |
| 1884.5 MHz ≤ f ≤ 1919.6 MHz | 300 kHz | -41 dBm (note 2) |
| NOTE 1: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in Table 6.7E are permitted for each UARFCN used in the measurement.<br>NOTE 2: Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a). | | |

NOTE: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.2.3a are permitted for each UARFCN used in the measurement.

## 5.5.3.3 Test purpose

### 5.5.3.3.1 3,84 Mcps Option

The test purpose is to verify the ability of the UE to limit the interference caused by unwanted transmitter effects to other systems operating at frequencies which are more than 12,5 MHz away from of the UE's carrier frequency.

### 5.5.3.3.2 1,28 Mcps Option

The test purpose is to verify the ability of the UE to limit the interference caused by unwanted transmitter effects to other systems operating at frequencies which are more than 4 MHz away from of the UE's carrier frequency.

### 5.5.3.3.3 7,68 Mcps Option

The test purpose is to verify the ability of the UE to limit the interference caused by unwanted transmitter effects to other systems operating at frequencies which are more than 25 MHz away from of the UE's carrier frequency.

## 5.5.3.4 Method of test

### 5.5.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.

3) Enter the UE into loopback test mode and start the loopback test.

### 5.5.3.4.2 Procedure

Measure the power of the spurious emissions applying measurement filters with bandwidths as specified in the relevant tables of 5.5.3.2.1 for 3,84 Mcps TDD Option, tables 5.5.3.2.2 for 1,28 Mcps TDD Option and tables 5.5.3.2.3 for 7,68 Mcps TDD option, respectively. The characteristic of the filters shall be approximately Gaussian (typical spectrum analyzer filters). The center frequency of the filter shall be swept over the frequency bands as given in the tables. The sweep time shall be sufficiently low to capture the active time slots.

## 5.5.3.5 Test requirements

### 5.5.3.5.1 3,84 Mcps TDD Option

The spurious emissions measured according to clause 5.5.3.4.2 shall not exceed the limits specified in the relevant tables of 5.5.3.5.1a and 5.5.3.5.1b.

**Table 5.5.3.5.1a: General Spurious emissions requirements**

| Frequency Bandwidth | Resolution Bandwidth | Test requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1GHz ≤ f < 12.75GHZ | 1MHz | -30 dBm |

**Table 5.5.3.5.1b: Additional Spurious emissions requirements**

| Frequency Bandwidth | Resolution Bandwidth | Test requirement |
|---|---|---|
| 925 MHz ≤ f ≤ 935 MHz | 100 kHz | -67 dBm* |
| 935 MHz < f ≤ 960 MHz | 100 kHz | -79 dBm* |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 kHz | -71 dBm* |
| 1884.5 MHz ≤ f ≤ 1919.6 MHz | 300kHz | -41 dBm** |

NOTE 1: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.5.1a for the 3,84 Mcps TDD Option are permitted for each UARFCN used in the measurement.

NOTE 2: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

** Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a).

### 5.5.3.5.2 1,28 Mcps TDD Option

The spurious emissions measured according to clause 5.5.3.4.2 shall not exceed the limits specified in the relevant tables of 5.5.3.5.2a and 5.5.3.5.2b.

**Table 5.5.3.5.2a: General Spurious emissions requirements (1,28 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Test requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1 GHz ≤ f < 12.75 GHz | 1 MHz | -30 dBm |

**Table 5.5.3.5.2b: Additional Spurious emissions requirements (1,28 Mcps TDD Option)**

| Frequency Bandwidth | Resolution Bandwidth | Test requirement |
|---|---|---|
| 925 MHz ≤ f ≤ 935 MHz | 100 KHz | -67 dBm* |
| 935 MHz < f ≤ 960 MHz | 100 KHz | -79 dBm* |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 KHz | -71 dBm* |

NOTE 1: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.5.2a for the 1,28 Mcps TDD Option are permitted for each UARFCN used in the measurement.

NOTE 2: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

#### 5.5.3.5.3 7,68 Mcps TDD Option

The spurious emissions measured according to clause 5.5.3.4.2 shall not exceed the limits specified in the relevant tables of 5.5.3.5.3a and 5.5.3.5.3b.

**Table 5.5.3.5.3a: General Spurious emissions requirements**

| Frequency Bandwidth | Measurement Bandwidth | Minimum requirement |
|---|---|---|
| 9 kHz ≤ f < 150 kHz | 1 kHz | -36 dBm |
| 150 kHz ≤ f < 30 MHz | 10 kHz | -36 dBm |
| 30 MHz ≤ f < 1000 MHz | 100 kHz | -36 dBm |
| 1 GHz ≤ f < 12.75 GHz | 1 MHz | -30 dBm |

**Table 5.5.3.5.3b: Additional Spurious emissions requirements**

| Frequency Bandwidth | Measurement Bandwidth | Minimum requirement |
|---|---|---|
| 921 MHz ≤ f < 925 MHz | 100 kHz | -60 dBm (note 1) |
| 925 MHz ≤ f ≤ 935 MHz | 100 kHz | -67 dBm (note 1) |
| 935 MHz < f ≤ 960 MHz | 100 kHz | -79 dBm (note 1) |
| 1805 MHz ≤ f ≤ 1880 MHz | 100 kHz | -71 dBm (note 1) |
| 2620 MHz ≤ f ≤ 2690 MHz | 3.84 MHz | -37 dBm (note 1) |
| 1884.5 MHz ≤ f ≤ 1919.6 MHz | 300 kHz | -41 dBm (note 2) |
| NOTE 1: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in Table 6.7E are permitted for each UARFCN used in the measurement. | | |
| NOTE 2: Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a). | | |

NOTE: The measurements are made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the applicable requirements defined in table 5.5.3.5.3a are permitted for each UARFCN used in the measurement.

NOTE 2: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 5.6 Transmit Intermodulation

## 5.6.1 Definition and applicability

The transmit intermodulation performance is a measure of the capability of the transmitter to inhibit the generation of signals in its non linear elements caused by the presence of the wanted signal and an interfering signal reaching the transmitter via the antenna.

User Equipment(s) transmitting in close vicinity of each other can produce intermodulation products, which can fall into the UE, or BS receive band as an unwanted interfering signal. The UE intermodulation attenuation is defined by the ratio of the RRC filtered mean power of the wanted signal to the RRC filtered mean power of the intermodulation product when an interfering CW signal is added at a level below the wanted signal.

The requirements of this test shall apply for all UTRA-UE.

## 5.6.2 Minimum Requirements

### 5.6.2.1 3,84 Mcps TDD Option

The requirement of transmitting intermodulation for carrier spacing 5 MHz is prescribed in the table below.

The normative reference for this requirement is TS 25.102 [1] clause 6.7.1.1.

**Table 5.6.2.1: Transmit Intermodulation (3,84 Mcps TDD Option)**

| Interference Signal Frequency Offset | 5MHz | 10MHz |
|---|---|---|
| Interference Signal Level | -40 dBc | |
| Interferer Modulation | CW<br>Note: BS Test uses a CDMA modulated signal | |
| Minimum Requirement | -31dBc | -41dBc |

### 5.6.2.2 1,28 Mcps TDD Option

The requirement of transmitting intermodulation for carrier spacing 1,6 MHz is prescribed in table 5.6.2.2.

The normative reference for this requirement is TS 25.102 [1] clause 6.7.1.1.

**Table 5.6.2.2: Transmit Intermodulation (1,28 Mcps TDD Option)**

| Interference Signal Frequency Offset | 1.6 MHz | 3.2 MHz |
|---|---|---|
| Interference Signal Level | -40 dBc | |
| Interferer Modulation | CW<br>Note: BS Test uses a CDMA modulated signal | |
| Minimum Requirement | -31dBc | -41dBc |

### 5.6.2.3 7,68 Mcps TDD Option

The requirement of transmitting intermodulation for carrier spacing 10 MHz is prescribed in the table 5.6.2.3.

The normative reference for this requirement is TS 25.102 [1] clause 6.7.1.3.

**Table 5.6.2.3: Transmit Intermodulation (7,68 Mcps TDD Option)**

| Interference Signal Frequency Offset | 10MHz | 20MHz |
|---|---|---|
| Interference Signal Level | -40 dBc | |
| Interferer Modulation | CW<br>Note: BS Test uses a CDMA modulated signal | |
| Minimum Requirement | -31dBc | -41dBc |

## 5.6.3 Test purpose

User Equipment(s) transmitting in close vicinity of each other can produce intermodulation products, which can fall into other UE, or BS receive band as an unwanted interfering signal.

It is the purpose of this test to limit interferences to the own and other systems due to intermodulation products.

## 5.6.4 Method of test

### 5.6.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and the interferer to the UE antenna connector as shown in figure A.2.
2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.
3) Enter the UE into loopback test mode and start the loopback test.

Configure parameters of the interferer according to table 5.6.2.1 for 3,84 Mcps TDD Option, table 5.6.2.2 for 1,28 Mcps TDD Option and table 5.6.2.3 for 7,68 Mcps TDD Option, respectively.

### 5.6.4.2 Procedure

#### 5.6.4.2.1 3,84 Mcps TDD Option

1) Measure the unwanted emissions according to 5.6.2. in a carrier offset spacing of 5 MHz and in a frequency range [5 MHz to 12.75 GHz], using an interferer +5MHz offset.

   The frequency occupied by the interferer is excluded from the measurement.

2) Repeat 1) with the other 3 interferer-configurations (-5Mz. +10 MHz, -10 MHz).

3) Measure the wanted power according to annex B.

4) Display 1) and 2) in dBc with respect to 3).

#### 5.6.4.2.2 1,28 Mcps TDD Option

1) Measure the unwanted emissions according to 5.6.2.2 in a carrier offset spacing of 1.6 MHz and in a frequency range [1.6 MHz to 12.75GHz], using an interferer +1.6MHz offset.

   The frequency occupied by the interferer is excluded from the measurement.

2) Repeat 1) with the other 3 interferer-configurations (-1.6 MHz, +3.2 MHz, -3.2 MHz).

3) Measure the wanted power according to annex B.

4) Display 1) and 2) in dBc with respect to 3).

#### 5.6.4.2.3 7,68 Mcps TDD Option

1) Measure the unwanted emissions according to 5.6.2.3 in a carrier offset spacing of 10 MHz and in a frequency range [10 MHz to 12.75 GHz], using an interferer +10MHz offset.

   The frequency occupied by the interferer is excluded from the measurement.

2) Repeat 1) with the other 3 interferer-configurations (-10Mz. +20 MHz, -20 MHz).

3) Measure the wanted power according to annex B.

4) Display 1) and 2) in dBc with respect to 3).

## 5.6.5 Test requirements

### 5.6.5.1 3,84 Mcps TDD Option

The results in 4) from clause 5.6.4.2.1 shall not exceed the predescribed values in table 5.6.5.1.

**Table 5.6.5.1: Transmit Intermodulation (3,84 Mcps TDD Option)**

| Interference Signal Frequency Offset | 5MHz | 10MHz |
|---|---|---|
| Interference Signal Level | -40 dBc | |
| Interferer Modulation | CW<br>Note: BS Test uses a CDMA modulated signal | |
| Minimum Requirement | [-31+TT] dBc | [-41+TT] dBc |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 5.6.5.2 1,28 Mcps TDD Option

The results in 4) from clause 5.6.4.2.2 shall not exceed the predescribed values in table 5.6.5.2.

**Table 5.6.5.2 : Transmit Intermodulation (1,28Mcps TDD Option)**

| Interference signal frequency offset | 1.6MHz | 3.2MHz |
|---|---|---|
| Interference signal level | -40dBc | |
| Minimum requirement of intermodulation products | -31dBc | -41 dBc |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 5.6.5.3 7,68 Mcps TDD Option

The results in 4) from clause 5.6.4.2.3 shall not exceed the predescribed values in table 5.6.5.3.

**Table 5.6.5.3: Transmit Intermodulation (7,68 Mcps TDD Option)**

| Interference Signal Frequency Offset | 10MHz | 20MHz |
|---|---|---|
| Interference Signal Level | -40 dBc | |
| Interferer Modulation | CW<br>Note: BS Test uses a CDMA modulated signal | |
| Minimum Requirement | [-31+TT] dBc | [-41+TT] dBc |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.7 Transmit Modulation

### 5.7.1 Error Vector Magnitude

#### 5.7.1.1 Definition and applicability

The Error Vector Magnitude (EVM) is a measure of the difference between the measured waveform and the theoretical modulated waveform (the error vector). Both waveforms pass through a matched Root Raised Cosine filter with bandwidth corresponding to the considered chip rate and roll-off $\alpha$ =0,22. Both waveforms are then further modified by selecting the frequency, absolute phase, absolute amplitude and chip clock timing so as to minimise the error vector. The EVM result is defined as the square root of the ratio of the mean error vector power to the mean reference signal power expressed as a %. The measurement interval is one timeslot excluding the guard period.

The requirement of this clause shall apply to all types of UTRA-UE.

#### 5.7.1.2 Minimum Requirements

The Error Vector Magnitude shall not exceed 17,5 % for the parameters specified in table 5.7.1.2.

**Table 5.7.1.2.: Test parameters for Error Vector Magnitude/Peak Code Domain Error**

| Parameter | Level | Unit |
|---|---|---|
| UE Output Power | ≥-20 | dBm |
| Operating conditions | Normal conditions | |
| Power control step size | 1 | dB |

The normative reference for this requirement is TS 25.102 [1] clause 6.8.2.1.

### 5.7.1.3 Test purpose

The transmitter shall generate a sufficient precise waveform, to enable the receiver to achieve the specified receiver performances.

### 5.7.1.4 Method of test

#### 5.7.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH, vibration; see clauses G.2.1, G.2.2 and G.2.3.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) A call is set up according to the generic call setup procedure using parameters as specified in table E.3.1.2.
3) Enter the UE into loopback test mode and start the loopback test.

#### 5.7.1.4.2 Procedure

1) Starting from the initial conditions, measure EVM (Error Vector Magnitude) of the UE according to annex B.
2) Set SS-level and signalling values such that the power level of the UE is between –20 and –19 dBm.
3) Measure EVM of the UE according to annex B.

### 5.7.1.5 Test requirements

The results in step 1) and 2) shall not exceed 17.5 % for parameters specified in table 5.7.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.7.1A Error Vector Magnitude with E-DCH16QAM

### 5.7.1A.1 Definition and applicability

The Error Vector Magnitude (EVM) is a measure of the difference between the measured waveform and the theoretical modulated waveform (the error vector). Both waveforms pass through a matched Root Raised Cosine filter with bandwidth corresponding to the considered chip rate and roll-off $\alpha$ =0,22. Both waveforms are then further modified by selecting the frequency, absolute phase, absolute amplitude and chip clock timing so as to minimise the error vector. The EVM result is defined as the square root of the ratio of the mean error vector power to the mean reference signal power expressed as a %. The measurement interval is one timeslot excluding the guard period.

The requirement of this clause shall apply to UTRA-UE which supports E-DCH with 16QAM.

### 5.7.1A.2 Minimum Requirements

The Error Vector Magnitude shall not exceed 14 % for the parameters specified in table 5.7.1.2.

**Table 5.7.1.2A.: Test parameters for Error Vector Magnitude/Peak Code Domain Error**

| Parameter | Level | Unit |
|---|---|---|
| UE Output Power | ≥-20 | dBm |
| Operating conditions | Normal conditions | |
| Power control step size | 1 | dB |

The normative reference for this requirement is TS 25.102 [1] clause 6.8.2.1.

## 5.7.1A.3 Test purpose

The transmitter shall generate a sufficient precise waveform, to enable the receiver to achieve the specified receiver performances.

### 5.7.1A.4 Method of test

#### 5.7.1A.4.1 Initial conditions

##### 5.7.1A.4.1.1 3.84Mcps TDD option

Void

##### 5.7.1A.4.1.2 1.28Mcps TDD option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH, vibration; see clauses G.2.1, G.2.2 and G.2.3.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.
2) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.2 and DL Reference Measurement Channel parameters are defined in Annex C4.2.1.1
4) The value of TRRI shall be set to '11100' and CRRI on E-AGCH shall be set to 1.
5) The value of PRRI is same. This ensures that the UL datarate remains constant.
6) The UE is switched on.
7) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.

##### 5.7.1A.4.1.3 7.68Mcps TDD option

Void

#### 5.7.1A.4.2 Procedure

##### 5.7.1A.4.2.1 3.84Mcps TDD Option

Void

##### 5.7.1A.4.2.2 1.28Mcps TDD Option

1) Starting from the initial conditions, measure EVM (Error Vector Magnitude) of the UE according to annex B.

2) Set SS level and signalling values such that the power level of the UE is between –20 and –19 dBm.

3) Measure EVM of the UE according to annex B.

#### 5.7.1A.4.2.3 7.68Mcps TDD Option

Void

### 5.7.1A.5 Test requirements

The results in step 1) and 2) shall not exceed 14 % for parameters specified in table 5.7.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 5.7.2 Peak code domain error

### 5.7.2.1 Definition and applicability

The code domain error is computed by projecting the error vector power onto the code domain at a specific spreading factor. The error power for each code is defined as the ratio to the mean power of the projetion onto the code, to the mean power of the composite reference waveform expressed in dB. And the Peak Code Domain Error is defined as the maximum value for Code Domain Error. The measurement interval is one timeslot.

The present document is applicable for multi-code transmission only.

The requirement of this test applies to all UTRA-UE, applicable for multi-code transmission.

### 5.7.2.2 Minimum Requirement

The peak code domain error shall not exceed -21dB at spreading factor 16 for 3,84 Mcps TDD Option, and 1,28 Mcps TDD Option . The peak code domain error shall not exceed -24dB at spreading factor 32 for 7,68 Mcps TDD Option.

The normative reference for this requirement is TS 25.102 [1] clause 6.8.3.1.

### 5.7.2.3 Test purpose

It is the purpose of this test to limit crosstalk among codes.

### 5.7.2.4 Method of test

#### 5.7.2.4.1 Initial conditions

##### 5.7.2.4.1.1 3,84 Mcps TDD Option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table 5.7.2.4.1.1 for the 3,84 Mcps TDD Option.

3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.7.2.4.1.1: Test parameters for Peak code Domain Error (3,84 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2kbps, according to annex C.2.2.1 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

5.7.2.4.1.2 1,28 Mcps TDD Option

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table 5.7.2.4.1.2 for the 1,28 Mcps TDD Option.

3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.7.2.4.1.2: Test parameters for Peak code Domain Error (1,28 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2kbps, according to annex C.2.2.2 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

5.7.2.4.1.3 7,68 Mcps TDD Option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.1.

2) A call is set up according to the generic call setup procedure using parameters as specified in table 5.7.2.4.1.3 for the 7,68 Mcps TDD Option.

3) Enter the UE into loopback test mode and start the loopback test.

**Table 5.7.2.4.1.3: Test parameters for Peak code Domain Error (7,68 Mcps TDD Option)**

| Parameter | Value/description |
|---|---|
| Reference measurement channel | Multicode 12,2kbps, according to annex C.2.2.3 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power |
| Data content | real life (sufficient irregular) |

5.7.2.4.2 Procedure

1) Starting from the initial conditions, measure peak code error(PCDE)of the UE according to annex B.

2) Set SS-level and signalling values such that the power level of the UE is between –20 and –19 dBm.

3) Measure PCDE of the UE according to annex B.

#### 5.7.2.5 Test requirements

The results in step 1) and 2) shall nor exceed -20 dB. for parameters specified in table 5.7.1.2 for 3,84 Mcps TDD Option, and 1,28 Mcps TDD Option.

The results in step 1) and 2) shall nor exceed -23 dB. for parameters specified in table 5.7.1.2 for 7,68 Mcps TDD Option.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 6 Receiver Characteristics

## 6.1 General

Receiving performance test of the UE is implemented during communicating with the SS via air interface. The procedure uses normal call protocol until the UE is communicating on traffic channel basically. (Refer to TS 34.108 [3] Common Test Environments for User Equipment (UE) Conformance Testing.) On the traffic channel, the UE provides special function for testing that is described in Logical Test Interface and the UE is tested using this function. (Refer to TS 34.109 [3] Logical Test Interface (FDD/TDD) Special conformance testing functions.)

Unless otherwise stated the receiver characteristics are specified at the antenna connector of the UE. For UE(s) with an integral antenna only, a reference antenna with a gain of 0 dBi is assumed. UE with an integral antenna may be taken into account by converting these power levels into field strength requirements, assuming a 0 dBi gain antenna. Receiver characteristics for UE(s) with multiple antennas/antenna connectors are for further study.

The UE antenna performance has a significant impact on system performance, and minimum requirements on the antenna efficiency are therefore intended to be included in future versions of the present document. It is recognized that different requirements and test methods are likely to be required for the different types of UE.

All the parameters in clause 6 are defined using the DL reference measurement channel (12,2 kbps) specified in clause C.3.3.

All Bit Error ratio (BER) measurements in clause 6 shall be performed according to the general rules for statistical testing in Annex F.6.

## 6.2 Reference sensitivity level

### 6.2.1 Definition and applicability

The reference sensitivity level is the minimum mean power received at the UE antenna connector at which the BER shall not exceed the specific value.

The requirements in this clause shall apply to all types of UTRA UE.

### 6.2.2 Minimum Requirements

#### 6.2.2.1 3,84 Mcps TDD Option

For the DL reference measurement channel 12,2 kBit/s specified in annex C, the BER shall not exceed 0.001 for the parameters specified in table 6.2.2.1.

**Table 6.2.2.1: Test parameters for reference sensitivity (3,84 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/3,84 MHz |

The normative reference for this requirement is TS 25.102 [1] clause 7.3.1.1.

### 6.2.2.2 1,28 Mcps TDD Option

The BER shall not exceed 0.001 for the parameters specified in table 6.2.2.2.

**Table 6.2.2.2: Test parameters for reference sensitivity (1,28Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -108 | dBm/1,28 MHz |

The normative reference for this requirement is TS 25.102 [1] clause 7.3.1.2.

### 6.2.2.3 7,68 Mcps TDD Option

For the DL reference measurement channel 12,2 kBit/s specified in annex C.3.1.3, the BER shall not exceed 0.001 for the parameters specified in table 6.2.2.3.

**Table 6.2.2.3: Test parameters for reference sensitivity (7,68 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/7.68 MHz |

The normative reference for this requirement is TS 25.102 [1] clause 7.3.1.3.

## 6.2.3 Test purpose

The test purpose is to verify the ability of the UE to receive a prescribed test signal at the lower end of the dynamic range under defined conditions (no interference, no multipath propagation) with a BER not exceeding a specified level. This test is also used as a reference case for other tests to allow the assessment of degradations due to various sources of interference.

## 6.2.4 Method of test

### 6.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: low range, mid range, high range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.3.

2) A call is set up according to the Generic call setup procedure.

3) Enter the UE into loopback test mode and start the loopback test.

4) The level of SS output signal measured at the UE antenna connector shall be –105 dBm for the 3,84 Mcps TDD Option and 7,68 Mcps TDD option and –108 dBm for the 1,28 Mcps TDD Option, respectively.

### 6.2.4.2 Procedure

1) Measure the BER of DCH received from the UE at the SS.

## 6.2.5 Test requirements

### 6.2.5.1 3,84 Mcps TDD Option

The measured BER, derived in step 1), shall not exceed 0.001 under conditions described in table 6.2.5.1 for the 3,84 Mcps TDD Option.

**Table 6.2.5.1: Test parameters for reference sensitivity (3,84 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -104.3 | dBm/3,84 MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.2.5.2 1,28 Mcps TDD Option

The measured BER, derived in step 1), shall not exceed 0.001 under conditions described in table 6.2.5.2 for the 1,28 Mcps TDD Option.

**Table 6.2.5.2: Test parameters for reference sensitivity (1,28 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -107.3 | dBm/1,28 MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.2.5.3 7,68 Mcps TDD Option

The measured BER, derived in step 1), shall not exceed 0.001 under conditions described in table 6.2.5.3 for the 7,68 Mcps TDD Option.

**Table 6.2.5.3: Test parameters for reference sensitivity (7,68 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -104.3 | dBm/7,68 MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 6.3 Maximum Input Level

## 6.3.1 Definition and applicability

The maximum input level is defined as the maximum mean power received at the UE antenna connector, which does not degrade the specified BER performance.

The requirements in this clause shall apply to all types of UTRA UE.

## 6.3.2 Minimum requirements

### 6.3.2.1 3,84 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.3.2.1.

**Table 6.3.2.1: Maximum input level (3,84 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | -7 | dB |
| $\hat{I}_{or}$ | -25 | dBm/3,84 MHz |

The reference for this requirement is TS 25.102 [1] clause 7.4.1.1.

### 6.3.2.2 1,28 Mcps TDD Option

The BER shall not exceed 0.001 for the parameters specified in table 6.3.2.2.

**Table 6.3.2.2: Maximum input level (1,28Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | -7 | dB |
| $\hat{I}_{or}$ | -25 | dBm/1,28 MHz |

The reference for this requirement is TS 25.102 [1] clause 7.4.1.2.

### 6.3.2.3 7,68 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.3.2.3.

**Table 6.3.2.3: Maximum input level (7,68 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | -10 | dB |
| $\hat{I}_{or}$ | -25 | dBm/7,68 MHz |

The reference for this requirement is TS 25.102 [1] clause 7.4.1.3.

## 6.3.3 Test purpose

The test purpose is to verify the ability of the UE to receive a prescribed test signal at the upper end of the dynamic range under defined conditions (no interference, no multipath propagation) with BER not exceeding a specified value.

## 6.3.4 Method of test

### 6.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS to the UE antenna connector as shown in figure A.3.
2) A call is set up according to the Generic call setup procedure.
3) Enter the UE into loopback test mode and start the loopback test.
4) The level of SS output signal measured at the UE antenna connector shall be according to table 6.3.2.1 (3,84 Mcps TDD Option), table 6.3.2.2(1,28 Mcps TDD Option) and table 6.3.2.3 (7,68 Mcps TDD Option), respectively.

### 6.3.4.2 Procedure

Measure the BER of DCH received from the UE at the SS.

## 6.3.5 Test requirements

The measured BER, derived in step 1), shall not exceed 0,001.

# 6.3A Maximum Input Level for HS-PDSCH Reception (16QAM)

## 6.3A.1 Definition and applicability

Maximum input level for HS-PDSCH reception is defined as the maximum power received at the UE antenna port, which shall not degrade the specified HSDPA throughput performance. The requirements and this test apply to all types of 1.28Mcps TDD option UE that support HSDPA (16QAM).

## 6.3A.2 Minimum requirements

The requirements are specified in terms of a minimum information bit throughput R for the DL reference channel specified in Annex C.4.2.2.2 with the addition of the parameters in Table 6.3A.1.

Using this configuration the throughput shall meet or exceed 500kbps

The reference for this requirement is TS 25.102 [1] clause 7.4.2.

**Table 6.3A.1 Minimum requirement parameters for 16QAM Maximum Input Level**

| Parameter | Unit | Value |
|---|---|---|
| $\frac{\sum HS-PDSCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/1.28 MHz | -25 |
| Redundancy and constellation version | - | 6 |
| Maximum number of HARQ transmissions | - | 1 |

## 6.3A.3 Test purpose

To verify that the UE HSDPA throughput meets the minimum requirements for the DL reference channel specified in Annex C.4.2.2.2 with the addition of the parameters specified in table 6.3A.2.

An inadequate maximum input level causes loss of coverage near the Node B.

## 6.3A.4 Method of test

### 6.3A.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

### 6.3A.4.2 Procedure

Connect the SS to the UE antenna connector as shown in figure A.1.

1) The UE is switched on.

2) An RRC connection is set-up according to the generic HSDPA set-up procedure specified in TS 34.108 [3] and SS RF parameters are given in tables 6.3A.1.

3) Measure the HS-PDSCH throughput *R* received by the UE by counting the number of NACK, ACK and statDTX on the UL HS-SICH (Throughput = blocksize*number of blocks acknowledged/time).

4) The UE is switched off.

## 6.3A.5 Test requirements

The measured throughput shall meet or exceed 500kbit/second. The minimum number of measurements required for a statistically significant result to this test is clarified in annex F.6.3, Table F.6.3.5.1.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

# 6.4 Adjacent Channel Selectivity (ACS)

## 6.4.1 Definition and applicability

Adjacent Channel Selectivity is a measure of a receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel. ACS is the ratio of the receive filter attenuation on the assigned channel frequency to the receiver filter attenuation on the adjacent channel(s).

The requirements of this test apply to all UTRA UE.

## 6.4.2 Minimum Requirements

### 6.4.2.1 3,84 Mcps TDD Option

For the UE of power class 2 and 3, the BER shall not exceed 0,001 for parameters specified in table 6.4.2.1. This test condition is equivalent to the ACS value 33 dB.

**Table 6.4.2.1: Test parameters for Adjacent Channel Selectivity (3,84 Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/3,84 MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -52 |
| $F_{uw}$ offset | MHz | +5 or -5 |

Explanatory note:

Within the reference sensitivity BER= 0.001 corresponds to a testsignal = -105 dBm/3,84 MHz and a noise level -99 dBm /3,84 MHz BW (S/I -6 dB).

Within ACS BER=0.001 is directly verified.

Known from the reference sensitivity, this corresponds to S/I –6dB in the wanted BW.

As a wanted signal of –91 dBm applied, an in-channel-interfering-signal of –85 dBm can be assumed.

Verifying a filter suppression of 33 dB indirectly, an adjacent-channel-interferer of –52 dBm is needed

The normative reference of this requirement is TS 25.102 [1] clause 7.5.

### 6.4.2.2 1,28 Mcps TDD Option

For the UE of power class 2 and 3, the BER shall not exceed 0,001 for parameters specified in table 6.4.2.2. This test condition is equivalent to the ACS value 33 dB.

**Table 6.4.2.2: Test parameters for Adjacent Channel Selectivity (1,28Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/1,28MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -54 |
| $F_{uw}$ offset | MHz | +1.6 or –1.6 |

Explanatory note:

Within the reference sensitivity BER= 0.001 corresponds to a testsignal = -108 dBm/1,28 MHz and a noise level -104 dBm /1,28 MHz BW (S/I -4 dB).

Within ACS BER=0.001 is directly verified.

Known from the reference sensitivity, this corresponds to S/I -4dB in the wanted BW.

As a wanted signal of –91 dBm applied, an in-channel-interfering-signal of -87 dBm can be assumed.

Verifying a filter suppression of 33 dB indirectly, an adjacent-channel-interferer of 54 dBm is needed

The normative reference of this requirement is TS 25.102 [1] clause 7.5.1.2.

### 6.4.2.3 7,68 Mcps TDD Option

For the UE of power class 2 and 3, the BER shall not exceed 0,001 for parameters specified in table 6.4.2.3. This test condition is equivalent to the ACS value 33 dB.

**Table 6.4.2.3: Test parameters for Adjacent Channel Selectivity (7,68 Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/7.68 MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -52 |
| $F_{uw}$ offset (3.84 Mcps Modulated) | MHz | +7.5 or -7.5 |
| $F_{uw}$ offset (7.68 Mcps Modulated) | MHz | +10 or -10 |

Explanatory note:

Within the reference sensitivity BER= 0.001 corresponds to a testsignal = -105 dBm/7,68 MHz and a noise level -99 dBm /7,68 MHz BW (S/I -6 dB).

Within ACS BER=0.001 is directly verified.

Known from the reference sensitivity, this corresponds to S/I –6dB in the wanted BW.

As a wanted signal of –91 dBm applied, an in-channel-interfering-signal of –85 dBm can be assumed.

Verifying a filter suppression of 33 dB indirectly, an adjacent-channel-interferer of –52 dBm is needed

The normative reference of this requirement is TS 25.102 [1] clause 7.5.1.3.

## 6.4.3 Test purpose

The test purpose is to verify the ability of the UE-receiver to sufficiently suppress the interfering signal in the channel adjacent to the wanted channel.

## 6.4.4 Method of test

### 6.4.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and the interferer to the UE antenna connector as shown in figure A.4.
2) A call is set up according to the Generic call setup procedure.
3) Enter the UE into loopback test mode and start the loopback test.

4) Set the signal generators to produce wanted and interference signals according table 6.4.2.1 for 3,84 Mcps TDD option, table 6.4.2.1.2 for 1,28 Mcps TDD option and table 6.4.2.1.3 for 7,68 Mcps TDD option. The interference signal shall be equivalent to a continuously running wideband CDMA signal with one code and chip frequency 3,84 Mchip/s for the 3,84 Mcps TDD Option,a 1,28 Mchp/s for the 1,28 Mcps TDD Option and 7,68 Mchip/s for the 7,68 Mcps TDD Option, respectively and rolloff 0.22.

### 6.4.4.2 Procedure

1) Set the interference signal 5 MHz for the 3,84 Mcps TDD Option,1.6 MHz for the 1,28 Mcps TDD Option and 10 MHz for the 7,68 Mcps TDD Option, respectively above the assigned channel frequency of the wanted signal.

2) Measure the BER of the wanted signal received from the UE at the SS.

3) Set the interference signal 5 MHz for the 3,84 Mcps TDD Option, 1.6 MHz for the 1,28 Mcps TDD Option and 10 MHz for the 7,68 Mcps TDD Option, respectively, below the assigned channel frequency of the wanted signal and repeat 2).

4) For the 7,68Mcps TDD option repeat steps 1 to 3 for an interfering 3.84Mcps signal with offsets 7.5MHz above and below the assigned channel frequency of the wanted signal.

## 6.4.5 Test Requirements

### 6.4.5.1 3,84 Mcps TDD Option

The measured BER, derived in step 2), shall not exceed 0,001 under conditions described in table 6.4.5.1 for the 3,84 Mcps TDD Option.

**Table 6.4.5.1: Test parameters for Adjacent Channel Selectivity (3,84 Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/3,84 MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -52 |
| $F_{uw}$ offset | MHz | +5 or -5 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

### 6.4.5.2 1,28 Mcps TDD Option

The measured BER, derived in step 2), shall not exceed 0,001 under conditions described in table 6.4.5.2 for the 1,28 Mcps TDD Option.

**Table 6.4.5.2: Test parameters for Adjacent Channel Selectivity (1,28Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/1,28MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -54 |
| $F_{uw}$ offset | MHz | +1.6 or –1.6 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.4.5.3 7,68 Mcps TDD Option

The measured BER, derived in step 2), shall not exceed 0,001 under conditions described in table 6.4.5.3 for the 7,68 Mcps TDD Option.

**Table 6.4.5.3: Test parameters for Adjacent Channel Selectivity (7,68 Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | dB | 0 |
| $\hat{I}_{or}$ | dBm/3,84 MHz | -91 |
| $I_{oac}$ mean power (modulated) | dBm | -52 |
| $F_{uw}$ offset (3.84 Mcps Modulated) | MHz | +7.5 or -7.5 |
| $F_{uw}$ offset (7.68 Mcps Modulated) | MHz | +10 or -10 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

## 6.5 Blocking Characteristics

### 6.5.1 Definition and applicability

The blocking characteristics is a measure of the receiver ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit.. The blocking performance shall apply at all frequencies except those at which a spurious response occur.

The requirements of this test apply to all UTRA UE.

### 6.5.2 Minimum Requirements

#### 6.5.2.1 3,84 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.5.2.1a and table 6.5.2.1b. For table 6.5.2.1b up to 24 exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size for the interference signal.

The normative reference for this requirement is TS 25.102 clause 7.6.1.1.

**Table 6.5.2.1a: In-band blocking (3,84 Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -102 | | dBm/3,84 MHz |
| $I_{ouw}$ mean power (modulated) | -56 (for $F_{uw\ offset}$ ±10 MHz) | -44 (for $F_{uw\ offset}$ ±15 MHz) | dBm |

**Table 6.5.2.1b: Out of band blocking (3,84 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/3,84 MHz |
| $I_{ouw}$ | -44 | -30 | -15 | dBm |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(a) | 1840 <f <1885<br>1935 <f <1995<br>2040 <f <2085 | 1815 <f <1840<br>2085 <f <2110 | 1< f <1815<br>2110< f <12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(b) | 1790 < f < 1835<br>2005 < f < 2050 | 1765 < f < 1790<br>2050 < f < 2075 | 1 < f < 1765<br>2075 < f < 12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(c) | 1850 < f < 1895<br>1945 < f < 1990 | 1825 < f < 1850<br>1990 < f < 2015 | 1 < f < 1825<br>2015 < f < 12750 | MHz |

NOTE 1: For operation referenced in 4.2(a), from 1885 <f< 1900 MHz, 1920 <f< 1935 MHz, 1995 <f< 2010 MHz and 2025<f< 2040 MHz , the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.1 shall be applied.

NOTE 2: For operation referenced in 4.2(b), from 1835 < f < 1850 MHz and 1990< f < 2005 MHz, the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.1 shall be applied.

NOTE 3: For operation referenced in 4.2(c), from 1895 < f < 1910 MHz and 1930< f < 1945 MHz, the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.1 shall be applied.

### 6.5.2.2 1,28 Mcps TDD Option

The BER shall not exceed 0.001 for the parameters specified in table 6.5.2.2a and table 6.5.2.2b. For table 6.5.2.2b up to 24 exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size.

The normative reference for this requirement is 3G TS 25.102 [1] clause 7.6.1.2.

**Table 6.5.2.2a: In-band blocking (1,28Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -105 | | dBm/1,28 MHz |
| $I_{ouw}$ mean power (modulated) | -61<br>(for $F_{uw}$ offset ±3.2 MHz) | -49<br>(for $F_{uw}$ offset ±4.8 MHz) | dBm |

**Table 6.5.2.2b: Out of band blocking (1,28Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -105 | -105 | -105 | dBm/1,28 MHz |
| $I_{ouw}$ (CW) | -44 | -30 | -15 | dBm |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(a) | 1840 <f <1895.2<br>1924.8 <f <2005.2<br>2029.8 <f <2085 | 1815 <f <1840<br>2085 <f <2110 | 1< f <1815<br>2110< f <12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(b) | 1790 < f < 1845.2<br>1994.8 < f < 2050 | 1765 < f < 1790<br>2050 < f < 2075 | 1 < f < 1765<br>2075 < f < 12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in clause 4.2(c) | 1850 < f < 1905.2<br>1934.8 < f < 1990 | 1825 < f < 1850<br>1990 < f < 2015 | 1 < f < 1825<br>2015 < f < 12750 | MHz |

NOTE 1: For operation referenced in 4.2(a), from 1895.2 <f< 1900 MHz, 1920 <f< 1924.8 MHz, 2005.2 <f< 2010 MHz and 2025<f< 2029.8 MHz , the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.2shall be applied.

NOTE 2: For operation referenced in 4.2(b), from 1845.2 < f < 1850 MHz and 1990< f < 1994.8 MHz, the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.2 shall be applied.

NOTE 3: For operation referenced in 4.2(c), from 1905.2 < f < 1910 MHz and 1930< f < 1934.8 MHz, the appropriate in-band blocking or adjacent channel selectivity in clause 6.4.2.2 shall be applied.

### 6.5.2.3 7,68 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.5.2.3a and table 6.5.2.3b. For table 6.5.2.3b up to 24 exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size for the interference signal.

The normative reference for this requirement is TS 25.102 clause 7.6.1.3.

**Table 6.5.2.3a: In-band blocking (7,68 Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -102 | | dBm/7.68 MHz |
| $I_{ouw}$ mean power (modulated) | -53 (for $F_{uw}$ offset ±20 MHz) | -41 (for $F_{uw}$ offset ±30 MHz) | dBm |

**Table 6.5.2.3b: Out of band blocking (7,68 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/7.68 MHz |
| $I_{ouw}$ (CW) | -44 | -30 | -15 | dBm |
| $F_{uw}$ For operation in frequency bands as definded in subclause 4.2(a) | 1840 <f <1870<br>1950 <f <1980<br>2055 <f <2085 | 1815 <f <1840<br>2085 <f <2110 | 1< f <1815<br>2110< f <12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in subclause 4.2(b) | 1790 < f < 1820<br>2020 < f < 2050 | 1765 < f < 1790<br>2050 < f < 2075 | 1 < f < 1765<br>2075 < f < 12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in subclause 4.2(c) | 1850 < f < 1880<br>1960 < f < 1990 | 1825 < f < 1850<br>1990 < f < 2015 | 1 < f < 1825<br>2015 < f < 12750 | MHz |
| $F_{uw}$ For operation in frequency bands as definded in subclause 4.2(d) | 2510 <f< 2540<br>2650 <f< 2680 | 2485 <f< 2510<br>2680 <f< 2705 | 1 <f< 2485<br>2705 <f< 12750 | MHz |
| 1. For operation referenced in 4.2(a), from 1870 <f< 1900 MHz, 1920 <f< 1950 MHz, 1980 <f< 2010 MHz and 2025<f< 2055 MHz , the appropriate in-band blocking or adjacent channel selectivity in section 6.4.2.3 shall be applied. | | | | |
| 2. For operation referenced in 4.2(b), from 1820 < f < 1850 MHz and 1990< f < 2020 MHz, the appropriate in-band blocking or adjacent channel selectivity in section 6.4.2.3 shall be applied. | | | | |
| 3. For operation referenced in 4.2(c), from 1880 < f < 1910 MHz and 1930< f < 1960 MHz, the appropriate in-band blocking or adjacent channel selectivity in section 6.4.2.3 shall be applied. | | | | |
| 4. For operation referenced in 4.2(d), from 2540 < f < 2570 MHz and 2620< f < 2650 MHz, the appropriate in-band blocking or adjacent channel selectivity in section 6.4.2.3 shall be applied. | | | | |

# Table 6.5.2.3c: Additional Out of band blocking (7,68 Mcps TDD Option)

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/7.68 MHz |
| $I_{ouw}$ (CW) | -44 | -30 | -15 | dBm |
| $F_{uw}$ For operation in frequency bands in 2010-2025 MHz as definded in subclause 5.2(a) | 1840 <f <1995<br>2040 <f <2085 | 1815 <f <1840<br>2085 <f <2110 | 1< f <1815<br>2110< f <12750 | MHz |
| NOTE 1: Additional requirement is applied for Band a) UE operating on 2010-2025MHz. | | | | |

## 6.5.3 Test purpose

"The test stresses the ability of the UE receiver to withstand high-level interference from unwanted signals at frequency offsets of 10 MHz or more, without undue degradation of its sensitivity."

## 6.5.4 Method of test

### 6.5.4.1 Initial conditions

For in-band case:

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

For out-of-band case:

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequency to be tested: 1 arbitrary frequency chosen from the low, mid or high range; see clause G.2.4.

1) Connect the SS and the interfering Signal generator to the UE antenna connector as shown in figure A.5.

2) A call is set up according to the Generic call setup procedure.

3) Enter the UE into loopback test mode and start the loopback test.

### 6.5.4.2 Procedure

1) The wanted signal frequency channel is set to mid range frequency. The wanted signal power level shall be set according to Table 6.5.5.1a for the 3,84 Mcps TDD option,Table 6.5.5.2a for the 1,28 Mcps TDD option and Table 6.5.5.3a for the 7,68 Mcps TDD option.

2) The interfering Signal Generator is stepped through the frequency range indicated in table 6.5.2.1a for the 3,84 Mcps TDD Option, table 6.5.2.2a for the 1,28 Mcps TDD Option and table 6.5.2.3a for the 7,68 Mcps TDD Option, respectively with a step size of 1 MHz. The interfering signal level shall be set according to Table 6.5.5.1a for the 3,84 Mcps TDD option, Table 6.5.5.2a for the 1,28 Mcps TDD option and Table 6.5.5.3a for the 7,68 Mcps TDD option.

3) The interference signal shall be equivalent to a continuously running wideband CDMA signal with one code and chip frequency 3,84 Mchip/s for the 3,84 Mcps TDD Option, 1,28 Mchp/s for the 1,28 Mcps TDD Option and 7,68 Mchip/s for the 7,68 Mcps TDD Option, respectively and rolloff 0.22.

4) Measure the BER of the wanted signal received from the UE at the SS for each step of the interferer.

5) The wanted signal frequency channel is set to an arbitrary frequency chosen from the low, mid or high range. The level of the wanted signal shall be set according to Table 6.5.5.1b for the 3,84 Mcps TDD option, table 6.5.5.2b for the 1,28 Mcps TDD option and table 6.5.5.3b for the 7,68 Mcps TDD option.

6) The interfering Signal Generator is stepped through the frequency range indicated in table 6.5.2.1b for the 3,84 Mcps TDD Option, table 6.5.2.2b for the 1,28 Mcps TDD Option and table 6.5.2.3b for the 7,68 Mcps TDD Option, respectively with a step size of 1 MHz. The interfering signal level shall be set according to Table 6.5.5.1b for the 3,84 Mcps TDD option, Table 6.5.5.2b for the 1,28 Mcps TDD option and Table 6.5.5.3b for the 7,68 Mcps TDD option.

7) The interference signal is a CW signal.

8) Measure the BER of the wanted signal received from the UE at the SS for each step of the interferer.

9) Record the frequencies for which BER exceed the test requirements in Table 6.5.5.1b for the 3,84 Mcps TDD option, table 6.5.5.2b for the 1,28 Mcps TDD option and table 6.5.5.3b for the 7,68 Mcps TDD option. These frequencies are further proceeding in subclause 6.6 Spurious Response.

10) For 7,68Mcps TDD option repeat steps 1 to 9 except use the channel power levels defined for the addition test conditions specigfied in 6.5.5.3c and the frequency ranges specified in 6.5.2.3c.

NOTE: Due to the large amount of time-consuming BER tests it is recommended to speed up a single BER test by reducing the 0.001-BER confidence level [10 000 bits under test or 10 errors] for screening the critical frequencies. Critical frequencies must be identified using standard BER confidence level. [30 000 bits or 30 errors].

## 6.5.5 Test requirements

### 6.5.5.1 3,84 Mcps TDD Option

The measured BER, derived in step 4) shall not exceed 0,001 (without exception) under test conditions described in table 6.5.5.1a.

The measured BER, derived in step 8), shall not exceed 0,001 except for up to 24 different frequencies of the interfering signal under test conditions described in table 6.5.5.1b.

These frequencies are further processed in clause 6.6 Spurious response.

**Table 6.5.5.1a: Test conditions In-band blocking (3,84 Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -102 | | dBm/3,84 MHz |
| $I_{ouw}$ mean power (modulated) | -56 (for $F_{uw\ offset}$ ±10 MHz) | -44 (for $F_{uw\ offset}$ ±15 MHz) | dBm |

**Table 6.5.5.1b: Test conditions Out of band blocking (3,84 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/3,84 MHz |
| $I_{ouw\ (CW)}$ | -44 | -30 | -15 | dBm |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.5.5.2 1,28 Mcps TDD Option

The measured BER, derived in step 4), shall not exceed 0,001 (without exception) under test conditions described in table 6.5.5.2a.

The measured BER, derived in step 8), shall not exceed 0,001 except for up to 24 different frequencies of the interfering signal under test conditions described in table 6.5.5.2b.

These frequencies are further processed in clause 6.6 Spurious response.

**Table 6.5.5.2a: Test conditions In-band blocking (1,28 Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -105 | | dBm/1,28 MHz |
| $I_{ouw}$ mean power (modulated) | -61 (for $F_{uw}$ offset ±3.2 MHz) | -49 (for $F_{uw}$ offset ±4.8 MHz) | dBm |

**Table 6.5.5.2b: Test conditions Out of band blocking (1,28 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -105 | -105 | -105 | dBm/1,28 MHz |
| $I_{ouw}$ | -44 | -30 | -15 | dBm |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4

### 6.5.5.3 7,68 Mcps TDD Option

The measured BER, derived in step 4) shall not exceed 0,001 (without exception) under test conditions described in table 6.5.5.3a.

The measured BER, derived in step 8), shall not exceed 0,001 except for up to 24 different frequencies of the interfering signal under test conditions described in table 6.5.5.3b and 6.5.5.3c.

These frequencies are further processed in clause 6.6 Spurious response.

**Table 6.5.5.3a: Test conditions In-band blocking (7,68 Mcps TDD Option)**

| Parameter | Level | | Unit |
|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | | dB |
| $\hat{I}_{or}$ | -102 | | dBm/7,68 MHz |
| $I_{ouw}$ mean power (modulated) | -53 (for $F_{uw\ offset}$ ±20 MHz) | -41 (for $F_{uw\ offset}$ ±30 MHz) | dBm |

**Table 6.5.5.3b: Test conditions Out of band blocking (7,68 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/7,68 MHz |
| $I_{ouw\ (CW)}$ | -44 | -30 | -15 | dBm |

**Table 6.5.5.3c: Test conditions additional Out of band blocking (7,68 Mcps TDD Option)**

| Parameter | Band 1 | Band 2 | Band 3 | Unit |
|---|---|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | 0 | 0 | dB |
| $\hat{I}_{or}$ | -102 | -102 | -102 | dBm/7.68 MHz |
| $I_{ouw}$ (CW) | -44 | -30 | -15 | dBm |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 6.6 Spurious Response

## 6.6.1 Definition and applicability

Spurious response is a measure of the receiver's ability to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency at which a response is obtained i.e. for which the blocking limit is not met.

The requirements of this test apply to all types of UTRA for the UE.

## 6.6.2 Minimum Requirements

### 6.6.2.1 3,84 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.6.2.1.

The normative reference for this requirement is TS 25.102 clause 7.7.1.1.

**Table 6.6.2.1: Spurious Response (3,84 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -102 | dBm/3,84 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

### 6.6.2.2 1,28 Mcps TDD Option

The BER shall not exceed 0.001 for the parameters specified in table 6.6.2.2.

**Table 6.6.2.2: Spurious Response (1,28Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/1,28 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

### 6.6.2.3 7,68 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.6.2.3.

The normative reference for this requirement is TS 25.102 clause 7.7.1.3.

**Table 6.6.2.3: Spurious Response (7,68 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -102 | dBm/7.68 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

## 6.6.3 Test purpose

Spurious response frequencies, identified in the blocking test, are measured against a less stringent test requirement. The test stresses the ability of the receiver to withstand high level interference signals without undue degradation of its sensitivity due to the receiver's frequency conversion concept.

## 6.6.4 Method of test

### 6.6.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequency to be tested: the same frequency as chosen in subclause 6.5.4.1 for Blocking characteristics out-of-band case.

1) Connect the SS and the unwanted signal to the UE antenna connector as shown in figure A.6.

2) A call is set up according to the Generic call setup procedure.

3) Enter the UE into loopback test mode and start the loopback test.

### 6.6.4.2 Procedure

1) Set the wanted signal frequency to the frequency used for the out-of-band blocking test. Set the power level of the wanted signal according to table 6.6.2.1 for the 3,84 Mcps TDD Option, table 6.6.2.2 for the 1,28 Mcps TDD Option and table 6.6.2.3 for the 7,68 Mcps TDD Option, respectively.

2) Set the frequency of the interferer signal according the recorded spurious response frequency values obtained from the out-of-band blocking test as described in 6.5.4.2, at which the blocking test failed. Set the power level of the interferer according to table 6.6.5.1 for the 3,84 Mcps TDD Option, table 6.6.5.2 for the 1,28 Mcps TDD Option and table 6.6.5.3 for the 7,68 Mcps TDD Option, respectively.

3) Measure the BER of DCH received from the UE at the SS.

## 6.6.5 Test requirements

### 6.6.5.1 3,84 Mcps TDD

The measured BER, derived in step 3), shall not exceed 0,001 under. test conditions described in table 6.6.5.1 for the 3,84 Mcps TDD Option.

**Table 6.6.5.1: Test Parameters Spurious Response (3,84 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |

| | | |
|---|---|---|
| $\hat{I}_{or}$ | -102 | dBm/3,84 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.6.5.2 1,28 Mcps TDD

The measured BER, derived in step 3), shall not exceed 0,001 under test conditions described in table 6.6.5.2 for the 1,28 Mcps TDD Option.

**Table 6.6.5.2: Test Parameters Spurious Response (1,28 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/1,28 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.6.5.3 7,68 Mcps TDD

The measured BER, derived in step 3), shall not exceed 0,001 under. test conditions described in table 6.6.5.3 for the 7,68 Mcps TDD Option.

**Table 6.6.5.3: Test Parameters Spurious Response (7,68 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -102 | dBm/3,84 MHz |
| $I_{ouw}$ (CW) | -44 | dBm |
| $F_{uw}$ | Spurious response frequencies | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 6.7 Intermodulation Characteristics

## 6.7.1 Definition and applicability

Third and higher order mixing of the two interfering RF signals can produce an interfering signal in the band of the desired channel. Intermodulation response rejection is a measure of the capability of the receiver to receiver a wanted signal on its assigned channel frequency in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal.

The requirements of this test shall apply to all UTRA UE.

## 6.7.2 Minimum Requirements

### 6.7.2.1 3,84 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.7.2.1.

The normative reference for this requirement is TS 25.102 clause 7.8.1.1.

**Table 6.7.2.1: Receive intermodulation characteristics (3,84 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -102 | dBm/3,84 MHz |
| $I_{ouw1(CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±10 | MHz |
| $F_{uw2}$ (Modulated) | ±20 | MHz |

### 6.7.2.2 1,28 Mcps TDD Option

The BER shall not exceed 0.001 for the parameters specified in table 6.7.2.2.

The normative reference for this requirement is TS 25.102 clause 7.8.1.2.

**Table 6.7.2.2: Receive intermodulation characteristics (1,28 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/1,28 MHz |
| $I_{ouw1\ (CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±3.2 | MHz |
| $F_{uw2}$ (Modulated) | ±6.4 | MHz |

### 6.7.2.3 7,68 Mcps TDD Option

The BER shall not exceed 0,001 for the parameters specified in table 6.7.2.3.

The normative reference for this requirement is TS 25.102 clause 7.8.1.3.

**Table 6.7.2.3: Receive intermodulation characteristics (7,68 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -102 | dBm/7.68 MHz |
| $I_{ouw1\ (CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±20 | MHz |
| $F_{uw2}$ (modulated) | ±40 | MHz |

## 6.7.3 Test purpose

The test stresses the ability of the receiver to withstand two or more high level interference signals without undue degradation of its sensitivity due to the receiver's non-linear elements.

## 6.7.4 Method of test

### 6.7.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and the unwanted signals to the UE antenna connector as shown in figure A.7.
2) A call is set up according to the Generic call setup procedure.
3) Enter the UE into loopback test mode and start the loopback test.

### 6.7.4.2 Procedure

1) Set the wanted and interfering signals as indicated in table 6.7.2.1 for the 3,84 Mcps TDD Option, table 6.7.2.2 for the 1,28 Mcps TDD Option and table 6.7.2.3 for the 7,68 Mcps TDD Option with positive offset with respect to the wanted signal.
2) Measure the BER of DCH received from the UE at the SS.
3) Set the interfering signals as indicated in table 6.7.2.1 for the 3,84 Mcps TDD Option, table 6.7.2.2 for the 1,28 Mcps TDD Option and table 6.7.2.3 for the 7,68 Mcps TDD Option with negative offset with respect to the wanted signal and repeat 2).

## 6.7.5 Test requirements

### 6.7.5.1 3,84 Mcps TDD Option

The measured BER, derived in step 2) and 3), shall not exceed 0,001 under test conditions described in table 6.7.5.1 for the 3,84 Mcps TDD Option.

**Table 6.7.5.1: Test parameters Receive intermodulation characteristics (3,84 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ Wanted Signal Level | -102 | dBm/3,84 MHz |
| $I_{ouw1(CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±10 | MHz |
| $F_{uw2}$ (Modulated) | ±20 | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.7.5.2 1,28 Mcps TDD Option

The measured BER, derived in step 2) and 3), shall not exceed 0,001 under test conditions described in table 6.7.5.2 for the 1,28 Mcps TDD Option.

**Table 6.7.5.2: Test parameters Receive intermodulation characteristics (1,28 Mcps TDD Option)**

| Parameter | Level | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ | -105 | dBm/1,28 MHz |
| $I_{ouw1\ (CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±3.2 | MHz |
| $F_{uw2}$ (Modulated) | ±6.4 | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in Annex F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.4.

### 6.7.5.3 7,68 Mcps TDD Option

The measured BER, derived in step 2) and 3), shall not exceed 0,001 under test conditions described in table 6.7.5.3 for the 7,68 Mcps TDD Option.

**Table 6.7.5.3: Test parameters Receive intermodulation characteristics (7,68 Mcps TDD Option)**

| Parameter | Value | Unit |
|---|---|---|
| $\frac{\Sigma DPCH_Ec}{I_{or}}$ | 0 | dB |
| $\hat{I}_{or}$ Wanted Signal Level | -102 | dBm/3,84 MHz |
| $I_{ouw1(CW)}$ | -46 | dBm |
| $I_{ouw2}$ mean power (modulated) | -46 | dBm |
| $F_{uw1}$ (CW) | ±20 | MHz |
| $F_{uw2}$ (Modulated) | ±40 | MHz |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

## 6.8 Spurious Emissions

### 6.8.1 Definition and applicability

The Spurious Emissions Power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector.

The requirements of this test are applicable for all UTRA UE.

### 6.8.2 Minimum Requirements

#### 6.8.2.1 3,84 Mcps TDD Option

The power of any spurious emission shall not exceed:

**Table 6.8.2.1: Receiver spurious emission requirements (3,84 Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz – 1,9 GHz and 1,92 GHz – 2,01 GHz and 2,025 GHz – 2,11 GHz | -47 dBm | 1 MHz | |
| 1,9 GHz – 1,92 GHz and 2,01 GHz – 2,025 GHz and 2,11 GHz – 2,170 GHz | -60 dBm | 3,84 MHz | |
| 2,170 GHz – 12,75 GHz | -47 dBm | 1 MHz | |

The normative reference for this requirement is TS 25.102 [1] clause 7.9.1.1.

#### 6.8.2.2 1,28 Mcps TDD Option

The power of any spurious emission shall not exceed.

**Table 6.8.2.2: Receiver spurious emission requirements (1,28Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz – 1.9 GHz and<br>1.92 GHz – 2.01 GHz and<br>2.025 GHz – 2.11 GHz | -47 dBm | 1 MHz | |
| 1.9 GHz – 1.92 GHz and<br>2.01 GHz – 2.025 GHz and<br>2.11 GHz – 2.170 GHz | -64 dBm | 1,28 MHz | |
| 2.170 GHz – 12.75 GHz | -47 dBm | 1 MHz | |

The normative reference for this requirement is TS 25.102 [1] clause 7.9.1.2.

### 6.8.2.3 7,68 Mcps TDD Option

The power of any spurious emission shall not exceed:

**Table 6.8.2.3: Receiver spurious emission requirements (7,68 Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz - 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz - 1.9 GHz and<br>1.92 GHz - 2.01 GHz and<br>2.025 GHz - 2.11 GHz<br>2.17 GHz - 2.57 GHz | -47 dBm | 1 MHz | |
| 1.9 GHz - 1.92 GHz and<br>2.01 GHz - 2.025 GHz and<br>2.11 GHz - 2.170 GHz<br>2.57 Ghz - 2.69 GHz | -57 dBm | 7.68 MHz | |
| 2.69 GHz - 12.75 GHz | -47 dBm | 1 MHz | |

The normative reference for this requirement is TS 25.102 [1] clause 7.9.1.3.

## 6.8.3 Test purpose

The test purpose is to verify the UE's ability to limit interference caused by receiver spurious emissions to the own and the other systems. The test requirements are tighter than in clause 5.5.3 ((TX) Spurious Emissions) because the time of Receive–Only-Operation is generally much longer than RX-TX-Operation.

## 6.8.4 Method of test

### 6.8.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: 1 arbitrary frequency selected between low and high range; see clause G.2.4.

1) Connect the measurement equipment to the UE antenna connector according to figure A.8.

2) RF parameters are setup according to table 6.8.4.1a for 3,84 Mcps TDD option, 6.8.4.1b for 1,28 Mcps TDD and table 6.8.4.1a for 7,68 Mcps TDD option respecively.

3) A call is set up according to the setup procedure specified in TS34.108 [3] sub-clause 7.3.3, with the following exceptions for information elements in System Information Block type3.

| Information Element | Value/Remark |
|---|---|
| - Cell selection and re-selection info | |
| - CHOICE mode | TDD |
| - Sintrasearch | 0 dB |
| - Sintersearch | 0 dB |
| - RAT List | This parameter is configurable |
| - Ssearch,RAT | 0 dB |
| - Maximum allowed UL TX power | Power level where Pcompensation=0 |

Note 1: The setup procedure (3) sets the UE into CELL_FACH state. With this state and the SS level (2) it is ensured that UE continuously monitors the S-CCPCH and no cell reselections are performed [see 25.304, subcl. 5.2.3.and 5.2.6]. No transmission of the UE will interfere with the measurement.

**Table 6.8.4.1a: RF parameters for receiver spurious test (3,84Mcps TDD Option and 7.68Mcps TDD option)**

| Parameter | Unit | Level |
|---|---|---|
| PCCPCH_Ec/Ior | dB | -3 |
| SCH_Ec/Ior | dB | -9 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 |
| PCCPCH RSCP | dBm | -64 |

**Table 6.8.4.1b: RF parameters for receiver spurious test (1,28Mcps TDD Option)**

| Parameter | Unit | Level |
|---|---|---|
| PCCPCH_Ec/Ior | dB | -3 |
| DwPCH_Ec/Ior | dB | 0 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 |
| PCCPCH RSCP | dBm | -64 |

### 6.8.4.2 Procedure

Measure the average power of spurious emissions by covering the frequency ranges of table 6.8.2.1 for the 3,84 Mcps TDD Option, table 6.8.2.2 for the 1,28 Mcps TDD Option and table 6.8.2.3 for the 7,68 Mcps TDD Option.

## 6.8.5 Test requirements

### 6.8.5.1 3,84 Mcps TDD Option

The spurious emissions shall be according to table 6.8.5.1 for the 3,84 Mcps TDD Option.

**Table 6.8.5.1: Receiver spurious emission test requirements (3,84 Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz – 1,9 GHz and 1,92 GHz – 2,01 GHz and 2,025 GHz – 2,11 GHz | -47 dBm | 1 MHz | |
| 1,9 GHz – 1,92 GHz and 2,01 GHz – 2,025 GHz and 2,11 GHz – 2,170 GHz | -60 dBm | 3,84 MHz | |
| 2,170 GHz – 12,75 GHz | -47 dBm | 1MHz | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.8.5.2 1,28 Mcps TDD Option

The spurious emissions shall be according to table 6.8.5.2 for the 1,28 Mcps TDD Option.

**Table 6.8.5.2: Receiver spurious emission requirements (1,28Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz – 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz – 1,9 GHz and<br>1,92 GHz – 2,01 GHz and<br>2,025 GHz – 2,11 GHz | -47 dBm | 1 MHz | |
| 1,9 GHz – 1,92 GHz and<br>2,01 GHz – 2,025 GHz and<br>2,11 GHz – 2,170 GHz | -64 dBm | 1,28 MHz | |
| 2,170 GHz – 12,75 GHz | -47 dBm | 1 MHz | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

### 6.8.5.3 7,68 Mcps TDD Option

The spurious emissions shall be according to table 6.8.5.3 for the 7,68 Mcps TDD Option.

**Table 6.8.5.3: Receiver spurious emission test requirements (7,68 Mcps TDD Option)**

| Band | Maximum level | Measurement Bandwidth | Note |
|---|---|---|---|
| 30 MHz - 1 GHz | -57 dBm | 100 kHz | |
| 1 GHz - 1.9 GHz and<br>1.92 GHz - 2.01 GHz and<br>2.025 GHz - 2.11 GHz<br>2.17 GHz - 2.57 GHz | -47 dBm | 1 MHz | |
| 1.9 GHz - 1.92 GHz and<br>2.01 GHz - 2.025 GHz and<br>2.11 GHz - 2.170 GHz<br>2.57 Ghz - 2.69 GHz | -57 dBm | 7.68 MHz | |
| 2.69 GHz - 12.75 GHz | -47 dBm | 1 MHz | |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in annex F clause F.4.

# 7 Performance Requirements

## 7.1 General

The performance requirements for the UE in this clause is specified for the measurement channels specified in annex C and the test environments specified in annex D.

All Block Error ratio (BLER) measurements in clause 7 shall be performed according to the general rules for statistical testing in Annex F.6.

### 7.1.2 Definition of Additive White Gaussian Noise (AWGN) Interferer

The minimum bandwidth of the AWGN interferer shall be 1.5 times chip rate of the radio access mode. (e.g. 5.76 MHz for a chip rate of 3,84 Mcps). The flatness across this minimum bandwidth shall be less than ±0.5 dB and the peak to average ratio at a probability of 0.001% shall exceed 10 dB.

## 7.2 Demodulation in static propagation conditions

### 7.2.1 Demodulation of DCH

#### 7.2.1.1 Definition and applicability

The performance requirement of DCH in static propagation conditions is determined by the maximum Block Error Ratio (BLER ). The BLER is specified for each individual data rate of the DCH. DCH is mapped into the Dedicated Physical Channel (DPCH).

The UE shall be tested only according to the datarates, supported. The data-rate-corresponding requirements shall apply to the UE.

#### 7.2.1.2 Minimum requirements

##### 7.2.1.2.1 3,84 Mcps TDD Option

For the parameters specified in table 7.2.1.2.1a the BLER shall not exceed the piece-wise linear BLER curve specified in table 7.2.1.2.1b. These requirements are applicable for TFCS size 16.

The reference for this requirement is TS 25.102 [1] clause 8.2.1.1.1.

**Table 7.2.1.2.1a: DCH parameters in static propagation conditions (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 |
|---|---|---|---|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | -6 | -3 | 0 | 0 | 0 |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | | |
| Cell Parameter* | | 0,1 | | | | |
| DPCH Channelization Codes* | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1 . .5 | C(i,16) i=1 . .9 | C(i,16) i=1 . .8 | - |
| OCNS Channelization Code* | C(k,Q) | C(3,16) | C(6,16) | - | - | - |
| Information Data Rate | kbps | 12,2 | 64 | 144 | 384 | 2048 |
| Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | | | | | |

**Table 7.2.1.2.1b: Performance requirements in AWGN channel (3,84 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 1,1 | $10^{-2}$ |
| 2 | 3,5 | $10^{-1}$ |
| | 3,8 | $10^{-2}$ |
| 3 | 3,4 | $10^{-1}$ |
| | 3,6 | $10^{-2}$ |
| 4 | 2,7 | $10^{-1}$ |
| | 3,0 | $10^{-2}$ |
| 5 | 3,5 | $10^{-1}$ |
| | 3,6 | $10^{-2}$ |

### 7.2.1.2.2 1,28 Mcps TDD Option

For the parameters specified in table 7.2.1.2.2a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.2.1.2.2b. The reference for this requirement is TS 25.102 [1] clause 8.2.1.1.2.

**Table 7.2.1.2.2a: DCH parameters in static propagation conditions (1,28Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| Number of $DPCH_o$ | | 8 | 2 | 2 | 0 |
| Scrambling code and basic midamble code number* | | 0 | 0 | 0 | 0 |
| DPCH Channelization Codes* | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1…8 | C(i,16) i=1…8 | C(i,16) i=1…10 |
| $DPCH_o$ Channelization Codes* | C(k,Q) | C(i,16) 3≤ i ≤10 | C(i,16) 9≤ i ≤10 | C(i,16) 9≤ i ≤10 | - |
| $\frac{DPCH_o_E_c}{I_{or}}$ | dB | -10 | -10 | -10 | 0 |
| $I_{oc}$ | dBm/1,28MHz | -60 | | | |
| Midamble | | Common midamble | | | |
| Information Data Rate | Kbps | 12.2 | 64 | 144 | 384 |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 7.2.1.2.2b: Performance requirements in AWGN channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 3.6 | $10^{-2}$ |
| 2 | 2.4 | $10^{-1}$ |
| | 2.7 | $10^{-2}$ |
| 3 | 2.8 | $10^{-1}$ |
| | 3.2 | $10^{-2}$ |
| 4 | 3.2 | $10^{-1}$ |

### 7.2.1.2.3 7,68 Mcps TDD Option

For the parameters specified in table 7.2.1.2.3a the BLER shall not exceed the piece-wise linear BLER curve specified in table 7.2.1.2.3b. These requirements are applicable for TFCS size 16.

The reference for this requirement is TS 25.102 [1] clause 8.2.1.1.3.

**Table 7.2.1.2.3a: DCH parameters in static propagation conditions (7,68 Mcps TDD Option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | -9 |
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| Cell Parameter* | - | 0,1 |
| DPCH Channelization Codes* | C(k, Q) | C(i, 32), i = 1,2 |
| OCNS Channelization Code* | C(k, Q) | C(3, 32) |
| Information Data Rate | kbps | 12.2 |
| *Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | |

**Table 7.2.1.2.3b: Performance requirements in AWGN channel (7,68 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 1.1 | $10^{-2}$ |

## 7.2.1.3 Test purpose

While the receiver tests in clause 6 aims for the RF hardware, this performance requirement aims for the receiver's signal processing.

The test purpose is to verify the ability of the receiver to receive a predefined test signal ,representing a static propagation channel for the wanted and for the co-channel signals from serving and adjacent cells, with a block error ratio (BLER) not exceeding a specified value.

## 7.2.1.4 Method of test

### 7.2.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS, AWGN Generator and additional components to the UE antenna connector as shown in figure A.9.

2) A call is set up according to the Generic call setup procedure. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) (64 kbit/s), (144 kbit/s), and (384 kbit/s) specified in annex C for 3.84 Mcps TDD option and 1.28Mcps TDD option. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) for 7.68 Mcps TDD option.

3) Enter the UE into loopback test mode and start the loopback test. (test 1) and/or activate the Ack/Nack test mode (test 1 to test 4).

4) The levels of the wanted signal and the co-channel signals are set according to table 7.2.1.2.1a and b for the 3,84 Mcps TDD Option, table 7.2.1.2.2a and b for the 1,28 Mcps TDD Option and table 7.2.1.2.3a and b for the 7,68 Mcps TDD Option, respectively.

### 7.2.1.4.2 Procedure

Measure the BLER of DCH received from the UE at the SS for all tests specified in table 7.2.1.2.1a for the 3,84 Mcps TDD Option, table 7.2.1.2.2a for the 1,28 Mcps TDD Option and table 7.2.1.2.3a for the 7,68 Mcps TDD Option, respectively.

## 7.2.1.5 Test requirements

### 7.2.1.5.1 3.84Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.2.1.2.1b for the 3,84 Mcps TDD Option

#### 7.2.1.5.2 1.28Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.2.1.5.2 for the1.28 Mcps TDD Option

**Table 7.2.1.5.2: Performance requirements in AWGN channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 3.9 | $10^{-2}$ |
| 2 | 2.7 | $10^{-1}$ |
| | 3.0 | $10^{-2}$ |
| 3 | 3.1 | $10^{-1}$ |
| | 3.5 | $10^{-2}$ |
| 4 | 3.5 | $10^{-1}$ |

#### 7.2.1.5.3 7.68Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.2.1.2.3b for the 7.68 Mcps TDD Option

NOTE: If the Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 7.3 Demodulation of DCH in multipath fading conditions

### 7.3.1 Multipath fading Case 1

#### 7.3.1.1 Definition and applicability

The performance requirement of DCH is determined by the maximum Block Error Ratio (BLER). The BLER is specified for each individual data ratio of the DCH. DCH is mapped into the Dedicated Physical Channel (DPCH).

The UE shall be tested only according to the dataratios, supported. The data-ratio-corresponding requirements shall apply to the UE.

#### 7.3.1.2 Minimum requirements

##### 7.3.1.2.1 3,84 Mcps TDD Option

For the parameters specified in table 7.3.1.2.1a the BLER shall not exceed the piece-wise linear BLER curve specified in table 7.3.1.2.1b. These requirements are applicable for TFCS size 16.

The reference for this requirement is TS 25.102 clause 8.3.1.1.1.

**Table 7.3.1.2.1a: DCH parameters in multipath Case 1 channel (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 |
|---|---|---|---|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | DB | -6 | -3 | 0 | 0 | 0 |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | | |
| Cell Parameter (note) | | 0,1 | | | | |
| DPCH Channelization Codes (note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1 . .5 | C(i,16) i=1 . .9 | C(i,16) i=1 . .8 | - |
| OCNS Channelization Code (note) | C(k,Q) | C(3,16) | C(6,16) | - | - | - |
| Information Data Rate | kbps | 12,2 | 64 | 144 | 384 | 2048 |
| Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | | | | | |

**Table 7.3.1.2.1b: Performance requirements in multipath Case 1 channel (3,84 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 13,9 | $10^{-2}$ |
| 2 | 13,7 | $10^{-1}$ |
| | 19,8 | $10^{-2}$ |
| 3 | 14,1 | $10^{-1}$ |
| | 20,6 | $10^{-2}$ |
| 4 | 13,8 | $10^{-1}$ |
| | 20,0 | $10^{-2}$ |
| 5 | 13,2 | $10^{-1}$ |
| | 17,8 | $10^{-2}$ |

### 7.3.1.2.2 1,28 Mcps TDD Option

For the parameters specified in table 7.3.1.2.2a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.1.2.2b. The reference for this requirement is TS 25.102 [1] clause 8.3.1.1.2.

**Table 7.3.1.2.2a: DCH parameters in a multipath Case 1 channel (1,28 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| Number of $DPCH_o$ | | 8 | 2 | 2 | 0 |
| Scrambling code and basic midamble code number (see note) | | 0 | 0 | 0 | 0 |
| DPCH Channelization Codes (see note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1…8 | C(i,16) i=1…8 | C(i,16) i=1…10 |
| $DPCH_o$ Channelization Codes (see note) | C(k,Q) | C(i,16) 3≤ i ≤10 | C(i,16) 9≤ i ≤10 | C(i,16) 9≤ i ≤10 | - |
| $\frac{DPCH_o_E_c}{I_{or}}$ | DB | -10 | -10 | -10 | 0 |
| $I_{oc}$ | dBm/1,28MHz | -60 | | | |
| Midamble | | Common midamble | | | |
| Information Data Rate | Kbps | 12.2 | 64 | 144 | 384 |
| Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 7.3.1.2.2b: Performance requirements in a multipath Case 1 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 22.4 | $10^{-2}$ |
| 2 | 15.8 | $10^{-1}$ |
| | 22.9 | $10^{-2}$ |
| 3 | 16.6 | $10^{-1}$ |
| | 23.9 | $10^{-2}$ |
| 4 | 16.5 | $10^{-1}$ |
| | 23.5 | $10^{-2}$ |

### 7.3.1.2.3 7,68 Mcps TDD Option

For the parameters specified in table 7.3.1.2.3a the BLER shall not exceed the piece-wise linear BLER curve specified in table 7.3.1.2.3b. This requirements is applicable for TFCS size 16.

The reference for this requirement is TS 25.102 clause 8.3.1.1.3.

**Table 7.3.1.2.3a: DCH parameters in multipath Case 1 channel (7,68 Mcps TDD Option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | -9 |
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| Cell Parameter* | - | 0,1 |
| DPCH Channelization Codes* | C(k, Q) | C(i, 32), i = 1,2 |
| OCNS Channelization Code* | C(k, Q) | C(3, 32) |
| Information Data Rate | kbps | 12.2 |
| *Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | |

**Table 7.3.1.2.3b: Performance requirements in multipath Case 1 channel (7,68 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 13.9 | $10^{-2}$ |

## 7.3.1.3 Test purpose

While the receiver tests in clause 6 aims for the RF hardware, this performance requirement aims for the receiver's signal processing.

The test purpose is to verify the ability of the receiver to receive a predefined test signal, representing a multipath propagation channel for the wanted and for the co-channel signals from serving and adjacent cells, with a block error ratio (BLER) not exceeding a specified value.

## 7.3.1.4 Method of test

### 7.3.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS, , the fading simulator, the AWGN generator and additional components to the UE antenna connector as shown in figure A.10.

2) A call is set up according to the Generic call setup procedure. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s), (64 kbit/s), (144 kbit/s), and (384 kbit/s) specified in

annex C for 3.84 Mcps TDD option and 1.28Mcps TDD option. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) for 7.68 Mcps TDD option.

3) Enter the UE into loopback test mode and start the loopback test. (test 1) and/or activate the Ack/Nack test mode (test 1 to test 4).

4) The levels of the wanted signal and the co-channel signals are set according to table 7.3.1.2.1a and b for the 3,84 Mcps TDD Option, table 7.3.1.2.2a and b for the 1,28 Mcps TDD Option and table 7.3.1.2.3a and b for the 7,68 Mcps TDD Option, respectively.

#### 7.3.1.4.2 Procedure

Measure the BLER of DCH received from the UE at the SS for all tests specified in table 7.3.1.2.1a for the 3,84 Mcps TDD Option, table 7.3.1.2.2a for the 1,28 Mcps TDD Option and table 7.3.1.2.3a for the 7,68 Mcps TDD Option, respectively.

### 7.3.1.5 Test requirements

#### 7.3.1.5.1 3.84Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.1.2.1b for the 3,84 Mcps TDD Option

#### 7.3.1.5.2 1.28Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.1.5.2b for the1.28 Mcps TDD Option

**Table 7.3.1.5.2b: Performance requirements in a multipath Case 1 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 23.0 | $10^{-2}$ |
| 2 | 16.4 | $10^{-1}$ |
| | 23.5 | $10^{-2}$ |
| 3 | 17.2 | $10^{-1}$ |
| | 24.5 | $10^{-2}$ |
| 4 | 17.1 | $10^{-1}$ |
| | 24.1 | $10^{-2}$ |

#### 7.3.1.5.3 7.68Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.1.2.3b for the 7.68 Mcps TDD Option

NOTE: If the Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 7.3.2 Multipath fading Case 2

### 7.3.2.1 Definition and applicability

The performance requirement of DCH is determined by the maximum Block Error Ratio (BLER ). The BLER is specified for each individual data rate of the DCH. DCH is mapped into the Dedicated Physical Channel (DPCH).

The UE shall be tested only according to the datarates, supported. The data-rate-corresponding requirements shall apply to the UE.

### 7.3.2.2 Minimum requirement

#### 7.3.2.2.1 3,84 Mcps TDD Option

For the parameters specified in table 7.3.2.2.1a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.2.2.1b. These requirements are applicable for TFCS size 16.

The reference for this requirement is TS 25.102 [1] clause 8.3.2.1.

**Table 7.3.2.2.1a: DCH parameters in multipath Case 2 channel (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 |
|---|---|---|---|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | DB | -3 | 0 | 0 | 0 | 0 |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | | |
| Cell Parameter (note) | | 0,1 | | | | |
| DPCH Channelization Codes (note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1 . .5 | C(i,16) i=1 . .9 | C(i,16) i=1 . .8 | - |
| OCNS Channelization Code (note) | C(k,Q) | C(3,16) | - | - | - | - |
| Information Data Rate | kbps | 12,2 | 64 | 144 | 384 | 2048 |
| Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | | | | | |

**Table 7.3.2.2.1b: Performance requirements in multipath Case 2 channel (3,84 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 5,8 | $10^{-2}$ |
| 2 | 5,7 | $10^{-1}$ |
| | 9,2 | $10^{-2}$ |
| 3 | 9,3 | $10^{-1}$ |
| | 12,7 | $10^{-2}$ |
| 4 | 8,8 | $10^{-1}$ |
| | 12,0 | $10^{-2}$ |
| 5 | 10,3 | $10^{-1}$ |
| | 12,7 | $10^{-2}$ |

### 7.3.2.2.2 1,28 Mcps TDD Option

For the parameters specified in table 7.3.2.2.2a: the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.2.2.2b. The reference for this requirement is TS 25.102 [1] clause 8.3.2.1.2.

**Table 7.3.2.2.2a: DCH parameters in multipath Case 2 channel (1,28 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| Number of $DPCH_o$ | | 8 | 2 | 2 | 0 |
| Scrambling code and basic midamble code number (see note) | | 0 | 0 | 0 | 0 |
| DPCH Channelization Codes (see note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1…8 | C(i,16) i=1…8 | C(i,16) i=1…10 |
| $DPCH_o$ Channelization Codes (see note) | C(k,Q) | C(i,16) 3≤ i ≤10 | C(i,16) 9≤ i ≤10 | C(i,16) 9≤ i ≤10 | - |
| $\frac{DPCH_o_E_c}{I_{or}}$ | dB | -10 | -10 | -10 | 0 |
| $I_{oc}$ | dBm/1,28MHz | -60 | | | |
| Midamble | | Common midamble | | | |
| Information Data Rate | Kbps | 12.2 | 64 | 144 | 384 |
| Note Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 7.3.2.2.2b: Performance requirements in multipath Case 2 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 13.6 | $10^{-2}$ |
| 2 | 9.8 | $10^{-1}$ |
| | 13.9 | $10^{-2}$ |
| 3 | 10.3 | $10^{-1}$ |
| | 14.4 | $10^{-2}$ |
| 4 | 10.5 | $10^{-1}$ |
| | 14.4 | $10^{-2}$ |

### 7.3.2.2.3 7,68 Mcps TDD Option

For the parameters specified in table 7.3.2.2.3a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.2.2.3b. These requirements are applicable for TFCS size 16.

The reference for this requirement is TS 25.102 [1] clause 8.3.2.1.3.

**Table 7.3.2.2.3a: DCH parameters in multipath Case 2 channel (7,68 Mcps TDD Option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | -6 |
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| Cell Parameter (note) | - | 0,1 |
| DPCH Channelization Codes (note) | C(k, Q) | C(i, 32), i = 1,2 |
| OCNS Channelization Code (note) | C(k, Q) | C(3, 32) |
| Information Data Rate | kbps | 12.2 |
| NOTE: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | |

**Table 7.3.2.2.3b: Performance requirements in multipath Case 2 channel (7,68 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 5.8 | $10^{-2}$ |

## 7.3.2.3 Test purpose

While the receiver tests in clause 6 aims for the RF hardware, this performance requirement aims for the receiver's signal processing.

The test purpose is to verify the ability of the receiver to receive a predefined test signal, representing a multipath propagation channel for the wanted and for the co-channel signals from serving and adjacent cells, with a block error ratio (BLER) not exceeding a specified value.

## 7.3.2.4 Method of test

### 7.3.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS, the fading simulator, the AWGN generator and additional components to the UE antenna connector as shown in figure A.10.

2) A call is set up according to the Generic call setup procedure. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) (64 kbit/s), (144 kbit/s), and (384 kbit/s) specified in

annex C for 3.84 Mcps TDD option and 1.28Mcps TDD option. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) for 7.68 Mcps TDD option.

3) Enter the UE into loopback test mode and start the loopback test. (test 1) and/or activate the Ack/Nack test mode (test 1 to test 4).

4) The levels of the wanted signal and the co-channel signals are set according to table 7.3.2.2.1a and b for the 3,84 Mcps TDD Option, table 7.3.2.2.2a and b for the 1,28 Mcps TDD Option and table 7.3.2.2.3a and b for the 7,68 Mcps TDD Option, respectively.

#### 7.3.2.4.2 Procedure

Measure the BLER of DCH received from the UE at the SS for all tests specified in table 7.3.2.2.1a for the 3,84 Mcps TDD Option, table 7.3.2.2.2a for the 1,28 Mcps TDD Option and table 7.3.2.2.3a for the 7,68 Mcps TDD Option, respectively.

### 7.3.2.5 Test requirements

#### 7.3.2.5.1 3.84Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.2.2.1b for the 3,84 Mcps TDD Option

#### 7.3.2.5.2 1.28Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.2.5.2b for the1.28 Mcps TDD Option

**Table 7.3.2.5.2b: Performance requirements in multipath Case 2 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 14.2 | $10^{-2}$ |
| 2 | 10.4 | $10^{-1}$ |
| | 14.5 | $10^{-2}$ |
| 3 | 10.9 | $10^{-1}$ |
| | 15.0 | $10^{-2}$ |
| 4 | 11.1 | $10^{-1}$ |
| | 15.0 | $10^{-2}$ |

#### 7.3.2.5.3 7.68Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.2.2.3b for the 7.68 Mcps TDD Option

NOTE: If the Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 7.3.3 Multipath fading Case 3

### 7.3.3.1 Definition and applicability

The performance requirement of DCH is determined by the maximum Block Error Ratio (BLER ). The BLER is specified for each individual data rate of the DCH. DCH is mapped into the Dedicated Physical Channel (DPCH).

The UE shall be tested only according to the datarates, supported. The data-rate-corresponding requirements shall apply to the UE.

### 7.3.3.2 Minimum requirements

#### 7.3.3.2.1 3,84 Mcps TDD Option

For the parameters specified in table 7.3.3.2.1a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.3.2.1b. These requirements are applicable for TFCS size 16.

The reference for this requirement is 3G TS 25.102 clause 8.3.3.1.1.

**Table 7.3.3.2.1a: DCH parameters in multipath Case 3 channel (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 |
|---|---|---|---|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | DB | -3 | 0 | 0 | 0 | 0 |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | | |
| Cell Parameter (note) | | 0,1 | | | | |
| DPCH Channelization Codes (note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1 . .5 | C(i,16) i=1 . .9 | C(i,16) i=1 . .8 | - |
| OCNS Channelization Code (note) | C(k,Q) | C(3,16) | - | - | - | - |
| Information Data Rate | kbps | 12,2 | 64 | 144 | 384 | 2048 |
| Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | | | | | |

**Table 7.3.3.2.1b: Performance requirements in multipath Case 3 channel (3,84 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 4,8 | $10^{-2}$ |
| 2 | 5,8 | $10^{-1}$ |
| | 8,5 | $10^{-2}$ |
| | 10,7 | $10^{-3}$ |
| 3 | 10,3 | $10^{-1}$ |
| | 13,3 | $10^{-2}$ |
| | 16,0 | $10^{-3}$ |
| 4 | 8,9 | $10^{-1}$ |
| | 11,5 | $10^{-2}$ |
| | 13,6 | $10^{-3}$ |
| 5 | 9,4 | $10^{-1}$ |
| | 11,5 | $10^{-2}$ |
| | 13,6 | $10^{-3}$ |

### 7.3.3.2.2 1,28 Mcps TDD Option

For the parameters specified in table 7.3.3.2.2a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.3.2.2b. The reference for this requirement is TS 25.102 [1] clause 8.3.3.1.2.

**Table 7.3.3.2.2a: DCH parameters in multipath Case 3 channel (1,28 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| Number of $DPCH_o$ | | 8 | 2 | 2 | 0 |
| Scrambling code and basic midamble code number (see note) | | 0 | 0 | 0 | 0 |
| DPCH Channelization Codes (see note) | C(k,Q) | C(i,16) i=1,2 | C(i,16) i=1…8 | C(i,16) i=1…8 | C(i,16) i=1…10 |
| $DPCH_o$ Channelization Codes (see note) | C(k,Q) | C(i,16) 3≤ i ≤10 | C(i,16) 9≤ i ≤10 | C(i,16) 9≤ i ≤10 | - |
| $\frac{DPCH_o_E_c}{I_{or}}$ | dB | -10 | -10 | -10 | 0 |
| $I_{oc}$ | dBm/1,28MHz | -60 | | | |
| Midamble | | Common midamble | | | |
| Information Data Rate | Kbps | 12.2 | 64 | 144 | 384 |
| Note Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 7.3.3.2.2b: Performance requirements in multipath Case 3 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 11.7 | $10^{-2}$ |
| 2 | 9.0 | $10^{-1}$ |
| | 11.7 | $10^{-2}$ |
| | 14.3 | $10^{-3}$ |
| 3 | 9.1 | $10^{-1}$ |
| | 11.2 | $10^{-2}$ |
| | 12.7 | $10^{-3}$ |
| 4 | 9.3 | $10^{-1}$ |
| | 10.8 | $10^{-2}$ |
| | 12.0 | $10^{-3}$ |

### 7.3.3.2.3 7,68 Mcps TDD Option

For the parameters specified in table 7.3.3.2.3a the BLER should not exceed the piece-wise linear BLER curve specified in table 7.3.3.2.3b. These requirements are applicable for TFCS size 16.

The reference for this requirement is 3G TS 25.102 clause 8.3.3.1.3.

**Table 7.3.3.2.3a: DCH parameters in multipath Case 3 channel (7,68 Mcps TDD Option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | -6 |
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| Cell Parameter* | - | 0,1 |
| DPCH Channelization Codes* | C(k, Q) | C(i, 32), i = 1,2 |
| OCNS Channelization Code* | C(k, Q) | C(3, 32) |
| Information Data Rate | kbps | 12.2 |
| *Note: Refer to TS 25.223 for definition of channelization codes and cell parameter. | | |

**Table 7.3.3.2.3b: Performance requirements in multipath Case 3 channel (7,68 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 4.8 | $10^{-2}$ |

## 7.3.3.3 Test purpose

While the receiver tests in clause 6 aims for the RF hardware, this performance requirement aims for the receiver's signal processing.

The test purpose is to verify the ability of the receiver to receive a predefined test signal ,representing a multipath propagation channel for the wanted and for the co-channel signals from serving and adjacent cells, with a block error ratio (BLER) not exceeding a specified value.

## 7.3.3.4 Method of test

### 7.3.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS, the fading simulator, the AWGN generator and additional components to the UE antenna connector as shown in figure A.10.

2) A call is set up according to the Generic call setup procedure. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s)(64 kbit/s), (144 kbit/s), and (384 kbit/s) specified in annex C for 3.84 Mcps TDD option and 1.28Mcps TDD option. The characteristic of the call shall be according to the DL reference measurement channels (12,2 kbit/s) for 7.68 Mcps TDD option.

3) Enter the UE into loopback test mode and start the loopback test. (test 1) and/or activate the Ack/Nack test mode (test 1 to test 4).

4) The levels of the wanted signal and the co-channel signals are set according to table 7.3.3.2.1a and b for the 3,84 Mcps TDD Option, table 7.3.3.2.2a and b for the 1,28 Mcps TDD Option and table 7.3.3.2.3a and b for the 7,68 Mcps TDD Option, respectively.

#### 7.3.3.4.2 Procedure

Measure the BLER of DCH received from the UE at the SS for all tests specified in table 7.3.3.2.1a for the 3,84 Mcps TDD Option, table 7.3.3.2.2a for the 1,28 Mcps TDD Option and table 7.3.3.2.3a for the 7,68 Mcps TDD Option, respectively.

### 7.3.3.5 Test requirements

#### 7.3.3.5.1 3.84Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.3.2.1b for the 3,84 Mcps TDD Option

#### 7.3.3.5.2 1.28Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.3.5.2b for the1.28 Mcps TDD Option

**Table 7.3.3.5.2b: Performance requirements in multipath Case 3 channel (1,28 Mcps TDD Option)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | BLER |
|---|---|---|
| 1 | 12.3 | $10^{-2}$ |
| 2 | 9.6 | $10^{-1}$ |
| | 12.3 | $10^{-2}$ |
| | 14.9 | $10^{-3}$ |
| 3 | 10.0 | $10^{-1}$ |
| | 11.8 | $10^{-2}$ |
| | 13.3 | $10^{-3}$ |
| 4 | 9.9 | $10^{-1}$ |
| | 11.4 | $10^{-2}$ |
| | 12.6 | $10^{-3}$ |

#### 7.3.3.5.3 7.68Mcps TDD Option

The measured BLER shall not exceed the values indicated in table 7.3.3.2.3b for the 7.68Mcps TDD Option.

NOTE: If the Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

# 7.4 Base station transmit diversity mode for 3,84 Mcps TDD Option

## 7.4.1 Demodulation of BCH in SCTD mode

This is not tested.

# 7.5 Power control in downlink

## 7.5.1 Definition and applicability

Power control in the uplink is the ability of the UE to converge to the required link quality set by the network while using minimum uplink power. The requirements of this test shall apply to the UTRA-TDD UE.

## 7.5.2 Power control in downlink for 3,84 Mcps TDD option, constant BLER Target

### 7.5.2.1 Minimum requirements

For the parameters specified in table 7.5.2.1 the downlink $\hat{I}_{or}/I_{oc}$ averaged over one timeslot shall be below the specified value in table 7.5.2 more than 90% of the time. BLER shall be as shown in table 7.5.2.2 Downlink power control is ON during the test.

**Table 7.5.2.1: Test parameters for downlink power control - constant BLER Target**

| Parameter | Unit | Test 1 |
|---|---|---|
| $\frac{DPCH_E_c}{I_{or}}$ | dB | 0 |
| $I_{oc}$ | dBm/3,84 MHz | -60 |
| Information Data Rate | kbps | 12,2 |
| Target quality value on DTCH | BLER | 0,01 |
| Propagation condition | | Case 1 |
| DL Power Control step size, $\Delta_{TPC}$ | dB | 1 |
| Maximum_DL_power (note) | dB | 0 |
| Minimum_DL_power (note) | dB | -27 |
| NOTE: Refer to TS 25.224 for description and definition | | |

NOTE: DL power is relative to P-CCPCH power.

**Table 7.5.2.2: Requirements for downlink power control - constant BLER Target**

| Parameter | Unit | Test 1 |
|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 8,5 |
| Measured quality on DTCH | BLER | 0,01±30% |

The reference for this requirement is TS 25.102 [1] clause 8.5.1.

### 7.5.2.2 Test purpose

To verify that the UE receiver is capable of converging to the required link quality set by the network while using as low power as possible.

### 7.5.2.3 Method of test

#### 7.5.2.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set up a call according to the Generic call setup procedure.

3) RF parameters are set up according to table 7.5.2.1

4) Enter the UE into loopback test mode and start the loopback test.

5) SS signals to UE target quality value on DTCH as specified in table 7.5.2.1. SS will vary the physical channel power in downlink according to the TPC commands from UE, and at the same time measure BLER. This is continued until the target quality value on DTCH is met, within the minimum accuracy requirement.

See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure and loopback test.

#### 7.5.2.3.2 Procedure

1) After the target quality on DTCH is met, BLER is measured. Simultaneously the downlink $\hat{I}_{or}/I_{oc}$ power ratio averaged over one slot is measured. This is repeated until adequate amount of measurements is done to reach the required confidence level.

2) The measured quality on DTCH (BLER) and the measured downlink $\hat{I}_{or}/I_{oc}$ power ratio values averaged over one slot are compared to the limits in table 7.5.2.2.

### 7.5.2.4 Test Requirements

a) The measured quality on DTCH does not exceed the values in table 7.5.2.2.

b) The downlink $\hat{I}_{or}/I_{oc}$ power ratio values, which are averaged over one slot, shall be below the values in table 7.5.2.2 more than 90 % of the time.

## 7.5.3 Power control in downlink for 1,28 Mcps TDD option, constant BLER Target

### 7.5.3.1 Definition and applicability

Power control in the downlink is the ability of the UE receiver to converge to required link quality set by the network while using as low power as possible in downlink. If a BLER target has been assigned to a DCCH (See clause C.3), then it has to be such that outer loop is based on DTCH and not on DCCH. The requirements and this test apply to all types of UTRA for the 1,28 Mcps TDD UE.

### 7.5.3.2 Minimum requirements

For the parameters specified in table 7.5.3.1 the downlink $\hat{I}_{or}/I_{oc}$ averaged over one timeslot power shall be below the specified value in table 7.5.3.2 more than 90% of the time. BLER shall be as shown in table 7.5.3.2. Downlink power control is ON during the test.

**Table 7.5.3.1: Test parameters for downlink power control – constant BLER Target**

| Parameter | Unit | Value |
|---|---|---|
| $\frac{\Sigma DPCH_E_c}{I_{or}}$ | dB | 0 |
| $I_{oc}$ | dBm/1.28 MHz | -60 |
| Information data rate | kbps | 12.2 |
| Target quality on DTCH | BLER | 0.01 |
| Propagation condition | Case 1 | |
| DL Power Control step size, $\Delta_{TPC}$ | dB | 1 |
| Maximum_DL_power * | dB | 0 |
| Minimum_DL_power * | dB | -27 |

Note: DL power is compared to P-CCPCH power.

**Table 7.5.3.2: Requirements for downlink power control – constant BLER Target**

| Parameter | Unit | Test 1 | |
|---|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 7.5 | |
| Measured quality on DTCH | BLER | 0,01±30% | |

The reference for this requirement is TS 25.102 [1] clause 8.5.1.

### 7.5.3.3 Test purpose

To verify that the UE receiver is capable of converging to the required link quality set by the network while using as low power as possible.

### 7.5.3.4 Method of test

#### 7.5.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set up a call according to the Generic call setup procedure.

3) RF parameters are set up according to table 7.5.3.1

4) Enter the UE into loopback test mode and start the loopback test.

5) SS signals to UE target quality value on DTCH as specified in table 7.5.3.1. SS will vary the physical channel power in downlink according to the TPC commands from UE, and at the same time measure BLER. This is continued until the target quality value on DTCH is met, within the minimum accuracy requirement.

See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure and loopback test.

#### 7.5.3.4.2 Procedure

1) After the target quality on DTCH is met, BLER is measured. Simultaneously the downlink $\hat{I}_{or}/I_{oc}$ power ratio averaged over one slot is measured. This is repeated until adequate amount of measurements is done to reach the required confidence level.

2) The measured quality on DTCH (BLER) and the measured downlink $\hat{I}_{or}/I_{oc}$ power ratio values averaged over one slot are compared to the limits in table 7.5.3.2.

### 7.5.3.5 Test Requirements

a) The measured quality on DTCH does not exceed the values in table 7.5.3.2.

b) The downlink $\hat{I}_{or}/I_{oc}$ power ratio values, which are averaged over one slot, shall be below the values in table 7.5.3.2.

## 7.5.4 Power control in downlink for 7,68 Mcps TDD option, constant BLER Target

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# 7.6 Uplink Power Control

## 7.6.1 Definition and applicability

Power control in the uplink is the ability of the UE to converge to the required link quality set by the network while using minimum uplink power. The requirements of this test shall apply to all types of the UTRA-TDD UE.

## 7.6.2 Minimum requirements

During period T1, the PCCPCH and a second Beacon Channel are transmitted in the DL in designated slots within each frame and at the same power level.

The UE transmits, using the channel of TS25.105, Annex A.2.1 UL reference measurement channel (12.2 kbps) in one UL slot. For different parts of the test, different UL slots will be designated.

The values of table 7.6.1, period T1 shall be selected. Then, with the received PCCPCH and Beacon power set at –60 dBm, the value of DPCH constant value shall be adjusted so that the mean UE output power is 5 dBm. These conditions are held steady during period T1.

Periods T1 and T2 are each 5 seconds long.

**Table 7.6.1: UL Power Control Test Conditions**

| | | Period T1 | Period T2 |
|---|---|---|---|
| $I_{BTS}$ all slots | dBm | -60 | |
| PCCPCH Power -Broadcast | dBm | 18 | |
| PCCPCH Power - Received | dBm | -60 | -70 |
| Mean UE transmit power | dBm | 5 | According to tables 7.6.2 and 7.6.3 |
| $SIR_{TARGET}$ | dB | 6 | |
| $I_{oc}$ in PCCPCH and Beacon Slots | dBm | -60 | |
| IE (information element) Alpha | As defined in 25.331 | 1.0 | |
| PCCPCH slot position | Integer 0 -14 | 0 | |
| Beacon slot position | Integer 0 -14 | 8 | |

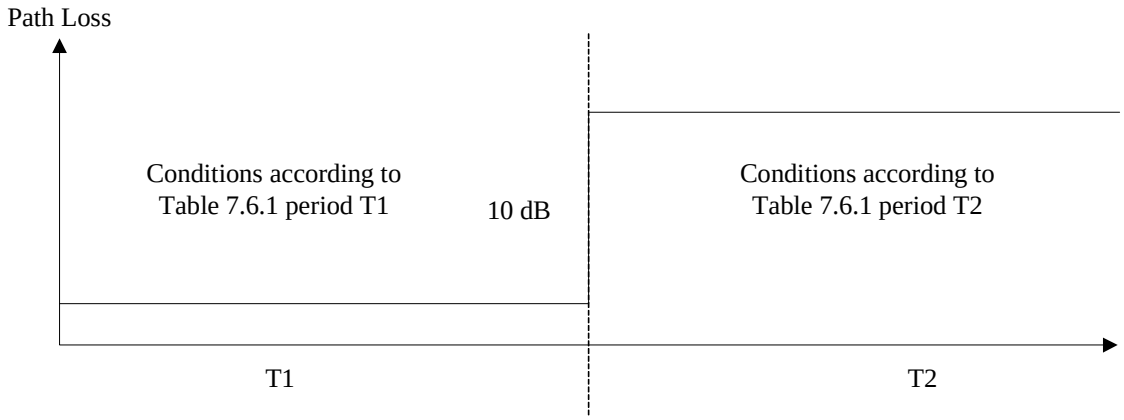



**Figure 7.6.1**

At the end of period T1, the PCCPCH and Beacon Received power shall be simultaneously decreased by 10 dB. These conditions are summarized in table 7.6.1, period T2.

For the first frame including the change in received power the UE output power shall satisfy the values in table 7.6.2.

For the 20th frame after the change in received power the UE output power shall satisfy the values in table 7.6.3.

**Table 7.6.2: Required UE Output Power, Frame Containing Power Level Change**

| Parameter | Units | Value | |
|---|---|---|---|
| UL transmission slot position | | 1,9 | 7,14 |
| UE output power | dBm | 15 ±4.0 | 5 ±0.5 |

**Table 7.6.3: Required UE Output Power, 20 Frames after Power Level Change**

| Parameter | Units | Value | |
|---|---|---|---|
| UL transmission slot position | | 1,9 | 7,14 |
| UE output power | dBm | 15 ±4.0 | 15 ±4.0 |

## 7.6.3 Test purpose

To verify the ability of the UE to converge to the required link quality set by the network while using minimum uplink power.

## 7.6.4 Method of test

### 7.6.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS and an AWGN source to the UE antenna connector as shown in figure A.9.

2) Set up a call according to the Generic call setup procedure using SS levels and signalling values as specified in table 7.6.1 for Period P1 and table 7.6.4 . The UE shall be signalled to transmit in timeslot position 1.

**Table 7.6.4: Test parameters for Uplink Power Control Test**

| Parameter | Value/description |
|---|---|
| UL Reference measurement channel | 12,2kbps, according to annex C.2.1 |
| DPCH constant value | 0 |
| Data content | real life (sufficient irregular) |

3) Enter the UE into loopback test mode and start the loopback test.

See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure and loopback test.

### 7.6.4.2 Procedure

1) The SS adjusts the DPCH constant value until the UE transmit power is 5 dBm.

2) After the UE output power has has been held constant at 5 dBm for at least 5 seconds, the received PCCPCH power shall be decreased by 10 dB to -70 dBm as shown in figure 7.6.1.

3) Measure the transmit power according to annex B for the first frame including the pathloss change and the 20$^{th}$ frame after the pathloss change.

4) Set the received PCCPCH power to -60 dBm.

5) SS signals UE to transmit in timeslot 7. Repeat step 1 - 4.

6) SS signals UE to transmit in timeslot 9. Repeat step 1 - 4.

7) SS signals UE to transmit in timeslot 14. Repeat step 1 - 4.

## 7.6.5 Test Requirements

The measured transmit power shall not exceed the prescribed tolerance in tables 7.6.5 and 7.6.6.

**Table 7.6.5: Required UE Output Power, Frame Containing Power Level Change**

| Parameter | Units | Value | |
|---|---|---|---|
| UL transmission slot position | | 1,9 | 7,14 |
| UE output power | dBm | 15 ±5.5 | 5 ±0.5 |

**Table 7.6.6: Required UE Output Power, 20 Frames after Power Level Change**

| Parameter | Units | Value | |
|---|---|---|---|
| UL transmission slot position | | 1,9 | 7,14 |
| UE output power | dBm | 15 ±5.5 | 15 ±5.5 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in annex F clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F clause F.4.

# 8 Requirements for Support of RRM

## 8.1 General

Void.

## 8.2 Idle Mode Tasks

### 8.2.1 RF Cell Selection Scenario

#### 8.2.1.1 Introduction

After a UE has switched on and a PLMN has been selected, the Cell selection process takes place, as described in TS 25.304. This process allows the UE to select a suitable cell where to camp on in order to access available services. In this process the UE can use stored information (*Stored information cell selection*) or not (*Initial cell selection*).

### 8.2.2 Cell Re-Selection

#### 8.2.2.1 Scenario 1: Cell re-selection to intra frequency TDD cell

##### 8.2.2.1.1 Definition and applicability

###### 8.2.2.1.1.1 3,84 Mcps TDD Option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

###### 8.2.2.1.1.2 1,28 Mcps TDD Option

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST to perform a Location Registration on cell 2.

This test is applicable to 1,28 Mcps TDD UE.

###### 8.2.2.1.1.3 7,68 Mcps TDD Option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

##### 8.2.2.1.2 Minimum requirement

###### 8.2.2.1.2.1 3,84 Mcps TDD Option

The cell re-selection delay shall be less than 8 s when the DRX cycle length is 1,28 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to table 4.1 in clause 4.2.2.7 in [2].

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.1.

#### 8.2.2.1.2.2 1,28 Mcps TDD Option

The cell re-selection delay shall be less than 8 s when the DRX cycle length is 1,28 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$: A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ NTDD}$ of 6.4s according to table 4.1A in clause 4.2 in [2].

$T_{SI}$: Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.1.

#### 8.2.2.1.2.3 7,68 Mcps TDD Option

The cell re-selection delay shall be less than 8 s when the DRX cycle length is 1,28 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to table 4.1B in clause 4.2.2.7.3 in [2].

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.1.

### 8.2.2.1.3 Test purpose

This test is to verify the requirement for the cell re-selection delay in the single carrier case

### 8.2.2.1.4 Method of test

#### 8.2.2.1.4.1 3,84 Mcps TDD Option

##### 8.2.2.1.4.1.1 Initial conditions

This scenario implies the presence of 1 carrier and 6 cells as given in table 8.2.2.1.1 and table 8.2.2.1.2. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.1.1: General test parameters for Cell Re-selection to intra-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.2.2.1.2: Cell re-selection single carrier to intra-frequency TDD cell**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.2.2.1.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

8.2.2.1.4.2 1,28 Mcps TDD Option

8.2.2.1.4.2.1 Initial conditions

This scenario implies the presence of 1 carrier and 6 cells as given in table 8.2.2.1.1A and 8.2.2.1.2A.

**Table 8.2.2.1.1A: General test parameters for Cell Re-selection to intra-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) -– Persistence value | | 0..1 | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.2.2.1.2A: Cell re-selection to intra-frequency TDD cell**

| 0 Parameter | 0 Unit | 0 Cell 1 | | | | 0 Cell 2 | | | | 0 Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 Timeslot Number | 1 | 1 0 | | 1 DWPTS | | 1 0 | | 0 DWPTS | | 0 0 | | 0 DWPTS | |
| 2 | 2 | 2 T1 | 2 T2 | 2 T1 | 1 T2 | 1 T1 | 1 T2 | 0 T1 | 0 T2 | 0 T1 | 0 T2 | 0 T1 | 0 T2 |
| 3 UTRA RF Channel Number | 3 | 3 Channel 1 | | | | 3 Channel 1 | | | | 3 Channel 1 | | | |
| PCCPCH_Ec/Ior | 4 dB | 4 -3 | 4 -3 | 4 | 0 | 2 -3 | 2 -3 | 1 | 1 | 1 -3 | 1 -3 | 1 | 1 |
| 5 DwPCH_Ec/Ior | 5 dB | 5 | 5 | 5 0 | 1 0 | 3 | 3 | 2 0 | 2 0 | 2 | 2 | 2 0 | 2 0 |
| 6 OCNS_Ec/Ior | 6 dB | 6 -3 | 6 -3 | 6 | 2 | 4 -3 | 4 -3 | 3 | 3 | 3 -3 | 3 -3 | 3 | 3 |
| 7 $\hat{I}_{or}/I_{oc}$ | 7 dB | 7 9 | 7 7 | 7 9 | 3 7 | 5 7 | 5 9 | 4 7 | 4 9 | 4 -1 | 4 -1 | 4 -1 | 4 -1 |
| PCCPCH RSCP | 8 dBm | 8 -64 | 8 -66 | 8 | 4 | 6 -66 | 6 -64 | 5 | 5 | 5 -74 | 5 -74 | 5 | 5 |
| 9 Qoffset1$_{s,n}$ | 9 dB | 9 C1, C2: 0; C1, C3:0; C1,C4:0 10 C1, C5:0; C1,C6:0 | | | | 9 C2, C1: 0; C2, C3:0; C2,C4:0 10 C2, C5: 0; C2, C6:0 | | | | 9 C3, C1: 0; C3, C2:0; C3,C4:0 10 C3, C5: 0; C3, C6:0 | | | |
| 10 Qhyst1$_s$ | 10 dB | 11 0 | | | | 11 0 | | | | 11 0 | | | |
| 11 Treselection | 11 s | 12 0 | | | | 12 0 | | | | 12 0 | | | |
| 12 Sintrasearch | 12 dB | 13 not sent | | | | 13 not sent | | | | 13 not sent | | | |
| 13 | 13 | 14 Cell 4 | | | | 14 Cell 5 | | | | 14 Cell 6 | | | |
| 14 Timeslot | 14 | 15 0 | | 15 DWPTS | | 15 0 | | 0 DWPTS | | 7 0 | | 7 DWPTS | |
| 15 | 15 | 16 T1 | 16 T2 | 16 T1 | 1 T2 | 8 T1 | 8 T2 | 0 T1 | 6 T2 | 6 T1 | 6 T2 | 6 T1 | 6 T2 |
| 16 UTRA RF Channel Number | 16 | 17 Channel 1 | | | | 17 Channel 1 | | | | 17 Channel 1 | | | |
| 17 PCCPCH_Ec/Ior | 17 dB | 18 -3 | 18 -3 | 18 | 0 | 9 -3 | 9 -3 | 1 | 7 | 7 -3 | 7 -3 | 7 | 7 |
| 18 DwPCH_Ec/Ior | 18 dB | 19 | 19 | 19 0 | 1 0 | 10 | 10 | 2 0 | 8 0 | 8 | 8 | 8 0 | 8 0 |
| 19 OCNS_Ec/Ior | 19 dB | 20 -3 | 20 -3 | 20 | 2 | 11 -3 | 11 -3 | 3 | 9 | 9 -3 | 9 -3 | 9 | 9 |
| 20 $\hat{I}_{or}/I_{oc}$ | 20 dB | 21 -1 | 21 -1 | 21 -1 | 3 -1 | 12 -1 | 12 -1 | 4 -1 | 10 -1 | 10 -1 | 10 -1 | 10 -1 | 10 -1 |
| 21 PCCPCH RSCP | 21 dBm | 22 -74 | 22 -74 | 22 | 4 | 13 -74 | 13 -74 | 5 | 11 | 11 -74 | 11 -74 | 11 | 11 |
| 22 Qoffset1$_{s,n}$ | 22 dB | 23 C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | 23 C5, C1: 0; C5, C2:0; C5,C3:0 24 C5, C4:0; C5, C6:0 | | | | 23 C6, C1: 0; C6, C2:0; C6,C3:0 24 C6, C4:0; C6, C5:0 | | | |
| 23 Qhyst1$_s$ | 23 dB | 24 0 | | | | 25 0 | | | | 25 0 | | | |
| 24 Treselection | 24 s | 25 0 | | | | 26 0 | | | | 26 0 | | | |
| 25 Sintrasearch | 25 dB | 26 not sent | | | | 27 not sent | | | | 27 not sent | | | |
| 26 $I_{oc}$ | 26 dBm/1,28 MHz | 27 -70 | | | | | | | | | | | |
| 27 Propagation Condition | 27 | 28 AWGN | | | | | | | | | | | |
| 28 * Note:In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | | | | | | | | | |

8.2.2.1.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for SYNCH-UL sequence in the UpPTS for sending for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps b)-g) until the confidence level according to annex F.6.2 is achieved.

8.2.2.1.4.3 7,68 Mcps TDD Option

8.2.2.1.4.3.1 Initial conditions

This scenario implies the presence of 1 carrier and 6 cells as given in table 8.2.2.1.1B and table 8.2.2.1.2B. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.1.1B: General test parameters for Cell Re-selection to intra-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.2.2.1.2B: Cell re-selection to intra-frequency TDD cell**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2,C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3,C6:0 | | | |
| Qhyst 1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| *OCNS_Ec/Ior* | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.2.2.1.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved.

### 8.2.2.1.5 Test Requirements

#### 8.2.2.1.5.1 3,84 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.1.5.2 1,28 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.1.5.3 7,68 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.2 Scenario 2: Cell re-selection to inter-frequency TDD cell

#### 8.2.2.2.1 Definition and applicability

##### 8.2.2.2.1.1 3,84 Mcps Option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.2.2.2.1.2 1,28 Mcps Option

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send the SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST to perform a Location Registration on cell 2.

This test is applicable to 1,28 Mcps TDD UE.

##### 8.2.2.2.1.3 7,68 Mcps Option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell, and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.2.2.2.2 Minimum requirement

##### 8.2.2.2.2.1 3,84 Mcps Option

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to table 4.1 in clause 4.2.2.7 in [2].

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.2.

##### 8.2.2.2.2.2 1,28 Mcps Option

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ NTDD}$ of 6.4s according to table 4.1A in clause 4.2 in [2].

$T_{SI}$ Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.2.

#### 8.2.2.2.2.3 7,68 Mcps Option

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to table 4.1B in clause 4.2.2.7.3 in [2].

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.2.

### 8.2.2.2.3 Test purpose

This test is to verify the requirement for the cell re-selection delay in the multi carrier case.

### 8.2.2.2.4 Method of test

#### 8.2.2.2.4.1 3,84 Mcps Option

##### 8.2.2.2.4.1.1 Initial conditions

This scenario implies the presence of 2 carriers and 6 cells as given in table 8.2.2.2.1 and table 8.2.2.2.2. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.2.1: General test parameters for Cell Re-selection to inter-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.2.2.2.2: Cell re-selection to inter-frequency TDD cell**

| 0 Parameter | 0 Unit | 0 Cell 1 | | | | | | | | 0 Cell 2 | | | | | | | | 0 Cell 3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 Timeslot Number | 1 | 1 0 | | | | 1 DWPTS | | | | 1 0 | | | | 0 DWPTS | | | | 0 0 | | | | 0 DWPTS | | | |
| 2 | 2 | 2 T1 | | 2 T2 | | 2 T1 | | 1 T2 | | 1 T1 | | 1 T2 | | 0 T1 | | 0 T2 | | 0 T1 | | 0 T2 | | 0 T1 | | 0 T2 | |
| 3 UTRA RF Channel Number | 3 | 3 Channel 1 4 | | | | | | | | 3 Channel 2 4 | | | | | | | | 3 Channel 1 4 | | | | | | | |
| 4 PCCPCH_Ec/Ior | 4 dB | 5 | -3 | 5 | -3 | 5 | | 0 | | 2 | -3 | 2 | -3 | 1 | | 1 | | 1 | -3 | 1 | -3 | 1 | | 1 | |
| 5 DwPCH_Ec/Ior | 5 dB | 6 | | 6 | | 6 | 0 | 1 | 0 | 3 | | 3 | | 2 | 0 | 2 | 0 | 2 | | 2 | | 2 | 0 | 2 | 0 |
| 6 $\hat{I}_{or}/I_{oc}$ | 6 dB | 7 | 10 | 7 | 7 | 7 | 10 | 2 | 7 | 4 | 7 | 4 | 10 | 3 | 7 | 3 | 10 | 3 | -1 | 3 | -1 | 3 | -1 | 3 | -1 |
| 7 PCCPCH RSCP | 7 dBm | 8 | -63 | 8 | -66 | 8 | | 3 | | 5 | -66 | 5 | -63 | 4 | | 4 | | 4 | -74 | 4 | -74 | 4 | | 4 | |
| 8 Qoffset1$_{s,n}$ | 8 dB | 9 C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | | | | | 9 C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | | | | | 9 C3, C1: 0; C3, C2:0; C3,C4:0 10 C3, C5:0; C3, C6:0 | | | | | | | |
| 9 Qhyst1$_s$ | 9 dB | 10 0 | | | | | | | | 10 0 | | | | | | | | 11 0 | | | | | | | |
| 10 Treselection | 10 s | 11 0 | | | | | | | | 11 0 | | | | | | | | 12 0 | | | | | | | |
| 11 Sintrasearch | 11 dB | 12 not sent | | | | | | | | 12 not sent | | | | | | | | 13 not sent | | | | | | | |
| 12 Sintersearch | 12 dB | 13 not sent | | | | | | | | 13 not sent | | | | | | | | 14 not sent | | | | | | | |
| 13 | 13 | 14 Cell 4 | | | | | | | | 14 Cell 5 | | | | | | | | 15 Cell 6 | | | | | | | |
| 14 Timeslot | 14 | 15 0 | | | | 15 DWPTS | | | | 16 0 | | | | 0 DWPTS | | | | 6 0 | | | | 6 DWPTS | | | |
| 15 | 15 | 16 T1 | | 16 T2 | | 17 T1 | | 1 T2 | | 7 T1 | | 7 T2 | | 0 T1 | | 5 T2 | | 5 T1 | | 5 T2 | | 5 T1 | | 5 T2 | |
| 16 UTRA RF Channel Number | 16 | 17 Channel 18 | | | | | | | | 17 Channel 2 18 | | | | | | | | 18 Channel 19 | | | | | | | |
| 17 PCCPCH_Ec/Ior | 17 dB | 19 | -3 | 19 | -3 | 20 | | 0 | | 8 | -3 | 8 | -3 | 1 | | 6 | | 6 | -3 | 6 | -3 | 6 | | 6 | |
| 18 DwPCH_Ec/Ior | 18 dB | 20 | | 20 | | 21 | 0 | 1 | 0 | 9 | | 9 | | 2 | 0 | 7 | 0 | 7 | | 7 | | 7 | 0 | 7 | 0 |
| 19 $\hat{I}_{or}/I_{oc}$ | 19 dB | 21 | -1 | 21 | -1 | 22 | -1 | 2 | -1 | 10 | -1 | 10 | -1 | 3 | -1 | 8 | -1 | 8 | -1 | 8 | -1 | 8 | -1 | 8 | -1 |
| 20 PCCPCH RSCP | 20 dBm | 22 | -74 | 22 | -74 | 23 | | 3 | | 11 | -74 | 11 | -74 | 4 | | 9 | | 9 | -74 | 9 | -74 | 9 | | 9 | |
| 21 Qoffset1$_{s,n}$ | 21 dB | 23 C4, C1: 0; C4, C2:0; C4,C3:0 24 C4, C5:0; C4, C6:0 | | | | | | | | 23 C5, C1: 0; C5, C2:0; C5,C3:0 24 C5, C4:0; C5, C6:0 | | | | | | | | 24 C6, C1: 0; C6, C2:0; C6,C3:0 25 C6, C4:0; C6, C5:0 | | | | | | | |
| 22 Qhyst1$_s$ | 22 dB | 25 0 | | | | | | | | 25 0 | | | | | | | | 26 0 | | | | | | | |
| 23 Treselection | 23 s | 26 0 | | | | | | | | 26 0 | | | | | | | | 27 0 | | | | | | | |
| 24 Sintrasearch | 24 dB | 27 not sent | | | | | | | | 27 not sent | | | | | | | | 28 not sent | | | | | | | |
| 25 Sintersearch | 25 dB | 28 not sent | | | | | | | | 28 not sent | | | | | | | | 29 not sent | | | | | | | |
| 26 $I_{oc}$ | 26 dBm/3,84 MHz | 29 -70 | | | | | | | | | | | | | | | | | | | | | | | |
| 27 Propagation Condition | 27 | 30 AWGN | | | | | | | | | | | | | | | | | | | | | | | |

#### 8.2.2.2.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

NOTE: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

### 8.2.2.2.4.2 1,28 Mcps Option

#### 8.2.2.2.4.2.1 Initial conditions

This scenario implies the presence of 2 carriers and 6 cells as given in table 8.2.2.2.1A and table 8.2.2.2.2A. For this test purpose the broadcast repetition period of the target cell shall be 1.28s. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.2.1A: General test parameters for Cell Re-selection to inter-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.2.2.2.2A: Cell re-selection to inter-frequency TDD cell**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 7 | 10 | 7 | 7 | 10 | 7 | 10 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -63 | -66 | | | -66 | -63 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | Cell 4 | | | | Cell 5 | | | | Cell 6 | | | |
| Timeslot | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| * Note: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | | | | | | | | | |

8.2.2.2.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps b)-g) until the confidence level according to annex F.6.2 is achieved.

NOTE: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

8.2.2.2.4.3 7,68 Mcps Option

8.2.2.2.4.3.1 Initial conditions

This scenario implies the presence of 2 carriers and 6 cells as given in table 8.2.2.2.1B and table 8.2.2.2.2B. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.2.1B: General test parameters for Cell Re-selection to intra-frequency TDD cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.2.2.2.2B: Cell re-selection to intra-frequency TDD cell**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 0 | 6 | 0 | 0 | 6 | 0 | 6 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -67 | -73 | | | -73 | -67 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -76 | -76 | | | -76 | -76 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.2.2.2.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved.

NOTE: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

### 8.2.2.2.5 Test Requirements

#### 8.2.2.2.5.1 3,84 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 8 s.

3) In step g), the UE shall respond on cell 2 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.2.5.2 1,28 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 8 s.

3) In step g), the UE shall respond on cell 2 within 8 s.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.2.5.3 7,68 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 8 s.

3) In step g), the UE shall respond on cell 2 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.2A Scenario 2A: 3,84 Mcps TDD cell re-selection for 1,28 Mcps TDD UE

[FFS]

### 8.2.2.2B Scenario 2B: 1,28 Mcps TDD cell re-selection for 3,84 Mcps TDD UE

FFS

### 8.2.2.2C Scenario 2C: 3,84 Mcps TDD cell re-selection for 7,68 Mcps TDD UE

#### 8.2.2.2C.1 Definition and applicability

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test applies to  to UEs supporting both 3.84 Mcps TDD and 7.68Mcps TDD.

#### 8.2.2.2C.2 Minimum requirement

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE:

$T_{evaluateTDD} + T_{SI}$

where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to Table 4.1B in section 4.2.

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.2.

#### 8.2.2.2C.3 Test purpose

This test is to verify the requirement for the cell re-selection delay in the 7.68Mcps TDD to 3.84 Mcps TDD case.

#### 8.2.2.2C.4 Method of test

##### 8.2.2.2C.4.1. Initial conditions

This scenario implies the presence of 1 higher chip rate (7.68 Mcps TDD OPTION) and 1 high chip rate (3.84 Mcps TDD OPTION)) cell as given in Table 8.2.2.2C.1 and 8.2.2.2C.1.

Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.2C.1: General test parameters for TDD higher chip rate (7.68 Mcps) to TDD high chip rate (3.84 Mcps) cell re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | 7.68 Mcps TDD OPTION cell |
| | Neighbour cell | | Cell2 | 3.84 Mcps TDD OPTION cell |
| Final condition | Active cell | | Cell2 | 3.84 Mcps TDD OPTION cell |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | S | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | S | 1,28 | The value shall be used for all cells in the test. |
| T1 | | S | 30 | |
| T2 | | S | 15 | |

**Table 8.2.2.2C.2: Test parameters for TDD higher chip rate (7.68 Mcps) to TDD high chip rate (3.84 Mcps) cell re-selection**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T 1 | T 2 | T1 | T2 | T 1 | T 2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 |
| *OCNS_Ec/Ior* | | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 7 | 10 | 7 | 7 | 10 | 7 | 10 |
| PCCPCH_RSCP | dBm | -63 | -66 | | | -66 | -63 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0 | | | | C2, C1: 0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | |
| Sintersearch | dB | not sent | | | | not sent | | | |
| $I_{oc}$ | | -70 dBm/7.68 MHz | | | | -70 dBm/3.84 MHz | | | |
| Propagation Condition | | AWGN | | | | | | | |

8.2.2.2C.4.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

NOTE: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

#### 8.2.2.2C.5 Test Requirements

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 8 s.

3) In step g), the UE shall respond on cell 2 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.2D Scenario 2D: 7,68 Mcps TDD cell re-selection for 3,84 Mcps TDD UE

#### 8.2.2.2D.1 Definition and applicability

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

The requirements and this test applies to to UEs supporting both 3.84 Mcps TDD and 7.68Mcps TDD.

#### 8.2.2.2D.2 Minimum requirement

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE:

$T_{evaluateTDD} + T_{SI}$

where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to Table 4.1B in section 4.2.

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.2.

#### 8.2.2.2D.3 Test purpose

This test is to verify the requirement for the cell re-selection delay in the 3.84Mcps TDD to 7.68 Mcps TDD case.

#### 8.2.2.2D.4 Method of test

##### 8.2.2.2D.4.1. Initial conditions

This scenario implies the presence of one 3.84 Mcps TDD serving cell, and one 7.68 Mcps TDD cell to be re-selected. The UE is requested to monitor neighbouring cells on one 3.84Mcps TDD carrier and one 7.68 Mcps TDD carrier. Test parameters are given in Table 8.2.2.2D.1and 8.2.2.2D.2.

Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.2D.1: General test parameters for 3.84 Mcps /7.68 Mcps TDD cell re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | 3.84 Mcps TDD OPTION cell |
| | Neighbour cell | | Cell 2 | 7.68 Mcps TDD OPTION cell |
| Final condition | Active cell | | Cell 2 | 7.68 Mcps TDD OPTION cell |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| $Q_{rxlevmin}$ | | dBm | -102 | |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | S | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | S | 1,28 | The value shall be used for all cells in the test. |
| T1 | | S | 30 | |
| T2 | | S | 15 | |

**Table 8.2.2.2D.2: Cell 1 specific test parameters for 3.84 Mcps TDD/7.68 Mcps TDD cell re-selection**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T 1 | T 2 | T1 | T2 | T 1 | T 2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 |
| *OCNS_Ec/Ior* | | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 7 | 10 | 7 | 7 | 10 | 7 | 10 |
| PCCPCH_RSCP | dBm | -63 | -66 | | | -66 | -63 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0 | | | | C2, C1: 0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | |
| Sintersearch | dB | not sent | | | | not sent | | | |
| $I_{oc}$ | | -70 dBm/3.84 MHz | | | | -70 dBm/7.68 MHz | | | |
| Propagation Condition | | AWGN | | | | | | | |

8.2.2.2D.4.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

NOTE: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

#### 8.2.2.2D.5 Test Requirements

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 8 s.

3) In step g), the UE shall respond on cell 2 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.3 Scenario 3: TDD/FDD Cell re-selection

#### 8.2.2.3.1 Definition and applicability

##### 8.2.2.3.1.1 3,84 Mcps Option

The cell re-selection delay is defined as the time from when the cell quality levels change to the moment when this change makes the UE reselect a better ranked cell, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

This test is for the case where the UE camps on a 3,84 Mcps TDD cell and reselects to an FDD cell.

The requirements and this test apply to UEs supporting both 3,84 Mcps TDD and FDD.

##### 8.2.2.3.1.2 1,28 Mcps Option

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Location Registration on cell 2.

This test is for the case where the UE camps on a 1,28 Mcps TDD cell and reselects to an FDD cell.

The requirements and this test apply to UEs supporting both 1,28 Mcps TDD and FDD.

##### 8.2.2.3.1.3 7,68 Mcps Option

The cell re-selection delay is defined as the time from when the cell quality levels change to the moment when this change makes the UE reselect a better ranked cell, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Location Registration on the new cell.

This test is for the case where the UE camps on a 7,68 Mcps TDD cell and reselects to an FDD cell.

The requirements and this test apply to UEs supporting both 7,68 Mcps TDD and FDD.

#### 8.2.2.3.2 Minimum requirements

##### 8.2.2.3.2.1 3,84 Mcps TDD Option

The cell re-selection delay shall be less than 8 s when the DRX cycle length is 1,28 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateFDD} + T_{SI}$, where:

$T_{evaluateFDD}$ See TS 25.123 [2] table 4.1 in clause 4.2.2.

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.3.

#### 8.2.2.3.2.2 1,28 Mcps TDD Option

The minimum requirement for the cell re-selection delay is less than 8 s with a DRX cycle length of 1280ms.

This shall be verified in more than 90% of the cases with a confidence level of 95%.

The normative reference for this requirement is TS 25.123 [2]clauses 4.2.2 and A4.2.3.

NOTE: The cell re-selection delay can be expressed as: $T_{evaluateFDD} + T_{SI}$, where:

$T_{evaluateFDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluateFDD}$ of 6.4s according to table F2.4.1.A in Annex F clause F.2.4.

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7,68 s, allow 8 s in the test case.

#### 8.2.2.3.2.3 7,68 Mcps TDD Option

The cell re-selection delay shall be less than 8 s when the DRX cycle length is 1,28 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateFDD} + T_{SI}$, where:

$T_{evaluateFDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s. See TS 25.123 [2] table 4.1 in clause 4.2.2.

$T_{SI}$ Maximum repetition rate of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.3.

### 8.2.2.3.3 Test purpose

The test purpose is to verify the requirement for the cell re-selection.

### 8.2.2.3.4 Method of test

#### 8.2.2.3.4.1 3,84 Mcps Option

##### 8.2.2.3.4.1.1 Initial conditions

This scenario implies the presence of 1 TDD and 1 FDD cell as given in table 8.2.2.3.4.1 and table 8.2.2.3.4.2. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.3.4.1: General test parameters for the TDD/FDD cell re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | TDD cell |
| | Neighbour cells | | Cell2 | FDD cell |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.2.2.3.4.2: TDD/FDD cell re-selection**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **n.a** | **n.a.** |
| | | **T1** | **T2** | **T 1** | **T 2** | **T 1** | **T 2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | -12 |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -12 | -12 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | n.a. | n.a. |
| PICH_Ec/Ior | dB | | | -3 | -3 | -15 | -15 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -0,941 | -0,941 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -2 | 3 | -2 | -2 | 3 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | |
| CPICH_RSCP | dBm | n.a. | | n.a. | | -82 | -77 |
| PCCPCH_RSCP | dBm | -70 | -75 | | | n.a. | n.a. |
| Cell_selection and reselectionquality _measure | | CPICH_RSCP | | | | CPICH_RSCP | |
| Qrxlevmin | dBm | -102 | | | | -115 | |
| Qoffset1$_{s,n}$ | dB | C1, C2: -12 | | | | C2, C1: +12 | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | |
| Treselection | s | 0 | | | | 0 | |
| Sintersearch | dB | not sent | | | | not sent | |
| Propagation Condition | | AWGN | | | | AWGN | |

8.2.2.3.4.1.2 Procedure

a) The SS activates cell 1 and cell 2 with T1 defined parameters and monitors them for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 30 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

8.2.2.3.4.2 1,28 Mcps Option

8.2.2.3.4.2.1 Initial Conditions

This test is to verify the requirement for the 1,28 Mcps TDD OPTION/FDD cell re-selection delay reported in clause 4.2. in [2]

This scenario implies the presence 1.28Mps TDD serving cell, and 1 FDD cell to be selected. The UE is requested to monitor neighbouring cells on 1 1.28Mcps TDD carrier and 1 FDD carrier. Test parameters are given in table 8.2.2.3.4.1A and table 8.2.2.3.4.2A.

Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.3.4.1A: General test parameters for the TDD/FDD cell re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | 1,28 Mcps TDD OPTION cell |
| | Neighbour cells | | Cell2 | FDD cell |
| Final condition | Active cell | | Cell2 | FDD cell |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.2.2.3.4.2A: Test parameters for the 1,28 Mcps TDD OPTION/FDD cell re-selection**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPts | | n.a. | |
| | | | | | | | |
| | | T1 | T2 | T 1 | T 2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | -12 |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | n.a. | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | -10 |
| SCH_Ec/Ior | dB | n.a. | | n.a. | | -12 | -12 |
| PICH_Ec/Ior | dB | | | | | -15 | -15 |
| *OCNS_Ec/Ior* | dB | -3 | | -3 | | -0,941 | -0,941 |
| $\hat{I}_{or}/I_{oc}$ | dB | 8 | 2 | 8 | 2 | -3 | -3 |
| $I_{oc}$ | | -70 dBm/1.28MHz | | | | -70 dBm/3.84MHz | |
| PCCPCH_RSCP | dBm | -65 | -71 | | | n.a. | n.a. |
| CPICH_RSCP | | n.a. | | | | -83 | -77 |
| Cell_selection_and_re selection quality _measure | | CPICH_RSCP | | | | CPICH_RSCP | |
| Qrxlevmin | dBm | -103 | | | | -115 | |
| $Qoffset1_{s,n}$ | dB | C1, C2: -12 | | | | C2, C1: +12 | |
| $Qhyst1_{s}$ | dB | 0 | | | | 0 | |
| Treselection | s | 0 | | | | 0 | |
| Sintersearch | dB | not sent | | | | | |
| Propagation Condition | | AWGN | | | | | |
| * Note: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | | | |

8.2.2.3.4.2.2 Procedure

a) The SS activates cell 1 and cell 2 with T1 defined parameters and monitors them for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 30 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps b)-g) until the confidence level according to annex F.6.2 is achieved.

#### 8.2.2.3.4.3 7,68 Mcps Option

##### 8.2.2.3.4.3.1 Initial conditions

This scenario implies the presence of 1 TDD and 1 FDD cell as given in table 8.2.2.3.4.1B and table 8.2.2.3.4.2B. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.3.4.1B: General test parameters for the TDD/FDD cell re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | TDD cell |
| | Neighbour cells | | Cell2 | FDD cell |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | During T1 cell 1 better ranked than cell 2 |
| T2 | | s | 15 | During T2 cell 2 better ranked than cell 1 |

**Table 8.2.2.3.4.2B: TDD/FDD cell re-selection**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **n.a** | **n.a.** |
| | | T1 | T2 | T 1 | T 2 | T 1 | T 2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | -12 |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -12 | -12 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | n.a. | n.a. |
| PICH_Ec/Ior | dB | | | -3 | -3 | -15 | -15 |
| *OCNS_EcIor* | dB | -3,12 | -3,12 | -3,12 | -3,12 | -0,941 | -0,941 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -2 | 3 | -2 | -2 | 3 |
| $I_{oc}$ | | -70 dBm/7.68 MHz | | | | -70 dBm/3.84 MHz | |
| CPICH_RSCP | dBm | n.a. | | n.a. | | -82 | -77 |
| PCCPCH_RSCP | dBm | -70 | -75 | | | n.a. | n.a. |
| Cell_selection_and reselection_quality _measure | | CPICH_RSCP | | | | *CPICH_RSCP* | |
| Qrxlevmin | dBm | -102 | | | | -115 | |
| Qoffset1$_{s,n}$ | dB | C1, C2: -12 | | | | C2, C1: +12 | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | |
| Treselection | s | 0 | | | | 0 | |
| Propagation Condition | | AWGN | | | | AWGN | |

#### 8.2.2.3.4.3.2 Procedure

a) The SS activates cell 1 and cell 2 with T1 defined parameters and monitors them for RRC CONNECTION REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 30 s, the parameters are changed as described for T2.

e) The SS waits for RRC CONNECTION REQUEST messages from the UE.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved.

### 8.2.2.3.5 Test requirements

#### 8.2.2.3.5.1 3,84 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.3.5.2 1,28 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.3.5.3 7,68 Mcps TDD Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 8 s.

3) In step g), the UE shall respond on cell 1 within 8 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.4 Scenario 4: inter RAT cell re-selection

#### 8.2.2.4.1 Definition and applicability

##### 8.2.2.4.1.1 3,84 Mcps Option

The cell re-selection delay is defined as the time from when the cell quality levels change to the moment when this change makes the UE reselect a better ranked cell, and starts to send LOCATION UPDATING REQUEST message to perform a Location update to the new cell.

This test is for the case where the UE camps on a 3,84 Mcps TDD cell and reselects to a GSM cell.

The requirements and this test apply to UEs supporting both 3,84 Mcps TDD and GSM.

##### 8.2.2.4.1.2 1,28 Mcps Option

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send LOCATION UPDATING REQUEST message to perform a Location update.

This test is for the case where the UE camps on a 1,28 Mcps TDD cell and reselects to a GSM cell.

The requirements and this test apply to UEs supporting both 1,28 Mcps TDD and GSM.

##### 8.2.2.4.1.3 7,68 Mcps Option

The cell re-selection delay is defined as the time from when the cell quality levels change to the moment when this change makes the UE reselect a better ranked cell, and starts to send LOCATION UPDATING REQUEST message to perform a Location update to the new cell.

This test is for the case where the UE camps on a 7,68 Mcps TDD cell and reselects to a GSM cell.

The requirements and this test apply to UEs supporting both 7,68 Mcps TDD and GSM.

#### 8.2.2.4.2 Minimum requirement

##### 8.2.2.4.2.1 3,84 Mcps Option

The cell re-selection delay shall be less than 26 s + $T_{BCCH}$, where $T_{BCCH}$ is the maximum time allowed to read BCCH data in the GSM cell.

NOTE: The cell re-selection delay can be expressed as:

$$4 * T_{measureGSM} + T_{BCCH}$$

where:

$T_{measureGSM}$ Specified in Table 4.1 in sub clause 4.2.2.7.1 in [2].

$T_{BCCH}$ Equal to 1.9 s, i.e. the maximum time allowed to read BCCH data when synchronised to a BCCH carrier from a GSM cell (specified in TS 45.005).

The rate of correct cell reselections observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.4

##### 8.2.2.4.2.2 1,28 Mcps Option

The cell re-selection delay shall be less than 8 s + $T_{BCCH}$ where $T_{BCCH}$ is the maximum time allowed to read BCCH data from GSM cell.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay can be expressed as: Max(3*$T_{measureNTDD}$ , $T_{measure\ GSM}$ +1DRX)+ $T_{BCCH}$, where:

$T_{measureTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{measureNTDD}$ of 1.28s according to Table 4.1A in [2].

$T_{measureGSM}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{measureGSM}$ of 6.4s according to Table 4.1A in [2].

DRX cycle length 1.28s is assumed, see Table A.4.1A in [2].

$T_{BCCH}$ Maximum time allowed to read BCCH data from GSM cell in TS 45.005.

According to[20], the maximum time allowed to read the BCCH data, when being synchronized to a BCCH carrier, is 1.9 s.

This gives a total of 7.68s +$T_{BCCH}$, thus allow 8s +$T_{BCCH.}$

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.4

#### 8.2.2.4.2.3 7,68 Mcps Option

The cell re-selection delay shall be less than 26 s + $T_{BCCH}$, where $T_{BCCH}$ is the maximum time allowed to read BCCH data in the GSM cell [21].

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay can be expressed as:

$$4 * T_{measureGSM} + T_{BCCH}$$

where:

$T_{measureGSM}$ Specified in Table 4.1B in sub clause 4.2.2.7.3 in [2].

$T_{BCCH}$ Equal to 1.9 s, i.e. the maximum time allowed to read BCCH data when synchronised to a BCCH carrier from a GSM cell [21].

This gives a total of 25.6 s + $T_{BCCH}$, allow 26 s + $T_{BCCH}$ in the test case.

### 8.2.2.4.3 Test purpose

This test verifies the UE meets the minimum requirement for the case where the UE camps on a TDD cell and reselects to a GSM cell.

### 8.2.2.4.4 Method of Test

#### 8.2.2.4.4.1 3,84 Mcps Option

##### 8.2.2.4.4.1.1 Initial conditions

This scenario implies the presence of 1 3,84 Mcps TDD and 1 GSM cell as given in table 8.2.2.4.1, 8.2.2.4.2, and 8.2.2.4.3. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.4.1: General test parameters for UTRAN to GSM Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | TDD Cell |
| | Neighbour cell | | Cell2 | GSM Cell |
| Final condition | Active cell | | Cell2 | |
| DRX cycle length | | s | 1,28 | UTRAN cell |
| BCCH repetition period (GSM cell) | | s | 1,87 | In GSM the system information is scheduled according to an 8 x (51 x 8) cycle (i.e. a system information message is transmitted every 235 ms). The cell selection parameters in system info 3 and 4 are transmitted at least every second. (GSM 05.02) |
| T1 | | s | 45 | |
| T2 | | s | 35 | |

**Table 8.2.2.4.2: Cell re-selection UTRAN to GSM cell case (cell 1)**

| Parameter | Unit | Cell 1 (UTRA) | | | |
|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 |
| PICH_Ec/Ior | dB | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -2 | 3 | -2 |
| $I_{oc}$ | dBm/3,84 MHz | –70 | | –70 | |
| PCCPCH RSCP | dBm | -70 | -75 | | |
| Propagation Condition | | AWGN | | AWGN | |
| Treselection | s | 0 | | | |
| $Ssearch_{RAT}$ | dB | not sent | | | |

**Table 8.2.2.4.3: Cell re-selection UTRAN to GSM cell case (cell 2)**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | **T1** | **T2** |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -90 | -75 |
| RXLEV_ACCESS_MIN | dBm | -100 | |
| MS_TXPWR_MAX_CCH | dBm | 30 | |

8.2.2.4.4.1.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST and LOCATION UPDATING REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 45 s, the parameters are changed as described for T2.

e) The SS waits for LOCATION UPDATING REQUEST messages from the UE.

f) After 35 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) [TBD] times.

8.2.2.4.4.2 1,28 Mcps Option

8.2.2.4.4.2.1 Initial conditions

This scenario implies the presence of 1 1,28 Mcps TDD serving cell, and 1 GSM cell to be re-selected. Test parameters are given in table 8.2.2.4.1A, 8.2.2.4.2A, and 8.2.2.4.3A.

The ranking of the cells shall be made according to the cell reselection criteria specified in TS25.304. Cell 1 and cell 2 shall belong to different location areas.

**Table 8.2.2.4.1A: General test parameters for UTRAN (1,28 Mcps TDD OPTION) to GSM Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cell | | Cell2 | |
| Final condition | Active cell | | Cell2 | |
| DRX cycle length | | s | 1,28 | |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.2.2.4.2A: Cell re-selection UTRAN to GSM cell case (cell 1)**

| Parameter | Unit | Cell 1 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| $\hat{I}_{or}/I_{oc}$ | dB | 13 | -12 | 13 | -12 |
| $I_{oc}$ | dBm/1,28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -70 | -95 | n.a. | n.a. |
| Propagation Condition | | AWGN | | AWGN | |
| Treselection | s | 0 | | | |
| $Ssearch_{RAT}$ | dB | Not sent | | | |
| Qrxlevmin | dBm | -103 | | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0 | | | |
| $Qhyst1_{s}$ | dB | 0 | | | |
| * Note: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | |

**Table 8.2.2.4.3A: Cell re-selection UTRAN to GSM cell case (cell 2)**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -75 | -75 |
| RXLEV_ACCESS_MIN | dBm | -104 | |
| MS_TXPWR_MAX_CCH | dBm | 33 | |

8.2.2.4.4.2.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST and LOCATION UPDATING REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for LOCATION UPDATING REQUEST messages from the UE.

f) After 15 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps b)-g) until the confidence level according to annex F.6.2 is achieved.

#### 8.2.2.4.4.3 7,68 Mcps Option

##### 8.2.2.4.4.3.1 Initial conditions

This scenario implies the presence of 1 7,68 Mcps TDD and 1 GSM cell as given in table 8.2.2.4.1B, 8.2.2.4.2B, and 8.2.2.4.3B. Cell 1 and cell 2 shall belong to different Location Areas.

**Table 8.2.2.4.1B: General test parameters for UTRAN to GSM Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | UTRA TDD Cell |
| | Neighbour cell | | Cell2 | GSM Cell |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| DRX cycle length | | s | 1,28 | UTRA TDD cell |
| T1 | | s | 45 | |
| T2 | | s | 35 | |

**Table 8.2.2.4.2B: Cell re-selection UTRAN to GSM cell case (cell 1)**

| Parameter | Unit | Cell 1 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 |
| PICH_Ec/Ior | dB | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -2 | 3 | -2 |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | -70 | |
| PCCPCH RSCP | dBm | -70 | -75 | n.a. | n.a. |
| Propagation Condition | | AWGN | | AWGN | |
| Qrxlevmin | dBm | -102 | | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0 | | | |
| Qhyst1 | dB | 0 | | | |
| Treselection | s | 0 | | | |
| Ssearch$_{RAT}$ | dB | not sent | | | |

**Table 8.2.2.4.3B: Cell re-selection UTRAN to GSM cell case (cell 2)**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | **T1** | **T2** |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -90 | -75 |
| RXLEV_ACCESS_MIN | dBm | -104 | |
| MS_TXPWR_MAX_CCH | dBm | 33 | |

##### 8.2.2.4.4.3.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST and LOCATION UPDATING REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 45 s, the parameters are changed as described for T2.

e) The SS waits for LOCATION UPDATING REQUEST messages from the UE.

f) After 35 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved.

#### 8.2.2.4.5 Test Requirements

##### 8.2.2.4.5.1 3,84 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 27.9 s.

3) In step g), the UE shall respond on cell 1.

For the test to pass, the total number of fulfilled test requirements in step 2) shall be at least 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

##### 8.2.2.4.5.2 1,28 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 9.7 s.

3) In step g), the UE shall respond on cell 1.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

##### 8.2.2.4.5.3 7,68 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 27.9 s.

3) In step g), the UE shall respond on cell 1.

For the test to pass, the total number of fulfilled test requirements in step 2) shall be at least 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.2.2.5 Scenario 4A: inter RAT cell acquisition and re-selection

#### 8.2.2.5.1 Definition and applicability

##### 8.2.2.5.1.1 3,84 Mcps Option

Void

Note: Scenario 4A does not apply for 3,84 Mcps TDD, this section numbering is purely for consistency with TS 25.123, where corresponding sections are also void.

#### 8.2.2.5.1.2 1,28 Mcps Option

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send RR Channel Request message for location update to Cell 2. ,

Unlike 1,28 Mcps scenario 4 the initial GSM RX_LEV is set below monitoring threshold and GSM measurements do not start until the second phase of the test. The requirements and this test apply to UEs supporting both 1,28 Mcps TDD and GSM.

#### 8.2.2.5.1.3 7,68 Mcps Option

Void

Note: Scenario 4A does not apply for 7,68 Mcps TDD, this section numbering is purely for consistency with TS 25.123, where corresponding sections are also void.

### 8.2.2.5.2 Minimum requirement

#### 8.2.2.5.2.1 3,84 Mcps Option

Void

#### 8.2.2.5.2.2 1,28 Mcps Option

The cell re-selection delay shall be less than 26 s+ $T_{BCCH}$, where $T_{BCCH}$ is the maximum time allowed to read BCCH data from GSM cell [21].

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The UE shall keep a running average of 4 measurements, thus gives 4* $T_{measureGSM}$ +$T_{BCCH}$, where:

$T_{measureGSM}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{measureGSM}$ of 6.4s according to Table 4.1A in section 4.2. in [2].

$T_{BCCH}$ Maximum time allowed to read BCCH data from GSM cell in TS 45.005 [20].

According to [21], the maximum time allowed to read the BCCH data, when being synchronized to a BCCH carrier, is 1.9 s.

This gives a total of 25,6s +$T_{BCCH}$, thus allow 26s +$T_{BCCH.}$

The normative reference for this requirement is TS 25.123 [2] clauses 4.2.2 and A.4.2.4.3

#### 8.2.2.5.2.3 7,68 Mcps Option

Void

### 8.2.2.5.3 Test purpose

This test verifies the UE meets the minimum requirement for the case where the UE camps on a TDD cell and then acquires and reselects to a GSM cell

### 8.2.2.5.4 Method of Test

#### 8.2.2.5.4.1 3,84 Mcps Option

Void

#### 8.2.2.5.4.2 1,28 Mcps Option

##### 8.2.2.5.4.2.1 Initial conditions

This scenario implies the presence of 1 1,28 Mcps TDD serving cell, and 1 GSM cell to be re-selected. Test parameters are given in table 8.2.2.5.1A, 8.2.2.5.2A, and 8.2.2.5.3A.

The ranking of the cells shall be made according to the cell reselection criteria specified in TS25.304. Cell 1 and cell 2 shall belong to different location areas.

**Table 8.2.2.5.1A: General test parameters for UTRAN (1,28 Mcps TDD OPTION) to GSM Cell Re-selection Scenario 4A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | 1,28 Mcps TDD OPTION cell |
| | Neighbour cell | | Cell2 | GSM cell |
| Final condition | Active cell | | Cell2 | GSM cell |
| DRX cycle length | | s | 1,28 | |
| HCS | | | Not Used | |
| T1 | | s | 45 | |
| T2 | | s | 45 | |

**Table 8.2.2.5.2A: Cell re-selection UTRAN to GSM cell case (cell 1) Scenario 4A**

| Parameter | Unit | Cell 1 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 6 | 6 | 6 |
| $I_{oc}$ | dBm/1,28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -77 | -77 | | |
| Propagation Condition | | AWGN | | AWGN | |
| Treselection | s | 0 | | | |
| $Ssearch_{RAT}$ | dB | Not sent | | | |
| Qrxlevmin | dBm | -103 | | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0 | | | |
| $Qhyst1_{s}$ | dB | 0 | | | |
| * Note: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | |

**Table 8.2.2.5.3A: Cell re-selection UTRAN to GSM cell case (cell 2) Scenario 4A**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -90 | -70 |
| RXLEV_ACCESS_MIN | dBm | -104 | |
| MS_TXPWR_MAX_CCH | dBm | 33 | |

8.2.2.5.4.2.2 Procedure

a) The SS activates cell 1 and 2 with T1 defined parameters and monitors cell 1 and 2 for RRC CONNECTION REQUEST and LOCATION UPDATING REQUEST messages from the UE.

b) The UE is switched on.

c) The SS waits until the UE camps on Cell 1 and sends the RRC CONNECTION REQUEST message.

d) After 45 s, the parameters are changed as described for T2.

e) The SS waits for RR Channel Request message for a location update to Cell 2 from the UE.

f) After 45 s, the parameters are changed as described for T1.

g) The SS waits for RRC CONNECTION REQUEST messages from the UE.

h) Repeat steps b)-g) until the confidence level according to annex F.6.2 is achieved.

#### 8.2.2.5.4.3 7,68 Mcps Option

Void

### 8.2.2.5.5 Test Requirements

#### 8.2.2.5.5.1 3,84 Mcps Option

void

#### 8.2.2.5.5.2 1,28 Mcps Option

1) In step c), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).
2) In step e), the UE shall respond on cell 2 within 28 s.
3) In step g), the UE shall respond on cell 1.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.2.2.5.5.3 7,68 Mcps Option

void

# 8.3 UTRAN Connected Mode Mobility

## 8.3.1 TDD/TDD Handover for 3,84 Mcps Option

### 8.3.1.1 Handover to intra-frequency cell

#### 8.3.1.1.1 Definition and applicability

Handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCH.

The requirements and this test apply to the UTRA TDD UE.

#### 8.3.1.1.2 Minimum requirement

The interruption time shall be less than 40 ms in the single carrier case when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/TDD intra-frequency handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 * KC + 180 * UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel |
| $T_{UL}$ | Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell |
| $F_{SFN}$ | Equal to 1 if SFN decoding is required and equal to 0 otherwise |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |

UC Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise

An intra-frequency TDD target cell shall be considered as known by the UE, if either or both of the following conditions are true:

- the target cell has been measured during the last 5 seconds
- the UE has had a radio link connected to the target cell during the last 5 seconds.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.1.

#### 8.3.1.1.3 Test purpose

The purpose of this test is to verify the requirement for the intra-frequency handover delay in CELL_DCH state in the single carrier case.

#### 8.3.1.1.4 Method of test

##### 8.3.1.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1.1.1 and 8.3.1.1.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP and SFN-CFN observed timed difference shall be reported together with Event 1G. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at the beginning of T3 with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The UL DPCH shall be transmitted in timeslot 12.

**Table 8.3.1.1.1: General test parameters for Handover to intra-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 10 | |

**Table 8.3.1.1.2: Cell specific test parameters for Handover to intra-frequency cell**

| Parameter | Unit | Cell 1 | | | | | | Cell 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | | | 4 | | | 0 | | | 5 | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | | 5 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. | n.a. | -3,12 | | n.a. | | Note 2 |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | | -Inf. | 3 | | -Inf. | | 3 |
| PCCPCH RSCP | dBm | -72 | | | n.a. | | | -Inf. | -70 | | n.a. | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | | | | | | | |

8.3.1.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 10 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCH to cell 2 less than 40 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 2 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 2) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| Note 1: The SFN-CFN observed time difference is calculated from the OFF and Tm parameters contained in the IE "Cell synchronisation information ", TS 25.331, clause 10.3.7.6. According to TS 25.331, 8.6.7.7, this IE is included in MEASUREMENT REPORT if IE "Cell synchronisation information reporting indicator" in IE "Cell reporting quantities" TS 25.331, clause 10.3.7.5 is set to TRUE in MEASUREMENT CONTROL.<br>Note 2: Reporting interval = 0 ms means no periodical reporting | |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -UL Target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -Indivdual Timeslot interference info | 1 |
| -Individual timeslot interference (10.3.6.38) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -Timeslot number | 12 |
| -UL Timeslot Interference | -90 dBm |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | TBD dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 3,84 Mcps |
| -Timeslot number | 12 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 3,84 Mcps |
| -Choice Burst Type | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 3,84 Mcps |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -Choice more timeslots | No more timeslots |

| Information Element | Value/Remark |
|---|---|
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps |
| -TX Diversity mode (10.3.6.86) | None |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps |
| -CHOICE *sync case* | Case 2 |
| -Timeslot | 0 |
| -Cell parameters ID | 20 |
| -SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes (10.3.6.32) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 3,84 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 3,84 Mcps |
| -CHOICE *Burst Type* | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 3,84 Mcps |
| -First timeslot channelisation codes (10.3.6.17) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/2 |
| -CHOICE *more timeslots* | No more timeslots |
| -SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Checked that this IE is present |
| -CHOICE mode | TDD |
| -Cell parameters ID | 20 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | 40 |

#### 8.3.1.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.1.2 Handover to inter-frequency cell

#### 8.3.1.2.1 Definition and applicability

The handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCH.

The requirements and this test apply to the UTRA TDD UE.

#### 8.3.1.2.2 Minimum requirement

The interruption time shall be less than 40 ms in the dual carrier case when the cell is known by the UE and the SFN of the target cell needs to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/TDD inter-frequency handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 * KC + 180 * UC \text{ ms}$$

where,

$T_{offset}$ Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

| | |
|---|---|
| $T_{UL}$ | Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell |
| $F_{SFN}$ | Equal to 1 if SFN decoding is required and equal to 0 otherwise |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |

An inter-frequency TDD target cell shall be considered as known by the UE, if either or both of the following conditions are true:

- the target cell has been measured during the last 5 seconds
- the UE has had a radio link connected to the target cell during the last 5 seconds.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.2.

#### 8.3.1.2.3 Test purpose

The purpose of this test is to verify the requirement for the inter-frequency handover delay in CELL_DCH state in the dual carrier case.

#### 8.3.1.2.4 Method of test

##### 8.3.1.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1.2.1 and 8.3.1.2.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. The PCCPCH RSCP and SFN-CFN observed time difference of the best cell on the unused frequency shall be reported together with Event 2C reporting. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at beginning of T3 with one active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. The UL DPCH shall be transmitted in timeslot 12.

**Table 8.3.1.2.1: General test parameters for Handover to inter-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | Hysteresis parameter for event 2C |
| Time to Trigger | | ms | 0 | |
| Threshold non-used frequency | | dBm | -80 | Applicable for Event 2C |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 TDD neighbours on Channel 2 | |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 10 | |

**Table 8.3.1.2.2: Cell Specific parameters for Handover to inter-frequency cell**

| Parameter | Unit | Cell 1 | | | | | | Cell 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | | | 4 | | | 2 | | | 5 | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | Channel 2 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | | 5 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. | n.a. | -3,12 | | n.a. | | Note 2 |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | | -Inf. | 7 | | -Inf | | 7 |
| PCCPCH RSCP | dBm | -72 | | | n.a. | | | -Inf. | -66 | | n.a. | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | | | | | | | |

8.3.1.2.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 10 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCH to cell 2 less than 40 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

MEASUREMENT CONTROL message, event 2C (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No Report |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2C |
| -Threshold used frequency | Not Present |
| -W used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -80 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -UL Target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -Indivdual Timeslot interference info | 1 |
| -Individual timeslot interference (10.3.6.38) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -Timeslot number | 12 |
| - UL Timeslot Interference | -90 dBm |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | TBD dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 3,84 Mcps |
| -Timeslot number | 12 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 3,84 Mcps |
| -CHOICE *Burst Type* | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 3,84 Mcps |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -CHOICE *more timeslots* | No more timeslots |

| Information Element | Value/Remark |
|---|---|
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps |
| -TX Diversity mode (10.3.6.86) | None |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| - CHOICE *mode* | TDD |
| - CHOICE *TDD option* | 3,84 Mcps |
| - CHOICE *sync case* | Case 2 |
| - Timeslot | 2 |
| - Cell parameters ID | 20 |
| - SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| - DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| - Downlink DPCH timeslots and codes (10.3.6.32) | |
| - First individual timeslot info (10.3.6.37) | |
| - Timeslot Number (10.3.6.84) | |
| - CHOICE *TDD option* | 3,84 Mcps |
| - Timeslot number | 5 |
| - TFCI existence | True |
| - Midamble shift and burst type (10.3.6.41) | |
| - CHOICE *TDD option* | 3,84 Mcps |
| - CHOICE *Burst Type* | Type 1 |
| - Midamble Allocation Mode | Default |
| - Midamble configuration burst type 1 and 3 | 16 |
| - Midamble shift | Not present |
| - CHOICE *TDD option* | 3,84 Mcps |
| - First timeslot channelisation codes (10.3.6.17) | |
| - CHOICE *codes representation* | Consecutive codes |
| - First channelisation code | 16/1 |
| - Last channelisation code | 16/2 |
| - CHOICE *more timeslots* | No more timeslots |
| - SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 in Table 8.6.2.4.1.2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Checked that this IE is present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 in Table 8.6.2.4.1.2 |
| -CHOICE mode | TDD |
| -Primary CCPCH Info | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 3,84 Mcps TDD |
| -CHOICE Sync Case | Not Present |
| -Cell Parameters ID | Set to cell parameters ID of Cell 2 |
| -SCTD Indicator | FALSE |

#### 8.3.1.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.1A TDD/TDD Handover for 1,28 Mcps Option

### 8.3.1A.1 Handover to intra-frequency cell

#### 8.3.1A.1.1 Scenario 1

This test case is applicable for UE handovers in single frequency network and UE handovers from primary frequency to primary frequency in multi-frequency network

##### 8.3.1A.1.1.1 Definition and applicability

Handover delay of the UE is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission of the new uplink DPCH or the SYNC-UL in case that a handover with SYNCH uplink exchange is recommended, including the RRC procedure delay as defined in [9].

The requirements and this test apply to the UTRA TDD UE 1,28 Mcps option.

#### 8.3.1A.1.1.2 Minimum requirement

The hard handover delay shall be less than 160 ms when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.2.

#### 8.3.1A.1.1.3 Test purpose

The purpose of this test is to verify the requirement for the intra-frequency handover delay in CELL_DCH state.

#### 8.3.1A.1.1.4 Method of test

##### 8.3.1A.1.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1A.1.1.1 and 8.3.1A.1.1.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP and SFN-CFN observed timed difference shall be reported together with Event 1G. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time ”now” with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE during period T2, after the UE has reported event 1G. The starting point of T3 is defined as the end of the last TTI containing the physical channel reconfiguration message.

**Table 8.3.1A.1.1.1: General test parameters for Handover to intra-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2.2 and A.2.1.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | In the case of multi-frequency network, the UE has a RRC connection on the primary frequency in cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | In the case of multi-frequency network, UE will handover to the primary frequency in cell 2. |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 5 | |

**Table 8.3.1A.1.1.2: Cell specific test parameters for Handover to intra-frequency cell 1,28Mcps option**

| Parameter | Unit | Cell 1 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | Note1 | | n.a. |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | 3 | | | 3 | | |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |
| **Parameter** | **Unit** | **Cell 2** | | | | | | | | |
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(NOTE3) | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | n.a. | | Note1 |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Inf. | 5 | | -Inf. | 5 | | -Inf. | | 5 |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -Inf. | -68 | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: The DPCH level is controlled by the power control loop
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note3: In the case of multi-frequency network, the UTRA RF Channel Number is the primary frequency's channel number.

8.3.1A.1.4.1.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 5 seconds, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G.
7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time "now".
8) After 5 seconds, the SS shall switch the power settings from T2 to T3
9) If the UE transmits the UL DPCH to cell 2 less than 160 ms from the beginning of time period T3 then the number of successful tests is increased by oneUE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2.
10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 2 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 2) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| Note 1: The SFN-CFN observed time difference is calculated from the OFF and Tm parameters contained in the IE "Cell synchronisation information ", TS 25.331, clause 10.3.7.6. According to TS 25.331, 8.6.7.7, this IE is included in MEASUREMENT REPORT if IE "Cell synchronisation information reporting indicator" in IE "Cell reporting quantities" TS 25.331, clause 10.3.7.5 is set to TRUE in MEASUREMENT CONTROL.<br>Note 2: Reporting interval = 0 ms means no periodical reporting | |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 30 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -*TDD* | 1,28 Mcps TDD |
| -UL target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | |
| --1,28 Mcps TDD | 1 |
| -TPC step size | 1 |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | 9 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 2 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| - SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -Choice more timeslots | No more timeslots |
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |

| Information Element | Value/Remark |
|---|---|
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| - *TSTD indicator* (10.3.6.85a) | TRUE |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| *TSTD indicator* | TRUE |
| -Cell parameters ID | 0 |
| -SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes (10.3.6.32) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| -SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot channelisation codes (10.3.6.17) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/2 |
| -CHOICE *more timeslots* | No more timeslots |
| -SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message for Intra frequency test cases

This message is common for all intra frequency test cases in clause 8.7 and is described in Annex I.

### 8.3.1A.1.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.1A.1.2 Scenario 2

This test case is applicable for UE handovers from primary frequency to secondary frequency in the case of multi-frequency network.

#### 8.3.1A.1.2.1 Definition and applicability

Handover delay of the UE is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission of the new uplink DPCH or the SYNC-UL in case that a handover with SYNCH uplink exchange is recommended, including the RRC procedure delay as defined in [9].

The requirements and this test apply to the UTRA TDD UE 1,28 Mcps option.

#### 8.3.1A.1.2.2 Minimum requirement

The hard handover delay shall be less than 160 ms when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.2.

#### 8.3.1A.1.2.3 Test purpose

The purpose of this test is to verify the requirement for the intra-frequency handover delay in CELL_DCH state.

#### 8.3.1A.1.2.4 Method of test

##### 8.3.1A.1.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1A.1.2.1 and 8.3.1A.1.2.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP and SFN-CFN observed timed difference shall be reported together with Event 1G. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time ”now” with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE during period T2, after the UE has reported event 1G. The starting point of T3 is defined as the end of the last TTI containing the physical channel reconfiguration message.

**Table 8.3.1A.1.2.1: General test parameters for Handover to intra-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2.2 and A.2.1.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | The UE has a RRC connection on the secondary frequency in cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | The UE will handover to the primary frequency in cell 2. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 5 | |

**Table 8.3.1A.1.2.2: Cell specific test parameters for Handover to intra-frequency cell 1,28Mcps option**

| Parameter | Unit | Cell 1 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | Note1 | | n.a. |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | 3 | | | 3 | | |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |
| **Parameter** | **Unit** | **Cell 2** | | | | | | | | |
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(NOTE3) | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Inf. | 5 | | -Inf. | 5 | | -Inf. | | 5 |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -Inf. | -68 | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: The DPCH level is controlled by the power control loop
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note3: In the case of multi-frequency network, the UTRA RF Channel Number is the primary frequency's channel number.

8.3.1A.1.2.4.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 5 seconds, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G.
7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time "now".
8) After 5 seconds, the SS shall switch the power settings from T2 to T3
9) If the UE transmits the UL DPCH to cell 2 less than 160 ms from the beginning of time period T3 then the number of successful tests is increased by oneUE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2.
10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 2 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 2) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| Note 1: The SFN-CFN observed time difference is calculated from the OFF and Tm parameters contained in the IE "Cell synchronisation information ", TS 25.331, clause 10.3.7.6. According to TS 25.331, 8.6.7.7, this IE is included in MEASUREMENT REPORT if IE "Cell synchronisation information reporting indicator" in IE "Cell reporting quantities" TS 25.331, clause 10.3.7.5 is set to TRUE in MEASUREMENT CONTROL. | |
| Note 2: Reporting interval = 0 ms means no periodical reporting | |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 30 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -*TDD* | 1,28 Mcps TDD |
| -UL target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | |
| --1,28 Mcps TDD | 1 |
| -TPC step size | 1 |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | 9 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 2 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| - SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -Choice more timeslots | No more timeslots |
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |

| Information Element | Value/Remark |
|---|---|
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| - *TSTD indicator* (10.3.6.85a) | TRUE |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| *TSTD indicator* | TRUE |
| -Cell parameters ID | 0 |
| -SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes (10.3.6.32) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| -SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot channelisation codes (10.3.6.17) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/2 |
| -CHOICE *more timeslots* | No more timeslots |
| -SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message for Intra frequency test cases

This message is common for all intra frequency test cases in clause 8.7 and is described in Annex I.

### 8.3.1A.1.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.1A.2 Handover to inter-frequency cell

### 8.3.1A.2.1 Scenario 1

This test case is applicable for UE handovers in single frequency network and UE handovers from primary frequency to the primary frequency in multi-frequency network.

#### 8.3.1A.2.1.1 Definition and applicability

Handover delay of the UE is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission of the new uplink DPCH or the SYNC-UL in case that a handover with SYNCH uplink exchange is recommended, including the RRC procedure delay as defined in [9].

The requirements and this test apply to the UTRA TDD UE 1,28 Mcps option.

#### 8.3.1A.2.1.2 Minimum requirement

The hard handover delay shall be less than 160ms when the cell is known by the UE and the SFN of the target cell needs to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.2.

#### 8.3.1A.2.1.3 Test purpose

The purpose of this test is to verify the requirement for the inter-frequency handover delay in CELL_DCH state.

#### 8.3.1A.2.1.4 Method of test

##### 8.3.1A.2.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1A.2.1.1 and 8.3.1A.2.1.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. The PCCPCH RSCP and SFN-CFN observed time difference of the best cell on the unused frequency shall be reported together with Event 2C reporting. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time"now" with one active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE during period T2, after the UE has reported event 2C , The starting point of T3 is defined as the end of the last TTI containing the physical channel reconfiguration message.

**Table 8.3.1A.2.1.1: General test parameters for Handover to inter-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2.2 and A.2.1.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | In the case of multi-frequency network, the UE has a RRC connection on the primary frequency in cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | In the case of multi-frequency network, the UE will handover to the primary frequency in cell 2. |
| Threshold non-used frequency | | dBm | -75 | Absolute threshold RSCP for Event 2C |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 TDD neighbours on Channel 2 | |
| T1 | | s | 5 | |
| T2 | | s | 10 | |
| T3 | | s | 5 | |

**Table 8.3.1A.2.1.2: Cell Specific parameters for Handover to inter-frequency cell**

| Parameter | Unit | Cell 1 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(Note 3) | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | Note1 | | n.a. |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | 3 | | | 3 | | |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |
| **Parameter** | **Unit** | **Cell 2** | | | | | | | | |
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(Note 3) | | Channel 2 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Inf. | 9 | | -Inf. | 9 | | -Inf. | | 9 |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -Inf. | -64 | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: The DPCH level is controlled by the power control loop
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note3: In the case of multi-frequency network, the UTRA RF Channel Number is the primary frequency's channel number.

8.3.1A.2.1.4.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 5 seconds, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C.
7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time "now".
8) After 10 seconds, the SS shall switch the power settings from T2 to T3
9) If the UE transmits the UL DPCH to cell 2 less than 160 ms from the beginning of time period T3 then the number of successful tests is increased by one.UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2.
10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
11) Repeat steps1-10 until the confidence level according to annex F.6.2 is achieved.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message, event 2C (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | Type 1 |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2C |
| -Threshold used frequency | Not Present |
| -W used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -75 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | now |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 30 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | |
| -UL target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | |
| -1,28 Mcps TDD | |
| -TPC stepsize | 1 |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | 9 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 2 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| - SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -CHOICE *more timeslots* | No more timeslots |
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |
| -Downlink information common for all radio links (10.3.6.24) | |

| Information Element | Value/Remark |
|---|---|
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| - TSTD indicator (10.3.6.85a) | TRUE |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| - CHOICE *mode* | TDD |
| - CHOICE *TDD option* | 1,28 Mcps |
| *TSTD indicator* | TRUE |
| - Cell parameters ID | 0 |
| - SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| - DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| - Downlink DPCH timeslots and codes (10.3.6.32) | |
| - First individual timeslot info (10.3.6.37) | |
| - Timeslot Number (10.3.6.84) | |
| - CHOICE *TDD option* | 1,28 Mcps |
| - Timeslot number | 5 |
| - TFCI existence | True |
| - Midamble shift and burst type (10.3.6.41) | |
| - CHOICE *TDD option* | 1,28 Mcps |
| - Midamble Allocation Mode | Default |
| - Midamble configuration | 16 |
| - Midamble shift | Not present |
| - CHOICE *TDD option* | 1,28 Mcps |
| - First timeslot channelisation codes (10.3.6.17) | |
| -Modulation | QPSK |
| -SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| - CHOICE *codes representation* | Consecutive codes |
| - First channelisation code | 16/1 |
| - Last channelisation code | 16/2 |
| - CHOICE *more timeslots* | No more timeslots |
| - SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message for Inter frequency test cases

This message is common for all inter frequency TDD test cases in clause 8.7 and is described in Annex I.

### 8.3.1A.2.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.1A.2.2 Scenario 2

This test case is applicable for UE handovers in single frequency network and UE handovers from primary frequency to the secondary frequency in multi-frequency network.

#### 8.3.1A.2.2.1 Definition and applicability

Handover delay of the UE is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission of the new uplink DPCH or the SYNC-UL in case that a handover with SYNCH uplink exchange is recommended, including the RRC procedure delay as defined in [9].

The requirements and this test apply to the UTRA TDD UE 1,28 Mcps option.

#### 8.3.1A.2.2.2 Minimum requirement

The hard handover delay shall be less than 160ms when the cell is known by the UE and the SFN of the target cell needs to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.2.

#### 8.3.1A.2.2.3 Test purpose

The purpose of this test is to verify the requirement for the inter-frequency handover delay in CELL_DCH state.

#### 8.3.1A.2.2.4 Method of test

##### 8.3.1A.2.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1A.2.2.1 and 8.3.1A.2.2.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. The PCCPCH RSCP and SFN-CFN observed time difference of the best cell on the unused frequency shall be reported together with Event 2C reporting. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time"now" with one active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE during period T2, after the UE has reported event 2C , The starting point of T3 is defined as the end of the last TTI containing the physical channel reconfiguration message.

**Table 8.3.1A.2.2.1: General test parameters for Handover to inter-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2.2 and A.2.1.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | In the case of multi-frequency network, the UE has a RRC connection on the secondary frequency in cell 1. |
| | Neighbour cell | | Cell 2 | |

| | | | | |
|---|---|---|---|---|
| Final condition | Active cell | | Cell 2 | In the case of multi-frequency network, the UE will handover to the primary frequency in cell 2. |
| Threshold non-used frequency | | dBm | -75 | Absolute threshold RSCP for Event 2C |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 TDD neighbours on Channel 2 | |
| T1 | | s | 5 | |
| T2 | | s | 10 | |
| T3 | | s | 5 | |

**Table 8.3.1A.2.2.2: Cell Specific parameters for Handover to inter-frequency cell**

| **Parameter** | **Unit** | **Cell 1** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(Note 3) | | Channel 1 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | Note1 | | n.a. |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | 3 | | | 3 | | |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |
| **Parameter** | **Unit** | **Cell 2** | | | | | | | | |
| **Timeslot Number** | | **0** | | | **DwPTS** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number(Note 3) | | Channel 2 | | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | n.a. | | |
| DwPCH_Ec/Ior | | | | | 0 | | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3 | | | | | | Note2 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Inf. | 9 | | -Inf. | 9 | | -Inf. | | 9 |
| $I_{oc}$ | dBm/1, 28 MHz | -70 | | | | | | | | |
| PCCPCH_RSCP | dBm | -Inf. | -64 | | n.a. | | | n.a. | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: The DPCH level is controlled by the power control loop
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note3: In the case of multi-frequency network, the UTRA RF Channel Number is the primary frequency's channel number.

8.3.1A.2.2.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 5 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time “now”.

8) After 10 seconds, the SS shall switch the power settings from T2 to T3

9) If the UE transmits the UL DPCH to cell 2 less than 160 ms from the beginning of time period T3 then the number of successful tests is increased by one.UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2.

10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat steps1-10 until the confidence level according to annex F.6.2 is achieved.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message, event 2C (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | Type 1 |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2C |
| -Threshold used frequency | Not Present |
| -W used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -75 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 30 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -*TDD* | 1,28 Mcps TDD |
| -UL target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | |
| --1,28 Mcps TDD | 1 |
| -TPC step size | 1 |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| -UL Target SIR | 9 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 2 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| - SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot code list | 1 |
| -Channelisation code | 8/1 |
| -Choice more timeslots | No more timeslots |
| **Downlink radio resources** | |
| -CHOICE *mode* | TDD |

| Information Element | Value/Remark |
|---|---|
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| - *TSTD indicator* (10.3.6.85a) | TRUE |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps |
| *TSTD indicator* | TRUE |
| -Cell parameters ID | 0 |
| -SCTD indicator | False |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | Not Present |
| -Time Info (10.3.6.83) | |
| -Activation Time | now |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes (10.3.6.32) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Midamble Allocation Mode | Default |
| -Midamble configuration | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 1,28 Mcps |
| -Modulation | QPSK |
| -SS-TPC Symbols | |
| -Additional TPC-SS Symbols | |
| -First timeslot channelisation codes (10.3.6.17) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/2 |
| -CHOICE *more timeslots* | No more timeslots |
| -SCCPCH information for FACH (10.3.6.70) | Not Present |

MEASUREMENT REPORT message for Inter frequency test cases

This message is common for all inter frequency TDD test cases in clause 8.7 and is described in Annex I.

### 8.3.1A.2.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

# 8.3.1B TDD/TDD Handover for 7,68 Mcps Option

## 8.3.1B.1 Handover to intra-frequency cell

### 8.3.1B.1.1 Definition and applicability

Handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCH.

The requirements and this test apply to the UTRA TDD UE 7,68 Mcps option.

### 8.3.1B.1.2 Minimum requirement

The interruption time shall be less than 40 ms in the single carrier case when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/TDD intra-frequency handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 * KC + 180 * UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel |
| $T_{UL}$ | Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell |
| $F_{SFN}$ | Equal to 1 if SFN decoding is required and equal to 0 otherwise |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |

An intra-frequency TDD target cell shall be considered as known by the UE, if either or both of the following conditions are true:

- the target cell has been measured during the last 5 seconds
- the UE has had a radio link connected to the target cell during the last 5 seconds.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.3.

### 8.3.1B.1.3 Test purpose

The purpose of this test is to verify the requirement for the intra-frequency handover delay in CELL_DCH state in the single carrier case.

### 8.3.1B.1.4 Method of test

#### 8.3.1B.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1B.1.1 and 8.3.1B.1.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP and SFN-CFN observed timed difference shall be reported together with Event 1G. The test consists of three successive time periods, with a

time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at the beginning of T3 with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The UL DPCH shall be transmitted in timeslot 12.

**Table 8.3.1B.1.1: General test parameters for Handover to intra-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 and A.2.1 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 10 | |

**Table 8.3.1B.1.2: Cell specific test parameters for Handover to intra-frequency cell**

| Parameter | Unit | Cell 1 | | | | | | Cell 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | | | 4 | | | 0 | | | 5 | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | | 5 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. | n.a. | -3,12 | | n.a. | | Note 2 |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | | -Inf. | 3 | | -Inf. | | 3 |
| PCCPCH RSCP | dBm | -72 | | | n.a. | | | -Inf. | -70 | | n.a. | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | | | | | | | |

8.3.1B.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 10 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCH to cell 2 less than 40 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE (Note 1) |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 2 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 2) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| Note 1: The SFN-CFN observed time difference is calculated from the OFF and Tm parameters contained in the IE "Cell synchronisation information ", TS 25.331, clause 10.3.7.6. According to TS 25.331, 8.6.7.7, this IE is included in MEASUREMENT REPORT if IE "Cell synchronisation information reporting indicator" in IE "Cell reporting quantities" TS 25.331, clause 10.3.7.5 is set to TRUE in MEASUREMENT CONTROL.<br>Note 2: Reporting interval = 0 ms means no periodical reporting | |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -Delay restriction flag | Not present |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -New DSCH-RNTI | Not Present |
| -New H-RNTI | Not Present |
| -CHOICE *mode* | TDD |
| -New E-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -UL Target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Indivdual Timeslot interference info | 1 |
| -Individual timeslot interference (10.3.6.38) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Timeslot number | 12 |
| -UL Timeslot Interference | -90 dBm |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| - TFCS ID | 1 |
| -UL Target SIR | 20 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -Timeslot number | 12 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 7,68 Mcps |
| -Choice Burst Type | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |

| Information Element | Value/Remark |
|---|---|
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 7,68 Mcps |
| -First timeslot code list | 1 |
| -Channelisation code | 16/1 |
| -Choice more timeslots | No more timeslots |
| -E-DCH info | Not Present |
| **Downlink radio resources** | |
| -Downlnk HS-PDSCH information | Not Present |
| -Downlink information common for all radio links (10.3.6.24) | |
| - Choice DPCH info | Downlink DPCH info common for all RL |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -MAC-d HFN initial value | Not present |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps |
| -TX Diversity mode (10.3.6.86) | None |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -MAC-hs reset indicator | Not present |
| -Post-verification period | Not present |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps |
| -CHOICE *sync case* | Case 2 |
| -Timeslot | 0 |
| -Cell parameters ID | 20 |
| -SCTD indicator | False |
| -CHOICE *DPCH info* | Downlink DPCH info for each RL |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | 7.68 Mcps TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | 1 |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes VHCR (10.3.6.32a) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -CHOICE *Burst Type* | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 7,68 Mcps |
| -First timeslot channelisation codes VHCR (10.3.6.17a) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/1 |
| -CHOICE *more timeslots* | No more timeslots |
| -UL CCTrCH TPC List | Not Present |
| - DL CCTrCH List to Remove | Not Present |
| - E-AGCH Info | Not Present |
| - CHOICE *mode* | TDD |
| - E=HICH information | Not Present |
| -MBMS PL Service Restriction Information | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Checked that this IE is present |
| -CHOICE mode | TDD |
| -Cell parameters ID | 20 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD option | 7.68Mcps TDD |
| -CHOICE SyncCase | Not Present |
| -Cell parameters ID | 40 |
| - SCTD indicator | FALSE |

#### 8.3.1B.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.1B.2 Handover to inter-frequency cell

#### 8.3.1B.2.1 Definition and applicability

The handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCH.

The requirements and this test apply to the UTRA TDD UE.

#### 8.3.1B.2.2 Minimum requirement

The interruption time shall be less than 40 ms in the dual carrier case when the cell is known by the UE and the SFN of the target cell needs to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/TDD inter-frequency handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 * KC + 180 * UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel |
| $T_{UL}$ | Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell |
| $F_{SFN}$ | Equal to 1 if SFN decoding is required and equal to 0 otherwise |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/TDD handover and equal to 0 otherwise |

An inter-frequency TDD target cell shall be considered as known by the UE, if either or both of the following conditions are true:

- the target cell has been measured during the last 5 seconds
- the UE has had a radio link connected to the target cell during the last 5 seconds.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.1.2 and A.5.1.3.

### 8.3.1B.2.3 Test purpose

The purpose of this test is to verify the requirement for the inter-frequency handover delay in CELL_DCH state in the dual carrier case.

### 8.3.1B.2.4 Method of test

#### 8.3.1B.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.1B.2.1 and 8.3.1B.2.2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. The PCCPCH RSCP and SFN-CFN observed time difference of the best cell on the unused frequency shall be reported together with Event 2C reporting. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at beginning of T3 with one active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. The UL DPCH shall be transmitted in timeslot 12.

**Table 8.3.1B.2.1: General test parameters for Handover to inter-frequency cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 and A.2.1 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | Hysteresis parameter for event 2C |
| Time to Trigger | | ms | 0 | |
| Threshold non-used frequency | | dBm | -80 | Applicable for Event 2C |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 TDD neighbours on Channel 2 | |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 10 | |

**Table 8.3.1B.2.2: Cell Specific parameters for Handover to inter-frequency cell**

| Parameter | Unit | Cell 1 | | | | | | Cell 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **4** | | | **2** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | Channel 2 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | | 5 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. | n.a. | | | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. | n.a. | -3,12 | | n.a. | | Note 2 |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | | -Inf. | 7 | | -Inf. | | 7 |
| PCCPCH RSCP | dBm | -72 | | | n.a. | | | -Inf. | -66 | | n.a. | | |
| $I_{oc}$ | dBm/ 7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | | | | | | | |

8.3.1B.2.4.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 10 seconds, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 10 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCH to cell 2 less than 40 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved..

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

MEASUREMENT CONTROL message, event 2C (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No Report |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2C |
| -Threshold used frequency | Not Present |
| -W used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -80 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -Delay restriction flag | Not present |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -New DSCH-RNTI | Not Present |
| -New H-RNTI | Not Present |
| -CHOICE *mode* | TDD |
| -New E-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN (Nt) | Same UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | TDD |
| -UL Target SIR | Not Present |
| -CHOICE *UL OL PC info* | Individually signalled |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Indivdual Timeslot interference info | 1 |
| -Individual timeslot interference (10.3.6.38) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Timeslot number | 12 |
| -UL Timeslot Interference | -90 dBm |
| -CHOICE *mode* | TDD |
| -Uplink timing advance control (10.3.6.96) | |
| -CHOICE *Timing Advance* | Disabled |
| -UL CCTrCH list | 1 |
| - TFCS ID | 1 |
| -UL Target SIR | 20 dB |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -Uplink DPCH timeslots and codes (10.3.6.94) | |
| -Dynamic SF Usage | False |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -Timeslot number | 12 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -Choice *TDD option* | 7,68 Mcps |
| -Choice Burst Type | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |

| Information Element | Value/Remark |
|---|---|
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 7,68 Mcps |
| -First timeslot code list | 1 |
| -Channelisation code | 16/1 |
| -Choice more timeslots | No more timeslots |
| -E-DCH info | Not Present |
| **Downlink radio resources** | |
| -Downlnk HS-PDSCH information | Not Present |
| -Downlink information common for all radio links (10.3.6.24) | |
| - Choice DPCH info | Downlink DPCH info common for all RL |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -Downlink DPCH power control information (10.3.6.23) | |
| -CHOICE *mode* | TDD |
| -TPC Step size | 1 dB |
| -MAC-d HFN initial value | Not present |
| -CHOICE *mode* | TDD |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps |
| -TX Diversity mode (10.3.6.86) | None |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -MAC-hs reset indicator | Not present |
| -Post-verification period | Not present |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps |
| -CHOICE *sync case* | Case 2 |
| -Timeslot | 0 |
| -Cell parameters ID | 20 |
| -SCTD indicator | False |
| -CHOICE *DPCH info* | Downlink DPCH info for each RL |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | 7.68 Mcps TDD |
| -DL CCTrCH list | 1 |
| -TFCS ID | 1 |
| -Time Info (10.3.6.83) | |
| -Activation Time | T3 |
| -Duration | Infinite |
| -Common timeslot info | Not Present |
| -Downlink DPCH timeslots and codes VHCR (10.3.6.32a) | |
| -First individual timeslot info (10.3.6.37) | |
| -Timeslot Number (10.3.6.84) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -Timeslot number | 5 |
| -TFCI existence | True |
| -Midamble shift and burst type (10.3.6.41) | |
| -CHOICE *TDD option* | 7,68 Mcps |
| -CHOICE *Burst Type* | Type 1 |
| -Midamble Allocation Mode | Default |
| -Midamble configuration burst type 1 and 3 | 16 |
| -Midamble shift | Not present |
| -CHOICE *TDD option* | 7,68 Mcps |
| -First timeslot channelisation codes VHCR (10.3.6.17a) | |
| -CHOICE *codes representation* | Consecutive codes |
| -First channelisation code | 16/1 |
| -Last channelisation code | 16/1 |
| -CHOICE *more timeslots* | No more timeslots |
| -UL CCTrCH TPC List | Not Present |
| - DL CCTrCH List to Remove | Not Present |
| - E-AGCH Info | Not Present |
| - CHOICE *mode* | TDD |
| - E=HICH information | Not Present |
| -MBMS PL Service Restriction Information | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 in Table 8.6.2.4.1.2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Checked that this IE is present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 in Table 8.6.2.4.1.2 |
| -CHOICE mode | TDD |
| -Primary CCPCH Info | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 7,68 Mcps TDD |
| -CHOICE Sync Case | Not Present |
| -Cell Parameters ID | Set to cell parameters ID of Cell 2 |
| -SCTD Indicator | FALSE |

### 8.3.1B.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.2 TDD/FDD Handover for 3,84 Mcps Option

### 8.3.2.1 Definition and applicability

The handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCCH.

The requirements and this test apply to the UTRA TDD / FDD UE.

### 8.3.2.2 Minimum requirement

The interruption time shall be less than 100 ms in the single carrier case when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/FDD handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + 40 + 50*KC + 150*UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell. |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |

An inter-frequency FDD target cell shall be considered known by the UE, if the target cell has been measured by the UE during the last 5 seconds.

The phase reference is the Primary CPICH.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.2 and A.5.2.

### 8.3.2.3 Test purpose

The purpose of this test is to verify the requirement for the TDD/FDD handover delay in CELL_DCH state.

### 8.3.2.4 Method of test

#### 8.3.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.2.1, 8.3.2.2 and 8.3.2.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G and 2B shall be used. The CPICH_RSCP of the best cell on the unused frequency shall be reported together with Event 2B reporting. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at the beginning of T3 with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

**Table 8.3.2.1: General test parameters for TDD/FDD handover**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 clause A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | TDD cell |
| | Neighbour cell | | Cell 2 | FDD cell |
| Final condition | Active cell | | Cell 2 | FDD cell |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 3 | Hysteresis parameter for event 2B |
| Time to Trigger | | ms | 0 | |
| Absolute threshold used frequency | | dBm | -71 | Applicable for Event 2B |
| Threshold non-used frequency | | dBm | -80 | Applicable for Event 2B |
| W non-used frequency | | | 1 | Applicable for Event 2B |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 FDD neighbours on Channel 2 | |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 5 | |
| T2 | | s | 15 | |
| T3 | | s | 5 | |

**Table 8.3.2.2: Cell 1 specific test parameters for TDD/FDD handover**

| Parameter | Unit | Cell 1 | | | | | |
|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **2** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | -1 | | 5 | -1 | |
| PCCPCH RSCP | dBm | -68 | -74 | | n.a. | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior. | | | | | | | |

**Table 8.3.2.3: Cell 2 specific test parameters for TDD/FDD handover**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1, T2 | T3 |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | n.a. | Note 1 |
| OCNS_Ec/Ior | dB | -0,941 | Note 2 |
| CPICH_RSCP | dBm | -83 | -77 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | 3 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | |
| Propagation Condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2 : The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$ | | | |

8.3.2.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 5 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2B.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 15 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCCH to cell 2 less than 100 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 [TBD] times.

Specific Message Contents

All messages indicated belowabove shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

MEASUREMENT CONTROL message, event 2B (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No Report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2B |
| -Threshold used frequency | -71 dBm |
| -W used frequency | 1 |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -80 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink(Nu) | Same uplink UARFCN as used for cell 2 |
| -UARFCN downlink(Nd) | Same downlink UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | FDD |
| -DPCCH power offset | -6dB |
| - PC Preamble | 1 frame |
| - SRB delay | 7 frames |
| - Power Control Algorithm | Algorithm1 |
| - TPC step size | 1dB |
| -CHOICE *mode* | FDD |
| -Scrambling code type | Long |
| -Scrambling code number | 0 (0 to 16777215) |
| -Number of DPDCH | Not Present(1) |
| -Spreading factor | SF is reference to TS34.108 clause 6.10 Parameter Set |
| -TFCI existence | TRUE |
| -Number of FBI bit | Not Present(0) |
| -Puncturing Limit | Reference to TS34.108 clause 6.10 Parameter Set |
| **Downlink radio resources** | |
| -CHOICE *mode* | FDD |
| -Downlink PDSCH information | Not Present |
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -DPC mode | 0 (single) |
| -CHOICE *mode* | FDD |
| -Power offset $P_{Pilot-DPDCH}$ | TBD |
| -DL rate matching restriction information | Not Present |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Fixed or Flexible Position | Flexible |
| -TFCI existence | TRUE |
| -CHOICE *SF* | Not Present |
| -Number of bits for Pilot bits(SF=128,256) | Not Present |
| -CHOICE *mode* | FDD |
| -DPCH compressed mode info (10.3.6.33) | Not Present (Note 1) |

| Information Element | Value/Remark |
|---|---|
| -TX Diversity mode (10.3.6.86) | None |
| -SSDT information (10.3.6.77) | Not Present |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH info (10.3.6.60) | |
| -Primary scrambling code | 350 |
| -PDSCH with SHO DCH info (10.3.6.47) | Not Present |
| -PDSCH code mapping (10.3.6.43) | Not Present |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH usage for channel estimation | Primary CPICH may be used |
| -DPCH frame offset | 0 chips |
| -Secondary CPICH info | Not Present |
| -DL channelisation code | |
| -Secondary scrambling code | 1 |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Code number | SF-1(SF is reference to TS34.108 clause 6.10 Parameter Set) |
| -Scrambling code change | No change |
| -TPC combination index | 0 |
| - SSDT Cell Identity | -a |
| - Closed loop timing adjustment mode | Not Present |
| - SCCPCH information for FACH (10.3.6.70) | Not Present |
| Note 1: IE "DPCH compressed mode info" is not needed as default values are applied that have previously been received in RADIO BEARER SETUP or RRC CONNECTION SETUP | |

MEASUREMENT REPORT message for Inter frequency test cases

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Not Present |
| -CPICH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2B |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.3.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]%of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.2A TDD/FDD Handover for 1,28 Mcps Option

### 8.3.2A.1 Definition and applicability

The handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCCH.

The requirements and this test apply to the UTRA 1,28Mcps TDD / FDD UE.

### 8.3.2A.2 Minimum requirement

The interruption time shall be less than 100 ms in the single carrier case when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/FDD handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + 40 + 50*KC + 150*UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell. |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |

An inter-frequency FDD target cell shall be considered known by the UE, if the target cell has been measured by the UE during the last 5 seconds.

The phase reference is the Primary CPICH.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.2.2 and A.5.2.2

## 8.3.2A.3 Test purpose

The purpose of this test is to verify the requirement for the TDD/FDD handover delay in CELL_DCH state.

## 8.3.2A.4 Method of test

### 8.3.2A.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.2.1, 8.3.2.2 and 8.3.2.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G and 2B shall be used. The CPICH_RSCP of the best cell on the unused frequency shall be reported together with Event 2B reporting. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at the beginning of T3 with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

**Table 8.3.2A.1: General test parameters for 1,28 Mcps TDD/FDD handover**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channels 12.2 kbps | As specified in TS 25.102 annex A and TS 25.101 annex A |
| Power Control | | | On | |
| Initial conditions | Active cell | | Cell 1 | TDD cell |
| | Neighbour cell | | Cell 2 | FDD cell |
| Final condition | Active cell | | Cell 2 | FDD cell |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 3 | Hysteresis parameter for event 2B |
| Time to Trigger | | ms | 0 | |
| Absolute threshold used frequency | | dBm | -71 | Applicable for Event 2B |
| Threshold non-used frequency | | dBm | -80 | Applicable for Event 2B |
| W non-used frequency | | | 1 | Applicable for Event 2B |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 FDD neighbours on Channel 2 | |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 5 | |
| T2 | | s | 15 | |
| T3 | | s | 5 | |

**Table 8.3.2A.2: Cell 1 (1,28 Mcps TDD cell) specific test parameters for TDD/FDD handover**

| Parameter | Unit | Cell 1 | | | | | |
|---|---|---|---|---|---|---|---|
| **Timeslot number** | | **0** | | | **5** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. |
| OCNS_Ec/Ior | dB | -3 | | | Note 2 | | n.a. |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | -1 | | 5 | -1 | |
| PCCPCH RSCP | dBm | -68 | -74 | | n.a. | | |
| $I_{oc}$ | dBm/ 1,28 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | |

**Table 8.3.2A.3: Cell 2 (FDD cell) specific test parameters for TDD/FDD handover**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| CPICH_Ec/Ior | dB | -10 | | |
| PCCPCH_Ec/Ior | dB | -12 | | |
| SCH_Ec/Ior | dB | -12 | | |
| PICH_Ec/Ior | dB | -15 | | |
| DPCH_Ec/Ior | dB | n.a. | | Note 1 |
| OCNS_Ec/Ior | dB | -0.941 | | Note 2 |
| CPICH_RSCP | dBm | -Inf | -75 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Inf | 5 | |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | |
| Propagation Condition | | AWGN | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2 : The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$. | | | | |

### 8.3.2A.4.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the test procedure to be specified in TS 34.108 [3] subclause 7.3.4.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 5 seconds, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2B.
7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.
8) After 15 seconds, the SS shall switch the power settings from T2 to T3
9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCCH to cell 2 less than 100 ms from the beginning of time period T3 then the number of successful tests is increased by one.
10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved..

Specific Message Contents

All messages indicated belowabove shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

MEASUREMENT CONTROL message, event 2B (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No Report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2B |
| -Threshold used frequency | -71 dBm |
| -W used frequency | 1 |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -80 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink(Nu) | Same uplink UARFCN as used for cell 2 |
| -UARFCN downlink(Nd) | Same downlink UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | FDD |
| -DPCCH power offset | -6dB |
| - PC Preamble | 1 frame |
| - SRB delay | 7 frames |
| - Power Control Algorithm | Algorithm1 |
| - TPC step size | 1dB |
| -CHOICE *mode* | FDD |
| -Scrambling code type | Long |
| -Scrambling code number | 0 (0 to 16777215) |
| -Number of DPDCH | Not Present(1) |
| -Spreading factor | SF is reference to TS34.108 clause 6.10 Parameter Set |
| -TFCI existence | TRUE |
| -Number of FBI bit | Not Present(0) |
| -Puncturing Limit | Reference to TS34.108 clause 6.10 Parameter Set |
| **Downlink radio resources** | |
| -CHOICE *mode* | FDD |
| -Downlink PDSCH information | Not Present |
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -DPC mode | 0 (single) |
| -CHOICE *mode* | FDD |
| -Power offset $P_{Pilot-DPDCH}$ | TBD |
| -DL rate matching restriction information | Not Present |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Fixed or Flexible Position | Flexible |
| -TFCI existence | TRUE |
| -CHOICE *SF* | Not Present |
| -Number of bits for Pilot bits(SF=128,256) | Not Present |
| -CHOICE *mode* | FDD |
| -DPCH compressed mode info (10.3.6.33) | Not Present (Note 1) |

| Information Element | Value/Remark |
|---|---|
| -TX Diversity mode (10.3.6.86) | None |
| -SSDT information (10.3.6.77) | Not Present |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH info (10.3.6.60) | |
| -Primary scrambling code | 350 |
| -PDSCH with SHO DCH info (10.3.6.47) | Not Present |
| -PDSCH code mapping (10.3.6.43) | Not Present |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH usage for channel estimation | Primary CPICH may be used |
| -DPCH frame offset | 0 chips |
| -Secondary CPICH info | Not Present |
| -DL channelisation code | |
| -Secondary scrambling code | 1 |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Code number | SF-1(SF is reference to TS34.108 clause 6.10 Parameter Set) |
| -Scrambling code change | No change |
| -TPC combination index | 0 |
| - SSDT Cell Identity | -a |
| - Closed loop timing adjustment mode | Not Present |
| - SCCPCH information for FACH (10.3.6.70) | Not Present |
| Note 1: IE "DPCH compressed mode info" is not needed as default values are applied that have previously been received in RADIO BEARER SETUP or RRC CONNECTION SETUP | |

MEASUREMENT REPORT message for Inter frequency test cases

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Not Present |
| -CPICH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2B |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.3.2A.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95%of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.2B TDD/FDD Handover for 7,68 Mcps Option

### 8.3.2B.1 Definition and applicability

The handover interruption time is defined as the time between the end of the last TTI containing a transport block on the old DPCH and the time the UE starts transmission of the new uplink DPCCH.

The requirements and this test apply to the UTRA TDD / FDD UE.

### 8.3.2B.2 Minimum requirement

The interruption time shall be less than 100 ms in the single carrier case when the cell is known by the UE and the SFN of the target cell does not need to be decoded. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The interruption time is dependent on whether the target cell is known for the UE or not.

If TDD/FDD handover is commanded, the interruption time shall be less than,

$$T_{interrupt} = T_{offset} + 40 + 50*KC + 150*UC \text{ ms}$$

where,

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell. |
| KC | Equal to 1 if a known target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |
| UC | Equal to 1 if an unknown target cell is indicated in the RRC message implying TDD/FDD handover and equal to 0 otherwise |

An inter-frequency FDD target cell shall be considered known by the UE, if the target cell has been measured by the UE during the last 5 seconds.

The phase reference is the Primary CPICH.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 25.123 [2] clauses 5.2 and A.5.3.

### 8.3.2B.3 Test purpose

The purpose of this test is to verify the requirement for the TDD/FDD handover delay in CELL_DCH state.

### 8.3.2B.4 Method of test

#### 8.3.2B.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.3.2B.1, 8.3.2B.2 and 8.3.2B.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G and 2B shall be used. The CPICH_RSCP of the best cell on the unused frequency shall be reported together with Event 2B reporting. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a Physical Channel reconfiguration message with activation time at the beginning of T3 with a new active cell, cell 2. The Physical Channel reconfiguration message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [16].

**Table 8.3.2B.1: General test parameters for TDD/FDD handover**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 annex A.2.2 and TS 25.101 annex A |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | TDD cell |
| | Neighbour cell | | Cell 2 | FDD cell |
| Final condition | Active cell | | Cell 2 | FDD cell |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 3 | Hysteresis parameter for event 2B |
| Time to Trigger | | ms | 0 | |
| Absolute threshold used frequency | | dBm | -71 | Applicable for Event 2B |
| Threshold non-used frequency | | dBm | -80 | Applicable for Event 2B |
| W non-used frequency | | | 1 | Applicable for Event 2B |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 FDD neighbours on Channel 2 | |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 5 | |
| T2 | | s | 15 | |
| T3 | | s | 5 | |

**Table 8.3.2B.2: Cell 1 specific test parameters for TDD/FDD handover**

| **Parameter** | **Unit** | **Cell 1** | | | | | |
|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **2** | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | -1 | | 5 | -1 | |
| PCCPCH RSCP | dBm | -68 | -74 | | n.a. | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |
| The DPCH level is controlled by the power control loop<br>The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$ | | | | | | | |

**Table 8.3.2B.3: Cell 2 specific test parameters for TDD/FDD handover**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1, T2 | T3 |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | n.a. | Note 1 |
| OCNS_Ec/Ior | dB | -0.941 | Note 2 |
| CPICH_RSCP | dBm | -83 | -77 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | 3 |
| $I_{oc}$ | dBm/7.68 MHz | -70 | |
| Propagation Condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2 : The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$. | | | |

8.3.2B.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 5 seconds, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2B.

7) SS shall transmit a PHYSICAL CHANNEL RECONFIGURATION message with activation time at T3.

8) After 15 seconds, the SS shall switch the power settings from T2 to T3

9) UE shall transmit a PHYSICAL CHANNEL RECONFIGURATION COMPLETE message on the UL DCCH of cell 2. If the UE transmits the UL DPCCH to cell 2 less than 100 ms from the beginning of time period T3 then the number of successful tests is increased by one.

10) After 5 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat step 1-10 until the confidence level according to annex F.6.2 is achieved..

Specific Message Contents

All messages indicated belowabove shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

MEASUREMENT CONTROL message, event 2B (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| -Inter-frequency reporting criteria | |
| -Filter coefficient | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No Report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -Inter-frequency set update (10.3.7.22) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Inter-frequency event identity (10.3.7.14) | Event 2B |
| -Threshold used frequency | -71 dBm |
| -W used frequency | 1 |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report cells within monitored set on non-used frequency |
| -Maximum number of reported cells per reported non-used frequency | 1 |
| -Parameters required for each non-used frequency | 1 |
| -Threshold non-used frequency | -80 dBm |
| -W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

PHYSICAL CHANNEL RECONFIGURATION message (step 7):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| **UE Information Elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Integrity protection mode info | Not Present |
| -Ciphering mode info | Not Present |
| -Activation time | At T3 |
| -New U-RNTI | Not Present |
| -New C-RNTI | Not Present |
| -RRC State Indicator | CELL_DCH |
| -UTRAN DRX cycle length coefficient | Not Present |
| **CN Information Elements** | |
| -CN Information info | Not Present |
| **UTRAN mobility information elements** | |
| -URA identity | Not Present |
| **RB information elements** | |
| -Downlink counter synchronisation info | Not Present |
| -RB with PDCP information list | Not Present |
| -RB with PDCP information | Not Present |
| **PhyCH information elements** | |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink(Nu) | Same uplink UARFCN as used for cell 2 |
| -UARFCN downlink(Nd) | Same downlink UARFCN as used for cell 2 |
| **Uplink radio resources** | |
| -Maximum allowed UL TX power | 33 dBm |
| -CHOICE *channel requirement* | Uplink DPCH info |
| -Uplink DPCH info (10.3.6.88) | |
| -Uplink DPCH power control info (10.3.6.91) | |
| -CHOICE *mode* | FDD |
| -DPCCH power offset | -6dB |
| - PC Preamble | 1 frame |
| - SRB delay | 7 frames |
| - Power Control Algorithm | Algorithm1 |
| - TPC step size | 1dB |
| -CHOICE *mode* | FDD |
| -Scrambling code type | Long |
| -Scrambling code number | 0 (0 to 16777215) |
| -Number of DPDCH | Not Present(1) |
| -Spreading factor | SF is reference to TS34.108 clause 6.10 Parameter Set |
| -TFCI existence | TRUE |
| -Number of FBI bit | Not Present(0) |
| -Puncturing Limit | Reference to TS34.108 clause 6.10 Parameter Set |
| **Downlink radio resources** | |
| -CHOICE *mode* | FDD |
| -Downlink PDSCH information | Not Present |
| -Downlink information common for all radio links (10.3.6.24) | |
| -Downlink DPCH info common for all RL (10.3.6.18) | |
| -Timing indicator | Initialise |
| -CFN-targetSFN frame offset | Not Present |
| -Downlink DPCH power control information (10.3.6.23) | |
| -DPC mode | 0 (single) |
| -CHOICE *mode* | FDD |
| -Power offset $P_{Pilot-DPDCH}$ | TBD |
| -DL rate matching restriction information | Not Present |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Fixed or Flexible Position | Flexible |
| -TFCI existence | TRUE |
| -CHOICE *SF* | Not Present |
| -Number of bits for Pilot bits(SF=128,256) | Not Present |
| -CHOICE *mode* | FDD |
| -DPCH compressed mode info (10.3.6.33) | Not Present (Note 1) |

| Information Element | Value/Remark |
|---|---|
| -TX Diversity mode (10.3.6.86) | None |
| -SSDT information (10.3.6.77) | Not Present |
| -Default DPCH Offset Value (10.3.6.16) | 0 |
| -Downlink information per radio link list | 1 |
| -Downlink information for each radio link (10.3.6.27) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH info (10.3.6.60) | |
| -Primary scrambling code | 350 |
| -PDSCH with SHO DCH info (10.3.6.47) | Not Present |
| -PDSCH code mapping (10.3.6.43) | Not Present |
| -Downlink DPCH info for each RL (10.3.6.21) | |
| -CHOICE *mode* | FDD |
| -Primary CPICH usage for channel estimation | Primary CPICH may be used |
| -DPCH frame offset | 0 chips |
| -Secondary CPICH info | Not Present |
| -DL channelisation code | |
| -Secondary scrambling code | 1 |
| -Spreading factor | Reference to TS34.108 clause 6.10 Parameter Set |
| -Code number | SF-1(SF is reference to TS34.108 clause 6.10 Parameter Set) |
| -Scrambling code change | No change |
| -TPC combination index | 0 |
| - SSDT Cell Identity | -a |
| - Closed loop timing adjustment mode | Not Present |
| - SCCPCH information for FACH (10.3.6.70) | Not Present |
| Note 1: IE "DPCH compressed mode info" is not needed as default values are applied that have previously been received in RADIO BEARER SETUP or RRC CONNECTION SETUP | |

MEASUREMENT REPORT message for Inter frequency test cases

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Checked that this IE is present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Not Present |
| -CPICH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2B |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 in Table 8.3.2.3 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.3.2B.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95%of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.3 TDD/GSM Handover

### 8.3.3.1 Definition and applicability

#### 8.3.3.1.1 3,84 Mcps option

The UTRAN to GSM cell handover interruption time is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission on the channel of the new RAT.

The requirements and this test apply to the combined TDD (3,84 Mcps option) and GSM UE.

#### 8.3.3.1.2 1,28 Mcps option

The UTRAN to GSM cell handover delay is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission on the channel of the new RAT.

The requirements and this test apply to the combined TDD (1,28 Mcps option) and GSM UE

#### 8.3.3.1.3 7,68 Mcps option

The UTRAN to GSM cell handover delay is defined as the time from the end of the last TTI containing an RRC message implying hard handover to the transmission on the channel of the new RAT.

The requirements and this test apply to the combined TDD (7,68 Mcps option) and GSM UE.

### 8.3.3.2 Minimum requirement

#### 8.3.3.2.1 3,84 Mcps option

The interruption time shall be less than 40 ms in the case where the UE has synchronised to the GSM cell before the HANDOVER FROM UTRAN COMMAND is received. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of [FFS]%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.3.2 and A.5.3.

#### 8.3.3.2.2 1,28 Mcps option

The handover delay shall be less than 90 ms in the case where the UE has synchronised to the GSM cell before the HANDOVER FROM UTRAN COMMAND is received. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.3.2 and A.5.3.

#### 8.3.3.2.3 7,68 Mcps option

The handover delay shall be less than 90 ms in the case where the UE has synchronised to the GSM cell before the HANDOVER FROM UTRAN COMMAND is received. The rate of correct handovers observed during repeated tests shall be at least 90% with a confidence level of 95%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.3.2 and A.5.3.

### 8.3.3.3 Test purpose

#### 8.3.3.3.1 3,84 Mcps option

To verify that the UE meets the minimum requirement.

#### 8.3.3.3.2 1,28 Mcps option

To verify that the UE meets the minimum requirement.

### 8.3.3.4 Method of test

#### 8.3.3.4.1 3,84 Mcps option

##### 8.3.3.4.1.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4 for TDD conditions, and clause A1.2 of TS 51.010-1 [23] for the corresponding GSM conditions.

The test parameters are given in Table 8.3.3.1, 8.3.3.2 and 8.3.3.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3C shall be used. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a HANDOVER FROM UTRAN COMMAND message with activation time at beginning of T3 with one active cell, cell 2. The HANDOVER FROM UTRAN COMMAND message shall be sent to the UE such that the delay between the last the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [9]. In the GSM Handover command contained in this message, IE starting time shall not be included.

Cell 1 is a UTRA TDD cell and cell 2 is a GSM cell. The Beacon timeslot shall be transmitted in timeslot 0 for cell 1 and no second Beacon timeslot shall be provided for cell 1. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3.

**Table 8.3.3.1: General test parameters for TDD/GSM handover**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Initial conditions | Active cell | Cell 1 | UTRA TDD cell |
| | Neighbour cell | Cell 2 | GSM cell |
| Final condition | Active cell | Cell 2 | GSM cell |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for Event 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before the start of time period T1. |
| $T_{identify\ abort}$ | s | 5 | |
| $T_{reconfirm\ abort}$ | s | 5 | |
| T1 | s | 10 | |
| T2 | s | 10 | |
| T3 | s | 10 | |

**Table 8.3.3.2: Cell Specific Parameters for Handover UTRAN to GSM cell case (cell 1)**

| **Parameter** | **Unit** | **Cell 1** | | | | | |
|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **1** | | |
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | | | 6 | | |
| PCCPCH RSCP | dBm | -68 | | | n.a. | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | |

**Table 8.3.3.3: Cell Specific Parameters for Handover UTRAN to GSM cell case (cell 2)**

| **Parameter** | **Unit** | **Cell 2** | |
|---|---|---|---|
| | | **T1** | **T2, T3** |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -85 | -75 |

#### 8.3.3.4.1.2 Procedure

1) The RF parameters for cell 1 are set up according to T1.
2) The UE is switched on
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4
4) The RF parameters for cell 2 are set up according to T1 and the SS configures a traffic channel
5) SS shall transmit a MEASUREMENT CONTROL message to cell 1
6) After 10 seconds, the SS shall switch the power settings from T1 to T2
7) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C
8) SS shall transmit a HANDOVER FROM UTRAN COMMAND message with activation time at T3 and indicating the traffic channel of the target GSM cell to the UE through DCCH of the serving UTRAN cell.
9) After 10 seconds, the SS shall switch the power settings from T2 to T3
10) UE shall transmit a burst on the traffic channel of cell 2 implying that it has switched to the GSM cell. The UE sends a HANDOVER ACCESS message. If the UE transmits access bursts on the new DCCH of the target cell less than 40 ms from the beginning of time period T3, then the number of successful tests is increased by one.

[Editor's note: TS 34.108, 7.3.4 shall specify the messages HANDOVER ACCESS, PHYSICAL INFORMATION, SABM, UA and HANDOVER COMPLETE]

11) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
12) Repeat step 1-11 [TBD] times

#### Specific Message Contents

All messages indicated below above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 5):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 4 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Required |
| -Inter-RAT reporting quantity (10.3.7.32) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 1 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

HANDOVER FROM UTRAN COMMAND message (step 8):

| Information Element | Value/remark |
|---|---|
| Message Type | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Activation time | At T3 |
| **RB information elements** | |
| -RAB information list | 1 |
| -RAB Info | Not present |
| **Other information elements** | |
| -CHOICE System type | GSM |
| -Frequency Band | GSM/DCS 1800 Band |
| -GSM message | |
| -Single GSM message | [TBD] |
| -GSM message List | GSM HANDOVER COMMAND formatted as BIT STRING(1..512). The contents of the HANDOVER COMMAND see next table. |

HANDOVER COMMAND

| |
|---|
| Same as the HANDOVER COMMAND for M = 2 in clause 26.6.5.1 of TS 51.010, except that the CHANNEL MODE IE is included with value = speech full rate or half rate version 3 |

MEASUREMENT REPORT message (step 7)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-RAT Measured results list |
| -Inter-RAT-frequency measured results (10.3.7.26) | 1 |
| -CHOICE System | GSM |
| -Measured GSM cells | 1 |
| -GSM Carrier RSSI | Checked that this IE is present |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |
| -Observed Time difference to GSM cell | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-RAT measurement event results |
| -Inter-RAT event identity | 3C |
| -Cells to report | 1 |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |

### 8.3.3.4.2 1,28 Mcps option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4 for TDD conditions, and clause A1.2 of TS 51.010-1 [24] for the corresponding GSM conditions.

The test parameters are given in Table 8.3.4.1, 8.3.4.2 and 8.3.4.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3C shall be used. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a HANDOVER FROM UTRAN COMMAND message with activation time "now" with one active cell, cell 2. The HANDOVER FROM UTRAN COMMAND message shall be sent to the UE during period T2. The starting point of T3 is defined as the end of the last TTI containing the HANDOVER COMMAND. In the GSM Handover command contained in this message, IE starting time shall not be included.

Cell 1 is a UTRA TDD cell and cell 2 is a GSM cell. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 2.

**Table 8.3.3.4.1: General test parameters for 1,28Mcps TDD/GSM handover**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Initial conditions | Active cell | Cell 1 | UTRA TDD cell |
| | Neighbour cell | Cell 2 | GSM cell |
| Final condition | Active cell | Cell 2 | GSM cell |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for Event 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before the start of time period T1. |
| $T_{identify\ abort}$ | s | 5 | As specified in section 8.1A.2.5 |
| $T_{reconfirm\ abort}$ | s | 5 | As specified in section 8.1A.2.5 |
| T1 | s | 10 | |
| T2 | s | 10 | |
| T3 | s | 10 | |

**Table 8.3.3.4.2: Cell 1 (1,28Mcps )specific test parameters for TDD/GSM handover**

| **Parameter** | **Unit** | **Cell 1** | | | | | |
|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **DwPTS** | | |
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | |
| DwPCH_Ec/Ior | dB | | | | 0 | | |
| OCNS_Ec/Ior | dB | -3 | | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | | 5 | | |
| $I_{oc}$ | dBm/1.28 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |

**Table 8.3.3.4.3: Cell 2 (GSM )specific test parameters for TDD/GSM handover**

| **Parameter** | **Unit** | **Cell 2** | |
|---|---|---|---|
| | | **T1** | **T2, T3** |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -85 | -75 |

8.3.3.4.2.2 Procedure

1) The RF parameters for cell 1 are set up according to T1.
2) The UE is switched on
3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4
4) The RF parameters for cell 2 are set up according to T1 and the SS configures a traffic channel

5) SS shall transmit a MEASUREMENT CONTROL message to cell 1

6) After 10 seconds, the SS shall switch the power settings from T1 to T2

7) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C

8) SS shall transmit a HANDOVER FROM UTRAN COMMAND message with activation time “now” and indicating the traffic channel of the target GSM cell to the UE through DCCH of the serving UTRAN cell. The start of T3 is defined as the end of the last TTI, containing the HANDOVER COMMAND.

10) UE shall transmit a burst on the traffic channel of cell 2 implying that it has switched to the GSM cell. The UE sends a HANDOVER ACCESS message. If the UE transmits access bursts on the new DCCH of the target cell less than 90 ms from the beginning of time period T3, then the number of successful tests is increased by one.

11) At the end of T3 the UE is switched off. Any timing information of cell 2 is deleted in the UE.

12) Repeat step 1-11 until the confidence level according to annex F.6.2A is achieved

### Specific Message Contents

All messages shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: 10.x.y.z in the IE description refers to clauses in TS 25.331 [9].

MEASUREMENT CONTROL message (step 5):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 4 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Required |
| -Inter-RAT reporting quantity (10.3.7.32) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 1 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

HANDOVER FROM UTRAN COMMAND message (step 8):

| Information Element | Value/remark |
|---|---|
| Message Type(10.2.15) | |
| **UE information elements**<br>-RRC transaction identifier<br>-Integrity check info<br>-Activation time | <br>0<br>Not Present<br>now |
| **RB information elements**<br>-RAB information list<br>-RAB Info | <br>1<br>Not present |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. |
| - CN domain identity | CS domain |
| - NAS Synchronization Indicator | Not present |
| - Re-establishment timer | Use T315 |
| **Other information elements**<br>-CHOICE System type<br>-Frequency Band<br><br><br>-CHOICE GSM message<br>- Single GSM message | <br>GSM<br>Set to "GSM/ PCS 1900" if GSM/ PCS 1900 is used in this test. Otherwise set to "GSM/DCS 1800 Band"<br><br>GSM HANDOVER COMMAND formatted and coded according to GSM specifications as BIT STRING (1..512). The first/ l*eftmost/ most significant* bit of the bit string contains bit 8 of the first octet of the GSM message. The contents of the HANDOVER COMMAND see next table. |

HANDOVER COMMAND

| Information Element (GSM) | Value/remark | Version |
|---|---|---|
| Protocol Discriminator | RR Management. | |
| Skip Indicator<br>Message Type<br>Cell Description<br>- Network Colour Code<br>- Base station Colour Code<br>- BCCH Carrier Number | 0000<br>00101011<br><br>1<br>5<br>1 | |
| Channel Description 2<br>- Channel Type and TDMA offset<br>- Timeslot Number | <br>TCH/F + ACCHs<br>Chosen arbitrarily by the test house, but not Zero. | |
| - Training Sequence Code | Chosen arbitrarily by the test house. | |
| - Hopping | Single RF channel. | |
| - ARFCN | 1 | |
| Handover Reference | | |
| - Handover Reference Value<br>Power Command and ACCESS Type<br>- ATC<br>- EPC_mode<br>- FPC<br><br>- EPC_FPC<br>- Power level<br>Synchronization Indication<br>Channel Mode<br>All other information elements | Chosen arbitrarily by the test house.<br><br>0<br>0<br>0<br><br>0<br>Chosen arbitrarily by the test house.<br>Not present.<br>speech full rate or half rate version 1<br>Not present. | <br><br><br>REL-5<br>R99 and REL-4 only<br>REL-5 |

MEASUREMENT REPORT message (step 7)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-RAT Measured results list |
| -Inter-RAT-frequency measured results (10.3.7.26) | 1 |
| -CHOICE System | GSM |
| -Measured GSM cells | 1 |
| -GSM Carrier RSSI | Checked that this IE is present |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |
| -Observed Time difference to GSM cell | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-RAT measurement event results |
| -Inter-RAT event identity | 3C |
| -Cells to report | 1 |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |

### 8.3.3.4.3 7,68 Mcps option

#### 8.3.3.4.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4 for TDD conditions, and clause A1.2 of TS 51.010-1 [23] for the corresponding GSM conditions.

The test parameters are given in Table 8.3.3.4.1B, 8.3.3.4.2B and 8.3.3.4.3B below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3C shall be used. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of cell 2.

UTRAN shall send a HANDOVER FROM UTRAN COMMAND message with activation time at beginning of T3 with one active cell, cell 2. The HANDOVER FROM UTRAN COMMAND message shall be sent to the UE such that the delay between the last the end of the last received TTI containing the message and the beginning of T3 is at least equal to the RRC procedure delay as defined in [9]. In the GSM Handover command contained in this message, IE starting time shall not be included.

Cell 1 is a UTRA TDD cell and cell 2 is a GSM cell. The Beacon timeslot shall be transmitted in timeslot 0 for cell 1 and no second Beacon timeslot shall be provided for cell 1. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3.

**Table 8.3.3.4.1B: General test parameters for TDD/GSM handover**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Initial conditions | Active cell | Cell 1 | UTRA TDD cell |
| | Neighbour cell | Cell 2 | GSM cell |
| Final condition | Active cell | Cell 2 | GSM cell |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for Event 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before the start of time period T1. |
| $T_{identify\ abort}$ | s | 5 | As specified in section 8.1.2.5 |
| $T_{reconfirm\ abort}$ | s | 5 | As specified in section 8.1.2.5 |
| T1 | s | 10 | |
| T2 | s | 10 | |
| T3 | s | 10 | |

**Table 8.3.3.4.2B: Cell Specific Parameters for Handover UTRAN to GSM cell case (cell 1)**

| **Parameter** | **Unit** | **Cell 1** | | | | | |
|---|---|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | | **1** | | |
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| UTRA RF Channel Number | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | n.a. | | |
| SCH_Ec/Ior | dB | -9 | | | n.a. | | |
| SCH_$t_{offset}$ | dB | 0 | | | n.a. | | |
| DPCH_Ec/Ior | dB | n.a. | | | Note 1 | | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | | | Note 2 | | n.a. |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | | | 6 | | |
| PCCPCH RSCP | dBm | -68 | | | n.a. | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | |
| Propagation Condition | | AWGN | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | | | |

**Table 8.3.3.4.3B: Cell Specific Parameters for Handover UTRAN to GSM cell case (cell 2)**

| **Parameter** | **Unit** | **Cell 2** | |
|---|---|---|---|
| | | **T1** | **T2, T3** |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -85 | -75 |

8.3.3.4.3.2 Procedure

1) The RF parameters for cell 1 are set up according to T1.

2) The UE is switched on

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4

4) The RF parameters for cell 2 are set up according to T1 and the SS configures a traffic channel

5) SS shall transmit a MEASUREMENT CONTROL message to cell 1

6) After 10 seconds, the SS shall switch the power settings from T1 to T2

7) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C

8) SS shall transmit a HANDOVER FROM UTRAN COMMAND message with activation time at T3 and indicating the traffic channel of the target GSM cell to the UE through DCCH of the serving UTRAN cell.

9) After 10 seconds, the SS shall switch the power settings from T2 to T3

10) UE shall transmit a burst on the traffic channel of cell 2 implying that it has switched to the GSM cell. The UE sends a HANDOVER ACCESS message. If the UE transmits access bursts on the new DCCH of the target cell less than 40 ms from the beginning of time period T3, then the number of successful tests is increased by one.

[Editor's note: TS 34.108, 7.3.4 shall specify the messages HANDOVER ACCESS, PHYSICAL INFORMATION, SABM, UA and HANDOVER COMPLETE]

11) After 10 seconds, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

12) Repeat step 1-11 until the confidence level according to annex F.6.2 is achieved.

### Specific Message Contents

All messages indicated below above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 5):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 4 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Required |
| -Inter-RAT reporting quantity (10.3.7.32) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 1 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

HANDOVER FROM UTRAN COMMAND message (step 8):

| Information Element | Value/remark |
|---|---|
| Message Type | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| -Activation time | At T3 |
| **RB information elements** | |
| -RAB information list | 1 |
| -RAB Info | Not present |
| **Other information elements** | |
| -CHOICE System type | GSM |
| -Frequency Band | GSM/DCS 1800 Band |
| -GSM message | |
| -Single GSM message | [TBD] |
| -GSM message List | GSM HANDOVER COMMAND formatted as BIT STRING(1..512). The contents of the HANDOVER COMMAND see next table. |

HANDOVER COMMAND

| |
|---|
| Same as the HANDOVER COMMAND for M = 2 in clause 26.6.5.1 of TS 51.010, except that the CHANNEL MODE IE is included with value = speech full rate or half rate version 3 |

MEASUREMENT REPORT message (step 7)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-RAT Measured results list |
| -Inter-RAT-frequency measured results (10.3.7.26) | 1 |
| -CHOICE System | GSM |
| -Measured GSM cells | 1 |
| -GSM Carrier RSSI | Checked that this IE is present |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |
| -Observed Time difference to GSM cell | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-RAT measurement event results |
| -Inter-RAT event identity | 3C |
| -Cells to report | 1 |
| -CHOICE BSIC | Verified BSIC |
| -inter-RAT cell id | Checked that this IE is present |

### 8.3.3.5 Test requirements

#### 8.3.3.5.1 3,84 Mcps option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.3.5.2 1,28 Mcps option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.3.5.3 7,68 Mcps option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.4 Cell Re-selection in CELL_FACH

### 8.3.4.1 Scenario 1: TDD/TDD cell re-selection single carrier case

#### 8.3.4.1.1 Definition and applicability

##### 8.3.4.1.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.3.4.1.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send SYNCH-UL sequence in the UpPTS for sending the CELL UPDATE with cause value "cell reselection".

The requirements and this test apply to the 1,28 Mcps TDD UE.

##### 8.3.4.1.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.4.1.2 Minimum requirement

##### 8.3.4.1.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 2.5 s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.2 and A.5.4.1.

##### 8.3.4.1.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 1.6s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.3 and A.5.4.2.

##### 8.3.4.1.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 2.5 s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.4 and A.5.4.3.

#### 8.3.4.1.3 Test purpose

This test verifies that the UE meets the minimum requirement for the cell re-selection delay in CELL_FACH for the single carrier case

#### 8.3.4.1.4 Method of test

##### 8.3.4.1.4.1 3,84 Mcps TDD option

###### 8.3.4.1.4.1.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.4.1.1.1, 8.3.4.1.1.2, 8.3.4.1.1.3, and 8.3.4.1.1.4.

**Table 8.3.4.1.1.1: General test parameters for Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.4.1.1.2: Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | Kbps | 24,4 |
| Channel symbol rate | Ksps | 12,2 |
| Slot Format # | - | 0 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Default Midamble |

**Table 8.3.4.1.1.3: Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolutional Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.1.1.4: Cell specific test parameters for Cell Re-selection in CELL_FACH**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0<br>C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0<br>C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0<br>C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset**1**$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0<br>C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0<br>C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0<br>C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |

| | | | | |
|---|---|---|---|---|
| FACH measurement occasion info | | not sent | not sent | not sent |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | |
| Propagation Condition | | AWGN | | |

Note: S-CCPCH shall not be located in TS0.

8.3.4.1.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclauses 7.3.3 and 7.4.2 to place the UE in CELL_FACH.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15 s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15 s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

h) Repeat steps d) to g) [TBD] times.

8.3.4.1.4.2 1,28 Mcps TDD option

8.3.4.1.4.2.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.4.1.4.2.1, 8.3.4.1.4.2.2, 8.3.4.1.4.2.3, and 8.3.4.1.4.2.4.

**Table 8.3.4.1.4.2.1: General test parameters for 1,28 Mcps Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 (initial), 5 (repetition) | |
| T2 | | s | 5 | |

**Table 8.3.4.1.4.2.2: Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | kbps | 35.2 |
| Channel symbol rate | ksps | 17.6 |
| Slot Format # | - | 0; 2 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Common Midamble |

**Table 8.3.4.1.4.2.3: Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolution Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.1.4.2.4: Cell specific test parameters for 1,28 Mcps Cell Re-selection in CELL_FACH**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset**1**$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

NOTE: S-CCPCH is located in an other downlink TS than TS0. Void.

8.3.4.1.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause s 7.3.3 and 7.4.2 to place the UE in CELL_FACH.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15 s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues

h) Repeat steps until the confidence level according to annex F.6.2 is achieved.

#### 8.3.4.1.4.3 7,68 Mcps TDD option

##### 8.3.4.1.4.3.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.4.1.4.3.1, 8.3.4.1.4.3.2, 8.3.4.1.4.3.3 and 8.3.4.1.4.3.4.

**Table 8.3.4.1.4.3.1: General test parameters for Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 (initial), 5 (repetition) | |
| T2 | | s | 5 | |

**Table 8.3.4.1.4.3.2: Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | Kbps | 24,4 |
| Channel symbol rate | Ksps | 12,2 |
| Slot Format # | - | 0 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Default Midamble |

**Table 8.3.4.1.4.3.3: Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolutional Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.1.4.3.4: Cell specific test parameters for Cell Re-selection in CELL_FACH**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |

| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_t$_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset**1**$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

Note: S-CCPCH shall not be located in TS0.

#### 8.3.4.1.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclauses 7.3.3 and 7.4.2 to place the UE in CELL_FACH.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15 s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15 s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved..

### 8.3.4.1.5 Test Requirements

#### 8.3.4.1.5.1 3,84 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 2.5 s.

3) In step g), the UE shall respond on cell 1 within 2.5 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.4.1.5.2 1,28 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 1.6 s.

3) In step g), the UE shall respond on cell 1 within 1.6 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.4.1.5.3 7,68 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 2 within 2.5 s.

3) In step g), the UE shall respond on cell 1 within 2.5 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4. 8.3.4.2 Scenario 2: TDD/TDD cell re-selection multi carrier case

#### 8.3.4.2.1 Definition and applicability

##### 8.3.4.2.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.3.4.2.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send SYNCH-UL sequence in the UpPTS for sending the CELL UPDATE with cause value "cell reselection".

The requirements and this test apply to the 1,28 Mcps TDD UE.

##### 8.3.4.2.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.4.2.2 Minimum requirement

##### 8.3.4.2.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 3 s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.2 and A.5.4.2.

##### 8.3.4.2.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 2 s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.3 and A.5.4.2.2.

##### 8.3.4.2.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 3 s. The rate of correct cell re-selections observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 5.4.4 and A.5.4.3.2.

#### 8.3.4.2.3 Test purpose

This test verifies that the UE meets the requirement for the cell re-selection delay in CELL_FACH for the multi carrier case.

#### 8.3.4.2.4 Method of test

##### 8.3.4.2.4.1 3,84 Mcps TDD option

###### 8.3.4.2.4.1.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.4.2.1.1, 8.3.4.2.1.2, 8.3.4.2.1.3, and 8.3.4.2.1.4.

**Table 8.3.4.2.1.1: General test parameters for Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.4.2.1.2: Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | Kbps | 24,4 |
| Channel symbol rate | Ksps | 12,2 |
| Slot Format # | - | 0 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Default Midamble |

**Table 8.3.4.2.1.3: Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolutional Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.2.1.4: Cell specific test parameters for Cell Re-selection in CELL_FACH**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 3 | 9 | 3 | 3 | 9 | 3 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -70 | | | -70 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| Inter-frequency TDD measurement indicator | | TRUE | | | | TRUE | | | | TRUE | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset**1**$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |

| | | | | |
|---|---|---|---|---|
| Treselection | | 0 | 0 | 0 |
| Sintrasearch | dB | not sent | not sent | not sent |
| Sintersearch | dB | not sent | not sent | not sent |
| FACH measurement occasion info | | not sent | not sent | not sent |
| Inter-frequency TDD measurement indicator | | TRUE | TRUE | TRUE |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | |
| Propagation Condition | | AWGN | | |

NOTE: S-CCPCH shall not be located in TS0.

8.3.4.2.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclauses 7.3.3 and 7.4.2 to place the UE in CELL_FACH. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE.

f) After another 15 s, the parameters are changed as described for T1. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

g) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE.

h) Repeat steps d) to g) [TBD] times.

8.3.4.2.4.2 1,28 Mcps TDD option

8.3.4.2.4.2.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.4.2.4.2.1, 8.3.4.2.4.2.2, 8.3.4.2.4.2.3, and 8.3.4.2.4.2.4.

**Table 8.3.4.2.4.2.1: General test parameters for 1,28Mcps Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | DBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | DBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | S | 1,28 | The value shall be used for all cells in the test. |
| T1 | | S | 15(initial), 5 (repetition) | |
| T2 | | S | 5 | |

**Table 8.3.4.2.4.2.2Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | kbps | 35.2 |
| Channel symbol rate | ksps | 17.6 |
| Slot Format # | - | 0; 2 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Common Midamble |

**Table 8.3.4.2.4.2.3 Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolution Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.2.4.2.4: Cell specific test parameters for 1,28Mcps Cell re-selection in CELL_FACH state**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 4 | 10 | 4 | 4 | 10 | 4 | 10 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -63 | -69 | | | -69 | -63 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2:C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3:C6:0 | | | |
| Qhyst1$_s$ | dBm | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion cycle length | | 4 | | | | 4 | | | | 4 | | | |
| Inter-frequency TDD measurement indicator | | TRUE | | | | TRUE | | | | TRUE | | | |
| Inter-frequency FDD measurement indicator | | FALSE | | | | FALSE | | | | FALSE | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0C4, C5:0; C4:C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5:C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6:C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion cycle length | | 4 | | | | 4 | | | | 4 | | | |
| Inter-frequency TDD measurement indicator | | TRUE | | | | TRUE | | | | TRUE | | | |
| Inter-frequency FDD measurement indicator | | FALSE | | | | FALSE | | | | FALSE | | | |
| $I_{oc}$ | dBm/ 1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

NOTE: S-CCPCH is located in an other downlink TS than TS0.

8.3.4.2.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclauses 7.3.3 7.4.2 to place the UE in CELL_FACH.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues

f) After another 15 s, the parameters are changed as described for T1.

g) The SS waits for SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message with cause value "cell reselection" from the UE. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues

h) Repeat steps d) to g)until the confidence level according to annex F.6.2 is achieved.

#### 8.3.4.2.4.3 7,68 Mcps TDD option

##### 8.3.4.2.4.3.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.4.2.4.3.1, 8.3.4.2.4.3.2, 8.3.4.2.4.3.3, and 8.3.4.2.4.3.4.

**Table 8.3.4.2.4.3.1: General test parameters for Cell Re-selection in CELL_FACH**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 (initial), 5 (repetition) | |
| T2 | | s | 5 | |

**Table 8.3.4.2.4.3.2: Physical channel parameters for S-CCPCH.**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | Kbps | 24,4 |
| Channel symbol rate | Ksps | 12,2 |
| Slot Format # | - | 0 |
| Frame allocation | - | Continuous frame allocation |
| Midamble allocation | - | Default Midamble |

**Table 8.3.4.2.4.3.3: Transport channel parameters for S-CCPCH**

| Parameter | FACH |
|---|---|
| Transport Channel Number | 1 |
| Transport Block Size | 240 |
| Transport Block Set Size | 240 |
| Transmission Time Interval | 20 ms |
| Type of Error Protection | Convolutional Coding |
| Coding Rate | ½ |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

**Table 8.3.4.2.4.3.4: Cell specific test parameters for Cell Re-selection in CELL_FACH**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 3 | 9 | 3 | 3 | 9 | 3 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -70 | | | -70 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| FACH measurement occasion info | | not sent | | | | not sent | | | | not sent | | | |
| Inter-frequency TDD measurement indicator | | TRUE | | | | TRUE | | | | TRUE | | | |
| $I_{oc}$ | dBm/ 7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| Timeslot | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |

| | | | | |
|---|---|---|---|---|
| Qoffset**1**$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0<br>C4, C5:0; C4, C6:0 | C5, C1: 0; C5, C2:0; C5,C3:0<br>C5, C4:0; C5, C6:0 | C6, C1: 0; C6, C2:0; C6,C3:0<br>C6, C4:0; C6, C5:0 |
| Qhyst1$_s$ | dB | 0 | 0 | 0 |
| Treselection | | 0 | 0 | 0 |
| Sintrasearch | dB | not sent | not sent | not sent |
| Sintersearch | dB | not sent | not sent | not sent |
| FACH measurement occasion info | | not sent | not sent | not sent |
| Inter-frequency TDD measurement indicator | | TRUE | TRUE | TRUE |
| $I_{oc}$ | dBm/ 7,68 MHz | -70 | | |
| Propagation Condition | | AWGN | | |

NOTE: S-CCPCH shall not be located in TS0.

#### 8.3.4.2.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclauses 7.3.3 and 7.4.2 to place the UE in CELL_FACH. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

d) After 15 s, the parameters are changed as described for T2.

e) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE.

f) After another 15 s, the parameters are changed as described for T1. If the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s and if no response is received the UE shall be switched off and the procedure returns to step 1. Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues.

g) The SS waits for CELL UPDATE message with cause value "cell reselection" from the UE.

h) Repeat steps d) to g) until the confidence level according to annex F.6.2 is achieved.

### 8.3.4.2.5 Test Requirements

#### 8.3.4.2.5.1 3,84 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 3 s.

3) In step g), the UE shall respond on cell 2 within 3 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.4.2.5.2 1,28 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 2 s.

3) In step g), the UE shall respond on cell 2 within 2 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.4.2.5.3 7,68 Mcps TDD option

1) In step d), after the UE has responded on cell 1, it shall not respond on any other cell (cell selection).

2) In step e), the UE shall respond on cell 1 within 3 s.

3) In step g), the UE shall respond on cell 2 within 3 s.

For the test to pass, the total number of fulfilled test requirements 2) and 3) shall be more than 90% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.5 Cell Re-selection in CELL_PCH

### 8.3.5.1 Scenario 1: TDD/TDD cell re-selection single carrier case

#### 8.3.5.1.1 Definition and applicability

##### 8.3.5.1.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.3.5.1.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send SYNCH-UL sequence in the UpPTS for sending the CELL UPDATE with cause value "cell reselection".

The requirements and this test apply to the 1,28 Mcps TDD UE.

##### 8.3.5.1.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.5.1.2 Minimum requirement

##### 8.3.5.1.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1 in clause 4.2.2.7.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5 and A.5.5.1.

##### 8.3.5.1.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1a in clause 4.2.2.

$T_{SI}$ Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allowing 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5.2.2 and A.5.5.1.

#### 8.3.5.1.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1B in clause 4.2.2.7.3.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5.2.3 and A.5.5.1.

### 8.3.5.1.3 Test purpose

This test verifies that the UE meets the minimum requirement for the cell re-selection delay in CELL_PCH for the single carrier case

### 8.3.5.1.4 Method of test

#### 8.3.5.1.4.1 3,84 Mcps TDD option

##### 8.3.5.1.4.1.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.5.1.1.1, and 8.3.5.1.1.2.

**Table 8.3.5.1.1.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.5.1.1.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.5.1.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded, the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) [TBD] times.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.1.4.2 1,28 Mcps TDD option

#### 8.3.5.1.4.2.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.5.1.4.2.1 and 8.3.5.1.4.2.2.

**Table 8.3.5.1.4.2.1: General test parameters for 1,28 Mcps Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3, Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) -– Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.5.1.4.2.2: 1,28Mcps Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.5.1.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message cause "cell reselection" within 8s, then a success is recorded.After receiving the CELL UPDATE message cause “cell reselection”,the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) If the UE has failed to respond with a SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the CELL UPDATE message cause “cell reselection”,the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message cause "cell reselection" within 8s, then a success is recorded and the procedure moves to step j).

i) If the UE has failed to respond with SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the CELL UPDATE message cause “cell reselection”,the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), so allow 8s in the test case.

##### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

##### RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.1.4.3 7,68 Mcps TDD option

#### 8.3.5.1.4.3.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.5.1.4.3.1, and 8.3.5.1.4.3.2.

**Table 8.3.5.1.4.3.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.5.1.4.3.2: Cell re-selection single carrier multi-cell case**

| **Parameter** | **Unit** | **Cell 1** | | | | **Cell 2** | | | | **Cell 3** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | 0 | | 8 | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.5.1.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded, the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved..

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.1.5 Test Requirements

#### 8.3.5.1.5.1 3,84 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.5.1.5.2 1,28 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.5.1.5.3 7,68 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.5.2 Scenario 2: TDD/TDD cell re-selection multi carrier case

#### 8.3.5.2.1 Definition and applicability

##### 8.3.5.2.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 3,84 Mcps option TDD UE.

##### 8.3.5.2.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send SYNCH-UL sequence in the UpPTS for sending the CELL UPDATE with cause value "cell reselection".

The requirements and this test apply to the 1,28 Mcps TDD UE.

##### 8.3.5.2.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the CELL UPDATE message with cause value "cell reselection" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.5.2.2 Minimum requirement

##### 8.3.5.2.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1 in clause 4.2.2.7.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5 and A.5.5.2.

##### 8.3.5.2.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1a in clause 4.2.2.

$T_{SI}$ Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allowing 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5.2 and A.5.5.2.2.

#### 8.3.5.2.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1B in clause 4.2.2.7.3.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.5.2 and A.5.5.2.

### 8.3.5.2.3 Test purpose

This test verifies that the UE meets the requirement for the cell re-selection delay in CELL_PCH for the multi carrier case.

### 8.3.5.2.4 Method of test

#### 8.3.5.2.4.1 3,84 Mcps TDD option

##### 8.3.5.2.4.1.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.5.2.1.1 and 8.3.5.2.1.2.

**Table 8.3.5.2.1.1: General test parameters for Cell Re-selection in Multi carrier case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.3.5.2.1.2: Cell re-selection multi carrier multi cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 0 | 6 | 0 | 0 | 6 | 0 | 6 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -67 | -73 | | | -73 | -67 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -76 | -76 | | | -76 | -76 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.5.2.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded, the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) [TBD] times.

NOTE 1: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

NOTE 2: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.2.4.2 1,28 Mcps TDD option

#### 8.3.5.2.4.2.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.5.2.4.2.1 and 8.3.5.2.4.2.2.

**Table 8.3.5.2.4.2.1: General test parameters for 1,28Mcps Cell Re-selection in Multi carrier case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3, Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.3.5.2.4.2.2: Cell re-selection 1,28 Mcps multi carrier multi cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 7 | 10 | 7 | 7 | 10 | 7 | 10 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -63 | -66 | | | -66 | -63 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | Cell 4 | | | | Cell 5 | | | | Cell 6 | | | |
| Timeslot | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/ 1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.3.5.2.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message cause "cell reselection" within 8s, then a success is recorded.After receiving the CELL UPDATE message cause "cell reselection",the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) If the UE has failed to respond with SYNCH-UL sequence in the UpPTSwithin the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the CELL UPDATE message cause "cell reselection",the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a SYNCH-UL sequence in the UpPTS for sending CELL UPDATE message cause "cell reselection" within 8s, then a success is recorded and the procedure moves to step j).

i) If the UE has failed to respond with SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the CELL UPDATE message cause "cell reselection",the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved

NOTE 1: T1 should initially be 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

NOTE 2: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.2.4.3 7,68 Mcps TDD option

#### 8.3.5.2.4.3.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.5.2.4.3.1 and 8.3.5.2.4.3.2.

**Table 8.3.5.2.4.3.1: General test parameters for Cell Re-selection in Multi carrier case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.3.5.2.4.3.2: Cell re-selection multi carrier multi cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 0 | 6 | 0 | 0 | 6 | 0 | 6 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -67 | -73 | | | -73 | -67 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -76 | -76 | | | -76 | -76 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.5.2.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.1 to place the UE in the CELL_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 2 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded, the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (CELL UPDATE message cause "cell reselection") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a CELL UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved..

NOTE 1: T1 is initially 30 s to allow enough time for the UE to search for cells as it has no prior knowledge of these.

NOTE 2: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | CELL PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.5.2.5 Test Requirements

#### 8.3.5.2.5.1 3,84 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.5.2.5.2 1,28 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.5.2.5.3 7,68 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.3.6 Cell Re-selection in URA_PCH

### 8.3.6.1 Scenario 1: TDD/TDD cell re-selection single carrier case

#### 8.3.6.1.1 Definition and applicability

##### 8.3.6.1.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.3.6.1.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the SYNCH-UL sequence in the UpPTS for sending URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 1,28 Mcps TDD UE. The two cells shall belong to different UTRAN Registration Areas (URAs).

##### 8.3.6.1.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.6.1.2 Minimum requirement

##### 8.3.6.1.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS25.123 [2] table 4.1 in clause 4.2.2.7.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6 and A.5.6.1.

##### 8.3.6.1.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1a in clause 4.2.2.

$T_{SI}$ Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allowing 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6 and A.5.6.1.

#### 8.3.6.1.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS25.123 [2] table 4.1B in clause 4.2.2.7.3.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6 and A.5.6.1.

### 8.3.6.1.3 Test purpose

This test verifies that the UE meets the minimum requirement for the cell re-selection delay in URA_PCH for the single carrier case.

### 8.3.6.1.4 Method of test

#### 8.3.6.1.4.1 3,84 Mcps TDD option

##### 8.3.6.1.4.1.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.6.1.1.1, and 8.3.6.1.1.2.

**Table 8.3.6.1.1.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.6.1.1.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.3.6.1.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded, the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) [TBD] times.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

8.3.6.1.4.2 1,28 Mcps TDD option

8.3.6.1.4.2.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.6.1.4.2.1, and 8.3.6.1.4.2.2. Cell 1 and Cell 2 shall belong to different UTRAN Registration Areas (URAs).

**Table 8.3.6.1.4.2.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) -– Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.6.1.4.2.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | | **0** | | **DWPTS** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.3.6.1.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a SYNCH-UL sequence in the UpPTS for sending URA UPDATE message cause "change of URA" within 8s, then a success is recorded.After receiving the URA UPDATE message cause "change of URA" ,the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) If the UE has failed to respond with a SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the URA UPDATE message cause "change of URA", the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a a SYNCH-UL sequence in the UpPTS for sending URA UPDATE message cause "change of URA" message within 8s, then a success is recorded and the procedure moves to step j).

i) If the UE has failed to respond with a SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the URA UPDATE message cause "change of URA",the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

8.3.6.1.4.3 7,68 Mcps TDD option

8.3.6.1.4.3.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.6.1.4.3.1, and 8.3.6.1.4.3.2.

**Table 8.3.6.1.4.3.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.6.1.4.3.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.6.1.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded, the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved..

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s(Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.6.1.5 Test Requirements

#### 8.3.6.1.5.1 3,84 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.6.1.5.2 1,28 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.6.1.5.3 7,68 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.3.6.2 Scenario 2: TDD/TDD cell re-selection multi carrier case

#### 8.3.6.2.1 Definition and applicability

##### 8.3.6.2.1.1 3,84 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 3,84 Mcps TDD UE.

##### 8.3.6.2.1.2 1,28 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send a SYNCH-UL sequence in the UpPTS for sending the URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 1,28 Mcps TDD UE. The cells shall belong to different UTRAN Registration Areas (URAs).

##### 8.3.6.2.1.3 7,68 Mcps TDD option

The cell re-selection delay is defined as the time from a change of cell levels to the moment when this change causes the UE to camp on a new cell , and starts to send the URA UPDATE message with cause value "change of URA" in the new cell.

The requirements and this test apply to the 7,68 Mcps TDD UE.

#### 8.3.6.2.2 Minimum requirement

##### 8.3.6.2.2.1 3,84 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS25.123 [2] table 4.1 in clause 4.2.2.7.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6 and A.5.6.1.

##### 8.3.6.2.2.2 1,28 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateNTDD} + T_{SI}$, where:

$T_{evaluateNTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS 25.123 [2] table 4.1a in clause 4.2.2.

$T_{SI}$ Time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allowing 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6.2 and A.5.6.2.1.2

#### 8.3.6.2.2.3 7,68 Mcps TDD option

The cell re-selection delay shall be less than 8 s.

NOTE:

The cell re-selection delay can be expressed as: $T_{evaluateTDD} + T_{SI}$, where:

$T_{evaluateTDD}$ A DRX cycle length of 1280ms is assumed for this test case, this leads to a $T_{evaluate\ TDD}$ of 6.4s according to TS25.123 [2] table 4.1B in clause 4.2.2.7.3.

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell. 1280 ms is assumed in this test case.

This gives a total of 7.68 s, allow 8s in the test case.

The normative reference for this requirement is TS 25.123 [2] clauses 5.6 and A.5.6.1.

### 8.3.6.2.3 Test purpose

This test verifies that the UE meets the minimum requirement for the cell re-selection delay in URA_PCH for the Multiple carrier case

### 8.3.6.2.4 Method of test

#### 8.3.6.2.4.1 3,84 Mcps TDD option

##### 8.3.6.2.4.1.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.6.2.1.1, and 8.3.6.2.1.2.

**Table 8.3.6.2.1.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | |
| T2 | | s | 15 | |

**Table 8.3.6.2.1.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | 7 | 9 | 7 | 7 | 9 | 7 | 9 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -64 | -66 | | | -66 | -64 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1,C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0 C2, C5: 0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5: 0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.3.6.2.4.1.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded, the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) [TBD] times.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s (Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

8.3.6.2.4.2 1,28 Mcps TDD option

8.3.6.2.4.2.1 Initial conditions

This scenario contains 6 cells and 2 carrier frequencies. The test parameters are given in Tables 8.3.6.2.4.2.1, and 8.3.6.2.4.2.2.

Cell1 and Cell2 shall belong to different UTRAN Registration Areas (URA).

**Table A.5.6.7: General test parameters for 1,28Mcps Cell Re-selection in Multi carrier case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -103 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1,28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1,28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table A.5.6.8: Cell re-selection 1,28Mcps multi carrier multi cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 10 | 7 | 10 | 7 | 7 | 10 | 7 | 10 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -63 | -66 | | | -66 | -63 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | Cell 4 | | | | Cell 5 | | | | Cell 6 | | | |
| Timeslot | | 0 | | DWPTS | | 0 | | DWPTS | | 0 | | DWPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | | | 0 | 0 | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| PCCPCH RSCP | dBm | -74 | -74 | | | -74 | -74 | | | -74 | -74 | | |
| $Qoffset1_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| $Qhyst1_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/1,28 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

8.3.6.2.4.2.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a SYNCH-UL sequence in the UpPTS for sending URA UPDATE message cause "change of URA" within 8s, then a success is recorded.After receiving the URA UPDATE message cause "change of URA", the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) If the UE has failed to respond with SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the URA UPDATE message cause "change of URA",the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a SYNCH-UL sequence in the UpPTS for sending URA UPDATE message cause "change of URA" within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with SYNCH-UL sequence in the UpPTS within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise after receiving the URA UPDATE message cause "change of URA",the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s (Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

8.3.6.2.4.3 7,68 Mcps TDD option

8.3.6.2.4.3.1 Initial conditions

This scenario contains 6 cells operating on the same carrier frequency. The test parameters are given in Tables 8.3.6.2.4.3.1, and 8.3.6.2.4.3.2.

**Table 8.3.6.2.4.3.1: General test parameters for Cell Re-selection single carrier multi-cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2, Cell3,Cell4, Cell5, Cell6 | |
| Final condition | Active cell | | Cell2 | |
| HCS | | | Not used | |
| UE_TXPWR_MAX_RACH | | dBm | 21 | The value shall be used for all cells in the test. |
| Qrxlevmin | | dBm | -102 | The value shall be used for all cells in the test. |
| Access Service Class (ASC#0) - Persistence value | | | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 30 | |
| T2 | | s | 15 | |

**Table 8.3.6.2.4.3.2: Cell re-selection single carrier multi-cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 10 | 10 | 10 | 10 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 0 | 6 | 0 | 0 | 6 | 0 | 6 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -67 | -73 | | | -73 | -67 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C1, C2: 0; C1, C3:0; C1,C4:0 C1, C5:0; C1, C6:0 | | | | C2, C1: 0; C2, C3:0; C2,C4:0C2, C5:0; C2, C6:0 | | | | C3, C1: 0; C3, C2:0; C3,C4:0 C3, C5:0; C3, C6:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| | | **Cell 4** | | | | **Cell 5** | | | | **Cell 6** | | | |
| **Timeslot** | | **0** | | **8** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 15 | 15 | 15 | 15 | 20 | 20 | 20 | 20 | 25 | 25 | 25 | 25 |
| PICH_Ec/Ior | dB | | | -3 | -3 | | | -3 | -3 | | | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 | -3 |
| PCCPCH RSCP | dBm | -76 | -76 | | | -76 | -76 | | | -76 | -76 | | |
| Qoffset1$_{s,n}$ | dB | C4, C1: 0; C4, C2:0; C4,C3:0 C4, C5:0; C4, C6:0 | | | | C5, C1: 0; C5, C2:0; C5,C3:0 C5, C4:0; C5, C6:0 | | | | C6, C1: 0; C6, C2:0; C6,C3:0 C6, C4:0; C6, C5:0 | | | |
| Qhyst1$_s$ | dB | 0 | | | | 0 | | | | 0 | | | |
| Treselection | s | 0 | | | | 0 | | | | 0 | | | |
| Sintrasearch | dB | not sent | | | | not sent | | | | not sent | | | |
| Sintersearch | dB | not sent | | | | not sent | | | | not sent | | | |
| $I_{oc}$ | dBm/7,68 MHz | -70 | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | |

#### 8.3.6.2.4.3.2 Procedure

a) The SS activates cell 1-6 with T1 defined parameters.

b) The UE is switched on.

c) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.4.2.7.2 to place the UE in the URA_PCH state on Cell 1 and then the SS waits for this process to complete.

d) After 15 s from the completion of step c) or the beginning of T1, the parameters are changed as described for T2.

e) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded, the SS shall transmit a URA UPDATE CONFIRM message and then the procedure moves to step g).

f) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T2 and if no response is received, the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step g).

g) After a total of 15 s from the beginning of T2, the parameters are changed as described for T1.

h) If the UE responds on Cell 1 with a PRACH (URA UPDATE message cause "change of URA") within 8s, then a success is recorded and the procedure moves to step j).

i) Since the UE has failed to respond with the correct message within the allowed time, a failure is recorded. The SS shall then wait for a total of 15s from the beginning of T1 and if no response is received the UE shall be switched off and the procedure returns to step a). Otherwise the SS shall transmit a URA UPDATE CONFIRM message and then the procedure continues with step j).

j) Repeat steps d) to i) until the confidence level according to annex F.6.2 is achieved..

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 7.78s (Minimum requirement + 100ms), allow 8s in the test case.

RADIO BEARER SETUP (Step 3)

| Information Element | Value/remark |
|---|---|
| RRC State Indicator | URA PCH |
| UTRAN DRX cycle length coefficient | 7 |

### 8.3.6.2.5 Test Requirements

#### 8.3.6.2.5.1 3,84 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.6.2.5.2 1,28 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.3.6.2.5.3 7,68 Mcps TDD option

For the test to pass, the total number of successful attempts shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.8.4 RRC Connection Control

## 8.4.1 RRC re-establishment delay

### 8.4.1.1 3,84 Mcps TDD option

#### 8.4.1.1.1 RRC re-establishment delay to a known target cell

##### 8.4.1.1.1.1 Definition and applicability

For UTRA TDD, the UE re-establishment delay $T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is defined as the time between the moment when radio link failure is considered by the UE to when the UE starts sending the RRC CELL UPDATE message to the UTRAN on RACH.

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is known if either or both of the following conditions are true:

- the UE has had a radio link connected to the cell during the last 5 seconds
- the cell has been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD (3,84 Mcps option) UE.

##### 8.4.1.1.1.2 Minimum requirement

The RRC re-establishment delay $T_{RE\text{-}ESTABLISH}$ to a known target cell shall be less than 2 s.

The rate of successful RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in this test case can be expressed as,

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN}.$$

where,

$T_{RRC\text{-}RE\text{-}ESTABLISH} = 160ms+(N_{313}-1)*10ms+T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN} = 50ms+T_{SEARCH\text{-}KNOWN} + T_{SI} + T_{RA}$,

and,

$N_{313}$ Equal to 20 and therefore resulting in 200 ms delay.

$T_{313}$ Equal to 0 s.

$T_{SEARCH\text{-}KNOWN}$ Equal to 100 ms

$T_{SI}$ Equal to 1280 ms, the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure performance value of system information blocks defined in TS 25.331 [9] for a UTRAN cell.

$T_{RA}$ Equal to 40 ms, the additional delay caused by the random access procedure.

This gives a total of 1820ms, allow 2 s in the test case.

##### 8.4.1.1.1.3 Test purpose

The test purpose is to verify that the RRC re-establishment delay to a known target cell is within the specified limits.

8.4.1.1.1.4 Method of test

8.4.1.1.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.4.1.1 and table 8.4.1.2 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE in order to camp on a cell shall be 1280 ms. DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consists of 2 successive time periods, with time durations of T1 and T2 respectively.

During T1, the DL DPCH in cell 1 shall be transmitted in timeslot 2 and the UL DPCH in cell 1 shall be transmitted in timeslot 10. At the beginning of time period T2, the DPCH shall be removed.

Cell 1 and cell shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.4.1.1: General test parameters for RRC re-establishment delay, known target cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | Cell 2 shall be included in the monitored set in Cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| N313 | | | 20 | |
| N315 | | | 1 | |
| T313 | | Seconds | 0 | |
| $T_{SI}$ | | ms | 1280 | |
| Monitored cell list size | | | 24 TDD neighbours on Channel 1 | |
| Reporting frequency | | Seconds | 4 | |
| T1 | | | 10 | |
| T2 | | | 6 | |

**Table 8.4.1.2: Cell specific parameters for RRC re-establishment delay test, known target cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | n.a. | n.a. | -3 | -3 | n.a. | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | dB | n.a. | n.a. | -3 | -3 | n.a. | n.a. | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -13 | 3 | -13 | 5 | 5 | 5 | 5 |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | | | | | | |
| P-CCPCH_RSCP | dB | -70 | -86 | n.a. | n.a. | -68 | -68 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | | | | |

#### 8.4.1.1.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 [3] (Message sequence chart for Handover Test procedure) is not yet specified.

4) The SS waits for random access requests from the UE on cell 2.

5) 10 s after step3 has completed, the parameters are changed to that as described for T2.

6) If the UE responds on cell 2 within 2.0 s from the beginning of time period T2 with a CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds from the beginning of time period T2, the RF parameters are set up according to T1.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 [TBD] times.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 25.331 [9] for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 1920ms (Minimum requirement + 100ms), allow 2 s in the test case.

#### 8.4.1.1.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.4.1.1.2 RRC re-establishment delay to an unknown target cell

#### 8.4.1.1.2.1 Definition and applicability

For UTRA TDD, the UE re-establishment delay $T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is defined as the time between the moment when radio link failure is considered by the UE to when the UE starts sending the RRC CELL UPDATE message to the UTRAN on RACH.

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is known if either or both of the following conditions are true:

- the UE has had a radio link connected to the cell during the last 5 seconds
- the cell has been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD (3,84 Mcps option) UE.

#### 8.4.1.1.2.2 Minimum requirement

The RRC re-establishment delay $T_{RE\text{-}ESTABLISH}$ to an unknown target cell shall be less than 3,7 s.

The rate of successful RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in this test case can be expressed as,

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}UNKNOWN}.$$

where,

$T_{RRC\text{-}RE\text{-}ESTABLISH}$= 160ms+($N_{313}$-1)*10ms+$T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN}$=50ms+$T_{SEARCH\text{-}UNKNOWN}$ *NF + $T_{SI}$ + $T_{RA}$,

and,

| | |
|---|---|
| $N_{313}$ | Equal to 20 and therefore resulting in 200 ms delay. |
| $T_{313}$ | Equal to 0 s. |
| $T_{SEARCH\text{-}UNKNOWN}$ | Equal to 800 ms |
| NF | Equal to 2, the number of different frequencies in the monitored set of cell 1. |
| $T_{SI}$ | Equal to 1280 ms, the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure performance value of system information blocks defined in TS 25.331 [9] for a UTRAN cell. |
| $T_{RA}$ | Equal to 40 ms, the additional delay caused by the random access procedure. |

This gives a total of 3320ms, allow 3,7 s in the test case.

#### 8.4.1.1.2.3 Test purpose

The test purpose is to verify that the RRC re-establishment delay to an unknown target cell is within the specified limits.

#### 8.4.1.1.2.4 Method of test

##### 8.4.1.1.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.4.1.3 and table 8.4.1.4 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE in order to camp on a cell shall be 1280 ms. DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consists of 2 successive time periods, with time durations of T1 and T2 respectively.

During T1, the DL DPCH in cell 1 shall be transmitted in timeslot 2 and the UL DPCH in cell 1 shall be transmitted in timeslot 10. At the beginning of time period T2, the DPCH shall be removed.

Cell 1 and cell shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.4.1.3: General test parameters for RRC re-establishment delay, unknown target cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | Cell 2 shall not be included in the monitored set in Cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| N313 | | | 20 | |
| N315 | | | 1 | |
| T313 | | Seconds | 0 | |
| $T_{SI}$ | | ms | 1280 | |
| Monitored cell list size | | | 16 TDD neighbours on Channel 1<br>16 TDD neighbours on Channel 2 | |
| Reporting frequency | | Seconds | 4 | |
| T1 | | | 10 | |
| T2 | | | 6 | |

**Table 8.4.1.4: Cell specific parameters for RRC re-establishment delay test, unknown target cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | n.a. | n.a. | -3 | -3 | n.a. | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | dB | n.a. | n.a. | -3 | -3 | n.a. | n.a. | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -13 | 3 | -13 | 5 | 5 | 5 | 5 |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | | | | | | |
| P-CCPCH_RSCP | dB | -70 | -86 | n.a. | n.a. | -68 | -68 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | | | | |

8.4.1.1.2.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) The SS waits for random access requests from the UE on cell 2.

5) 10 s after step3 has completed, the parameters are changed to that as described for T2.

6) If the UE responds on cell 2 within 3.7 s from the beginning of time period T2 with a CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds the RF parameters are set up according to T1.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 [TBD] times

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 25.331 [9] for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 3420ms (Minimum requirement + 100ms), allow 3.7s in the test case.

#### 8.4.1.1.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.4.1.2 1,28 Mcps TDD option

#### 8.4.1.2.1 RRC re-establishment delay to an known target cell

##### 8.4.1.2.1.1 Definition and applicability

The UE Re-establishment delay requirement ($T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$) is defined as the time between the moment when radio link failure is considered by the UE, to when the UE starts to send SYNC-UL in the UpPTS for sending a CELL UPDATE message using the cause “radio link failure”.

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is known if either or both of the following conditions are true:

- the UE has had a dedicated connection to the cell during the last 5 seconds.
- the cell has been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD UE, 1,28 Mcps option..

##### 8.4.1.2.1.2 Minimum requirement

The Re-establishment delay $T_{RE\text{-}ESTABLISH}$ to a known cell shall be less than 1.9 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The Re-establishment delay in this case can be expressed as

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN}.$$

where

$T_{RRC\text{-}RE\text{-}ESTABLISH}$= 160ms+$(N_{313}-1)$*10ms+$T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH_REQ\text{-}KNOWN}$=50ms+$T_{search}$ + $T_{SI}$ + $T_{RA}$,

$N_{313}$= 20

$T_{313}$= 0s

$T_{search}$= 100ms

$T_{RA}$ = The additional delay caused by the random access procedure. 35 ms is assumed in this test case.

$T_{SI}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms). 1280 ms is assumed in this test case.

This gives a total of 1815ms, allow 1.9s in the test case.

#### 8.4.1.2.1.3 Test purpose

To verify that the UE meets the minimum requirement.

#### 8.4.1.2.1.4 Method of test

##### 8.4.1.2.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G2.2

Frequencies to be tested: mid range; see clause G.2.4

The test parameters are given in table 8.4.1.5 and table 8.4.1.6 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell shall be 1280 ms. And DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consist of 2 successive time periods, with a time duration of T1 and T2 respectively. At the start of time period T2, the dedicated channel is removed.

**Table 8.4.1.5: General test parameters for RRC re-establishment delay, Test 1**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH Parameters | | DL and UL Reference measurement channel 12.2 kbps | As specified in clause C.3.1 and C.2.1 |
| Power Control | | On | |
| Active cell, Initial condition | | Cell 1 | |
| Active cell, Final condition | | Cell 2 | |
| N313 | | 20 | |
| N315 | | 1 | |
| T313 | Seconds | 0 | |
| Monitored cell list size | | 24 | Monitored set shall only include intra frequency neighbours. |
| Cell 2 | | | Included in the monitored set |
| Reporting frequency | Seconds | 4 | |
| T1 | s | 10 | |
| T2 | s | 6 | |

**Table 8.4.1.6: Cell specific parameters for RRC re-establishment delay test, Test 1**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | 5 | 0 | 5 |
| | | T0 | T0 | T0 | T0 |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| DPCH_Ec/Ior | dB | Not applicable | Note 2 | Not applicable | |
| OCNS_Ec/Ior | dB | Note 2 | Note 2 | Note 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | [3] | 3 | -infinity | |
| $I_{oc}$ | dBm/ 1.28 MHz | -70 | | | |
| PCCPCH_RSCP | dBm | -70 | Not applicable | -infinity | |
| Propagation Condition | | AWGN | | | |
| NOTE 1: The DPCH level is controlled by the power control loop.<br>NOTE 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$. | | | | | |

**Table 8.4.1.7: Cell specific parameters for RRC re-establishment delay test, Test 1**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 5 | | 0 | | 5 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | |
| DPCH_Ec/Ior | dB | Not applicable | | Note 1 | -infinity | Not applicable | | | |
| OCNS_Ec/Ior | dB | Note 2 | | Note 2 | | Note 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | -3 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | [3] | -infinity | 3 | -infinity | 6 | 6 | | |
| $I_{oc}$ | dBm/ 1.28 MHz | -70 | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | -infinity | Not applicable | | -67 | -67 | | |
| Propagation Condition | | AWGN | | | | | | | |
| NOTE 1: The DPCH level is controlled by the power control loop.<br>NOTE 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$. | | | | | | | | | |

8.4.1.2.1.4.2 Procedure

1) The RF parameters are set up according to column T0 in table 8.4.1.6.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4

4) The RF parameters are set up according to column T1 in table 8.4.1.7.

5) 10 s after step4 has completed, the parameters are changed to that as described for column T2.

6) If the UE responds on cell 2 within 2.0 s from the beginning of time period T2 with a SYNC-UL in the UpPTS for sending CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds from the beginning of time period T2, the RF parameters are set up according to T0.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 until the confidence level according to annex F.6.2A is achieved.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 1915ms(Minimum requirement + 100ms), allow 2s in the test case.

#### 8.4.1.2.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.4.1.2.2 RRC re-establishment delay to an unknown target cell

#### 8.4.1.2.2.1 Definition and applicability

The UE Re-establishment delay requirement ($T_{UE\text{-}E\text{-}ESTABLISH\text{-}REQ}$) is defined as the time between the moment when radio link failure is considered by the UE, to when the UE starts to send SYNC-UL in the UpPTS for sending a CELL UPDATE message using the cause “radio link failure”

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is NOT known if both of the following conditions are true:

- the UE has NOT had a dedicated connection to the cellduring the last 5 conds.
- the cell has NOT been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD UE, 1,28 Mcps option..

#### 8.4.1.2.2.2 Minimum requirement

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The Re-establishment delay in this case can be expressed as

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}UNKNOWN}.$$

where

$T_{RRC\text{-}RE\text{-}ESTABLISH}$= 160ms+$(N_{313}-1)$*10ms+$T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}UNKNOWN}$=50ms+$T_{search}$*NF + $T_{SI}$ + $T_{RA}$,

$N_{313}$= 20

$T_{313}$= 0s

$T_{search}$= 800ms

*NF* is the number of different frequencies in the monitored set. 3 frequencies are assumed in this test case.

$T_{RA}$ = The additional delay caused by the random access procedure. 35 ms is assumed in this test case.

$T_{SI}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell (ms).1280 ms is assumed in this test case.

This gives a total of 4115ms, allow 4.2s in the test case.

#### 8.4.1.2.2.3 Test purpose

To verify that the UE meets the minimum requirement

8.4.1.2.2.4 Method of test

8.4.1.2.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G2.2

Frequencies to be tested: mid range; see clause G.2.4

The test parameters are given in table 8.4.1.7 and table 8.4.1.8 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell shall be 1280 ms. And DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consists of 2 successive time periods, with a time duration of T1 and T2 respectively. At the start of time period T2, the dedicated channel is removed.

**Table 8.4.1.7. General test parameters for RRC re-establishment delay, Test 2**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH Parameters | | DL and UL Reference measurement channel 12.2 kbps | As specified in clause A.3.1 and A.2.1 |
| Power Control | | On | |
| Active cell, initial condition | | Cell 1 | |
| Active cell, final condition | | Cell 2 | |
| N313 | | 20 | |
| N315 | | 1 | |
| T313 | Seconds | 0 | |
| Monitored cell list size | | 24 | Monitored set shall include 2 additional frequencies. |
| Cell 2 | | | Cell 2 is not included in the monitored set. Cell 2 is located on one of the 2 additional frequencies of the monitored set. |
| Reporting frequency | Seconds | 4 | |
| T1 | s | 10 | |
| T2 | s | 6 | |

**Table 8.4.1.8 Cell specific parameters for RRC re-establishment delay test, Test 2**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 5 | | 0 | | | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | | |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | -3 | | | |
| DPCH_Ec/Ior | dB | Not applicable | | Note 1 | -infinity | Not applicable | | | |
| OCNS_Ec/Ior | dB | Note 2 | | Note 2 | | Note 2 | | | |
| | | | | | | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -infinity | 3 | -infinity | -infinity | 6 | | |
| $I_{oc}$ | dBm/ 1.28 MHz | -70 | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | -infinity | Not applicable | | -- infinity | -67 | | |
| Propagation Condition | | AWGN | | | | | | | |

NOTE 1: The DPCH level is controlled by the power control loop.
NOTE 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.

8.4.1.2.2.4.2 Procedure

1) The RF parameters are set up according to column T1 in table 8.4.1.4.2.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4

4) The SS waits for random access requests from the UE on cell 2.

5) 10 s after step3 has completed, the parameters are changed to that as described for column T2.

6) If the UE responds on cell 2 within 4.3 s from the beginning of time period T2 with a SYNC-UL in the UpPTS for sending CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds the RF parameters are set up according to T1.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 until the confidence level according to annex F.6.2A is achieved.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in 25.331 for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 4215ms(Minimum requirement + 100ms), allow 4.3s in the test case.

#### 8.4.1.2.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.4.1.3 7,68 Mcps TDD option

#### 8.4.1.3.1 RRC re-establishment delay to a known target cell

##### 8.4.1.3.1.1 Definition and applicability

For UTRA TDD, the UE re-establishment delay $T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is defined as the time between the moment when radio link failure is considered by the UE to when the UE starts sending the RRC CELL UPDATE message to the UTRAN on RACH.

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is known if either or both of the following conditions are true:

- the UE has had a radio link connected to the cell during the last 5 seconds
- the cell has been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD (7,68 Mcps option) UE.

##### 8.4.1.3.1.2 Minimum requirement

The RRC re-establishment delay $T_{RE\text{-}ESTABLISH}$ to a known target cell shall be less than 2 s.

The rate of successful RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in this test case can be expressed as,

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN}.$$

where,

$T_{RRC\text{-}RE\text{-}ESTABLISH} = 160ms + (N_{313}-1)*10ms + T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN} = 50ms + T_{SEARCH\text{-}KNOWN} + T_{SI} + T_{RA}$,

and,

$N_{313}$ Equal to 20 and therefore resulting in 200 ms delay.

$T_{313}$ Equal to 0 s.

$T_{SEARCH-KNOWN}$ Equal to 100 ms

$T_{SI}$ Equal to 1280 ms, the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure performance value of system information blocks defined in TS 25.331 [9] for a UTRAN cell.

$T_{RA}$ Equal to 40 ms, the additional delay caused by the random access procedure.

This gives a total of 1820ms, allow 2 s in the test case.

#### 8.4.1.3.1.3 Test purpose

The test purpose is to verify that the RRC re-establishment delay to a known target cell is within the specified limits.

#### 8.4.1.3.1.4 Method of test

##### 8.4.1.3.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.4.1.3.1.4.1 and table 8.4.1.3.1.4.2 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE in order to camp on a cell shall be 1280 ms. DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consists of 2 successive time periods, with time durations of T1 and T2 respectively.

During T1, the DL DPCH in cell 1 shall be transmitted in timeslot 2 and the UL DPCH in cell 1 shall be transmitted in timeslot 10. At the beginning of time period T2, the DPCH shall be removed.

Cell 1 and cell shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.4.1.3.1.4.1: General test parameters for RRC re-establishment delay, known target cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | Cell 2 shall be included in the monitored set in Cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| N313 | | | 20 | |
| N315 | | | 1 | |
| T313 | | Seconds | 0 | |
| $T_{SI}$ | | ms | 1280 | |
| Monitored cell list size | | | 24 TDD neighbours on Channel 1 | |
| Reporting frequency | | Seconds | 4 | |
| T1 | | | 10 | |
| T2 | | | 6 | |

**Table 8.4.1.3.1.4.2: Cell specific parameters for RRC re-establishment delay test, known target cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | n.a. | n.a. | -3 | -3 | n.a. | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | dB | n.a. | n.a. | -3 | -3 | n.a. | n.a. | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -13 | 3 | -13 | 5 | 5 | 5 | 5 |
| $I_{oc}$ | dBm/ 7.68 MHz | -70 | | | | | | | |
| P-CCPCH_RSCP | dB | -70 | -86 | n.a. | n.a. | -68 | -68 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | | | | |

### 8.4.1.3.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.4.

[Editor's note: subclause 7.3.4 in TS 34.108 [3] (Message sequence chart for Handover Test procedure) is not yet specified.

4) The SS waits for random access requests from the UE on cell 2.

5) 10 s after step3 has completed, the parameters are changed to that as described for T2.

6) If the UE responds on cell 2 within 2.0 s from the beginning of time period T2 with a CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds from the beginning of time period T2, the RF parameters are set up according to T1.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 until the confidence level according to annex F.6.2 is achieved.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 25.331 [9] for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 1920ms (Minimum requirement + 100ms), allow 2 s in the test case.

### 8.4.1.3.1.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.4.1.3.2 RRC re-establishment delay to an unknown target cell

#### 8.4.1.3.2.1 Definition and applicability

For UTRA TDD, the UE re-establishment delay $T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is defined as the time between the moment when radio link failure is considered by the UE to when the UE starts sending the RRC CELL UPDATE message to the UTRAN on RACH.

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ}$ is depending on whether the target cell is known by the UE or not. A cell is known if either or both of the following conditions are true:

- the UE has had a radio link connected to the cell during the last 5 seconds
- the cell has been measured by the UE during the last 5 seconds.

The requirements of this test apply to the TDD (7,68 Mcps option) UE.

#### 8.4.1.3.2.2 Minimum requirement

The RRC re-establishment delay $T_{RE\text{-}ESTABLISH}$ to an unknown target cell shall be less than 3,7 s.

The rate of successful RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in this test case can be expressed as,

$$T_{RE\text{-}ESTABLISH} = T_{RRC\text{-}RE\text{-}ESTABLISH} + T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}UNKNOWN}.$$

where,

$T_{RRC\text{-}RE\text{-}ESTABLISH} = 160ms+(N_{313}-1)*10ms+T_{313}$

$T_{UE\text{-}RE\text{-}ESTABLISH\text{-}REQ\text{-}KNOWN} = 50ms+T_{SEARCH\text{-}UNKNOWN} *NF + T_{SI} + T_{RA}$,

and,

$N_{313}$ Equal to 20 and therefore resulting in 200 ms delay.

$T_{313}$ Equal to 0 s.

$T_{SEARCH\text{-}UNKNOWN}$ Equal to 800 ms

NF Equal to 2, the number of different frequencies in the monitored set of cell 1.

$T_{SI}$ Equal to 1280 ms, the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure performance value of system information blocks defined in TS 25.331 [9] for a UTRAN cell.

$T_{RA}$ Equal to 40 ms, the additional delay caused by the random access procedure.

This gives a total of 3320ms, allow 3,7 s in the test case.

#### 8.4.1.3.2.3 Test purpose

The test purpose is to verify that the RRC re-establishment delay to an unknown target cell is within the specified limits.

#### 8.4.1.3.2.4 Method of test

##### 8.4.1.3.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.4.1.3.2.4.1 and table 8.4.1.3.2.4.2 below. The maximum repetition period of the relevant system info blocks that needs to be received by the UE in order to camp on a cell shall be 1280 ms. DRX cycle length shall be 1280ms. In the measurement control information it is indicated to the UE that periodic reporting shall be used. The test consists of 2 successive time periods, with time durations of T1 and T2 respectively.

During T1, the DL DPCH in cell 1 shall be transmitted in timeslot 2 and the UL DPCH in cell 1 shall be transmitted in timeslot 10. At the beginning of time period T2, the DPCH shall be removed.

Cell 1 and cell shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.4.1.3.2.4.1: General test parameters for RRC re-establishment delay, unknown target cell case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | Cell 2 shall not be included in the monitored set in Cell 1. |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Access Service Class (ASC#0) - Persistence value | | - | 1 | Selected so that no additional delay is caused by the random access procedure. The value shall be used for all cells in the test. |
| N313 | | | 20 | |
| N315 | | | 1 | |
| T313 | | Seconds | 0 | |
| $T_{SI}$ | | ms | 1280 | |
| Monitored cell list size | | | 16 TDD neighbours on Channel 1<br>16 TDD neighbours on Channel 2 | |
| Reporting frequency | | Seconds | 4 | |
| T1 | | | 10 | |
| T2 | | | 6 | |

**Table 8.4.1.3.2.4.2: Cell specific parameters for RRC re-establishment delay test, unknown target cell case**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Timeslot Number** | | **0** | | **8** | | **0** | | **8** | |
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | n.a. | n.a. | -3 | -3 | n.a. | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | dB | n.a. | n.a. | -3 | -3 | n.a. | n.a. | -3 | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | -13 | 3 | -13 | 5 | 5 | 5 | 5 |
| $I_{oc}$ | dBm/ 7.68 MHz | -70 | | | | | | | |
| P-CCPCH_RSCP | dB | -70 | -86 | n.a. | n.a. | -68 | -68 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | | | | |

8.4.1.3.2.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.

[Editor's note: subclause 7.3.4 in TS 34.108 (Message sequence chart for Handover Test procedure) is not yet specified]

4) The SS waits for random access requests from the UE on cell 2.

5) 10 s after step3 has completed, the parameters are changed to that as described for T2.

6) If the UE responds on cell 2 within 3.7 s from the beginning of time period T2 with a CELL_UPDATE command then the number of successful tests is increased by one.

7) SS shall transmit a RRC CONNECTION RELEASE message to make the UE transit to idle mode.

8) After 6 seconds the RF parameters are set up according to T1.

9) The SS shall wait for 30s to make the UE complete cell reselection to cell1.

10) Repeat step 3-9 until the confidence level according to annex F.6.2 is achieved.

NOTE: The time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 25.331 [9] for a UTRAN cell. Since the maximum repetition period of the relevant system info blocks that needs to be received by the UE to camp on a cell is 1280ms and the maximum RRC procedure delay for reception system information block is 100ms, 1380 ms is assumed in this test case. Therefore this gives a total of 3420ms (Minimum requirement + 100ms), allow 3.7s in the test case.

#### 8.4.1.3.2.5 Test requirements

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.4.2 Transport Format Combination selection in UE

### 8.4.2.1 Interactive or Background, PS, UL: 64 kbps

#### 8.4.2.1.1 Definition and applicability

##### 8.4.2.1.1.1 3,84 Mcps TDD option

When the UE estimates that a certain TFC would require more power than the maximum transmit power, it shall limit the usage of transport format combinations for the assigned transport format combination set. This in order to make it possible for the network operator to maximise the coverage. Transport format combination selection is described in section 11.4 of TS 25.321 [14].

##### 8.4.2.1.1.1A 1,28 Mcps TDD option

When the UE estimates that a certain TFC would require more power than the maximum transmit power, it shall limit the usage of transport format combinations for the assigned transport format combination set. This allows the network operator to maximise the coverage. Transport format combination selection is described in section 11.4 of TS 25.321 [14].

##### 8.4.2.1.1.2 7,68 Mcps TDD option

When the UE estimates that a certain TFC would require more power than the maximum transmit power, it shall limit the usage of transport format combinations for the assigned transport format combination set. This in order to make it possible for the network operator to maximise the coverage. Transport format combination selection is described in section 11.4 of TS 25.321 [14].

#### 8.4.2.1.2 Minimum requirements

##### 8.4.2.1.2.1 3,84 Mcps TDD option

The UE shall continuously evaluate based on the *Elimination*, *Recovery* and *Blocking* criteria defined below, how TFCs can be used for the purpose of TFC selection. The evaluation shall be performed using the estimated UE transmit power of a given CCTrCH in its associated timeslots.

In the case of a single CCTrCH or multiple CCTrCHs having mutually exclusive timeslot assignments, the UE shall consider the *Eliminiation* criterion for a given TFC of a CCTrCH to be fulfilled if for 3 successive frames the estimated UE transmit power is greater than the Maximum UE transmitter power for at least one timeslot associated with the CCTrCH in each frame.

In the case of multiple CCTrCHs not having mutually exclusive timeslot assignments, if for a given CCTrCH for 3 successive frames the estimated UE transmit power is greater than the Maximum UE transmitter power for at least one timeslot associated with the CCTrCH in each frame, the UE shall consider the *Elimination* criterion for a given TFC to be fulfilled if the use of this TFC will cause the estimated UE transmit power to continue to be greater than the Maximum UE transmitter power in at least one timeslot associated with the CCTrCH.

In the case of multi-frame operation of UL Physical Channels, the UE shall only consider active frames for the evaluation of the *Elimination* criterion.

If the *Elimination* criterion for a given TFC is fulfilled, the MAC in the UE shall consider that the TFC is in Excess-Power state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within $T_{notify}$ from the moment the *Elimination* criterion was fulfilled.

The UE shall not consider the *Recovery* criterion for a given TFC to be fulfilled until the use of this TFC will not cause the estimated UE transmit power to be greater than the Maximum UE transmitter power for all UL timeslots associated with the TFC for a minimum of 3 successive frames.

In the case of multi-frame operation of UL Physical Channels, the UE shall only consider active frames for the evaluation of the *Recovery* criterion.

If the *Recovery* criterion for a given TFC is fulfilled, the MAC in the UE shall consider that the TFC is in Supported state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within $T_{notify}$ from the moment the *Recovery* criterion was fulfilled.

The UE shall consider the *Blocking* criterion for a given TFC to be fulfilled at the latest at the start of the longest uplink TTI after the moment at which the TFC will have been in Excess-Power state for a duration of

$$(T_{notify} + T_{modify} + T_{L1_proc}).$$

where:

$T_{notify}$ equals 15 ms, and

$T_{modify}$ equals $MAX(T_{adapt_max}, T_{TTI})$, and

$T_{L1\ proc}$ equals 35 ms, and

$T_{adapt_max}$ equals $MAX(T_{adapt_1}, T_{adapt_2}, ..., T_{adapt_N})$, and

N equals the number of logical channels that need to change rate, and

$T_{adapt_n}$ equals the time it takes for higher layers to provide data to MAC in a new supported bitrate, for logical channel n. Table 6A.1 defines $T_{adapt}$ times for different services. For services where no codec is used $T_{adapt}$ shall be considered to be equal to 0 ms.

**Table 8.4.2.1.1: $T_{adapt}$**

| Service | $T_{adapt}$ [ms] |
|---|---|
| UMTS AMR | 40 |
| UMTS AMR 2 | 60 |

$T_{TTI}$ equals the longest uplink TTI of the selected TFC (ms).

The Maximum UE transmitter power is defined as follows

Maximum UE transmitter power = MIN(Maximum allowed UL TX Power, UE maximum transmit power)

where

Maximum allowed UL TX Power is set by SS and defined in TS 25.331 [9], and

UE maximum transmit power is defined by the UE power class, and specified in TS 25.102 [1].

The normative reference for these requirements is TS 25.123 [2] clauses 6A.2 and A.6A.2.

8.4.2.1.2.1A 1,28 Mcps TDD option

The UE shall continuously evaluate based on the *Elimination*, *Recovery* and *Blocking* criteria defined below, how TFCs can be used for the purpose of TFC selection. The evaluation shall be performed using the estimated UE transmit power of a given TFC. The UE transmit power estimation shall be made using the UE transmitted power measured over the measurement period and the gain factors of the corresponding TFC.

The UE shall consider the *Eliminiation* criterion for a given TFC to be fulfilled if the estimated UE transmit power needed for this TFC is greater than the Maximum UE transmitter power for at least X out of Y successive measurement periods. The MAC in the UE shall consider that the TFC is in Excess-Power state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within [15 ms] from the moment the *Elimination* criterion was fulfilled.

The UE shall consider the *Recovery* criterion for a given TFC to be fulfilled if the estimated UE transmit power needed for this TFC has not been greater than the Maximum UE transmitter power for at least Y successive measurement periods. The MAC in the UE shall consider that the TFC is in Supported state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within $T_{notify}$ from the moment the *Recovery* criterion was fulfilled.

The UE shall consider the *Blocking* criterion for a given TFC to be fulfilled at the latest at the start of the longest uplink TTI after the moment at which the TFC will have been in Excess-Power state for a duration of ($T_{notify}$ + $T_{modify}$+ $T_{L1_proc}$).

where:

$T_{notify}$ equals [15] ms, and

$T_{modify}$ equals MAX($T_{adapt_max}$,$T_{TTI}$), and

$T_{L1\ proc}$ equals 15 ms, and

$T_{adapt_max}$ equals MAX($T_{adapt_1}$, $T_{adapt_2}$, ..., $T_{adapt_N}$), and

N equals the number of logical channels that need to change rate, and

$T_{adapt_n}$ equals the time it takes for higher layers to provide data to MAC in a new supported bitrate, for logical channel n. Table 8.4.2.1.1A defines $T_{adapt}$ times for different services. For services where no codec is used $T_{adapt}$ shall be considered to be equal to 0 ms.

**Table 8.4.2.1.1A $T_{adapt}$ (1,28Mcps)**

| Service | $T_{adapt}$ [ms] |
|---|---|
| AMR | 40 |

$T_{TTI}$ equals the longest uplink TTI of the selected TFC (ms).

The Maximum UE transmitter power is defined as follows

Maximum UE transmitter power = MIN(Maximum allowed UL TX Power, UE maximum transmit power)

Where

Maximum allowed UL TX Power is set by SS and defined in TS 25.331 [9], and

UE maximum transmit power is defined by the UE power class, and specified in TS 25.102 [1].

The normative reference for these requirements is TS 25.123 [2] clauses 6A.2.2.2 and A.6.A2.2.2

#### 8.4.2.1.2.2 7,68 Mcps TDD option

The UE shall continuously evaluate based on the *Elimination*, *Recovery* and *Blocking* criteria defined below, how TFCs can be used for the purpose of TFC selection. The evaluation shall be performed using the estimated UE transmit power of a given CCTrCH in its associated timeslots.

In the case of a single CCTrCH or multiple CCTrCHs having mutually exclusive timeslot assignments, the UE shall consider the *Eliminiation* criterion for a given TFC of a CCTrCH to be fulfilled if for 3 successive frames the estimated UE transmit power is greater than the Maximum UE transmitter power for at least one timeslot associated with the CCTrCH in each frame.

In the case of multiple CCTrCHs not having mutually exclusive timeslot assignments, if for a given CCTrCH for 3 successive frames the estimated UE transmit power is greater than the Maximum UE transmitter power for at least one timeslot associated with the CCTrCH in each frame, the UE shall consider the *Elimination* criterion for a given TFC to be fulfilled if the use of this TFC will cause the estimated UE transmit power to continue to be greater than the Maximum UE transmitter power in at least one timeslot associated with the CCTrCH.

In the case of multi-frame operation of UL Physical Channels, the UE shall only consider active frames for the evaluation of the *Elimination* criterion.

If the *Elimination* criterion for a given TFC is fulfilled, the MAC in the UE shall consider that the TFC is in Excess-Power state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within $T_{notify}$ from the moment the *Elimination* criterion was fulfilled.

The UE shall not consider the *Recovery* criterion for a given TFC to be fulfilled until the use of this TFC will not cause the estimated UE transmit power to be greater than the Maximum UE transmitter power for all UL timeslots associated with the TFC for a minimum of 3 successive frames.

In the case of multi-frame operation of UL Physical Channels, the UE shall only consider active frames for the evaluation of the *Recovery* criterion.

If the *Recovery* criterion for a given TFC is fulfilled, the MAC in the UE shall consider that the TFC is in Supported state for the purpose of TFC selection.

MAC in the UE shall indicate the available bitrate for each logical channel to upper layers within $T_{notify}$ from the moment the *Recovery* criterion was fulfilled.

The UE shall consider the *Blocking* criterion for a given TFC to be fulfilled at the latest at the start of the longest uplink TTI after the moment at which the TFC will have been in Excess-Power state for a duration of

$$(T_{notify} + T_{modify} + T_{L1_proc}).$$

where:

$T_{notify}$ equals 15 ms, and

$T_{modify}$ equals $MAX(T_{adapt_max}, T_{TTI})$, and

$T_{L1\ proc}$ equals 35 ms, and

$T_{adapt_max}$ equals $MAX(T_{adapt_1}, T_{adapt_2}, ..., T_{adapt_N})$, and

N equals the number of logical channels that need to change rate, and

$T_{adapt_n}$ equals the time it takes for higher layers to provide data to MAC in a new supported bitrate, for logical channel n. Table 6A.1 defines $T_{adapt}$ times for different services. For services where no codec is used $T_{adapt}$ shall be considered to be equal to 0 ms.

**Table 8.4.2.1.1: $T_{adapt}$**

| Service | $T_{adapt}$ [ms] |
|---|---|
| UMTS AMR | 40 |
| UMTS AMR 2 | 60 |

$T_{TTI}$ equals the longest uplink TTI of the selected TFC (ms).

The Maximum UE transmitter power is defined as follows

Maximum UE transmitter power = MIN(Maximum allowed UL TX Power, UE maximum transmit power)

where

Maximum allowed UL TX Power is set by SS and defined in TS 25.331 [9], and

UE maximum transmit power is defined by the UE power class, and specified in TS 25.102 [1].

The normative reference for these requirements is TS 25.123 [2] clauses 6A.2 and A.6A.2.

### 8.4.2.1.3 Test purpose

#### 8.4.2.1.3.1 3,84 Mcps TDD option

The purpose is to verify the UE blocks (stops using) a currently used TFC when the UE output power is not sufficient to support that TFC. This test will verify the general requirement on TFC selection in section 8.4.2.1.2 for a RAB intended for packet data services, i.e. Interactive or Background, PS, UL: 64kbps as defined in TS 34.108 [3].

#### 8.4.2.1.3.1A 1,28 Mcps TDD option

The purpose is to verify the UE blocks (stops using) a currently used TFC when the UE output power is not sufficient to support that TFC. This test will verify the general requirement on TFC selection in section 8.4.2.1.2A for a RAB intended for packet data services, i.e. Interactive or Background, PS, UL: 64kbps as defined in TS 34.108 [3].

#### 8.4.2.1.3.2 7,68 Mcps TDD option

The purpose is to verify the UE blocks (stops using) a currently used TFC when the UE output power is not sufficient to support that TFC. This test will verify the general requirement on TFC selection in section 8.4.2.1.2 for a RAB intended for packet data services, i.e. Interactive or Background, PS, UL: 64kbps as defined in TS 34.108 [3].

### 8.4.2.1.4 Method of test

#### 8.4.2.1.4.1 Initial conditions

##### 8.4.2.1.4.1.1 3,84 Mcps TDD option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.4.2.1.2, 8.4.2.1.3, Table 8.4.2.1.4 and Table 8.4.2.1.5 below. The test consists of 2 successive time periods, with a time duration of T1 and T2 respectively.

Details on the UL reference RAB in table 8.4.2.1.3 can be found in TS 34.108 [3] section "Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH".

**Table 8.4.2.1.2: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| TFCS size | | 10 | |
| TFCS | | UL_TFC0, UL_TFC1, UL_TFC2, UL_TFC3, UL_TFC4, UL_TFC5, UL_TFC6, UL_TFC7, UL_TFC8, UL_TFC9 | Gain factors for TFC0 to TFC9 shall be set to 1. |
| Power Control | | On | |
| Active cell | | Cell 1 | |
| Maximum allowed UL TX power | dBm | 0 | Value of IE "Maximum allowed UL Tx power |
| Primary CCPCH Tx power | dBm | 18 | Value of IE "Primary CCPCH Tx power" |
| UL timeslot interference | dBm | -80 | Value of IE "UL timeslot interference"<br>This value shall apply to all timeslots |
| $\alpha$ | | 1 | IE "Alpha" either not sent or explicitly set to value |
| UL target SIR | dB | 6 | |
| DPCH constant offset | dB | adjustable | Value of IE "DPCH constant power |
| T1 | s | 10 | |
| T2 | s | 10 | |

**Table 8.4.2.1.3: Transport channel parameters for UL reference RAB, Interactive or Background and DCCH**

| Parameter | Unit | 64 kbps RAB | DCCH 3.4kbps |
|---|---|---|---|
| Transport Channel Number | | 1 | 2 |
| Transmission Time Interval | ms | 20 | 40 |
| Type of Error Protection | | Turbo coding | Convolutional coding |
| Coding Rate | | 1/3 | |
| Size of CRC | bits | 16 | |
| Transport Block Size | bits | 336 | 148 |
| Transport Block Set Size | bits | 336*B (B=0,1,2,3,4) | 148*B (B=0,1) |
| Transport Format Set | bits | | |
| TF0 | | 0x336 | 0x148 |
| TF1 | | 1x336 | 1x148 |
| TF2 | | 2x336 | N/A |
| TF3 | | 3x336 | N/A |
| TF4 | | 4x336 | N/A |

**Table 8.4.2.1.4: UL TFCI**

| TFCI | (64 kbps RAB, DCCH) |
|---|---|
| UL_TFC0 | (TF0, TF0) |
| UL_TFC1 | (TF0, TF1) |
| UL_TFC2 | (TF1, TF0) |
| UL_TFC3 | (TF1, TF1) |
| UL_TFC4 | (TF2, TF0) |
| UL_TFC5 | (TF2, TF1) |
| UL_TFC6 | (TF3, TF0) |
| UL_TFC7 | (TF3, TF1) |
| UL_TFC8 | (TF4, TF0) |
| UL_TFC9 | (TF4, TF1) |

**Table 8.4.2.1.5: Physical channel parameters**

| Parameter | Unit | Value |
|---|---|---|
| UL timeslot | | 7 |
| Burst type | | 1 |
| Resource units | | {(spreading factor 16 x 1 code) + (spreading factor 4 x 1 code)} x 1 time slot |
| TFCI | Bits | 16 |
| TPC | Bits | 2 |
| Frame allocation | | Continuous |

The test shall be performed in AWGN channel propagation conditions. The P-CCPCH in the DL shall be transmitted in timeslot 0.

The amount of available user data shall be sufficient to allow uplink transmission at the highest bit rate (UL_TFC8 or UL_TFC9) during the entire test and it shall be ensured that the UE is using UL_TFC8 or UL_TFC9 at the end of T1.

8.4.2.1.4.1.1A 1,28 Mcps TDD option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.4.2.1.5A , 8.4.2.1.5B and Table 8.4.2.1.5C below. The test consists of 2 successive time periods, with a time duration of T1 and T2 respectively.

Details on the UL reference RAB in table 8.4.2.1.3 can be found in TS 34.108 [3] section "Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH".

**Table 8.4.2.1.5A General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| TFCS size | | 10 | |
| TFCS | | UL_TFC0, UL_TFC1, UL_TFC2, UL_TFC3, UL_TFC4, UL_TFC5, UL_TFC6, UL_TFC7, UL_TFC8, UL_TFC9 | |
| Power Control | | On | |
| TPC step size | dB | 1 | |
| Maximum allowed UL TX power | dBm | 21 | |
| T1 | s | 30 | |
| T2 | s | 10 | |

**Table 8.4.2.1.5B: UL Reference RAB interactive or background**

| | TFI | 64 kbps RAB (20ms TTI) | DCCH 3.4kbps (40ms TTI) |
|---|---|---|---|
| TFS | TF0, bits | 0x336 | 0x148 |
| | TF1, bits | 1x336 | 1x148 |
| | TF2, bits | 2x336 | N/A |
| | TF3, bits | 3x336 | N/A |
| | TF4, bits | 4x336 | N/A |

**Table 8.4.2.1.5C: UL TFCI**

| TFCI | (64 kbps RAB, DCCH) |
|---|---|
| UL_TFC0 | (TF0, TF0) |
| UL_TFC1 | (TF0, TF1) |
| UL_TFC2 | (TF1, TF0) |
| UL_TFC3 | (TF1, TF1) |
| UL_TFC4 | (TF2, TF0) |
| UL_TFC5 | (TF2, TF1) |
| UL_TFC6 | (TF3, TF0) |
| UL_TFC7 | (TF3, TF1) |
| UL_TFC8 | (TF4, TF0) |
| UL_TFC9 | (TF4, TF1) |

The test shall be performed in AWGN channel propagation conditions.

The amount of available user data shall be sufficient to allow uplink transmission at the highest bit rate (UL_TFC8 or UL_TFC9) during the entire test and it shall be ensured that the UE is using UL_TFC8 or UL_TFC9 at the end of T1

8.4.2.1.4.1.2 7,68 Mcps TDD option

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.4.2.1.4.1.2.1, 8.4.2.1.4.1.2.2, 8.4.2.1.4.1.2.3 and 8.4.2.1.4.1.2.4 below. The test consists of 2 successive time periods, with a time duration of T1 and T2 respectively.

Details on the UL reference RAB in table 8.4.2.1.4.1.2.2 can be found in TS 34.108 [3] section "Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH".

**Table 8.4.2.1.4.1.2.1: General test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| TFCS size | | 10 | |
| TFCS | | UL_TFC0, UL_TFC1, UL_TFC2, UL_TFC3, UL_TFC4, UL_TFC5, UL_TFC6, UL_TFC7, UL_TFC8, UL_TFC9 | Gain factors for TFC0 to TFC9 shall be set to 1. |
| Power Control | | On | |
| Active cell | | Cell 1 | |
| Maximum allowed UL TX power | dBm | 0 | Value of IE "Maximum allowed UL Tx power |
| Primary CCPCH Tx power | dBm | 18 | Value of IE "Primary CCPCH Tx power" |
| UL timeslot interference | dBm | -80 | Value of IE "UL timeslot interference"<br>This value shall apply to all timeslots |
| $\alpha$ | | 1 | IE "Alpha" either not sent or explicitly set to value |
| UL target SIR | dB | 6 | |
| DPCH constant offset | dB | adjustable | Value of IE "DPCH constant power |
| T1 | s | 10 | |
| T2 | s | 10 | |

**Table 8.4.2.1.4.1.2.2: Transport channel parameters for UL reference RAB, Interactive or Background and DCCH**

| Parameter | Unit | 64 kbps RAB | DCCH 3.4kbps |
|---|---|---|---|
| Transport Channel Number | | 1 | 2 |
| Transmission Time Interval | ms | 20 | 40 |
| Type of Error Protection | | Turbo coding | Convolutional coding |
| Coding Rate | | 1/3 | |
| Size of CRC | bits | 16 | |
| Transport Block Size | bits | 336 | 148 |
| Transport Block Set Size | bits | 336*B (B=0,1,2,3,4) | 148*B (B=0,1) |
| Transport Format Set<br>TF0<br>TF1<br>TF2<br>TF3<br>TF4 | bits | <br>0x336<br>1x336<br>2x336<br>3x336<br>4x336 | <br>0x148<br>1x148<br>N/A<br>N/A<br>N/A |

**Table 8.4.2.1.4.1.2.3: UL TFCI**

| TFCI | (64 kbps RAB, DCCH) |
|---|---|
| UL_TFC0 | (TF0, TF0) |
| UL_TFC1 | (TF0, TF1) |
| UL_TFC2 | (TF1, TF0) |
| UL_TFC3 | (TF1, TF1) |
| UL_TFC4 | (TF2, TF0) |
| UL_TFC5 | (TF2, TF1) |
| UL_TFC6 | (TF3, TF0) |
| UL_TFC7 | (TF3, TF1) |
| UL_TFC8 | (TF4, TF0) |
| UL_TFC9 | (TF4, TF1) |

**Table 8.4.2.1.4.1.2.4: Physical channel parameters**

| Parameter | Unit | Value |
|---|---|---|
| UL timeslot | | 7 |
| Burst type | | 1 |
| Resource units | | {(spreading factor 32 x 1 code) + (spreading factor 8 x 1 code)} x 1 time slot |
| TFCI | Bits | 16 |
| TPC | Bits | 2 |
| Frame allocation | | Continuous |

The test shall be performed in AWGN channel propagation conditions. The P-CCPCH in the DL shall be transmitted in timeslot 0.

The amount of available user data shall be sufficient to allow uplink transmission at the highest bit rate (UL_TFC8 or UL_TFC9) during the entire test and it shall be ensured that the UE is using UL_TFC8 or UL_TFC9 at the end of T1.

8.4.2.1.4.2 Procedure

8.4.2.1.4.2.1 3,84 Mcps TDD option

1) The UE is switched on.

2) The SS shall signal to the UE the allowed TFCS according to table 8.4.2.1.2.

3) For T1=30 secs the SS shall ensure that the received P-CCPCH power level in the UE is set to –60dBm and that the value of the DPCH constant value is adjusted such that the mean UE output power is –10dBm.

4) The SS shall decrease the received P-CCPCH power level in the UE by 20 dB.

5) The time from the beginning of T2 until the UE blocks (stops using) UL_TFC8 and UL_TFC9 shall be measured by the SS. The UE shall stop using UL_TFC8 and UL_TFC9 within 170 ms from beginning of time period T2.

6) Repeat steps 3-5 [50] times.

8.4.2.1.4.2.1A 1,28 Mcps TDD option

1. The UE is switched on.

2. The SS shall signal to the UE the allowed TFCS according to table above.

3. For T1=30 secs the SS shall ensure that the received P-CCPCH power level in the UE is set to –60dBm and that the value of the DPCH constant value is adjusted such that the mean UE output power is 10dB below the UE Maximum allowed UL TX power..

4. The system simulator shall continously send TPC_cmd=Up to the UE from the beginning of T2 until the end of T2.

NOTE: This will guarantee that UL_TFC8 to UL_TFC9 can not be supported beacuse the UE reaches the maximum UL Tx power and the UTRAN SS continues sending power-up commands.

1. The time from the beginning of T2 until the UE blocks (stops using) UL_TFC8 and UL_TFC9 shall be measured by the SS. The UE shall stop using UL_TFC8 and UL_TFC9 within [250] ms from beginning of time period T2.

2. Repeat steps 3-5 until the confidence level according to annex F.6.2 is achieved.

NOTE: The delay from the begining of T2 can be expressed as: $T_{ramp} + T_{detect_block} + T_{notify} + T_{modify} + T_{L1_proc} + T_{align_TTI}$, where:

$T_{ramp}$ Margin added for the increase of UE output power to the UE maximum power. A margin of 7 frames (70ms) is used, i.e. 14 TPC commands.

$T_{detect_block}$ The time needed to detect that UL_TFC8 and UL_TFC9 can no longer be supported, i.e. defines the maximum time to detect that the *Limited TFC Set* criterion is fulfilled for UL_TFC8 and UL_TFC9. This figure is currently TBD as X and Y in the general requirement, see section 6.4.2 of TS25.123 [2], are not finalised yet.

$T_{notify}$ Equal to [15] ms, the time allowed for MAC to indicate to higher layers that UL_TFC8 and UL_TFC9 can no longer be supported.

$T_{modify}$ Equal to $MAX(T_{adapt_max}, T_{TTI}) = MAX(0, 40)=40ms$

$T_{adapt_max}$ Equals to 0ms for the case without codec.

$T_{L1_proc}$ Equals 15ms.

$T_{align_TTI}$ Align with the longest uplink TTI where the new TFC can be selected. The worst case equals 40ms in this test case.

$T_{TTI}$ Equals 40 ms in the test case.

This gives a maximum delay of $(70 + T_{detect_block} + [15] + 40 + 15 + 40)$ ms from the beginning of T2.

8.4.2.1.4.2.2 7.68 Mcps TDD option

1) The UE is switched on.

2) The SS shall signal to the UE the allowed TFCS according to table 8.4.2.1.4.1.2.1.

3) For T1=30 secs the SS shall ensure that the received P-CCPCH power level in the UE is set to –60dBm and that the value of the DPCH constant value is adjusted such that the mean UE output power is –10dBm.

4) The SS shall decrease the received P-CCPCH power level in the UE by 20 dB.

5) The time from the beginning of T2 until the UE blocks (stops using) UL_TFC8 and UL_TFC9 shall be measured by the SS. The UE shall stop using UL_TFC8 and UL_TFC9 within 170 ms from beginning of time period T2.

6) Repeat steps 3-5 [50] times.

#### 8.4.2.1.5 Test requirements

##### 8.4.2.1.5.1 3,84 Mcps TDD option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

##### 8.4.2.1.5.1A 1,28 Mcps TDD option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.

##### 8.4.2.1.5.2 7,68 Mcps TDD option

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

Note: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

# 8.5 Timing Characteristics

## 8.5.1 UE Timing Advance

### 8.5.1.1 3,84 Mcps TDD Option

#### 8.5.1.1.1 Definition and applicability

Timing advance is the correction to UE transmit timing required in order to avoid large delay spread at the Node B. The timing advance value is provided to the UE by UTRAN.

The requirements and this test apply to the TDD (3,84 Mcps option) UE.

#### 8.5.1.1.2 Minimum requirement

The UE shall adjust the timing of its transmissions with an accuracy better than or equal to ±0.5 chip to the signalled timing advance value.

The normative reference for this requirement is TS 25.123 [2] clauses 7.1.1. and A.7.1.1

#### 8.5.1.1.3. Test purpose

To verify that the UE meets the minimum requirement.

#### 8.5.1.1.4 Method of test

##### 8.5.1.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.5.1.1 and table 8.5.1.2. The test consists of two successive time periods, with a time duration of T1and T2 respectively. At the start of time duration T1, the UE shall transmit with the Uplink Timing Advance value set to zero, i.e. Timing Advance disabled.

During time period T1, UTRAN shall send an Uplink Physical Channel control message with activation time at the beginning of T2. The Uplink Physical Channel Control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T2 is greater than or equal to the RRC procedure delay as defined in [9].

**Table 8.5.1.1: General test parameters for Timing Advance test**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Timing Advance value | | 0 | IE "Uplink timing advance control" value disabled. |
| Final condition | Timing Advance value | | 5 | IE "Uplink timing advance" value set to 5. |
| Monitored cell list size | | | 6 TDD neighbors on Channel 1 | |
| $T_{SI}$ | | S | 1,28 | The value shall be used for all cells in the test. |
| T1 | | S | 5 | |
| T2 | | S | 5 | |

**Table 8.5.1.2: Cell specific test parameters for Timing Advance test**

| Parameter | Unit | Cell 1 | | | |
|---|---|---|---|---|---|
| **DL timeslot number** | | **0** | | **2** | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | n.a. | |
| SCH_Ec/Ior | dB | -9 | | n.a. | |
| SCH_$t_{offset}$ | dB | 0 | | n.a. | |
| DPCH_Ec/Ior | dB | n.a. | | Note 1 | |
| OCNS_Ec/Ior | dB | -3,12 | | Note 2 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | | |
| Propagation Condition | | AWGN | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | |

8.5.1.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) At the start of time interval T1, the SS shall transmit an UPLINK PHYSICAL CHANNEL CONTROL message with timing advance disabled

5) During the interval T1, the SS shall transmit an UPLINK PHYSICAL CHANNEL CONTROL message with timing advance enabled and the timing advance value set to 5.

6) UE shall apply the signalled timing advance value.

7) After 10 seconds, the UE is switched off.

8) Repeat Step 1-7 [TBD] times

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

UPLINK PHYSICAL CHANNEL CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **PhyCH Information elements** | |
| -CCTrCH Power Control Info | Not Present |
| -Choice *TDD Option* | 3,84 Mcps TDD |
| -Alpha | Not Present |
| -Special Burst Scheduling | Not Present |
| -Timing Advance Control (10.3.6.96) | |
| -Choice *Timing Advance* | Disabled |
| -PRACH Constant Value | Not Present |
| -PUSCH Constant Value | Not Present |
| -UE positioning related parameters | Not Present |

UPLINK PHYSICAL CHANNEL CONTROL message (step 5):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **PhyCH Information elements** | |
| -CCTrCH Power Control Info | Not Present |
| -Choice *TDD Option* | 3,84 Mcps TDD |
| -Alpha | Not Present |
| -Special Burst Scheduling | Not Present |
| -Timing Advance Control (10.3.6.96) | |
| -Choice *Timing Advance* | Enabled |
| -Choice *TDD Option* | 3,84 Mcps TDD |
| -UL Timing Advance (10.3.6.9.95) | 5 |
| -Activation Time | At T2 |
| -PRACH Constant Value | Not Present |
| -PUSCH Constant Value | Not Present |
| -UE positioning related parameters | Not Present |

#### 8.5.1.1.5 Test requirements

The UE shall apply the signalled Timing Advance value to the UL DPCH transmission timing at the designated activation time, i.e the beginning of time period T2. The Timing Advance adjustement accuracy shall be within ±0.5 chip.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

### 8.5.1.2 1,28 Mcps TDD Option

#### 8.5.1.2.1 Definition and applicability

Timing advance is applied to adjust the UE transmit time in order to ensure that all the signal received by the BS is synchronised. The timing advance value is provided to the UE by UTRAN.

The requirements and this test apply to the TDD (1.28 Mcps TDD option) UE.

#### 8.5.1.2.2 Minimum requirement

The UE transmitter shall have the capability of changing the transmission timing with a step size of 1/8, 2/8, 3/8, …, 1 chip according to the value of $\Delta_{SS}$, within n=(1,2,…,6) time slots excluding special timeslots (DwPTS, GP, UpPTS) after the SS_cmd arrived (closed loop). For the open loop any step being a multiple of 1/8 chip has to be allowed.

The minimum transmission timing step $\Delta_{SS,min}$ due to closed loop uplink synchronization control shall be within the range shown in Table 8.5.1.1A.

**Table 8.5.1.1A: Uplink synchronisation control range**

| SS_cmd | Uplink synchronisation control range for minimum step | |
|---|---|---|
| | 1/8 chip step size | |
| | Lower | Upper |
| Up | 1/9 chip | 1/7 chip |
| Down | 1/9 chip | 1/7 chip |

#### 8.5.1.2.3. Test purpose

The purpose of this test is to verify the ability of the UE to adjust its timing advance according to the SS commands within the specified accuracy defined in TS 25.123 [2] section 7.1.2.

#### 8.5.1.2.4 Method of test

##### 8.5.1.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.5.1.1A and table 8.5.1.2A.

**Table 8.5.1.2A: General test parameters for Timing Advance test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Monitored cell list size | | 6 TDD neighbours on Channel 1 | |

**Table 8.5.1.3A: Cell specific test parameters for Timing Advance test**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| Timeslot Number | | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | |
| DwPCH_Ec/Ior | dB | | 0 |
| OCNS_Ec/Ior | dB | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 6 |
| $I_{oc}$ | dBm/ 1.28 MHz | -80 | |
| PCCPCH RSCP | dBm | -77 | |
| Propagation Condition | | AWGN | |

##### 8.5.1.2.4.2 Procedure

1) The RF parameters are set up according to table 8.5.1.2A..
2) Connect the SS to the UE antenna connector as shown in figure A.1.
3) A call is set up according to the Generic call setup procedure.

4) Enter the UE into loopback test mode and start the loopback test.
See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure and loopback test.

5) Configure the uplink channel to set UE transmitter to enable synchronisation steps of size 1/8 chip.

6) Send a synchronisation shift (SS) commands with the value “00”(Down )to the UE under test

7) Send a synchronisation shift (SS) commands with the value “11”(Up) to the UE under test

8) Repeat steps 6-7 until [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

#### 8.5.1.2.5 Test requirements

In step 6 and step7, The UE transmitter timing due to closed loop uplink synchronization control shall be within the range shown in 8.5.1.4A.

**Table 8.5.1.4A: Uplink synchronisation control range**

| SS_cmd | **Lower** | **Upper** |
|---|---|---|
| Up | 1/9 chip-[TBD]ns | 1/7 chip+[TBD]ns |
| Down | 1/9 chip-[TBD]ns | 1/7 chip+[TBD]ns |

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

### 8.5.1.3 7,68 Mcps TDD Option

#### 8.5.1.3.1 Definition and applicability

Timing advance is the correction to UE transmit timing required in order to avoid large delay spread at the Node B. The timing advance value is provided to the UE by UTRAN.

The requirements and this test apply to the TDD (7,68 Mcps option) UE.

#### 8.5.1.3.2 Minimum requirement

The UE shall adjust the timing of its transmissions with an accuracy better than or equal to ±0.5 chip to the signalled timing advance value.

The normative reference for this requirement is TS 25.123 [2] clauses 7.1.3. and A.7.1.3

#### 8.5.1.3.3. Test purpose

To verify that the UE meets the minimum requirement.

#### 8.5.1.3.4 Method of test

##### 8.5.1.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in table 8.5.1.3.4.1.1 and table 8.5.1.3.4.1.2. The test consists of two successive time periods, with a time duration of T1and T2 respectively. At the start of time duration T1, the UE shall transmit with the Uplink Timing Advance value set to zero, i.e. Timing Advance disabled.

During time period T1, UTRAN shall send an Uplink Physical Channel control message with activation time at the beginning of T2. The Uplink Physical Channel Control message shall be sent to the UE such that the delay between the

end of the last received TTI containing the message and the beginning of T2 is greater than or equal to the RRC procedure delay as defined in [9].

**Table 8.5.1.3.4.1.1: General test parameters for Timing Advance test**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Timing Advance value | | 0 | IE "Uplink timing advance" value zero or IE "Uplink timing advance control" value disabled. |
| Final condition | Timing Advance value | | 5 | IE "Uplink timing advance" value set to 5. |
| Monitored cell list size | | | 6 TDD neighbors on Channel 1 | |
| $T_{SI}$ | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 5 | |
| T2 | | s | 5 | |

**Table 8.5.1.3.4.1.2: Cell specific test parameters for Timing Advance test**

| Parameter | Unit | Cell 1 | | | |
|---|---|---|---|---|---|
| DL timeslot number | | 0 | | 2 | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | n.a. | |
| SCH_Ec/Ior | dB | -9 | | n.a. | |
| SCH_$t_{offset}$ | dB | 0 | | n.a. | |
| DPCH_Ec/Ior | dB | n.a. | | Note 1 | |
| OCNS_Ec/Ior | dB | -3,12 | | Note 2 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | |
| $I_{oc}$ | dBm/ 7,68 MHz | -70 | | | |
| Propagation Condition | | AWGN | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior . | | | | | |

8.5.1.3.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) At the start of time interval T1, the SS shall transmit an UPLINK PHYSICAL CHANNEL CONTROL message with timing advance disabled

5) During the interval T1, the SS shall transmit an UPLINK PHYSICAL CHANNEL CONTROL message with timing advance enabled and the timing advance value set to 5.

6) UE shall apply the signalled timing advance value.

9) After 10 seconds, the UE is switched off.

10) Repeat Step 1-7 until the confidence level according to annex F.6.2 is achieved.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

UPLINK PHYSICAL CHANNEL CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **PhyCH Information elements** | |
| -CCTrCH Power Control Info | Not Present |
| -Choice *TDD Option* | 7,68 Mcps TDD |
| -Alpha | Not Present |
| -Special Burst Scheduling | Not Present |
| -Timing Advance Control (10.3.6.96) | |
| -Choice *Timing Advance* | Disabled |
| -PRACH Constant Value | Not Present |
| -PUSCH Constant Value | Not Present |
| -UE positioning related parameters | Not Present |
| - HS-SICH power control info | Not Present |

UPLINK PHYSICAL CHANNEL CONTROL message (step 5):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **PhyCH Information elements** | |
| -CCTrCH Power Control Info | Not Present |
| -Choice *TDD Option* | 7,68 Mcps TDD |
| -Alpha | Not Present |
| -Special Burst Scheduling | Not Present |
| -Timing Advance Control (10.3.6.96) | |
| -Choice *Timing Advance* | Enabled |
| -Choice *TDD Option* | 7,68 Mcps TDD |
| -Extended UL Timing Advance (10.3.6.9.95a) | 10 |
| -Activation Time | At T2 |
| -PRACH Constant Value | Not Present |
| -PUSCH Constant Value | Not Present |
| -UE positioning related parameters | Not Present |

#### 8.5.1.3.5 Test requirements

The UE shall apply the signalled Timing Advance value to the UL DPCH transmission timing at the designated activation time, i.e the beginning of time period T2. The Timing Advance adjustement accuracy shall be within ±0.5 chip.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

## 8.5.2 UE Transmit Timing

Void.

# 8.6 UE Measurements Procedures

## 8.6.1 TDD intra frequency measurements

### 8.6.1.1 Event 1G triggered reporting in AWGN propagation conditions

#### 8.6.1.1.1 Definition and applicability

##### 8.6.1.1.1.1 3,84 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (3,84 Mcps option) UE.

##### 8.6.1.1.1.2 1,28 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the 1.28 Mcps option TDD UE.

##### 8.6.1.1.1.3 7,68 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (7,68 Mcps option) UE.

#### 8.6.1.1.2 Minimum requirement

##### 8.6.1.1.2.1 3,84 Mcps TDD Option

The UE shall be able to identify a new detectable intra-frequency TDD cell belonging to the monitored set within $T_{identify\ intra}$ ms, where $T_{identify\ intra}$ = 800 ms.

When L3 filtering is used, an additional delay can be expected.

In CELL_DCH state, the UE shall be capable of performing P-CCPCH RSCP measurements for $X_{measurement\ intra}$ identified intra-frequency TDD cells of the monitored set with a measurement period for intra-frequency P-CCPCH RSCP measurements $T_{measurement\ period\ intra,}$ where

$X_{measurement\ intra}$ = 6 (cells)

$T_{measurement\ period\ intra}$ = 200 ms

The UE physical layer shall be capable of reporting these measurements to higher layers with the measurement period $T_{measurement\ period\ intra}$.

If the UE has identified more than $X_{measurement\ intra}$ intra-frequency TDD cells, the UE shall perform measurements of all identified cells but the reporting rate of P-CCPCH RSCP measurements of cells from the UE physical layer to higher layers may be decreased.. The measurement accuracy for all measured cells shall be as specified in the section 9.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1.2.2.1, 8.1.2.2.2 and A.8.1.1

#### 8.6.1.1.2.2 1,28 Mcps TDD Option

The UE shall be able to identify a new intra frequency TDD cell belonging to the monitored set within

$$T_{identify\ intra} = T_{basic\ identify\ TDD,\ intra} \cdot \frac{N_{Period,\ Intra}}{N_{Intra}} \text{ ms}$$

$T_{basic_identify_TDD,\ intra}$ = 800 ms. This is the time period used in the intra frequency equation where the maximum allowed time for the UE to identify a new TDD cell is defined.

$N_{Period,Intra:}$ = 40 Number of subframes in $T_{Measurement_Period,\ Intra.}$

$N_{Intra}$ : This is the minimum number of sub-frame in that the period of TS0, DwPTS and main guard period is available for intra frequency measurements, during the measurement period.

$T_{Measurement_Period,\ Intra}$ = 200 ms. The measurement period for Intra frequency P-CCPCH RSCP measurements.

If a cell belonging to monitored set has been detectable at least for the time period $T_{identify_intra}$ and then enters or leaves the reporting range, the event triggered measurement reporting delay shall be less than 200ms when the L3 filter has not been used and the UE P-CCPCH measurement capabilities of Section 8.1A.2.2.2 in [2] are valid.

The event triggered measurement reporting delay on cells not belonging to monitored set, measured without L3 filtering, shall be less than the above defined $T_{identify\ detected\ set}$.

#### 8.6.1.1.2.3 7,68 Mcps TDD Option

The UE shall be able to identify a new detectable intra-frequency TDD cell belonging to the monitored set within $T_{identify\ intra}$ ms, where $T_{identify\ intra}$ = 800 ms.

When L3 filtering is used, an additional delay can be expected.

In CELL_DCH state, the UE shall be capable of performing P-CCPCH RSCP measurements for $X_{measurement\ intra}$ identified intra-frequency TDD cells of the monitored set with a measurement period for intra-frequency P-CCPCH RSCP measurements $T_{measurement\ period\ intra,}$ where

$X_{measurement\ intra}$ = 6 (cells)

$T_{measurement\ period\ intra}$ = 200 ms

The UE physical layer shall be capable of reporting these measurements to higher layers with the measurement period $T_{measurement\ period\ intra}$.

If the UE has identified more than $X_{measurement\ intra}$ intra-frequency TDD cells, the UE shall perform measurements of all identified cells but the reporting rate of P-CCPCH RSCP measurements of cells from the UE physical layer to higher layers may be decreased.. The measurement accuracy for all measured cells shall be as specified in the section 9.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1B.2.2.1, 8.1B.2.2.2 and A.8.1.1

### 8.6.1.1.3 Test purpose

#### 8.6.1.1.3.1 3,84 Mcps TDD Option

To verify that the UE meets the minimum requirement.

#### 8.6.1.1.3.2 1,28 Mcps TDD Option

To verify the UE meets the minimum requirements.

#### 8.6.1.1.3.3 7,68 Mcps TDD Option

To verify the UE meets the minimum requirements.

#### 8.6.1.1.4 Method of test

##### 8.6.1.1.4.1 3,84 Mcps TDD Option

###### 8.6.1.1.4.1.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.1.1.1 and 8.6.1.1.2. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. Three cells shall be present in the test, cell 1 being the serving cell and cell 2 and cell 3 being neighbour cells on the used frequency. All cells shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP shall be reported together with Event 1G. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The DL DPCH shall be transmitted in timeslot 2 and the UL DPCH shall be transmitted in timeslot 10. The TTI of the uplink DCCH shall be 20ms.

**Table 8.6.1.1.1: General test parameters for Event 1G triggered reporting in AWGN propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2, Cell 3 | |
| Final condition | Active cell | | Cell 1 | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Threshold used frequency | | dBm | -70 | Applicable for Event 1G |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 12 TDD neighbours on Channel 1 | |
| T1 | | s | 6 | |
| T2 | | s | 6 | |
| T3 | | s | 6 | |

**Table 8.6.1.1.2: Cell specific parameters for Event 1G triggered correct reporting in AWGN propagation condition**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| **DL timeslot number** | | **0** | | | **0** | | | **0** | | |
| UTRA RF Channel Number | | Channel 1 | | | Channel 1 | | | Channel 1 | | |
| PCCPCH_Ec/Ior | dB | -3 | | | -3 | | | -3 | | |
| SCH_Ec/Ior | dB | -9 | | | -9 | | | -9 | | |
| SCH_$t_{offset}$ | | 0 | | | 5 | | | 10 | | |
| OCNS_Ec/Ior | dB | -3,12 | | | -3,12 | | | -3,12 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | 5 | | 5 | 7 | -Inf | -Inf | | 7 |
| PCCPCH RSCP | dBm | -66 | -68 | | -68 | -66 | -Inf | -Inf | | -66 |
| $I_{oc}$ | dBm / 3,84 MHz | -70 | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | |

8.6.1.1.4.1.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 6 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 240 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 6 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 3. The measurement reporting delay from the beginning of T3 shall be less than 840 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

9) After 6 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

10) Repeat steps 1-9 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |

### 8.6.1.1.4.2 1,28 Mcps TDD Option

Test environment : normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested : mid range ;see clauses G.2.4.

The test parameters are given in Table 8.6.1.1.4A and 8.6.1.1.4B below. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. Three cells shall be present in the test, cell 1 being the serving cell and cell 2 and cell 3 being neighbour cells on the used frequency. All cells shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP shall be reported together with Event 1G reporting. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9]. The cell specific test parameters are given in Table 8.6.1.1B below.

The TTI of the uplink DCCH shall be 40ms.

**Table 8.6.1.1.4A: General test parameters for Event 1G triggered reporting in AWGN propagation condition 1,28Mcps option**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DPCH parameters active cell | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A. The DPCH is located in an other timeslot than 0 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2, Cell 3 | |
| O | DB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | DB | 0 | |
| Time to Trigger | Ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1 | |
| T1 | S | 6 | |
| T2 | S | 6 | |
| T3 | S | 6 | |

**Table 8.6.1.1.4B: Cell specific parameters for Event 1G triggered correct reporting in AWGN propagation condition 1,28 Mcps option**

| Parameter | Unit | Cell 1 | | | | | | Cell 2 | | | | | | Cell 3 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | | | DwPTS | | | 0 | | | DwPTS | | | 0 | | | DwPTS | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 1 | | | | | | Channel 1 | | | | | | Channel 1 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | -3 | | | | | | -3 | | | | | |
| DwPCH_Ec/Ior | dB | | | | 0 | | | | | | 0 | | | | | | 0 | | |
| OCNS_Ec/Ior | dB | -3 | | | | | | -3 | | | | | | -3 | | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 6 | 4 | | 6 | 4 | | 4 | 6 | -Inf | 4 | 6 | -Inf | -Inf | | 6 | -Inf | | 6 |
| PCCPCH RSCP | dBm | -67 | -69 | | | | | -69 | -67 | -Inf | | | | -Inf | | -67 | | | |
| $I_{oc}$ | dBm / 1,28 MHz | -70 | | | | | | | | | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | | | | | | | | | |

NOTE: The DPCH of all cells are located in a timeslot other than 0.

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 6 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 280 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

7) After 6 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 3. The measurement reporting delay from the beginning of T3 shall be less than 880 ms. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

9) After 6 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

10) Repeat steps 1-9 until the confidence level according to annex F.6.2 is achieved.

### Specific Message Contents

All messages indicated above shall use the default message content in clause 9 of 34.108 [3], with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 1.28 Mcps TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 1.28 Mcps TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |

8.6.1.1.4.3 7,68 Mcps TDD Option

8.6.1.1.4.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.1.1.4.3.1.1 and 8.6.1.1.4.3.1.2. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. Three cells shall be present in the test, cell 1 being the serving cell and cell 2 and cell 3 being neighbour cells on the used frequency. All cells shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP shall be reported together with Event 1G. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The DL DPCH shall be transmitted in timeslot 2 and the UL DPCH shall be transmitted in timeslot 10. The TTI of the uplink DCCH shall be 20ms.

**Table 8.6.1.1.4.3.1.1: General test parameters for Event 1G triggered reporting in AWGN propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2, Cell 3 | |
| Final condition | Active cell | | Cell 1 | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Threshold used frequency | | dBm | -70 | Applicable for Event 1G |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 12 TDD neighbours on Channel 1 | |
| T1 | | s | 6 | |
| T2 | | s | 6 | |
| T3 | | s | 6 | |

**Table 8.6.1.1.4.3.1.2: Cell specific parameters for Event 1G triggered correct reporting in AWGN propagation condition**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| **DL timeslot number** | | **0** | | | **0** | | | **0** | | |
| UTRA RF Channel Number | | Channel 1 | | | Channel 1 | | | Channel 1 | | |
| PCCPCH_Ec/Ior | dB | -3 | | | -3 | | | -3 | | |
| SCH_Ec/Ior | dB | -9 | | | -9 | | | -9 | | |
| SCH_$t_{offset}$ | | 0 | | | 5 | | | 10 | | |
| OCNS_Ec/Ior | dB | -3,12 | | | -3,12 | | | -3,12 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | 5 | | 5 | 7 | -Inf | -Inf | | 7 |
| PCCPCH RSCP | dBm | -66 | -68 | | -68 | -66 | -Inf | -Inf | | -66 |
| $I_{oc}$ | dBm / 7,68 MHz | -70 | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | |

#### 8.6.1.1.4.3.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 6 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 240 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
7) After 6 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.
8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1G for cell 3. The measurement reporting delay from the beginning of T3 shall be less than 840 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
9) After 6 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
10) Repeat steps 1-9 until the confidence level according to annex F.6.2 is achieved.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | 2 |
| -Timeslot | 0 |
| -Cell parameters ID | 0 |
| -SCTD indicator | FALSE |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |

| Information Element/Group name | Value/Remark |
|---|---|
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |
| -SCTD indicator | FALSE |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1G |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not Present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 3 |
| -SCTD indicator | FALSE |

### 8.6.1.1.5 Test requirements

#### 8.6.1.1.5.1 3,84 Mcps TDD Option

The UE shall send one Event 1G triggered measurement report for Cell 2 with a measurement reporting delay less than 240ms from the beginning of time period T2.

The UE shall send one Event 1G triggered measurement report for Cell 3 with a measurement reporting delay less than 840ms from the beginning of time period T3.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

#### 8.6.1.1.5.2 1,28 Mcps TDD Option

The UE shall send one Event 1G triggered measurement report for Cell 2 with a measurement reporting delay less than 280ms from the beginning of time period T2.

The UE shall send one Event 1G triggered measurement report for Cell 3 with a measurement reporting delay less than 880ms from the beginning of time period T3.

The UE shall not send any incorrect event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

#### 8.6.1.1.5.3 7,68 Mcps TDD Option

The UE shall send one Event 1G triggered measurement report for Cell 2 with a measurement reporting delay less than 240ms from the beginning of time period T2.

The UE shall send one Event 1G triggered measurement report for Cell 3 with a measurement reporting delay less than 840ms from the beginning of time period T3.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

### 8.6.1.2 Event 1H and 1I triggered reporting in AWGN propagation condition

#### 8.6.1.2.1 Definition and applicability

##### 8.6.1.2.1.1 3,84 Mcps TDD Option

In the event triggered reporting, the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (3,84 Mcps option) UE.

##### 8.6.1.2.1.2 1,28 Mcps TDD Option

The purpose of this test is to verify that the UE makes correct reporting of events 1H (timeslot ISCP below threshold) and 1I (timeslot ISCP above threshold). This test will partly verify the requirements in [2]

##### 8.6.1.2.1.3 7,68 Mcps TDD Option

In the event triggered reporting, the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (7,68 Mcps option) UE.

#### 8.6.1.2.2 Minimum requirement

##### 8.6.1.2.2.1 3,84 Mcps TDD Option

In CELL_DCH state the measurement period for intra frequency Timeslot ISCP measurements on arbitrary DL timeslots, including Beacon timeslots is 400 ms. When no inter frequency measurement is scheduled, the UE shall be capable of performing Timeslot ISCP measurements for a total of 10 different combinations of an arbitrary DL timeslot and an intra-frequency cell, including the current serving cell.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1.2.2.2.A and A.8.1.2

##### 8.6.1.2.2.2 1,28 Mcps TDD Option

The UE shall send one event 1I triggered measurement report, with a measurement reporting delay less than 400 ms from the beginning of time period T2 as defined in the test method below.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 400 ms from the beginning of time period T3 as defined in the test method below.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 400 ms from the beginning of time period T4 as defined in the test method below.

##### 8.6.1.2.2.3 7,68 Mcps TDD Option

In CELL_DCH state the measurement period for intra frequency Timeslot ISCP measurements on arbitrary DL timeslots, including Beacon timeslots is 400 ms. When no inter frequency measurement is scheduled, the UE shall be capable of performing Timeslot ISCP measurements for a total of 10 different combinations of an arbitrary DL timeslot and an intra-frequency cell, including the current serving cell.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1B.2.2.3 and A.8.1.2

#### 8.6.1.2.3 Test purpose

##### 8.6.1.2.3.1 3,84 Mcps TDD Option

To verify that the UE meets the minimum requirement.

##### 8.6.1.2.3.2 1,28 Mcps TDD Option

To verify that the UE meets the minimum requirements.

##### 8.6.1.2.3.3 7,68 Mcps TDD Option

To verify that the UE meets the minimum requirements.

#### 8.6.1.2.4 Method of test

##### 8.6.1.2.4.1 3,84 Mcps TDD Option

###### 8.6.1.2.4.1.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.6.1.2.1, 8.6.1.2.2 and 8.6.1.2.3. The test consists of five successive time periods, with a time duration of T1, T2, T3, T4 and T5 respectively. Two cells shall be present in the test, cell 1 being the current serving cell and cell 2 being a neighbour cell on the used frequency.

In the measurement control information it shall be indicated to the UE that event-triggered reporting with event 1H and event 1I shall be used and that Timeslot ISCP and P-CCPCH RSCP shall be reported together with event 1H and 1I. Measurement control information shall be sent to the UE before the beginning of time period T1.

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The UL DPCH shall be transmitted in timeslot 10. In addition, timeslots 3 and 4 shall be allocated as DL timeslots. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.6.1.2.1: General test parameters for correct event 1H and event 1I reporting in AWGN propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 1 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Timeslot list cell 1 | | | 2, 3, 4 | Timeslot numbers in IE "Cell info" for Cell 1 |
| Timeslot list cell 2 | | | 4 | Timeslot numbers in IE "Cell info" for Cell 2 |
| Threshold used frequency | | dBm | -68 | Threshold 1 applicable for event 1H, cell 1 timeslots 2, 4 and cell 2 timeslot 4 |
| Threshold used frequency | | dBm | -73 | Threshold 2 applicable for event 1H, cell 1 timeslots 2, 3, 4 and cell 2 timeslot 4 |
| Threshold used frequency | | dBm | -67 | Applicable for event 1I, cell 1 timeslots 2, 4 and cell 2 timeslot 4 |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | Cell 2 shall belong to the monitored set |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 5 | |
| T4 | | s | 5 | |
| T5 | | s | 5 | |

**Table 8.6.1.2.2: Cell 1 specific parameters for correct event 1H and 1I reporting in AWGN propagation condition**

| Parameter | Unit | Cell 1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | | |
| **DL timeslot number** | | **0** | | | | | **2** | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | -9 | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | 5 | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | Note 1 | | | | |
| OCNS_Ec/Ior | dB | -3,12 | | | | | Note 2 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 4 | | | | | 4 | | | | |
| PCCPCH RSCP | dBm | -69 | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 3,84 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| **DL timeslot number** | | **3** | | | | | **4** | | | | |
| PCCPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | n.a. | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | 0 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | | | 0 | | | | 6 |
| PCCPCH RSCP | dBm | n.a. | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 3,84 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior | | | | | | | | | | | |

**Table 8.6.1.2.3: Cell 2 specific parameters for correct event 1H and 1I reporting in AWGN propagation condition**

| Parameter | Unit | Cell 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | | |
| **DL timeslot number** | | **0** | | | | | **2** | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | -9 | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | 10 | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | -3,12 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | 0 | 6 | 0 | | |
| PCCPCH RSCP | dBm | -72 | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 3,84 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| **DL timeslot number** | | **3** | | | | | **4** | | | | |
| PCCPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | n.a. | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | 0 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | | | 6 | | | | 0 |
| PCCPCH RSCP | dBm | n.a. | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 3,84 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |

8.6.1.2.4.1.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 5 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1I for cell 1 in timeslot 2. The measurement reporting delay from the beginning of T2 shall be less than 480 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 5 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 1 in timeslot 2. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

9) After 5 seconds from the beginning of T3, the SS shall switch the power settings from T3 to T4.

10) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 1 in timeslot 4. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

11) After 5 seconds from the beginning of T4, the SS shall switch the power settings from T4 to T5.

12) UE shall transmit a MEASUREMENT REPORT message triggered by event 1I for cell 2 in timeslot 4. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

13) After 5 seconds from the beginning of T5, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

14) Repeat steps 1-13 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 2 |
| -Intra-frequency event identity | Event 1H |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -68 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| -Intra-frequency event identity | Event 1H |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -73 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| -Intra-frequency event identity | Event 1I |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -67 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1I |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |

MEASUREMENT REPORT message (step 10)

| **Information Element** | **Value/remark** |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |

MEASUREMENT REPORT message (step 12)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 1 |
| -Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1I |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 2 |

### 8.6.1.2.4.2 1,28 Mcps TDD Option

Test environment : normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested : mid range ;see clauses G.2.4.

The test parameters are given in Table 8.6.1.2.4.2A, Table 8.6.1.2.4.2B and Table 8.6.1.2.4.2C below. The test consists of four successive time periods, with a time duration of T1, T2, T3 and T4 respectively. Two cells shall be present in the test, cell 1 being the current serving cell and cell 2 being a neighbour cell on the used frequency.

In the measurement control information it shall be indicated to the UE that event-triggered reporting with event 1H and event 1I shall be used and that Timeslot ISCP and P-CCPCH RSCP shall be reported together with event 1H and 1I. Measurement control information shall be sent to the UE before the beginning of time period T1.

The UL DPCH shall be transmitted in timeslot 2. In addition, timeslots 5 and 6 shall be allocated as DL timeslots.

**Table 8.6.1.2.4.2A: General test parameters for correct event 1H and 1I reporting in AWGN propagation condition 1,28Mcps option**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters active cell | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A. |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 1 | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Timeslot list cell 1 | | | 5, 6 | Timeslot numbers in IE "Cell info" for Cell 1 |
| Timeslot list cell 2 | | | 6 | Timeslot numbers in IE "Cell info" for Cell 2 |
| Threshold used frequency | | dBm | -68 | Applicable for event 1H, cell 1 timeslots 5, 6 and cell 2 timeslot 6 |
| Threshold used frequency | | dBm | -66 | Applicable for event 1I, cell 1 timeslots 5, 6 and cell 2 timeslot 6 |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | Cell 2 shall belong to the monitored set |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 5 | |
| T4 | | s | 5 | |

**Table 8.6.1.2.4.2B: Cell 1 specific test parameters for correct event 1H and 1I reporting in AWGN propagation condition 1,28Mcps Option**

| Parameter | Unit | Cell 1 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | | | | |
| DL timeslot number | | 0 | | | | 5 | | | | 6 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | | | | | | |
| DPCH_Ec/Ior | dB | | | | | Note 1 | | | | | | | |
| OCNS_Ec/Ior | dB | -3 | | | | Note 2 | | | | 0 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 4 | | | | 3 | | | | 0 | 6 | | 0 |
| PCCPCH RSCP | dBm | -69 | | | | n.a. | | | | n.a. | | | |
| $I_{oc}$ | dBm / 1,28 MHz | -70 | | | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior | | | | | | | | | | | | | |

**Table 8.6.1.2.4.2C: Cell 2 specific test parameters for correct event 1H and 1I reporting in AWGN propagation condition 1,28Mcps option**

| Parameter | Unit | Cell 2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | |
| DL timeslot number | | 0 | | | | 6 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | | |
| DPCH_Ec/Ior | dB | | | | | | | | |
| OCNS_Ec/Ior | dB | -3 | | | | 0 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 4 | | | | 6 | | 0 | |
| PCCPCH RSCP | dBm | -69 | | | | n.a. | | | |
| $I_{oc}$ | dBm / 1,28 MHz | -70 | | | | | | | |

Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 5 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1I for cell 1 in timeslot 6. The measurement reporting delay from the beginning of T2 shall be less than 480 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 5 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 2 in timeslot 6 . The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

9) After 5 seconds from the beginning of T3, the SS shall switch the power settings from T3 to T4.

10) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 1 in timeslot 6. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

11) After 5 seconds from the beginning of T4, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

12) Repeat steps 1-13 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 2 |
| -Intra-frequency event identity | Event 1H |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -68 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| -Intra-frequency event identity | Event 1I |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -66 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 6 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1I |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 6 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 2 |

MEASUREMENT REPORT message (step 10)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 6 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |

8.6.1.2.4.3 7,68 Mcps TDD Option

8.6.1.2.4.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.6.1.2.4.3.1.1, 8.6.1.2.4.3.1.2 and 8.6.1.2.4.3.1.3. The test consists of five successive time periods, with a time duration of T1, T2, T3, T4 and T5 respectively. Two cells shall be present in the test, cell 1 being the current serving cell and cell 2 being a neighbour cell on the used frequency.

In the measurement control information it shall be indicated to the UE that event-triggered reporting with event 1H and event 1I shall be used and that Timeslot ISCP and P-CCPCH RSCP shall be reported together with event 1H and 1I. Measurement control information shall be sent to the UE before the beginning of time period T1.

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The UL DPCH shall be transmitted in timeslot 10. In addition, timeslots 3 and 4 shall be allocated as DL timeslots. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.

**Table 8.6.1.2.4.3.1.1: General test parameters for correct event 1H and event 1I reporting in AWGN propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 1 | |
| HCS | | | Not used | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Timeslot list cell 1 | | | 2, 3, 4 | Timeslot numbers in IE "Cell info" for Cell 1 |
| Timeslot list cell 2 | | | 4 | Timeslot numbers in IE "Cell info" for Cell 2 |
| Threshold used frequency | | dBm | -68 | Threshold 1 applicable for event 1H, cell 1 timeslots 2, 4 and cell 2 timeslot 4 |
| Threshold used frequency | | dBm | -73 | Threshold 2 applicable for event 1H, cell 1 timeslots 2, 3, 4 and cell 2 timeslot 4 |
| Threshold used frequency | | dBm | -67 | Applicable for event 1I, cell 1 timeslots 2, 4 and cell 2 timeslot 4 |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | Cell 2 shall belong to the monitored set |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 5 | |
| T4 | | s | 5 | |
| T5 | | s | 5 | |

**Table 8.6.1.2.4.3.1.2: Cell 1 specific parameters for correct event 1H and 1I reporting in AWGN propagation condition**

| Parameter | Unit | Cell 1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | | |
| **DL timeslot number** | | **0** | | | | | **2** | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | -9 | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | 5 | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | Note 1 | | | | |
| OCNS_Ec/Ior | dB | -3,12 | | | | | Note 2 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 4 | | | | | 4 | | | | |
| PCCPCH RSCP | dBm | -69 | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 7,68 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| **DL timeslot number** | | **3** | | | | | **4** | | | | |
| PCCPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | n.a. | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | 0 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | | | 0 | | | | 6 |
| PCCPCH RSCP | dBm | n.a. | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 7,68 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior | | | | | | | | | | | |

**Table 8.6.1.2.4.3.1.3: Cell 2 specific parameters for correct event 1H and 1I reporting in AWGN propagation condition**

| Parameter | Unit | Cell 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| UTRA RF Channel Number | | Channel 1 | | | | | | | | | |
| **DL timeslot number** | | **0** | | | | | **2** | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | -9 | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | 10 | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | -3,12 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 1 | | | | | 0 | 6 | 0 | | |
| PCCPCH RSCP | dBm | -72 | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 7,68 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |
| **DL timeslot number** | | **3** | | | | | **4** | | | | |
| PCCPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| SCH_$t_{offset}$ | dB | n.a. | | | | | n.a. | | | | |
| DPCH_Ec/Ior | dB | n.a. | | | | | n.a. | | | | |
| OCNS_Ec/Ior | dB | 0 | | | | | 0 | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | | | | 6 | | | | 0 |
| PCCPCH RSCP | dBm | n.a. | | | | | n.a. | | | | |
| $I_{oc}$ | dBm / 7,68 MHz | -70 | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | | | |

#### 8.6.1.2.4.3.2 Procedure

1) The RF parameters are set up according to T1.
2) The UE is switched on.
3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.
4) SS shall transmit a MEASUREMENT CONTROL message.
5) After 5 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.
6) UE shall transmit a MEASUREMENT REPORT message triggered by event 1I for cell 1 in timeslot 2. The measurement reporting delay from the beginning of T2 shall be less than 480 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
7) After 5 seconds from the beginning of T2, the SS shall switch the power settings from T2 to T3.
8) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 1 in timeslot 2. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
9) After 5 seconds from the beginning of T3, the SS shall switch the power settings from T3 to T4.
10) UE shall transmit a MEASUREMENT REPORT message triggered by event 1H for cell 1 in timeslot 4. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
11) After 5 seconds from the beginning of T4, the SS shall switch the power settings from T4 to T5.
12) UE shall transmit a MEASUREMENT REPORT message triggered by event 1I for cell 2 in timeslot 4. The measurement reporting delay from the beginning of T3 shall be less than 480 ms. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.
13) After 5 seconds from the beginning of T5, the UE is switched off. Any timing information of cell 2 is deleted in the UE.
14) Repeat steps 1-13 until the confidence level according to annex F.6.2 is achieved.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | TRUE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 2 |
| -Intra-frequency event identity | Event 1H |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -68 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| -Intra-frequency event identity | Event 1H |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |

| Information Element/Group name | Value/Remark |
|---|---|
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -73 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| -Intra-frequency event identity | Event 1I |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | -67 |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1I |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not Present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 1 |
| -SCTD indicator | FALSE |

MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not Present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 1 |
| -SCTD indicator | FALSE |

MEASUREMENT REPORT message (step 10)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 1 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 4 |
| -Timeslot ISCP | Checked that this IE is present for 4 timeslots |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1H |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not Present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 1 |
| -SCTD indicator | FALSE |

MEASUREMENT REPORT message (step 12)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Intra-frequency Measured results list |
| -Intra-frequency measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to Cell parameters ID of cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | 1 |
| -Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Intra-frequency measurement event results |
| -Intra-frequency event identity | 1I |
| -Cell measurement event results (10.3.7.4) | |
| -CHOICE mode | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | Not Present |
| -Cell parameters ID | Set to Cell parameters ID of Cell 2 |
| -SCTD indicator | FALSE |

### 8.6.1.2.5 Test requirements

#### 8.6.1.2.5.1 3,84 Mcps TDD Option

The UE shall send one event 1I triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T2.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T3.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T4.

The UE shall send one event 1I triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T5.

The UE shall not send event 1H or 1I triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

#### 8.6.1.2.5.2 1,28 Mcps TDD Option

The UE shall not send event 1H or 1I triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

#### 8.6.1.2.5.3 7,68 Mcps TDD Option

The UE shall send one event 1I triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T2.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T3.

The UE shall send one event 1H triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T4.

The UE shall send one event 1I triggered measurement report, with a measurement reporting delay less than 480 ms from the beginning of time period T5.

The UE shall not send event 1H or 1I triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

### 8.6.1.3 Correct reporting of neighbours in fading propagation conditions

#### 8.6.1.3.1 3,84 Mcps TDD Option

##### 8.6.1.3.1.1 Definition and applicability

In the event triggered reporting, the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (3,84 Mcps option) UE.

##### 8.6.1.3.1.2 Minimum requirement

The requirements are the same as in sub clause 8.6.1.1.2

The normative reference for this requirement is TS 25.123 [2] clauses 8.1.2.2.2.1, 8.1.2.2.2 and A.8.1.3

##### 8.6.1.3.1.3 Test purpose

To verify that the UE meets the minimum requirements and also verify that the UE performs sufficient layer 1 filtering of the measurements. The test is performed in fading propagation conditions.

##### 8.6.1.3.1.4 Method of test

###### 8.6.1.3.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.6.1.3.1 and 8.6.1.3.2. The test consists of one time period with time duration of T1. Two cells shall be present in the test, cell 1 being the current serving cell and cell 2 being a neighbour cell on the used frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP shall be reported together with Event 1G.

The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The TTI of the UL DCCH shall be 20ms.

**Table 8.6.1.3.1: General test parameters for correct reporting of neighbours in fading propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 1 | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 200 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | Sent before the beginning of time period T1 |
| T1 | | s | 200 | |

**Table 8.6.1.3.2: Cell specific test parameters for correct reporting of neighbours in fading propagation condition**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T1 | T1 | T1 |
| **DL timeslot number** | | **0** | **8** | **0** | **8** |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | n.a. | -3 | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 2 | 2 |
| PICH_Ec/Ior | dB | n.a. | -3 | n.a. | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | 7 | 2 | 2 |
| PCCPCH RSCP | dBm | -66 | n.a. | -71 | n.a. |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | | |
| Propagation Condition | | Case 4 as specified in Annex DTS25.102 Annex B | | | |

8.6.1.3.1.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 200 seconds from the beginning of T1, the UE is switched off.

6) Repeat steps 1-5 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

#### 8.6.1.3.1.5 Test requirements

The number of Event 1G triggered measurement reports during time period T1 shall be less than 60.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

### 8.6.1.3.2 1,28 Mcps TDD Option

Void.

### 8.6.1.3.3 7,68 Mcps TDD Option

#### 8.6.1.3.3.1 Definition and applicability

In the event triggered reporting, the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (7,68 Mcps option) UE.

#### 8.6.1.3.3.2 Minimum requirement

The requirements are the same as in sub clause 8.6.1.1.2

The normative reference for this requirement is TS 25.123 [2] clauses , 8.1B.2.2.2 and A.8.1.3.3

#### 8.6.1.3.3.3 Test purpose

To verify that the UE meets the minimum requirements and also verify that the UE performs sufficient layer 1 filtering of the measurements. The test is performed in fading propagation conditions.

#### 8.6.1.3.3.4 Method of test

##### 8.6.1.3.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Tables 8.6.1.3.3.4.1.1 and 8.6.1.3.3.4.1.2. The test consists of one time period with time duration of T1. Two cells shall be present in the test, cell 1 being the current serving cell and cell 2 being a neighbour cell on the used frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 1G shall be used, and that P-CCPCH RSCP shall be reported together with Event 1G.

The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The TTI of the UL DCCH shall be 20ms.

**Table 8.6.1.3.3.4.1.1: General test parameters for correct reporting of neighbours in fading propagation condition**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final condition | Active cell | | Cell 1 | |
| O | | dB | 0 | Cell individual offset. This value shall be used for all cells in the test. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 200 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1 | Sent before the beginning of time period T1 |
| T1 | | s | 200 | |

**Table 8.6.1.3.3.4.1.2: Cell specific test parameters for correct reporting of neighbours in fading propagation condition**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T1 | T1 | T1 |
| **DL timeslot number** | | **0** | **8** | **0** | **8** |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | n.a. | -3 | n.a. |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 2 | 2 |
| PICH_Ec/Ior | dB | n.a. | -3 | n.a. | -3 |
| OCNS_Ec/Ior | dB | -3,12 | -3,12 | -3,12 | -3,12 |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | 7 | 2 | 2 |
| PCCPCH RSCP | dBm | -66 | n.a. | -71 | n.a. |
| $I_{oc}$ | dBm/ 7,68 MHz | -70 | | | |
| Propagation Condition | | Case 4 as specified in TS25.102 Annex B | | | |

8.6.1.3.3.4.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 200 seconds from the beginning of T1, the UE is switched off.

6) Repeat steps 1-5 until the confidence level according to annex F.6.2 is achieved..

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Intra-frequency measurement reporting criteria |
| -Intra-frequency measurement reporting criteria (10.3.7.39) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 1G |
| -Triggering condition 2 | Not Present |
| -Reporting Range Constant | Not Present |
| -Cells forbidden to affect Reporting Range | Not Present |
| -W | Not Present |
| -Hysteresis | 0 dB |
| -Threshold used frequency | Not Present |
| -Reporting deactivation threshold | Not Present |
| -Replacement activation threshold | Not Present |
| -Time to trigger | 0 ms |
| -Amount of reporting | Infinity |
| -Reporting interval | 0 ms (Note 1) |
| -Reporting cell status | Not Present |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |
| Note 1: Reporting interval = 0 ms means no periodical reporting | |

#### 8.6.1.3.3.5 Test requirements

The number of Event 1G triggered measurement reports during time period T1 shall be less than 60.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

## 8.6.2 TDD inter frequency measurements

### 8.6.2.1 Correct reporting of neighbours in AWGN propagation condition

#### 8.6.2.1.1 Definition and applicability

##### 8.6.2.1.1.1 3,84 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (3,84 Mcps option) UE.

##### 8.6.2.1.1.2 1,28 Mcps TDD Option

The purpose of this test is to verify that the UE makes correct reporting of an event within the required times when doing inter frequency measurements on a neighbouring TDD cell. The test will partly verify the requirements in section 8.1A.2 and 9.1. of [2]. The requirements and this test apply to the TDD (1.28 Mcps option) UE.

##### 8.6.2.1.1.3 7,68 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the TDD (7,68 Mcps option) UE.

#### 8.6.2.1.2 Minimum requirement

##### 8.6.2.1.2.1 3,84 Mcps TDD Option

When idle intervals are used for inter-frequency TDD measurements, the UE shall be able to identify a new detectable inter-frequency TDD cell belonging to the monitored set within

$$T_{\text{identify inter}} = Max\left\{ 5000, N_{\text{basic identify TDD inter}} \cdot \frac{T_{\text{measurement period TDD inter}}}{N_{\text{TDD inter}}} \cdot N_{Freq} \right\} ms$$

If the UE does not require idle intervals to perform inter-frequency TDD measurements, the UE shall be able to identify a new detectable inter-frequency TDD cell belonging to the monitored set within 5000 ms.

When idle intervals are used for TDD inter frequency measurements, the UE shall be capable of performing P-CCPCH RSCP measurements for $X_{\text{measurement TDD inter}}$ inter-frequency TDD cells per TDD frequency of the monitored set.

The UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in section 9 and with a measurement period of $T_{\text{measurement inter}}$.

$$T_{\text{measurement inter}} = Max\left\{ T_{\text{measurement period TDD inter}}, N_{\text{basic measurement TDD inter}} \cdot \frac{T_{\text{measurement period TDD inter}}}{N_{\text{TDD inter}}} \cdot N_{Freq} \right\} ms$$

If the UE does not require idle intervals to perform TDD inter-frequency measurements, the measurement period for inter frequency P-CCPCH RSCP measurements shall be 480 ms.

Where,

$X_{measurement\ TDD\ inter}$ = 6 (cells)

$T_{measurement_period\ inter}$ = 480 ms. The time period used for calculating the measurement period $T_{measurement_inter}$ for inter frequency P-CCPCH RSCP measurements.

$N_{TDD\ inter}$: This is the available number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period $T_{TDD\ inter}$. The UE shall consider that a measurement opportunity on a Beacon timeslot of an inter-frequency TDD cell is provided if an idle interval of length equal to or greater than 3 timeslots less 2*0.5 ms implementation margin for frequency switching per idle interval completely overlaps in time with the Beacon timeslot of the inter-frequency TDD cell.

$N_{basic_identify_TDD\ inter}$ = 80. This is a number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period used in the inter frequency TDD equation where the maximum allowed time for the UE to identify a new detectable inter-frequency TDD cell is defined.

$N_{basic_measurement_TDD\ inter}$ = 5. This is a number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period $T_{TDD\ inter}$ used in the inter-frequency TDD equation where the measurement period for inter-frequency P-CCPCH RSCP measurements is defined.

$N_{Freq\ TDD}$: This is the number of TDD frequencies indicated in the inter frequency measurement control information.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ inter}$ defined in 25.123 [2]. When L3 filtering is used, an additional delay can be expected.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1.2.3.1, 8.1.2.3.2 and A.8.2.1

#### 8.6.2.1.2.2 1,28 Mcps TDD Option

When idle intervals are used for inter-frequency TDD measurements, the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify\ inter} = Max\left\{5000, N_{basic\ identify\ TDD,inter} \cdot \frac{T_{Measurement\ Period,Inter}}{N_{Inter}} \cdot N_{Freq}\right\} ms$$

when

$N_{Inter}$: This is the minimum number of sub-frame in that the signal of P-CCPCH and DwPCH can be received for inter frequency target cell during the period $T_{Measurement_Period\ inter}$ with an arbitrarily chosen timing. It depends on the channel allocation and is calculated by assuming 2*0.1ms for implementation margin (for the description of the idle intervals see Annex A of 25.225).

$T_{basic_identify_TDD,inter}$= 800ms. This is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD cell is defined.

$T_{Measurement_Period\ Inter}$= 480 ms. The period used for calculating the measurement period $T_{measurement_inter}$ for inter frequency P-CCPCH RSCP measurements.

$N_{Freq:}$ Number of TDD frequencies indicated in the inter frequency measurement control information.

If the UE does not require idle intervals to perform inter-frequency TDD measurements, the UE shall be able to identify a new detectable inter-frequency TDD cell belonging to the monitored set within 5000 ms.

A cell shall be considered detectable when P-CCPCH Ec/Io $\geq$ -8 dB and DwPCH_Ec/Io $\geq$ -5 dB. When L3 filtering is used an additional delay can be expected.

The test consist of 2 successive time periods, with a time duration T1 and T2. The test parameters are given in tables A.8.2C and A.8.2D below. Two cells shall be present in the test, cell 1 being the active cell and cell 2 being a 1.28Mcps TDD option neighbour cell on the unused frequency.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. P-CCPCH RSCP of the best cell on the unused frequency shall be reported together with Event 2C reporting. The

measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in 25.331 [9]

#### 8.6.2.1.2.3 7,68 Mcps TDD Option

When idle intervals are used for inter-frequency TDD measurements, the UE shall be able to identify a new detectable inter-frequency TDD cell belonging to the monitored set within

$$T_{\text{identify inter}} = Max\left\{ 5000, N_{\text{basic identify TDD inter}} \cdot \frac{T_{\text{measurement period TDD inter}}}{N_{\text{TDD inter}}} \cdot N_{Freq} \right\} ms$$

If the UE does not require idle intervals to perform inter-frequency TDD measurements, the UE shall be able to identify a new detectable inter-frequency TDD cell belonging to the monitored set within 5000 ms.

When idle intervals are used for TDD inter frequency measurements, the UE shall be capable of performing P-CCPCH RSCP measurements for $X_{\text{measurement TDD inter}}$ inter-frequency TDD cells per TDD frequency of the monitored set.

The UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in section 9 and with a measurement period of $T_{\text{measurement inter}}$.

$$T_{\text{measurement inter}} = Max\left\{ T_{\text{measurement period TDD inter}}, N_{\text{basic measurement TDD inter}} \cdot \frac{T_{\text{measurement period TDD inter}}}{N_{\text{TDD inter}}} \cdot N_{Freq} \right\} ms$$

If the UE does not require idle intervals to perform TDD inter-frequency measurements, the measurement period for inter frequency P-CCPCH RSCP measurements shall be 480 ms.

Where,

$X_{\text{measurement TDD inter}}$ = 6 (cells)

$T_{\text{measurement_period inter}}$ = 480 ms. The time period used for calculating the measurement period $T_{\text{measurement_inter}}$ for inter frequency P-CCPCH RSCP measurements.

$N_{\text{TDD inter}}$: This is the available number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period $T_{\text{TDD inter}}$. The UE shall consider that a measurement opportunity on a Beacon timeslot of an inter-frequency TDD cell is provided if an idle interval of length equal to or greater than 3 timeslots less 2*0.5 ms implementation margin for frequency switching per idle interval completely overlaps in time with the Beacon timeslot of the inter-frequency TDD cell.

$N_{\text{basic_identify_TDD inter}}$ = 80. This is a number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period used in the inter frequency TDD equation where the maximum allowed time for the UE to identify a new detectable inter-frequency TDD cell is defined.

$N_{\text{basic_measurement_TDD inter}}$ = 5. This is a number of measurement opportunities for a Beacon timeslot of an inter-frequency TDD cell during the time period $T_{\text{TDD inter}}$ used in the inter-frequency TDD equation where the measurement period for inter-frequency P-CCPCH RSCP measurements is defined.

$N_{\text{Freq TDD}}$: This is the number of TDD frequencies indicated in the inter frequency measurement control information.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{\text{identify inter}}$ defined in 25.123 [2]. When L3 filtering is used, an additional delay can be expected.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1B.2.3.1, 8.1B.2.3.2 and A.8.2.1

### 8.6.2.1.3 Test purpose

#### 8.6.2.1.3.1 3,84 Mcps TDD Option

To verify that the UE meets the minimum requirement.

8.6.2.1.3.2 1,28 Mcps TDD Option

To verify that the UE meets the minimum time requirements for identifying neighbouring TDD cells

8.6.2.1.3.3 7,68 Mcps TDD Option

To verify that the UE meets the minimum time requirements for identifying neighbouring interfrequency TDD cells.

8.6.2.1.4 Method of test

8.6.2.1.4.1 3,84 Mcps TDD Option

8.6.2.1.4.1.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.2.1.1 and 8.6.2.1.2. The test consists of 2 successive time periods, with a time duration T1 and T2. Two cells shall be present in the test, cell 1 being the serving cell and cell 2 being a UTRA TDD neighbour cell on the unused frequency. All cells shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. P-CCPCH RSCP of the best cell on the unused frequency shall be reported together with Event 2C reporting. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. The TTI of the uplink DCCH shall be 20 ms.

**Table 8.6.2.1.1: General test parameters for Correct reporting of TDD inter-frequency neighbours in AWGN propagation condition (3,84 Mcps option)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in Annex CTS 25.102 annex A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | UTRA TDD cell |
| | Neighbour cell | | Cell 2 | UTRA TDD cell |
| Threshold non used frequency | | dB | -71 | Applicable for event 2C |
| Hysteresis | | dB | 0 | Applicable for event 2C |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 24 on channel 1<br>16 on channel 2 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |

**Table 8.6.2.1.2: Cell specific parameters for Correct reporting of TDD inter-frequency neighbours in AWGN propagation condition (3,84 Mcps option)**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | | | | -3 | -3 | | | -3 | -3 |
| OCNS | | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | 3 | 3 | -Infinity | 9 | -Infinity | 9 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | | | | | |
| PCCPCH_RSCP | dB | -70 | -70 | | | -Infinity | -64 | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| Propagation Condition | | AWGN | | | | | | | |

8.6.2.1.4.1.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2c for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5080 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 [TBD] times.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD option | 3,84 Mcps TDD |
| -CHOICE Sync case | 2 |
| -Timeslot | 0 |
| -cell parameters ID | Set to cell parameters ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.6.2.1.2 |
| -Timesllot list | Not Present |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | |
| -Frequency quality estimate | |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored |

| Information Element/Group name | Value/Remark |
|---|---|
| | set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -71 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -CHOICE mode | TDD |
| -Primary CCPCH Info | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 3,84 Mcps TDD |
| -CHOICE Sync Case | Not Present |
| -Cell Parameters ID | Set to cell parameters ID of Cell 2 |
| -SCTD Indicator | FALSE |

### 8.6.2.1.4.2 1,28 Mcps TDD Option

Test environment : normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested : mid range ;see clauses G.2.4.

Cell 1 is the active cell, Cell 2 is a neighbour cell on the unused frequency. The power level on Cell 1 is kept constant and the power level of Cell 2 is changed using "change of best cell event". General test parameters are given in the table 8.6.2.1.3 below and they are signalled from test device. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. P-CCPCH RSCP of the best cell has to be reported together with Event 2C reporting. New measurement control information, which defines neighbour cells etc., is always sent before the event starts. The cell specific test parameters are shown in Table 8.6.2.1.4.

**Table 8.6.2.1.3: General test parameters for correct reporting of TDD inter frequency neighbours in AWGN propagation condition 1,28Mcps option**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters active cell | | | DL and UL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2. The DPCH is located in an other timeslot than 0 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | 1.28Mcps TDD cell |
| | Neighbour cell | | Cell 2 | 1.28Mcps TDD cell |
| Final condtions | Active cell | | Cell 1 | |
| Threshold non used frequency | | dBm | -71 | Absolute P-CCPCH RSCP threshold for event 2C |
| W non-used frequency | | | 1 | Applicable for event 2C |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 24 on channel 1<br>16 on channel 2 | Measurement control information is sent before T1 starts. |
| T1 | | s | 10 | |
| T2 | | s | 10 | |

**Table 8.6.2.1.4 Cell Specific Parameters for Correct Reporting of Neighbours in AWGN Propagation Condition 1,28Mcps option**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | -3 | | | |
| DwPCH_Ec/Ior | dB | | | 0 | | | | 0 | |
| OCNS_Ec/Ior | dB | -3 | | | | -3 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | | | -Infinity | 6 | | |
| $I_{oc}$ | dBm/1.28 MHz | -70 | | | | | | | |
| PCCPCH_RSCP | dBm | -70 | -70 | | | -Infinity | -67 | | |
| Propagation Condition | | AWGN | | | | | | | |

NOTE: The DPCH of all cells are located in a timeslot other than 0.

8.6.2.1.4.2.2 Test Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2c for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5080 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T2, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 until the confidence level according to annex F.6.2 is achieved.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD option | 1,28 Mcps TDD |
| -cell parameters ID | Set to cell parameters ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.6.2.1.2 |
| -Timesllot list | Not Present |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | FALSE |
| -Frequency quality estimate | FALSE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored |

| Information Element/Group name | Value/Remark |
|---|---|
| | set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -71 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -Non frequency related measurement event results | |
| -CHOICE mode | TDD |
| -Primary CCPCH Info | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 1.28 Mcps TDD |
| -TSTD indicator | TRUE |
| -Cell Parameters ID | Set to cell parameters ID of Cell 2 |
| -SCTD Indicator | FALSE |

### 8.6.2.1.4.3 7,68 Mcps TDD Option

#### 8.6.2.1.4.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.2.1.4.3.1.1 and 8.6.2.1.4.3.1.2. The test consists of 2 successive time periods, with a time duration T1 and T2. Two cells shall be present in the test, cell 1 being the serving cell and cell 2 being a UTRA TDD neighbour cell on the unused frequency. All cells shall be synchronised, i.e. share the same frame and timeslot timing.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used. P-CCPCH RSCP of the best cell on the unused frequency shall be reported together with Event 2C reporting. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. The TTI of the uplink DCCH shall be 20 ms.

**Table 8.6.2.1.4.3.1.1: General test parameters for Correct reporting of TDD inter-frequency neighbours in AWGN propagation condition (7,68 Mcps option)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 annex A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | UTRA TDD cell |
| | Neighbour cell | | Cell 2 | UTRA TDD cell |
| Threshold non used frequency | | dB | -71 | Applicable for event 2C |
| Hysteresis | | dB | 0 | Applicable for event 2C |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 24 on channel 1<br>16 on channel 2 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |

**Table 8.6.2.1.4.3.1.2: Cell specific parameters for Correct reporting of TDD inter-frequency neighbours in AWGN propagation condition (7,68 Mcps option)**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | 0 | | 8 | |
| | | T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -3 | -3 | | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | 15 | 15 | 15 | 15 |
| PICH_Ec/Ior | | | | -3 | -3 | | | -3 | -3 |
| OCNS | | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 | -4,28 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | 3 | 3 | -Infinity | 9 | -Infinity | 9 |
| $I_{oc}$ | dBm/7.68 MHz | -70 | | | | | | | |
| PCCPCH_RSCP | dB | -70 | -70 | | | -Infinity | -64 | | |
| Propagation Condition | | AWGN | | | | | | | |

8.6.2.1.4.3.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 10 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2c for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5080 ms. If the UE fails to report the event within the

required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 until the confidence level according to annex F.6.2 is achieved..

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE mode | TDD |
| -CHOICE TDD option | 7,68 Mcps TDD |
| -CHOICE Sync case | 2 |
| -Timeslot | 0 |
| -cell parameters ID | Set to cell parameters ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.6.2.1.2 |
| -Timesllot list | Not Present |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | |
| -Frequency quality estimate | |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored |

| Information Element/Group name | Value/Remark |
|---|---|
| | set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -71 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | TDD |
| -Cell parameters ID | Set to cell parameters ID of Cell 2 |
| -Proposed TGSN | Not Present |
| -Primary CCPCH RSCP | Checked that this IE is present |
| -Pathloss | Not Present |
| -Timeslot list | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | TDD |
| -UARFCN(Nt) | Same frequency as channel 2 |
| -CHOICE mode | TDD |
| -Primary CCPCH Info | |
| -CHOICE mode | TDD |
| -CHOICE TDD Option | 7,68 Mcps TDD |
| -CHOICE Sync Case | Not Present |
| -Cell Parameters ID | Set to cell parameters ID of Cell 2 |
| -SCTD Indicator | FALSE |

### 8.6.2.1.5 Test requirements

#### 8.6.2.1.5.1 3,84 Mcps TDD Option

The UE shall send one Event 2C triggered measurement report for Cell 2 with a measurement reporting delay less than 5080ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

#### 8.6.2.1.5.2 1,28 Mcps TDD Option

The UE shall send an EVENT 2C message of cell 2 within 5080ms from the beginning of the time T2

The UE shall not send any event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

#### 8.6.2.1.5.3 7,68 Mcps TDD Option

The UE shall send one Event 2C triggered measurement report for Cell 2 with a measurement reporting delay less than 5080ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

## 8.6.3 FDD measurements

### 8.6.3.1 Correct reporting of FDD neighbours in AWGN propagation condition

#### 8.6.3.1.1 Definition and applicability

##### 8.6.3.1.1.1 3,84 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the combined FDD and TDD (3,84 Mcps option) UE.

##### 8.6.3.1.1.2 1,28 Mcps TDD Option

The purpose of this test is to verify that the UE makes correct reporting of an event when measuring on UTRA FDD cells. This test will partly verify the requirements in section 8.1A.2 and 9.1. of [2]

The test consists of two successive time periods, with a time duration T1 and T2. Two cells shall be present in the test, cell 1 being current active 1.28Mcps TDD cell and cell 2 being a UTRA FDD neighbouring cell.

In the measurement control information it is indicated to the UE hat event-triggered reporting with Event 2C shall be used and the CPICH RSCP of the best cell on the unused frequency shall be reported together with Event 2C.The measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in 25.331 [9].

##### 8.6.3.1.1.3 7,68 Mcps TDD Option

In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH.

The requirements and this test apply to the combined FDD and TDD (7,68 Mcps option) UE.

#### 8.6.3.1.2 Minimum requirement

##### 8.6.3.1.2.1 3,84 Mcps TDD Option

When idle intervals are used for inter-frequency FDD measurements, the UE shall be able to identify a new detectable inter-frequency FDD cell belonging to the monitored set within

$$T_{\text{identify FDD inter}} = Max\left\{5000, T_{\text{basic identify FDD inter}} \cdot \frac{T_{\text{measurement period FDD inter}}}{T_{\text{FDD inter}}} \cdot N_{Freq}\right\} ms$$

If the UE does not require idle intervals to perform FDD inter-frequency measurements, the UE shall be able to identify a new detectable inter-frequency FDD cell belonging to the monitored set within 5000 ms.

When L3 filtering is used an additional delay can be expected.

An inter-frequency FDD cell shall be considered detectable, when CPICH Ec/Io $\geq$ -20 dB, SCH_Ec/Io $\geq$ -17 dB and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

When idle intervals are used for FDD inter frequency measurements, the UE physical layer shall be capable of reporting measurements to higher layers with measurement period given by

$$T_{\text{measurement FDD inter}} = Max\left\{T_{\text{measurement period FDD inter}}, T_{\text{basic measurement FDD inter}} \cdot \frac{T_{\text{measurement period FDD inter}}}{T_{\text{FDD inter available}}} \cdot N_{Freq,FDD}\right\} ms$$

If the UE does not require idle intervals to perform FDD inter-frequency measurements, the measurement period for inter frequency CPICH measurements shall be 480 ms.

The UE shall be capable of performing CPICH measurements for $X_{\text{measurement FDD inter}}$ inter-frequency FDD cells per frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{\text{measurement FDD inter}}$.

$X_{\text{basic measurement FDD inter}}$ = 6 (cells)

$T_{\text{measurement_period FDD inter}}$ = 480 ms. The time period used for calculating the measurement period $T_{\text{measurement_FDD inter}}$ for inter frequency CPICH measurements.

$T_{\text{FDD inter: available}}$: This is the available time for measurements on inter-frequency FDD cells. $T_{\text{FDD inter available}}$ shall be derived from $T_{\text{FDD inter}}$ by assuming 2*0.5 ms implementation margin for frequency switching per idle interval and by only taking into account the remaining number of full timeslots per idle interval. Idle intervals smaller than 3 timeslots shall not be taken into account for calculating $T_{\text{FDD inter available}}$.

$T_{\text{basic_identify_FDD inter}}$ = 800 ms. This is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new detectable inter-frequency FDD cell is defined.

$T_{\text{basic_measurement_FDD inter}}$ = 50 ms. This is the time period used in the inter-frequency equation for defining the measurement period for inter frequency CPICH measurements.

$N_{Freq}$: This is the number of FDD frequencies indicated in the inter frequency measurement control information.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{\text{identify inter}}$ defined in 25.123 [2]. When L3 filtering is used, an additional delay can be expected.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1.2.4 and A.8.3.1

#### 8.6.3.1.2.2 1,28 Mcps TDD Option

The UE shall send the Event 2C triggered measurement report, with a measurement reporting delay less than 5.08 s from the beginning of time period T2.

#### 8.6.3.1.2.3 7,68 Mcps TDD Option

When idle intervals are used for inter-frequency FDD measurements, the UE shall be able to identify a new detectable inter-frequency FDD cell belonging to the monitored set within

$$T_{\text{identify FDD inter}} = Max\left\{5000, T_{\text{basic identify FDD inter}} \cdot \frac{T_{\text{measurement period FDD inter}}}{T_{\text{FDD inter}}} \cdot N_{Freq}\right\} ms$$

If the UE does not require idle intervals to perform FDD inter-frequency measurements, the UE shall be able to identify a new detectable inter-frequency FDD cell belonging to the monitored set within 5000 ms.

When L3 filtering is used an additional delay can be expected.

An inter-frequency FDD cell shall be considered detectable, when CPICH Ec/Io $\geq$ -20 dB, SCH_Ec/Io $\geq$ -17 dB and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

When idle intervals are used for FDD inter frequency measurements, the UE physical layer shall be capable of reporting measurements to higher layers with measurement period given by

$$T_{\text{measurement FDD inter}} = Max\left\{ T_{\text{measurement period FDD inter}}, T_{\text{basic measurement FDD inter}} \cdot \frac{T_{\text{measurement period FDD inter}}}{T_{\text{FDD inter available}}} \cdot N_{Freq,FDD} \right\} ms$$

If the UE does not require idle intervals to perform FDD inter-frequency measurements, the measurement period for inter frequency CPICH measurements shall be 480 ms.

The UE shall be capable of performing CPICH measurements for $X_{\text{measurement FDD inter}}$ inter-frequency FDD cells per frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{\text{measurement FDD inter}}$.

$X_{\text{basic measurement FDD inter}}$ = 6 (cells)

$T_{\text{measurement_period FDD inter}}$ = 480 ms. The time period used for calculating the measurement period $T_{\text{measurement_FDD inter}}$ for inter frequency CPICH measurements.

$T_{\text{FDD inter: available}}$: This is the available time for measurements on inter-frequency FDD cells. $T_{\text{FDD inter available}}$ shall be derived from $T_{\text{FDD inter}}$ by assuming 2*0.5 ms implementation margin for frequency switching per idle interval and by only taking into account the remaining number of full timeslots per idle interval. Idle intervals smaller than 3 timeslots shall not be taken into account for calculating $T_{\text{FDD inter available}}$.

$T_{\text{basic_identify_FDD inter}}$ = 800 ms. This is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new detectable inter-frequency FDD cell is defined.

$T_{\text{basic_measurement_FDD inter}}$ = 50 ms. This is the time period used in the inter-frequency equation for defining the measurement period for inter frequency CPICH measurements.

$N_{\text{Freq}}$: This is the number of FDD frequencies indicated in the inter frequency measurement control information.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{\text{identify inter}}$ defined in 25.123 [2]. When L3 filtering is used, an additional delay can be expected.

The normative reference for this requirement is TS 25.123 [2] clauses 8.1B.2.4 and A.8.3.1

### 8.6.3.1.3 Test purpose

#### 8.6.3.1.3.1 3,84 Mcps TDD Option

To verify that the UE meets the minimum requirement.

#### 8.6.3.1.3.2 1,28 Mcps TDD Option

To verify that the UE meets the minimum time requirements for identifying neighbouring FDD cells.

#### 8.6.3.1.3.3 7,68 Mcps TDD Option

To verify that the UE meets the minimum time requirements for identifying neighbouring FDD cells.

### 8.6.3.1.4 Method of test

#### 8.6.3.1.4.1 3,84 Mcps TDD Option

##### 8.6.3.1.4.1.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.3.1.1 and 8.6.3.1.2. The test consists of two successive time periods, with time durations of T1 and T2 respectively. Two cells shall be present in the test, cell 1 being the serving UTRA TDD cell and cell 2 being a UTRA FDD neighbour cells on the unused frequency.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used and that CPICH Ec/I0 of the best cell on the unused frequency shall be reported together with Event 2C. The

Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for cell 1. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. The TTI of the uplink DCCH shall be 20 ms.

**Table 8.6.3.1.1: General test parameters for Correct reporting of FDD inter-frequency neighbours in AWGN propagation condition (3,84 Mcps option)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | UTRA TDD cell |
| | Neighbour cell | | Cell 2 | UTRA FDD cell |
| Final conditions | Active cell | | Cell 1 | |
| Threshold non used frequency | | dB | -18 | Applicable for event 2C |
| W non-used frequency | | | 1 | Applicable for event 2C |
| Hysteresis | | dB | 0 | Applicable for event 2C |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on channel 1<br>6 FDD neighbours on channel 2 | |
| T1 | | s | 15 | |
| T2 | | s | 10 | |

**Table 8.6.3.1.2: Cell specific parameters for Correct reporting of FDD inter-frequency neighbours in AWGN propagation condition (3,84 Mcps option)**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | n.a | |
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -12 | |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | n.a. | |
| PICH_Ec/Ior | | | | -3 | -3 | -15 | |
| | | | | | | | |
| OCNS | dB | -4,28 | -4,28 | -4,28 | -4,28 | -0,941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | 3 | 3 | -infinity | -1.8 |
| $I_{oc}$ | dBm/ 3,84 MHz | -70 | | | | -70 | |
| CPICH_Ec/Io | | n.a. | | | | -infinity | -14 |
| | | | | | | | |
| PCCPCH_RSCP | dB | -70 | -70 | -70 | -70 | n.a. | |
| Propagation Condition | | AWGN | | | | AWGN | |

8.6.3.1.4.1.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 15 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5040 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T2, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 [TBD] times.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: 10.x.y.z in the IE description refers to clauses in TS 25.331 [9].

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | FDD |
| -Primary CPICH info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -Primary CPICH Tx Power | Set to Primary CPICH Tx Power of Cell2 described in Table 8.6.3.1.2 |
| -Tx Diversity Indicator | FALSE |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH_Ec/No |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | |
| -Frequency quality estimate | |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | TRUE |
| -CPICH RSCP reporting indicator | FALSE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |

| Information Element/Group name | Value/Remark |
|---|---|
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -18 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Checked that this IE is present |
| -CPICH RSCP | Not Present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.6.3.1.4.2 1,28 Mcps TDD Option

Cell 1 is current active TDD cell, Cell 2 is a FDD cell. The power level of CPICH RSCP of cell 2 and the P-CCPCH RSCP of cell 1 is changed. General test parameters are given in the table 8.6.3.1.3 below and they are signalled from test device. New measurement control information, which defines neighbour cells etc., is always sent before the handover starts. The test parameters are given in Table 8.6.3.1.4 below.

**Table 8.6.3.1.3: General test parameters for Correct reporting of FDD neighbours in AWGN propagation condition 1,28Mcps option**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters active cell | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A. The DPCH is located in an other timeslot than 0. |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Active cell Initial conditions | Active cell | | Cell 1 | 1.28Mcps TDD cell |
| | Neighbour cell | | Cell2 | FDD cell |
| Final conditions | Active cell | | Cell 1 | 1.28Mcps TDD cell |
| Threshold non used frequency | | dBm | -86 | Absolute CPICH RSCP threshold for event 2C |
| Hysteresis | | dB | 0 | |
| W non-used frequency | | | 1 | Applicable for event 2C |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on Channel 1<br>6 FDD neighbours on Channel 2 | Measurement control information is sent before T1 starts. |
| T1 | | s | 10 | |
| T2 | | s | 10 | |

**Table 8.6.3.1.4 Cell Specific parameters for Correct reporting of FDD neighbours in AWGN propagation condition: 1,28Mcps option**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | | n.a | n.a. |
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | -12 |
| SCH_Ec/Ior | dB | n.a. | n.a. | n.a. | n.a. | -12 | -12 |
| PICH_Ec/Ior | dB | n.a. | n.a. | n.a. | n.a. | -15 | -15 |
| DwPCH_Ec/Ior | dB | | | 0 | 0 | n.a. | n.a. |
| OCNS | dB | -3 | -3 | | | -0,941 | -0,941 |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | 3 | 3 | -Infinity | -2 |
| $I_{oc}$ | dBm/1.28 MHz | -70 | | | | | |
| $I_{oc}$ | dBm/3.84 MHz | | | | | -70 | |
| CPICH_RSCP | dBm | n.a. | | | | -Infinity | -82 |
| PCCPCH_RSCP | dBm | -70 | -70 | | | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | AWGN | |

Note: The DPCH of cell 1 is located in a timeslot other than 0.

8.6.3.1.4.2.1 Test Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 15 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5040 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T2, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 [TBD] times.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex A of 34.123-1 [21], with the following exceptions:

Note: 10.x.y.z in the IE description refers to clauses in TS 25.331 [9].

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | FDD |
| -Primary CPICH info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -Primary CPICH Tx Power | Set to Primary CPICH Tx Power of Cell2 described in Table 8.6.3.1.2 |
| -Tx Diversity Indicator | FALSE |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH_RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | |
| -Frequency quality estimate | |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | TRUE |
| -CPICH RSCP reporting indicator | FALSE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |

| Information Element/Group name | Value/Remark |
|---|---|
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -86 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Checked that this IE is present |
| -CPICH RSCP | Not Present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.6.3.1.4.3 7,68 Mcps TDD Option

#### 8.6.3.1.4.3.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

The test parameters are given in Table 8.6.3.1.4.3.1.1 and 8.6.3.1.4.3.1.2. The test consists of two successive time periods, with time durations of T1 and T2 respectively. Two cells shall be present in the test, cell 1 being the serving UTRA TDD cell and cell 2 being a UTRA FDD neighbour cells on the unused frequency.

In the measurement control information it is indicated to the UE that event-triggered reporting with Event 2C shall be used and that CPICH Ec/I0 of the best cell on the unused frequency shall be reported together with Event 2C. The Measurement control message shall be sent to the UE such that the delay between the end of the last received TTI containing the message and the beginning of T1 is at least equal to the RRC procedure delay as defined in [9].

The second Beacon timeslot shall be provided in timeslot 8 for cell 1. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. The TTI of the uplink DCCH shall be 20 ms.

**Table 8.6.3.1.4.3.1.1: General test parameters for Correct reporting of FDD inter-frequency neighbours in AWGN propagation condition (7,68 Mcps option)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| DPCH parameters | | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | | On | |
| Target quality value on DTCH | | BLER | 0.01 | |
| Initial conditions | Active cell | | Cell 1 | UTRA TDD cell |
| | Neighbour cell | | Cell 2 | UTRA FDD cell |
| Final conditions | Active cell | | Cell 1 | |
| Threshold non used frequency | | dB | -18 | Applicable for event 2C |
| W non-used frequency | | | 1 | Applicable for event 2C |
| Hysteresis | | dB | 0 | Applicable for event 2C |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| Monitored cell list size | | | 6 TDD neighbours on channel 1<br>6 FDD neighbours on channel 2 | |
| T1 | | s | 15 | |
| T2 | | s | 10 | |

**Table 8.6.3.1.4.3.1.2: Cell specific parameters for Correct reporting of FDD inter-frequency neighbours in AWGN propagation condition (7,68 Mcps option)**

| Parameter | Unit | Cell 1 | | | | Cell 2 | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | 8 | | n.a | |
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number | | Channel 1 | | | | Channel 2 | |
| CPICH_Ec/Ior | dB | n.a. | | n.a. | | -10 | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | | -12 | |
| SCH_Ec/Ior | dB | -9 | -9 | -9 | -9 | -12 | |
| SCH_$t_{offset}$ | | 0 | 0 | 0 | 0 | n.a. | |
| PICH_Ec/Ior | | | | -3 | -3 | -15 | |
| | | | | | | | |
| OCNS | dB | -4,28 | -4,28 | -4,28 | -4,28 | -0,941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | 3 | 3 | 3 | -infinity | -1.8 |
| $I_{oc}$ | dBm/ 7.68 MHz | -70 | | | | -70 | |
| CPICH_Ec/Io | | n.a. | | | | -infinity | -14 |
| | | | | | | | |
| PCCPCH_RSCP | dB | -70 | -70 | -70 | -70 | n.a. | |
| Propagation Condition | | AWGN | | | | AWGN | |

8.6.3.1.4.3.2 Procedure

1) The RF parameters are set up according to T1.

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit a MEASUREMENT CONTROL message.

5) After 15 seconds from the beginning of T1, the SS shall switch the power settings from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 2C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 5040 ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of succesfull tests is increased by one.

7) After 10 seconds from the beginning of T2, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

8) Repeat steps 1-7 until the confidence level according to annex F.6.2 is achieved..

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions:

Note: 10.x.y.z in the IE description refers to clauses in TS 25.331 [9].

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | No inter-frequency cells removed |
| -New inter-frequency cells | 1 |
| -Inter-frequency cell id | 1 |
| -Frequency info (10.3.6.36) | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -Cell info (10.3.7.2) | |
| -Cell individual offset | Not Present |
| -Reference time difference to cell | Not Present |
| -Read SFN indicator | False |
| -CHOICE *mode* | FDD |
| -Primary CPICH info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -Primary CPICH Tx Power | Set to Primary CPICH Tx Power of Cell2 described in Table 8.6.3.1.2 |
| -Tx Diversity Indicator | FALSE |
| -Cell selection and re-selection info | Not Present |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | FDD |
| -Measurement quantity for frequency quality estimate | CPICH_Ec/No |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | |
| -Frequency quality estimate | |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | TRUE |
| -CPICH RSCP reporting indicator | FALSE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | Not Present |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* | Inter-frequency measurement reporting criteria |
| -Inter-frequency measurement reporting criteria (10.3.7.19) | |
| -Parameters required for each event | 1 |
| -Intra-frequency event identity | Event 2C |
| -Threshold used frequency | Not Present |
| -W Used frequency | Not Present |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting Cell Status (10.3..61) | |
| -CHOICE reported cell | Report cells within active and/or monitored set on used frequency or within virtual active and/or monitored set on non-used frequency |
| -Maximum number of reported cells | 3 |

| Information Element/Group name | Value/Remark |
|---|---|
| -Parameters required for each non-used frequenc | |
| - Threshold non-used frequency | -18 |
| - W non-used frequency | 1 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| -Inter-frequency cell measured results | 1 |
| -Cell measured results (10.3.7.3) | |
| -Cell identity | Not Present |
| -SFN-SFN observed time difference | Not Present |
| -Cell synchronisation info | Not Present |
| -CHOICE mode | FDD |
| -Primary CPICH Info | |
| -Primary scrambling code | Set to Primary scrambling code of Cell2 |
| -CPICH Ec/No | Checked that this IE is present |
| -CPICH RSCP | Not Present |
| -Pathloss | Not Present |
| Measured results on RACH | Not Present |
| Additional measured results | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 2C |
| -Inter-frequency cells | 1 |
| -Frequency Info | |
| -CHOICE *mode* | FDD |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -CHOICE mode | FDD |
| -Primary CPICH info | |
| -Primary Scrambling Code | Set to Primary scrambling code of Cell2 |

### 8.6.3.1.5 Test requirements

#### 8.6.3.1.5.1 3,84 Mcps TDD Option

The UE shall send one Event 2C triggered measurement report for Cell 2 with a measurement reporting delay less than 5040ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of [FFS]% of the cases.

#### 8.6.3.1.5.2 1,28 Mcps TDD Option

The UE shall send one Event 2C triggered measurement report, with a measurement reporting delay less than 5080ms from the beginning of time period T2.

The UE shall not send any measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

#### 8.6.3.1.5.3 7,68 Mcps TDD Option

The UE shall send one Event 2C triggered measurement report for Cell 2 with a measurement reporting delay less than 5040ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

For the test to pass, the total number of successful tests shall be more than 90% with a confidence level of 95% of the cases.

## 8.6.4 GSM measurements

### 8.6.4.1 Correct reporting of GSM neighbours in AWGN propagation condition

#### 8.6.4.1.1 3,84 Mcps TDD option

FFS

#### 8.6.4.1.2 1,28 Mcps TDD option

##### 8.6.4.1.2.1 Definition and applicability

.In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertaint\y resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. The purpose of this test is to verify that the UE makes correct reporting of an event when doing inter frequency measurements on a neighbouring GSM cell. The test will partly verify the requirements in section 8.4 of [2].

The requirements and this test apply to the combined GSM and TDD (1,28Mcps option) UE.

##### 8.6.4.1.2.2 Minimum requirement

The purpose of this test is to verify that the UE makes correct reporting of an event when doing inter-RAT GSM measurements.

##### 8.6.4.1.2.3 Test purpose

To verify that the UE meets the minimum time requirements for identifying neighbouring GSM cells.

##### 8.6.4.1.2.4 Method of test

###### 8.6.4.1.2.4.1 Initial condition of Test 1

Test 1 is with BSIC verification required case.

Two cells shall be present in the test, Cell 1 is current active TDD cell, cell 2 is a GSM cell. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. The test parameters are given in Tables 8.6.4.1.2.1 and 8.6.4.1.2.2 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3B and 3C shall be used. At the start of time duration T1, the UE may not have any timing information of cell 2. The power level of cell 2 is increased and then reduced, and a measurement report should be triggered at each transition.

**Table 8.6.4.1.2.1 General test parameters for Correct reporting of GSM neighbours in AWGN propagation condition**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2. The DPCH is located in an other timeslot than 0. |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Active cell | | Cell 1 | |
| Inter-RAT measurement quantity | | GSM Carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for event 3B and 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before T1 starts. |
| T Identify abort | s | 5.0 | |
| T Reconfirm abort | s | 5.0 | |
| T1 | s | 5 | |
| T2 | s | 7 | |
| T3 | s | 5 | |

**Table 8.6.4.1.2.2 Cell specific test parameters for Correct reporting of GSM neighbours in AWGN propagation condition (cell 1) LCR TDD**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1, T2, T3 | |
| Timeslot Number | | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | |
| DwPCH_Ec/Ior | dB | | 0 |
| OCNS_Ec/Ior | dB | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | |
| $I_{oc}$ | dBm/1.28 MHz | -70 | |
| PCCPCH_RSCP | dBm | -70 | |
| Propagation Condition | | AWGN | |
| Note 1: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior.<br>Note 2: PCCPCH RSCP levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 8.6.4.1.2.3 Cell specific test parameters for Correct reporting of GSM neighbours in AWGN propagation condition (cell 2) GSM**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| Absolute RF Channel Number | | ARFCN 1 | | |
| RXLEV | dBm | -infinity | -75 | -85 |

8.6.4.1.2.4.1 Initial condition of Test 2

Test 2 is without BSIC verification required case.

Two cells shall be present in the test, Cell 1 is current active TDD cell, cell 2 is a GSM cell. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. The test parameters are given in Tables 8.6.4.1.2.4 and 8.6.4.1.2.5 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3B and 3C shall be used. At the start of time duration T1, the UE may not have any timing information of cell 2. The power level of cell 2 is increased and then reduced, and a measurement report should be triggered at each transition.

**Table 8.6.4.1.2.4 General test parameters for Correct reporting of GSM neighbours in AWGN propagation condition**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2. The DPCH is located in an other timeslot than 0. |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Active cell | | Cell 1 | |
| Inter-RAT measurement quantity | | GSM Carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for event 3B and 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before T1 starts. |
| T Identify abort | s | 5.0 | |
| T Reconfirm abort | s | 5.0 | |
| T1 | s | 5 | |
| T2 | s | 2 | |
| T3 | s | 5 | |

**Table 8.6.4.1.2.5 Cell specific test parameters for Correct reporting of GSM neighbours in AWGN propagation condition (cell 1) LCR TDD**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1, T2, T3 | |
| Timeslot Number | | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | |
| DwPCH_Ec/Ior | dB | | 0 |
| OCNS_Ec/Ior | dB | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | |
| $I_{oc}$ | dBm/1.28 MHz | -70 | |
| PCCPCH_RSCP | dBm | -70 | |
| Propagation Condition | | AWGN | |
| Note 1: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior.<br>Note 2: PCCPCH RSCP levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 8.6.4.1.2.6 Cell specific test parameters for Correct reporting of GSM neighbours in AWGN propagation condition (cell 2) GSM**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| Absolute RF Channel Number | | ARFCN 1 | | |
| RXLEV | dBm | -infinity | -75 | -85 |

8.6.4.1.2.5 Test Procedure

8.6.4.1.2.5.1 Test 1

1) The RF parameters are set up according to test T1.( see Table 8.6.4.1.2.2 and Table 8.6.4.1.2.3)

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit the MEASUREMENT CONTROL message.

5) After 5seconds from the beginning of T1, the SS shall switch the GSM level setting from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 6.04s. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

7) After 7seconds from the beginning of T2, the SS shall switch the GSM level setting from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 3B for cell 2. The measurement reporting delay from the beginning of T3 shall be less than 1040ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

9) After the SS receivers the measurement report message in step 8 ) or 5 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat steps 1-9according to Annex F.6.2 Table F.6.2.8.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions.:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity | PCcPCH_RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Required |
| -Inter-RAT reporting quantity (10.3.7.32) | |
| -Reporting cell status (10.3.7.61) | Not Present |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 2 |
| -Inter-RAT event identity (10.3.7.24 | Event 3B |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

First MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3C |
| -Inter-frequency cells | 1 |

Second MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3B |
| -Inter-frequency cells | 1 |

#### 8.6.4.1.2.5.2 Test 2

1) The RF parameters are set up according to test T1.( see Table 8.6.4.1.2.5 and Table 8.6.4.1.2.6)

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit the MEASUREMENT CONTROL message.

5) After 5 seconds from the beginning of T1, the SS shall switch the GSM level setting from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 1040ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

7) After 2 seconds from the beginning of T2, the SS shall switch the GSM level setting from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 3B for cell 2. The measurement reporting delay from the beginning of T3 shall be less than 1040ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

9) After the SS receivers the measurement report message in step 8 ) or 5 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

10) Repeat steps 1-9according to Annex F.6.2 Table F.6.2.8.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions.:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity | PCcPCH_RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Not Required |
| -Inter-RAT reporting quantity (10.3.7.32) | |
| -Reporting cell status (10.3.7.61) | Not Present |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 2 |
| -Inter-RAT event identity (10.3.7.24 | Event 3B |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

First MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |

| | |
|---|---|
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3C |
| -Inter-frequency cells | 1 |

Second MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3B |
| -Inter-frequency cells | 1 |

#### 8.6.4.1.2.6 Test requirements

##### 8.6.4.1.2.6.1 Test 1

The UE shall send one Event 3C triggered measurement report for cell 2, with a measurement reporting delay less than 6.04s from the beginning of time period T2.

The UE shall send one Event 3B triggered measurement report for cell 2, with a measurement reporting delay less than 1040 ms from the beginning of time period T3.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of events correctly reported during repeated tests shall be at least 90%.

##### 8.6.4.1.2.6.2 Test 2

The UE shall send one Event 3C triggered measurement report for cell 2, with a measurement reporting delay less than 1040ms from the beginning of time period T2.

The UE shall send one Event 3B triggered measurement report for cell 2, with a measurement reporting delay less than 1040 ms from the beginning of time period T3.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of events correctly reported during repeated tests shall be at least 90%.

### 8.6.4.1.3 7,68 Mcps TDD option

#### 8.6.4.1.3.1 Definition and applicability

.In the event triggered reporting period the measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertaint\y resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. The purpose of this test is to verify that the UE makes correct reporting of an event when doing inter frequency measurements on a neighbouring GSM cell. The test will partly verify the requirements in section 8.4 of [2].

The requirements and this test apply to the combined GSM and TDD (7,68Mcps option) UE.

8.6.4.1.3.2 Minimum requirement

The purpose of this test is to verify that the UE makes correct reporting of an event when doing inter-RAT GSM measurements.

8.6.4.1.3.3 Test purpose

To verify that the UE meets the minimum time requirements for identifying neighbouring GSM cells.

8.6.4.1.3.4 Method of test

8.6.4.1.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

Two cells shall be present in the test, Cell 1 is current active TDD cell, cell 2 is a GSM cell. The test consists of three successive time periods, with a time duration of T1, T2 and T3 respectively. The test parameters are given in Tables 8.6.4.1.3.4.1.1, 8.6.4.1.3.4.1.2 and 8.6.4.1.3.4.1.3 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event 3B and 3C shall be used. At the start of time duration T1, the UE may not have any timing information of cell 2. The power level of cell 2 is increased and then reduced, and a measurement report should be triggered at each transition.

**Table 8.6.4.1.3.4.1.1: General test parameters for correct reporting of GSM neighbours in AWGN propagation condition**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL Reference Measurement Channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Active cell | | Cell 1 | |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | Required | |
| Threshold other system | dBm | -80 | Absolute GSM carrier RSSI threshold for Events 3B and 3C. |
| Hysteresis | dB | 0 | |
| Time to Trigger | ms | 0 | |
| Filter coefficient | | 0 | |
| Monitored cell list size | | 12 TDD neighbours on Channel 1<br>6 GSM neighbours including ARFCN 1 | Measurement control information is sent before the start of time period T1. |
| $T_{identify\ abort}$ | s | 5 | As specified in section 8.1.2B.5 of [2] |
| $T_{reconfirm\ abort}$ | s | 5 | As specified in section 8.1.2B.5 of [2] |
| T1 | s | 10 | |
| T2 | s | 10 | |
| T3 | s | 10 | |

**Table 8.6.4.1.3.4.1.2: Cell specific parameters for correct reporting of GSM neighbours in AWGN propagation condition (cell 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1, T2, T3 | |
| DL timeslot number | | 0 | 1 |
| UTRA RF Channel number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | n.a. |
| SCH_Ec/Ior | dB | -9 | n.a. |
| SCH_$t_{offset}$ | | 0 | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | Note 2 |
| DPCH_Ec/Ior | dB | n.a. | Note 1 |
| $\hat{I}$or/Ioc | dB | 6 | 6 |
| Io, Note 1 | dBm / 7.68 MHz | -70 | |
| Propagation condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior. | | | |

**Table 8.6.4.1.3.4.1.3: Cell specific parameters for correct reporting of GSM neighbours in AWGN propagation condition (cell 2)**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| Absolute RF Channel Number | | ARFCN 1 | | |
| RXLEV | dBm | -85 | -75 | -85 |

8.6.4.1.3.4.2 Test Procedure

1) The RF parameters are set up according to test T1.( see Table 8.6.4.1.3.4.1.2 and Table 8.6.4.1.3.4.1.3)

2) The UE is switched on.

3) A call is set up according to the generic set-up procedure specified in TS 34.108 [3] subclause 7.4.2 to place the UE in CELL_DCH.

4) SS shall transmit the MEASUREMENT CONTROL message.

5) After 5seconds from the beginning of T1, the SS shall switch the GSM level setting from T1 to T2.

6) UE shall transmit a MEASUREMENT REPORT message triggered by event 3C for cell 2. The measurement reporting delay from the beginning of T2 shall be less than 1040ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

7) After 7seconds from the beginning of T2, the SS shall switch the GSM level setting from T2 to T3.

8) UE shall transmit a MEASUREMENT REPORT message triggered by event 3B for cell 2. The measurement reporting delay from the beginning of T3 shall be less than 1040ms. If the UE fails to report the event within the required delay, then a failure is recorded. If the reporting delay for this event is within the required limit, the number of successful tests is increased by one.

9) After the SS receivers the measurement report message in step 8 ) or 5 seconds from the beginning of T3, the UE is switched off. Any timing information of cell 2 is deleted in the UE.

11) Repeat steps 1-until the confidence level according to annex F.6.2 is achieved.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] with the following exceptions.:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message (step 4):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Event trigger |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-RAT measurement |
| -Inter-RAT measurement (10.3.7.27) | |
| -Inter-RAT measurement objects list (10.3.7.23) | Not Present |
| -Inter-RAT measurement quantity (10.3.7.29) | |
| -Measurement quantity for UTRAN quality estimate (10.3.7.38) | |
| -Filter coefficient | 0 |
| -CHOICE mode | TDD |
| -Measurement quantity | PCCPCH_RSCP |
| -CHOICE system | GSM |
| -Measurement quantity | GSM Carrier RSSI |
| -Filter coefficient | 0 |
| -BSIC verification required | Required |
| -Inter-RAT reporting quantity (10.3.7.32) | Not Present |
| -Reporting cell status (10.3.7.61) | Not Present |
| -CHOICE report criteria | Inter-RAT measurement reporting criteria |
| -Inter-RAT measurement reporting criteria (10.3.7.30) | |
| -Parameters required for each event | 2 |
| -Inter-RAT event identity (10.3.7.24 | Event 3B |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| -Inter-RAT event identity (10.3.7.24) | Event 3C |
| -Threshold own system | Not Present |
| -W | Not Present |
| -Threshold other system | -80 dBm |
| -Hysteresis | 0 dB |
| -Time to trigger | 0 ms |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report cells within active set or within virtual active set or of the other RAT |
| -Maximum number of reported cells | 2 |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

First MEASUREMENT REPORT message (step 6)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3C |
| -Inter-frequency cells | 1 |
| Inter-RAT cell info indication | Not present |

Second MEASUREMENT REPORT message (step 8)

| Information Element | Value/remark |
|---|---|
| Message Type (10.2.17) | |
| Integrity check info | Not Present |
| Measurement identity | 1 |
| Measured Results (10.3.7.44) | |
| -CHOICE Measurement | Inter-frequency Measured results list |
| -Inter-frequency measured results | 1 |
| -Frequency info | |
| -CHOICE *mode* | GSM |
| -UARFCN uplink (Nu) | Not Present |
| -UARFCN downlink (Nd) | Same frequency as channel 2 |
| -UTRA carrier RSSI | Not Present |
| Event results (10.3.7.7) | |
| -CHOICE event result | Inter-frequency measurement event results |
| -Inter-frequency event identity | 3B |
| -Inter-frequency cells | 1 |
| Inter-RAT cell info indication | Not present |

#### 8.6.4.1.3.4 Method of test

The UE shall send one Event 3C triggered measurement report for cell 2, with a measurement reporting delay less than 1040 ms from the start of time period T2.

The UE shall send one Event 3B triggered measurement report for cell 2, with a measurement reporting delay less than 1040 ms from the start of time period T3.

The UE shall not send any Event 3B or 3C triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

## 8.7 Measurements Performance Requirements

Unless explicitly stated:

- Reported measurements shall be within defined range in 90 % of the cases.
- Measurement channel is 12.2 kbps as defined in annex C, sub-clause C.3.1. This measurement channel is used in active cell and cells to be measured.
- Cell 1 is the active cell.
- Single task reporting.

– Power control is active.

# 8.7.1 P-CCPCH RSCP

## 8.7.1.1 Intra frequency measurement accuracy for 3,84 Mcps TDD Option

### 8.7.1.1.1 Absolute accuracy requirement

#### 8.7.1.1.1.1 Definition and applicability

The absolute accuracy of P-CCPCH RSCP is defined as the P-CCPCH RSCP meaasured from one cell compared to the actual P-CCPCH RSCP power from the same cell.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.1.1.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.1.1.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Big|_{in\ dB} \geq -8dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Big|_{in\ dB} \geq -13dB$$

**Table 8.7.1.1.1.1: P-CCPCH_RSCP absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3.84MHz ] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.1 and A.9.1.1.1.1.

#### 8.7.1.1.1.3 Test Purpose

The purpose of this test is to verify that the absolute P-CCPCH RSCP measurement accuracy is within the specified limits.

#### 8.7.1.1.1.4 Method of test

##### 8.7.1.1.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. P-CCPCH RSCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.1.1.1.2.

**Table 8.7.1.1.1.2: P-CCPCH RSCP intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 0 | 0 | 0 | 0 | 0 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.7 | | -59.8 | | -98.7 | |
| Îor/Ioc | dB | 5 | 2 | 9 | 2 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -73.7 | -76.7 | -53.8 | -60.8 | -98.7 | -101.7 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1.1.2.

#### 8.7.1.1.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value in MEASUREMENT REPORT messages. PCCPCH_RSCP power of Cell 1 reported by UE is compared to actual PCCPCH_RSCP power for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>1<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

### 8.7.1.1.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.1.1.2 Relative accuracy requirement

#### 8.7.1.1.2.1 Definition and applicability

The relative accuracy of PCCPCH RSCP is defined as the PCCPCH RSCP measured from one cell compared to the PCCPCH RSCP measured from another cell on the same frequency.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.1.1.2.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.1.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left( \frac{P-CCPCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \geq -8dB$$

$$\left( \frac{SCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \geq -13dB$$

$$\left| \text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB} \right| \leq 20dB$$

Relative Io difference [dB] ≤ relative RSCP difference [dB]

It is assumed that the measurements of P-CCPCH RSCP1 and P-CCPCH RSCP2 can be performed within 20ms due to slot allocations in the cells concerned.

**Table 8.7.1.1.2.1: P-CCPCH_RSCP intra-frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions | |
|---|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3.84MHz] | relative RSCP difference [dB] |
| P-CCPCH_RSCP | dBm | ±1 | ±1 | -94...-50 | <2 |
| | | ±2 | ±2 | | 2...14 |
| | | ±3 | ± 3 | | >14 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2 and A.9.1.1.1.1.

#### 8.7.1.1.2.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits.

#### 8.7.1.1.2.4 Method of test

##### 8.7.1.1.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. P-CCPCH RSCP intra frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.1.1.2.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1.1.2.

#### 8.7.1.1.2.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 3) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for intra frequency measurement in clause 8.7.1.1.1.4.2 shall be used.

#### 8.7.1.1.2.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1.2.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.1.1A Intra frequency measurement accuracy for 1,28 Mcps TDD Option

### 8.7.1.1A.1 Absolute accuracy requirement

#### 8.7.1.1A.1.1 Definition and applicability

The absolute accuracy of P-CCPCH RSCP is defined as the P-CCPCH RSCP measured from one cell compared to the actual P-CCPCH RSCP power from the same cell.

The requirements and this test apply to all types of UTRA TDD UE 1,28 Mcps option.

#### 8.7.1.1A.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.1.1A.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left( \frac{P-CCPCH_E_c}{I_o} \right) \Bigg|_{in\ dB} \geq -8dB$$

$$\left( \frac{DwPCH_E_c}{I_o} \right) \Bigg|_{in\ dB} \geq -5dB$$

**Table 8.7.1.1A.1.1: P-CCPCH_RSCP absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1.28MHz] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The rate of correct measurements observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.1 and A.9.2.1.1.1.

### 8.7.1.1A.1.3 Test Purpose

The purpose of this test is to verify that the absolute P-CCPCH RSCP measurement accuracy is within the specified limits.

### 8.7.1.1A.1.4 Method of test

#### 8.7.1.1A.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 5 and the UL DPCH shall be transmitted in timeslot 2. P-CCPCH RSCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.1.1A.1.2.

**Table 8.7.1.1A.1.2: P-CCPCH RSCP Intra frequency test parameters**

| | Test 1 | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 2 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -76.6 | | | |
| PCCPCH RSCP, Note 1 | dBm | -74.6 | | -77.6 | |
| Io, Note 1 | dBm/ 1,28 MHz | -69 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 2** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | | 2 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -60.2 | | | |
| PCCPCH RSCP, Note 1 | dBm | -54.2 | | -61.2 | |
| Io, Note 1 | dBm/ 1,28 MHz | -50 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 3** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 3 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -101.9 | | | |
| PCCPCH RSCP, Note 1 | dBm | -99.9 | | -101.9 | |
| Io, Note 1 | dBm/ 1,28 MHz | -94 | | | |
| Propagation condition | | AWGN | | | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1A.1.2.

8.7.1.1A.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value in MEASUREMENT REPORT messages. PCCPCH_RSCP power of Cell 1 reported by UE is compared to actual PCCPCH_RSCP power for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1A.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1A.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>setup<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>1<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

### 8.7.1.1A.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1A.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.1.1A.2 Relative accuracy requirement

#### 8.7.1.1A.2.1 Definition and applicability

The relative accuracy of PCCPCH RSCP is defined as the PCCPCH RSCP measured from one cell compared to the PCCPCH RSCP measured from another cell on the same frequency.

The requirements and this test apply to all types of UTRA TDD UE 1,28 Mcps option.

#### 8.7.1.1A.2.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.1A.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -8dB$$

$$\left(\frac{DwPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -5dB$$

$$\left|\text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB}\right| \leq 20dB$$

Relative Io difference [dB] ≤ relative RSCP difference [dB]

It is assumed that the measurements of P-CCPCH RSCP1 and P-CCPCH RSCP2 can be performed within 20ms due to slot allocations in the cells concerned.

**Table 8.7.1.1A.2.1: P-CCPCH_RSCP intra-frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions | |
|---|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1.28MHz] | relative RSCP difference [dB] |
| P-CCPCH_RSCP | dBm | ±1 | ±1 | -94...-50 | <2 |
| | | ±2 | ±2 | | 2...14 |
| | | ±3 | ± 3 | | >14 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2 and A.9.2.1.1.1.

#### 8.7.1.1A.2.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits.

#### 8.7.1.1A.2.4 Method of test

##### 8.7.1.1A.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 5 and the UL DPCH shall be transmitted in timeslot 2. P-CCPCH RSCP intra frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.1A.1.2.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1A.1.2.

#### 8.7.1.1A.2.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 3) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1A.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1A.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for intra frequency measurement in clause 8.7.1.1A.1.4.2 shall be used.

#### 8.7.1.1A.2.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1A.2.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.1.1B Intra frequency measurement accuracy for 7,68 Mcps TDD Option

### 8.7.1.1B.1 Absolute accuracy requirement

#### 8.7.1.1B.1.1 Definition and applicability

The absolute accuracy of P-CCPCH RSCP is defined as the P-CCPCH RSCP meaasured from one cell compared to the actual P-CCPCH RSCP power from the same cell.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.1.1B.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.1.1B.1.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left( \frac{P-CCPCH_E_c}{I_o} \right) \Bigg|_{in\ dB} \geq -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -13dB$$

**Table 8.7.1.1B.1.2.1: P-CCPCH_RSCP absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3.84MHz ] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.1.3 and A.9.3.1.1.1.

### 8.7.1.1B.1.3 Test Purpose

The purpose of this test is to verify that the absolute P-CCPCH RSCP measurement accuracy is within the specified limits.

### 8.7.1.1B.1.4 Method of test

#### 8.7.1.1B.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. P-CCPCH RSCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.1.1B.1.4.1.1.

**Table 8.7.1.1B.1.4.1.1: P-CCPCH RSCP intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 0 | 0 | 0 | 0 | 0 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.7 | | -59.8 | | -98.7 | |
| Îor/Ioc | dB | 5 | 2 | 9 | 2 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -73.7 | -76.7 | -53.8 | -60.8 | -98.7 | -101.7 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1B.1.4.1.1.

#### 8.7.1.1B.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value in MEASUREMENT REPORT messages. PCCPCH_RSCP power of Cell 1 reported by UE is compared to actual PCCPCH_RSCP power for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria*<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>1<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

### 8.7.1.1B.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1B.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.1.1B.2 Relative accuracy requirement

#### 8.7.1.1B.2.1 Definition and applicability

The relative accuracy of PCCPCH RSCP is defined as the PCCPCH RSCP measured from one cell compared to the PCCPCH RSCP measured from another cell on the same frequency.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.1.1B.2.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.1B.2.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -13dB$$

$$\left|\text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB}\right| \leq 20dB$$

Relative Io difference [dB] ≤ relative RSCP difference [dB]

It is assumed that the measurements of P-CCPCH RSCP1 and P-CCPCH RSCP2 can be performed within 20ms due to slot allocations in the cells concerned.

**Table 8.7.1.1B.2.2.1: P-CCPCH_RSCP intra-frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions | |
|---|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 7.68MHz] | relative RSCP difference [dB] |
| P-CCPCH_RSCP | dBm | ±1 | ±1 | -94...-50 | <2 |
| | | ±2 | ±2 | | 2...14 |
| | | ±3 | ± 3 | | >14 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2.3 and A.9.3.1.1.1.

#### 8.7.1.1B.2.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits.

#### 8.7.1.1B.2.4 Method of test

##### 8.7.1.1B.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. P-CCPCH RSCP intra frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.1B.1.4.1.1.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.1.1.2.

#### 8.7.1.1B.2.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 3) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for intra frequency measurement in clause 8.7.1.1B.1.4.2 shall be used.

#### 8.7.1.1B.2.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.1B.2.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.1.2 Inter frequency measurement accuracy for 3,84 Mcps TDD Option

#### 8.7.1.2.1 Relative accuracy requirement

##### 8.7.1.2.1.1 Definition and applicability

The P-CCPCH_RSCP inter-frequency relative accuracy is defined as the P-CCPCH_RSCP measured from one cell compared to the P-CCPCH_RSCP measured from another cell on a different frequency.

The requirements and this test apply to all types of UTRA TDD UE.

##### 8.7.1.2.1.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.2.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left| \text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB} \right| \leq 20dB$$

$$\left( \frac{P-CCPCH_E_c}{I_o} \right)\Big|_{in\ dB} \geq -8dB$$

$$\left( \frac{SCH_E_c}{I_o} \right)\Big|_{in\ dB} \geq -13dB$$

**Table 8.7.1.2.1.1 P-CCPCH_RSCP inter-frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3.84MHz ] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 6 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2 and A.9.1.1.

8.7.1.2.1.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits for the inter frequency case.

8.7.1.2.1.4 Method of test

8.7.1.2.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. P-CCPCH RSCP inter frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.2.1.2.

**Table 8.7.1.2.1.2: P-CCPCH RSCP inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.2 | -75.2 | -57.8 | -54.1 | -98.7 | -97 |
| Îor/Ioc | dB | 5 | 5 | 7 | 2 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -73.2 | -73.2 | -54.8 | -55.1 | -98.7 | -100 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.2.1.2.

#### 8.7.1.2.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL messages for intra frequency and inter frequency measurements.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 5) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.2.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

First MEASUREMENT CONTROL message for intra frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | Not Present |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 250 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

Second MEASUREMENT CONTROL message for inter frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

### 8.7.1.2.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.2.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.1.2A Inter frequency measurement accuracy for 1,28 Mcps TDD Option

### 8.7.1.2A.1 Relative accuracy requirement

#### 8.7.1.2A.1.1 Definition and applicability

The P-CCPCH_RSCP inter-frequency relative accuracy is defined as the P-CCPCH_RSCP measured from one cell compared to the P-CCPCH_RSCP measured from another cell on a different frequency.

The requirements and this test apply to all types of UTRA TDD UE 1,28 Mcps option.

#### 8.7.1.2A.1.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.2A.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left| \text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB} \right| \le 20dB$$

$$\left( \frac{P-CCPCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \ge -8dB$$

$$\left( \frac{DwPCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \ge -5dB$$

**Table 8.7.1.2A.1.1 P-CCPCH_RSCP inter-frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1.28MHz] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 6 | -94...-50 |

The rate of correct measurements observed during repeated tests shall be at least 90%.

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2 and A.9.2.1.

#### 8.7.1.2A.1.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits for the inter frequency case.

#### 8.7.1.2A.1.4 Method of test

##### 8.7.1.2A.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. P-CCPCH RSCP inter frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.2A.1.2.

**Table 8.7.1.2A.1.2: P-CCPCH RSCP Inter frequency test parameters**

| | **Test 1** | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 5 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -75.2 | | -75.2 | |
| PCCPCH RSCP, Note 1 | dBm | -73.2 | | -73.2 | |
| Io, Note 1 | dBm/ 1,28 MHz | -69 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 2** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | | 2 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -57.8 | | -54.1 | |
| PCCPCH RSCP, Note 1 | dBm | -53.8 | | -55.1 | |
| Io, Note 1 | dBm/ 1,28 MHz | -50 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 3** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | 0 | |
| $I_{oc}$ | dBm/ 1,28 MHz | -98.7 | | -97 | |
| PCCPCH RSCP, Note 1 | dBm | -98.7 | | -100 | |
| Io, Note 1 | dBm/ 1,28 MHz | -94 | | | |
| Propagation condition | | AWGN | | | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.2A.1.2.

8.7.1.2A.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 3) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.2A.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

First MEASUREMENT CONTROL message for Intra frequency measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| UE information elements<br>-RRC transaction identifier | <br>0 |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command<br>-Measurement Reporting Mode<br>- Measurement Report Transfer Mode<br>- Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurement list<br>-CHOICE Measurement Type<br>-Intra-frequency measurement<br>- Intra-frequency measurement objects list<br>-Intra-frequency cell info list<br>-Intra-frequency measurement quantity<br>-Filter coefficient<br>-CHOICE mode<br>-Measurement quantity<br>-Intra-frequency reporting quantity<br>-Reporting quantities for active set cells<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE mode<br>- Timeslot ISCP reporting indicator<br>- Proposed TGSN Reporting required<br>- Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator<br>-Reporting quantities for monitored set cells<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE mode<br>- Timeslot ISCP reporting indicator<br>- Proposed TGSN Reporting required<br>PCCPCH RSCP reporting indicator<br>-Pathloss reporting indicator<br>-Reporting quantities for detected set cells<br>-Reporting cell status<br>-CHOICE reported cell<br><br>-Maximum number of reported cells<br>-Measurement validity<br>-CHOICE *report criteria*<br>-Amount of reporting<br>-Reporting interval | <br>1<br>setup<br>Acknowledged mode RLC<br>Periodical reporting<br><br>Not Present<br>Intra-frequency measurement<br><br><br>Not Present<br><br>0<br>TDD<br>PCCPCH RSCP<br><br><br>TRUE<br><br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE<br><br>FALSE<br><br>TRUE<br>FDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE<br>Not Present<br><br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br>Periodical reporting criteria<br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info | <br>Not Present |

Second MEASUREMENT CONTROL message for Inter frequency measurement (step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| UE information elements<br>-RRC transaction identifier | <br>0 |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command<br>-Measurement Reporting Mode<br>- Measurement Report Transfer Mode<br>- Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurement list<br>-CHOICE Measurement Type<br>-Inter-frequency measurement object list<br>-CHOICE Inter-frequency cell removal<br>-New inter-frequency cells<br>-Cell for measurement<br>-Inter-frequency measurement quantity<br>-CHOICE reporting criteria<br>-Filter coefficient<br>-CHOICE mode<br>-Measurement quantity for frequency quality estimate<br>-Inter-frequency reporting quantity<br>-UTRA Carrier RSSI<br>-Frequency quality estimate<br>-Non frequency related cell reporting quantities<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE mode<br>-Timeslot ISCP reporting indicator<br>- Proposed TGSN Reporting required<br>-PCCPCH RSCP reporting indicator<br>-Pathloss reporting indicator<br>-Reporting cell status<br>-CHOICE reported cell<br><br>-Maximum number of reported cells<br>-Measurement validity<br>-Inter-frequency set update<br>-CHOICE report criteria<br>-Amount of reporting<br>-Reporting interval | <br>2<br>Setup<br><br>Acknowledged mode RLC<br>Periodical reporting<br><br>Not Present<br>Inter-frequency measurement<br><br>Not Present<br>Cell 2 information is included<br>Not Present<br><br>Inter-frequency reporting criteria<br>0<br>TDD<br>PCCPCH RSCP<br><br><br>TRUE<br>TRUE<br><br>TRUE<br><br>TRUE<br>FDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE<br>Report cells within monitored set on non-used frequency<br><br>2<br>Not Present<br>Not Present<br><br>Periodical reporting criteria<br>Infinity<br>500 ms |
| Physical channel information elements<br>-DPCH compressed mode status info | <br>Not Present |

MEASUREMENT REPORT message for Inter frequency test cases

This message is common for all inter frequency test cases in clause 8.7 and is described in Annex I.

### 8.7.1.2A.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.2A.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.1.2B Inter frequency measurement accuracy for 7,68 Mcps TDD Option

### 8.7.1.2B.1 Relative accuracy requirement

#### 8.7.1.2B.1.1 Definition and applicability

The P-CCPCH_RSCP inter-frequency relative accuracy is defined as the P-CCPCH_RSCP measured from one cell compared to the P-CCPCH_RSCP measured from another cell on a different frequency.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.1.2B.1.2 Minimum Requirements

The relative accuracy requirements in table 8.7.1.2B.1.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left| \text{P - CCPCH RSCP1}\big|_{in\ dB} - \text{P - CCPCH RSCP2}\big|_{in\ dB} \right| \le 20dB$$

$$\left( \frac{P-CCPCH_E_c}{I_o} \right)\Big|_{in\ dB} \ge -11dB$$

$$\left( \frac{SCH_E_c}{I_o} \right)\Big|_{in\ dB} \ge -13dB$$

Table 8.7.1.2B.1.2.1 P-CCPCH_RSCP inter-frequency relative accuracy

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/7.68 MHz] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 6 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.1.2.3 and A.9.3.1.

#### 8.7.1.2B.1.3 Test Purpose

The purpose of this test is to verify that the relative P-CCPCH RSCP measurement accuracy is within the specified limits for the inter frequency case.

#### 8.7.1.2B.1.4 Method of test

##### 8.7.1.2B.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. P-CCPCH RSCP inter frequency relative accuracy requirements are tested by using test parameters in table 8.7.1.2B.1.4.1.1.

**Table 8.7.1.2B.1.4.1.1: P-CCPCH RSCP inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.2 | -75.2 | -57.8 | -54.1 | -98.7 | -97 |
| Îor/Ioc | dB | 5 | 5 | 7 | 2 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -73.2 | -73.2 | -54.8 | -55.1 | -98.7 | -100 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.1.2B.1.4.1.1.

### 8.7.1.2B.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL messages for intra frequency and inter frequency measurements.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check PCCPCH_RSCP value of Cell 1 and Cell 2 in MEASUREMENT REPORT messages. PCCPCH RSCP power value measured from Cell 1 is compared to PCCPCH RSCP power value measured from Cell 2 for each MEASUREMENT REPORT message.

4) The result of step 5) is compared to actual power level difference of PCCPCH RSCP of Cell 1 and Cell 2.

5) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.1.2.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 3) and 4) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

First MEASUREMENT CONTROL message for intra frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>1<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

Second MEASUREMENT CONTROL message for inter frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

### 8.7.1.2B.1.5 Test requirements

The PCCPCH RSCP measurement accuracy shall meet the requirements in clause 8.7.1.2B.1.2 for at least 900 of the measurement reports at each input level in step 4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.2 CPICH measurements (FDD)

### 8.7.2.1 CPICH RSCP

#### 8.7.2.1.1 Absolute measurement accuracy for 3,84 Mcps TDD Option

##### 8.7.2.1.1.1 Definition and applicability

The absolute accuracy of CPICH RSCP is defined as the CPICH RSCP measured in an UTRA FDD cell on one frequency compared to the actual CPICH RSCP power of that cell on the same frequency.

The requirements and this test apply only to UE supporting both UTRA TDD and UTRA FDD.

##### 8.7.2.1.1.2 Minimum Requirements

The accuracy requirements in table 8.7.2.1.1.1 are valid under the following conditions:

- $CPICH_RSCP1,2|_{dBm} \geq -114$ dBm.

- $\left| \frac{I_o}{\left(\hat{I}_{or}\right)} \right|_{in\ dB} - \left| \left( \frac{CPICH_E_c}{I_{or}} \right) \right|_{in\ dB} \leq 20dB$.

**Table 8.7.2.1.1.1: CPICH RSCP inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 3,84 MHz] |
| CPICH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.2.1 and A.9.1.2.1.

##### 8.7.2.1.1.3 Test purpose

The purpose of this test is to verify that the CPICH RSCP absolute measurement accuracy is within the specified limits.

##### 8.7.2.1.1.4 Method of test

###### 8.7.2.1.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 is a UTRA TDD cell and cell 2 is a UTRA FDD cell. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. No second Beacon timeslot shall be provided for cell 1. CPICH RSCP inter frequency absolute accuracy requirements are tested by using test parameters in table 8.7.2.1.1.2.

**Table 8.7.2.1.1.2: CPICH RSCP inter frequency tests parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | n.a. | 0 | n.a. |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| CPICH_Ec/Ior | dB | n.a. | -10 | n.a. | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -12 | -3 | -12 |
| SCH_Ec/Ior | dB | -9 | -12 | -9 | -12 |
| SCH_$t_{offset}$ | | 5 | n.a. | 5 | n.a. |
| PICH_Ec/Ior | dB | n.a. | -15 | n.a. | -15 |
| OCNS_Ec/Ior | dB | -3.12 | -0.94 | -3.12 | -0.94 |
| Ioc | dBm/ 3,84 MHz | -57.7 | -60 | -84.7 | -84 |
| $\hat{I}$or/Ioc | dB | 7 | 9.54 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -53.7 | n.a. | -84.7 | n.a. |
| CPICH RSCP, Note 1 | dBm | n.a. | -60.46 | n.a. | -94 |
| Io, Note 1 | dBm/ 3,84 MHz | -50 | -50 | -80 | -81 |
| Propagation condition | - | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP, CPICH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.2.1.1.2.

#### 8.7.2.1.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurement.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check CPICH RSCP value of Cell 2 in the MEASUREMENT REPORT messages. CPICH RSCP levels of Cell 2 reported by the UE is compared to the actual CPICH RSCP value of Cell 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.2.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 2 |
| -Measurement Command | Setup |
| -Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurement list | Not Present |
| -CHOICE Measurement Type | Inter-frequency measurement |
| -Inter-frequency measurement object list | |
| -CHOICE Inter-frequency cell removal | Not Present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity | |
| -CHOICE reporting criteria | Inter-frequency reporting criteria |
| -Filter coefficient | 0 |
| -CHOICE mode | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity | Not Present |
| -Inter-frequency set update | Not Present |
| -CHOICE report criteria | Periodical reporting criteria |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info | Not Present |

#### 8.7.2.1.1.5 Test requirements

The CPICH RSCP measurement accuracy shall meet the requirements in clause 8.7.2.1.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.2.1A.1 Absolute measurement accuracy for 1,28 Mcps TDD Option

#### 8.7.2.1A.1.1 Definition and applicability

The absolute accuracy of CPICH RSCP is defined as the CPICH RSCP measured in an UTRA FDD cell on one frequency compared to the actual CPICH RSCP power of that cell on the same frequency.

The requirements and this test apply only to UE supporting both UTRA TDD and UTRA FDD.

8.7.2.1A.1.2 Minimum Requirements

The accuracy requirements in table 8.7.2.1A.1.1 are valid under the following conditions:

- CPICH_RSCP1,2$|_{dBm} \geq$ -114 dBm.

- $\left.\frac{I_o}{\left(\hat{I}_{or}\right)}\right|_{in\ dB} - \left.\left(\frac{CPICH_E_c}{I_{or}}\right)\right|_{in\ dB} \leq 20dB$.

**Table 8.7.2.1A.1.1: FDD CPICH RSCP inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 3,84 MHz] |
| CPICH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.2.1 and A.9.1.2.1.

8.7.2.1A.1.3 Test purpose

The purpose of this test is to verify that the CPICH RSCP absolute measurement accuracy is within the specified limits.

8.7.2.1A.1.4 Method of test

8.7.2.1A.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 is a UTRA 1,28Mcps TDD cell and cell 2 is a UTRA FDD cell. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. No second Beacon timeslot shall be provided for cell 1. CPICH RSCP inter frequency absolute accuracy requirements are tested by using test parameters in table 8.7.2.1A.1.2.

**Table 8.7.2.1A.1.2: CPICH RSCP inter frequency tests parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 DwPTS | n.a. | 0 DwPTS | n.a. |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| CPICH_Ec/Ior | dB | n.a. | -10 | n.a. | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -12 | -3 | -12 |
| DwPCH_Ec/Ior | dB | 0 | n.a. | 0 | n.a. |
| SCH_Ec/Ior | dB | n.a. | -12 | n.a. | -12 |
| PICH_Ec/Ior | dB | n.a. | -15 | n.a. | -15 |
| OCNS_Ec/Ior | dB | -3 | -0.94 | -3 | -0.94 |
| Ioc, Note 2 | dBm/ 3.84 MHz | n.a. | -60 | n.a. | -84 |
| Ioc, Note 2 | dBm/ 1.28 MHz | -57.7 | n.a. | -84.7 | n.a. |
| | | | | | |
| Îor/Ioc | dB | 7 | 9.54 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -53.7 | n.a. | -84.7 | n.a. |
| CPICH RSCP, Note 1 | dBm | n.a. | -60.46 | n.a. | -94 |
| Io, Notes 1, 2 | dBm/3.84 MHz | n.a. | -50 | n.a. | -81 |
| Io, Notes 1, 2 | dBm/1.28 MHz | -50 | n.a. | -80 | n.a. |
| Propagation condition | - | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP, CPICH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 2: Ioc and Io are given independently for TDD and FDD cells. | | | | | |

A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.2.1A.1.2.

#### 8.7.2.1A.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurement.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check CPICH RSCP value of Cell 2 in the MEASUREMENT REPORT messages. CPICH RSCP levels of Cell 2 reported by the UE is compared to the actual CPICH RSCP value of Cell 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE.

5) After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.2.1A.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 2 |
| -Measurement Command | Setup |
| -Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurement list | Not Present |
| -CHOICE Measurement Type | Inter-frequency measurement |
| -Inter-frequency measurement object list | |
| -CHOICE Inter-frequency cell removal | Not Present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity | |
| -CHOICE reporting criteria | Inter-frequency reporting criteria |
| -Filter coefficient | 0 |
| -CHOICE mode | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity | Not Present |
| -Inter-frequency set update | Not Present |
| -CHOICE report criteria | Periodical reporting criteria |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info | Not Present |

#### 8.7.2.1.1.5 Test requirements

The CPICH RSCP measurement accuracy shall meet the requirements in clause 8.7.2.1A.1.2 in at least 900 of the measurements in each test.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.2.1B.1 Absolute measurement accuracy for 7,68 Mcps TDD Option

#### 8.7.2.1B.1.1 Definition and applicability

The absolute accuracy of CPICH RSCP is defined as the CPICH RSCP measured in an UTRA FDD cell on one frequency compared to the actual CPICH RSCP power of that cell on the same frequency.

The requirements and this test apply only to UE supporting both UTRA TDD and UTRA FDD.

### 8.7.2.1B.1.2 Minimum Requirements

The accuracy requirements in table 8.7.2.1B.1.2.1 are valid under the following conditions:

- CPICH_RSCP1,2$|_{dBm} \geq$ -114 dBm.

- $\left.\frac{I_o}{\left(\hat{I}_{or}\right)}\right|_{in\ dB} - \left.\left(\frac{CPICH_E_c}{I_{or}}\right)\right|_{in\ dB} \leq 20dB$.

**Table 8.7.2.1B.1.2.1: CPICH RSCP inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 3,84 MHz] |
| CPICH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.2.1 and A.9.3.2.1.

### 8.7.2.1B.1.3 Test purpose

The purpose of this test is to verify that the CPICH RSCP absolute measurement accuracy is within the specified limits.

### 8.7.2.1B.1.4 Method of test

#### 8.7.2.1B.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 is a UTRA TDD cell and cell 2 is a UTRA FDD cell. The DL DPCH shall be transmitted in timeslot 1 and the UL DPCH shall be transmitted in timeslot 3. No second Beacon timeslot shall be provided for cell 1. CPICH RSCP inter frequency absolute accuracy requirements are tested by using test parameters in table 8.7.2.1B.1.4.1.1.

**Table 8.7.2.1B.1.4.1.1: CPICH RSCP inter frequency tests parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | n.a. | 0 | n.a. |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| CPICH_Ec/Ior | dB | n.a. | -10 | n.a. | -10 |
| PCCPCH_Ec/Ior | dB | -3 | -12 | -3 | -12 |
| SCH_Ec/Ior | dB | -9 | -12 | -9 | -12 |
| SCH_$t_{offset}$ | | 5 | n.a. | 5 | n.a. |
| PICH_Ec/Ior | dB | n.a. | -15 | n.a. | -15 |
| OCNS_Ec/Ior | dB | -3.12 | -0.94 | -3.12 | -0.94 |
| Ioc | dBm/ 3,84 MHz | -57.7 | -60 | -84.7 | -84 |
| Îor/Ioc | dB | 7 | 9.54 | 3 | 0 |
| PCCPCH RSCP, Note 1 | dBm | -53.7 | n.a. | -84.7 | n.a. |
| CPICH RSCP, Note 1 | dBm | n.a. | -60.46 | n.a. | -94 |
| Io, Note 1 | dBm/ 3,84 MHz | -50 | -50 | -80 | -81 |
| Propagation condition | - | AWGN | | AWGN | |
| NOTE 1: PCCPCH RSCP, CPICH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.2.1B.1.4.1.1.

#### 8.7.2.1B.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurement.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check CPICH RSCP value of Cell 2 in the MEASUREMENT REPORT messages. CPICH RSCP levels of Cell 2 reported by the UE is compared to the actual CPICH RSCP value of Cell 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.2.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

7) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 2 |
| -Measurement Command | Setup |
| -Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurement list | Not Present |
| -CHOICE Measurement Type | Inter-frequency measurement |
| -Inter-frequency measurement object list | |
| -CHOICE Inter-frequency cell removal | Not Present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity | |
| -CHOICE reporting criteria | Inter-frequency reporting criteria |
| -Filter coefficient | 0 |
| -CHOICE mode | FDD |
| -Measurement quantity for frequency quality estimate | CPICH RSCP |
| -Inter-frequency reporting quantity | |
| -UTRA Carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE mode | FDD |
| -CPICH Ec/N0 reporting indicator | FALSE |
| -CPICH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity | Not Present |
| -Inter-frequency set update | Not Present |
| -CHOICE report criteria | Periodical reporting criteria |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info | Not Present |

#### 8.7.2.1B.1.5 Test requirements

The CPICH RSCP measurement accuracy shall meet the requirements in clause 8.7.2.1B.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.2.2 CPICH Ec/Io

Void

NOTE: This section is included for consistency with numbering in TS 25.123 [2] currently no test covering requirements in section 9.1.1.3 of [2]exists.

## 8.7.3 Timeslot ISCP

### 8.7.3.1 Intra frequency measurement accuracy for 3,84 Mcps TDD Option

#### 8.7.3.1.1 Absolute accuracy requirement

##### 8.7.3.1.1.1 Definition and applicability

The absolute accuracy of Timeslot ISCP is defined as the Timeslot ISCP meaasured from one cell / timeslot combination compared to the actual Timeslot ISCP level for the same cell / timeslot combination.

The requirements and this test apply to all types of UTRA TDD UE.

##### 8.7.3.1.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.3.1.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left( \frac{P-CCPCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \geq -8dB$$

$$\left( \frac{SCH_E_c}{I_o} \right)\Bigg|_{in\ dB} \geq -13dB$$

**Table 8.7.3.1.1.1: UE Timeslot ISCP intra frequency absolute accuracy (3,84Mcps option)**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 3.84 MHz] |
| Timeslot_ISCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.2.1.2.1.1 and A.9.1.3.

##### 8.7.3.1.1.3 Test Purpose

The purpose of this test is to verify that the Timeslot ISCP measurement accuracy is within the specified limits.

##### 8.7.3.1.1.4 Method of test

###### 8.7.3.1.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The Timeslot ISCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.3.1.1.2.

**Table 8.7.3.1.1.2: Timeslot ISCP intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 0 | 0 | 0 | 0 | 0 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.7 | | -59.8 | | -98.7 | |
| Îor/Ioc | dB | 5 | 2 | 9 | 2 | 3 | 0 |
| Timeslot ISCP, Note 1 | dBm | -73.7 | -70.7 | -57.8 | -50.8 | -98.7 | -95.7 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Timeslot ISCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.3.1.1.2.

#### 8.7.3.1.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check Timeslot ISCP values for Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations in MEASUREMENT REPORT messages. These Timeslot ISCP values reported by the UE are compared to the actual Timeslot ISCP levels for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.3.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.3.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | |
| -CHOICE *Intra-frequency cell removal* | Not present |
| -New intra-frequency cells | 2 |
| -Intra-frequency cell id | 1 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -CHOICE *Sync case* | 2 |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 1 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 1 as described in Table 8.7.3.1.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency cell id | 2 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 3,84 Mcps TDD |
| -CHOICE *Sync case* | 2 |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.7.3.1.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |

| Information Element/Group name | Value/Remark |
|---|---|
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual / active set cells + 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

#### 8.7.3.1.1.5 Test requirements

The Timeslot ISCP measurement accuracy shall meet the requirements in clause 8.7.3.1.1.2 for at least 900 of the reported Timeslot ISCP levels at each input level in step 4 for both Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.3.1A Intra frequency measurement accuracy for 1,28 Mcps TDD Option

### 8.7.3.1A.1 Absolute accuracy requirement

#### 8.7.3.1A.1.1 Definition and applicability

The absolute accuracy of Timeslot ISCP is defined as the Timeslot ISCP meaasured from one cell / timeslot combination compared to the actual Timeslot ISCP level for the same cell / timeslot combination.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.3.1A.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.3.1A.1.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left( \frac{P-CCPCH_E_c}{I_o} \right) \Big|_{in\ dB} \geq -8dB$$

$$\left( \frac{DwPCH_E_c}{I_o} \right) \Big|_{in\ dB} \geq -5dB$$

**Table 8.7.3.1A.1.1: UE TDD Timeslot ISCP intra frequency absolute accuracy (1,28 Mcps option)**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 1.28MHz] |
| Timeslot_ISCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.2.1.2.1.2

### 8.7.3.1A.1.3 Test Purpose

The purpose of this test is to verify that the Timeslot ISCP measurement accuracy is within the specified limits.

### 8.7.3.1A.1.4 Method of test

#### 8.7.3.1A.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The Timeslot ISCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.3.1A.1.2.

**Table 8.7.3.1A.1.2: Timeslot ISCP intra frequency test parameters**

| | Test 1 | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 2 | |
| $I_{oc}$ | dBm/1.28 MHz | -76.6 | | | |
| TS ISCP, Note 1 | dBm | -74.6 | | -71.6 | |
| Io, Note 1 | dBm/1.28 MHz | -69 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 2** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 9 | | 2 | |
| $I_{oc}$ | dBm/1.28 MHz | -60.2 | | | |
| TS ISCP, Note 1 | dBm | -58.2 | | -51.2 | |
| Io, Note 1 | dBm/1.28 MHz | -50 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 3** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 3 | |
| $I_{oc}$ | dBm/1.28 MHz | -101.9 | | | |
| TS ISCP, Note 1 | dBm | -98.9 | | -96.9 | |
| Io, Note 1 | dBm/1.28 MHz | -94 | | | |
| Propagation condition | | AWGN | | | |
| NOTE 1: TS ISCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.3.1A.1.2.

8.7.3.1A.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check Timeslot ISCP values for Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations in MEASUREMENT REPORT messages. These Timeslot ISCP values reported by the UE are compared to the actual Timeslot ISCP levels for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.3.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.3.1A.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | |
| -CHOICE *Intra-frequency cell removal* | Not present |
| -New intra-frequency cells | 2 |
| -Intra-frequency cell id | 1 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps TDD |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 1 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 1 as described in Table 8.7.3.1A.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency cell id | 2 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 1,28 Mcps TDD |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.7.3.1A.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |

| Information Element/Group name | Value/Remark |
|---|---|
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual / active set cells + 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

#### 8.7.3.1A.1.5 Test requirements

The Timeslot ISCP measurement accuracy shall meet the requirements in clause 8.7.3.1A.1.2 for at least 900 of the reported Timeslot ISCP levels at each input level in step 4 for both Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.3.1B Intra frequency measurement accuracy for 7,68 Mcps TDD Option

### 8.7.3.1B.1 Absolute accuracy requirement

#### 8.7.3.1B.1.1 Definition and applicability

The absolute accuracy of Timeslot ISCP is defined as the Timeslot ISCP meaasured from one cell / timeslot combination compared to the actual Timeslot ISCP level for the same cell / timeslot combination.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.3.1B.1.2 Minimum Requirements

The absolute accuracy requirements in table 8.7.3.1B.1.2.1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm.

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -13dB$$

**Table 8.7.3.1B.1.2.1: UE Timeslot ISCP intra frequency absolute accuracy (7,68Mcps option)**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 7.68 MHz] |
| Timeslot_ISCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clauses 9.1.1.3.1.3. and A.9.3.3.

8.7.3.1B.1.3 Test Purpose

The purpose of this test is to verify that the Timeslot ISCP measurement accuracy is within the specified limits.

8.7.3.1B.1.4 Method of test

8.7.3.1B.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for both cell 1 and cell 2. The Timeslot ISCP intra frequency absolute accuracy requirements are tested by using test parameters in table 8.7.3.1B.1.4.1.1.

**Table 8.7.3.1B.1.4.1.1: Timeslot ISCP intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 0 | 0 | 0 | 0 | 0 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.7 | | -59.8 | | -98.7 | |
| Îor/Ioc | dB | 5 | 2 | 9 | 2 | 3 | 0 |
| Timeslot ISCP, Note 1 | dBm | -73.7 | -70.7 | -57.8 | -50.8 | -98.7 | -95.7 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Timeslot ISCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.3.1B.1.4.1.1.

8.7.3.1B.1.4.2 Procedure

1) SS shall transmit MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check Timeslot ISCP values for Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations in MEASUREMENT REPORT messages. These Timeslot ISCP values reported by the UE are compared to the actual Timeslot ISCP levels for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.3.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have

been received from UE, the RF parameters are set up according to table 8.7.3.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| Measurement Information elements | |
| -Measurement Identity | 1 |
| -Measurement Command (10.3.7.46) | Modify |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Intra-frequency measurement |
| -Intra-frequency measurement (10.3.7.36) | |
| -Intra-frequency measurement objects list (10.3.7.33) | |
| -CHOICE *Intra-frequency cell removal* | Not present |
| -New intra-frequency cells | 2 |
| -Intra-frequency cell id | 1 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | 2 |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 1 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 1 as described in Table 8.7.3.1.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency cell id | 2 |
| -Cell info | |
| -Cell individual offset | 0 |
| -Reference time difference to cell | Not present |
| -Read SFN indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Primary CCPCH info (10.3.6.57) | |
| -CHOICE *mode* | TDD |
| -CHOICE *TDD option* | 7,68 Mcps TDD |
| -CHOICE *Sync case* | 2 |
| -Timeslot | 0 |
| -Cell parameters ID | Set to cell parameter ID of cell 2 |
| -SCTD indicator | FALSE |
| -Primary CCPCH Tx power | Set to Primary CCPCH Tx power of cell 2 as described in Table 8.7.3.1.2. |
| -Timeslot number | 0 |
| -Burst type | 1 |
| -Intra-frequency measurement quantity (10.3.7.38) | |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity list | 1 |
| -Measurement quantity | Timeslot ISCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for monitored set cells (10.3.7.5) | |

| Information Element/Group name | Value/Remark |
|---|---|
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell Identity reporting indicator | FALSE |
| -CHOICE *mode* | TDD |
| -Timeslot ISCP reporting indicator | TRUE |
| -Proposed TGSN reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE *reported cell* | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual / active set cells + 1 |
| -Measurement validity (10.3.7.51) | Not Present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| Physical channel information elements | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

#### 8.7.3.1B.1.5 Test requirements

The Timeslot ISCP measurement accuracy shall meet the requirements in clause 8.7.3.1B.1.2 for at least 900 of the reported Timeslot ISCP levels at each input level in step 4 for both Cell 1 / Timeslot 0 and Cell 2 / Timeslot 0 combinations.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4 UTRA carrier RSSI

### 8.7.4.1 Absolute measurement accuracy for 3,84 Mcps TDD Option

#### 8.7.4.1.1 Definition and applicability

The absolute accuracy of UTRA carrier RSSI is defined as the UTRA carrier RSSI measured from one frequency compared to the actual UTRA carrier RSSI power of that same frequency.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.4.1.2 Minimum Requirements

**Table 8.7.4.1.1: UTRA carrier RSSI inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 3,84 MHz] |
| UTRA carrier RSSI | dBm | ± 4 | ± 7 | -94...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.

#### 8.7.4.1.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

#### 8.7.4.1.4 Method of test

##### 8.7.4.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1.2.

**Table 8.7.4.1.2: UTRA carrier RSSI inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.2 | -75.2 | -57.8 | -54.1 | -98.7 | -97 |
| $\hat{I}$or/Ioc | dB | 5 | 5 | 7 | 2 | 3 | 0 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.1.2.

#### 8.7.4.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | TRUE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

### 8.7.4.1.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.1.2. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.1.2 as shown in table 8.7.4.1.3.

**Table 8.7.4.1.3: UTRA carrier RSSI absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3,84 MHz] |
| UTRA carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.1.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4.1A Absolute measurement accuracy for 1,28 Mcps TDD Option

### 8.7.4.1A.1 Definition and applicability

The absolute accuracy of UTRA carrier RSSI is defined as the UTRA carrier RSSI measured from one frequency compared to the actual UTRA carrier RSSI power of that same frequency.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.4.1A.2 Minimum Requirements

**Table 8.7.4.1A.1: 1,28Mcps UTRA carrier RSSI inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 1,28 MHz] |
| UTRA carrier RSSI | dBm | ± 4 | ± 7 | -94...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |
| | | | | |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.

### 8.7.4.1A.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

### 8.7.4.1A.4 Method of test

#### 8.7.4.1A.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1A.2.

**Table 8.7.4.1A.2: 1,28Mcps UTRA carrier RSSI inter frequency test parameters**

| | Test 1 | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 5 | |
| $I_{oc}$ | dBm/1.28 MHz | -75.2 | | -75.2 | |
| Io, Note 1 | dBm/1.28 MHz | -69 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 2** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | | 2 | |
| $I_{oc}$ | dBm/1.28 MHz | -57.8 | | -54.1 | |
| Io, Note 1 | dBm/1.28 MHz | -50 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 3** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | 0 | |
| $I_{oc}$ | dBm/1.28 MHz | -98.7 | | -97 | |
| Io, Note 1 | dBm/1.28 MHz | -94 | | | |
| Propagation condition | | AWGN | | | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.1A.2.

8.7.4.1A.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE.

5) After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1A.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT

messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1A.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

7) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

8) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3]

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | TRUE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

## 8.7.4.1A.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.1A.2 for at least 900 of the reported RSSI levels at each input level. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.1A.2 as shown in table 8.7.4.1A.3. (only relevent for the lowest power test 3)

**Table 8.7.4.1A.3: 1,28Mcps UTRA carrier RSSI absolute accuracy (corrected for RX noise)**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1,28 MHz] |
| UTRA carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.1.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4.1B Absolute measurement accuracy for 7,68 Mcps TDD Option

### 8.7.4.1B.1 Definition and applicability

The absolute accuracy of UTRA carrier RSSI is defined as the UTRA carrier RSSI measured from one frequency compared to the actual UTRA carrier RSSI power of that same frequency.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.4.1B.2 Minimum Requirements

**Table 8.7.4.1B.2.1: UTRA carrier RSSI inter frequency absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 7.68 MHz] |
| UTRA Carrier RSSI | dBm | ± 4 | ± 7 | -94...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.1.3 and A9.3.4

### 8.7.4.1B.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

### 8.7.4.1B.4 Method of test

#### 8.7.4.1B.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1B.4.1.1.

**Table 8.7.4.1B.4.1.1: UTRA carrier RSSI inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.2 | -75.2 | -57.8 | -54.1 | -98.7 | -97 |
| Îor/Ioc | dB | 5 | 5 | 7 | 2 | 3 | 0 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.1B.4.1.1.

### 8.7.4.1B.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1B.4.1.1 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1B.4.1.1 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

#### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurement (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | TRUE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | FALSE |
| -Cell identity reporting indicator | FALSE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

## 8.7.4.1B.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.1B.2. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.1B.2 as shown in table 8.7.4.1B.3.

Table 8.7.4.1B.5.1: UTRA carrier RSSI absolute accuracy

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/7.68 MHz] |
| UTRA Carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.3.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4.2 Relative measurement accuracy for 3,84 Mcps TDD Option

### 8.7.4.2.1 Definition and applicability

The relative accuracy requirement is defined as the UTRA carrier RSSI measured from one frequency compared to the UTRA Carrier RSSI measured from another frequency.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.4.2.2 Minimum Requirements

The accuracy requirements in table 8.7.4.2.1 are valid under the following condition:

$$| \text{Channel 1_Io}|_{\text{dBm/3,84 MHz}} - \text{Channel 2_Io}|_{\text{dBm/3,84 MHz}} | < 20 \text{ dB}.$$

Table 8.7.4.2.1: UTRA carrier RSSI inter frequency relative accuracy

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3,84 MHz] |
| UTRA carrier RSSI | dBm | ± 7 | ± 11 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.

### 8.7.4.2.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

### 8.7.4.2.4 Method of test

#### 8.7.4.2.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1.2.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.1.2.

#### 8.7.4.2.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for Inter-frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated. After further 1000 MEASUREMENT REPORT messages have been

received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurements in clause 8.7.4.1.4.2 is used.

### 8.7.4.2.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.2.2. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.2.2 as shown in table 8.7.4.2.2.

**Table 8.7.4.2.2: UTRA carrier RSSI relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3,84 MHz] |
| UTRA carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.1.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4.2A Relative measurement accuracy for 1,28 Mcps TDD Option

### 8.7.4.2A.1 Definition and applicability

The relative accuracy requirement is defined as the UTRA carrier RSSI measured from one frequency compared to the UTRA Carrier RSSI measured from another frequency.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.4.2A.2 Minimum Requirements

The accuracy requirements in table 8.7.4.2A.1 are valid under the following condition:

$| \text{Channel 1_Io}|_{\text{dBm/1.28 MHz}} - \text{Channel 2_Io}|_{\text{dBm/1.28 MHz}} | < 20$ dB.

**Table 8.7.4.2A.1: UTRA carrier RSSI inter frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1,28 MHz] |
| UTRA carrier RSSI | dBm | ± 7 | ± 11 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.

### 8.7.4.2A.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

### 8.7.4.2A.4 Method of test

#### 8.7.4.2A.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing.. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1A.2.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.2A.2.

**Table 8.7.4.2A.2: 1,28Mcps UTRA carrier RSSI inter frequency test parameters**

| | **Test 1** | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 5 | | 5 | |
| $I_{oc}$ | dBm/1.28 MHz | -75.2 | | -75.2 | |
| Io, Note 1 | dBm/1.28 MHz | -69 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 2** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 7 | | 2 | |
| $I_{oc}$ | dBm/1.28 MHz | -57.8 | | -54.1 | |
| Io, Note 1 | dBm/1.28 MHz | -50 | | | |
| Propagation condition | | AWGN | | | |
| | **Test 3** | | | | |
| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number | | Channel 1 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | | 0 | |
| $I_{oc}$ | dBm/1.28 MHz | -98.7 | | -97 | |
| Io, Note 1 | dBm/1.28 MHz | -94 | | | |
| Propagation condition | | AWGN | | | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

#### 8.7.4.2A.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for Inter-frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE.

5) After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

The same MEASUREMENT CONTROL message for absolute inter frequency measurements in clause 8.7.4.1A.4.2 is used.

### 8.7.4.2A.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.2A.2 for at least 900 of the reported RSSI levels at each input level. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.2A.2 as shown in table 8.7.4.2A.3. (only relevent for the lowest power test 3)

**Table 8.7.4.2A.3: UTRA carrier RSSI relative accuracy (corrected for RX noise)**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/1,28 MHz] |
| UTRA carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.1.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.4.2B Relative measurement accuracy for 7,68 Mcps TDD Option

### 8.7.4.2B.1 Definition and applicability

The relative accuracy requirement is defined as the UTRA carrier RSSI measured from one frequency compared to the UTRA Carrier RSSI measured from another frequency.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.4.2B.2 Minimum Requirements

The accuracy requirements in table 8.7.4.2B.2.1 are valid under the following condition:

| Channel 1_Io|$_{dBm/3,84\ MHz}$ -Channel 2_Io|$_{dBm/3,84\ MHz}$ | < 20 dB.

**Table 8.7.4.2B.2.1: UTRA carrier RSSI inter frequency relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/7.68 MHz] |
| UTRA Carrier RSSI | dBm | ± 7 | ± 11 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.4.2 and A9.3.4

### 8.7.4.2B.3 Test Purpose

The purpose of this test is to verify that the UTRA carrier RSSI measurement accuracy is within the specified limits.

### 8.7.4.2B.4 Method of test

#### 8.7.4.2B.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case both cells are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. UTRA carrier RSSI absolute accuracy requirements are tested by using test parameters in table 8.7.4.1B.4.1.1.

1) A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.4.1B.4.1.1.

#### 8.7.4.2B.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for Inter-frequency measurements.

2) UE shall transmit periodically the MEASUREMENT REPORT messages.

3) SS shall check UTRA carrier RSSI value of Channel 2 in MEASUREMENT REPORT messages. UTRA carrier RSSI power of Channel 2 reported by UE is compared to actual UTRA carrier RSSI value of Channel 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1B.4.1.1for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.4.1B.4.1.1 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, step 3) above is repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurements in clause 8.7.4.1B.4.2 is used.

#### 8.7.4.2B.5 Test requirements

The UTRA carrier RSSI absolute measurement accuracy shall meet the requirements in clause 8.7.4.2B.2. The effect of assumed thermal noise and noise generated in the receiver (–99 dBm) shall be added into the required accuracy defined in subclause 8.7.4.2B.2 as shown in table 8.7.4.2B.5.1.

**Table 8.7.4.2B.5.1: UTRA carrier RSSI relative accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/3,84 MHz] |
| UTRA carrier RSSI | dBm | -4…5.2 | -7…8.2 | -94…-87 |
| | dBm | ± 4 | ± 7 | -87...-70 |
| | dBm | ± 6 | ± 9 | -70...-50 |

The normative reference for this requirement is TS 25.123 [2] clause A.9.3.4.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.5 GSM carrier RSSI

#### 8.7.5.1.1 RSSI (RX_LEV) 3,84 Mcps TDD Option

Void

#### 8.7.5.1A.1 RSSI (RX_LEV) 1,28 Mcps TDD Option

##### 8.7.5.1A.1.1 Definition and applicability

The absolute accuracy of GSM RSSI is defined as the RX_LEV measured in a GSM cell on one frequency compared to the actual power of that cell.

The requirements and this test apply only to UE supporting both 1,28Mcps UTRA TDD and GSM.

##### 8.7.5.1A.1.2 Minimum Requirements

**Table 8.7.5.1A.1.1: GSM RX_LEV absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Input level dBm |
| RX_LEV | dBm | ± 4 | ± 6 | -110…-70 |
| | dBm | ± 6 | ± 6 | -70…-48 |
| | dBm | ± 9 | ± 9 | -48..-38 |

RXLEV 0 = less than -110 dBm.
RXLEV 1 = -110 dBm to -109 dBm
RXLEV 2 = -109 dBm to -108 dBm
:
:
RXLEV 62 = -49 dBm to -48 dBm
RXLEV 63 = greater than -48 dBm

The normative reference for this requirement is TS 45.008 [20] clause 8.1.2

##### 8.7.5.1A.1.3 Test purpose

The purpose of this test is to verify that the GSM RSSI absolute measurement accuracy is within the specified limits.

This test will verify the requirements in section 9.1.1.5 and A.9.1.5 of TS25.123.

8.7.5.1A.1.4 Method of test

8.7.5.1A.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

Cell 1 is a UTRA 1,28Mcps TDD cell and cell 2 is a GSM cell

A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test are set up according to table 8.7.5.1A.1.2.

**Table 8.7.5.1A.1.2. General GSM RSSI test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | No | |
| Monitored cell list size | | 6 GSM neighbours including ARFCN 1 | |

**Table 8.7.5.1A.1.3 Cell 1 (TDD Cell) specific test parameters**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| DL timeslot number | | 0 | DwPTS |
| UTRA RF Channel number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | |
| DwPCH_Ec/Ior | dB | | 0 |
| OCNS_Ec/Ior | dB | -3 | |
| Îor/Ioc | dB | 3 | |
| Ioc | dBm / 1.28MHz | -70 | |
| Propagation condition | | AWGN | |

**Table 8.7.5.1A.1.4 Cell 2 specific GSM Cell test parameters**

| Parameter | Unit | TEST1 | TEST2 | TEST3 |
|---|---|---|---|---|
| UTRA RF Channel number | | 2 | | |
| Cell Level | dBm/200KHz | -100 | -75 | -50 |
| Propagation condition | | AWGN | AWGN | AWGN |

8.7.5.1A.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter RAT measurement. In the measurement control information periodic reporting of the GSM carrier RSSI is requested to the UE.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check RX_LEV value of Cell 2 in the MEASUREMENT REPORT messages. Levels of Cell 2 reported by the UE are compared to the actual level of Cell 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the GSM cell RF parameters are set up according to table 8.7.5.1A.1.4 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the GSM cell RF parameters are set up according to table 8.7.5.1A.1.4 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

7) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

8) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

Note: Numbers in brackets after an item e.g "Message Type (10.2.17)" in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message for GSM RSSI measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command<br>-Measurement Reporting Mode<br>- Measurement Report Transfer Mode<br>- Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurement list<br>-CHOICE Measurement Type<br>-Inter-frequency measurement object list<br>-CHOICE Inter-frequency cell removal<br>-New inter-frequency cells<br>-Cell for measurement<br>-Inter-frequency measurement quantity<br>-CHOICE reporting criteria<br>-Filter coefficient<br>-CHOICE mode<br>-Measurement quantity<br>-Inter-frequency reporting quantity<br>-UTRA Carrier RSSI<br>-Frequency quality estimate<br>-Non frequency related cell reporting quantities<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Maximum number of reported cells<br>-Measurement validity<br>-Inter-frequency set update<br>-CHOICE report criteria<br>-Amount of reporting<br>-Reporting interval | <br>2<br>Setup<br><br>Acknowledged mode RLC<br>Periodical reporting<br><br>Not Present<br>Inter-RAT measurement<br><br>Not Present<br>Cell 2 information is included<br>Not Present<br><br>Inter-frequency reporting criteria<br>0<br>GSM<br>RX_LEV<br><br><br>FALSE<br>FALSE<br><br>No report<br><br><br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br>Not Present<br>Periodical reporting criteria<br>Infinity<br>500 ms |
| Physical channel information elements<br>-DPCH compressed mode status info | <br>Not Present |

8.7.2.1.1.5 Test requirements

The RX_LEV measurement accuracy shall meet the requirements in clause 8.7.5.1A.1.1 for at least 900 of the reported levels at each input level.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.5.1B.1 RSSI (RX_LEV) 7,68 Mcps TDD Option

### 8.7.5.1B.1.1 Definition and applicability

Cell 1 is a UTRA TDD cell and cell 2 is a GSM cell. In the measurement control information it is indicated to the UE that periodic reporting of the GSM carrier RSSI measurement is used.

The requirements and this test apply only to UE supporting both 7,68Mcps UTRA TDD and GSM.

### 8.7.5.1B.1.2 Minimum Requirements

**Table 8.7.5.1B.1.2.1: GSM RX_LEV absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Input level dBm |
| RX_LEV | dBm | ± 4 | ± 6 | -110…-70 |
| | dBm | ± 6 | ± 6 | -70…-48 |
| | dBm | ± 9 | ± 9 | -48..-38 |

RXLEV 0 = less than -110 dBm.
RXLEV 1 = -110 dBm to -109 dBm
RXLEV 2 = -109 dBm to -108 dBm
:
:
RXLEV 62 = -49 dBm to -48 dBm
RXLEV 63 = greater than -48 dBm

The normative reference for this requirement is TS 45.008 [20] clause 8.1.2

### 8.7.5.1B.1.3 Test purpose

The purpose of this test is to verify that the GSM RSSI absolute measurement accuracy is within the specified limits.

This test will verify the requirements in section 9.1.1.5 and A.9.3.5 of TS25.123.

### 8.7.5.1B.1.4 Method of test

#### 8.7.5.1B.1.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

Cell 1 is a UTRA 1,28Mcps TDD cell and cell 2 is a GSM cell

A call is set up according to the test procedure specified in TS 34.108 [3] clause 7.3.2.3. The RF parameters for Test are set up according to table 8.7.5.1B.1.4.1.1, 8.7.5.1B.1.4.1.2 and 8.7.5.1B.1.4.1.3.

**Table 8.7.5.1B.1.4.1.1 General GSM RSSI test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DCH parameters | | DL reference measurement channel 12.2 kbps | As specified in TS 25.102 section A.2.2 |
| Power Control | | On | |
| Target quality value on DTCH | BLER | 0.01 | |
| Inter-RAT measurement quantity | | GSM carrier RSSI | |
| BSIC verification required | | No | |
| Monitored cell list size | | 6 GSM neighbours including ARFCN 1 | |

**Table 8.7.5.1B.1.4.1.2 Cell 1 (TDD Cell) specific test parameters**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| DL timeslot number | | 0 | 1 |
| UTRA RF Channel number | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | n.a. |
| SCH_Ec/Ior | dB | -9 | n.a. |
| SCH_$t_{offset}$ | | 0 | n.a. |
| OCNS_Ec/Ior | dB | -3,12 | Note 2 |
| DPCH_Ec/Ior | dB | n.a. | Note 1 |
| Îor/Ioc | dB | 6 | 6 |
| Io, Note 1 | dBm / 7.68 MHz | -70 | |
| Propagation condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior. | | | |

**Table 8.7.5.1B.1.4.1.3 Cell 2 specific GSM Cell test parameters**

| Parameter | Unit | TEST1 | TEST2 | TEST3 |
|---|---|---|---|---|
| UTRA RF Channel number | | 2 | | |
| Cell Level | dBm/200KHz | -100 | -75 | -50 |
| Propagation condition | | AWGN | AWGN | AWGN |

### 8.7.5.1B.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message for inter RAT measurement. In the measurement control information periodic reporting of the GSM carrier RSSI is requested to the UE.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check RX_LEV value of Cell 2 in the MEASUREMENT REPORT messages. Levels of Cell 2 reported by the UE are compared to the actual level of Cell 2 for each MEASUREMENT REPORT message.

4) SS shall count number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the GSM cell RF parameters are set up according to table 8.7.5.1B.1.4.1.3 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

6) After further 1000 MEASUREMENT REPORT messages have been received from UE, the GSM cell RF parameters are set up according to table 8.7.5.1B.1.4.1.3 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

7) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

8) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3], with the following exceptions:

Note: Numbers in brackets after an item e.g “Message Type (10.2.17)” in the IE description are references to clause numbers in TS 25.331 [9] describing that item in more detail.

MEASUREMENT CONTROL message for GSM RSSI measurement (Step 1):

| Information Element | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command<br>-Measurement Reporting Mode<br>- Measurement Report Transfer Mode<br>- Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurement list<br>-CHOICE Measurement Type<br>-Inter-frequency measurement object list<br>-CHOICE Inter-frequency cell removal<br>-New inter-frequency cells<br>-Cell for measurement<br>-Inter-frequency measurement quantity<br>-CHOICE reporting criteria<br>-Filter coefficient<br>-CHOICE mode<br>-Measurement quantity<br>-Inter-frequency reporting quantity<br>-UTRA Carrier RSSI<br>-Frequency quality estimate<br>-Non frequency related cell reporting quantities<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Maximum number of reported cells<br>-Measurement validity<br>-Inter-frequency set update<br>-CHOICE report criteria<br>-Amount of reporting<br>-Reporting interval | <br>2<br>Setup<br><br>Acknowledged mode RLC<br>Periodical reporting<br><br>Not Present<br>Inter-RAT measurement<br><br>Not Present<br>Cell 2 information is included<br>Not Present<br><br>Inter-frequency reporting criteria<br>0<br>GSM<br>RX_LEV<br><br><br>FALSE<br>FALSE<br><br>No report<br><br><br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br>Not Present<br>Periodical reporting criteria<br>Infinity<br>500 ms |
| Physical channel information elements<br>-DPCH compressed mode status info | <br>Not Present |

### 8.7.5.1B.1.5 Test requirements

The RX_LEV measurement accuracy shall meet the requirements in clause 8.7.5.1B.1.2 for at least 900 of the reported levels at each input level.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.6 SIR

Void

## 8.7.7 Transport Channel BLER

Void

# 8.7.8 SFN-SFN observed time difference

## 8.7.8.1 SFN-SFN observed time difference type 1

### 8.7.8.1.1 Measurement accuracy for 3,84 Mcps TDD Option

#### 8.7.8.1.1.1 Definition and applicability

This measurement is specified in clause 5.1.10 of TS 25.225 [22]. The reference point for the SFN-SFN observed time difference type 1 shall be the antenna connector of the UE.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.8.1.1.2 Minimum requirements

The accuracy requirement in table 8.7.8.1.1.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\right|_{in\ dBm} - \text{P - CCPCH RSCP2}\left|_{in\ dBm}\right| \le 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \ge -8dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \ge -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.8.1.1.1: SFN-SFN observed time difference type 1 accuracy**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/3,84 MHz] |
| SFN-SFN observed time difference type 1 | chip | +/-0,5 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.8 and A.9.1.8.

#### 8.7.8.1.1.3 Test purpose

The purpose of this test is to verify that the measurement accuracy of SFN-SFN observed time difference type 1 is within the limit specified in clause 8.7.8.1.1.2.

8.7.8.1.1.4 Method of test

8.7.8.1.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…9830400 chip. The SFN-SFN observed time difference type 1 accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.8.1.1.2.

**Table 8.7.8.1.1.2: SFN-SFN observed time difference type 1 intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| $\hat{I}$or/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.8.1.1.2.

8.7.8.1.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "SFN-SFN observed time difference type 1" value in MEASUREMENT REPORT message. The reported value shall be compared to the actually set SFN-SFN observed time difference type 1 value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.8.1.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.8.1.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>Type 1<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>Type 1<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

8.7.8.1.1.5 Test requirements

The SFN-SFN observed time difference type 1 accuracy shall meet the requirements in clause 8.7.8.1.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.8.1.1A Measurement accuracy for 1,28 Mcps TDD Option

Void

### 8.7.8.1.1B Measurement accuracy for 7,68 Mcps TDD Option

#### 8.7.8.1.1B.1 Definition and applicability

This measurement is specified in clause 5.1.10 of TS 25.225 [22]. The reference point for the SFN-SFN observed time difference type 1 shall be the antenna connector of the UE.

The requirements and this test apply to all types of UTRA TDD UE.

#### 8.7.8.1.1B.2 Minimum requirements

The accuracy requirement in table 8.7.8.1.1B.2.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\right|_{in\ dBm} - \text{P - CCPCH RSCP2}\left|_{in\ dBm}\right| \le 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \ge -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \ge -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.8.1.1B.2.1: SFN-SFN observed time difference type 1 accuracy**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/7.68 MHz] |
| SFN-SFN observed time difference | chip | +/-1 for both type 1 and 2 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.8.1.3 and A.9.3.8.

#### 8.7.8.1.1B.3 Test purpose

The purpose of this test is to verify that the measurement accuracy of SFN-SFN observed time difference type 1 is within the limit specified in clause 8.7.8.1.1B.2.

8.7.8.1.1B.4 Method of test

8.7.8.1.1B.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2.During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…19660800 chip. The SFN-SFN observed time difference type 1 accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.8.1.1B.4.1.1.

**Table 8.7.8.1.1B.4.1.1: SFN-SFN observed time difference type 1 intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| $\hat{I}$or/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.8.1.1B.4.1.1.

8.7.8.1.1B.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "SFN-SFN observed time difference type 1" value in MEASUREMENT REPORT message. The reported value shall be compared to the actually set SFN-SFN observed time difference type 1 value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.8.1.1B.4.1.1 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.8.1.1B.4.1.1 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>Type 1<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>Type 1<br>FALSE<br>FALSE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

8.7.8.1.1B.5 Test requirements

The SFN-SFN observed time difference type 1 accuracy shall meet the requirements in clause 8.7.8.1.1B.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

#### 8.7.8.2 SFN-SFN observed time difference type 2

Void

### 8.7.9 Observed time difference to GSM cell

Void

### 8.7.10 UE GPS Timing of Cell Frames for UP

Void

### 8.7.11 SFN-CFN observed time difference

#### 8.7.11.1 Intra frequency measurement requirement for 3,84 Mcps option

##### 8.7.11.1.1 Definition and applicability

The intra frequency SFN-CFN observed time difference is defined as the SFN-CFN observed time difference from the active cell to a neighbour cell that is in the same frequency. This measurement is specified in clause 5.1.11 of TS 25.225 [22]. The reference point for the SFN-CFN observed time difference shall be the antenna connector of the UE.

The requirements and this test apply to all types of UTRA TDD UE.

##### 8.7.11.1.2 Minimum requirements

The accuracy requirement in table 8.7.11.1.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\right|_{in\ dBm} - \text{P - CCPCH RSCP2}\left|_{in\ dBm}\right| \leq 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -8dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.11.1.1: SFN-CFN observed time difference accuracy for an intra frequency UTRA TDD neighbour cell**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/3,84 MHz] |
| SFN-CFN observed time difference | chip | +/-0,5 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.11 and A.9.1.10.

### 8.7.11.1.3 Test Purpose

The purpose of this test is to verify that the measurement accuracy of SFN-CFN observed time difference is within the limit specified in clause 8.7.11.1.2.

### 8.7.11.1.4 Method of test

#### 8.7.11.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…255 frames. The SFN-CFN observed time difference accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.11.1.2.

**Table 8.7.11.1.2: SFN-CFN observed time difference intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| Îor/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.11.1.2.

#### 8.7.11.1.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "OFF" and "Tm" values in MEASUREMENT REPORT message and calculate SFN-CFN observed time difference value according to the definition in clause 5.1.11 of TS 25.225 [22]. This value shall be compared to the actually set SFN-CFN observed time difference value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to

table 8.7.11.1.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.1.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>TRUE<br>TRUE<br>TDD<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>TRUE<br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

### 8.7.11.1.5 Test requirements

The SFN-CFN observed time difference accuracy shall meet the requirements in clause 8.7.11.1.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.11.1A Intra frequency measurement requirement for 1,28 Mcps option

Void

## 8.7.11.1B Intra frequency measurement requirement for 7,68 Mcps option

### 8.7.11.1B.1 Definition and applicability

The intra frequency SFN-CFN observed time difference is defined as the SFN-CFN observed time difference from the active cell to a neighbour cell that is in the same frequency. This measurement is specified in clause 5.1.11 of TS 25.225 [22]. The reference point for the SFN-CFN observed time difference shall be the antenna connector of the UE.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.11.1B.2 Minimum requirements

The accuracy requirement in table 8.7.11.1B.2.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\right|_{in\ dBm} - \text{P - CCPCH RSCP2}\left|_{in\ dBm}\right| \leq 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} \geq -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Bigg|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Bigg|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Bigg|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.11.1B.2.1: SFN-CFN observed time difference accuracy for an intra frequency UTRA TDD neighbour cell**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/7.68 MHz] |
| SFN-CFN observed time difference | chip | +/-1 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.11.1 and A.9.3.10.

### 8.7.11.1B.3 Test Purpose

The purpose of this test is to verify that the measurement accuracy of SFN-CFN observed time difference is within the limit specified in clause 8.7.11.1B.2.

### 8.7.11.1B.4 Method of test

#### 8.7.11.1B.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case all cells are on the same frequency. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…255 frames. The SFN-CFN observed time difference accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.11.1B.4.1.1.

**Table 8.7.11.1B.4.1.1: SFN-CFN observed time difference intra frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | | Channel 1 | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| Îor/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.11.1B.4.1.1.

### 8.7.11.1B.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "OFF" and "Tm" values in MEASUREMENT REPORT message and calculate SFN-CFN observed time difference value according to the definition in clause 5.1.11 of TS 25.225 [22]. This value shall be compared to the actually set SFN-CFN observed time difference value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.1B.4.1.1 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.1B.4.1.1 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| UE information elements<br>-RRC transaction identifier<br>-Integrity check info | <br>0<br>Not Present |
| Measurement Information elements<br>-Measurement Identity<br>-Measurement Command (10.3.7.46)<br>-Measurement Reporting Mode (10.3.7.49)<br>-Measurement Report Transfer Mode<br>-Periodical Reporting / Event Trigger Reporting Mode<br>-Additional measurements list (10.3.7.1) | <br>1<br>Modify<br><br>AM RLC<br>Periodical reporting<br>Not Present |
| -CHOICE *Measurement type*<br>-Intra-frequency measurement (10.3.7.36)<br>-Intra-frequency measurement objects list (10.3.7.33)<br>-Intra-frequency measurement quantity (10.3.7.38)<br>-Filter coefficient (10.3.7.9)<br>-CHOICE *mode*<br>-Measurement quantity list | Intra-frequency measurement<br><br>Not Present<br><br>0<br>TDD<br>1 |
| -Measurement quantity | Primary CCPCH RSCP |
| -Intra-frequency reporting quantity (10.3.7.41) | |
| -Reporting quantities for active set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>TRUE<br>TRUE<br>TDD<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for monitored set cells (10.3.7.5)<br>-SFN-SFN observed time difference reporting indicator<br>-Cell synchronisation information reporting indicator<br>-Cell Identity reporting indicator<br>-CHOICE *mode*<br>-Timeslot ISCP reporting indicator<br>-Proposed TGSN reporting required<br>-Primary CCPCH RSCP reporting indicator<br>-Pathloss reporting indicator | <br>No report<br>TRUE<br>TRUE<br>TDD<br>FALSE<br>FALSE<br>TRUE<br>TRUE |
| -Reporting quantities for detected set cells (10.3.7.5) | Not Present |
| -Reporting cell status (10.3.7.61)<br>-CHOICE *reported cell*<br><br>-Maximum number of reported cells<br>-Measurement validity (10.3.7.51)<br>-CHOICE *report criteria* (10.3.7.<br>-Periodical reporting criteria (10.3.7.53)<br>-Amount of reporting<br>-Reporting interval | <br>Report all active set cells + cells within monitored set on used frequency<br>Virtual/active set cells + 2<br>Not Present<br><br><br>Infinity<br>250 ms |
| Physical channel information elements<br>-DPCH compressed mode status info (10.3.6.34) | <br>Not Present |

## 8.7.11.1B.5 Test requirements

The SFN-CFN observed time difference accuracy shall meet the requirements in clause 8.7.11.1B.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.11.2 Inter frequency measurement requirement for 3,84 Mcps option

### 8.7.11.2.1 Definition and applicability

The inter frequency SFN-CFN observed time difference is defined as the SFN-CFN time difference from the active cell to a UTRA TDD neighbour cell that is in a different frequency. This measurement is specified in clause 5.1.11 of TS 25.225 [22]. The reference point for the SFN-CFN observed time difference shall be the antenna connector of the UE.

The requirements and this test apply to all types of UTRA TDD UE.

### 8.7.11.2.2 Minimum requirements

The accuracy requirement in table 8.7.11.2.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\big|_{in\ dBm} - \text{P - CCPCH RSCP2}\big|_{in\ dBm}\right| \le 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Big|_{in\ dB} \ge -8dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Big|_{in\ dB} \ge -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Big|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Big|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Big|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Big|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Big|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Big|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.11.2.1: SFN-CFN observed time difference accuracy for an inter frequency UTRA TDD neighbour cell**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/3,84 MHz] |
| SFN-CFN observed time difference | chip | +/-0,5 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.11 and A.9.1.10.

### 8.7.11.2.3 Test purpose

The purpose of this test is to verify that the measurement accuracy of SFN-CFN observed time difference is within the limit specified in clause 8.7.11.2.2.

### 8.7.11.2.4 Method of test

#### 8.7.11.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case, UTRA TDD cell 1 and UTRA TDD cell 2 are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…256 frames. The SFN-CFN observed time difference accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.11.2.2.

**Table 8.7.11.2.2: SFN-CFN observed time difference inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 3,84 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| Îor/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 3,84 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.11.2.2.

### 8.7.11.2.4.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "OFF" and "Tm" values in MEASUREMENT REPORT message and calculate SFN-CFN observed time difference value according to the definition in clause 5.1.11 of TS 25.225 [22]. This value shall be compared to the actually set SFN-CFN observed time difference value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.2.2 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.2.2 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell identity reporting indicator | TRUE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

### 8.7.11.2.5 Test requirements

The SFN-CFN observed time difference accuracy shall meet the requirements in clause 8.7.11.2.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.11.2A Inter frequency measurement requirement for 1,28 Mcps option

Void

## 8.7.11.2B Inter frequency measurement requirement for 7,68 Mcps option

### 8.7.11.2B.1 Definition and applicability

The inter frequency SFN-CFN observed time difference is defined as the SFN-CFN time difference from the active cell to a UTRA TDD neighbour cell that is in a different frequency. This measurement is specified in clause 5.1.11 of TS 25.225 [22]. The reference point for the SFN-CFN observed time difference shall be the antenna connector of the UE.

The requirements and this test apply to 7.68Mcps TDD UE.

### 8.7.11.2B.2 Minimum requirements

The accuracy requirement in table 8.7.11.2B.2.1 is valid under the following conditions:

P-CCPCH_RSCP1,2 ≥ -102 dBm..

$$\left|\text{P - CCPCH RSCP1}\big|_{in\ dBm} - \text{P - CCPCH RSCP2}\big|_{in\ dBm}\right| \le 20dB$$

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Big|_{in\ dB} \ge -11dB$$

$$\left(\frac{SCH_E_c}{I_o}\right)\Big|_{in\ dB} \ge -13dB$$

where the received P-CCPCH Ec/Io is defined as,

$$\left(\frac{P-CCPCH_E_c}{I_o}\right)\Big|_{in\ dB} = \left(\frac{P-CCPCH_E_c}{I_{or}}\right)\Big|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Big|_{in\ dB}$$

and the received SCH Ec/Io is defined as,

$$\left(\frac{SCH_E_c}{I_o}\right)\Big|_{in\ dB} = \left(\frac{SCH_E_c}{I_{or}}\right)\Big|_{in\ dB} - \frac{I_o}{\left(\hat{I}_{or}\right)}\Big|_{in\ dB}$$

and SCH_Ec/Ior is equally divided between primary synchronisation code and the sum of all secondary synchronisation codes, where the secondary synchronisation codes are also equally divided.

**Table 8.7.11.2B.2.1: SFN-CFN observed time difference accuracy for an inter frequency UTRA TDD neighbour cell**

| Parameter | Unit | Accuracy [chip] | Conditions |
|---|---|---|---|
| | | | Io [dBm/7.68 MHz] |
| SFN-CFN observed time difference | chip | +/-1 | -94...-50 |

The normative reference for this requirement is TS 25.123 [2] clause 9.1.1.11.1 and A.9.3.10.

### 8.7.11.2B.3 Test purpose

The purpose of this test is to verify that the measurement accuracy of SFN-CFN observed time difference is within the limit specified in clause 8.7.11.2B.2.

### 8.7.11.2B.4 Method of test

#### 8.7.11.2B.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

In this case, UTRA TDD cell 1 and UTRA TDD cell 2 are on different frequencies. Cell 1 and cell 2 shall be synchronised, i.e. share the same frame and timeslot timing. The DL DPCH shall be transmitted in timeslot 4 and the UL DPCH shall be transmitted in timeslot 12. The second Beacon timeslot shall be provided in timeslot 8 for cell 1 and in timeslot 10 for cell 2. During the test, the timing difference between cell 1 and cell 2 can be set to any value from 0…256 frames. The SFN-CFN observed time difference accuracy requirements in the intra-frequency case are tested by using test parameters in Table 8.7.11.2B.4.1.1.

**Table 8.7.11.2B.4.1.1: SFN-CFN observed time difference inter frequency test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| DL timeslot number | | 0 | 2 | 0 | 2 | 0 | 2 |
| UTRA RF Channel number | | Channel 1 | Channel 2 | Channel 1 | Channel 2 | Channel 1 | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| SCH_Ec/Ior | dB | -9 | | -9 | | -9 | |
| SCH_$t_{offset}$ | | 0 | 5 | 0 | 5 | 0 | 5 |
| OCNS_Ec/Ior | dB | -3,12 | | -3,12 | | -3,12 | |
| Ioc | dBm / 7.68 MHz | -75.2 | -75.2 | -57.8 | -54.7 | -98.7 | -98.7 |
| Îor/Ioc | dB | 5 | 5 | 7 | 3 | 3 | 3 |
| Io, Note 1 | dBm / 7.68 MHz | -69 | | -50 | | -94 | |
| Propagation condition | | AWGN | | AWGN | | AWGN | |
| NOTE 1: Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

1) A call is set up according to the test procedure specified in TS 34.108 [3] subclause 7.3.2.3. The RF parameters for Test 1 are set up according to table 8.7.11.2B.4.1.1.

### 8.7.11.2.4B.2 Procedure

1) SS shall transmit the MEASUREMENT CONTROL message.

2) UE shall transmit periodically MEASUREMENT REPORT messages.

3) SS shall check "OFF" and "Tm" values in MEASUREMENT REPORT message and calculate SFN-CFN observed time difference value according to the definition in clause 5.1.11 of TS 25.225 [22]. This value shall be compared to the actually set SFN-CFN observed time difference value for each MEASUREMENT REPORT message.

4) SS shall count the number of MEASUREMENT REPORT messages transmitted by UE. After 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.2B.4.1.1 for Test 2. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated. After further 1000 MEASUREMENT REPORT messages have been received from UE, the RF parameters are set up according to table 8.7.11.2B.4.1.1 for Test 3. While RF parameters are being set up, MEASUREMENT REPORT messages from UE are ignored. SS shall wait for additional 1s and ignore the MEASUREMENT REPORT messages during this period. Then, steps 2) and 3) above are repeated.

5) After further 1000 MEASUREMENT REPORT messages have been received from UE, the SS shall transmit RRC CONNECTION RELEASE message.

6) UE shall transmit RRC CONNECTION RELEASE COMPLETE message.

### Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3] and in Annex I, with the following exceptions:

MEASUREMENT CONTROL message for inter frequency measurements (Step 1):

| Information Element/Group name | Value/Remark |
|---|---|
| Message Type (10.2.17) | |
| **UE information elements** | |
| -RRC transaction identifier | 0 |
| -Integrity check info | Not Present |
| **Measurement Information elements** | |
| -Measurement Identity | 2 |
| -Measurement Command (10.3.7.46) | Setup |
| -Measurement Reporting Mode (10.3.7.49) | |
| -Measurement Report Transfer Mode | AM RLC |
| -Periodical Reporting / Event Trigger Reporting Mode | Periodical reporting |
| -Additional measurements list (10.3.7.1) | Not Present |
| -CHOICE *Measurement type* | Inter-frequency measurement |
| -Inter-frequency measurement (10.3.7.16) | |
| -Inter-frequency measurement objects list (10.3.7.13) | |
| -CHOICE *inter-frequency cell removal* | Not present |
| -New inter-frequency cells | Cell 2 information is included |
| -Cell for measurement | Not Present |
| -Inter-frequency measurement quantity (10.3.7.18) | |
| -CHOICE *reporting critera* | Inter-frequency reporting criteria |
| -Filter coefficient (10.3.7.9) | 0 |
| -CHOICE *mode* | TDD |
| -Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| -Inter-frequency reporting quantity (10.3.7.21) | |
| -UTRA carrier RSSI | FALSE |
| -Frequency quality estimate | TRUE |
| -Non frequency related cell reporting quantities (10.3.7.5) | |
| -SFN-SFN observed time difference reporting indicator | No report |
| -Cell synchronisation information reporting indicator | TRUE |
| -Cell identity reporting indicator | TRUE |
| -CHOICE mode | TDD |
| -Timeslot ISCP reporting indicator | FALSE |
| -Proposed TGSN Reporting required | FALSE |
| -Primary CCPCH RSCP reporting indicator | TRUE |
| -Pathloss reporting indicator | FALSE |
| -Reporting cell status (10.3.7.61) | |
| -CHOICE reported cell | Report all active set cells + cells within monitored set on used frequency |
| -Maximum number of reported cells | Virtual/active set cells + 2 |
| -Measurement validity (10.3.7.51) | Not present |
| -Inter-frequency set update | Not present |
| -CHOICE *report criteria* (10.3.7. | |
| -Periodical reporting criteria (10.3.7.53) | |
| -Amount of reporting | Infinity |
| -Reporting interval | 500 ms |
| **Physical channel information elements** | |
| -DPCH compressed mode status info (10.3.6.34) | Not Present |

### 8.7.11.2B.5 Test requirements

The SFN-CFN observed time difference accuracy shall meet the requirements in clause 8.7.11.2B.2.

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 8.7.12 UE transmitted power

Void

# 8.7.13 UE Transmission Power Headroom

## 8.7.13.1 UPH for 3.84Mcps TDD Option

Void

## 8.7.13.1A UPH for 1.28Mcps TDD Option

### 8.7.13.1A.1 Definition and applicability

The accuracy requirements for the UE transmission power headroom depends on the total power transmitted by the UE as defined in the functionality in section 9.2.6.3 of TS 25.321[14], section 9.1.2.3.2 of TS 25.123 [2] and section 5.1.16 of TS 25.225 [22]. The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 1.28Mcps TDD UE that supports E-DCH and HSDPA.

### 8.7.13.1A.2 Minimum Requirements

The UE transmission power headroom (UPH) is defined in section 5.1.16 of TS 25.225 [22] as the ratio of the maximum UE transmission power and the product of P e-base power and serving cell path loss, and shall be calculated as following:

$$UPH = \frac{P_{\max,tx}}{P_{e-base} \cdot L_{Path_loss}}$$

Where:

P max, tx = min {Maximum allowed UL TX Power, P max} is the UE maximum transmission power;

Maximum allowed UL TX Power is set by UTRAN and defined in [9];

P max is the UE nominal maximum output power according to the UE power class and specified in [1] table 6.1;

P e-base is a closed-loop quantity defined in [9] and L Path_loss is the serving cell path loss.

The reference point for the UE transmission power headroom shall be the antenna connector of the UE.

The accuracy requirements for UE transmission power headroom depends on the total power transmitted by the UE. Table 8.7.13.1A defines the accuracy of the measured quantity as defined in section 9.1.2.3.2 of TS 25.123 [2].

**Table 8.7.13.1A.1: UPH reporting accuracy**

| Total UE putput power value (dBm) | UPH reporting Accuracy [dB]* | |
|---|---|---|
| | PUEMAX 24dBm | PUEMAX 21dBm |
| Total output power=PUEMAX | +1/-3 | ±2 |
| Total output power =PUEMAX-1 | +1,5/-3,5 | ±2,5 |
| Total output power =PUEMAX-2 | +2/-4 | ±3 |
| Total output power =PUEMAX-3 | +2,5/-4,5 | ±3,5 |
| PUEMAX-10≤ Total output power <PUEMAX-3 | +3/-5 | ±4 |
| *Note: UPH reporting accuracy is the difference between the UPH reported by the UE and the actual uplink power headroom<br>**Note: PUEMAX, is the maximum output power level without tolerance defined for the power class of the UE in 3GPP TS 25.102 | | |

### 8.7.13.1A.3 Test purpose

The purpose of this test case is to verify that the UE transmission power headroom measurement report accuracy is within the specified limits defined in section 9.1.2.3.2 of 25.123 [2] shown in table 8.7.13.1A.1.

### 8.7.13.1A.4 Method of test

#### 8.7.13.1A.4.1 Initial conditions

Test environment: normal, TL/VL, TL/VH, TH/VL, TH/VH; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS (node B emulator) to the UE antenna connector as shown in figure A.1.

2) HSDPA and E-DCH configurations are set as in table 8.7.13.1A.2.

**Table 8.7.13.1A.2: General test parameters for UE transmission power headroom**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| DL DCH configuration | | DL Reference Measurement Channel 12.2 kbps | As specified in Annex C.3.1.2 of the present document |
| DL configuration | | DL Fixed Reference Channel (FRC **1**) | As specified in Annex C.4.2.1 of the present document |
| E-DCH configuration | | 5 ms TTI E-DCH Transport Block Size Table 0 according to TS 25.321 [13] annex BC.1 | |
| DL Power Control | | Off | |
| Active cell | | Cell 1 | |
| Reference E-TFCI index | | 0 as per Table 0 according to TS 25.321 [13] annex BC.1. | |

3) The power levels and cell specific parameters are set as in table 8.7.13.1A.3.

**Table 8.7.13.1A.3: Cell Specific parameters for UE transmission power headroom**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| Timeslot Number | | 0 | DwPTS |
| UTRA RF Channel Number (NOTE 2) | | Channel 1 | |
| PCCPCH_Ec/Ior | dB | - 3 | |
| D wPCH _Ec/Ior | dB | | 0 |
| OCNS_Ec/Ior | dB | - 3 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 3 | |
| $I_{oc}$ | dBm/ 1.28 MHz | -70 | |
| PCCPCH RSCP, Note 1 | dB m | -70 | |
| Propagation Condition | | AWGN | |
| NOTE 1: PCCPCH RSCP level has been calculated from other parameters for information purposes. They are not settable parameters themselves.<br><br>NOTE2: In the case of multi-frequency network, the UTRA RF Channel Number can be set for the primary frequency in this test. | | | |

4) The UE is switched on.

5) An E-DCH call is set up according to TS 34.108 [3] 7.3.9 with the following exceptions in the RADIO BEARER SETUP message. These e xceptions are derived from Table 8.7.13.2, and in addition allow the beta values to be set and each UL physical channel to be at constant power during the measurement.

**Table 8.7.13.1A.4: Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)**

| Information Element/Group name | Value/Remark |
|---|---|
| E-PUCH Info(TDD only) | |
| >E-TFCS information | |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.2 |
| >>>Reference Beta | -14 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.3 |
| >>>Reference Beta | -13 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.4 |
| >>>Reference Beta | -12 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.5 |
| >>>Reference Beta | -10 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.6 |
| >>>Reference Beta | -9 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.7 |
| >>>Reference Beta | -8 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.8 |
| >>>Reference Beta | -7 |
| >>Reference Beta Information QPSK list | |
| >>>Reference Code Rate | 0.9 |
| >>>Reference Beta | -6 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.2 |
| >>>Reference Beta | -9 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.3 |
| >>>Reference Beta | -8 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.4 |
| >>>Reference Beta | -6 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.5 |
| >>>Reference Beta | -5 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.6 |
| >>>Reference Beta | -4 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.7 |
| >>>Reference Beta | -2 |
| >>Reference Beta Information 16QAM list | |
| >>>Reference Code Rate | 0.8 |
| >>>Reference Beta | -1 |
| >>Reference Beta Information 16QAM | |

| | |
|---|---|
| list | |
| >>>Reference Code Rate | 0.9 |
| >>>Reference Beta | 0 |
| >CHOICE TDD mode | |
| >>1.28 Mcps TDD | |
| | |
| >>>PRXdes_base | -85 |
| >>>Beacon PL Est. | |
| >>>TPC step size | 1 |
| >>>E-PUCH TS configuration list | |
| >>>>TS number | 1 |
| >>>E-PUCH TS configuration list | |
| >>>>TS number | 2 |
| >>>Minimum allowed code rate | 3(0.1) |
| >>>Maximum allowed code rate | 50(0.8) |

### 8.7.13.1A.4.2 Test procedure

1) The Scheduling Information configuration for the E-DCH indicates to the UE that it shall periodically report Scheduling Information, which contains UPH measurement every E-DCH TTI. During the test the system simulator shall not send any scheduling grant to the UE, and therefore the UE will not send any payload data on the E-DCH.

2) The SS shall set the CPRI value in E-AGCH to be 3 dB for a power class 3 UE, or 0 dB for a power class 4 UE by using uplink power control. CRRI and TRRI value in E-AGCH respectively is “00100”and “11000”.

3) The SS measures both the power transmitted by the UE on E-PUCH and the total output power of the UE in time slot which assigned E-PUCH transmitted by SS. The SS shall analyse the E-TFCI transmitted on the E-UCCH for each E-DCH TTI and estimates the βe. The SS averages both the E-PUCH output power βe and total output power of the UE over 100 times. Then, the SS calculates the difference between average E-PUCH power measured and βe.

4) The SS estimates the UE transmission power headroom as the difference between the maximum allowed uplink transmits power (P max) and the difference calculated in step 3.

5) The SS notes the UE transmission power headroom value reported in the Scheduling Information.

6) The SS calculates the difference between the UE transmission power headroom value estimated in step 4 and the reported UE transmission power headroom noted in step 5. The SS notes this as the UE transmission power headroom accuracy, and compares it to the applicable limit according to the total output power measured in step 3.

7) If the UE transmission power headroom accuracy exceeds the value in Table 8.7.13.1A.5 count a bad result, otherwise a good result with respect to the actually set TX power.

8) Repeat steps 3 to 7 in order to collect more good or bad results for the currently set power level. Continue the repetition, until statistical significance according to Annex F.6.2.8 is achieved.

9) The SS sends 5 up TPC commands at the nexting E-AGCH to bring the TX power of the UE up by a nominal 1 dB step, and then alternate UP/DOWN to maintain constant TX power.

10) Repeat steps 3 through 9 and note the UE transmission power headroom accuracy for each UE total power value until the UE stops reporting UPH or does not give lower UPH values for 8 consecutive repetitions of steps 3 through 9.

### 8.7.13.1A.5 Test requirements

The UE transmission power headroom measurement report accuracy recorded in steps 6, 9 and 10 above shall meet the requirements in table 8.7.13.1A.5. The rate of correct measurements observed during repeated tests shall be at least 90%. To pass the test, the UE transmission power headroom accuracy for each power level in the reporting range must pass. Once a power level is passed, no more results need be collected on this power level.

**Table 8.7.13.1A.5: Test requirement for UPH reporting accuracy**

| Total UE putput power value (dBm) | UPH reporting Accuracy [dB]* | |
|---|---|---|
| | PUEMAX 24dBm | PUEMAX 21dBm |
| Total output power=PUEMAX | +1.7/-3.7 | ±2.7 |
| Total output power =PUEMAX-1 | +2.2/-4.2 | ±3.2 |
| Total output power =PUEMAX-2 | +2.7/-4.7 | ±3.7 |
| Total output power =PUEMAX-3 | +3.2/-5.2 | ±4.2 |
| PUEMAX-10≤ Total output power <PUEMAX-3 | +3.7/-5.7 | ±4.7 |
| *Note: UPH reporting accuracy is the difference between the UPH reported by the UE and the actual uplink power headroom<br>**Note: PUEMAX, is the maximum output power level without tolerance defined for the power class of the UE in 3GPP TS 25.102 | | |

NOTE1: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

### 8.7.13.1B UPH for 7.68Mcps TDD Option

Void

# 9 Performance requirements for HSDPA

## 9.1 General

Receiving performance test of the UE is implemented during communicating with the SS via air interface. The procedure uses normal call protocol until the UE is communicating on traffic channel basically. (Refer to TS 34.108 [3] Common Test Environments for User Equipment (UE) Conformance Testing.) On the traffic channel, the UE provides special function for testing that is described in Logical Test Interface and the UE is tested using this function. (Refer to TS 34.109 [4] Logical Test Interface (FDD/TDD) Special conformance testing functions.)

Unless otherwise stated the receiver characteristics are specified at the antenna connector of the UE. For UE(s) with an integral antenna only, a reference antenna with a gain of 0 dBi is assumed. UE with an integral antenna may be taken into account by converting these power levels into field strength requirements, assuming a 0 dBi gain antenna. Receiver characteristics for UE(s) with multiple antennas/antenna connectors are for further study.

The UE antenna performance has a significant impact on system performance, and minimum requirements on the antenna efficiency are therefore intended to be included in future versions of the present document. It is recognized that different requirements and test methods are likely to be required for the different types of UE.

All Bit Error ratio (BER) measurements shall be performed according to the general rules for statistical testing in Annex F.6.

## 9.2 Performance requirement for 3.84 Mcps TDD option

During the Fixed Reference Channel tests the behaviour of the Node-B emulator in response to the ACK/NACK signalling field of the HS-DPCCH is specified in Table 9.2.1:

**Table 9.2.1: Node-B Emulator Behaviour in response to ACK/NACK/DTX**

| 0HS-DPCCH ACK/NACK Field State | 0Node-B Emulator Behaviour |
|---|---|
| 1ACK | 1ACK: new transmission using 1st redundancy and constellation version (RV) |
| 2NACK | 2NACK: retransmission using the next RV (up to the maximum permitted number or RV's) |
| 3DTX | 3DTX: retransmission using the RV previously transmitted to the same H-ARQ process |

### 9.2.1 HS-DSCH throughput for Fixed Reference Channels

#### 9.2.1.1 Definition and applicability

The HS-DSCH data throughput for fixed reference channels is defined by the capabilities of the UE as defined in [24] , and the throughput is measured by counting the amount of data succeesfully received at the UE by monitoring the ACK/NACK signalling field of the HS-SICH received at the SS, while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 3.84 Mcps TDD UE from release 5 and later that support HSDPA.

#### 9.2.1.2 Minimum requirements

The requirements are specified in terms of minimum information bit throughput R for the DL reference channels specified in Annex C.4.1 with the addition of the relevant parameters in Tables 9.2.1.1 and 9.2.1.3.

Using this configuration the throughput shall meet or exceed the minimum requirements specified in tables 9.2.1.2 and 9.2.1.4.

**Table 9.2.1.1: Test parameters for fixed reference measurement channel requirements for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option) QPSK**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | |
| Scrambling code and basic midamble code number* | - | 0, 1 | | | |
| Number of TS | - | 8 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..16 | | | C(i,16) i=1..14 |
| Number of Hybrid ARQ processes | - | 4 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence** | - | {0,0,0,0} s=1, R=0, b=0 | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -12,04 | | | -11.46 |
| $\frac{\sum HS-PDSCH_E_c}{I_{or}}$ | dB | 0 | | | |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |
| Note **: This sequence implies Chase combining | | | | | |

**Table 9.2.1.2: Performance requirements for fixed reference measurement channel requirement in multipath channels for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option) QPSK**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 8,5 | 1300 |
| 2 | PB3 | 9,0 | 1300 |
| 3 | VA30 | 9,75 | 1300 |
| 4 | VA120 | 11,5 | 1400 |

**Table 9.2.1.3: Test parameters for fixed reference measurement channel requirements for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option) 16QAM**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 0, 1 | | | |
| Number of TS | - | 8 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..16 | | | C(i,16) i=1..14 |
| Number of Hybrid ARQ processes | - | 4 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence** | - | {0,0,0,0} s=1, r=0 | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -12,04 | | | -11,46 |
| $\frac{\sum HS-PDSCH_E_c}{I_{or}}$ | dB | 0 | | | |
| $I_{oc}$ | dBm/3,84 MHz | -60 | | | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note **: This sequence implies Chase combining | | | | | |

**Table 9.2.1.4: Performance requirements for fixed reference measurement channel requirement in multipath channels for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option) 16QAM**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 16,0 | 2600 |
| 2 | PB3 | 17,5 | 2600 |
| 3 | VA30 | 18,5 | 2600 |
| 4 | VA120 | 14,5 | 1600 |

The reference for this requirement is TS 25.102 [1] clauses 9.1.1.1 and 9.1.1.2.

## 9.2.1.3 Test purpose

To verify the ability of the receiver to receive a predefined test signal, representing a multi-path fading channel with information bit throughput R not falling below a specified value.

## 9.2.1.4 Method of test

### 9.2.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS (node B emulator) and fader and AWGN noise source to the UE antenna connector as shown in figure A.10.

2) Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3 with levels according to Annex E.3.

3) Set the node B emulator ACK/NACK/DTX behaviour according to table 9.2.1. Set the test parameters and levels for tests 1-4 according to tables 9.2.1.1. The reference channel configuration is defined in section C.4.1. The configuration of the TX power for downlink physical channels is annex in E.3.

4) The information bit data shall be pseudo random and not repeated before 10 different information bit payload blocks are processed. (e.g. Fixed reference Channel Definition (16 QAM, test 1,2,3): The information bit payload block is 52996 bits long. Hence the PRBSequence must be at least 52996 * 10 bits long. ) Use a PRBS from ITU-T O.153 Ref [25].

5) Setup the fading simulator with fading conditions as described in table D.2.2.1A.

9.2.1.4.2 Procedure

a. Once the HSDPA connection is setup, start transmitting HSDPA data.

b. For all relevant propagation conditions, for all relevant Ioc levels, for all relevant Ec/Ior, for all relevant Îor/Ioc defined in Tables 9.2 and 9.4 count the number of NACK, ACK and statDTX on the UL HS-SICH during the test interval and decide pass or fail according to Annex F.6.3 tables F.6.3.5.3 and F.6.3.5.4.

### 9.2.1.5 Test Requirements

Tables 9.2.1.1 to 9.2.1.4 define the primary level settings including test tolerance for all relevant throughput tests. The pass / fail decision for throughput is done according to Annex F.6.3.

## 9.2.2 HS-DSCH throughput for Variable Reference Channels

### 9.2.2.1 Definition and applicability

The HS-DSCH data throughput for variable reference channels is defined by the capabilities of the UE as defined in [24] , and the throughput is measured by counting the amount of data succeesfully received at the UE by monitoring the ACK/NACK signalling field of the HS-SICH received at the SS, while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 3.84 Mcps TDD UE from release 5 and later that support HSDPA.

### 9.2.2.2 Minimum requirements

For the parameters specified in Table 9.2.2.1 the measured throughput R shall exceed the throughput specified in Table 9.2.2.2 for each radio condition. The variable Reference Channel is specified in Annex C.4.3.

**Table 9.2.2.1: Test parameters for variable reference measurement channel requirements for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| Scrambling code and basic midamble code number* | - | 0, 1 | | | |
| Number of TS | - | 8 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16)<br>i=1..16 | | | |
| Number of Hybrid ARQ processes** | - | 4 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 1 | | | |
| Redundancy and constellation version coding sequence | (Xrv, s, r, b) | (0, 1, 0, 0) | | | |
| $HS\text{-}PDSCH_i_Ec/Ior$ | dB | –12,04 | | | |
| $\frac{\sum_{1}^{i} HS-PDSCH_Ec_i}{Ior}$ | dB | 0 | | | |
| $I_{oc}$ | dBm/3,84MHz | -60 | | | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note **: For timing requirements, HARQ is not active | | | | | |

**Table 9.2.2.2: Performance requirements for variable reference measurement channel requirement in multipath channels for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option)**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 8,8 | 1240 |
| | | 14,8 | 2500 |
| | | 18,8 | 3600 |
| | | 24,8 | 5000 |
| 2 | PB3 | 8,8 | 1220 |
| | | 14,8 | 2430 |
| | | 20,8 | 4030 |
| | | 24,8 | 5080 |
| 3 | VA30 | 10,1 | 1190 |
| | | 16,1 | 2290 |
| | | 20,1 | 3220 |
| | | 24,1 | 4260 |
| 4 | VA120 | 7,1 | 590 |
| | | 11,1 | 1180 |
| | | 15,1 | 1840 |
| | | 19,1 | 2390 |

## 9.2.2.3 Test purpose

To verify that the UE receiver is capable meeting the minimum requirements for support of HSDPA in the conditions defined below, with the selection of QPSK and 16QAM modulation, and appropriate blocksize being determined by the SS based on the CQI reported by the UE..

## 9.2.2.4 Method of test

### 9.2.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS (node B emulator) and fader and AWGN noise source to the UE antenna connector as shown in figure A.10.

2) Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3 with levels according to Annex E.3.

3) Set the node B emulator ACK/NACK/DTX behaviour according to table 9.2.1. Set the test parameters and levels for tests 1-4 according to tables 9.2.2.1. The reference channel configuration is defined in section C.4.3. The configuration of the TX power for downlink physical channels is annex in E.3.

4) The information bit data shall be pseudo random and not repeated before 10 different information bit payload blocks are processed. Use a PRBS from ITU-T O.153 Ref [25]

5) Setup the fading simulator with fading conditions as described in table D.2.2.1A.

### 9.2.2.4.2 Procedure

1) Once the HSDPA connection is setup, start transmitting HSDPA Data.

2) The SS shall transmit a HSDPA block to UE selecting any TBS value,then SS will decode the CQI report and transmit a new block on the HS-DSCH with the transport block size and modulation scheme recommended by the UE.

3) For any HSDPA block transmitted by the SS, record transmitted block size and relevant received ACK, NACK and statDTX reported by UE. If UE reports ACK,the transmitted block is correctly received by UE.Continue transmission of the HS-PDSCH data and record transmitted block size until [2000] records have been reached.

4) For all relevant propagation conditions ,calculate the throughput, which is the ratio of the sum of correctly received transport bits over the testing time. [2000] multiplied by transmission time interval is the testing time.

### 9.2.2.5 Test Requirements

Table 9.2.2.1 define the primary level settings including test tolerance for all relevant throughput tests. the measured throughput shall be equal to or better than the the specified value in table 9.2.2.2.

## 9.2.3 Reporting of HS-DSCH Channel Quality Indicator

### 9.2.3.1 Definition and applicability

The reporting accuracy of channel quality indicator (CQI) under AWGN environments is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The CQI is measured while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 3.84 Mcps TDD UE from release 5 and later that support HSDPA.

### 9.2.3.2 Minimum requirements

For the parameters specified in Table 9.2.3.1, the reported CQI value shall be in the range of +/-10 of the reported median more than 90% of the time. The BLER for the reported median CQI shall be less than 10%.

**Table 9.2.3.1: Test parameters for variable reference measurement channel requirements for 7,3 Mbps – Category 8 - UE (3,84 Mcps TDD Option)**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| Scrambling code and basic midamble code number* | - | 0, 1 | |
| Number of TS | - | 8 | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16)<br>i=1..16 | |
| Number of Hybrid ARQ processes** | - | 4 | |
| Maximum number of Hybrid ARQ transmissions | - | 1 | |
| Redundancy and constellation version coding sequence | (Xrv, s, r, b) | (0, 1, 0, 0) | |
| HS-PDSCH$_i$_Ec/Ior | dB | –12,04 | |
| $\frac{\sum_{1}^{i} HS-PDSCH_Ec_i}{Ior}$ | dB | 0 | |
| $\hat{I}_{or} / I_{oc}$ | dB | 5 | 10 |
| $I_{oc}$ | dBm/3,84MHz | -60 | |
| Note*: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note**: For timing requirements, HARQ is not active | | | |

**Table 9.3.3.2 Performance requirements for CQI reporting measurement channel requirements for 7.3 Mbps – Category 9- UE (3.84 Mcps TDD Option)**

| Test | Permitted CQI range from median (x) | % of time that CQI must be within +/- x of median (Y) | Maximum BLER for median reported CQI |
|---|---|---|---|
| Test 1 | +/- 10 | 90 | 10% |
| Test 2 | | | |

### 9.2.3.3 Test purpose

To verify that the variance of the CQI reports in an AWGN channel is within the limits defined and that a BLER of better than 10% is obtained for the median reported CQI.

### 9.2.3.4 Method of test

#### 9.2.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS and an AWGN source to the UE antenna connector as shown in figure A.9.

2) Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3 with levels according to Annex E.3.

3) Set the node B emulator ACK/NACK/DTX behaviour according to table 9.2.1. Set the test parameters and levels for tests 1-2 according to tables 9.2.3.1. The reference channel configuration is defined in section C.4.1. The configuration of the TX power for downlink physical channels is annex in E.3.

4) The information bit data shall be pseudo random and not repeated before 10 different information bit payload blocks are processed. (e.g. Fixed reference Channel Definition (16 QAM, test 1,2,3): The information bit payload block is 52996 bits long. Hence the PRBSequence must be at least 52996 * 10 bits long. ) Use a PRBS from ITU-T O.153 Ref [25].

#### 9.2.3.4.2 Procedure

Note: the following part of the procedure will test, if the UE reports a limited range of CQI indicator under the predefined channel conditions.

1) The SS shall transmit a HSDPA block to UE selecting any TBS value excluding 0,then SS will decodes the CQI report and transmits a new block on the HS-DSCH with the transport block size and modulation scheme recommended by the UE.For any HSDPA block transmitted by the SS, record the received RTBS value of CQI indicator and the ACK, NACK and statDTX . Continue transmission of the HS-PDSCH data and collect RTBS value of CQI until [2000] reports have been gathered.

2) Set up a relative frequency distribution for the RTBS value of CQI indicator reported. Calculate the median value (Median RTBS is the RTBS that is at or crosses 50% distribution from the lower RTBS side). This RTBS value of CQI indicator is declared as Median CQI value,

3) If [1800] or more of the RTBS value of CQI indicator are in the range (Median CQI - 10) ≤ Median CQI ≤ ( Median CQI + 10) then continue with next step , otherwise fail the UE.

4) Calculate the the ratio ( NACK + statDTX / ACK + NACK + statDTX) when the SS transmit the TBS according to the median-CQI value.If the ratio< 0.1 ,then pass the UE,otherwise fail the UE.

5) Repeat the same procedure (steps 2 to 5 ) with test conditions according to the table 9.3.3.1 for Test 2, Test 3 and Test 4

### 9.2.3.5 Test Requirements

The pass fail decision as specified in the test procedure in 9.3.1.4.2.

No test tolerance is applied to the test parameters.

## 9.2.4 HS-SCCH Detection Performance

void

## 9.2.4A HS-SCCH Detection Performance

### 9.2.4.1 Definition and applicability

The detection performance of the HS-SCCH is determined by the probability of event $E_m$, which is declared when the UE is signaled on HS-SCCH, but DTX is observed in the corresponding HS-SICH ACK/NACK field. The probability of event $E_m$ is denoted $P(E_m)$.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 3.84 Mcps TDD UE from release 5 and later that support HSDPA.

## 9.2.4.2 Minimum requirements

For the test parameters in Table 9.2.4.1, for each value of HS-SCCH-1 $E_c/I_{or}$ specified in Table 9.2.4.2, the measured $P(E_m)$ shall be less than or equal to the corresponding specified value of $P(E_m)$.

**Table 9.2.4.1: Test parameters for HS-SCCH detection (3.84 Mcps TDD option)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| Number of TS under test | - | 1 | | |
| Number of HS-SCCH codes per timeslot | - | 4 | | |
| HS-SCCH UE Identity ( $x_{ue,1}$ , $x_{ue,2}$ , …, $x_{ue,16}$ ) | - | UE1 = 0000000000000000 (UE1 under test)<br>UE2 = 0101010101010101<br>UE3 = 1010101010101010<br>UE4 = 1111111111111111 | | |
| HS-SCCH Channelization Codes* | C(k,Q) | HS-SCCH-1 = C(1, 16), for UE1 (UE under test)<br>HS-SCCH-2 = C(2, 16) for UE2<br>HS-SCCH-3 = C(3, 16) for UE3<br>HS-SCCH-4 = C(4, 16) for UE4 | | |
| HS-SCCH $E_c/I_{or}$ | dB | HS-SCCH-2_$E_c/I_{or}$ = HS-SCCH-3_$E_c/I_{or}$ = HS-SCCH-4_$E_c/I_{or}$,<br>Where, $\sum$ HS-SCCH-X_$E_c/I_{or}$ = 1, where X = 1, 2, 3, 4 | | |

**Table 9.2.4.2: Minimum requirement for HS-SCCH detection (3.84 Mcps TDD option)**

| Test Number | Propagation Conditions | Reference value | | |
|---|---|---|---|---|
| | | HS-SCCH-1 $E_c / I_{or}$ (dB) | $\hat{I}_{or} / I_{oc}$ (dB) | $P(E_m)$ |
| 1 | PA3 | -1.6 | 0 | 0.05 |
| 2 | PA3 | -3.0 | 5 | 0.01 |
| 3 | VA30 | -2.5 | 0 | 0.01 |

The reference for this requirement is TS 25.102 [2] clause 9.1.4.

## 9.2.4.3 Test purpose

To verify that $P(E_m)$ does not exceed the limit in table 9.2.4.2.

## 9.2.4.4 Method of test

### 9.2.4.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1. Connect SS, multipath fading simulator and an AWGN noise source to the UE antenna connector as shown in figure A.10.
2. Setup fading simulators as fading condition, which are described in table D.2.2.1A.
3. Set the node B emulator ACK/NACK/DTX behaviour according to table 9.2.1. Set the test parameters and levels for tests 1-3 according to tables 9.2.4.1 and 9.2.4.2. The reference channel configuration is defined in section C.4.1. The configuration of the TX power for downlink physical channels is annex in E.3.

### 9.2.4.4.2 Procedure

1. The UE is switched on.
2. Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3.

3. Count the number of NACK, ACK and statDTX on the UL HS-DPCCH during the test interval and decide pass or fail according to Annex F.6.1 and table F.6.1.8. NACK and ACK are counted as a pass and statDTX is counted as a failure.

#### 9.2.4.5 Test Requirements

The probability of event $E_m$ denoted as $P(E_m)$ (test procedure step 3) shall not exceed the specified value in table 9.2.4.2.

No test tolerance is applied to the test parameters.

## 9.3 Performance requirements for 1.28 Mcps TDD option

### 9.3.1 HS-DSCH throughput for Fixed Reference Channels

The overall performance metric for HS-DSCH requirements is the throughput "R" measured on HS-DSCH.

#### 9.3.1.1 Definition and applicability

The HS-DSCH data throughput for fixed reference channels is defined by the capabilities of the UE as defined in [24] , and the throughput is measured by counting the amount of data succeesfully received at the UE by monitoring the ACK/NACK signalling field of the HS-SICH received at the SS, while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 1,28 Mcps TDD UE from release 5 and later that support HSDPA.

#### 9.3.1.2 Minimum requirements

For the parameters specified in below tables the measured throughput shall be equal to or better than the the specified value in belows table specified the performance requirement. The reference for this requirement is TS 25.102 [1] section 9.

#### 9.3.1.3 Test purpose

To verify that the UE receiver is capable meeting the minimum requirements for support of HSDPA in the conditions defined below, for QPSK and 16QAM .modulation.

**Table 9.3.1.1: Node-B Emulator Behaviour in response to ACK/NACK/DTX (Fixed reference channel)**

| HS-DPCCH ACK/NACK Field State | Node-B Emulator Behaviour |
|---|---|
| ACK | ACK: new transmission using 1st redundancy version (RV) |
| NACK | NACK: retransmission using the next RV (up to the maximum permitted number or RV's) |
| DTX | DTX: retransmission using the RV previously transmitted to the same H-ARQ process |

**Table 9.3.1.2: Test parameters for fixed reference measurement channel requirements for 0.5 Mbps UE class QPSK**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | |
| Scrambling code and basic midamble code number* | - | 1 | | | |
| Midamble | | Common midamble | | | |
| Number of TS | | 2 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..10 | | | |
| Number of Hybrid ARQ processes | - | 4 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence | - | {0,0,0,0} | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -10 | | | |
| $I_{oc}$ | dBm/1.28 MHz | -60 | | | |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 9.3.1.3: Minimum performance requirements for fixed reference measurement channel requirement in multipath channels for 0.5Mbps UE class QPSK**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10 | 160 |
| 2 | PB3 | 10 | 170 |
| 3 | VA30 | 10 | 161 |
| 4 | VA120 | 10 | 153 |

**Table 9.3.1.4: Test parameters for fixed reference measurement channel requirements for 1.1 Mbps UE class, 16QAM**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 1 | | | |
| Midamble | | Common midamble | | | |
| Number of TS | - | 2 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..12 | | | |
| Number of Hybrid ARQ processes | - | 4 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence | - | {6,2,1,5} | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -10.8 | | | |
| $I_{oc}$ | dBm/1.28 MHz | -60 | | | |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | |

**Table 9.3.1.5: Minimum performance requirements for fixed reference measurement channel requirement in multipath channels for 1.1Mbps UE class, 16QAM**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 388 |
| 2 | PB3 | 15 | 347 |
| 3 | VA30 | 15 | 316 |
| 4 | VA120 | 15 | 274 |

**Table 9.3.1.6: Test parameters for fixed reference measurement channel requirements for 1.6 Mbps UE class**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 | Test 6 | Test 7 | Test 8 |
|---|---|---|---|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 1 | | | | | | | |
| Midamble | | Common midamble | | | | | | | |
| Number of TS | - | 3 | | | | | | | |
| Number of Hybrid ARQ processes | - | 4 | | | | | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | | | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..10 | | | | C(i,16) i=1..12 | | | |
| Redundancy and constellation version coding sequence | - | {0,0,0,0} | | | | {6,2,1,5} | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -10 | | | | -10.8 | | | |
| $I_{oc}$ | dBm/ 1.28MHz | -60 | | | | | | | |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | | | | | |

**Table 9.3.1.7: Minimum performance requirements for fixed reference measurement channel requirement in multipath channels for 1.6Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10 | 270 |
| 2 | PB3 | 10 | 278 |
| 3 | VA30 | 10 | 259 |
| 4 | VA120 | 10 | 242 |
| 5 | PA3 | 15 | 488 |
| 6 | PB3 | 15 | 471 |
| 7 | VA30 | 15 | 431 |
| 8 | VA120 | 15 | 377 |

**Table 9.3.1.8: Test parameters for fixed reference measurement channel requirements for 2.2 Mbps UE class**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 | Test 6 | Test 7 | Test 8 |
|---|---|---|---|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 1 | | | | | | | |
| Minimum | | Common midamble | | | | | | | |
| Number of TS | - | 4 | | | | | | | |
| Number of Hybrid ARQ processes | - | 4 | | | | | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | | | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..10 | | | | C(i,16) i=1..12 | | | |
| Redundancy and constellation version coding sequence | - | {0,0,0,0} | | | | {6,2,1,5} | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -10 | | | | -10.8 | | | |
| $I_{oc}$ | dBm/ 1.28MHz | -60 | | | | | | | |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | | | | | |

**Table 9.3.1.9: Minimum performance requirements for fixed reference measurement channel requirement in multipath channels for 2.2Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10 | 360 |
| 2 | PB3 | 10 | 343 |
| 3 | VA30 | 10 | 320 |
| 4 | VA120 | 10 | 275 |
| 5 | PA3 | 15 | 615 |
| 6 | PB3 | 15 | 606 |
| 7 | VA30 | 15 | 554 |
| 8 | VA120 | 15 | 493 |

**Table 9.3.1.10: Test parameters for fixed reference measurement channel requirements for 2.8 Mbps UE class**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 | Test 6 | Test 7 | Test 8 |
|---|---|---|---|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 1 | | | | | | | |
| Midamble | | Common midamble | | | | | | | |
| Number of TS | - | 5 | | | | | | | |
| Number of Hybrid ARQ processes | - | 4 | | | | | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | | | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) i=1..10 | | | | C(i,16) i=1..12 | | | |
| Redundancy and constellation version coding sequence | - | {0,0,0,0} | | | | {6,2,1,5} | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -10 | | | | -10.8 | | | |
| $I_{oc}$ | dBm/ 1.28MHz | -60 | | | | | | | |
| *Note: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | | | | | | | |

**Table 9.3.1.11: Minimum performance requirements for fixed reference measurement channel requirement in multipath channels for 2.8Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10 | 461 |
| 2 | PB3 | 10 | 470 |
| 3 | VA30 | 10 | 438 |
| 4 | VA120 | 10 | 409 |
| 5 | PA3 | 15 | 890 |
| 6 | PB3 | 15 | 810 |
| 7 | VA30 | 15 | 730 |
| 8 | VA120 | 15 | 630 |

These RF channels are described in Appendix D

**Table 9.3.1.12: Test requirements for fixed reference measurement channel requirement in multipath channels for 0.5Mbps UE class QPSK**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10.6 | 160 |
| 2 | PB3 | 10.6 | 170 |
| 3 | VA30 | 10.6 | 161 |
| 4 | VA120 | 10.6 | 153 |

**Table 9.3.1.13: Test requirements for fixed reference measurement channel requirement in multipath channels for 1.1Mbps UE class, 16QAM**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15.6 | 388 |
| 2 | PB3 | 15.6 | 347 |
| 3 | VA30 | 15.6 | 316 |
| 4 | VA120 | 15.6 | 274 |

**Table 9.3.1.14: Test requirements for fixed reference measurement channel requirement in multipath channels for 1.6Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10.6 | 270 |
| 2 | PB3 | 10.6 | 278 |
| 3 | VA30 | 10.6 | 259 |
| 4 | VA120 | 10.6 | 242 |
| 5 | PA3 | 15.6 | 488 |
| 6 | PB3 | 15.6 | 471 |
| 7 | VA30 | 15.6 | 431 |
| 8 | VA120 | 15.6 | 377 |

**Table 9.3.1.15: Test requirements for fixed reference measurement channel requirement in multipath channels for 2.2Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10.6 | 360 |
| 2 | PB3 | 10.6 | 343 |
| 3 | VA30 | 10.6 | 320 |
| 4 | VA120 | 10.6 | 275 |
| 5 | PA3 | 15.6 | 615 |
| 6 | PB3 | 15.6 | 606 |
| 7 | VA30 | 15.6 | 554 |
| 8 | VA120 | 15.6 | 493 |

**Table 9.3.1.16: Test requirements for fixed reference measurement channel requirement in multipath channels for 2.8Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 10.6 | 461 |
| 2 | PB3 | 10.6 | 470 |
| 3 | VA30 | 10.6 | 438 |
| 4 | VA120 | 10.6 | 409 |
| 5 | PA3 | 15.6 | 890 |
| 6 | PB3 | 15.6 | 810 |
| 7 | VA30 | 15.6 | 730 |
| 8 | VA120 | 15.6 | 630 |

### 9.3.1.4 Method of test

#### 9.3.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set Ack/Nack/ DTX handling at the SS as table 9.3.1.1,Setup fading simulators as fading conditions, which are described in table D.2.2.1.A.

3) Set up a HSDPA connection according to the Generic HSDPA setup procedure. See TS 34.108 [3] and TS 34.109 [4] for details regarding generic HSDPA setup procedure.

4) Set the test parameters according to above tables of test parameters. The configuration of the downlink channels is defined in table C.4.2.

#### 9.3.1.4.2 Procedure

1) Once the HSDPA connection is setup, start transmitting HSDPA Data.

2) The information bit data shall be pseudo random and not repeated not before 10 different information bit payload blocks are processed. (e.g. Fixed reference Channel Definition (16 QAM): The information bit payload block is 3650 bits long. Hence the PRBSequence must be at least 3650 * 10 bits long. ) Use a PRBS from ITU-R O.153 Ref [26].

3) For all relevant propagation conditions,count the number of NACK, ACK and statDTX on the UL HS-SICH during the test interval and decide pass or fail according to Annex F.6.3 tables F.6.3.5.1 and F.6.3.5.2.

### 9.3.1.5 Test Requirements

Tables of test parameters define the primary level settings including test tolerance for all relevant throughput tests. the measured throughput shall be equal to or better than the the specified value in below tables of test requirement.

The pass / fail decision for throughput is done according to Annex F.6.3.

## 9.3.2 HS-DSCH throughput for Variable Reference Channels

The overall performance metric for HS-DSCH requirements is the throughput “R” measured on HS-DSCH.

### 9.3.2.1 Definition and applicability

The HS-DSCH data throughput for variable reference channels is defined by the capabilities of the UE as defined in [24] , and the throughput is measured by counting the amount of data succeesfully received at the UE by monitoring the ACK/NACK signalling field of the HS-SICH received at the SS, while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 1,28 Mcps TDD UE from release 5 and later that support HSDPA.

### 9.3.2.2 Minimum requirements

For the parameters specified in tables 9.3.2.1 the measured throughput shall be equal to or better than the the specified value in table 9.3.2.2. The reference for this requirement is TS 25.102 [1] section 9.

### 9.3.2.3 Test purpose

To verify that the UE receiver is capable meeting the minimum requirements for support of HSDPA in the conditions defined below, with the selection of QPSK and 16QAM modulation, and appropriate blocksize being determined by the SS based on the CQI reported by the UE..

**Table 9.3.2.1: Test parameters for variable reference measurement channel requirements for 0.5 Mbps UE class**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| HS-PDSCH Modulation and TBS | - | * | | |
| Scrambling code and basic midamble code Number ** | - | 1 | | |
| Midamble | | Common midamble | | |
| Number of TS | - | 2 | | |
| Number of $DPCH_0$ | - | 0 | | |
| Number of HARQ Process | - | 4 | | |
| Number of transmission | - | 1 | | |
| Redundancy and constellation version coding sequence | Xrv | 0 | | |
| HS-PDSCH Channelization Codes** | C(k,Q) | C(i,16)<br>1≤i≤10 | | |
| $HS\text{-}PDSCH_i$_Ec/Ior | dB | -10 | | |
| Ioc | dBm | -60 | | |

* Note 1 As requested by the last received CQI report
**Note 2 Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.
***Note 3 If the indicated CQI is 0, the Node-B emulator shall format the next HS-PDSCH transmission with the transport block size and the modulation scheme that were previously used.

**Table 9.3.2.2: Minimum performance requirements for variable reference measurement channel requirement in multipath channels for 0.5 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 242 |
| 2 | PB3 | 15 | 244 |
| 3 | VA30 | 15 | 211 |

**Table 9.3.2.3: Test parameters for variable reference measurement channel requirements for 1.1 Mbps UE class**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| HS-PDSCH Modulation and TBS | - | * | | |
| Scrambling code and basic midamble code Number ** | - | 1 | | |
| Midamble | | Common midamble | | |
| Number of TS | - | 2 | | |
| Number of $DPCH_0$ | - | 0 | | |
| Number of HARQ Process | - | 4 | | |
| Number of transmission | - | 1 | | |
| Redundancy and constellation version coding sequence | Xrv | 0 | | |
| HS-PDSCH Channelization Codes** | C(k,Q) | C(i,16)<br>1≤i≤10 | | |
| $HS\text{-}PDSCH_i$_Ec/Ior | dB | -10 | | |
| Ioc | dBm | -60 | | |

* Note 1 As requested by the last received CQI report
**Note 2 Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.
***Note 3 If the indicated CQI is 0, the Node-B emulator shall format the next HS-PDSCH transmission with the transport block size and the modulation scheme that were previously used.

**Table 9.3.2.4: Minimum performance requirements for variable reference measurement channel requirement in multipath channels for 1.1 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 318 |
| 2 | PB3 | 15 | 323 |
| 3 | VA30 | 15 | 213 |

**Table 9.3.2.5: Test parameters for variable reference measurement channel requirements for 1.6 Mbps UE class**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| HS-PDSCH Modulation and TBS | - | * | | |
| Scrambling code and basic midamble code Number ** | - | 1 | | |
| Midamble | | Common midamble | | |
| Number of TS | - | 3 | | |
| Number of $DPCH_0$ | - | 0 | | |
| Number of HARQ Process | - | 4 | | |
| Number of transmission | - | 1 | | |
| Redundancy and constellation version coding sequence | Xrv | 0 | | |
| HS-PDSCH Channelization Codes** | C(k,Q) | C(i,16)<br>1≤i≤10 | | |
| $HS\text{-}PDSCH_i$_Ec/Ior | dB | -10 | | |
| Ioc | dBm | -60 | | |

* Note 1 As requested by the last received CQI report
**Note 2 Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.
***Note 3 If the indicated CQI is 0, the Node-B emulator shall format the next HS-PDSCH transmission with the transport block size and the modulation scheme that were previously used.

**Table 9.3.2.6: Minimum performance requirements for variable reference measurement channel requirement in multipath channels for 1.6Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 480 |
| 2 | PB3 | 15 | 483 |
| 3 | VA30 | 15 | 323 |

**Table 9.3.2.7: Test parameters for variable reference measurement channel requirements for 2.2 Mbps UE class**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| HS-PDSCH Modulation and TBS | - | * | | |
| Scrambling code and basic midamble code Number ** | - | 1 | | |
| Midamble | | Common midamble | | |
| Number of TS | - | 4 | | |
| Number of $DPCH_0$ | - | 0 | | |
| Number of HARQ Process | - | 4 | | |
| Number of transmission | - | 1 | | |
| Redundancy and constellation version coding sequence | Xrv | 0 | | |
| HS-PDSCH Channelization Codes** | C(k,Q) | C(i,16)<br>1≤i≤10 | | |
| $HS\text{-}PDSCH_i$_Ec/Ior | dB | -10 | | |
| Ioc | dBm | -60 | | |

* Note 1 As requested by the last received CQI report
**Note 2 Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.
***Note 3 If the indicated CQI is 0, the Node-B emulator shall format the next HS-PDSCH transmission with the transport block size and the modulation scheme that were previously used.

**Table 9.3.2.8: Minimum performance requirements for variable reference measurement channel requirement in multipath channels for 2.2 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 625 |
| 2 | PB3 | 15 | 631 |
| 3 | VA30 | 15 | 418 |

**Table 9.3.2.9: Test parameters for variable reference measurement channel requirements for 2.8 Mbps UE class**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| HS-PDSCH Modulation and TBS | - | * | | |
| Scrambling code and basic midamble code Number ** | - | 1 | | |
| Midamble | | Common midamble | | |
| Number of TS | - | 2 | | |
| Number of $DPCH_0$ | - | 0 | | |
| Number of HARQ Process | - | 4 | | |
| Number of transmission | - | 1 | | |
| Redundancy and constellation version coding sequence | Xrv | 0 | | |
| HS-PDSCH Channelization Codes** | C(k,Q) | C(i,16) 1≤i≤10 | | |
| $HS\text{-}PDSCH_i$_Ec/Ior | dB | -10 | | |
| Ioc | dBm | -60 | | |

* Note 1 As requested by the last received CQI report
**Note 2 Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.
***Note 3 If the indicated CQI is 0, the Node-B emulator shall format the next HS-PDSCH transmission with the transport block size and the modulation scheme that were previously used.

**Table 9.3.2.10: Minimum performance requirements for variable reference measurement channel requirement in multipath channels for 2.8 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 783 |
| 2 | PB3 | 15 | 792 |
| 3 | VA30 | 15 | 544 |

These RF channels are described in Appendix D.

## 9.3.2.4 Method of test

### 9.3.2.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set Ack/Nack/DTX handling at the SS such that regardless of the response from the UE (Ack, Nack or DTX) new data is sent each time, this is because HARQ transmissions are set to one, i.e. no re-transmission of failed blocks.

3) Setup fading simulators as fading conditions , which are described in table D.2.2.1.A.

4) Set up a HSDPA connection according to the Generic HSDPA setup procedure,See TS 34.108 [3] and TS 34.109 [4] for details regarding generic HSDPA setup procedure.

5) Set the test parameters for tests according to above tables of test paramters

6) The information bit data shall be pseudo random and not repeated not before 10 different information bit payload blocks are processed. Use a PRBS from ITU-R O.153 Ref [26].

#### 9.3.2.4.2 Procedure

1) Once the HSDPA connection is setup, start transmitting HSDPA Data.

   The SS shall transmit a HSDPA block to UE selecting any TBS value excluding 0 or 63(the first HSDPA block size isn't recorded),then SS will decode the CQI report and transmit a new block on the HS-DSCH with the transport block size and modulation scheme recommended by the UE.

2) For any HSDPA block transmitted by the SS, record transmitted block size and relevant received ACK, NACK and statDTX reported by UE. If UE reports ACK,the transmitted block is correctly received by UE.Continue transmission of the HS-PDSCH data and record transmitted block size until [2000] records have been reached.

3) For all relevant propagation conditions ,calculate the throughput, which is the ratio of the sum of correctly received transport bits over the testing time. [2000] multiplied by transmission time interval is the testing time.

### 9.3.2.5 Test Requirements

Tables of test parameters define the primary level settings including test tolerance for all relevant throughput tests. the measured throughput shall be equal to or better than the the specified value in below tables of testrequirement.

**Table 9.3.2.11: Test requirements for variable reference measurement channel requirement in multipath channels for 0.5 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15.6 | 242 |
| 2 | PB3 | 15.6 | 244 |
| 3 | VA30 | 15.6 | 211 |

**Table 9.3.2.12: Test requirements for variable reference measurement channel requirement in multipath channels for 1.1 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15.6 | 318 |
| 2 | PB3 | 15.6 | 323 |
| 3 | VA30 | 15.6 | 213 |

**Table 9.3.2.13: Test requirements for variable reference measurement channel requirement in multipath channels for 1.6Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15.6 | 480 |
| 2 | PB3 | 15.6 | 483 |
| 3 | VA30 | 15.6 | 323 |

**Table 9.3.2.14: Test requirements for variable reference measurement channel requirement in multipath channels for 2.2 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15.6 | 625 |
| 2 | PB3 | 15.6 | 631 |
| 3 | VA30 | 15.6 | 418 |

**Table 9.3.2.15: Test requirements for variable reference measurement channel requirement in multipath channels for 2.8 Mbps UE class**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 15 | 783 |
| 2 | PB3 | 15 | 792 |
| 3 | VA30 | 15 | 544 |

The pass / fail decision for throughput is done according to Annex F.6.3.

## 9.3.3 Reporting of HS-DSCH Channel Quality Indicator

### 9.3.3.1 Definition and applicability

The reporting accuracy of channel quality indicator (CQI) under AWGN environments is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The CQI is measured while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 1,28 Mcps TDD UE from release 5 and later that support HSDPA.

### 9.3.3.2 Minimum requirements

For the parameters specified in tables 9.3.3.1 the UE shall report a CQI value within the limits of table 9.3.3.2.

### 9.3.3.3 Test purpose

To verify that the UE receiver is capable meeting the minimum requirements for support of HSDPA in the conditions defined below, such that CQI reported by the UE falls within the acceptable range.

**Table 9.3.3.1: Test parameters for CQI reporting measurement channel requirements**

| | | 0.5Mbps UE | 1.1Mbps UE | | 1.6Mbps UE | | 2.2Mbps UE | | 2.8Mbps UE | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Parameter** | **Unit** | **Test 1** | **Test 2** | **Test 3** | **Test 4** | **Test 5** | **Test 6** | **Test 7** | **Test 8** | **Test 9** |
| Number of TS | - | 2 | 2 | | 3 | | 4 | | 5 | |
| Number of HS-PDSCH codes per TS | - | 10 | 10 | | 10 | | 10 | | 10 | |
| HS-PDSCH$_i$_Ec/Ior | dB | -10 | -10 | | -10 | | -10 | | -10 | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,16) 1≤i≤10 | C(i,16) 1≤i≤10 | | C(i,16) 1≤i≤10 | | C(i,16) 1≤i≤10 | | C(i,16) 1≤i≤10 | |
| Number of DPCH$_o$ | - | 0 | | | | | | | | |
| Number of HARQ Process | - | 4 | | | | | | | | |
| Number of transmission | - | 1 | | | | | | | | |
| $\hat{I}_{or} / I_{oc}$ | dB | 1 | 1 | 8 | 1 | 8 | 1 | 8 | 1 | 8 |

**Table 9.3.3.2 Performance requirements for CQI reporting measurement channel requirements**

| Test | Permitted CQI range from median (x) | % of time that CQI must be within +/- x of median (Y) | Maximum BLER for median reported CQI |
|---|---|---|---|
| Test 1 | +/- 3 | 90 | 10% |
| Test 2 | +/- 3 | 90 | |
| Test 3 | +/- 2 | 90 | |
| Test 4 | +/- 2 | 90 | |

#### 9.3.3.4 Method of test

##### 9.3.3.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set Ack/Nack/DTX handling at the SS such that regardless of the response from the UE (Ack, Nack or DTX) new data is sent each time, this is because HARQ transmissions are set to one, i.e. no re-transmission of failed blocks.

3) Set up a call according to the Generic call setup procedure.

See TS 34.108 [3] and TS 34.109 [4] for details regarding generic call setup procedure.

Note: The following value of x is derived from Table 9.3.3.2

##### 9.3.3.4.2 Procedure

1) Set test conditions according to test 1 according table 9.3.3.1.

Note: the following part of the procedure will test, if the UE reports a limited range of CQI indicator under the predefined channel conditions.

2) The SS shall transmit a HSDPA block to UE selecting any TBS value excluding 0 or 63,then SS will decodes the CQI report and transmits a new block on the HS-DSCH with the transport block size and modulation scheme recommended by the UE.For any HSDPA block transmitted by the SS, record the received RTBS value of CQI indicator and the ACK, NACK and statDTX . Continue transmission of the HS-PDSCH data and collect RTBS value of CQI until [2000] reports have been gathered.

3) Set up a relative frequency distribution for the RTBS value of CQI indicator reported. Calculate the median value (Median RTBS is the RTBS that is at or crosses 50% distribution from the lower RTBS side). This RTBS value of CQI indicator is declared as Median CQI value,

4) If [1800] or more of the RTBS value of CQI indicator are in the range (Median CQI - x) ≤ Median CQI ≤ ( Median CQI + x) then continue with next step , otherwise fail the UE.

5) Calculate the the ratio ( NACK + statDTX / ACK + NACK + statDTX) when the SS transmit the TBS according to the median-CQI value.If the ratio< 0.1 ,then pass the UE,otherwise fail the UE.

6) Repeat the same procedure (steps 2 to 5) with test conditions according to the table 9.3.3.1 for Test 2 to Test 8.

#### 9.3.3.5 Test Requirements

The pass fail decision as specified in the test procedure in 9.3.3.4.2.

No test tolerance is applied to the test parameters.

### 9.3.4 HS-SCCH Detection Performance

#### 9.3.4.1 Definition and applicability

The detection performance of the HS-SCCH is determined by the probability of event $E_m$, which is declared when the UE is signaled on HS-SCCH, but DTX is observed in the corresponding HS-SICH ACK/NACK field. The probability of event $E_m$ is denoted $P(E_m)$.

This corresponds to case when the SS indicates to the UE via the HS-SCCH that HSDPA data is to be sent, but the UE fails to decode this correctly.

The normative reference for this test is 25.102 [2], section 9.

The requirements and this test apply to 1,28 Mcps TDD UE from release 5 and later that support HSDPA.

### 9.3.4.2 Minimum requirements

For the parameters specified in tables 9.3.4.1, for each value of HS-SCCH $\hat{I}_{or} / I_{oc}$ specified in Table 9.3.4.3 the measured $P(E_m)$ shall be less than or equal to the corresponding specified value of $P(E_m)$. The reference for this requirement is TS 25.102 [1] section 9.

### 9.3.4.3 Test purpose

To verify that the UE receiver is capable meeting the minimum requirements for support of HSDPA in the conditions defined below, and does not report a DTX when valid data was sent, more often than the performance limits allow.

**Table 9.3.4.1: Test parameters for HS-SCCH detection (1.28Mcps TDD option)**

| Parameter | Unit | Test 1 | Test2 |
|---|---|---|---|
| Number of TS under test | - | 1 | |
| Number of HS-SCCH codes per timeslot | - | 8 (4 x2) | |
| Scrambling code and basic midamble code number* | - | 0 | |
| Midamble | | Common midamble | |
| Number of $DPCH_o$ | - | 2 | |
| Number of H-ARQ process | - | 4 | |
| HS-SCCH UE Identity ( $x_{ue,1}$, $x_{ue,2}$, …, $x_{ue,16}$ ) | - | UE1 = 0000000000000000 (UE1 under test)<br>UE2 = 0101010101010101<br>UE3 = 1010101010101010<br>UE4 = 1111111111111111 | |
| HS-SCCH Channelization Codes* | C(k,Q) | C(i,16)<br>1≤i≤8 | |
| HS-SCCH Channelization Codes for UE under test | C(k,Q) | C(i,16)<br>1≤i≤2 | |
| $DPCH_o$ Channelization Codes | C(k,Q) | C(i,16)<br>9≤i≤10 | |
| $\frac{HS-SCCH_i_E_c}{I_{or}}$ | dB | -10 | |
| $I_{oc}$ | dBm/1.28MHz | -60 | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code. | | | |

**Table 9.3.4.2: Test parameters for HS-PDSCH (1.28Mcps TDD option)**

| Parameter | Unit | Test 1 | Test2 |
|---|---|---|---|
| Number of TS under test | - | 2 | |
| Number of HS-PDSCH codes per timeslot | - | 10 | |
| Scrambling code and basic midamble code number* | - | 0 | |
| HS-PDSCH Channelization Codes for UE under test | C(k,Q) | C(i,16)<br>1≤i≤10 | |
| $\frac{HS-\mathrm{PDS}CH_i_E_c}{I_{or}}$ | dB | -10 | |
| $I_{oc}$ | dBm/1.28MHz | -60 | |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | ** | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note **: Allocate a sufficient power to $\frac{\hat{I}_{or}}{I_{oc}}$ so that probability of missing HS-DPSCH is very low. | | | |

**Table 9.3.4.3: Minimum requirement for HS-SCCH detection (1.28Mcps TDD option)**

| Test Number | Propagation Conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | $P(E_m)$ |
|---|---|---|---|
| 1 | PA3 | 16 | 0.01 |
| 2 | VA30 | 12 | 0.01 |

The propagation conditions are described in Annex D.

### 9.3.4.4 Method of test

#### 9.3.4.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect SS, multipath fading simulator and an AWGN source to the UE antenna connector as shown in figure A.10.

2) Set Ack/Nack handling at the SS as table 9.3.1.1 (Fixed channel test)

3) Set up test parameter according to table 9.3.4.1 and able 9.3.4.2.

#### 9.3.4.4.2 Procedure

1. The UE is switched on.

2. An RRC connection is set-up according to the generic HSDPA set-up procedure specified in TS 34.108 [3].

3. Count the number of NACK, ACK and statDTX on the UL HS-SICH during the test interval and decide pass or fail according to Annex F.6.1 and table F.6.1.8. NACK and ACK are counted as a pass and statDTX is counted as a failure.

### 9.3.4.5 Test Requirements

The probability of event $E_m$ denoted as $P(E_m)$ (test procedure step 3) shall not exceed the specified value in table 9.3.4.4.

**Table 9.3.4.4: Test requirement for HS-SCCH detection (1.28Mcps TDD option)**

| Test Number | Propagation Conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | $P(E_m)$ |
|---|---|---|---|
| 1 | PA3 | 16.6 | 0.01 |
| 2 | VA30 | 12.6 | 0.01 |

## 9.4 Performance requirement for 7.68 Mcps TDD option

During the Fixed Reference Channel tests the behaviour of the Node-B emulator in response to the ACK/NACK signalling field of the HS-DPCCH is specified in Table 9.4.1:

**Table 9.4.1: Node-B Emulator Behaviour in response to ACK/NACK/DTX**

| HS-DPCCH ACK/NACK Field State | Node-B Emulator Behaviour |
|---|---|
| ACK | ACK: new transmission using 1st redundancy and constellation version (RV) |
| NACK | NACK: retransmission using the next RV (up to the maximum permitted number or RV's) |
| DTX | DTX: retransmission using the RV previously transmitted to the same H-ARQ process |

# 9.4.1 HS-DSCH throughput for Fixed Reference Channels

## 9.4.1.1 Definition and applicability

The HS-DSCH data throughput for fixed reference channels is defined by the capabilities of the UE as defined in [24] , and the throughput is measured by counting the amount of data succeesfully received at the UE by monitoring the ACK/NACK signalling field of the HS-SICH received at the SS, while random data is streamed from the SS to the UE.

The normative reference for this test is 25.102 [2], section 9.3.

The requirements and this test apply to 7.68 Mcps TDD UE from release 7 and later that support HSDPA.

## 9.4.1.2 Minimum requirements

The requirements are specified in terms of minimum information bit throughput R for the DL reference channels specified in Annex C.4.2A with the addition of the relevant parameters in Tables 9.4.1.1 and 9.4.1.3.

Using this configuration the throughput shall meet or exceed the minimum requirements specified in tables 9.4.1.2 and 9.4.1.4.

**Table 9.4.1.1: Test parameters for fixed reference measurement channel requirements for 5,3 Mbps – Category 8 - UE (7,68 Mcps TDD Option) QPSK**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | QPSK | | | |
| Scrambling code and basic midamble code number* | - | 0, 1 | | | |
| Number of TS | - | 4 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,32)<br>i=1..32 | | | |
| Number of Hybrid ARQ processes | - | 3 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence** | - | {0,0,0,0}<br>s=1, R=0, b=0 | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -15,05 | | | |
| $\frac{\sum HS-PDSCH_E_c}{I_{or}}$ | dB | 0 | | | |
| $I_{oc}$ | dBm/7,68 MHz | -60 | | | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note **: This sequence implies Chase combining | | | | | |

**Table 9.4.1.2: Performance requirements for fixed reference measurement channel requirement in multipath channels for 5,3 Mbps – Category 8 - UE (7,68 Mcps TDD Option) QPSK**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 5,2 | 880 |
| 2 | PB3 | 5,5 | 880 |
| 3 | VA30 | 6,2 | 880 |
| 4 | VA120 | 6,2 | 880 |

**Table 9.4.1.3: Test parameters for fixed reference measurement channel requirements for 5,3 Mbps – Category 8 - UE (7,68 Mcps TDD Option) 16QAM**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| HS-PDSCH Modulation | - | 16QAM | | | |
| Scrambling code and basic midamble code number* | - | 0, 1 | | | |
| Number of TS | - | 4 | | | |
| HS-PDSCH Channelization Codes* | C(k,Q) | C(i,32)<br>i=1..32 | | | |
| Number of Hybrid ARQ processes | - | 3 | | | |
| Maximum number of Hybrid ARQ transmissions | - | 4 | | | |
| Redundancy and constellation version coding sequence** | - | {0,0,0,0}<br>s=1, R=0, b=0 | | | |
| $\frac{HS-PDSCH_E_c}{I_{or}}$ | dB | -15,05 | | | |
| $\frac{\sum HS-PDSCH_E_c}{I_{or}}$ | dB | 0 | | | |
| $I_{oc}$ | dBm/7,68 MHz | -60 | | | |
| Note *: Refer to TS 25.223 for definition of channelization codes, scrambling code and basic midamble code.<br>Note **: This sequence implies Chase combining | | | | | |

**Table 9.4.1.4: Performance requirements for fixed reference measurement channel requirement in multipath channels for 5,3 Mbps – Category 8 - UE (7,68 Mcps TDD Option) 16QAM**

| Test Number | Propagation conditions | $\frac{\hat{I}_{or}}{I_{oc}}$ [dB] | R (Throughput) [kbps] |
|---|---|---|---|
| 1 | PA3 | 11,1 | 1765 |
| 2 | PB3 | 13,2 | 1765 |
| 3 | VA30 | 13,7 | 1765 |
| 4 | VA120 | 13,6 | 1765 |

The reference for this requirement is TS 25.102 [1] clauses 9.3.1.1 and 9.3.1.2.

### 9.4.1.3 Test purpose

To verify the ability of the receiver to receive a predefined test signal, representing a multi-path fading channel with information bit throughput R not falling below a specified value.

### 9.4.1.4 Method of test

#### 9.4.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS (node B emulator) and fader and AWGN noise source to the UE antenna connector as shown in figure A.10.

2) Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3 with levels according to Annex E.3.

3) Set the node B emulator ACK/NACK/DTX behaviour according to table 9.4.1. Set the test parameters and levels for tests 1-4 according to tables 9.4.1.1. The reference channel configuration is defined in section C.4.2A. The configuration of the TX power for downlink physical channels is annex in E.3.

4) The information bit data shall be pseudo random and not repeated before 10 different information bit payload blocks are processed. (e.g. Fixed reference Channel Definition (16 QAM, test 1,2,3): The information bit payload block is 52996 bits long. Hence the PRBSequence must be at least 52996 * 10 bits long. ) Use a PRBS from ITU-T O.153 Ref [25].

5) Setup the fading simulator with fading conditions as described in table D.2.2.3.2 or D.2.2.3.4 depending on the frequency band.

#### 9.4.1.4.2 Procedure

a. Once the HSDPA connection is setup, start transmitting HSDPA data.

b. For all relevant propagation conditions, for all relevant Ioc levels, for all relevant Ec/Ior, for all relevant $\hat{I}or/Ioc$ defined in Tables 9.4.1.2 and 9.4.1.4 count the number of NACK, ACK and statDTX on the UL HS-SICH during the test interval and decide pass or fail according to Annex F.6.3 tables F.6.3.5.5 and F.6.3.5.6.

### 9.4.1.5 Test Requirements

Tables 9.4.1.1 to 9.4.1.4 define the primary level settings including test tolerance for all relevant throughput tests. The pass / fail decision for throughput is done according to Annex F.6.3.

## 9.4.2 HS-DSCH throughput for Variable Reference Channels

FFS

## 9.4.3 Reporting of HS-DSCH Channel Quality Indicator

FFS

## 9.4.4 HS-SCCH Detection Performance

### 9.4.4.1 Definition and applicability

The detection performance of the HS-SCCH is determined by the probability of event $E_m$, which is declared when the UE is signaled on HS-SCCH, but DTX is observed in the corresponding HS-SICH ACK/NACK field. The probability of event $E_m$ is denoted $P(E_m)$.

The normative reference for this test is 25.102 [2], section 9.3.4.

The requirements and this test apply to 7.68 Mcps TDD UE from release 7 and later that support HSDPA.

### 9.4.4.2 Minimum requirements

For the test parameters in Table 9.4.4.1, for each value of HS-SCCH-1 $E_c/I_{or}$ specified in Table 9.4.4.2, the measured $P(E_m)$ shall be less than or equal to the corresponding specified value of $P(E_m)$.

**Table 9.4.4.1: Test parameters for HS-SCCH detection (7.68 Mcps TDD option)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| Number of TS under test | - | 1 | | |
| Number of HS-SCCH codes per timeslot | - | 4 | | |
| HS-SCCH UE Identity ( $x_{ue,1}$ , $x_{ue,2}$ , …, $x_{ue,16}$ ) | - | UE1 = 0000000000000000 (UE1 under test) UE2 = 0101010101010101 UE3 = 1010101010101010 UE4 = 1111111111111111 | | |
| HS-SCCH Channelization Codes* | C(k,Q) | HS-SCCH-1 = C(1, 32), for UE1 (UE under test) HS-SCCH-2 = C(2, 32) for UE2 HS-SCCH-3 = C(3, 32) for UE3 HS-SCCH-4 = C(4, 32) for UE4 | | |
| HS-SCCH $E_c/I_{or}$ | dB | HS-SCCH-2_$E_c/I_{or}$ = HS-SCCH-3_$E_c/I_{or}$ = HS-SCCH-4_$E_c/I_{or}$, Where, ∑ HS-SCCH-X_$E_c/I_{or}$ = 1, where X = 1, 2, 3, 4 | | |

**Table 9.4.4.2: Minimum requirement for HS-SCCH detection (7.68 Mcps TDD option)**

| Test Number | Propagation Conditions | Reference value | | |
|---|---|---|---|---|
| | | **HS-SCCH-1** $E_c / I_{or}$ **(dB)** | $\hat{I}_{or} / I_{oc}$ **(dB)** | $P(E_m)$ |
| 1 | PA3 | -6.0 | 0 | 0.05 |
| 2 | PA3 | -7.5 | 5 | 0.01 |
| 3 | VA30 | -6.0 | 0 | 0.01 |

The reference for this requirement is TS 25.102 [2] clause 9.3.4.

## 9.4.4.3 Test purpose

To verify that $P(E_m)$ does not exceed the limit in table 9.4.4.2.

## 9.4.4.4 Method of test

### 9.4.4.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1. Connect SS, multipath fading simulator and an AWGN noise source to the UE antenna connector as shown in figure A.10.
2. Setup fading simulators as fading condition, which are described in table D.2.2.3.1.
3. Set the node B emulator ACK/NACK/DTX behaviour according to table 9.2.1. Set the test parameters and levels for tests 1-3 according to tables 9.4.4.1 and 9.4.4.2. The reference channel configuration is defined in section C.4.2A. The configuration of the TX power for downlink physical channels is annex in E.3.

### 9.4.4.4.2 Procedure

1. The UE is switched on.
2. Set up an HSDPA call according to TS 34.108 [3] clause 7.3.6.3.
3. Count the number of NACK, ACK and statDTX on the UL HS-DPCCH during the test interval and decide pass or fail according to Annex F.6.1 and table F.6.1.8. NACK and ACK are counted as a pass and statDTX is counted as a failure.

## 9.4.4.5 Test Requirements

The probability of event $E_m$ denoted as $P(E_m)$ (test procedure step 3) shall not exceed the specified value in table 9.4.4.2.

No test tolerance is applied to the test parameters.

# 10 Performance requirements (MBMS)

## 10.1 Demodulation of MCCH

### 10.1.1 Definition and applicablility

#### 10.1.1.1 3.84 Mcps TDD Option

The receive characteristic of the MCCH is determined by the RLC SDU error rate (RLC_SDU_ER). The requirement is valid for all RRC states for which the UE has capabilities.

The requirements and this test apply to the 3,84 Mcps TDD UE.

#### 10.1.1.2 7.68 Mcps TDD Option

The receive characteristic of the MCCH is determined by the RLC SDU error rate (RLC_SDU_ER). The requirement is valid for all RRC states for which the UE has capabilities.

The requirements and this test apply to the 7,68 Mcps TDD UE.

### 10.1.2 Minimum requirement

#### 10.1.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 10.1.1, the measured average downlink S-CCPCH_$E_c$/$I_{or}$ power ratio shall be below the specified value for the RLC_SDU_ER shown in Table 10.1.2.

**Table 10.1.1: Test parameters for MCCH detection**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | -3 |
| Number of Interfering codes/timeslot | - | 7 × SF16 |
| MCCH Data Rate | kbps | 7.2 |
| Propagation condition | - | VA3 |

**Table 10.1.2: Test requirements for MCCH detection**

| Test Number | S-CCPCH_Ec/Ior (dB) | RLC_SDU_ER |
|---|---|---|
| 1 | -1.25 | 0.01 |

#### 10.1.2.2 7.68 Mcps TDD Option

For the parameters specified in Table 10.1.3, the measured average downlink S-CCPCH_$E_c$/$I_{or}$ power ratio shall be below the specified value for the RLC_SDU_ER shown in Table 10.1.4.

**Table 10.1.3: Test parameters for MCCH detection**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | -3 |
| Number of Interfering codes/timeslot | - | 15 × SF32 |
| MCCH Data Rate | kbps | 7.2 |
| Propagation condition | - | VA3 |

**Table 10.1.4: Test requirements for MCCH detection**

| Test Number | S-CCPCH_Ec/Ior (dB) | RLC_SDU_ER |
|---|---|---|
| 1 | -4.7 | 0.01 |

## 10.1.3 Test Purpose

The aim of the test is to verify that the RLC SDU error rate (RLC SDU ER) for the MCCH channel does not exceed 0.01.

## 10.1.4 Method of test

### 10.1.4.1 Initial conditions

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in Figure A.10.

2) The MCCH Reference Measurement Channel parameters are defined in Annex C.5.1.

3) The configuration for the downlink channel for each radio link is defined in Annex.E.

4) The UE is switched on.

5) Set up a call according to the generic call setup procedure in TS34.108 [3] clause 7.3.10.

6) Setup the test parameter for MCCH detection test as specified in Table 10.1.1 and 10.1.2 for 3.84 Mcps TDD and Table 10.1.3. and 10.1.4 for 7.68 Mcps TDD. Set up fading simulator as per the fading conditionVA3, which is described in table D.2.2.

7) Enter the UE into loopback test mode 3. See TS 34.108 and TS 34.109 for details regarding loopback test mode 3 for MBMS.

8) Switch on the fading simulator.

### 10.1.4.2 Procedure

1) SS shall start the test by sending data on the MCCH radio bearer and maintain the count of transmitted RLC SDU blocks on the MCCH.

2) SS shall send a “UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST” message and wait for the UE to response with a “UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE” reporting the received RLC SDU counter value.

3) SS shall compute the RLC SDU error rate based on the transmitted RLC SDUs count and received RLC SDU count reported by the UE.

4) The test shall be run until the statistical significance according to Annex to F.6.1.8 is achieved.

### 10.1.5 Test Requirements

The RLC SDU error rate (RLC SDU ER) for all the MTCH demodulation tests shall not exceed 0.1 for the specified $\frac{\hat{I}_{or}}{I_{oc}}$.

The RLC SDU error rate = the ratio of (transmitted RLC SDU count – received RLC SDU count) / (transmitted RLC SDU count) < 0.1

## 10.2 Demodulation of MTCH

### 10.2.1 Definition and applicablility

#### 10.2.1.1 3.84 Mcps TDD Option

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 3,84 Mcps TDD UE.

#### 10.2.1.2 1.28 Mcps TDD Option

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to Release 6 and later releases for all types of UTRA for the 1.28Mcps TDD UE that support MBMS.

#### 10.2.1.3 7.68 Mcps TDD Option

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 7,68 Mcps TDD UE.

### 10.2.2 Minimum requirement

#### 10.2.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 10.2.1 the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.2.2.

**Table 10.2.1: Parameters for MTCH detection for 3.84Mcps TDD**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 | | | |
| $\Sigma$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | 0 | 0 | 0 | 0 |
| MTCH Data Rate | Kbps | 128 | 256 | 256 | 256 |
| Propagation condition | - | VA3 | | Extended Delay Spread (see Table B.1D) | |
| Number of Radio Links | - | 2 | 3 | 1 | 1 |
| S-CCPCH Modulation | - | QPSK | QPKS | 16QAM | 16QAM |
| Number of UE Rx Branches | - | 1 | 1 | 1 | 2 |

**Table 10.2.2: Test requirements for MTCH detection for 3.84Mcps TDD**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 5.7 | 0.1 |
| 2 | 5.5 | 0.1 |
| 3 | 14.5 | 0.1 |
| 4 | 8.3 | 0.1 |

### 10.2.2.2 1.28 Mcps TDD Option

For the parameters specified in Table 10.2.3 the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.2.4.

**Table 10.2.3: Parameters for MTCH detection**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| $I_{oc}$ | dBm/1.28 MHz | -60 | |
| $\Sigma$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | 0 | |
| MTCH Data Rate | kbps | 64 | 128 |
| Propagation condition | - | VA3 | |
| Number of Radio Links | - | 3 | 3 |

**Table 10.2.4: Test requirements for MTCH detection**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 4.8 | 0.1 |
| 2 | 6.0 | 0.1 |

### 10.2.2.3 7.68 Mcps TDD Option

For the parameters specified in Table 10.2.5 the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.2.6.

**Table 10.2.5.: Parameters for MTCH detection for 7.68 Mcps TDD**

| Parameters | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 | | | |
| $\Sigma$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | -3 | -3 | -3 | -3 |
| MTCH Data Rate | Kbps | 128 | 256 | 256 | 256 |
| Number of interfering codes/timeslot | - | 16 × SF32 | 16 × SF32 | 16 × SF32 | 16 × SF32 |
| Propagation condition | - | VA3 | | Extended Delay Spread (see Table B.8) | |
| Number of Radio Links | - | 2 | 3 | 1 | 1 |
| S-CCPCH Modulation | - | QPSK | QPSK | 16QAM | 16QAM |
| Number of UE Rx Branches | - | 1 | 1 | 1 | 2 |

**Table 10.2.6: Test requirements for MTCH detection for 7.68 Mcps TDD**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 6.1 | 0.1 |
| 2 | 5.0 | 0.1 |
| 3 | 14.7 | 0.1 |
| 4 | 8.2 | 0.1 |

## 10.2.3 Test Purpose

The aim of the test is to verify that the RLC SDU error rate (RLC SDU ER) for each individual data rate of the MTCH channel does not exceed 0.1. The test shall be performed in CELL_PCH state only.

## 10.2.4 Method of test

### 10.2.4.1 Initial conditions

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in Figure A.16.

2) The DL Reference Measurement Channel parameters are defined in Annex C.5.2.

3) The same MTCH data shall be sent in all radio links during the test.

4) The UE is switched on.

5) Set up a call according to the generic call setup procedure in TS34.108 [3] clause 7.3.10 with transition to the CELL_PCH state.

6) Setup the test parameter for MTCH detection test as specified in Table 10.2.1 and 10.2.2 for 3.84 Mcps TDD , Table 10.2.3 and 10.2.4 for 1.28Mcps TDD and Table 10.2.5. and 10.2.6 for 7.68 Mcps TDD. Set up fading simulator as per the fading conditionVA3, which is described in table D.2.2.

7) Enter the UE into loopback test mode 3. See TS 34.108 and TS 34.109 for details regarding loopback test mode 3 for MBMS.

8) Switch on the fading simulator.

### 10.2.4.2 Procedure

1) SS shall start the test by sending data on the MTCH radio bearer and maintain the count of transmitted RLC SDU blocks on the MTCH.

2) SS shall send a “UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST” message and wait for the UE to response with a “UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE” reporting the received RLC SDU counter value.

3) SS shall compute the RLC SDU error rate based on the transmitted RLC SDUs count and received RLC SDU count reported by the UE.

4) The test shall be run until the statistical significance according to Annex to F.6.1.8 is achieved.

## 10.2.5 Test Requirements

The RLC SDU error rate (RLC SDU ER) for all the MTCH demodulation tests does not exceed 0.1.

The RLC SDU error rate = the ratio of (transmitted RLC SDU count – received RLC SDU count) / (transmitted RLC SDU count) < 0.1

# 10.2A Demodulation of MTCH-for MBSFN capable UE

## 10.2A.1 Definition and applicablility

### 10.2A.1.1 3.84 Mcps TDD Option

void

### 10.2A.1.2 1.28 Mcps TDD Option

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 1.28Mcps TDD UE that support MBSFN.

### 10.2A.1.3 7.68 Mcps TDD Option

void

## 10.2A.2 Minimum requirement

### 10.2A.2.1 3.84 Mcps TDD Option

void

### 10.2A.2.2 1.28 Mcps TDD Option

For the parameters specified in Table 10.2A.3A the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.2A.4A.

**Table 10.2A.3A: Parameters for MTCH detection**

| Parameters | Unit | Test 1[1] | Test 2[1] | Test 3[2] | Test 4[2] |
|---|---|---|---|---|---|
| MTCH Data rate | Kbps | 192 | 384 | 192 | 384 |
| Rx antenna | - | 1 | 2 | 1 | 2 |
| Modulation | - | QPSK | 16QAM | QPSK | 16QAM |
| $I_{oc}$ | dBm/1.28 MHz | -60 | -60 | -60 | -60 |
| $\sum$(S-CCPCH_$E_c$)/$I_{or}$ | dB | 0 | 0 | 0 | 0 |
| Propagation condition | - | MBSFN channel model 1 (Annex D) | MBSFN channel model 1 (Annex D) | MBSFN channel model 2 (Annex D) | MBSFN channel model 2 (Annex D) |
| Slot Format # | - | 0[4] | 2[4] | 4[4] | 7[4] |
| NOTE1: Test 1 and Test 2 are specified for the UE supporting normal delay spread.<br>NOTE2: Test 3 and Test 4 are specified for the UE supporting extended delay spread.<br>NOTE3: In the case of Rx diversity, the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.<br>NOTE4: See Table 8Ha in TS25.221. | | | | | |

**Table 10.2A.4A: Test requirements for MTCH detection**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 13.3 | 0.1 |
| 2 | 14.7 | 0.1 |
| 3 | 13.3 | 0.1 |
| 4 | 15.1 | 0.1 |

# 10.3 Demodulation of MTCH and cell identification

## 10.3.1 1.28Mcps TDD option

### 10.3.1.1 Definition and applicability

MBMS combining is not controlled by a network but instead it is autonomously handled by a terminal. UE has to be able to receive MTCH and identify intra-frequency neighbour cells according to the requirements. The receive characteristic of the MTCH combined with cell identification is determined by RLC SDU error rate (RLC SDU ER).

The requirements and this test apply to Release 6 and later releases for1.28Mcps TDD UE that supports MBMS.

### 10.3.1.2 Minimum requirements

For the parameters specified in Table 10.3.1.1 the average downlink S-CCPCH $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU error rate shown in Table 10.3.1.2. The cell reselection parameters are given in clause in Table 10.3.1.3. The different cells are assumed to be time aligned.

**Table 10.3.1.1: Parameters for MTCH demodulation requirements with cell identification**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Stage 1 | Stage 2 | Stage 3 |
| Time in each stage | s | 2s | 800ms | 2s |
| $I_{oc}$ | dBm/1.28MHz | -60 | | |
| Propagation condition | | VA 3 | | |
| MTCH Data Rate | kbps | 64kbps | | |
| Number of Radio Links | | Cell 1, Cell 2 | Cell 1, Cell2, Cell3 | Cell 1, Cell 3 |

**Table 10.3.1.2: Requirements for MTCH detection**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ | RLC SDU ER |
|---|---|---|
| 1 | 6.1 | 0.05 |

Parameters for combined MTCH demodulation and cell identification requirements are defined in Table 10.3.1.3.

**Table 10.3.1.3: Cell reselection parameters**

| Parameter | Unit | Value |
|---|---|---|
| Serving cell in the initial condition | | Cell1 |
| Neighbour cells | | 32 intra-frequency neighbour cells are indicated including Cell2 and Cell3 |
| Cell_selection_and_reselection_quality_measure | | PCCPCH RSCP |
| Qrxlevmin | dBm | -103 |
| UE_TXPWR_MAX_RACH | dB | 21 |
| Treselection | Seconds | 4 |
| Sintrasearch | dB | not sent |
| IE “FACH Measurement occasion info” | | not sent |

The reference for this requirement is TS 25.102 [1] clause 10.3.1 and TS 25.102 [1] annex A.4.2.2.

### 10.3.1.3 Test purpose

To verify that the RLC SDU error rate does not exceed the value at the S-CCPCH $\frac{\hat{I}_{or}}{I_{oc}}$ r specified in table 10.3.1.2.

### 10.3.1.4 Method of test

#### 10.3.1.4.1 Initial conditions

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulators to the UE antenna connector as shown in Figure A.16.
2) The DL Reference Measurement Channel parameters are defined in Annex C.5.1.2
3) The same MTCH data shall be sent in all active cells during the test.
4) The UE is switched on.
5) The UE selects the broadcast service provided by the SS (included in the MBMS_ACTIVATED_SERVICES variable).
6) Set up a call according to the generic call setup procedure specified in TS 34.108 [3] clause 7.3.10 to configure and activate an MBMS session with transition to CELL_PCH state. The SS broadcast the cell reselection parameters according to table 10.3.1.3.

#### 10.3.1.4.2 Procedures

1) The SS configures and activates cell 1 to 3 with RF parameters according to Stage 1 conditions in Table 10.3.1.1.
2) The SS transmits valid MAC headers and RLC SDUs on the MTCH radio bearer for 2 seconds. The SS counts the number of transmitted RLC SDUs on the MTCH with valid MAC headers ($N_{SS_Stage1}$).
3) The SS transmits invalid MAC headers sets the test parameters for Cell 1, Cell 2 and Cell 3 as specified in table 10.3.1.1 and Stage 2.
4) The SS transmits valid MAC headers and RLC SDUs on the MTCH radio bearer for 0.8 seconds. The SS counts the number of transmitted RLC SDUs on the MTCH with valid MAC headers ($N_{SS_Stage2}$).
5) The SS transmits invalid MAC headers sets the test parameters for Cell 1, Cell 2 and Cell 3 as specified in table 10.3.1.1 and Stage 3.
6) The SS transmits valid MAC headers and RLC SDUs on the MTCH radio bearer for 2 seconds. The SS counts the number of transmitted RLC SDUs on the MTCH with valid MAC headers ($N_{SS_Stage3}$).
7) Repeat step 9) to 14) until the confidence level for RLC SDU's according to annex F.6.1.8.
8) The SS transmits a PAGING TYPE 1 message to the UE on the PCH and the UE moves to CELL_FACH state, sending a CELL UPDATE message. The SS replies with a CELL UPDATE CONFIRM message.
9) The UE confirms the received C-RNTI to be used in CELL FACH state by transmitting a UTRAN MOBILITY INFORMATION CONFIRM message on the uplink DCCH.
10) The SS sends a "UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST" message.
11) The SS waits for the UE to respond by a "UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE" message reporting the number of received RLC SDUs on MTCH ($N_{UEOK}$).
12) The SS in Cell 1 calculates the SDU error rate as: (Number of SS transmitted RLC SDUs – UE received RLC SDUs) / Number of SS transmitted RLC SDUs; where Number of SS transmitted RLC SDUs = $N_{SS_Stage1}$ + $N_{SS_Stage2}$ + $N_{SS_Stage3}$; and UE received RLC SDUs = $N_{UEOK}$.
13) End the RRC connection according to the generic call setup procedure specified in TS 34.108 [3] clause 7.3.10.

Specific Message Contents

All messages indicated above shall use the same content as described in the default message content in clause 9 of 34.108 [3].

### 10.3.1.5 Test requirements

The test parameters are specified in table 10.3.1.4.

**Table 10.3.1.4: Parameters for MTCH demodulation requirements with cell identification**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Stage 1 | Stage 2 | Stage 3 |
| Time in each stage | s | 2s | 800ms | 2s |
| $I_{oc}$ | dBm/1.28MHz | -60 | | |
| Propagation condition | | VA 3 | | |
| MTCH Data Rate | kbps | 64kbps | | |
| Number of Radio Links | | Cell 1, Cell 2 | Cell 1, Cell2, Cell3 | Cell 1, Cell 3 |

The RLC SDU ER shall not exceed the value specified in table 10.3.1.5 for the specified $\frac{\hat{I}_{or}}{I_{oc}}$

**Table 10.3.1.5: Requirements for MTCH detection**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ | RLC SDU ER |
|---|---|---|
| 1 | 6.1 | 0.05 |

NOTE: If the above Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for this test is non-zero. The Test Tolerance for this test is defined in clause F.2 and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in clause F.4.

## 10.4 Demodulation of MCCH for a MBSFN capable UE

### 10.4.1 Definition and applicablility

#### 10.4.1.1 3.84 Mcps TDD Option

The test is only applicable for UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MCCH is determined by the RLC SDU error rate (RLC_SDU_ER). The requirement is valid for all RRC states for which the UE has capabilities.

The requirements and this test apply to the 3,84 Mcps TDD UE.

#### 10.4.1.2 7.68 Mcps TDD Option

The test is only applicable for UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MCCH is determined by the RLC SDU error rate (RLC_SDU_ER). The requirement is valid for all RRC states for which the UE has capabilities.

The requirements and this test apply to the 7,68 Mcps TDD UE.

## 10.1.2 Minimum requirement

### 10.1.4.1 3.84 Mcps TDD Option

For the parameters specified in table 10.4.1, the measured average downlink S-CCPCH_$E_c$/$I_{or}$ power ratio shall be below the specified value for the RLC_SDU_ER shown in table 10.4.2.

**Table 10.4.1: Test parameters for MCCH detection for MBSFN enabled UE**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | 12 |
| Number of Interfering codes/timeslot | - | 7 × SF16 |
| MCCH Data Rate | kbps | 7.2 |
| Propagation condition | - | Extended delay spread (see Appendix B) |
| Slot Format #i | - | 20 |

**Table 10.4.2: Test requirements for MCCH detection (at least two receiver antennas) for MBSFN enabled UE**

| Test Number | S-CCPCH_Ec/Ior (dB) | RLC_SDU_ER |
|---|---|---|
| 1 | -19.29 | 0.01 |

### 10.4.2.2 7.68 Mcps TDD Option

For the parameters specified in table 10.4.3, the measured average downlink S-CCPCH_$E_c$/$I_{or}$ power ratio shall be below the specified value for the RLC_SDU_ER shown in table 10.4.4.

**Table 10.4.3: Test parameters for MCCH detection for MBSFN enabled UE**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | 12 |
| Number of Interfering codes/timeslot | - | 15 × SF32 |
| MCCH Data Rate | kbps | 7.2 |
| Propagation condition | - | Extended delay spread (see Appendix B) |
| Slot Format #i | - | 20 |

**Table 10.4.4: Test requirements for MCCH detection (at least two receiver antennas) for MBSFN enabled UE**

| Test Number | S-CCPCH_Ec/Ior (dB) | RLC_SDU_ER |
|---|---|---|
| 1 | -22.71 | 0.01 |

## 10.4.3 Test Purpose

The aim of the test is to verify that the RLC SDU error rate (RLC SDU ER) for the MCCH channel does not exceed 0.01.

## 10.4.4 Method of test

### 10.4.4.1 Initial conditions

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in Figure A.17.

2) The MCCH Reference Measurement Channel parameters are defined in Annex C.5.1.

3) The configuration for the downlink channel for each radio link is defined in Annex.E.

4) The UE is switched on.

5) Set up a call according to the generic call setup procedure in TS34.108 [3] clause 7.3.10.

6) Setup the test parameter for MCCH detection test as specified in Table 10.4.1 and 10.4.2 for 3.84 Mcps TDD and Table 10.4.3. and 10.4.4 for 7.68 Mcps TDD. Set up fading simulator as per the fading condition EDS, which is described in table D.2.2.1B for 3.84Mcps and table D.2.2.3.5 for 3.84Mcps.

7) Enter the UE into loopback test mode 3. See TS 34.108 and TS 34.109 for details regarding loopback test mode 3 for MBMS.

8) Switch on the fading simulator.

### 10.4.4.2 Procedure

1) SS shall start the test by sending data on the MCCH radio bearer and maintain the count of transmitted RLC SDU blocks on the MCCH.

2) SS shall send a “UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST” message and wait for the UE to response with a “UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE” reporting the received RLC SDU counter value.

3) SS shall compute the RLC SDU error rate based on the transmitted RLC SDUs count and received RLC SDU count reported by the UE.

4) The test shall be run until the statistical significance according to Annex to F.6.1.8 is achieved.

## 10.4.5 Test Requirements

The RLC SDU error rate (RLC SDU ER) for all the MTCH demodulation tests shall not exceed 0.1 for the specified S-CCPCH_$E_c/I_{or}$ power ratio.

The RLC SDU error rate = the ratio of (transmitted RLC SDU count – received RLC SDU count) / (transmitted RLC SDU count) < 0.1

# 10.5 Demodulation of MTCH for a MBSFN capable UE

## 10.5.1 Definition and applicablility

### 10.5.1.1 3.84 Mcps TDD Option

The test is only applicable for UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 3,84 Mcps TDD UE.

### 10.2.1.2 7.68 Mcps TDD Option

The test is only applicable for UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 7,68 Mcps TDD UE.

## 10.5.2 Minimum requirement

### 10.5.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 10.5.1 the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.5.2.

**Table 10.5.1: Parameters for MTCH detection for 3.84Mcps TDD for MBSFN capable UE**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 |
| $\sum$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | 0 |
| MTCH Data Rate | kbps | 512 |
| Propagation condition | - | Extended delay spread (see Appendix B) |
| Number of Radio Links | - | 1 |
| S-CCPCH Modulation | - | 16QAM |

**Table 10.5.2: Test requirements for MTCH detection for 3.84Mcps TDD for MBSFN capable UE (at least two receiver antennas)**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 14.58 | 0.1 |

### 10.5.2.2 7.68 Mcps TDD Option

For the parameters specified in Table 10.5.3 the average downlink $\frac{\hat{I}_{or}}{I_{oc}}$ power ratio shall be below the specified value for the RLC SDU ER shown in Table 10.5.4.

**Table 10.5.3.: Parameters for MTCH detection for 7.68 Mcps TDD**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| $\sum$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | -3 |
| MTCH Data Rate | kbps | 512 |
| Number of interfering codes/timeslot | - | 16 × SF32 |
| Propagation condition | - | Extended delay spread (see Appendix B) |
| Number of Radio Links | - | 1 |
| S-CCPCH Modulation | - | 16QAM |

**Table 10.5.4: Test requirements for MTCH detection for 7.68 Mcps TDD**

| Test Number | $\frac{\hat{I}_{or}}{I_{oc}}$ (dB) | RLC SDU ER |
|---|---|---|
| 1 | 14.21 | 0.1 |

## 10.5.3 Test Purpose

The aim of the test is to verify that the RLC SDU error rate (RLC SDU ER) for each individual data rate of the MTCH channel does not exceed 0.1. The test shall be performed in CELL_PCH state only.

## 10.5.4 Method of test

### 10.5.4.1 Initial conditions

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in Figure A.17.

2) The DL Reference Measurement Channel parameters are defined in Annex C.5.2.

3) The configuration for the downlink channel for each radio link is defined in Annex.E..

4) The UE is switched on.

5) Set up a call according to the generic call setup procedure in TS34.108 [3] clause 7.3.10 with transition to the CELL_PCH state.

6) Setup the test parameter for MTCH detection test as specified in Table 10.5.1 and 10.5.2 for 3.84 Mcps TDD , and Table 10.5.5. and 10.5.6 for 7.68 Mcps TDD. Set up fading simulator as per the fading condition EDS, which is described in table D.2.2.1B for 3.84Mcps and table D.2.2.3.5 for 3.84Mcps.

7) Enter the UE into loopback test mode 3. See TS 34.108 and TS 34.109 for details regarding loopback test mode 3 for MBMS.

8) Switch on the fading simulator.

### 10.2.4.2 Procedure

1) SS shall start the test by sending data on the MTCH radio bearer and maintain the count of transmitted RLC SDU blocks on the MTCH.

2) SS shall send a “UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST” message and wait for the UE to response with a “UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE” reporting the received RLC SDU counter value.

3) SS shall compute the RLC SDU error rate based on the transmitted RLC SDUs count and received RLC SDU count reported by the UE.

4) The test shall be run until the statistical significance according to Annex to F.6.1.8 is achieved.

## 10.5.5 Test Requirements

The RLC SDU error rate (RLC SDU ER) for all the MTCH demodulation tests does not exceed 0.1.

The RLC SDU error rate = the ratio of (transmitted RLC SDU count – received RLC SDU count) / (transmitted RLC SDU count) < 0.1

# 10.6 MBSFN TDD and FDD same platform sharing

## 10.6.1 Definition and applicablility

### 10.6.1.1 3.84 Mcps TDD Option

This test case is to ensure that a simulataneous demodulation of MTCH and FDD transmission is possible for a MBSFN TDD UE sharing the same platform with a FDD UE. The test is only applicable for TDD UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 3,84 Mcps TDD UE.

### 10.6.1.2 7.68 Mcps TDD Option

This test case is to ensure that a simulataneous demodulation of MTCH and FDD transmission is possible for a MBSFN TDD UE sharing the same platform with a FDD UE. The test is only applicable for TDD UEs with at least two receiver antenna connectors where the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated.

The receive characteristic of the MTCH is determined by RLC SDU error rate (RLC SDU ER). RLC SDU ER is specified for each individual data rate of the MTCH. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

The requirements and this test apply to the 7,68 Mcps TDD UE.

## 10.6.2 Minimum requirement

### 10.6.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 10.6.1 the average downlink $\hat{I}_{or}$ power shall be below the specified value for the RLC SDU ER shown in Table 10.6.2.

**Table 10.6.1: Parameters for MTCH detection sharing same platform with FDD for 3.84Mcps TDD**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| FDD UE Tx Pwr | dBm/ 3.84 MHz | Nominal Maximum Output Power | Nominal Maximum Output Power |
| $I_{oc}$ | dBm/ 3.84 MHz | -infinity | -infinity |
| $\sum$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | 0 | 0 |
| MTCH Data Rate | kbps | 512 | 512 |
| Number of interfering codes/timeslot | - | 0 | 0 |
| Propagation condition | - | Extended Delay Spread (see Appendix B) | Extended Delay Spread (see Appendix B) |
| Number of Radio Links | - | 1 | 1 |
| S-CCPCH Modulation | - | 16QAM | 16QAM |
| TDD operating frequencies | MHz | 1900-1920 | 2570-2620 |
| FDD operating band | - | Band I | Band VII |
| TDD/FDD carrier frequencies | - | Applicable for all combinations of TDD and FDD carrier frequencies except for combinations where the carrier frequency separation is less than 15 MHz | Applicable for all combinations of TDD and FDD carrier frequencies except for combinations where the carrier frequency separation is less than 15 MHz |

**Table 10.6.2: Test requirements for MTCH detection sharing same platform with FDD for 3.84Mcps TDD UE (at least two receiver antennas)**

| Test Number | $\hat{I}_{or}$ (dBm) | RLC SDU ER |
|---|---|---|
| 1 | -83.42 | 0.1 |
| 2 | -83.42 | 0.1 |

### 10.6.2.3 7.68 Mcps TDD Option

For the parameters specified in Table 10.6.3 the average downlink $\hat{I}_{or}$ power shall be below the specified value for the RLC SDU ER shown in Table 10.6.4.

**Table 10.6.3: Parameters for MTCH detection sharing same platform with FDD for 7.68Mcps TDD**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| FDD UE Tx Pwr | dBm/ 3.84 MHz | Nominal Maximum Output Power | Nominal Maximum Output Power |
| $I_{oc}$ | dBm/ 7.68 MHz | -infinity | -infinity |
| $\Sigma$(S-CCPCH_$E_c$)/$I_{or}$ per active timeslot | dB | -3 | -3 |
| MTCH Data Rate | kbps | 512 | 512 |
| Number of interfering codes/timeslot | - | 16 × SF32 | 16 × SF32 |
| Propagation condition | - | Extended Delay Spread (see Appendix B) | Extended Delay Spread (see Appendix B) |
| Number of Radio Links | - | 1 | 1 |
| S-CCPCH Modulation | - | 16QAM | 16QAM |
| TDD operating frequencies | MHz | 1900-1920 | 2570-2620 |
| FDD operating band | - | Band I | Band VII |
| TDD/FDD carrier frequencies | - | Applicable for all combinations of TDD and FDD carrier frequencies except for combinations where the carrier frequency separation is less than 17.5 MHz | Applicable for all combinations of TDD and FDD carrier frequencies except for combinations where the carrier frequency separation is less than 17.5 MHz |

**Table 10.6.4: Test requirements for MTCH detection sharing same platform with FDD for 3.84Mcps TDD UE (at least two receiver antennas)**

| Test Number | $\hat{I}_{or}$ (dBm) | RLC SDU ER |
|---|---|---|
| 1 | -80.79 | 0.1 |
| 2 | -80.79 | 0.1 |

## 10.6.3 Test Purpose

The aim of the test is to verify that the RLC SDU error rate (RLC SDU ER) for each individual data rate of the MTCH channel does not exceed 0.1. The test shall be performed in CELL_PCH state only.

## 10.6.4 Method of test

### 10.6.4.1 Initial conditions

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in Figure A.17.

2) The DL Reference Measurement Channel parameters are defined in Annex C.5.2.

3) The configuration for the downlink channel for each radio link is defined in Annex.E..

4) The UE is switched on.

5) Set up a call according to the generic call setup procedure in TS34.108 [3] clause 7.3.10 with transition to the CELL_PCH state.

6) Setup the test parameter for MTCH detection test as specified in Table 10.5.1 and 10.5.2 for 3.84 Mcps TDD , and Table 10.5.5. and 10.5.6 for 7.68 Mcps TDD. Set up fading simulator as per the fading condition EDS, which is described in table D.2.2.1B for 3.84Mcps and table D.2.2.3.5 for 3.84Mcps.

7) Enter the UE into loopback test mode 3. See TS 34.108 and TS 34.109 for details regarding loopback test mode 3 for MBMS.

8) Use the SS to ensure that the UE is transmitting an FDD signal as defined in table 10.6.1 for 3.84Mcps TDD and table 10.6.3 for 7.68Mcps TDD.

9) Switch on the fading simulator.

#### 10.6.4.2 Procedure

1) SS shall start the test by sending data on the MTCH radio bearer and maintain the count of transmitted RLC SDU blocks on the MTCH.

2) SS shall send a “UE TEST LOOP MODE 3 RLC SDU COUNTER REQUEST” message and wait for the UE to response with a “UE TEST LOOP MODE 3 RLC SDU COUNTER RESPONSE” reporting the received RLC SDU counter value.

3) SS shall compute the RLC SDU error rate based on the transmitted RLC SDUs count and received RLC SDU count reported by the UE.

4) The test shall be run until the statistical significance according to Annex to F.6.1.8 is achieved.

### 10.6.5 Test Requirements

The RLC SDU error rate (RLC SDU ER) for all the MTCH demodulation tests does not exceed 0.1.

The RLC SDU error rate = the ratio of (transmitted RLC SDU count – received RLC SDU count) / (transmitted RLC SDU count) < 0.1

# 11 Performance requirement (E-DCH)

## 11.1 Detection of E-DCH HARQ ACK Indicator Channel (E-HICH)

### 11.1.1 Definition and applicablility

#### 11.1.1.1 3.84 Mcps TDD Option

The performance of the E-HICH detection is determined by the false ACK probability (probability of detecting an ACK given that a NACK was sent) and the false NACK probability (probability of detecting a NACK given that an ACK was sent).

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 3.84 Mcps TDD UE that support HSDPA and E-DCH.

#### 11.1.1.2 1.28 Mcps TDD Option

The performance of the E-HICH detection is determined by the false ACK probability (probability of detecting an ACK given that a NACK was sent) and the false NACK probability (probability of detecting a NACK given that an ACK was sent).

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 1.28 Mcps TDD UE that support HSDPA and E-DCH.

### 11.1.1.3 7.68 Mcps TDD Option

The performance of the E-HICH detection is determined by the false ACK probability (probability of detecting an ACK given that a NACK was sent) and the false NACK probability (probability of detecting a NACK given that an ACK was sent).

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 7.68 Mcps TDD UE that support HSDPA and E-DCH.

## 11.1.2 Minimum requirement

### 11.1.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 11.1.1 the average downlink E-HICH $E_c/I_{or}$ power ratio shall be below the specified value for the false ACK and false NACK probabilities shown in Table 11.1.2.

**Table 11.1.1: Test parameters for E-HICH detection (3.84 Mcps TDD option)**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 | |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | 0 | |
| Number of Interfering codes/timeslot | - | 7 × SF16 (all codes have equal powers) | |
| E-HICH signalling pattern | - | 100% NACK | 100% ACK |
| Propagation condition | - | VA30 | |

**Table 11.1.2: Test requirements for E-HICH detection (3.84 Mcps TDD option)**

| Test Number | E-HICH Ec/Ior (dB) | Parameter | Probability |
|---|---|---|---|
| 1 | -18.5 | False ACK | 2E-3 |
| 2 | -18.5 | False NACK | 2E-2 |

### 11.1.2.2 1.28 Mcps TDD Option

For the parameters specified in Table 11.1.3 the average downlink E-HICH $E_c/I_{or}$ power ratio shall be below the specified value for the false ACK and false NACK probabilities shown in Table 11.1.4.

**Table 11.1.3: Test parameters for E-HICH detection (1.28 Mcps TDD option)**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| $I_{oc}$ | dBm/1.28 MHz | -60 | |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | 0 | |
| Number of Interfering codes/timeslot | - | 7 × SF16 (all codes have equal powers) | |
| E-HICH signalling pattern | - | 100% NACK | 100% ACK |
| Propagation condition | - | VA30 | |

**Table 11.1.4: Test requirements for E-HICH detection (1.28 Mcps TDD option)**

| Test Number | E-HICH Ec/Ior (dB) | Parameter | Probability |
|---|---|---|---|
| 1 | -7.5 | False ACK | 2E-3 |
| 2 | -7.5 | False NACK | 2E-2 |

### 11.1.2.3 7.68 Mcps TDD Option

For the parameters specified in Table 11.3 the average downlink E-HICH $E_c/I_{or}$ power ratio shall be below the specified value for the false ACK and false NACK probabilities shown in Table 11.4.

**Table 11.1.5: Test parameters for E-HICH detection (7.68 Mcps TDD option)**

| Parameters | Unit | Test 1 | Test 2 |
|---|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 | |
| $\frac{\hat{I}_{or}}{I_{oc}}$ | dB | 0 | |
| Number of Interfering codes/timeslot | - | 15 × SF32 (all codes have equal powers) | |
| E-HICH signalling pattern | - | 100% NACK | 100% ACK |
| Propagation condition | - | VA30 | |

**Table 11.1.6: Test requirements for E-HICH detection (7.68 Mcps TDD option)**

| Test Number | E-HICH Ec/Ior (dB) | Parameter | Probability |
|---|---|---|---|
| 1 | -21.7 | False ACK | 2E-3 |
| 2 | -21.7 | False NACK | 2E-2 |

## 11.1.3 Test Purpose

The test verifies that the false ACK propability and false NACK propability of E-HICH meet the requirement.

## 11.1.4 Method of test

### 11.1.4.1 Initial conditions

#### 11.1.4.1.1 3.84Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.

2) Set the test parameters for the false ACK test and the false NACK test as specified in table 11.1.1.

3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.1 and DL Reference Measurement Channel parameters are defined in Annex C4.1.1.1

4) The value of TRRI shall be set to '111111' and CRRI on E-AGCH shall be set to 15. Note that the radio bearer reconfiguration message used to conifigure the UE will define 6 E-PUCH TS and therefore the length of the TRRI will be 6 bits.

5) The value of PRRI is set to 31. This ensures that the UL datarate remains constant.

6) The UE is switched on.

7) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.

8) Switch on the fading simulator.

#### 11.1.4.1.2 1.28Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.

2) Set the test parameters for the false ACK test and the false NACK test as specified in 11.1.3 for 1.28Mcps TDD option.

3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.2 and DL Reference Measurement Channel parameters are defined in Annex C4.2.1.1

4) The value of TRRI shall be set to '11000' and CRRI on E-AGCH shall be set to 3.

5) The value of PRRI is same. This ensures that the UL datarate remains constant.

6) The UE is switched on.

7) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.

8) Switch on the fading simulator.

#### 11.1.4.1.3 7.68Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.

2) Set the test parameters for the false ACK test and the false NACK test as specified in table 11.1.1.

3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.3 and DL Reference Measurement Channel parameters are defined in Annex C4.2A.1.1

4) The value of TRRI shall be set to '111111' and CRRI on E-AGCH shall be set to 31. Note that the radio bearer reconfiguration message used to conifigure the UE will define 6 E-PUCH TS and therefore the length of the TRRI will be 6 bits.

5) The value of PRRI is set to 31. This ensures that the UL datarate remains constant.

6) The UE is switched on.

7) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.

8) Switch on the fading simulator.

### 11.1.4.2 Procedure11.1.4.2.1 3.84Mcps TDD Option

The measurement principle for the false ACK test (step 2 to 5) and for the false NACK test (step 6 to 9) is as follows. Upon the UE transmission on E-UCCH and E-PUCH, the SS reacts with E-HICH = ACK or NACK. The UE transmits new data or retransmissions on the corresponding E-UCCH and E-PUCH. The SS shall discriminate between:

(1) new data is a sign for ACK, received by the UE

(2) retransmission is a sign for NACK, received by the UE . The later is interpreted as NACK by higher layer and causes retransmission.

. 1. The Nominal Avg. Information Bit Rate in the DL is set to 2649.6kbps according to Annex C4.1.1.1. The expected througput under this level and fading conditions is 1300 kbps. The UL datarate was configured by signalling TRRI=111111 and CRRI=15 on the E-AGCH. The expected UL datarate is 34.7 kbps corresponding to E-TFC Index 46.

2. In the test false ACK the SS responds with 100% ACK.

3. If the UE indicates on the E-UCCH a retransmission, the ACK from the SS was received as NACK by the UE. This is counted as false(ACK).

   If the UE indicates on the E-UCCH new data, the ACK from the SS was received as ACK by the UE. This is counted as correct ACK.

4. Continue until statistical significance according to Annex F.6.4 is achieved.

5. If the number of retransmissions reaches the maximum number of retransmissions due to several false ACK detections in series, the first new data on the E-PUCH with E-UCCH are not the consequence of ACK. This case is not counted as sample.

6. In the test false NACK the SS responds with 100% NACK.

7. If the UE indicates on the E-UCCH new data, the NACK from the SS was received as ACK by the UE. This is counted as false(NACK ). If the UE indicates on E-UCCH retransmission, the NACK from the SS was received as NACK by the UE. This is counted as correct reception.

8. Continue until statistical significance according to Annex F.6.4 is achieved.

9. The number of retransmissions will reach the maximum number of transmissions due to several retransmissions in series. The first new data on the E-PUCH with E-UCCH are not the consequence of ACK received by the UE. This case is not counted as sample.

#### 11.1.4.2.2 1.28Mcps TDD Option

The measurement principle for the false ACK test (step 2 to 5) and for the false NACK test (step 6 to 9) is as follows. Upon the UE transmission on E-UCCH and E-PUCH, the SS reacts with E-HICH = ACK or NACK. The UE transmits new data or retransmissions on the corresponding E-UCCH and E-PUCH. The SS shall discriminate between:

(1) new data is a sign for ACK, received by the UE

(2) retransmission is a sign for NACK, received by the UE . The later is interpreted as NACK by higher layer and causes retransmission.

. 1. The Nominal Avg. Information Bit Rate in the DL is set to 199.2 kbps according to FRC1in Annex C4.2.1.1. The expected througput under this level and fading conditions is 161 kbps. The UL datarate was configured by signalling TRRI=11000 and CRRI=3 on the E-AGCH. The expected UL datarate is 57.4 kbps corresponding to E-TFC Index 56.

2. In the test false ACK the SS responds with 100% ACK.

3. If the UE indicates on the E-UCCH a retransmission, the ACK from the SS was received as NACK by the UE. This is counted as false(ACK).

   If the UE indicates on the E-UCCH new data, the ACK from the SS was received as ACK by the UE. This is counted as correct ACK.

4. Continue until statistical significance according to Annex F.6.4 is achieved.

5. If the number of retransmissions reaches the maximum number of retransmissions due to several false ACK detections in series, the first new data on the E-PUCH with E-UCCH are not the consequence of ACK. This case is not counted as sample.

6. In the test false NACK the SS responds with 100% NACK.

7. If the UE indicates on the E-UCCH new data, the NACK from the SS was received as ACK by the UE. This is counted as false(NACK ). If the UE indicates on E-UCCH retransmission, the NACK from the SS was received as NACK by the UE. This is counted as correct reception.

8. Continue until statistical significance according to Annex F.6.4 is achieved.

9. The number of retransmissions will reach the maximum number of transmissions due to several retransmissions in series. The first new data on the E-PUCH with E-UCCH are not the consequence of ACK received by the UE. This case is not counted as sample.

#### 11.1.4.2.3 7.68Mcps TDD Option

The measurement principle for the false ACK test (step 2 to 5) and for the false NACK test (step 6 to 9) is as follows. Upon the UE transmission on E-UCCH and E-PUCH, the SS reacts with E-HICH = ACK or NACK. The UE transmits new data or retransmissions on the corresponding E-UCCH and E-PUCH. The SS shall discriminate between:

(1) new data is a sign for ACK, received by the UE

(2) retransmission is a sign for NACK, received by the UE . The later is interpreted as NACK by higher layer and causes retransmission.

. 1. The Nominal Avg. Information Bit Rate in the DL is set to 1761.2kbps according to Annex C4.2A.1.1. The expected througput under this level and fading conditions is 880 kbps. The UL datarate was configured by signalling TRRI=111111 and CRRI=15 on the E-AGCH. The expected UL datarate is 35.9 kbps corresponding to E-TFC Index 41.

2. In the test false ACK the SS responds with 100% ACK.

3. If the UE indicates on the E-UCCH a retransmission, the ACK from the SS was received as NACK by the UE. This is counted as false(ACK).

   If the UE indicates on the E-UCCH new data, the ACK from the SS was received as ACK by the UE. This is counted as correct ACK.

4. Continue until statistical significance according to Annex F.6.4 is achieved.

5. If the number of retransmissions reaches the maximum number of retransmissions due to several false ACK detections in series, the first new data on the E-PUCH with E-UCCH are not the consequence of ACK. This case is not counted as sample.

6. In the test false NACK the SS responds with 100% NACK.

7. If the UE indicates on the E-UCCH new data, the NACK from the SS was received as ACK by the UE. This is counted as false(NACK ). If the UE indicates on E-UCCH retransmission, the NACK from the SS was received as NACK by the UE. This is counted as correct reception.

8. Continue until statistical significance according to Annex F.6.4 is achieved.

9. The number of retransmissions will reach the maximum number of transmissions due to several retransmissions in series. The first new data on the E-PUCH with E-UCCH are not the consequence of ACK received by the UE. This case is not counted as sample.

### 11.1.5 Test Requirements

The false ACK and false NACK detection probability shall not exceed to the values specified in Table 11.1.2 for 3.84 Mcps TDD option, 11.1.4 for 1.28Mcps TDD option and Table 11.1.6 for 7.68 Mcps TDD option.

## 11.2 Demodulation of E-DCH Absolute Grant Channel (E-AGCH)

### 11.2.1 Definition and applicablility

#### 11.2.2.1 3.84 Mcps TDD Option

The performance of the E-AGCH detection is determined by the missed detection probability.

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 3.84 Mcps TDD UE that support HSDPA and E-DCH.

### 11.2.2.2 1.28 Mcps TDD Option

The performance of the E-AGCH detection is determined by the missed detection probability.

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 1.28 Mcps TDD UE that support HSDPA and E-DCH.

### 11.2.2.3 7.68 Mcps TDD Option

The performance of the E-AGCH detection is determined by the missed detection probability.

The requirements and this test apply to Release 7 and later releases for all types of UTRA for the 7.68 Mcps TDD UE that support HSDPA and E-DCH.

## 11.2.2 Minimum requirement

### 11.2.2.1 3.84 Mcps TDD Option

For the parameters specified in Table 11.2.1 the average downlink E-AGCH $\hat{I}_{or}/I_{oc}$ power ratio shall be below the specified value for the missed detection probability shown in Table 11.2.2.

**Table 11.2.1: Test parameters for E-AGCH detection (3.84 Mcps TDD option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/3.84 MHz | -60 |
| $\frac{E_c}{I_{or}}$ | dB | -6.02 |
| Number of Interfering codes/timeslot | - | 3 × SF16 |
| Total bits in Timeslot Resource Related Information (TRRI) | bits | 6 |
| Total bits in Resource Duration Indicator (RDI) | bits | 3 |
| Total bits in E-AGCH | bits | 38 |
| Propagation condition | - | VA30 |

**Table 11.2.2: Test requirements for E-AGCH detection (3.84 Mcps TDD option)**

| Test Number | E-AGCH $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 1.6 | 0.01 |

### 11.2.2.2 1.28Mcps TDD Option

For the parameters specified in Table 11.2.3 the average downlink $\hat{I}_{or}/I_{oc}$ power ratio shall be below the specified value for the missed detection probability shown in Table 11.2.4.

**Table 11.2.3: Test parameters for E-AGCH detection (1.28 Mcps TDD option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/1.28 MHz | -60 |
| $\frac{\text{E - AGCH_}E_c}{I_{or}}$ | dB | -3 |
| Number of Interfering codes/timeslot | - | 2 × SF16 |
| Total bits in Timeslot Resource Related Information (TRRI) | bits | 5 |
| Total bits in Resource Duration Indicator (RDI) | bits | 3 |
| Total bits in E-AGCH | bits | 26 |
| Propagation condition | - | VA30 |

**Table 11.2.4: Test requirements for E-AGCH detection (1.28 Mcps TDD option)**

| Test Number | $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 8 | 0.01 |

### 11.2.2.3 7.68 Mcps TDD Option

For the parameters specified in Table 11.2.5 the average downlink E-AGCH $\hat{I}_{or}/I_{oc}$ power ratio shall be below the specified value for the missed detection probability shown in Table 11.2.6.

**Table 11.2.5: Test parameters for E-AGCH detection (7.68 Mcps TDD option)**

| Parameters | Unit | Test 1 |
|---|---|---|
| $I_{oc}$ | dBm/7.68 MHz | -60 |
| $\frac{E_c}{I_{or}}$ | dB | -9.03 |
| Number of Interfering codes/timeslot | - | 7 × SF32 |
| Total bits in Timeslot Resource Related Information (TRRI) | bits | 6 |
| Total bits in Resource Duration Indicator (RDI) | bits | 3 |
| Total bits in E-AGCH | bits | 39 |
| Propagation condition | - | VA30 |

**Table 11.2.6: Test requirements for E-AGCH detection (7.68 Mcps TDD option)**

| Test Number | E-AGCH $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 1.2 | 0.01 |

## 11.2.3 Test Purpose

The aim of the test is to verify that the missed detection probability of the E-AGCH channel does not exceed 0.01.

## 11.2.4 Method of test

### 11.2.4.1 Initial conditions

#### 11.2.4.1.1 3.84Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.

2) Set the test parameters for the miss detection E-AGCH test as specified in table 11.2.1.

3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.1 and DL Reference Measurement Channel parameters are defined in Annex C4.1.1.1

4) The value of TRRI shall be set to '111111' and CRRI on E-AGCH shall be set to 15.

5) The UE is switched on.

6) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.

7) Switch on the fading simulator.

#### 11.2.4.1.2 1.28Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.
2) Set the test parameters for the miss detection E-AGCH test as specified in 11.2.3 for 1.28Mcps TDD option.
3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.2 and DL Reference Measurement Channel parameters are defined in Annex C4.2.1.1
4) The value of TRRI shall be set to '11000' and CRRI on E-AGCH shall be set to 3.
5) The UE is switched on.
6) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.
7) Switch on the fading simulator.

#### 11.2.4.1.3 7.68Mcps TDD Option

Test environment: normal; see clauses G.2.1 and G.2.2.

Frequencies to be tested: mid range; see clause G.2.4.

1) Connect the SS and AWGN noise source and fading simulator to the UE antenna connector as shown in figure Figure A.10.
2) Set the test parameters for the miss detection E-AGCH test as specified in table 11.2.1.
3) The UL Reference Measurement Channel parameters are defined in Annex C.6.1.3 and DL Reference Measurement Channel parameters are defined in Annex C4.2A.1.1
4) The value of TRRI shall be set to '111111' and CRRI on E-AGCH shall be set to 31.
5) The UE is switched on.
6) Enter the UE into loopback mode 1, looping back both the 12.2kbps RMC and HSDPA to E-DCH, and start the loopback test. See TS 34.108 [3] clause 7.3.9 and TS 34.109 [4] clauses 5.3.2.3 and 5.3.2.6. To fill the RLC transmit buffer, run the loopback for [3]s before starting the procedure.
7) Switch on the fading simulator.

### 11.2.4.2 Procedure

#### 11.2.4.2.1 3.84Mcps TDD Option

1) The SS shall signal 100% ACK on the E-HICH for all processes.
2) The SS shall signal the PRRI according to the E-AGCH information sequence as defined in Table 11.2.4.6A.
3) The SS shall analyse the E-TFCI transmitted on the E-UCCH for each E-DCH TTI to determine if a missed detection event has occurred by correlating the detected E-TFCIs with the expected E-TFCIs corresponding to

the PRRI sent on E-AGCH when the TRRI and CRRI is constant. If the expected E-TFC is not detected by the SS, record a missed detection event.

4) The test shall be run such that statistical significance according to Annex F.6.4 is achieved

#### 11.2.4.2.2 1.28Mcps TDD Option

1) The SS shall signal 100% ACK on the E-HICH for all processes.

2) The SS shall signal the PRRI according to the E-AGCH information sequence as defined in Table 11.2.4.7.

3) The SS shall analyse the E-TFCI transmitted on the E-UCCH for each E-DCH TTI to determine if a missed detection event has occurred by correlating the detected E-TFCIs with the expected E-TFCIs corresponding to the PRRI sent on E-AGCH when the TRRI and CRRI is constant. If the expected E-TFC is not detected by the SS, record a missed detection event.

4) The test shall be run such that statistical significance according to Annex F.6.4 is achieved

#### 11.2.4.2.3 7.68Mcps TDD Option

1) The SS shall signal 100% ACK on the E-HICH for all processes.

2) The SS shall signal the PRRI according to the E-AGCH information sequence as defined in Table 11.2.4.9.

3) The SS shall analyse the E-TFCI transmitted on the E-UCCH for each E-DCH TTI to determine if a missed detection event has occurred by correlating the detected E-TFCIs with the expected E-TFCIs corresponding to the PRRI sent on E-AGCH when the TRRI and CRRI is constant. If the expected E-TFC is not detected by the SS, record a missed detection event.

4) The test shall be run such that statistical significance according to Annex F.6.4 is achieved

## 11.2.5 Test Requirements

11.2.5.1 3.84Mcps TDD Option

The missed detection probability shall not exceed to the values specified in Table 11. 2.4.6B for 3.84 Mcps TDD.

The missed detection probability = the ratio of (missed detection event)/ (all detected E-TFCI event) $\leq 0.01$.

Tables 11.2.4.6A and 11.2.4.6B define the primary level settings including test tolerance and test parameters for the test.

**Table 11.2.4.6A: Test parameters for E-AGCH detection**

| Parameter | Unit | Missed detection |
|---|---|---|
| PRRI | | TBD |

**Table 11.2.4.6B : Test requirements for E-AGCH detection ( 3.84 Mcps TDD option)**

| Test Number | E-AGCH $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 2.2 | 0.01 |

11.2.5.2 1.28Mcps TDD Option

The missed detection probability shall not exceed to the values specified in Table 11.2.4.8.

The missed detection probability = the ratio of (missed detection event)/ (all detected E-TFCI event) $\leq 0.01$.

Tables 11.2.4.7 and 11.2.4.8 define the primary level settings including test tolerance and test parameters for the test.

**Table 11.2.4.7: Test parameters for E-AGCH detection**

| Parameter | Unit | Missed detection |
|---|---|---|
| PRRI | | TBD |

**Table 11.2.4.8 : Test requirements for E-AGCH detection ( 1.28 Mcps TDD option)**

| Test Number | E-AGCH $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 8.6 | 0.01 |

11.2.5.3 7.68Mcps TDD Option

The missed detection probability shall not exceed to the values specified in Table 11. 2.4.9 for 7.68 Mcps TDD option.

The missed detection probability = the ratio of (missed detection event)/ (all detected E-TFCI event) $\leq 0.01$.

Tables 11.2.4.9 and 11.2.4.10 define the primary level settings including test tolerance and test parameters for the test.

**Table 11.2.4.9: Test parameters for E-AGCH detection**

| Parameter | Unit | Missed detection |
|---|---|---|
| PRRI | | TBD |

**Table 11.2.4.10 : Test requirements for E-AGCH detection ( 7.68 Mcps TDD option)**

| Test Number | E-AGCH $\hat{I}_{or}/I_{oc}$ (dB) | Missed Detection Probability |
|---|---|---|
| 1 | 1.8 | 0.01 |

# Annex A (informative): Connection Diagrams

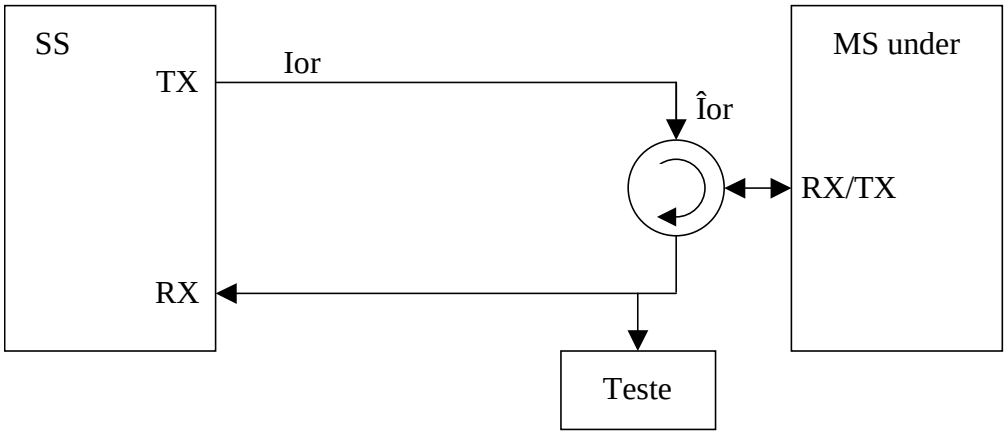



**Figure A.1: Connection for Basic TX Test**

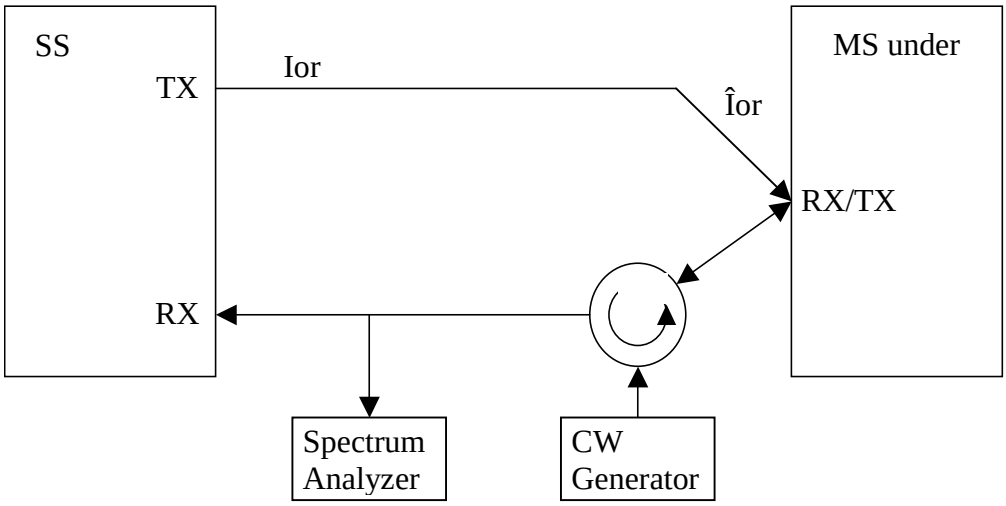



**Figure A.2: Connection for TX Intermodulation Test**

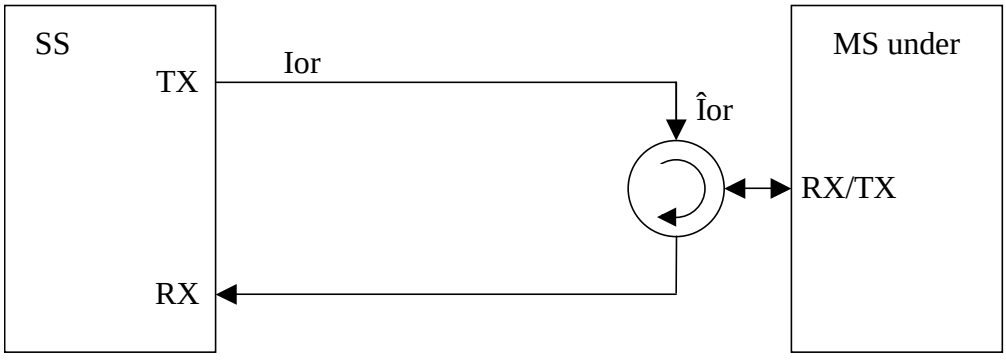



**Figure A.3: Connection for Basic RX Test**

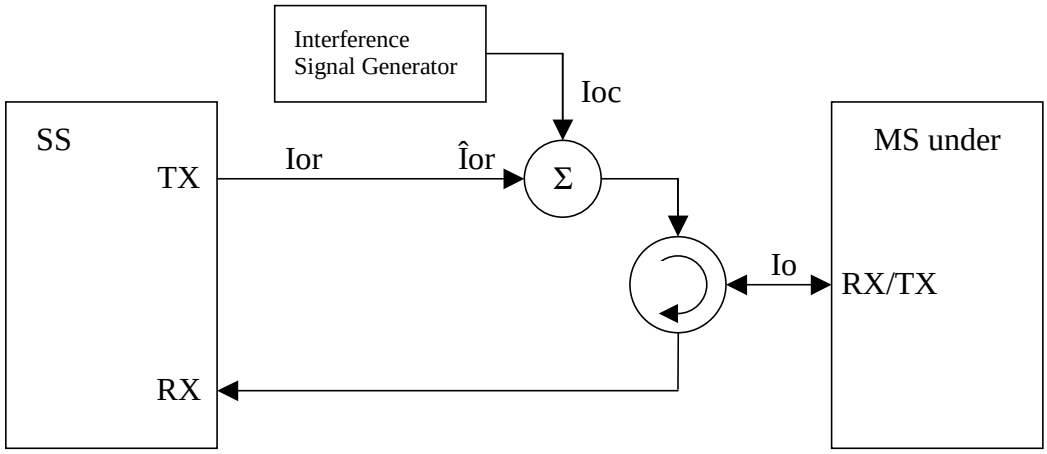



**Figure A.4: Connection for RX Test with Interference**

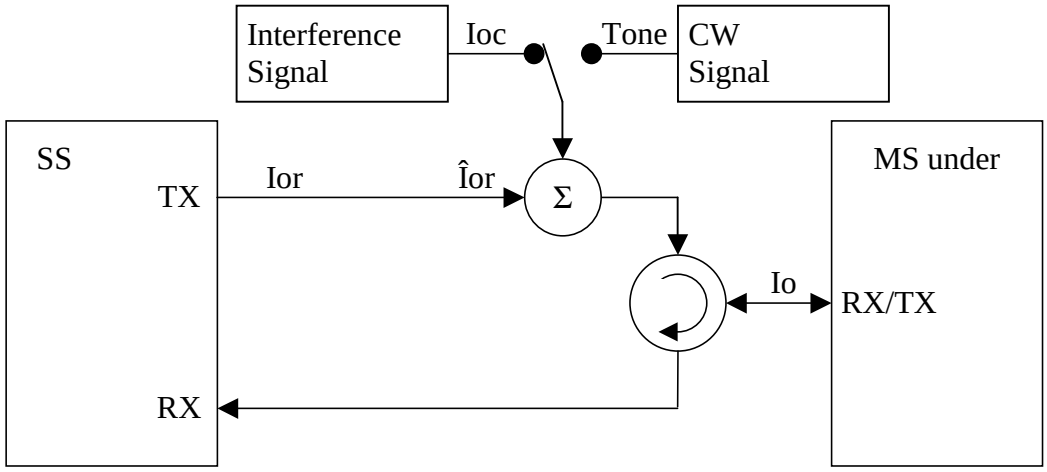



**Figure A.5: Connection for RX Test with Interference or additional CW**

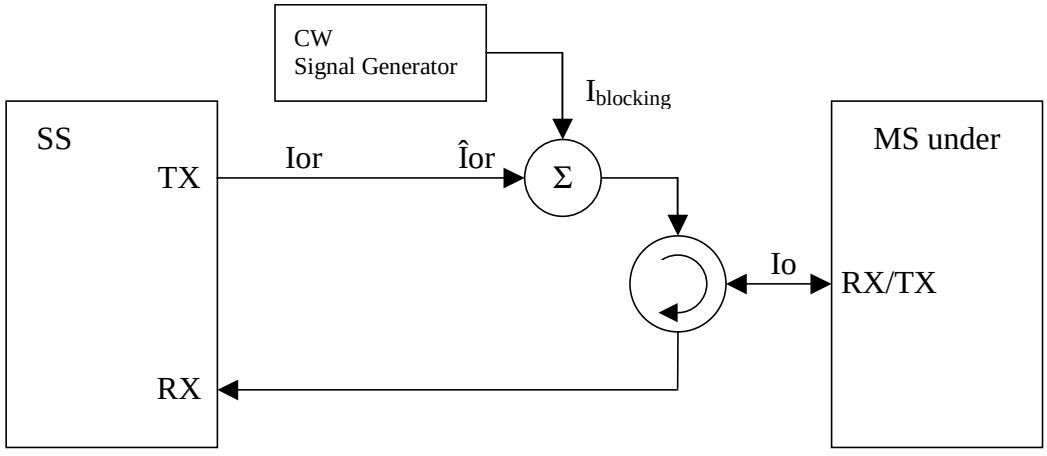



**Figure A.6: Connection for RX Test with additional CW**

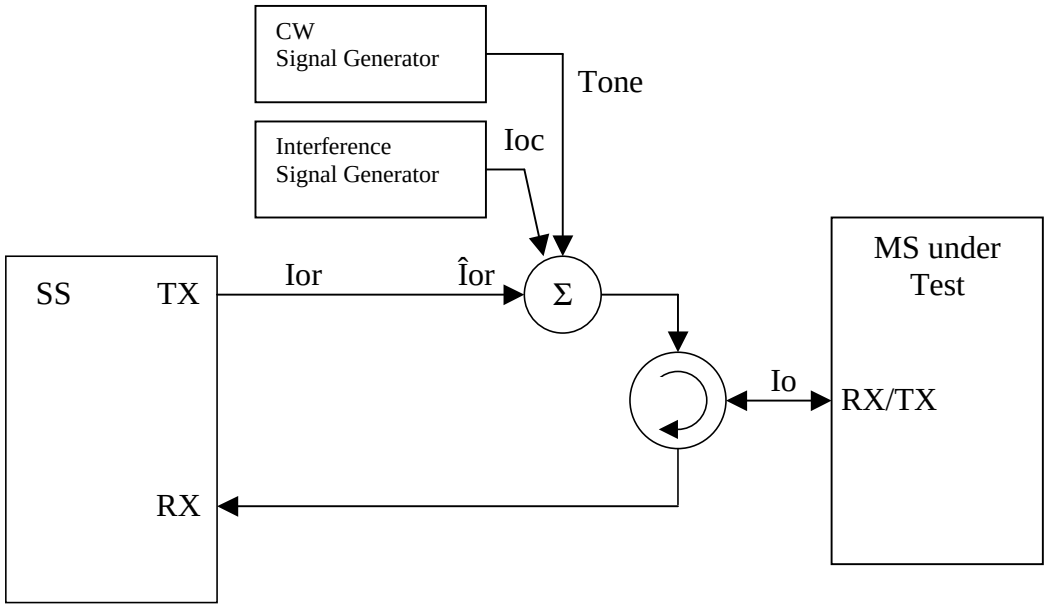



**Figure A.7: Connection for RX Test with both Interference and additional CW**




**Figure A.8: Connection for Spurious Emission Test**

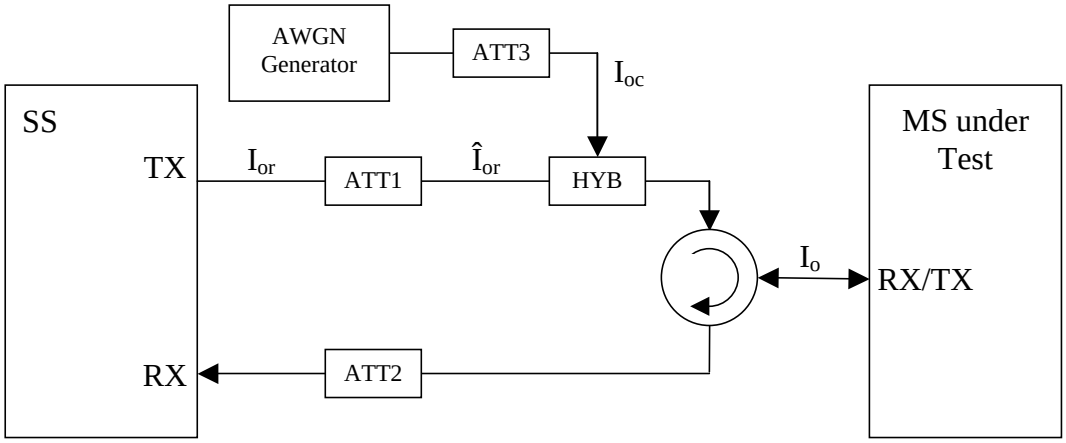



**Figure A.9: Connection for Static Channel Test**

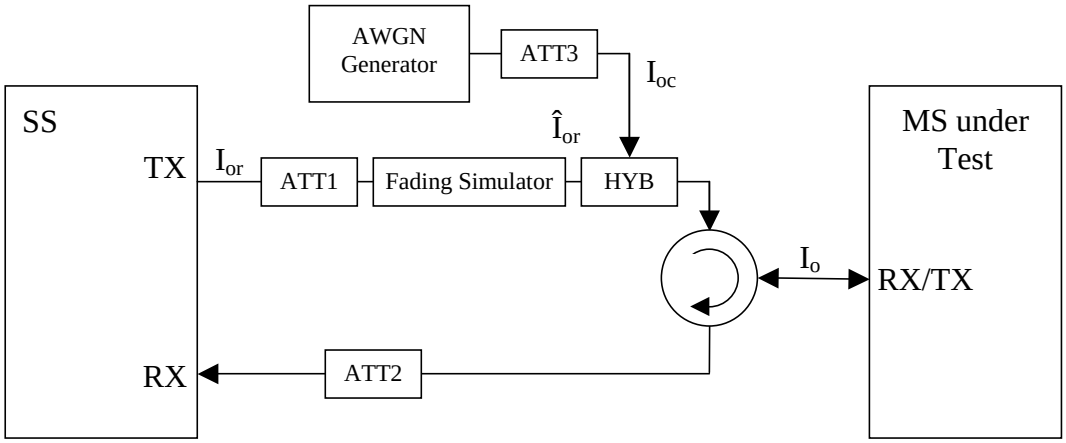



**Figure A.10: Connection for Multiple Fading Channel Test**

**Figure A.11: Void**

**Figure A.12: Void**

**Figure A.13: Void**

**Figure A.14: Void**

**Figure A.15: Void**

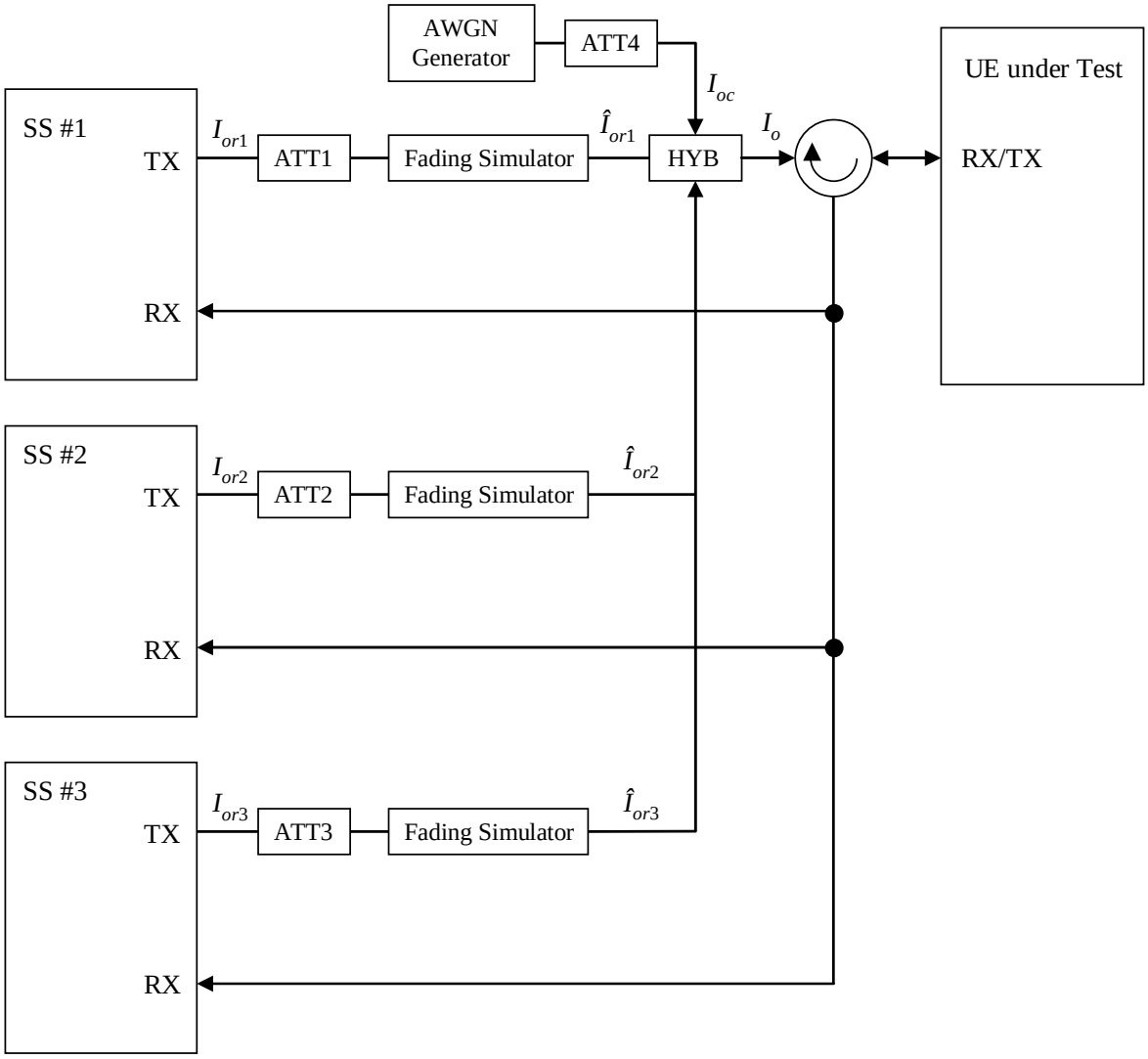



**Figure A.16: Connection for three cell tests with Multi-path Fading propagation**

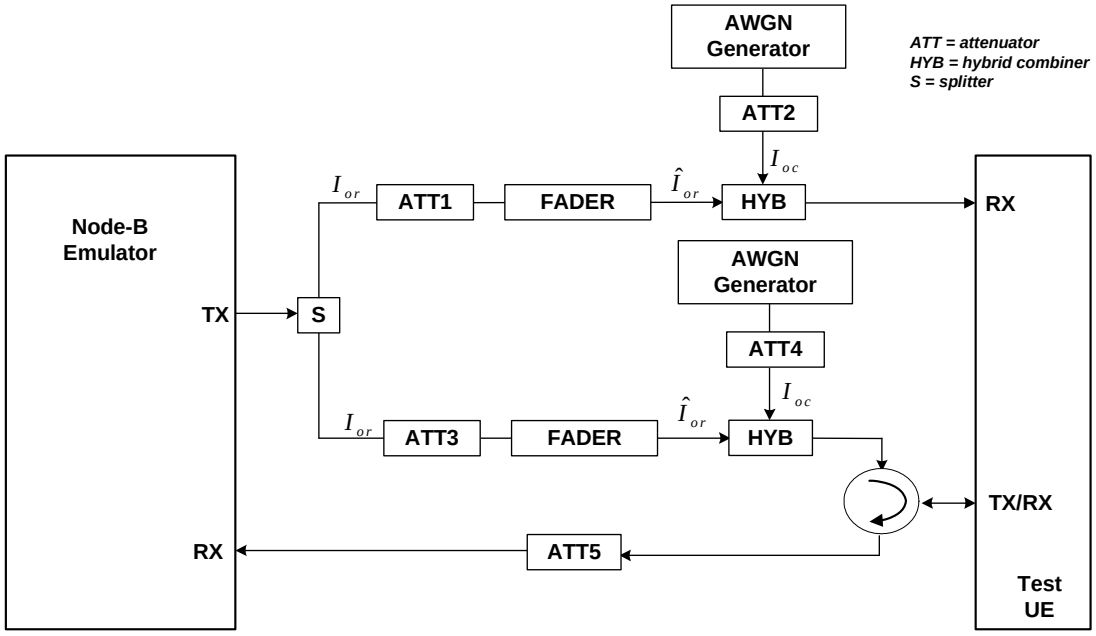



**Figure A.17: Connection for single cell tests with Multi-path Fading propagation and UE receive diversity**

# Annex B (normative): Global In-Channel TX-Test

## B.1 General

The global in-channel Tx test enables the measurement of all relevant parameters that describe the in-channel quality of the output signal of the Tx under test in a single measurement process.

The parameters describing the in-channel quality of a transmitter, however, are not necessarily independent. The algorithm chosen for description inside this annex places particular emphasis on the exclusion of all interdependencies among the parameters. Any other algorithm (e.g. having better computational efficiency) may be applied, as long as the results are the same within the accuracy limits.

All notes referred in the various clauses of B.2 are put together in B.3.

## B.2 Definition of the process

### B.2.1 Basic principle

The process is based on the comparison of the actual **output signal of the TX under test**, received by an ideal receiver, with a **reference signal**, that is generated by the measuring equipment and represents an ideal error free received signal. The reference signal shall be composed of the same number of codes at the correct spreading factors as contained in the test signal. Note, for simplification, the notation below assumes only codes of one spreading factor although the algorithm is valid for signals containing multiple spreading factors.

All signals are represented as equivalent (generally complex) baseband signals.

### B.2.2 Output signal of the TX under test

The output signal of the TX under test is acquired by the measuring equipment, filtered by a matched filter (RRC 0.22, correct in shape and in position on the frequency axis) and stored for further processing.

The following form represents the physical signal in the entire measurement interval:

one vector **Z**, containing N = ns x sf + ma complex samples;

with

ns: number of symbols in the measurement interval;

sf: number of chips per symbol. (sf: spreading factor) (see Note: Symbol length)

ma: number of midamble chips

### B.2.3 Reference signal

The reference signal is constructed by the measuring equipment according to the relevant TX specifications.

It is filtered by the same matched filter, mentioned in B.2.2., and stored at the Inter-Symbol-Interference free instants. The following form represents the reference signal in the entire measurement interval:

one vector **R**, containing N = ns x sf + ma complex samples;

ns, sf, ma: see B.2.2

## B.2.4 void

## B.2.5 Classification of measurement results

The measurement results achieved by the global in-channel TX test can be classified into two types:

Results of type "deviation", where the error-free parameter has a non-zero magnitude. (These are the parameters that quantify the integral physical characteristic of the signal).These parameters are:

- RF Frequency
- Power (in case of single code)
- Code Domain Power (in case of multi code)
- Timing

(Additional parameters: see Note: Deviation)

Results of type "residual", where the error-free parameter has value zero. (These are the parameters that quantify the error values of the measured signal, whose ideal magnitude is zero). These parameters are:

- Error Vector Magnitude (EVM);
- Peak Code Domain Error (PCDE).

(Additional parameters: see Note residual).

## B.2.6 Process definition to achieve results of type "deviation"

The reference signal (**R**; see clause B.2.3) and the signal under Test (Z; see subclause B.2.2) are varied with respect to the parameters mentioned in clause B.2.5 under "results of type deviation" in order to achieve best fit. Best fit is achieved when the RMS difference value between the varied signal under test and the varied reference signal is an absolute minimum.

Overview:

$$FCT\ \left[\ Z\left(\tilde{f},\tilde{t},\tilde{\varphi},g_1,g_2,...,\ g_{synch}\right) - R\left(f,t,\varphi,\tilde{g}_1,\tilde{g}_2,...,\ \tilde{g}_{synch}\right)\ \right] = Minimum\quad !$$

Z : Signal under test.

R: Reference signal,

with frequency f, the timing t, the phase φ, gain of code1 ($g_1$), gain of code2 ($g_2$) etc, and the gain of the synch channel $g_{synch}$

The parameters marked with a tilde in Z and R are varied in order to achieve a best fit.

Detailed formula: see Note: Formula for the minimum process

The varied reference signal, after the best fit process, will be called **R'.**

The varied signal under test, after the best fit process, will be called **Z'.**

The varying parameters, leading to **R' and Z'** represent directly the wanted results of type "deviation". These measurement parameters are expressed as deviation from the reference value with units same as the reference value.

In case of multi code, the type-"deviation"-parameters (frequency, timing and (RF-phase)) are varied commonly for all codes such that the process returns one frequency-deviation, one timing deviation, (one RF-phase –deviation).

(These parameters are not varied on the individual codes signals such that the process would return kr frequency errors... . (kr: number of codes in the reference signal)).

The only type-"deviation"-parameters varied individually are the code domain gain factors (g1, g2, …).

#### E.2.5.1 Decision Point Power

The mean-square value of the signal-under-test, sampled at the best estimate of the of Intersymbol-Interference-free points using the process defined in subclause 2.5, is referred to the *Decision Point Power* (DPP):

#### E.2.5.2 Code-Domain Power

The samples, Z', are separated into symbol intervals to create ns time-sequential vectors **z** with sf complex samples comprising one symbol interval. The *Code Domain Power* is calculated according to the following steps:

1) Take the vectors **z** defined above.

2) To achieve meaningful results it is necessary to descramble **z,** leading to **z'**

3) Take the orthogonal vectors of the channelization code set **C** (all codes belonging to one spreading factor) as defined in TS 25.213 and TS 25.223 (range +1, -1), and normalize by the norm of the vectors to produce **C**norm=**C**/sqrt(sf). (see Note: Symbol length)

4) Calculate the inner product of **z'** with **C**norm**.**. Do this for all symbols of the measurement interval and for all codes in the code space.
This gives an array of format k x ns, each value representing a specific symbol and a specific code, which can be exploited in a variety of ways.

   k: total number of codes in the code space

   ns: number of symbols in the measurement interval

5) Calculate k mean-square values, each mean-square value unifying ns symbols within one code.
(These values can be called "*Absolute CodeDomainPower* (CDP)" [Volt$^2$].) The sum of the k values of CDP is equal to DPP.

6) Normalize by the decision point power to obtain

$$Relative\ CodeDomain\ Power = \frac{Absolute\ CodeDomainPower}{DecisionPointPower}$$

## B.2.7 Process definition to achieve results of type "residual"

The difference between the varied reference signal (**R'**; see clauseB.2.6.) and the varied TX signal under test (**Z'**; see clause B.2.6) is the error vector **E** versus time:

**E** = **Z'** – **R'.**

Depending on the parameter to be evaluated, it is appropriate to represent **E** in one of the following two different forms:

Form EVM (representing the physical error signal in the entire measurement interval)

One vector **E**, containing N = ns x sf + ma complex samples;

ns, sf, ma: see B.2.2

Form PCDE (derived from Form EVM by separating the samples into symbol intervals)

ns time-sequential vectors **e** with sf complex samples comprising one symbol interval.

**E** gives results of type "residual" applying the two algorithms defined in clauses B 2.7.1 and B 2.7.2.

### B.2.7.1 Error Vector Magnitude (EVM)

The Error Vector Magnitude EVM is calculated according to the following steps:

1) Take the error vector **E** defined in clause B.2.7 (Form EVM) and calculate the RMS value of **E**; the result will be called RMS(**E**).

2) Take the varied reference vector **R'** defined in clause B.2.6 and calculate the RMS value of **R'**; the result will be called RMS(**R'**).

3) Calculate EVM according to:

$$EVM = \frac{RMS(E)}{RMS(R')} \times 100\%$$ (here, EVM is relative and expressed in %)

(see note TDD)

(see note: Formula for EVM)

### B.2.7.2 Peak Code Domain Error (PCDE)

The Peak Code Domain Error is calculated according to the following steps:

1) Take the error vectors **e** defined in clause B.2.7 (Form PCDE).

2) Take the orthogonal vectors of the channelisation - code set **C** (all codes belonging to one spreading factor) as defined in TS 25.213 and TS 25.223 (range +1, -1). (see Note: Symbol length) and normalize by the norm of the vectors to produce **C**norm= **C**/sqrt(sf). (see Note: Symbol length).

3) To achieve meaningful results it is necessary to descramble **e,** leading to **e'**.

4) Calculate the inner product of **e'** with **C**norm. Do this for all symbols of the measurement interval and for all codes in the code space.
This gives an array of format k x ns, each value representing an error-vector representing a specific symbol and a specific code, which can be exploited in a variety of ways.

k: total number of codes in the code space

ns: number of symbols in the measurement interval

5) Calculate k RMS values, each RMS value unifying ns symbols within one code.
(These values can be called "*Absolute CodeEVMs*" [Volt].)

6) Find the peak value among the k "*Absolute CodeEVMs*".
(This value can be called "*Absolute PeakCodeEVM*" [Volt].)

7) Calculate PCDE according to:

.

$$10*\lg \frac{(\text{"}Absolute\ PeakCodeEVM\text{"})^2}{(RMS(\mathbf{R}'))^2}\ dB$$ (a relative value in dB).

(see Note: Scrambling code)

(see Note TDD)

## B.3 Notes

**Note: Symbol length)**

A general code multiplexed signal is multicode and multirate. In order to avoid unnecessary complexity, the measurement applications use a unique symbol-length, corresponding to a spreading factor, regardless of the really intended spreading factor. Nevertheless the complexity with a multicode / multirate signal can be mastered by introducing appropriate definitions.

**Note: Deviation)**

It is conceivable to regard more parameters as type „deviation" e.g. Chip frequency and RF-phase.

As chip-frequency and RF-frequency are linked together by a statement in the core specifications [1] it is sufficient to process RF frequency only.

A parameter RF-phase must be varied within the best fit process (B 2.6.). Although necessary, this parameter-variation doesn't describe any error, as the modulation schemes used in the system don't depend on an absolute RF-phase.

**Note: Residual)**

It is conceivable to regard more parameters as type „residual" e.g. IQ origin offset. As it is not the intention of the test to separate for different error sources, but to quantify the quality of the signal, all such parameters are not extracted by the best fit process, instead remain part of EVM and PCDE.

**Note Scrambling Code)**

To interpret the measurement results in practice it should be kept in mind that erroneous code power on unused codes is generally de-scrambled differently under test conditions and under real life conditions, whereas erroneous code power on used codes is generally de-scrambled equally under test conditions and under real life conditions. It might be indicated if a used or unused code hits PCDE.

**Note TDD)**

EVM covers the midamble part as well as the data part; however PCDE disregards the midamble part.

**Note: Fomula for the minimum process**

$$L(\Delta\tilde{f}, \Delta\tilde{t}, \Delta\tilde{\varphi}, \Delta\tilde{g}_c, ..., \Delta\tilde{g}_{mid}) = \sum_{\nu=0}^{N-1} \left| Z(\nu) - R(\nu) \right|^2$$

Legend:

L : the function to be minimised

The parameters to be varied in order to minimize are:

$\Delta\tilde{f}$ : the RF frequency offset

$\Delta\tilde{t}$ : the timing offset

$\Delta\tilde{\varphi}$ : the phase offset

$\Delta\tilde{g}_c$ ... code power offsets (one offset for each code)

$\Delta\tilde{g}_{mid}$ : the power offset of the midamble

Z(ν): Samples of the signal under Test

R(ν): Samples of the reference signal

$\sum_{\nu=0}^{N-1}$ : counting index $\nu$ starting at the beginning of the measurement interval and ending at its end.

N = No of chips during the measurement interval.

Z(ν): Samples of the signal under Test. It is modelled as a sequence of complex baseband samples Z(γ) with a time-shift Δt, a frequency offset Δf, a phase offset Δφ, the latter three with respect to the reference signal.

$$Z(\nu) = Z(\nu - \Delta\tilde{t}) * e^{-j2\pi\Delta\tilde{f}\nu} * e^{-j\Delta\tilde{\varphi}}$$

R(ν): Samples of the reference signal:

$$R(\nu) = \sum_{c=1}^{No.of\ codes} (g_c + \Delta\tilde{g}_c) * Chip_c(\nu) + (g_{mid} + \Delta\tilde{g}_{mid}) * Chip_{mid}(\nu)$$

g : nominal gain of the code channel or midamble

$\Delta\tilde{g}$ : The gain offset to be varied in the minimum process

Chip(ν) is the chipsequence of the code channel or midamble

Indices at g, Δg and Chip:

The index indicates the code channel: c = 1,2,... No of code channels

Range for $Chip_c$ : +1,-1

Note: **Formula for EVM**

$$EVM = \sqrt{\frac{\sum_{\nu=0}^{N-1} \left| Z'(\gamma) - R'(\gamma) \right|^2}{\sum_{\nu=0}^{N-1} \left| R'(\gamma) \right|^2}} * 100\ \%$$

Z'(γ), R'(γ) are the varied measured and reference signals.

# Annex C (normative): Measurement channels

## C.1 General

Void.

## C.2 UL Reference measurement channels

### C.2.1 UL reference measurement channel (12,2 kbps)

#### C.2.1.1 3,84 TDD Option

**Table C.2.1.1: UL reference measurement channel physical parameters (12,2 kbps)**

| Parameter | |
|---|---|
| Information data rate | 12,2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 10% / 0% |

**Table C.2.1.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps)**

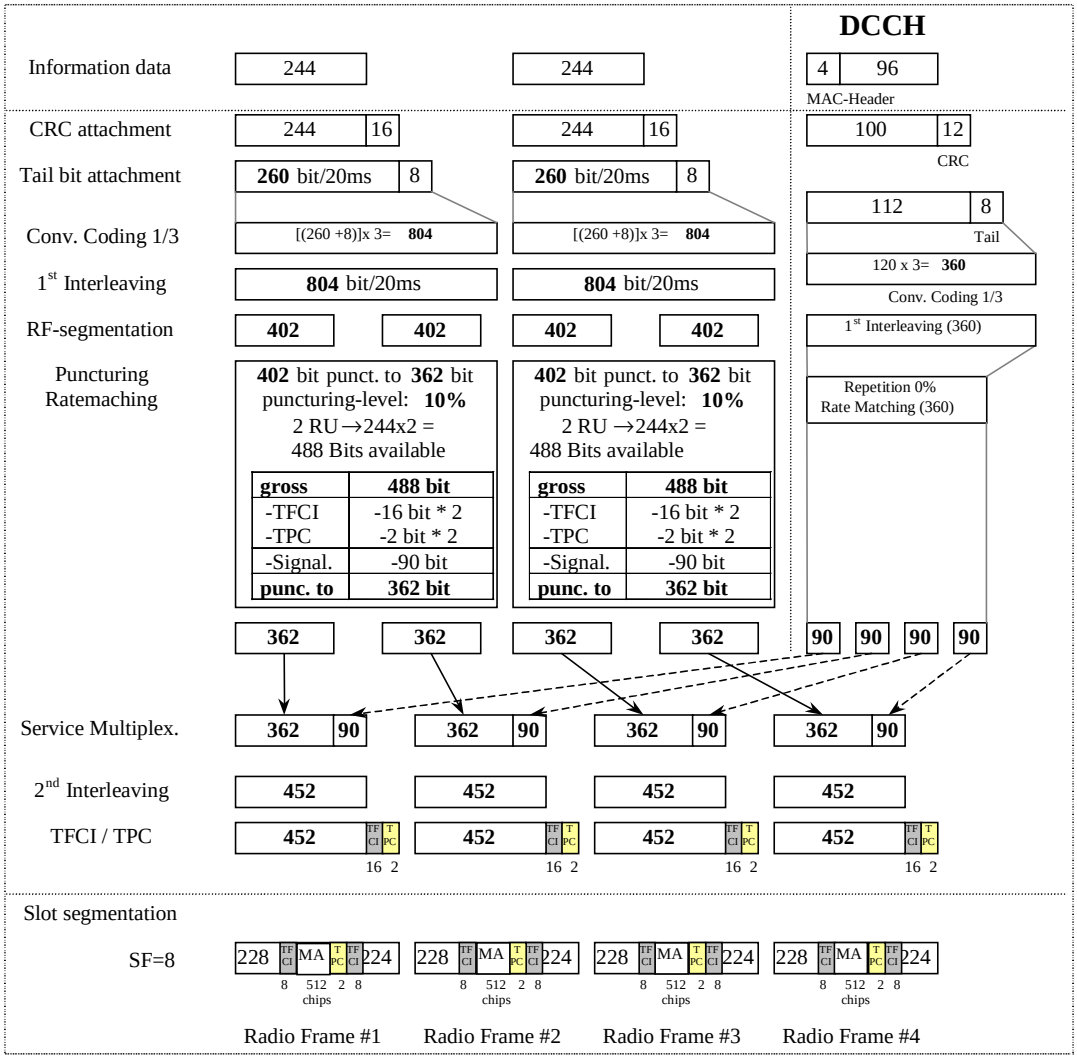
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 220 | 242 |

**Table C.2.1.3: UL reference measurement channel, TFCS (12.2 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

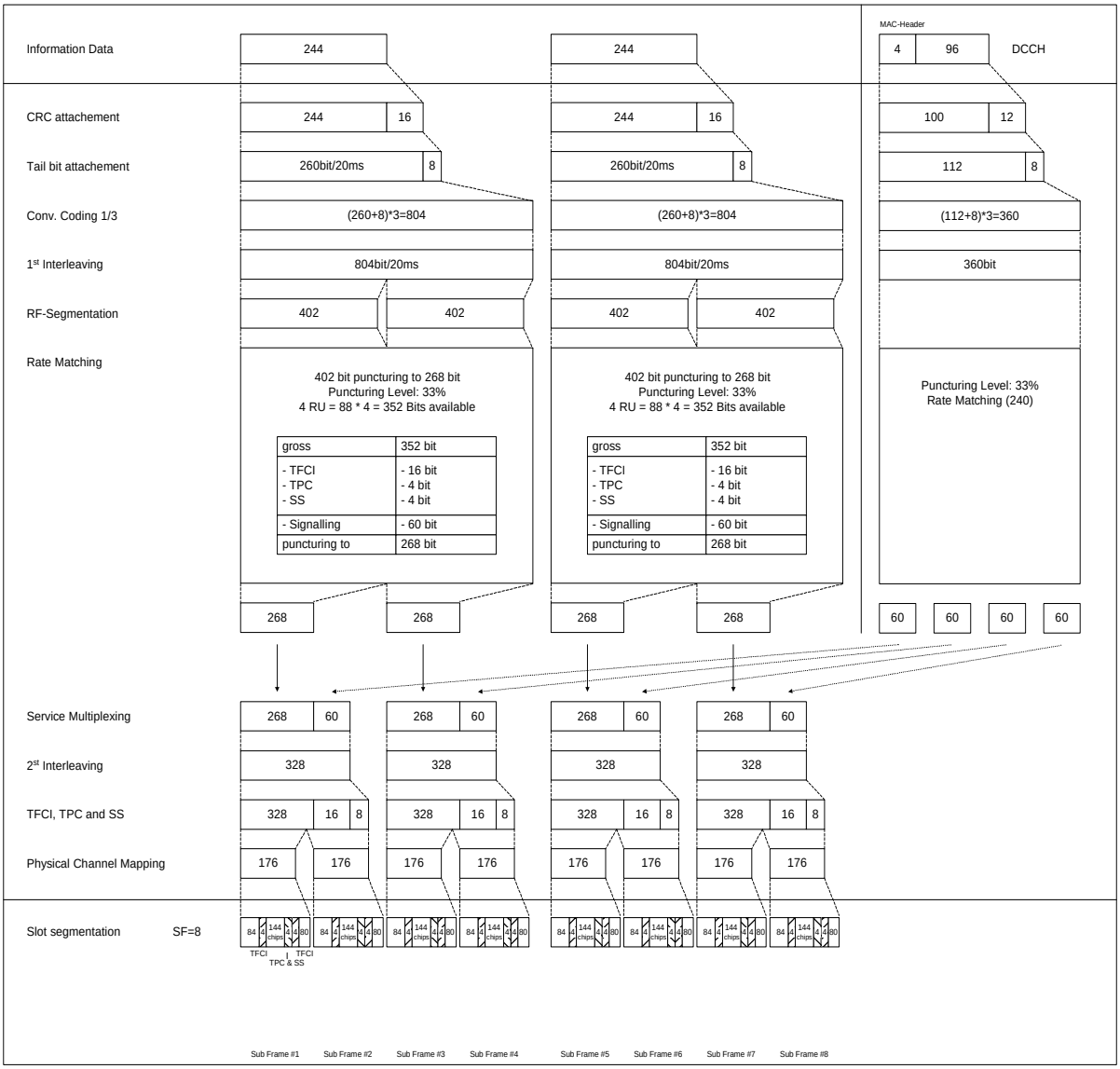
**Table C.2.1.4: UL reference measurement channel, puncturing limit (12.2 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.88 |



## C.2.1.2 UL reference measurement channel (12,2 kbps) for 1,28 Mcps TDD Option

| Parameter | |
|---|---|
| Information data rate | 12,2 kbps |
| RU's allocated | 1TS (1*SF8) = 2RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 4 Bit/user/10ms |
| TFCI | 16 Bit/user/10ms |
| Synchronisation Shift (SS) | 4 Bit/user/10ms |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3: DCH of the DTCH / DCH of the DCCH | 33% / 33% |



## C.2.1.3 7,68 TDD Option

| Parameter | Value |
|---|---|
| Information data rate | 12.2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 10% / 0% |

**Table C.2.1.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps)**

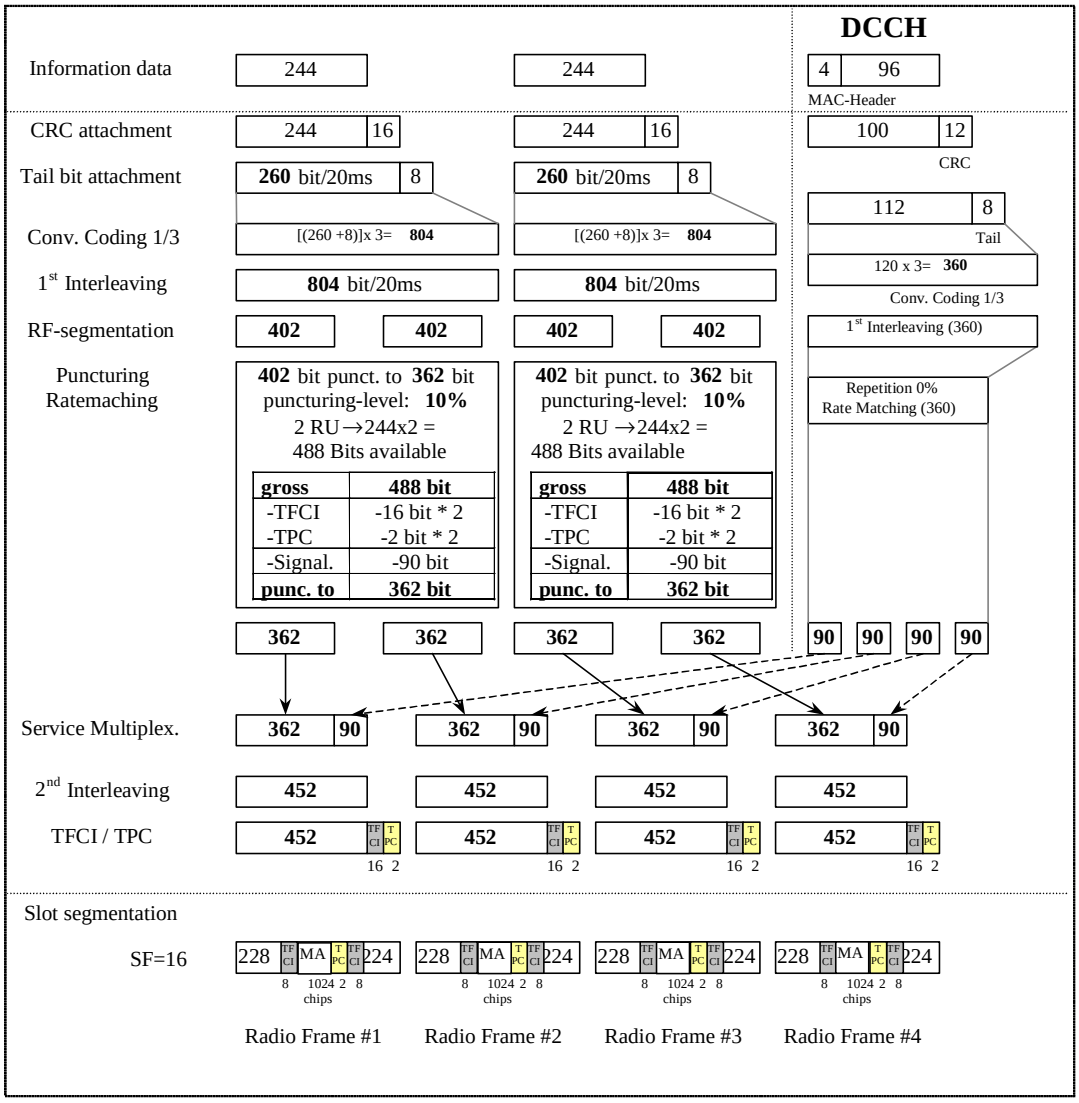
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 220 | 242 |

**Table C.2.1.3: UL reference measurement channel, TFCS (12.2 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.1.4: UL reference measurement channel, puncturing limit (12.2 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.88 |




# C.2.2 UL multi code reference measurement channel (12,2 kbps)

## C.2.2.1 3,84 Mcps TDD Option

**Table C.2.2.1: UL reference measurement channel physical parameters (12,2 kbps, multi code)**

| Parameter | |
|---|---|
| Information data rate | 12,2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 5% / 0 % |

**Table C.2.2.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps, multicode)**

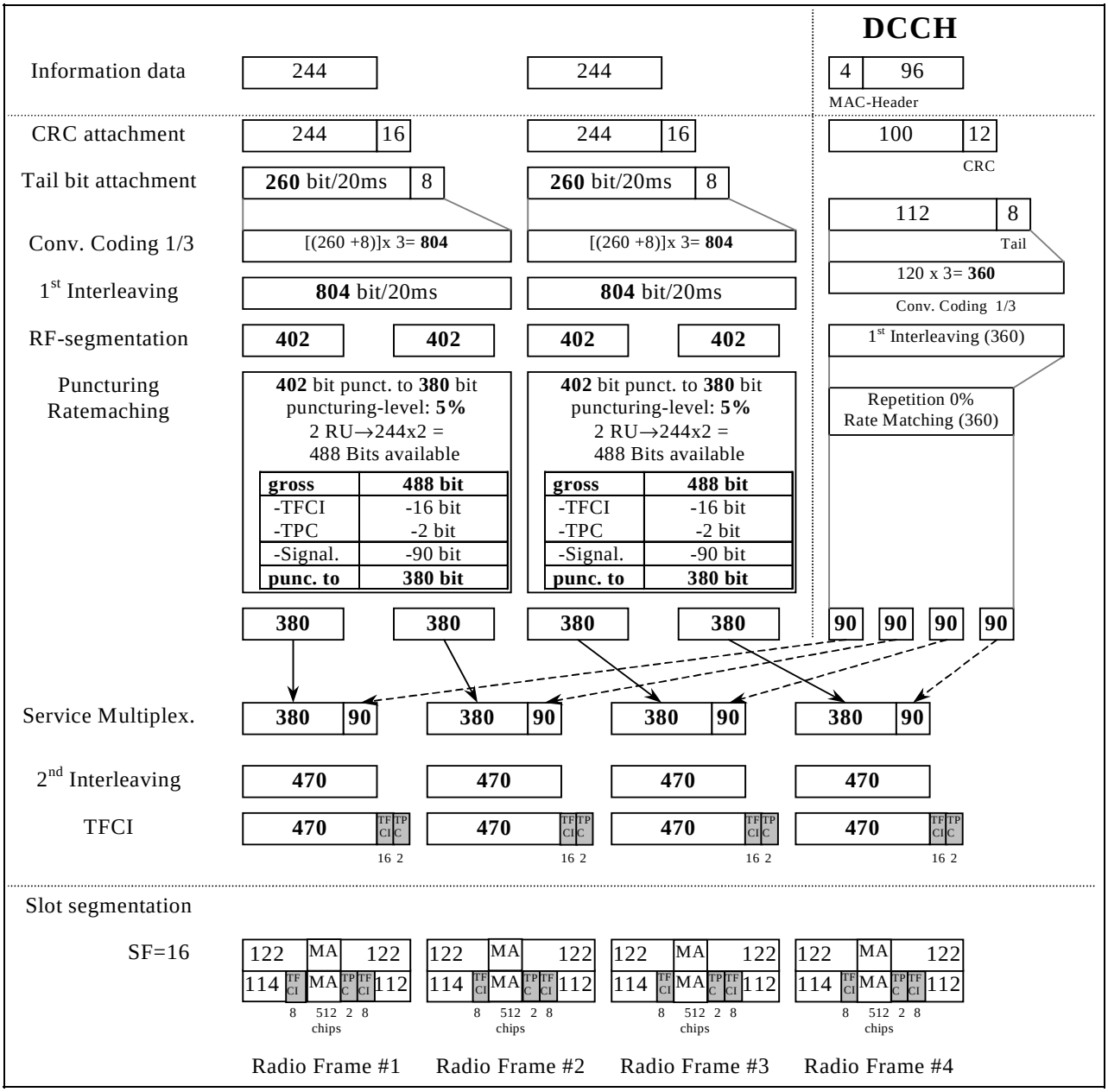
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 220 | 232 |

**Table C.2.2.3: UL reference measurement channel, TFCS (12.2 kbps, multicode)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.2.4: UL reference measurement channel, puncturing limit (12.2 kbps, mulitcode)**

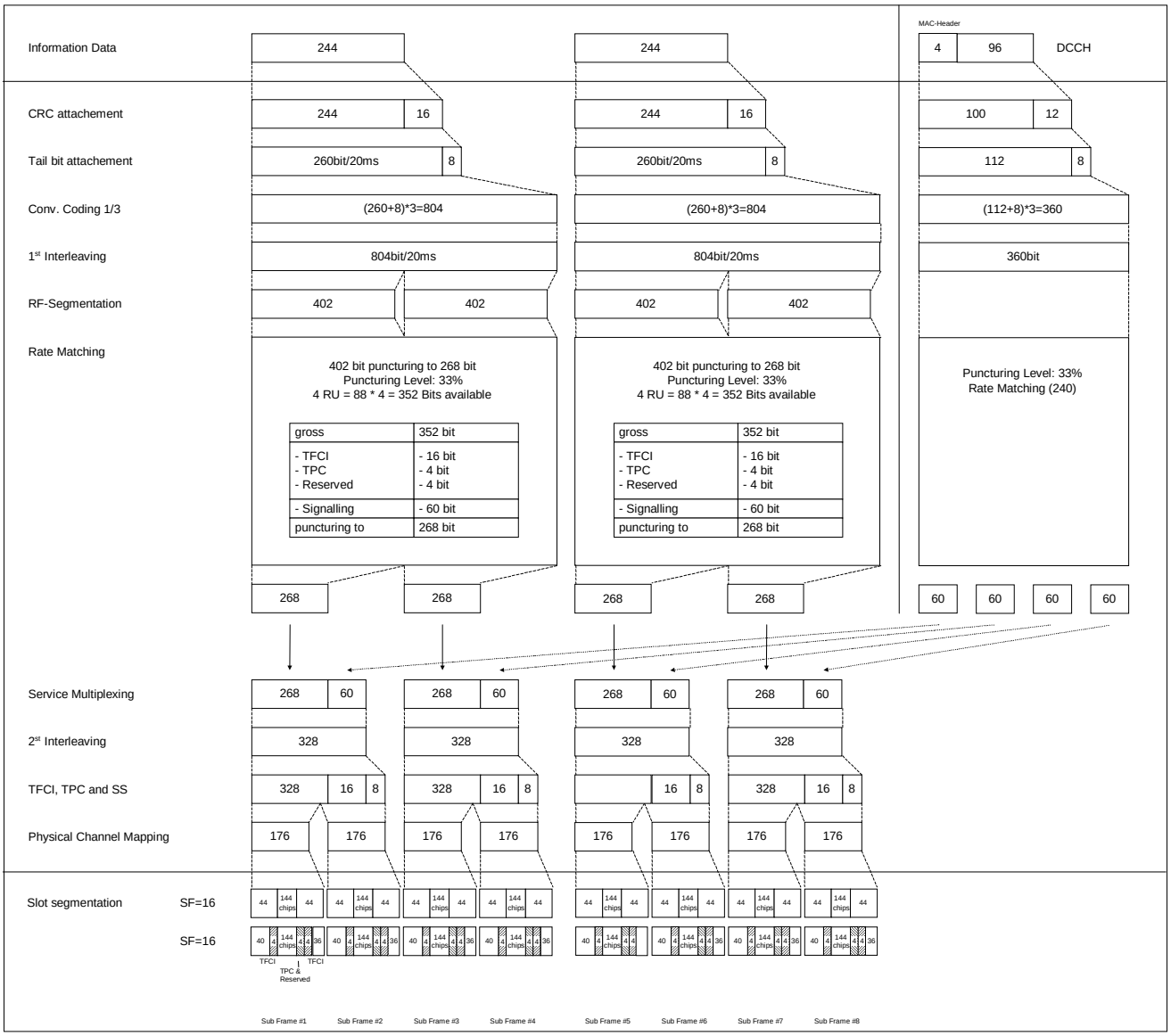
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.92 |




## C.2.2.2 UL multi code reference measurement channel (12,2 kbps) for 1,28 Mcps TDD Option

**Table C.2.2.2**

| Parameter | Value |
|---|---|
| Information data rate | 12,2 kbps |
| RU's allocated | 1TS (2*SF16) = 2RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 4 Bit/user/10ms |
| TFCI | 16 Bit/user/10ms |
| 4 Bit reserved for future use (place of SS) | 4 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate 1/3: DCH of the DTCH / DCH of the DCCH | 33% / 33% |




## C.2.2.3 7,68 Mcps TDD Option

| Parameter | Value |
|---|---|
| Information data rate | 12.2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 5% / 0 % |

**Table C.2.2.5: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps, multicode)**

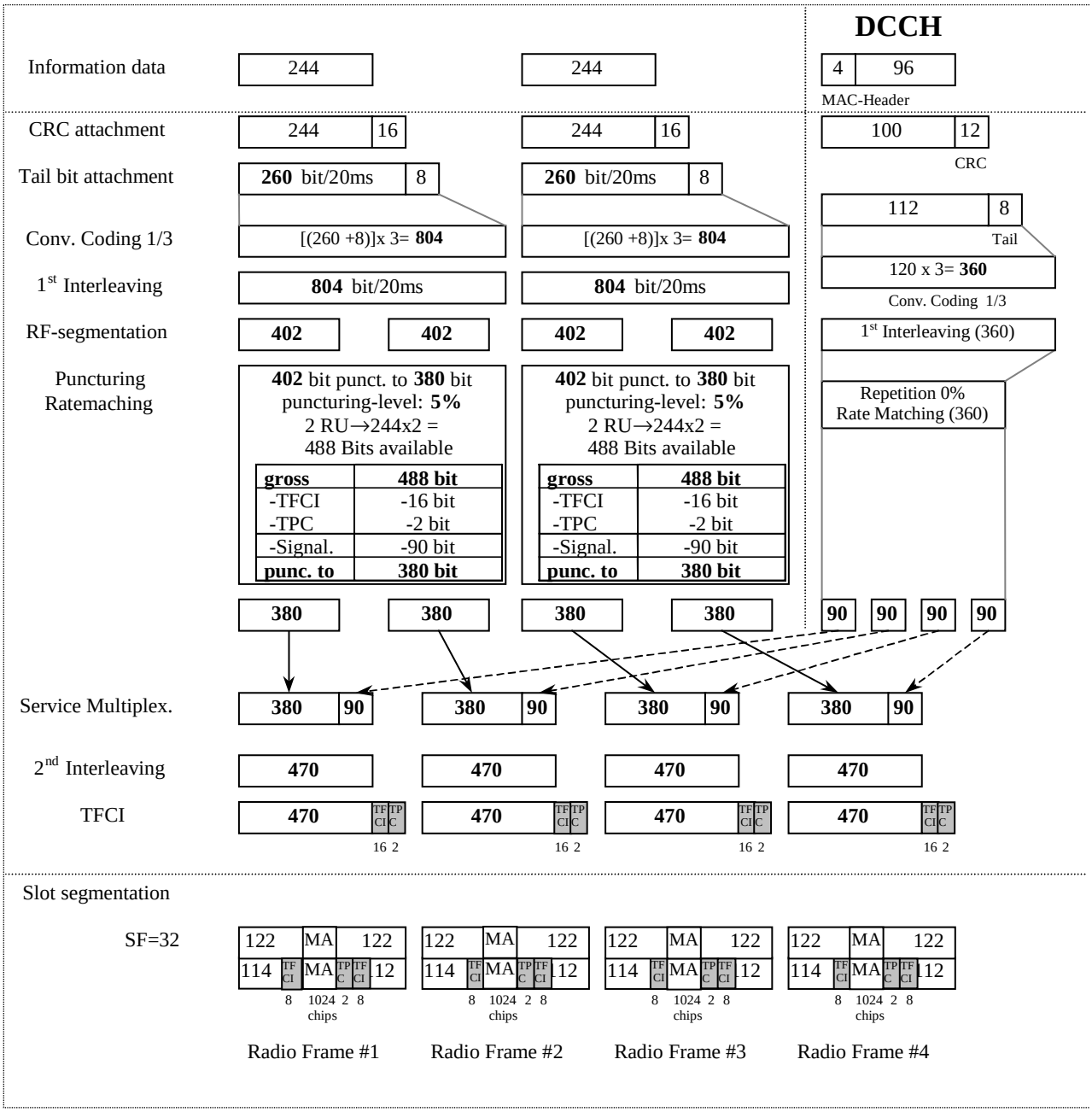
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 220 | 232 |

**Table C.2.2.6: UL reference measurement channel, TFCS (12.2 kbps, multicode)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.2.7: UL reference measurement channel, puncturing limit (12.2 kbps, mulitcode)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.92 |




# C.2.3 UL reference measurement channel (64 kbps)

## C.2.3.1 3.84 Mcps TDD Option

**Table C.2.3.1: UL reference measurement channel physical parameters (64kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 64 kbps |
| RU´s allocated | 1 SF4 + 1 SF16 = 5RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.8% / 13.3% |

**Table C.2.3.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (64 kbps)**

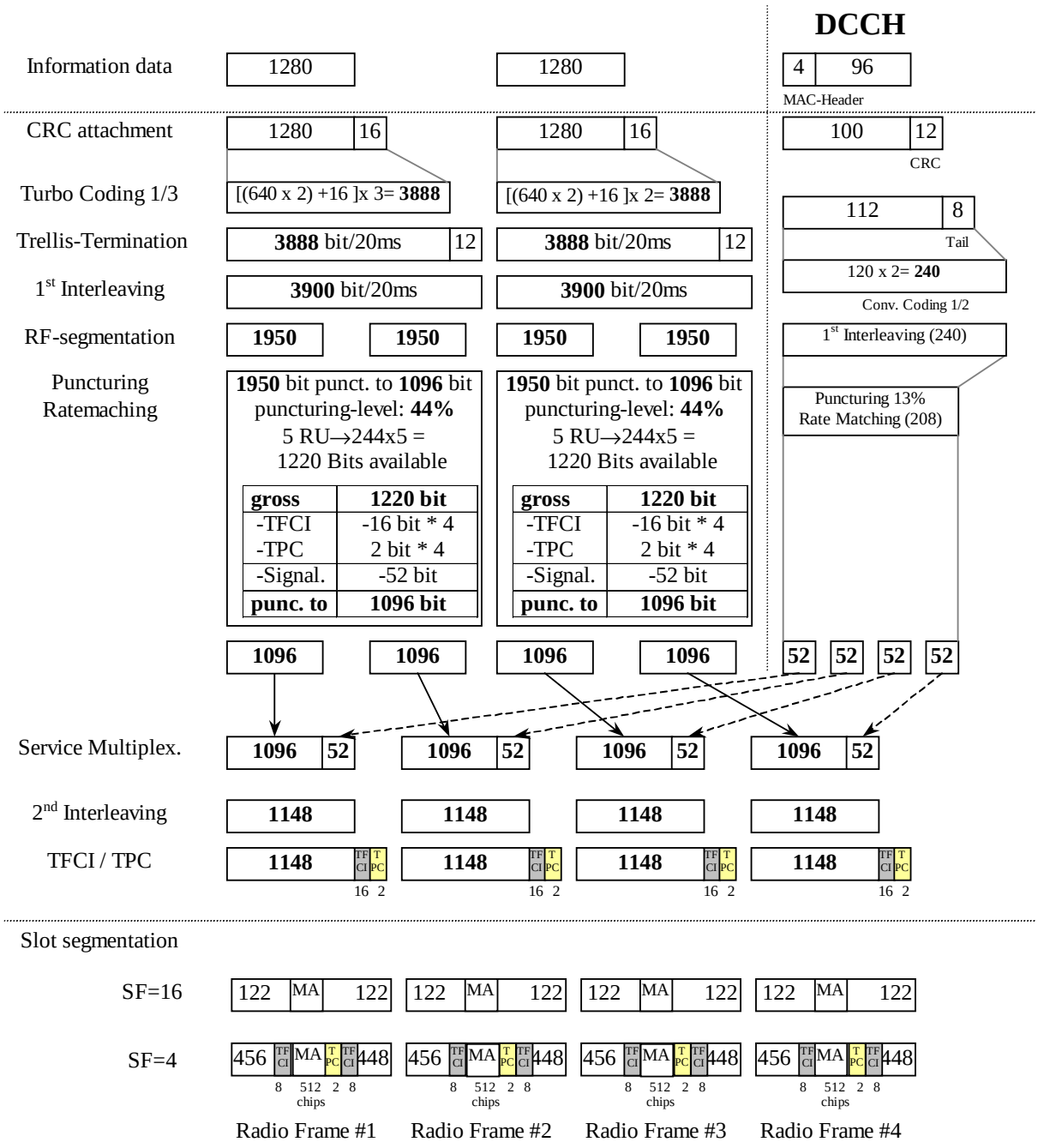
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 1280 | 88/80 |
| | Max data rate, bps | | 64000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 1280 | 100 |
| | TFS | TF0, bits | 0*1280 | 0*100 |
| | | TF1, bits | 1*1280 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 3900 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1950 | 60 |
| | RM attribute | | 158 | 240 |

**Table C.2.3.3: UL reference measurement channel, TFCS (64 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.3.4: UL reference measurement channel, puncturing limit (64 kbps)**

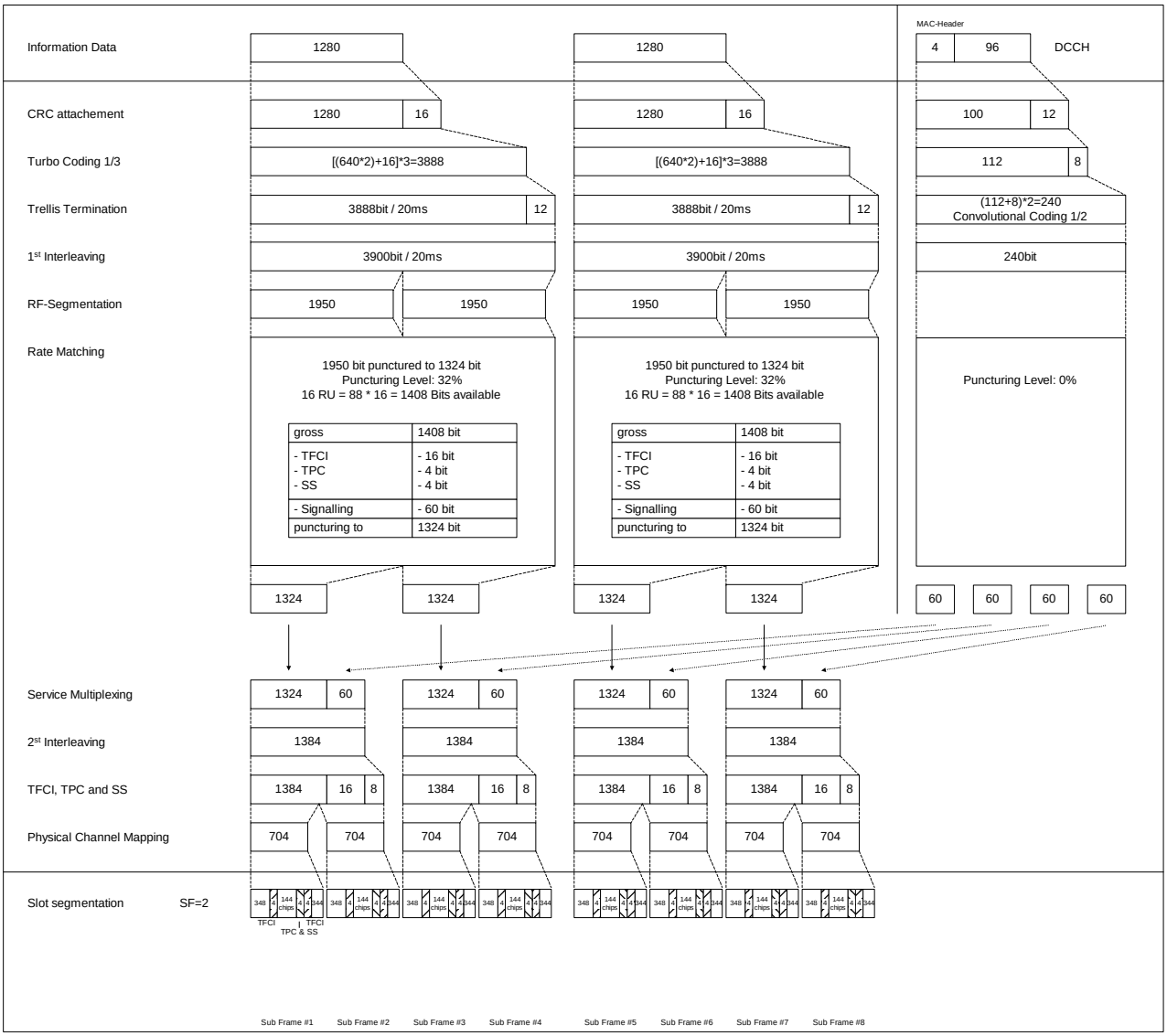
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |




## C.2.3.2 1.28 Mcps TDD Option

**Table C.2.3.2**

| Parameter | Value |
|---|---|
| Information data rate | 64 kbps |
| RU's allocated | 1TS (1*SF2) = 8RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 4 Bit/user/10ms |
| TFCI | 16 Bit/user/10ms |
| Synchronisation Shift (SS) | 4 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH / ½ DCH of the DCCH | 32% / 0 |




## C.2.3.3 7.68 Mcps TDD Option

**Table C.2.3.5: UL reference measurement channel physical parameters (64kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 64 kbps |
| RU´s allocated | 1 SF8 + 1 SF32 = 5RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.8% / 13.3% |

**Table C.2.3.6: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (64 kbps)**

| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 1280 | 88/80 |
| | Max data rate, bps | | 64000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 1280 | 100 |
| | TFS | TF0, bits | 0*1280 | 0*100 |
| | | TF1, bits | 1*1280 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 3900 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1950 | 60 |
| | RM attribute | | 158 | 240 |

**Table C.2.3.7: UL reference measurement channel, TFCS (64 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.3.8: UL reference measurement channel, puncturing limit (64 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |

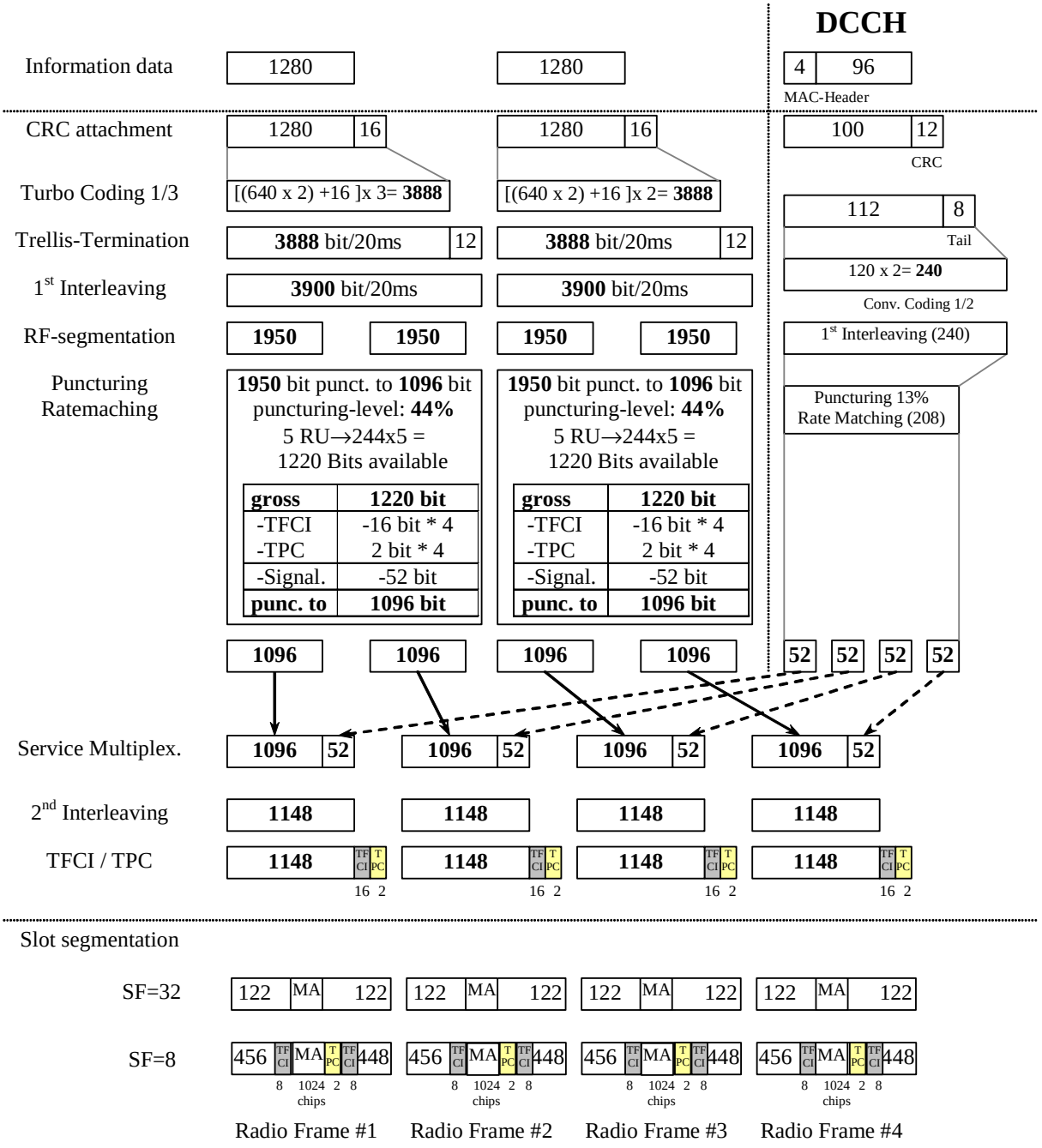

DCCH
Information data
1280
1280
4
96
MAC-Header
CRC attachment
1280
16
1280
16
100
12
CRC
Turbo Coding 1/3
[(640 x 2) +16 ]x 3= 3888
[(640 x 2) +16 ]x 2= 3888
112
8
Tail
Trellis-Termination
3888 bit/20ms
12
3888 bit/20ms
12
120 x 2= 240
Conv. Coding 1/2
1st Interleaving
3900 bit/20ms
3900 bit/20ms
RF-segmentation
1950
1950
1950
1950
1st Interleaving (240)
Puncturing
Ratemaching
1950 bit punct. to 1096 bit puncturing-level: 44%
5 RU→244x5 =
1220 Bits available
gross 1220 bit
-TFCI -16 bit * 4
-TPC 2 bit * 4
-Signal. -52 bit
punc. to 1096 bit
1950 bit punct. to 1096 bit puncturing-level: 44%
5 RU→244x5 =
1220 Bits available
gross 1220 bit
-TFCI -16 bit * 4
-TPC 2 bit * 4
-Signal. -52 bit
punc. to 1096 bit
Puncturing 13%
Rate Matching (208)
1096
1096
1096
1096
52
52
52
52
Service Multiplex.
1096
52
1096
52
1096
52
1096
52
2nd Interleaving
1148
1148
1148
1148
TFCI / TPC
1148
TFCI
TPC
16 2
1148
TFCI
TPC
16 2
1148
TFCI
TPC
16 2
1148
TFCI
TPC
16 2
Slot segmentation
SF=32
122
MA
122
122
MA
122
122
MA
122
122
MA
122
SF=8
456
TFCI
MA
TPC
TFCI
448
8 1024 chips 2 8
456
TFCI
MA
TPC
TFCI
448
8 1024 chips 2 8
456
TFCI
MA
TPC
TFCI
448
8 1024 chips 2 8
456
TFCI
MA
TPC
TFCI
448
8 1024 chips 2 8
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

**Figure A.2B**

# C.2.4 UL reference measurement channel (144 kbps)

## C.2.4.1 3.84 Mcps TDD Option

**Table C.2.4.1: UL reference measurement channel physical parameters (144kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 144 kbps |
| RU´s allocated | 1 SF2 + 1 SF16 = 9RU |
| Midamble | 256 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 47.3% / 20% |

**Table C.2.4.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (144 kbps)**

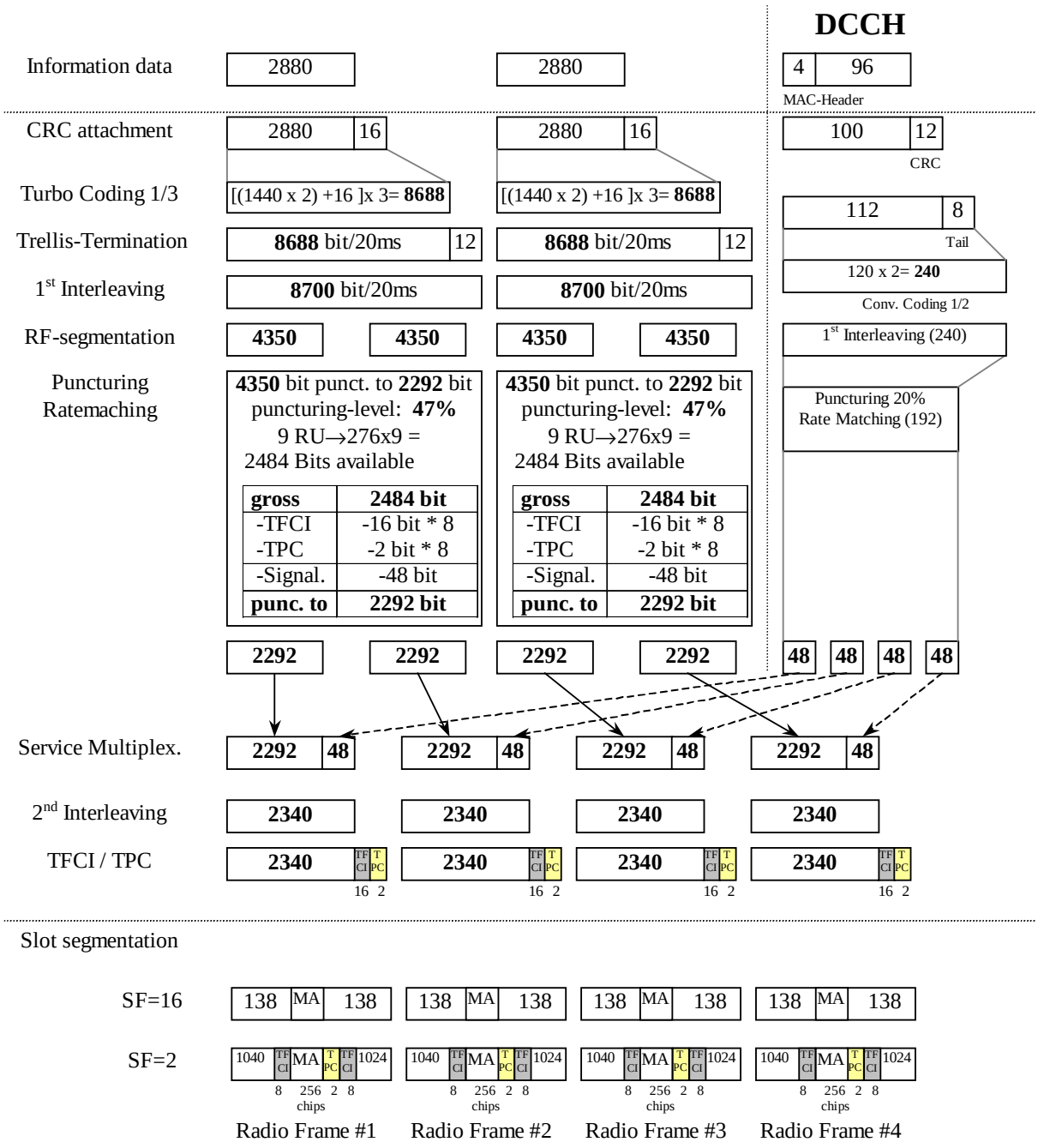
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 2880 | 88/80 |
| | Max data rate, bps | | 144000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*2880 | 0*100 |
| | | TF1, bits | 1*2880 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 8700 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 4350 | 60 |
| | RM attribute | | 160 | 240 |

**Table C.2.4.3: UL reference measurement channel, TFCS (144 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.4.4: UL reference measurement channel, puncturing limit (144 kbps)**

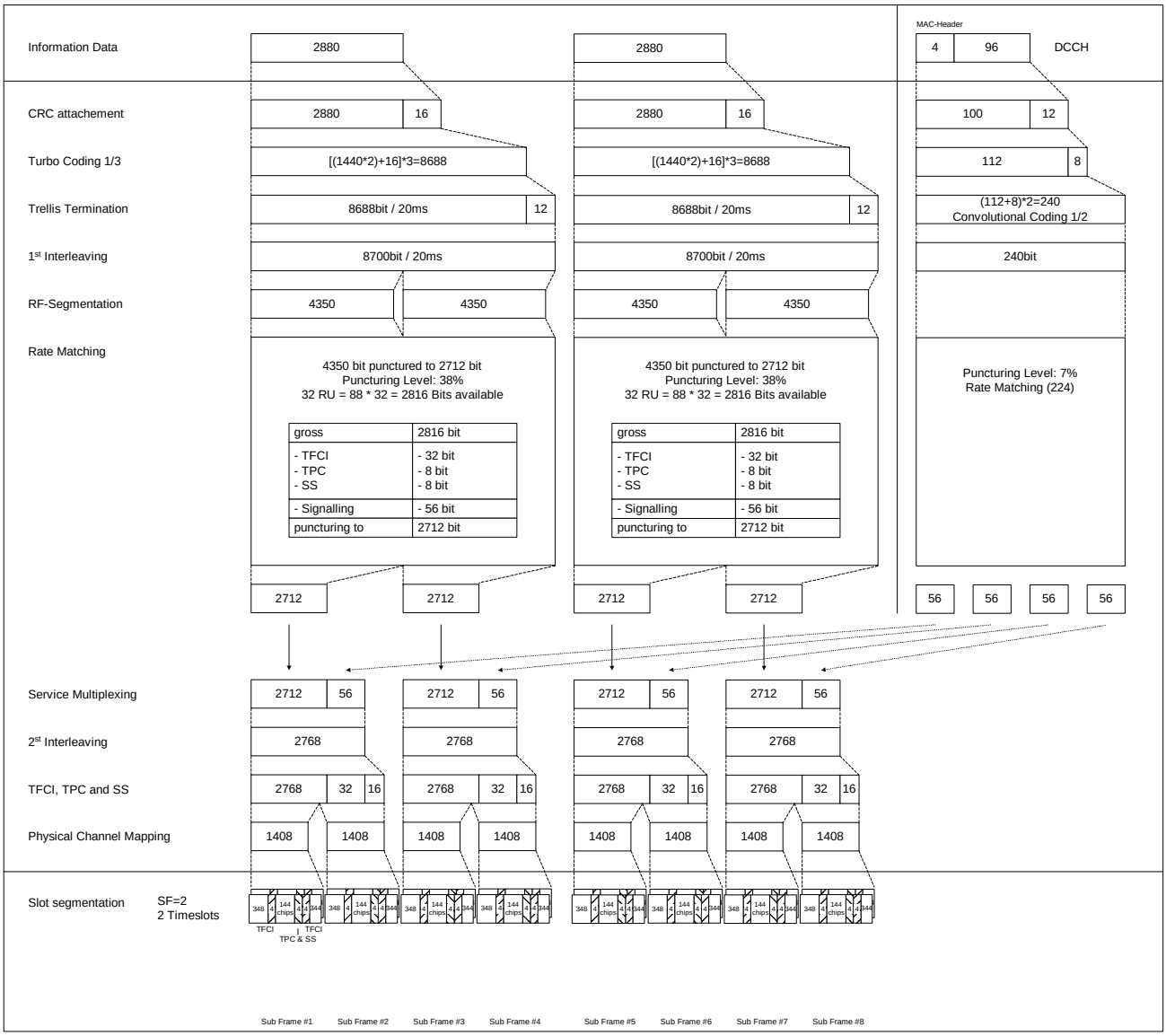
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |




## C.2.4.2 1.28 Mcps TDD Option

**Table C.2.4.2**

| Parameter | Value |
|---|---|
| Information data rate | 144 kbps |
| RU's allocated | 2TS (1*SF2) = 16RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 8 Bit/user/10ms |
| TFCI | 32 Bit/user/10ms |
| Synchronisation Shift (SS) | 8 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH / ½ DCH of the DCCH | 38% / 7% |




## C.2.4.3 7.68 Mcps TDD Option

**Table C.2.4.5: UL reference measurement channel physical parameters (144kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 144 kbps |
| RU´s allocated | 1 SF4 + 1 SF32 = 9RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 47.3% / 20% |

**Table C.2.4.6: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (144 kbps)**

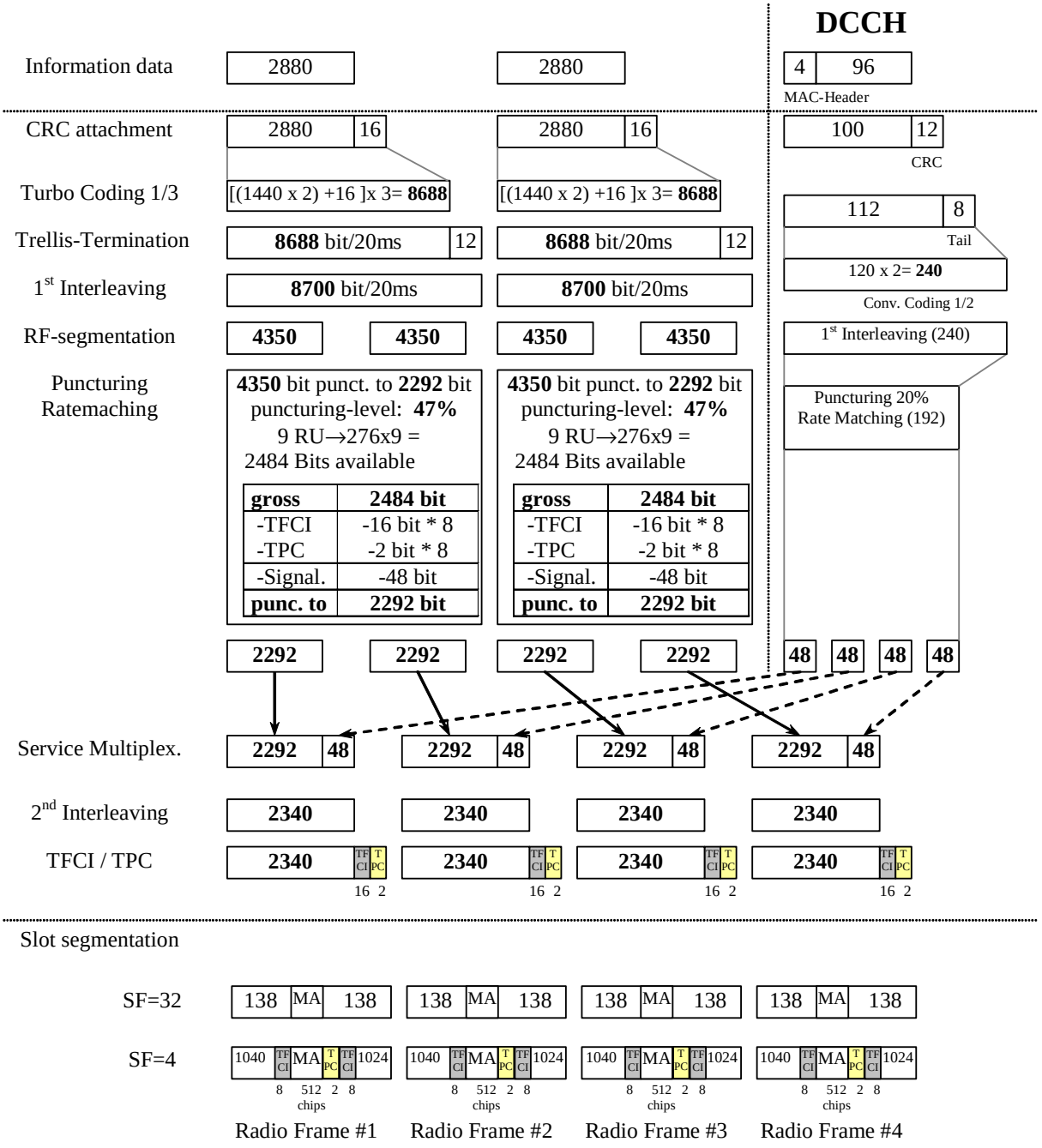
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 2880 | 88/80 |
| | Max data rate, bps | | 144000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*2880 | 0*100 |
| | | TF1, bits | 1*2880 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 8700 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 4350 | 60 |
| | RM attribute | | 160 | 240 |

**Table C.2.4.7: UL reference measurement channel, TFCS (144 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.4.8: UL reference measurement channel, puncturing limit (144 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |


DCCH
Information data
2880
2880
4
96
MAC-Header
CRC attachment
2880
16
2880
16
100
12
CRC
Turbo Coding 1/3
[(1440 x 2) +16 ]x 3= 8688
[(1440 x 2) +16 ]x 3= 8688
112
8
Tail
Trellis-Termination
8688 bit/20ms
12
8688 bit/20ms
12
120 x 2= 240
Conv. Coding 1/2
1st Interleaving
8700 bit/20ms
8700 bit/20ms
RF-segmentation
4350
4350
4350
4350
1st Interleaving (240)
Puncturing
Ratemaching
4350 bit punct. to 2292 bit
puncturing-level: 47%
9 RU→276x9 =
2484 Bits available
gross 2484 bit
-TFCI -16 bit * 8
-TPC -2 bit * 8
-Signal. -48 bit
punc. to 2292 bit
4350 bit punct. to 2292 bit
puncturing-level: 47%
9 RU→276x9 =
2484 Bits available
gross 2484 bit
-TFCI -16 bit * 8
-TPC -2 bit * 8
-Signal. -48 bit
punc. to 2292 bit
Puncturing 20%
Rate Matching (192)
2292
2292
2292
2292
48
48
48
48
Service Multiplex.
2292
48
2292
48
2292
48
2292
48
2nd Interleaving
2340
2340
2340
2340
TFCI / TPC
2340
TFCI
TPC
2340
TFCI
TPC
2340
TFCI
TPC
2340
TFCI
TPC
16 2
16 2
16 2
16 2
Slot segmentation
SF=32
138
MA
138
138
MA
138
138
MA
138
138
MA
138
SF=4
1040
TFCI
MA
TPC
TFCI
1024
8 512 2 8
chips
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

# C.2.5 UL reference measurement channel (384 kbps)

## C.2.5.1 3.84 Mcps TDD Option

**Table C.2.5.1: UL reference measurement channel physical parameters (384kbps)**

| Parameter | |
|---|---|
| Information data rate | 384 kbps |
| RU´s allocated | 8*3TS = 24RU |
| Midamble | 256 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.4% / 15.3% |

**Table C.2.5.2: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (384 kbps)**

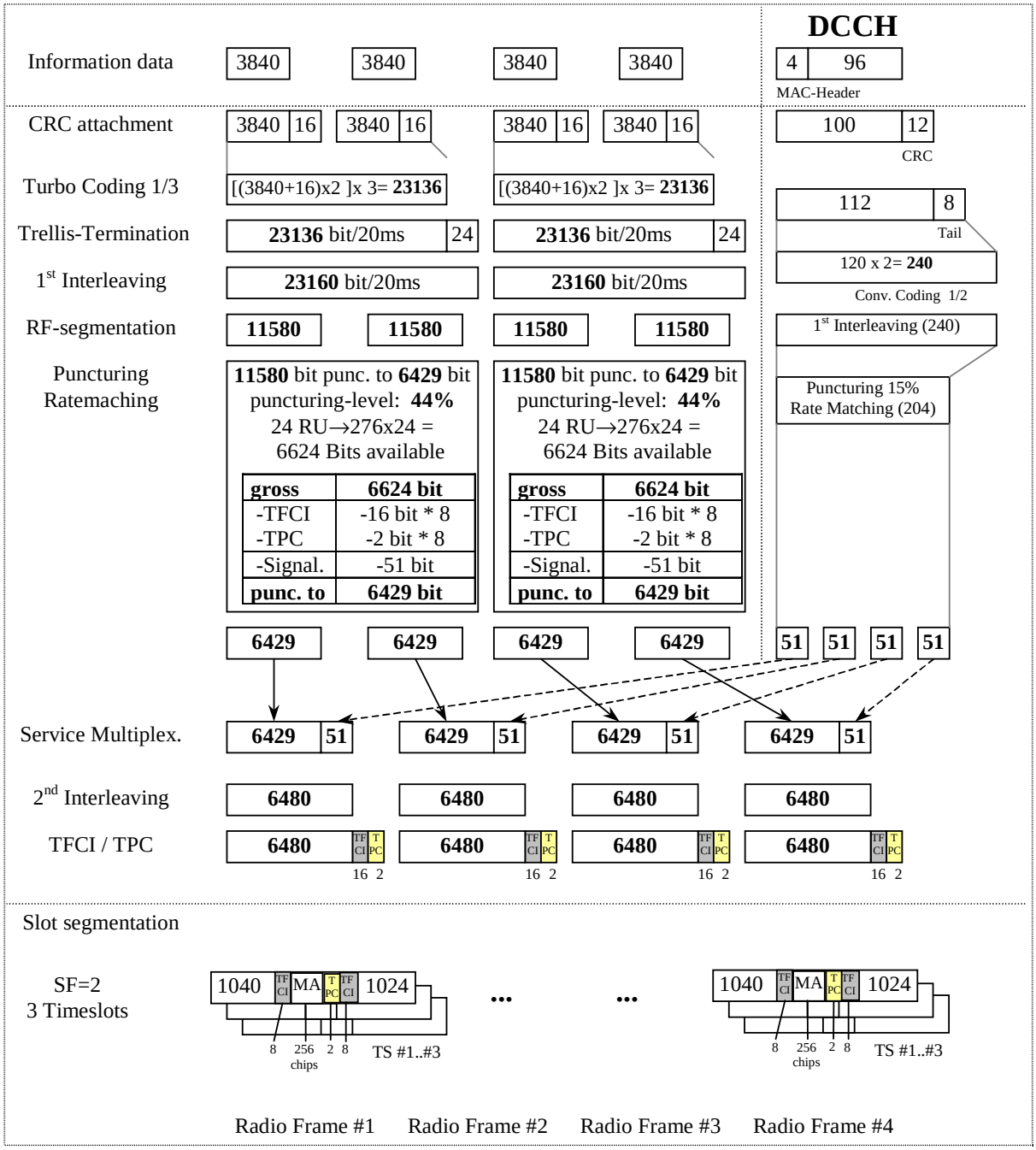
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 3840 | 88/80 |
| | Max data rate, bps | | 384000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*3840 | 0*100 |
| | | TF1, bits | 2*3840 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 23160 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 11580 | 60 |
| | RM attribute | | 158 | 240 |

**Table C.2.5.3: UL reference measurement channel, TFCS (384 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.5.4: UL reference measurement channel, puncturing limit (384 kbps)**

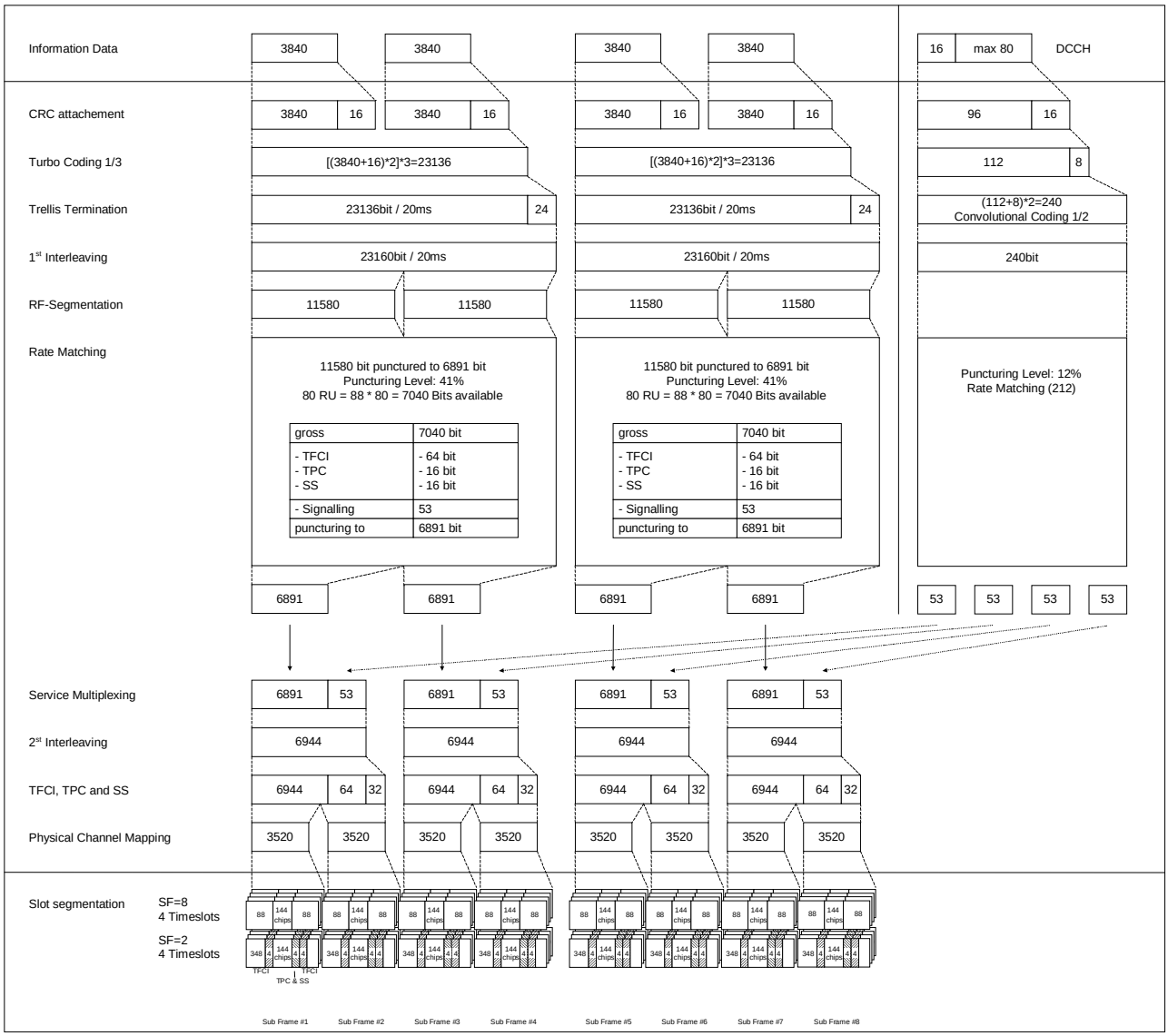
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |


DCCH
Information data
3840
3840
3840
3840
4
96
MAC-Header
CRC attachment
3840 16
3840 16
3840 16
3840 16
100
12
CRC
Turbo Coding 1/3
[(3840+16)x2 ]x 3= 23136
[(3840+16)x2 ]x 3= 23136
112
8
Tail
Trellis-Termination
23136 bit/20ms
24
23136 bit/20ms
24
120 x 2= 240
Conv. Coding 1/2
1st Interleaving
23160 bit/20ms
23160 bit/20ms
RF-segmentation
11580
11580
11580
11580
1st Interleaving (240)
Puncturing
Ratematching
11580 bit punc. to 6429 bit
puncturing-level: 44%
24 RU→276x24 =
6624 Bits available
11580 bit punc. to 6429 bit
puncturing-level: 44%
24 RU→276x24 =
6624 Bits available
Puncturing 15%
Rate Matching (204)
gross 6624 bit
-TFCI -16 bit * 8
-TPC -2 bit * 8
-Signal. -51 bit
punc. to 6429 bit
6429
6429
6429
6429
51
51
51
51
Service Multiplex.
6429 51
6429 51
6429 51
6429 51
2nd Interleaving
6480
6480
6480
6480
TFCI / TPC
6480 TFCI TPC
16 2
Slot segmentation
SF=2
3 Timeslots
1040 TFCI MA TPC TFCI 1024
8 256 chips 2 8
TS #1..#3
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

## C.2.5.2 1.28 Mcps TDD Option

**Table C.2.5.2**

| Parameter | Value |
|---|---|
| Information data rate | 384 kbps |
| RU's allocated | 4TS (1*SF2 + 1*SF8) = 40RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 16 Bit/user/10ms |
| TFCI | 64 Bit/user/10ms |
| Synchronisation Shift (SS) | 16 Bit/user/10ms |
| Inband signalling DCCH | max 2.0 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH / ½ DCH of the DCCH | 41% / 12% |



## C.2.5.3 7.68 Mcps TDD Option

**Table C.2.5.3: UL reference measurement channel physical parameters (384kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 384 kbps |
| RU´s allocated | 8*3TS = 24RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 2 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.4% / 15.3% |

**Table C.2.5.4: UL reference measurement channel using RLC-TM for DTCH, transport channel parameters (384 kbps)**

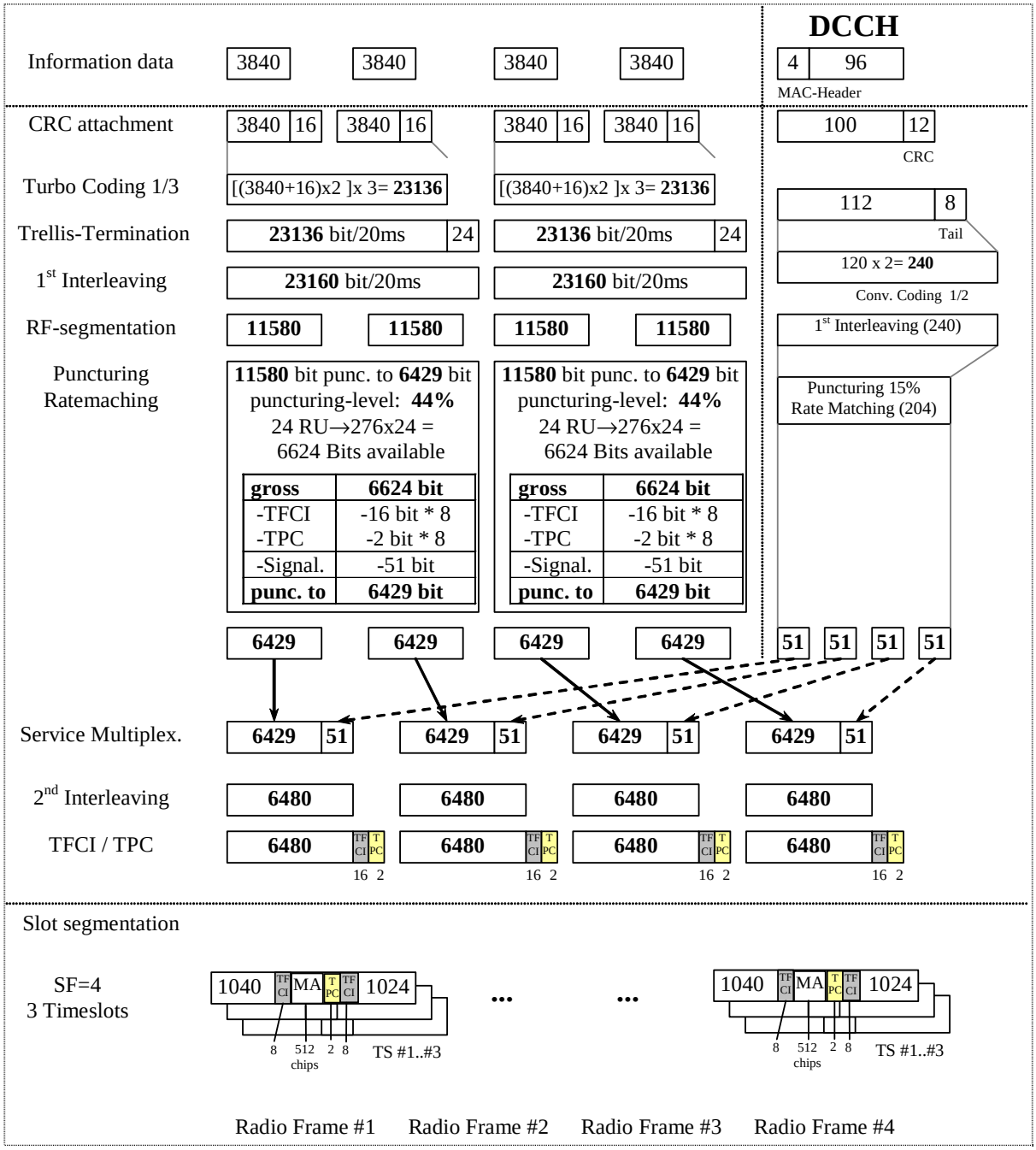
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 3840 | 88/80 |
| | Max data rate, bps | | 384000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*3840 | 0*100 |
| | | TF1, bits | 2*3840 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 23160 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 11580 | 60 |
| | RM attribute | | 158 | 240 |

**Table C.2.5.5: UL reference measurement channel, TFCS (384 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.2.5.6: UL reference measurement channel, puncturing limit (384 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |


DCCH
Information data
3840
3840
3840
3840
4
96
MAC-Header
CRC attachment
3840 16
3840 16
3840 16
3840 16
100
12
CRC
Turbo Coding 1/3
[(3840+16)x2 ]x 3= 23136
[(3840+16)x2 ]x 3= 23136
112
8
Tail
Trellis-Termination
23136 bit/20ms
24
23136 bit/20ms
24
120 x 2= 240
1st Interleaving
23160 bit/20ms
23160 bit/20ms
Conv. Coding 1/2
RF-segmentation
11580
11580
11580
11580
1st Interleaving (240)
Puncturing
Ratematching
11580 bit punc. to 6429 bit
puncturing-level: 44%
24 RU→276x24 =
6624 Bits available
gross 6624 bit
-TFCI -16 bit * 8
-TPC -2 bit * 8
-Signal. -51 bit
punc. to 6429 bit
11580 bit punc. to 6429 bit
puncturing-level: 44%
24 RU→276x24 =
6624 Bits available
gross 6624 bit
-TFCI -16 bit * 8
-TPC -2 bit * 8
-Signal. -51 bit
punc. to 6429 bit
Puncturing 15%
Rate Matching (204)
6429
6429
6429
6429
51
51
51
51
Service Multiplex.
6429 51
6429 51
6429 51
6429 51
2nd Interleaving
6480
6480
6480
6480
TFCI / TPC
6480 TFCI TPC
6480 TFCI TPC
6480 TFCI TPC
6480 TFCI TPC
16 2
16 2
16 2
16 2
Slot segmentation
SF=4
3 Timeslots
1040 TFCI MA TPC TFCI 1024
8 512 chips 2 8 TS #1..#3
...
...
1040 TFCI MA TPC TFCI 1024
8 512 chips 2 8 TS #1..#3
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

# C.3 DL Reference measurement channels

## C.3.1 DL reference measurement channel (12,2 kbps)

### C.3.1.1 3,84 Mcps TDD Option

**Table C.3.1.1: DL reference measurement channel physical parameters (12.2kbps)**

| Parameter | |
|---|---|
| Information data rate | 12,2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 5% / 0 % |

**Table C.3.1.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps)**

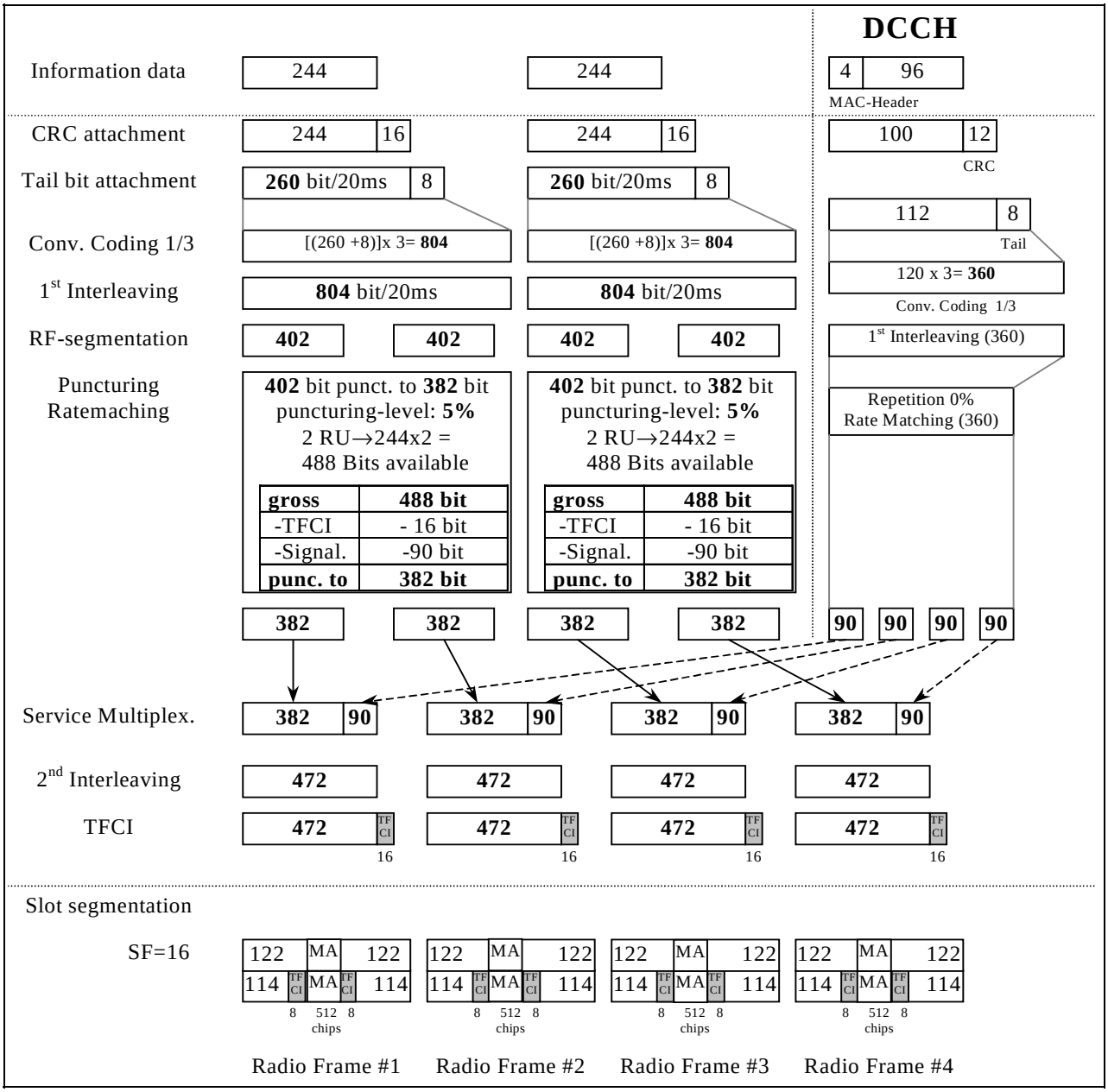
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Downlink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 228 | 240 |

**Table C.3.1.3: DL reference measurement channel, TFCS (12.2 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.1.4: DL reference measurement channel, puncturing limit (12.2 kbps)**

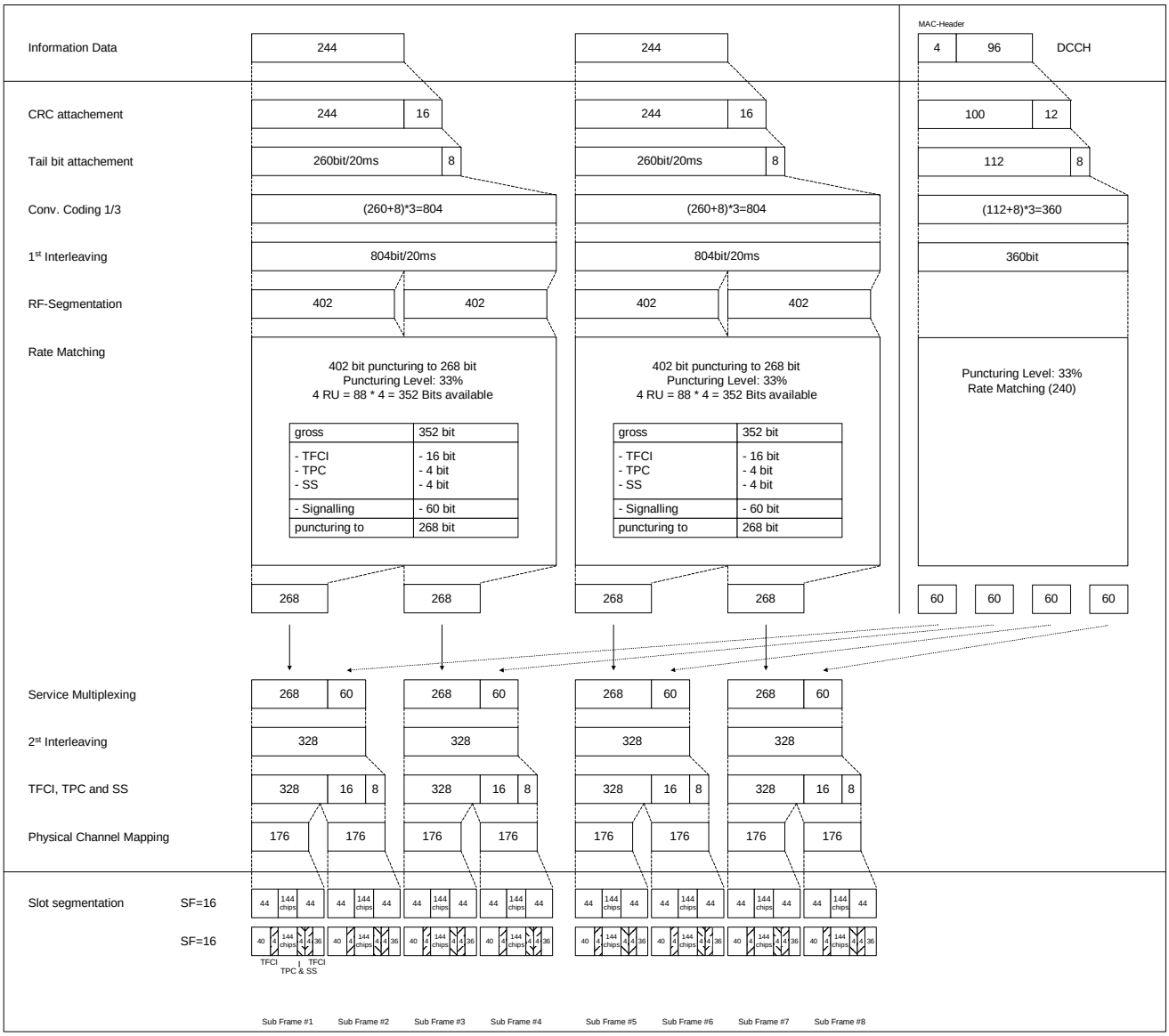
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.92 |




## C.3.1.2 DL reference measurement channel (12,2 kbps) for 1,28 Mcps TDD Option

**Table C.3.1.2**

| Parameter | Value |
|---|---|
| Information data rate | 12,2 kbps |
| RU's allocated | 1TS (2*SF16) = 2RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 4 Bit/user/10ms |
| TFCI | 16 Bit/user/10ms |
| Synchronisation Shift (SS) | 4 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 33% / 33% |




## C.3.1.3 7,68 Mcps TDD Option

**Table C.3.1.5: DL reference measurement channel physical parameters (12.2kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 12.2 kbps |
| RU´s allocated | 2 RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 5% / 0 % |

**Table C.3.1.6: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (12.2 kbps)**

| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 244 | 88/80 |
| | Max data rate, bps | | 12200 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 244 | 100 |
| | TFS | TF0, bits | 0*244 | 0*100 |
| | | TF1, bits | 1*244 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Convolution Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 804 | 360 |
| | Downlink: Max number of bits/radio frame before rate matching | | 402 | 90 |
| | RM attribute | | 228 | 240 |

**Table C.3.1.7: DL reference measurement channel, TFCS (12.2 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.1.8: DL reference measurement channel, puncturing limit (12.2 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.92 |

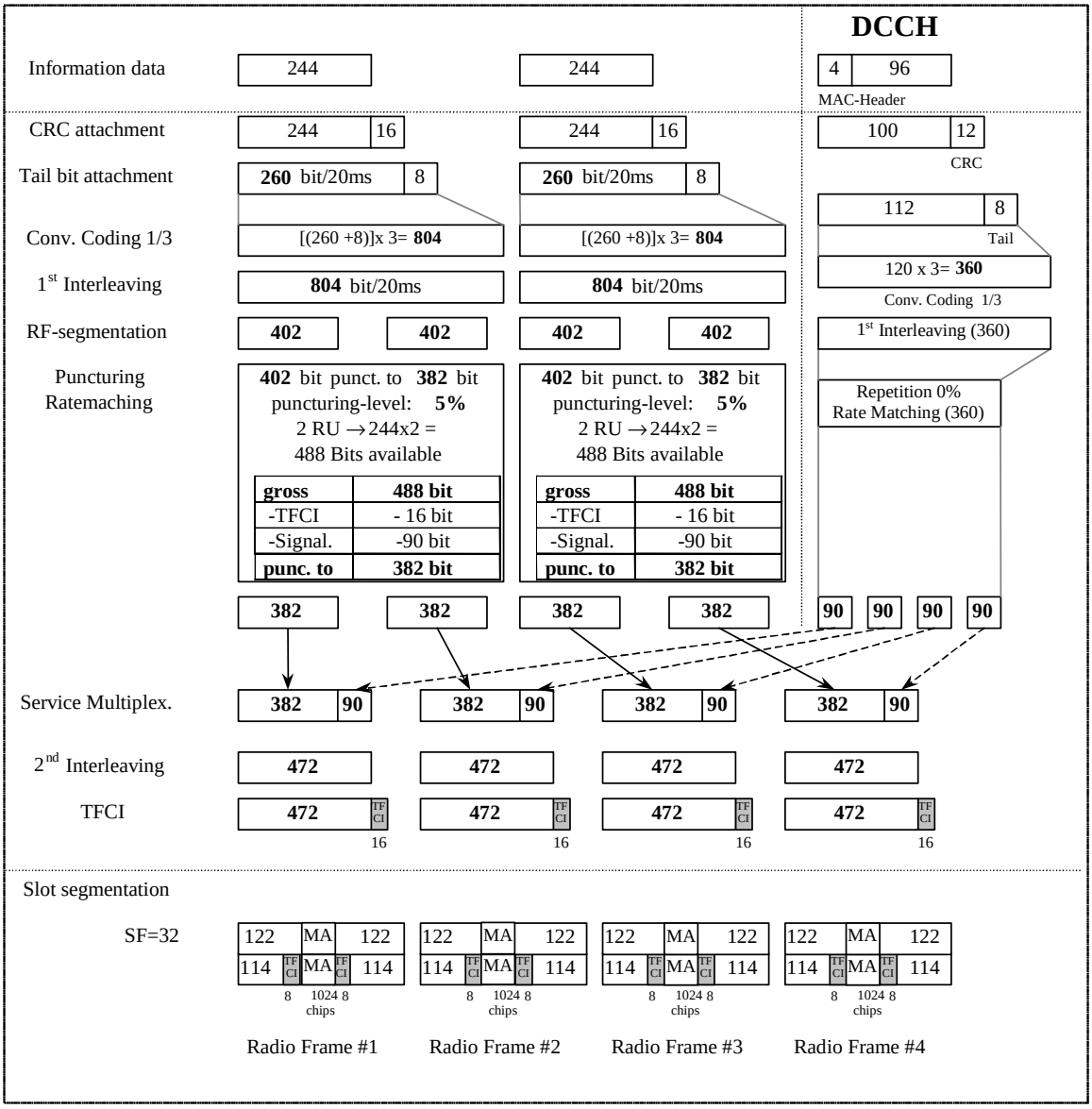



**Figure A.2B**

# C.3.2 DL reference measurement channel (64 kbps)

## C.3.2.1 3,84 Mcps TDD Option

**Table C.3.2.1: DL reference measurement channel physical parameters (64kbps)**

| Parameter | |
|---|---|
| Information data rate | 64 kbps |
| RU´s allocated | 5 codes SF16 = 5RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 41.1% / 10% |

**Table C.3.2.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (64 kbps)**

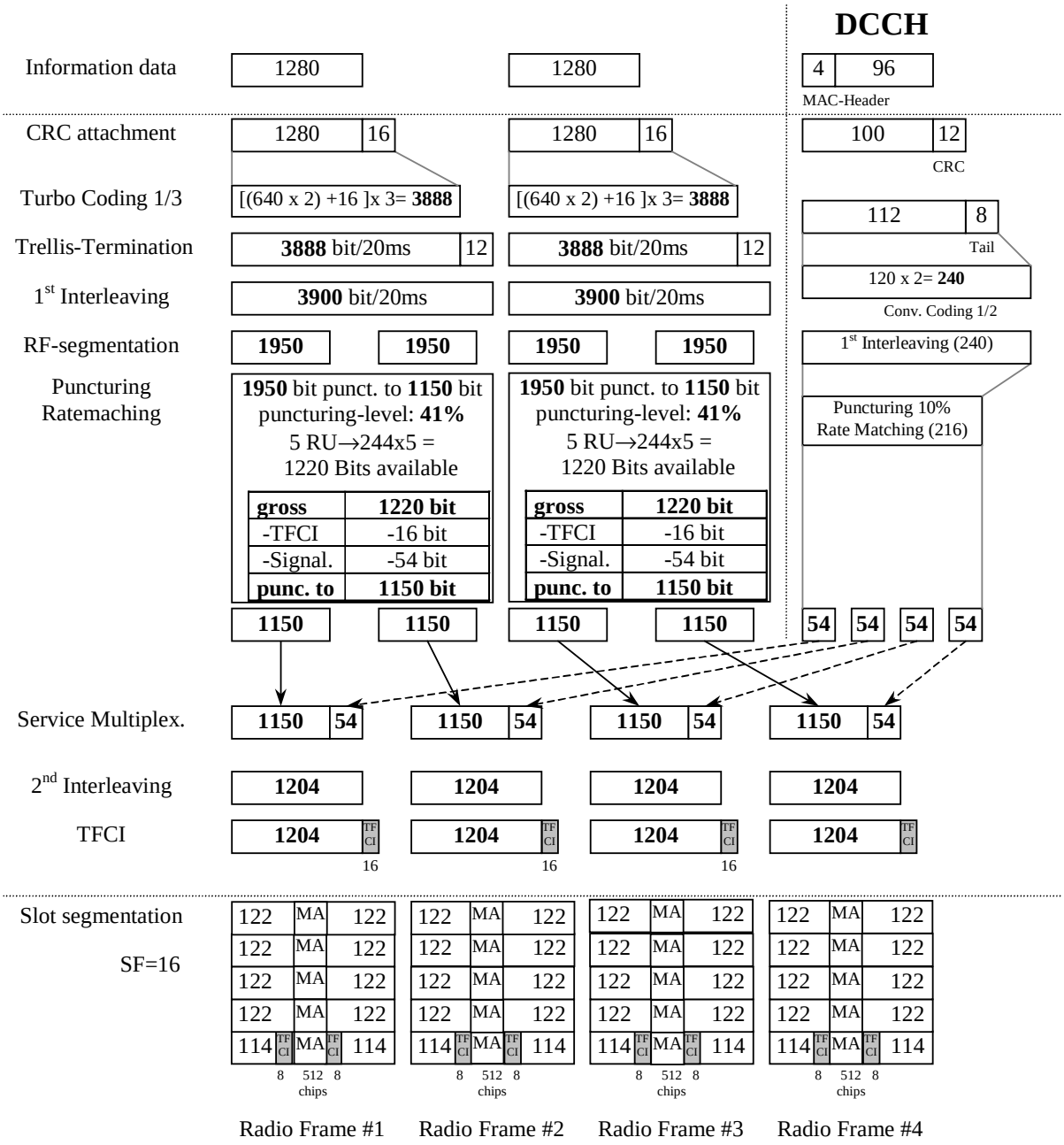
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 1280 | 88/80 |
| | Max data rate, bps | | 64000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 1280 | 100 |
| | TFS | TF0, bits | 0*1280 | 0*100 |
| | | TF1, bits | 1*1280 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 3900 | 240 |
| | Downlink: Max number of bits/radio frame before rate matching | | 1950 | 60 |
| | RM attribute | | 160 | 240 |

**Table C.3.2.3: DL reference measurement channel, TFCS (64 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.2.4: DL reference measurement channel, puncturing limit (64 kbps)**

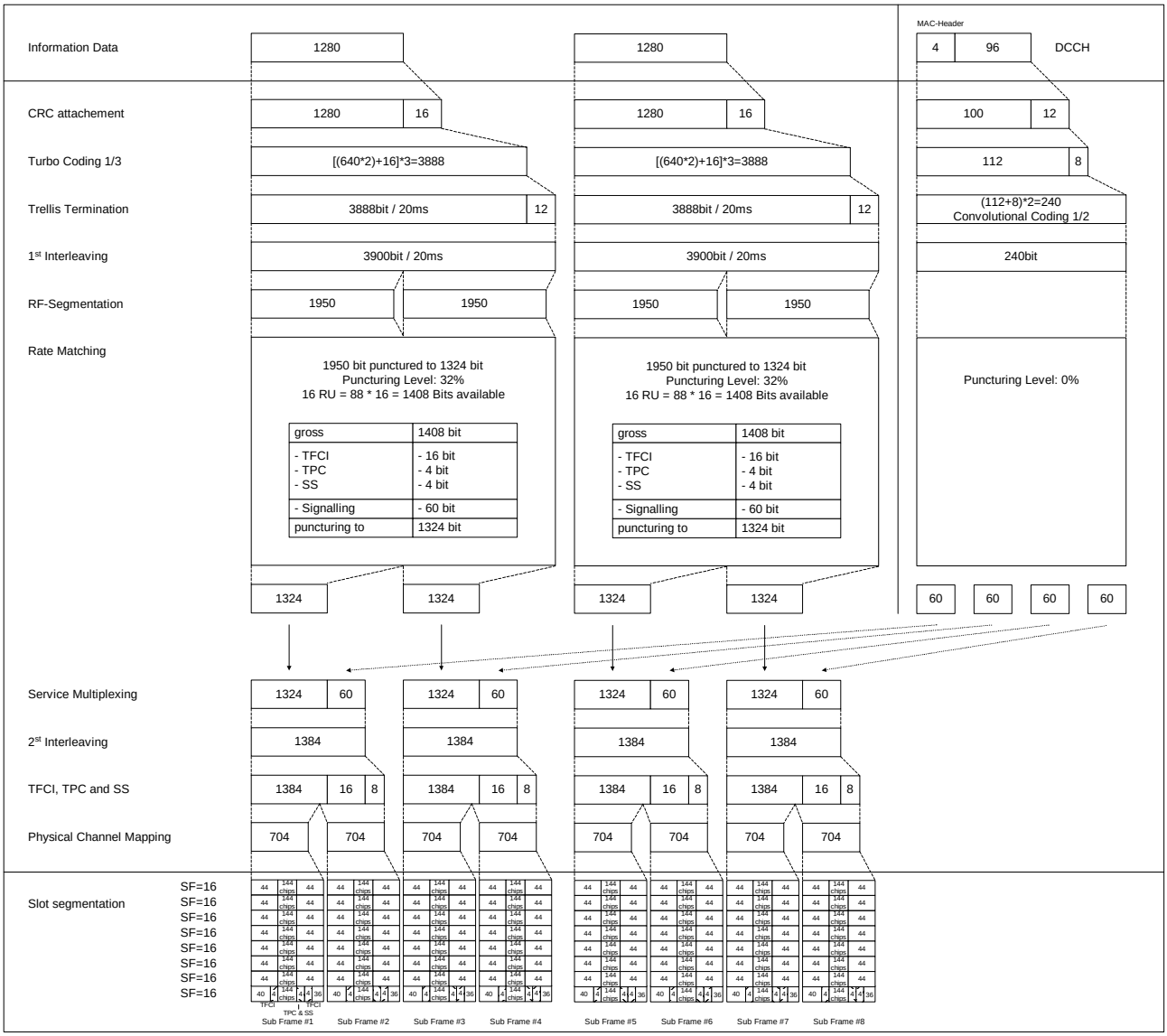
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |




## C.3.2.2 DL reference measurement channel (64 kbps) for 1,28 Mcps TDD Option

**Table C.3.2.2**

| Parameter | Value |
|---|---|
| Information data rate | 64 kbps |
| RU's allocated | 1TS (8*SF16) = 8RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 4 Bit/user/10ms |
| TFCI | 16 Bit/user/10ms |
| Synchronisation Shift (SS) | 4 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH/ ½ DCH of the DCCH | 32% / 0 |




## C.3.2.3 7,68 Mcps TDD Option

**Table C.3.2.5: DL reference measurement channel physical parameters (64kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 64 kbps |
| RU´s allocated | 5 codes SF32 = 5RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 41.1% / 10% |

**Table C.3.2.6: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (64 kbps)**

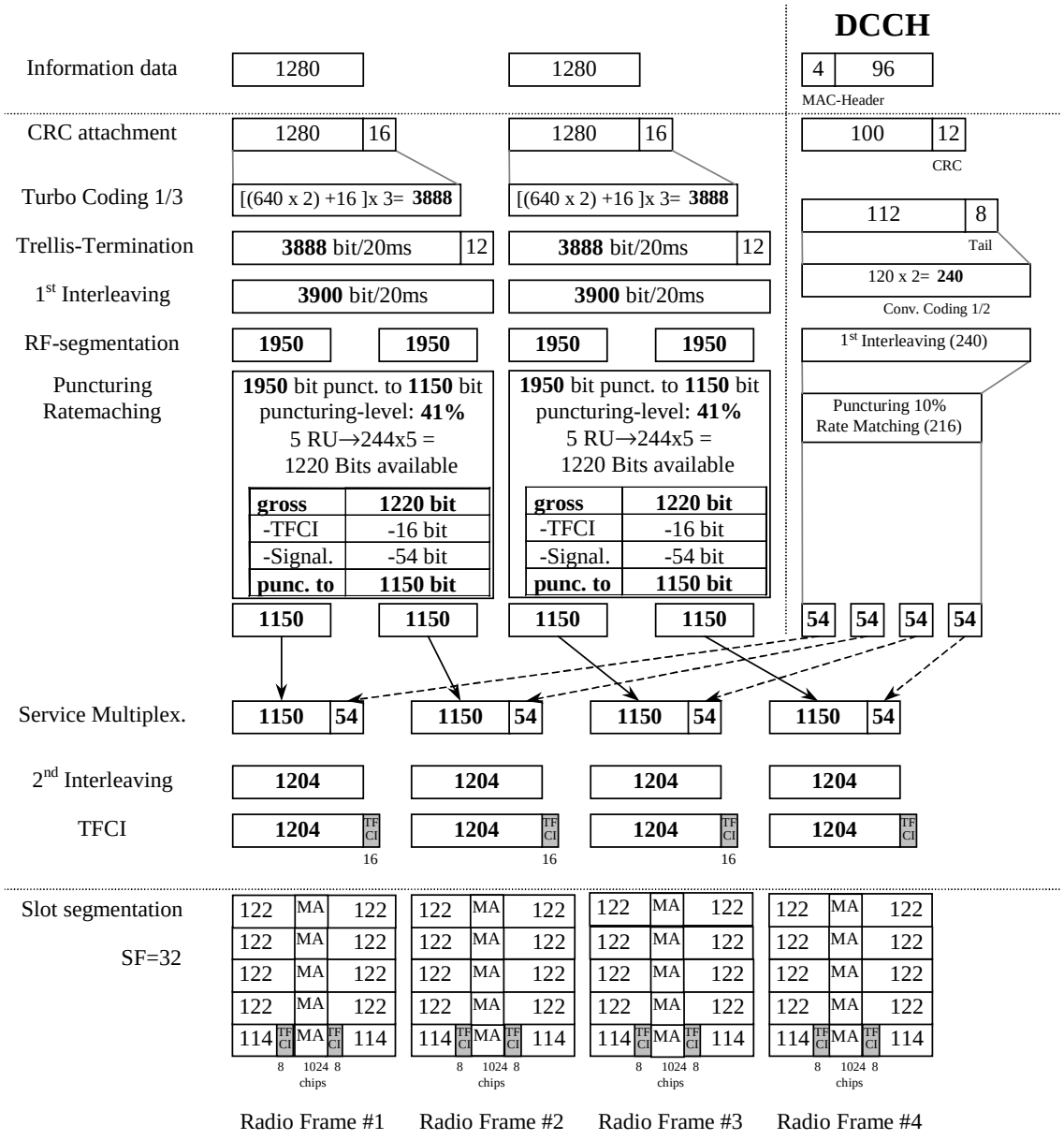
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 1280 | 88/80 |
| | Max data rate, bps | | 64000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 1280 | 100 |
| | TFS | TF0, bits | 0*1280 | 0*100 |
| | | TF1, bits | 1*1280 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 3900 | 240 |
| | Downlink: Max number of bits/radio frame before rate matching | | 1950 | 60 |
| | RM attribute | | 160 | 240 |

**Table C.3.2.7: DL reference measurement channel, TFCS (64 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.2.8: DL reference measurement channel, puncturing limit (64 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |




# C.3.3 DL reference measurement channel (144 kbps)

## C.3.3.1 3,84 Mcps TDD Option

**Table C.3.3.1: DL reference measurement channel physical parameters (144kbps)**

| Parameter | |
|---|---|
| Information data rate | 144 kbps |
| RU´s allocated | 9 codes SF16 = 9RU |
| Midamble | 256 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH / ½ DCH of the DCCH | 44.5% / 16.6% |

**Table C.3.3.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (144 kbps)**

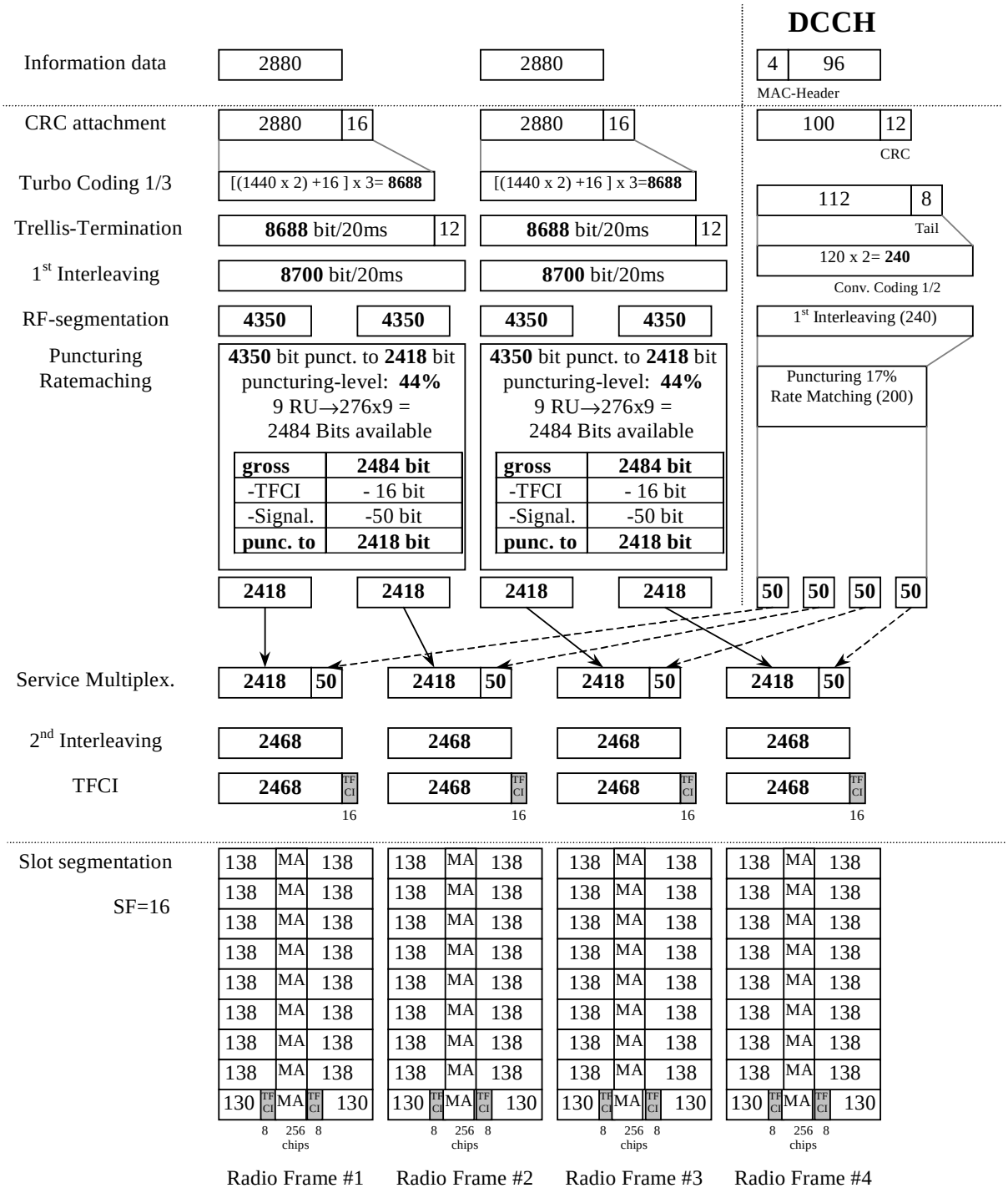
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 2880 | 88/80 |
| | Max data rate, bps | | 144000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*2880 | 0*100 |
| | | TF1, bits | 1*2880 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 8700 | 240 |
| | Downlink: Max number of bits/radio frame before rate matching | | 4350 | 60 |
| | RM attribute | | 162 | 240 |

**Table C.3.3.3: DL reference measurement channel, TFCS (144 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.3.4: DL reference measurement channel, puncturing limit (144 kbps)**

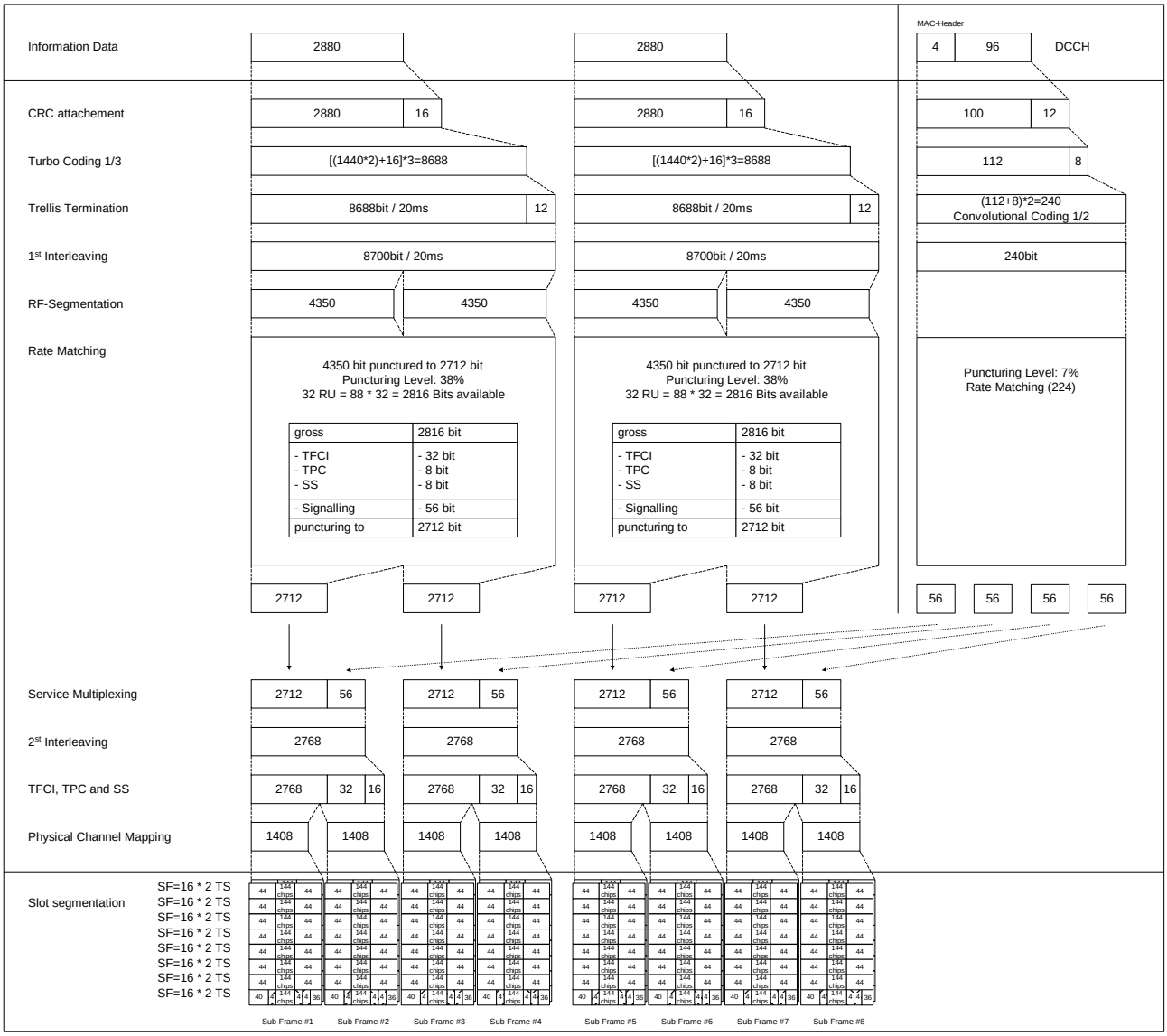
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |


DCCH
Information data
2880
2880
4
96
MAC-Header
CRC attachment
2880
16
2880
16
100
12
CRC
Turbo Coding 1/3
[(1440 x 2) +16 ] x 3= 8688
[(1440 x 2) +16 ] x 3=8688
112
8
Tail
Trellis-Termination
8688 bit/20ms
12
8688 bit/20ms
12
120 x 2= 240
1st Interleaving
8700 bit/20ms
8700 bit/20ms
Conv. Coding 1/2
RF-segmentation
4350
4350
4350
4350
1st Interleaving (240)
Puncturing
Ratemaching
4350 bit punct. to 2418 bit
puncturing-level: 44%
9 RU→276x9 =
2484 Bits available
gross 2484 bit
-TFCI - 16 bit
-Signal. -50 bit
punc. to 2418 bit
4350 bit punct. to 2418 bit
puncturing-level: 44%
9 RU→276x9 =
2484 Bits available
gross 2484 bit
-TFCI - 16 bit
-Signal. -50 bit
punc. to 2418 bit
Puncturing 17%
Rate Matching (200)
2418
2418
2418
2418
50
50
50
50
Service Multiplex.
2418
50
2418
50
2418
50
2418
50
2nd Interleaving
2468
2468
2468
2468
TFCI
2468
TFCI
16
2468
TFCI
16
2468
TFCI
16
2468
TFCI
16
Slot segmentation
SF=16
138
MA
138
130
TFCI
MA
TFCI
130
8
256 chips
8
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

## C.3.3.2 DL reference measurement channel (144 kbps) for 1,28 Mcps TDD Option

**Table C.3.3.2**

| Parameter | Value |
|---|---|
| Information data rate | 144 kbps |
| RU's allocated | 2TS (8*SF16) = 16RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 8 Bit/user/10ms |
| TFCI | 32 Bit/user/10ms |
| Synchronisation Shift (SS) | 8 Bit/user/10ms |
| Inband signalling DCCH | 2.4 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH/ ½ DCH of the DCCH | 38% / 7% |



## C.3.3.3 7,68 Mcps TDD Option

**Table C.3.3.5: DL reference measurement channel physical parameters (144kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 144 kbps |
| RU´s allocated | 9 codes SF32 = 9RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH / ½ DCH of the DCCH | 44.5% / 16.6% |

**Table C.3.3.6: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (144 kbps)**

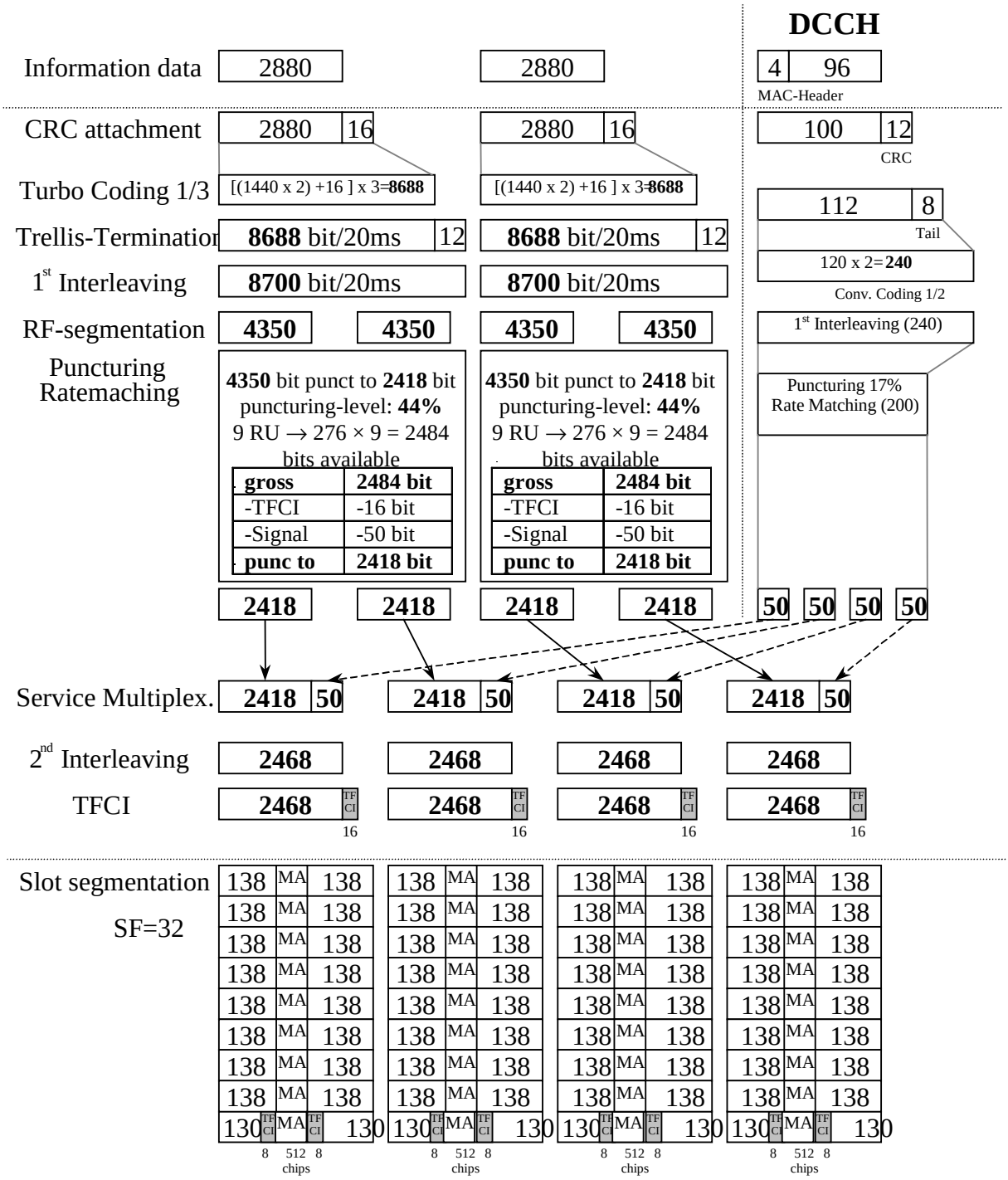
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 2880 | 88/80 |
| | Max data rate, bps | | 144000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 6 | 10 |
| | TB sizes, bit | | 2880 | 100 |
| | TFS | TF0, bits | 0*2880 | 0*100 |
| | | TF1, bits | 1*2880 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | 1/3 | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 8700 | 240 |
| | Downlink: Max number of bits/radio frame before rate matching | | 4350 | 60 |
| | RM attribute | | 162 | 240 |

**Table C.3.3.7: DL reference measurement channel, TFCS (144 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.3.8: DL reference measurement channel, puncturing limit (144 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.52 |


DCCH
Information data
2880
2880
4
96
MAC-Header
CRC attachment
2880
16
2880
16
100
12
CRC
Turbo Coding 1/3
[(1440 x 2) +16 ] x 3=8688
[(1440 x 2) +16 ] x 3=8688
112
8
Tail
Trellis-Termination
8688 bit/20ms
12
8688 bit/20ms
12
120 x 2=240
1st Interleaving
8700 bit/20ms
8700 bit/20ms
Conv. Coding 1/2
RF-segmentation
4350
4350
4350
4350
1st Interleaving (240)
Puncturing Ratematching
4350 bit punct to 2418 bit puncturing-level: 44%
9 RU → 276 × 9 = 2484 bits available
gross 2484 bit
-TFCI -16 bit
-Signal -50 bit
punc to 2418 bit
4350 bit punct to 2418 bit puncturing-level: 44%
9 RU → 276 × 9 = 2484 bits available
gross 2484 bit
-TFCI -16 bit
-Signal -50 bit
punc to 2418 bit
Puncturing 17%
Rate Matching (200)
2418
2418
2418
2418
50
50
50
50
Service Multiplex.
2418
50
2418
50
2418
50
2418
50
2nd Interleaving
2468
2468
2468
2468
TFCI
2468
TFCI
16
2468
TFCI
16
2468
TFCI
16
2468
TFCI
16
Slot segmentation
SF=32
138
MA
138
130
TFCI
MA
TFCI
130
8
512 chips
8
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

# C.3.4 DL reference measurement channel (384 kbps)

## C.3.4.1 3,84 Mcps TDD Option

**Table C.3.4.1: DL reference measurement channel physical parameters (384kbps)**

| Parameter | |
|---|---|
| Information data rate | 384 kbps |
| RU´s allocated | 8*3TS = 24RU |
| Midamble | 256 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.4% / 15.3% |

**Table C.3.4.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (384 kbps)**

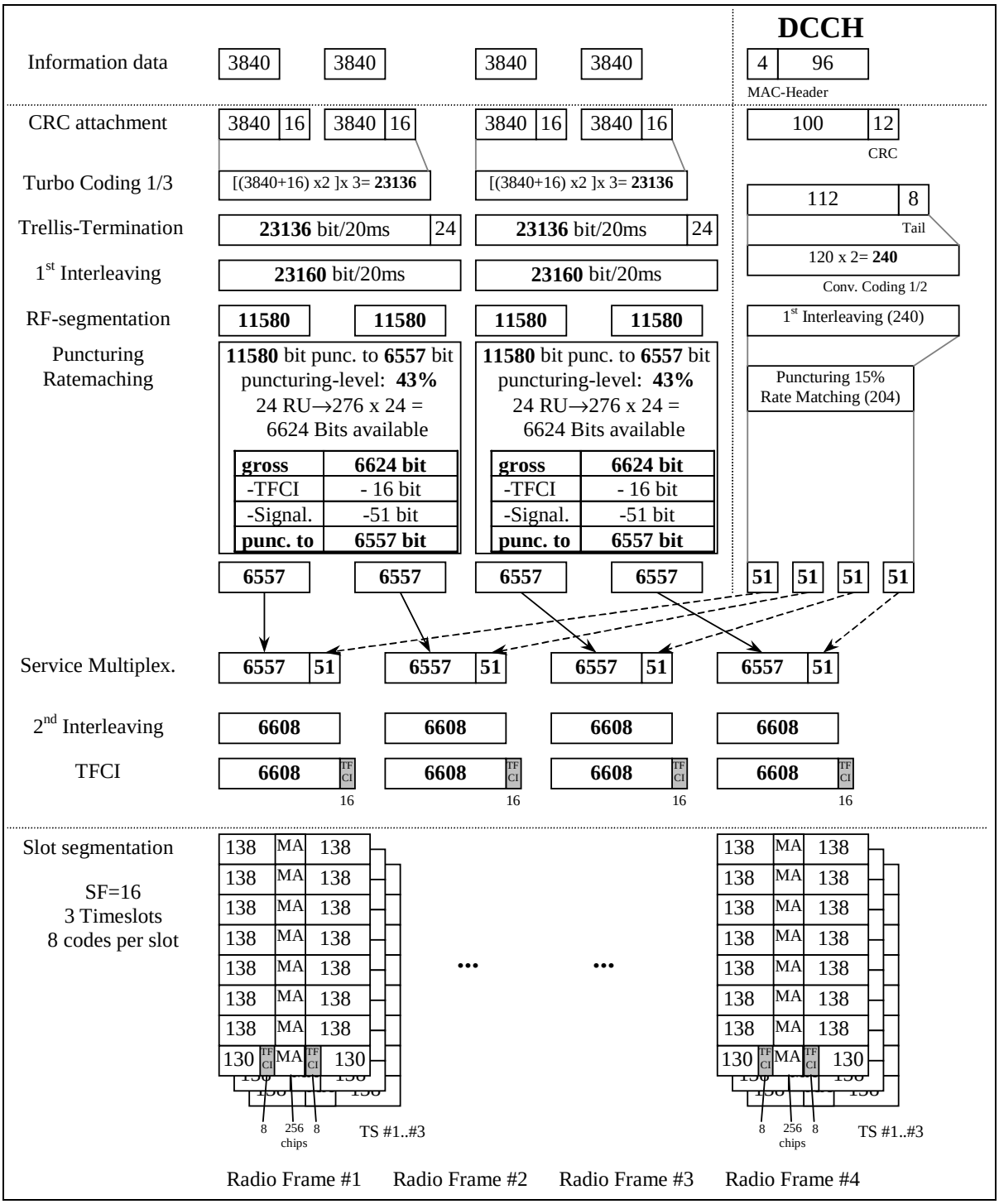
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 3840 | 88/80 |
| | Max data rate, bps | | 384000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 3840 | 100 |
| | TFS | TF0, bits | 0*3840 | 0*100 |
| | | TF1, bits | 2*3840 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 23160 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 11580 | 60 |
| | RM attribute | | 162 | 240 |

**Table C.3.4.3: DL reference measurement channel, TFCS (384 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.4.4: DL reference measurement channel, puncturing limit (384 kbps)**

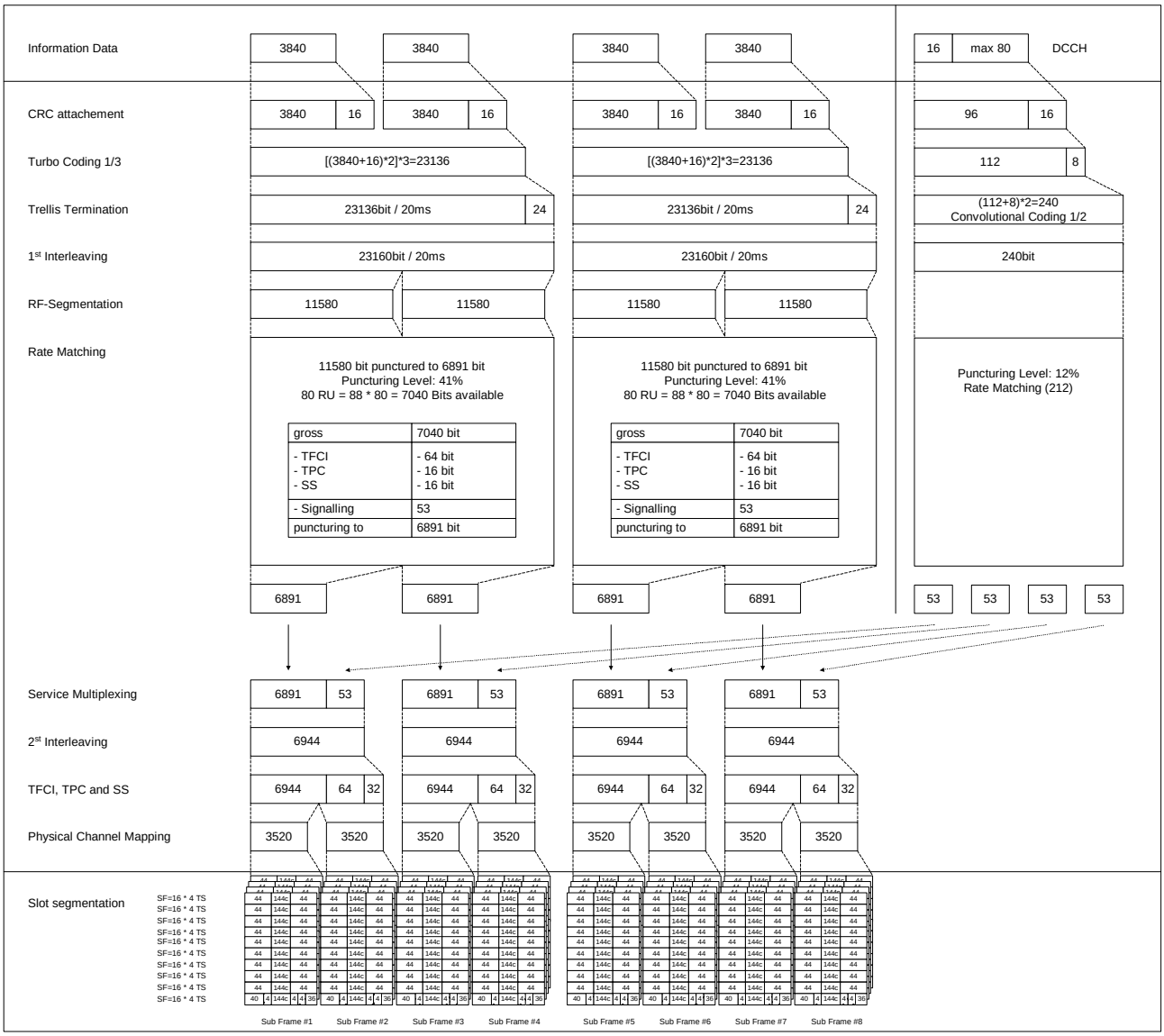
| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |


DCCH
Information data
3840
3840
3840
3840
4
96
MAC-Header
CRC attachment
3840 16
3840 16
3840 16
3840 16
100
12
CRC
Turbo Coding 1/3
[(3840+16) x2 ]x 3= 23136
[(3840+16) x2 ]x 3= 23136
112
8
Tail
Trellis-Termination
23136 bit/20ms
24
23136 bit/20ms
24
120 x 2= 240
Conv. Coding 1/2
1st Interleaving
23160 bit/20ms
23160 bit/20ms
RF-segmentation
11580
11580
11580
11580
1st Interleaving (240)
Puncturing
Ratemaching
11580 bit punc. to 6557 bit
puncturing-level: 43%
24 RU→276 x 24 =
6624 Bits available
11580 bit punc. to 6557 bit
puncturing-level: 43%
24 RU→276 x 24 =
6624 Bits available
Puncturing 15%
Rate Matching (204)
gross 6624 bit
-TFCI - 16 bit
-Signal. -51 bit
punc. to 6557 bit
6557
6557
6557
6557
51
51
51
51
Service Multiplex.
6557 51
6557 51
6557 51
6557 51
2nd Interleaving
6608
6608
6608
6608
TFCI
6608 TFCI
16
Slot segmentation
SF=16
3 Timeslots
8 codes per slot
138 MA 138
130 TFCI MA TFCI 130
8
256 chips
8
TS #1..#3
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

## C.3.4.2 DL reference measurement channel (384 kbps) for 1,28 Mcps TDD Option

**Table C.3.4.2**

| Parameter | Value |
|---|---|
| Information data rate | 384 kbps |
| RU's allocated | 4TS (10*SF16) = 40RU/5ms |
| Midamble | 144 |
| Interleaving | 20 ms |
| Power control (TPC) | 16 Bit/user/10ms |
| TFCI | 64 Bit/user/10ms |
| Synchronisation Shift (SS) | 16 Bit/user/10ms |
| Inband signalling DCCH | max.2 kbps |
| Puncturing level at Code rate: 1/3 DCH of the DTCH/ ½ DCH of the DCCH | 41% / 12% |



## C.3.4.3 7,68 Mcps TDD Option

**Table C.3.4.5: DL reference measurement channel physical parameters (384kbps)**

| Parameter | Value |
|---|---|
| Information data rate | 384 kbps |
| RU´s allocated | 8*3TS = 24RU |
| Midamble | 512 chips |
| Interleaving | 20 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate : 1/3 DCH of the DTCH / ½ DCH of the DCCH | 43.4% / 15.3% |

**Table C.3.4.6: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (384 kbps)**

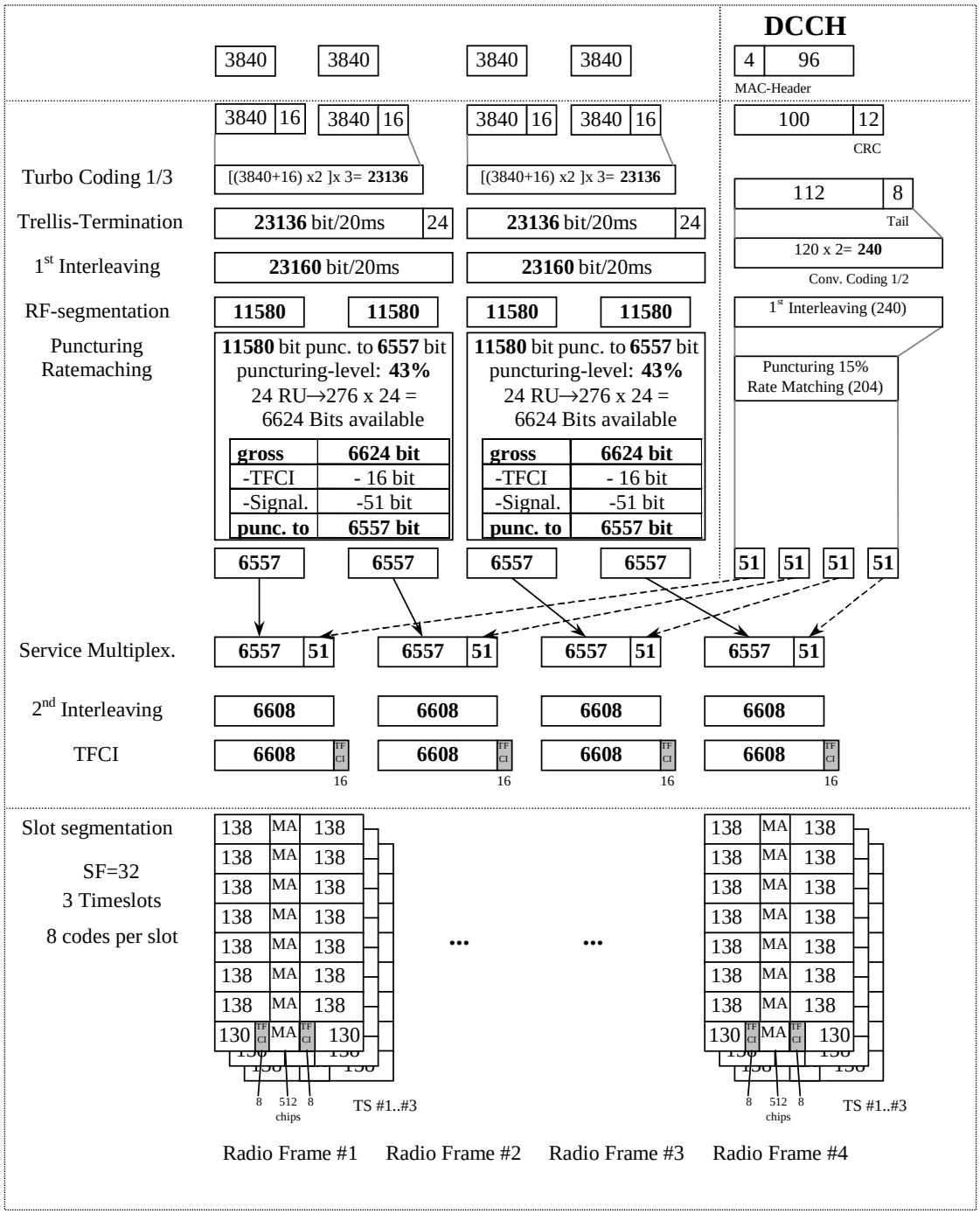
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 3840 | 88/80 |
| | Max data rate, bps | | 384000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 3840 | 100 |
| | TFS | TF0, bits | 0*3840 | 0*100 |
| | | TF1, bits | 2*3840 | 1*100 |
| | TTI, ms | | 20 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/2 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 23160 | 240 |
| | Uplink: Max number of bits/radio frame before rate matching | | 11580 | 60 |
| | RM attribute | | 162 | 240 |

**Table C.3.4.7: DL reference measurement channel, TFCS (384 kbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.4.8: DL reference measurement channel, puncturing limit (384 kbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.56 |

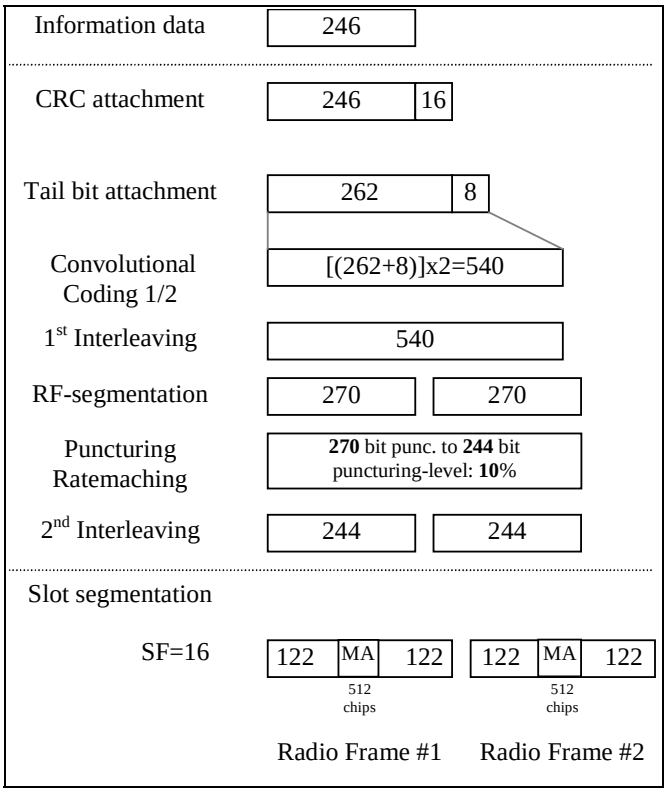

DCCH
3840
3840
3840
3840
4
96
MAC-Header
3840 16
3840 16
3840 16
3840 16
100
12
CRC
Turbo Coding 1/3
[(3840+16) x2 ]x 3= 23136
[(3840+16) x2 ]x 3= 23136
112
8
Tail
Trellis-Termination
23136 bit/20ms
24
23136 bit/20ms
24
120 x 2= 240
Conv. Coding 1/2
1st Interleaving
23160 bit/20ms
23160 bit/20ms
RF-segmentation
11580
11580
11580
11580
1st Interleaving (240)
Puncturing
Ratematching
11580 bit punc. to 6557 bit puncturing-level: 43% 24 RU→276 x 24 = 6624 Bits available
11580 bit punc. to 6557 bit puncturing-level: 43% 24 RU→276 x 24 = 6624 Bits available
Puncturing 15% Rate Matching (204)
gross 6624 bit
-TFCI - 16 bit
-Signal. -51 bit
punc. to 6557 bit
gross 6624 bit
-TFCI - 16 bit
-Signal. -51 bit
punc. to 6557 bit
6557
6557
6557
6557
51
51
51
51
Service Multiplex.
6557 51
6557 51
6557 51
6557 51
2nd Interleaving
6608
6608
6608
6608
TFCI
6608 TFCI
6608 TFCI
6608 TFCI
6608 TFCI
16
16
16
16
Slot segmentation
SF=32
3 Timeslots
8 codes per slot
138 MA 138
130 TFCI MA TFCI 130
8
512 chips
8
TS #1..#3
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

# C.3.5 BCH reference measurement channel

## C.3.5.1 3,84 Mcps TDD Option

[mapped to 1 code SF16]

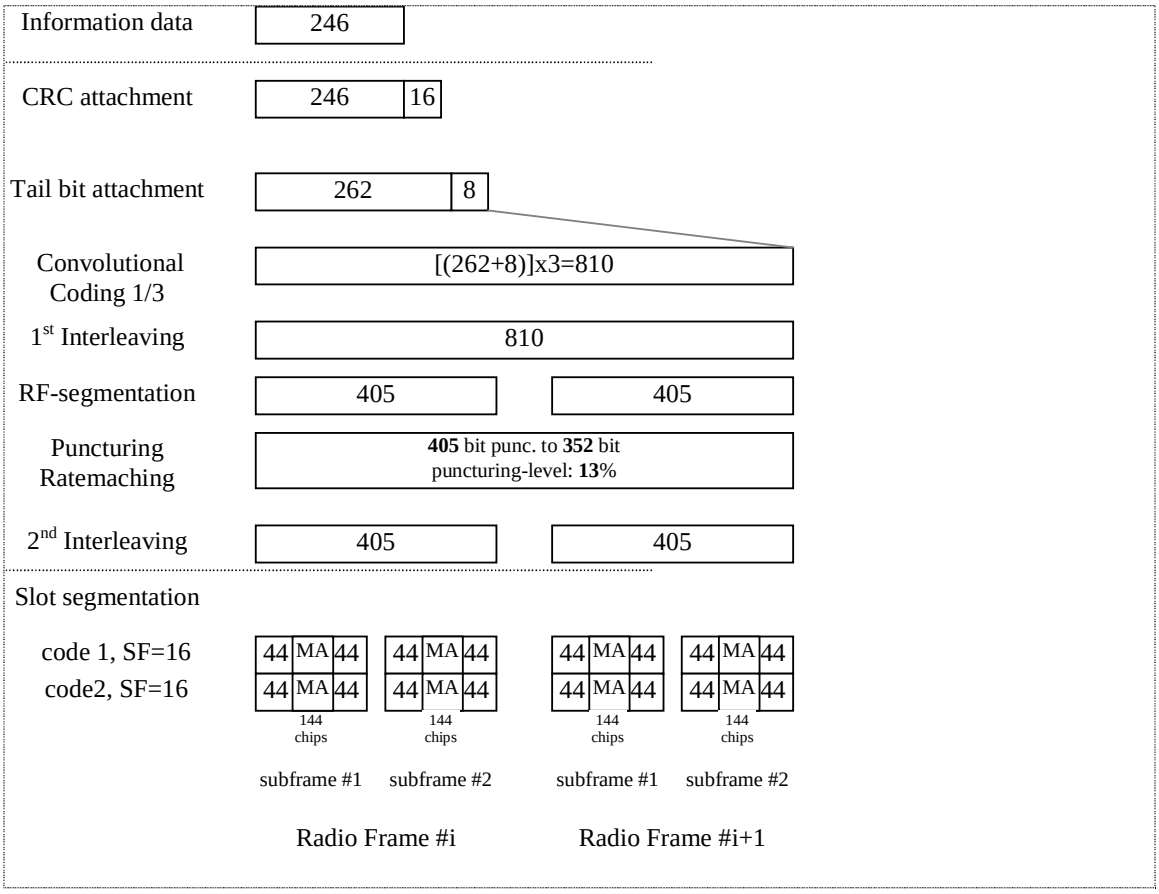
| Parameter | |
|---|---|
| Information data rate: | 12.3 kbps |
| RU´s allocated | 1 RU |
| Midamble | **512** chips |
| Interleaving | 20 ms |
| Power control | 0 bit |
| TFCI | 0 bit |
| Puncturing level | 10% |




## C.3.5.2 BCH reference measurement channel (12.3 kbps) for 1,28 Mcps TDD Option

**Table C.3.5.2**

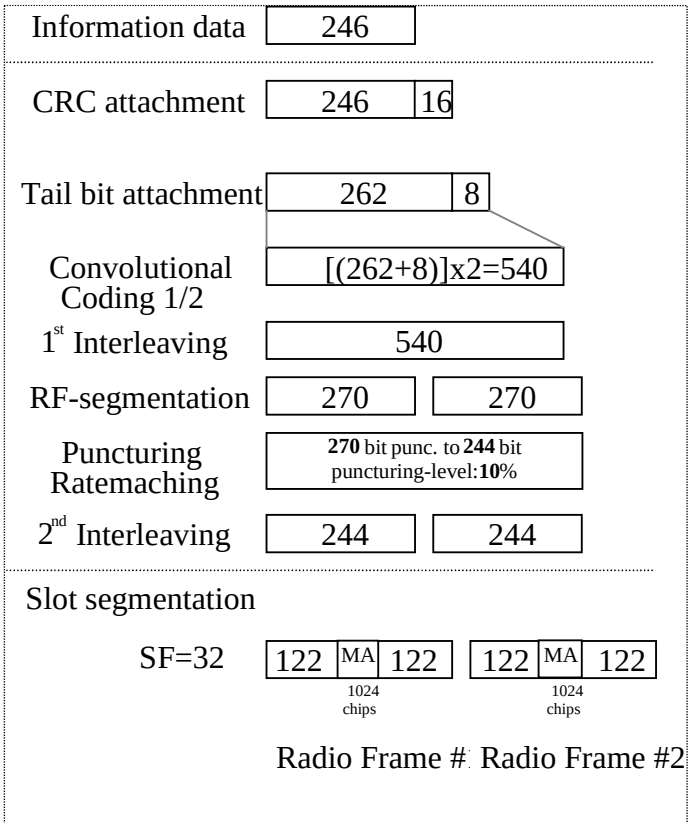
| Parameter | Value |
|---|---|
| Information data rate: | 12.3 kbps |
| RU´s allocated | 2 RU |
| Midamble | 144 chips |
| Interleaving | 20 ms |
| Power control | 0 bit |
| TFCI | 0 bit |
| Puncturing level | 13% |




## C.3.5.3 7,68 Mcps TDD Option

**Table C.3.5.3**

| Parameter | Value |
|---|---|
| Information data rate: | 12.3 kbps |
| RU´s allocated | 1 RU |
| Midamble | 1024 chips |
| Interleaving | 20 ms |
| Power control | 0 bit |
| TFCI | 0 bit |
| Puncturing level | 10% |




## C.3.6 DL reference measurement channel (2 Mbps) for 3,84 Mcps TDD Option

**Table C.3.6.1: DL reference measurement channel physical parameters (2 Mbps)**

| Parameter | Value |
|---|---|
| Information data rate | *2048 kbps* |
| RU´s allocated | *16*12TS = 192RU* |
| Midamble | **256** chips |
| Interleaving | 10 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH/ DCH of the DCCH | 13.9% / 0% |

**Table C.3.6.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (2 Mbps)**

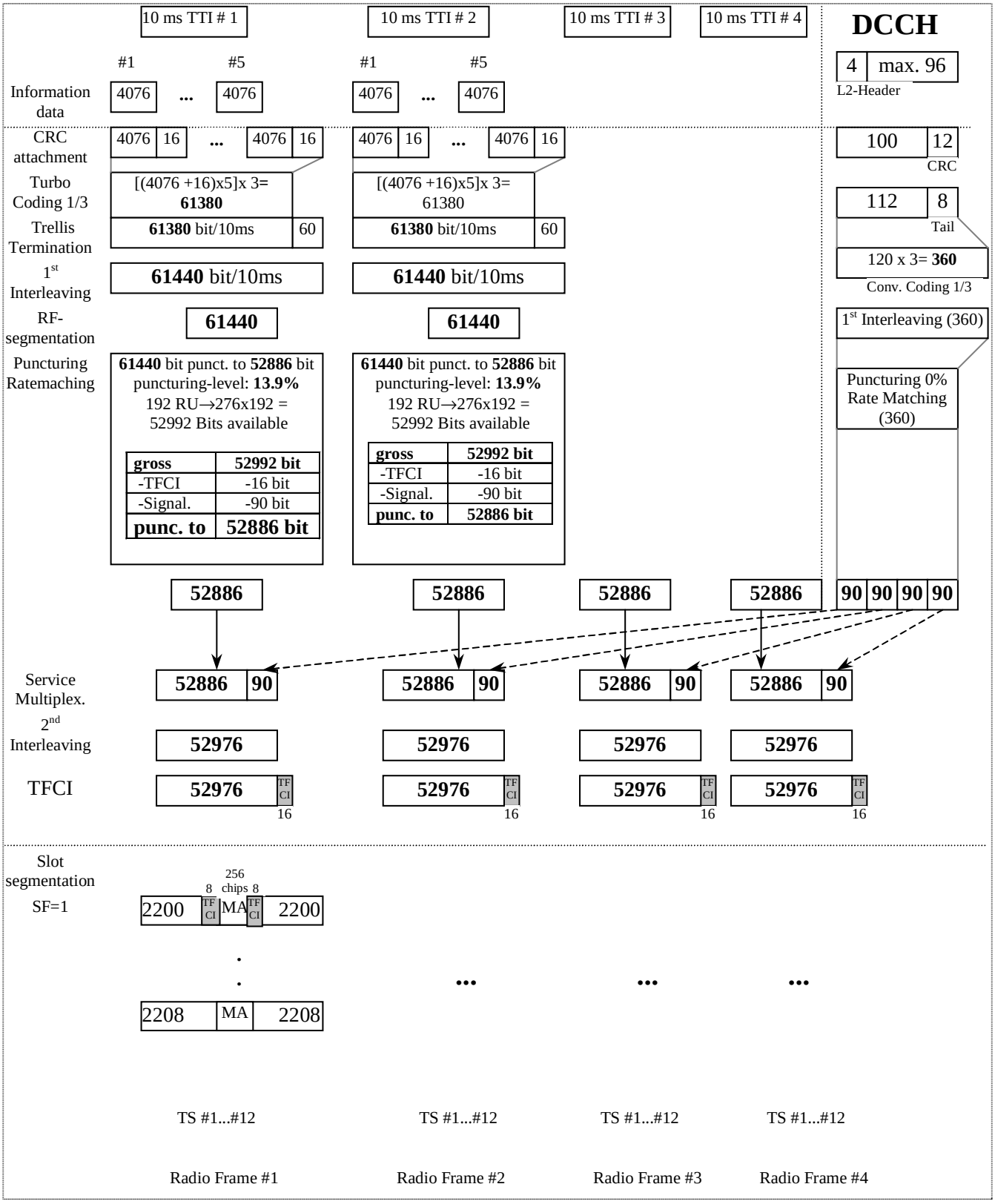
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 4076 | 88/80 |
| | Max data rate, bps | | 2038000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 4076 | 100 |
| | TFS | TF0, bits | 0*4076 | 0*100 |
| | | TF1, bits | 5*4076 | 1*100 |
| | TTI, ms | | 10 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 61440 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 61440 | 90 |
| | RM attribute | | 207 | 240 |

**Table C.3.6.3: DL reference measurement channel, TFCS (2 Mbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.6.4: DL reference measurement channel, puncturing limit (2 Mbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.84 |


10 ms TTI # 1
10 ms TTI # 2
10 ms TTI # 3
10 ms TTI # 4
DCCH
#1
#5
Information data
4076
...
CRC attachment
16
Turbo Coding 1/3
[(4076 +16)x5]x 3= 61380
Trellis Termination
61380 bit/10ms
60
1st Interleaving
61440 bit/10ms
RF-segmentation
61440
Puncturing Ratematching
61440 bit punct. to 52886 bit
puncturing-level: 13.9%
192 RU→276x192 =
52992 Bits available
gross
52992 bit
-TFCI
-16 bit
-Signal.
-90 bit
punc. to
52886 bit
4
max. 96
L2-Header
100
12
CRC
112
8
Tail
120 x 3= 360
Conv. Coding 1/3
1st Interleaving (360)
Puncturing 0% Rate Matching (360)
52886
90
Service Multiplex.
2nd Interleaving
52976
TFCI
TF CI
Slot segmentation
SF=1
256 chips
2200
MA
2208
TS #1...#12
Radio Frame #1
Radio Frame #2
Radio Frame #3
Radio Frame #4

# C.3.7 DL reference measurement channel (2 Mbps) for 7,68 Mcps TDD Option

**Table C.3.7.1: DL reference measurement channel physical parameters (2 Mbps)**

| Parameter | Value |
|---|---|
| Information data rate | 2048 kbps |
| RU´s allocated | 16*12TS = 192RU |
| Midamble | 512 chips |
| Interleaving | 10 ms |
| Power control | 0 Bit/user |
| TFCI | 16 Bit/user |
| Inband signalling DCCH | 2 kbps |
| Puncturing level at Code rate 1/3 : DCH of the DTCH / DCH of the DCCH | 13.9% / 0% |

**Table C.3.7.2: DL reference measurement channel using RLC-TM for DTCH, transport channel parameters (2 Mbps)**

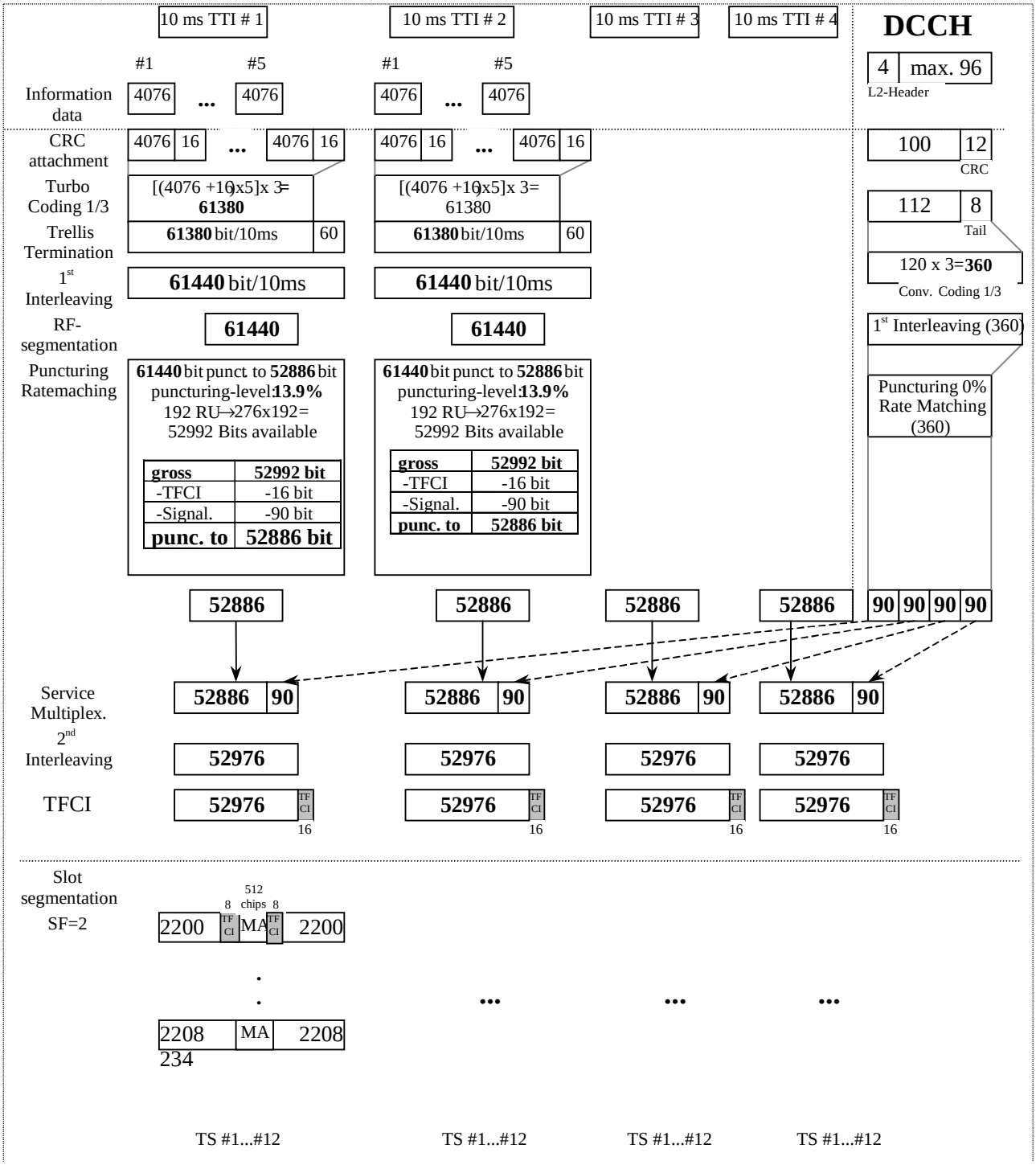
| Higher Layer | RAB/Signalling RB | | RAB | SRB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DCCH |
| | RLC mode | | TM | UM/AM |
| | Payload sizes, bit | | 4076 | 88/80 |
| | Max data rate, bps | | 2038000 | 2200/2000 |
| | PDU header, bit | | N/A | 8/16 |
| | TrD PDU header, bit | | 0 | N/A |
| MAC | MAC header, bit | | 0 | 4 |
| | MAC multiplexing | | N/A | Yes |
| Layer 1 | TrCH type | | DCH | DCH |
| | Transport Channel Identity | | 1 | 5 |
| | TB sizes, bit | | 4076 | 100 |
| | TFS | TF0, bits | 0*4076 | 0*100 |
| | | TF1, bits | 5*4076 | 1*100 |
| | TTI, ms | | 10 | 40 |
| | Coding type | | Turbo Coding | Convolution Coding |
| | Coding Rate | | N/A | 1/3 |
| | CRC, bit | | 16 | 12 |
| | Max number of bits/TTI after channel coding | | 61440 | 360 |
| | Uplink: Max number of bits/radio frame before rate matching | | 61440 | 90 |
| | RM attribute | | 207 | 240 |

**Table C.3.7.3: DL reference measurement channel, TFCS (2 Mbps)**

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (DTCH, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

**Table C.3.7.4: DL reference measurement channel, puncturing limit (2 Mbps)**

| | | |
|---|---|---|
| DPCH Downlink | Puncturing limit | 0.84 |


10 ms TTI # 1
10 ms TTI # 2
10 ms TTI # 3
10 ms TTI # 4
DCCH
#1
#5
Information data
4076
CRC attachment
16
Turbo Coding 1/3
[(4076 +16)x5]x 3= 61380
Trellis Termination
61380 bit/10ms
60
1st Interleaving
61440 bit/10ms
RF-segmentation
61440
Puncturing Ratematching
61440 bit punct. to 52886 bit puncturing-level:13.9%
192 RU→276x192= 52992 Bits available
gross 52992 bit
-TFCI -16 bit
-Signal. -90 bit
punc. to 52886 bit
4 max. 96
L2-Header
100 12
CRC
112 8
Tail
120 x 3=360
Conv. Coding 1/3
1st Interleaving (360)
Puncturing 0% Rate Matching (360)
52886
90
Service Multiplex.
2nd Interleaving
52976
TFCI
16
Slot segmentation
SF=2
512 chips
8
2200
MA
2208
234
TS #1...#12

# C.4 HSDPA reference measurement channels

## C.4.1 HSDPA reference measurement channels for 3,84 Mcps TDD option

### C.4.1.1 Reference measurement channels for 7,3 Mbps – Category 8 - UE

#### C.4.1.1.1 QPSK modulation scheme for test 1, 2, 3

**Table C.4.1.1: HS-PDSCH fixed reference channel for the PA3, PB3, and VA30 Channel models - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | Mbps | 2,6496 |
| Number of HARQ Processes | Processes | 4 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 26496 |
| Number Code Blocks | Blocks | 6 |
| Total Available of Soft Channel bits in UE | Bits | 353280 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 88320 |
| Number of coded bits per TTI | Bits | 35328 |
| Coding Rate | | 3/4 |
| Number of HS-PDSCH Timeslots | Slots | 8 |
| Number of HS-PDSCH codes per TS | Codes | 16 |
| Spreading factor | SF | 16 |

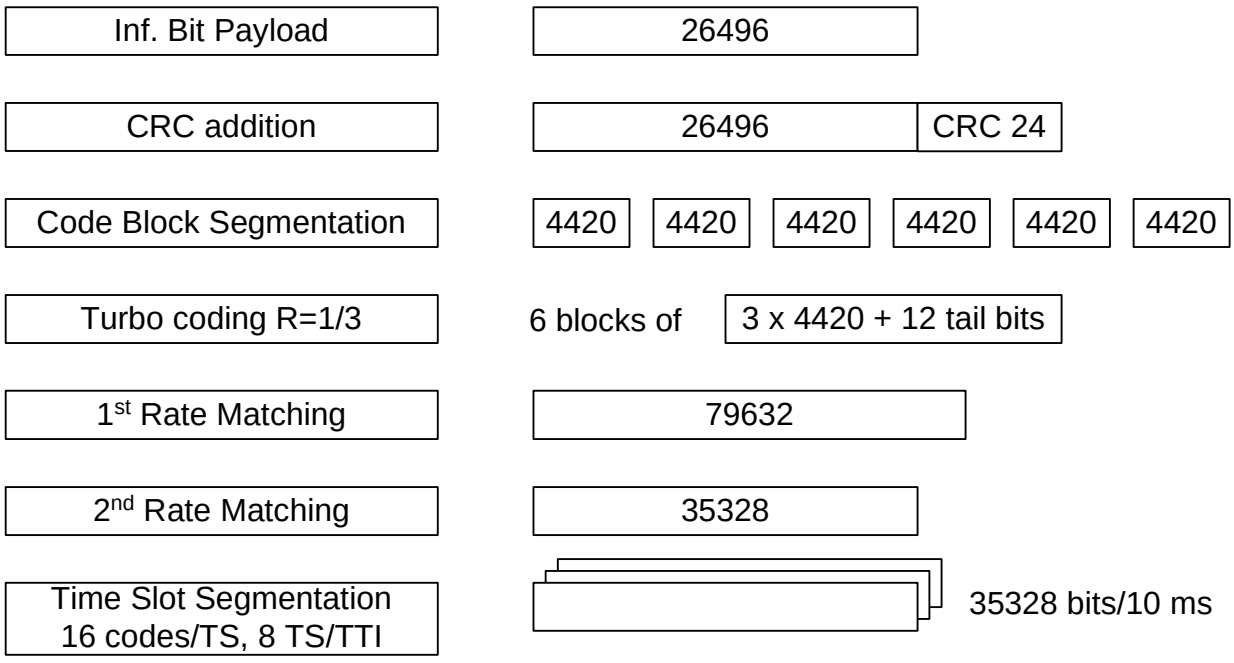



**Figure C.4.1.1: Coding for HS-PDSCH fixed reference channel with QPSK modulation for the PA3, PB3, and VA30 Channels – Category 8**

### C.4.1.1.2 QPSK modulation scheme for test 4

**Table C.4.1.2: HS-PDSCH fixed reference channel for the VA120 Channel model - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | Mbps | 2,3176 |
| Number of HARQ Processes | Processes | 4 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 23176 |
| Number Code Blocks | Blocks | 5 |
| Total Available of Soft Channel bits in UE | Bits | 353280 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 88320 |
| Number of coded bits per TTI | Bits | 30912 |
| Coding Rate | | 3/4 |
| Number of HS-PDSCH Timeslots | Slots | 8 |
| Number of HS-PDSCH codes per TS | Codes | 14 |
| Spreading factor | SF | 16 |

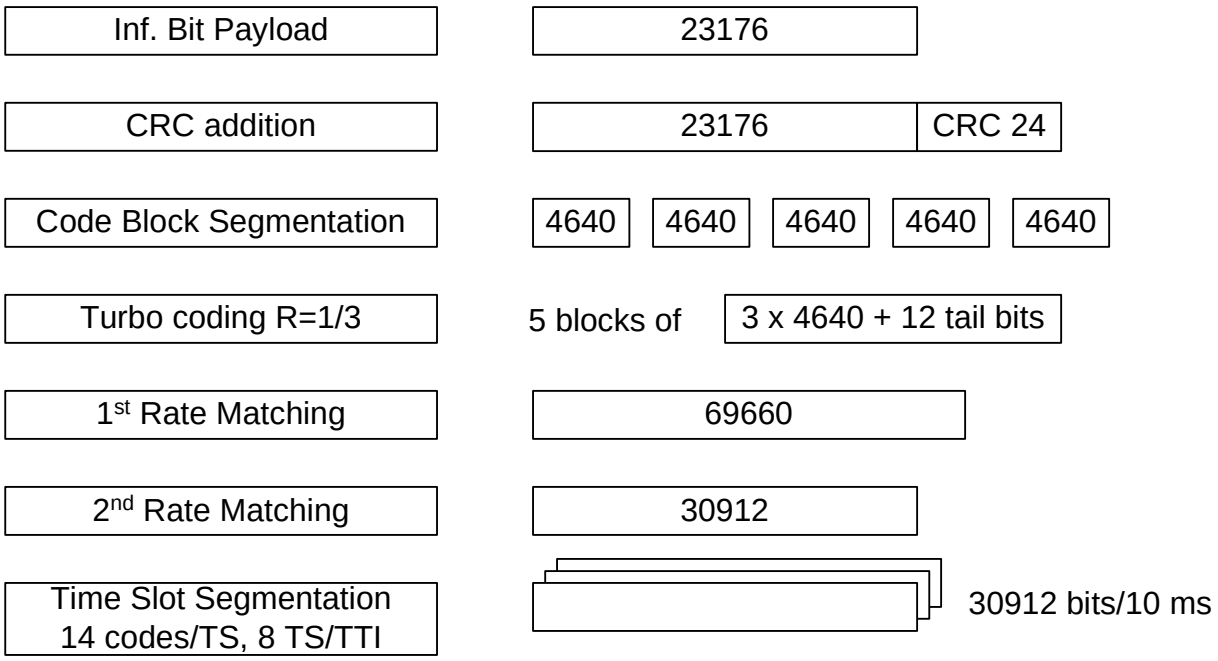



**Figure C.4.1.2: Coding for HS-PDSCH fixed reference channel with QPSK modulation for the VA120 Channel – Category 8**

### C.4.1.1.3 16QAM modulation scheme for test 1, 2, 3

**Table C.4.1.3: HS-PDSCH fixed reference channel for the PA3, PB3, and VA30 Channel models - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Modulation | | **16-QAM** |
| Maximum information bit throughput | Mbps | 5,2996 |
| Number of HARQ Processes | Processes | 4 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 52996 |
| Number Code Blocks | Blocks | 11 |
| Total Available of Soft Channel bits in UE | Bits | 353280 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 88320 |
| Number of coded bits per TTI | Bits | 70656 |
| Coding Rate | | 3/4 |
| Number of HS-PDSCH Timeslots | Slots | 8 |
| Number of HS-PDSCH codes per TS | Codes | 16 |
| Spreading factor | SF | 16 |

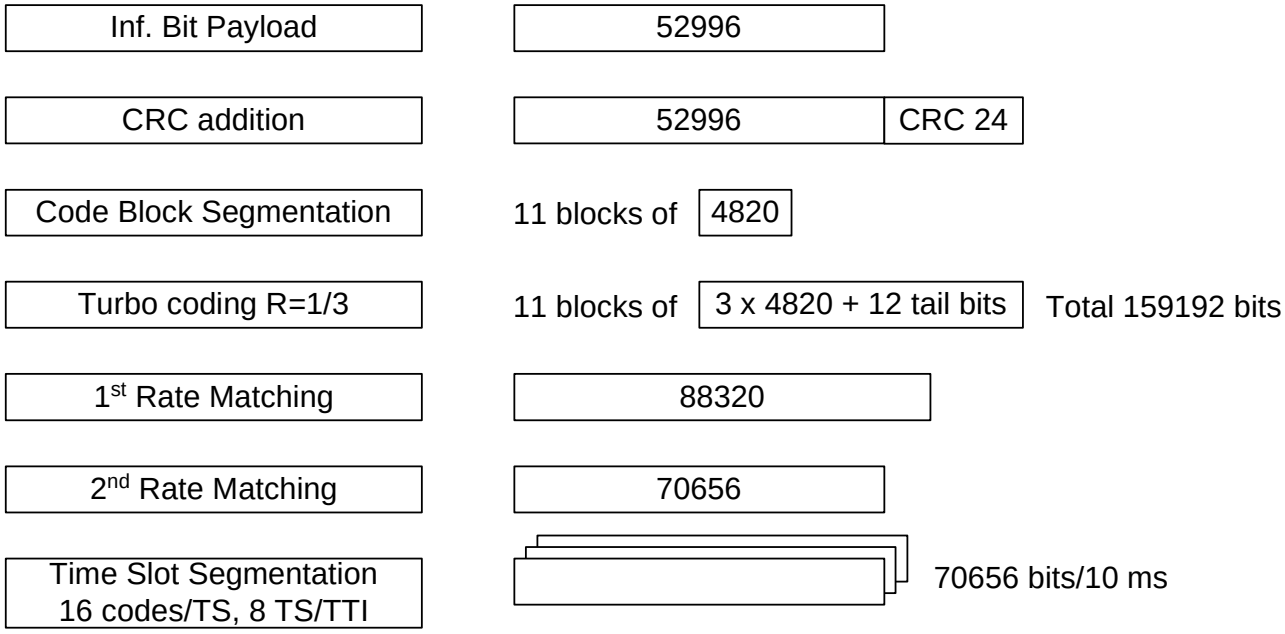



**Figure C.4.1.3: Coding for HS-PDSCH fixed reference channel with 16-QAM modulation for the PA3 PB3, and VA30 Channels – Category 8**

### C.4.1.1.4 16QAM modulation scheme for test 4

**Table C.4.1.4: HS-PDSCH fixed reference channel for the PA3, PB3, and VA30 Channel models - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Modulation | | **16-QAM** |
| Maximum information bit throughput | Mbps | 3,4773 |
| Number of HARQ Processes | Processes | 4 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 34773 |
| Number Code Blocks | Blocks | 7 |
| Total Available of Soft Channel bits in UE | Bits | 353280 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 88320 |
| Number of coded bits per TTI | Bits | 61824 |
| Coding Rate | | 9/16 |
| Number of HS-PDSCH Timeslots | Slots | 8 |
| Number of HS-PDSCH codes per TS | Codes | 14 |
| Spreading factor | SF | 16 |

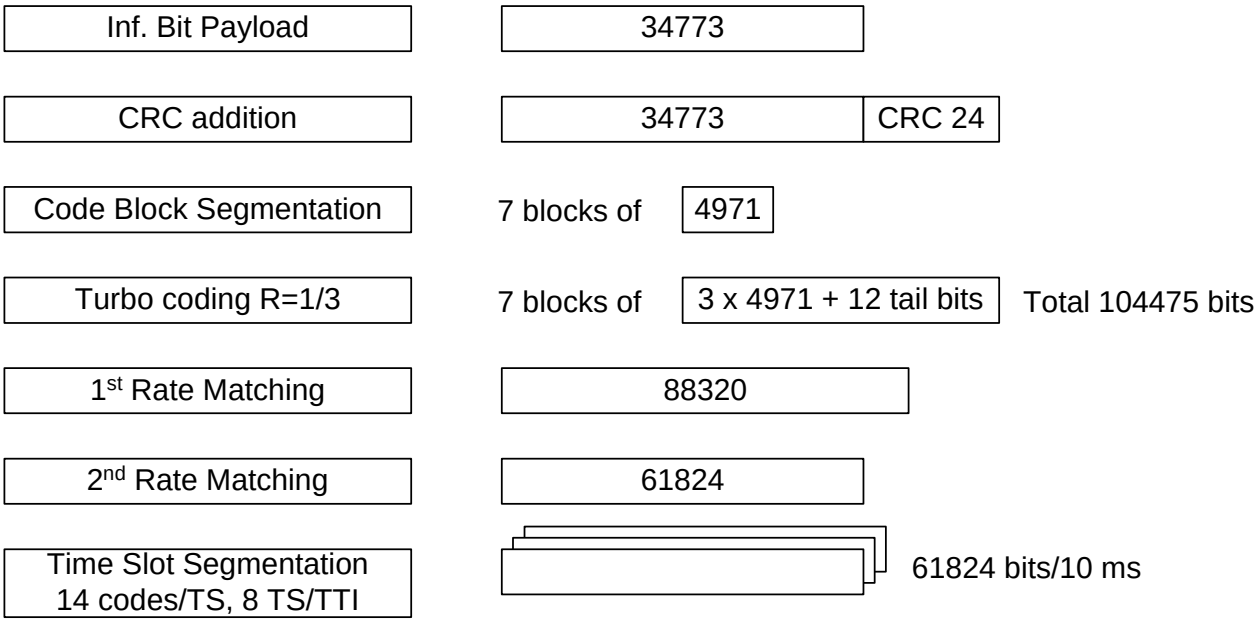


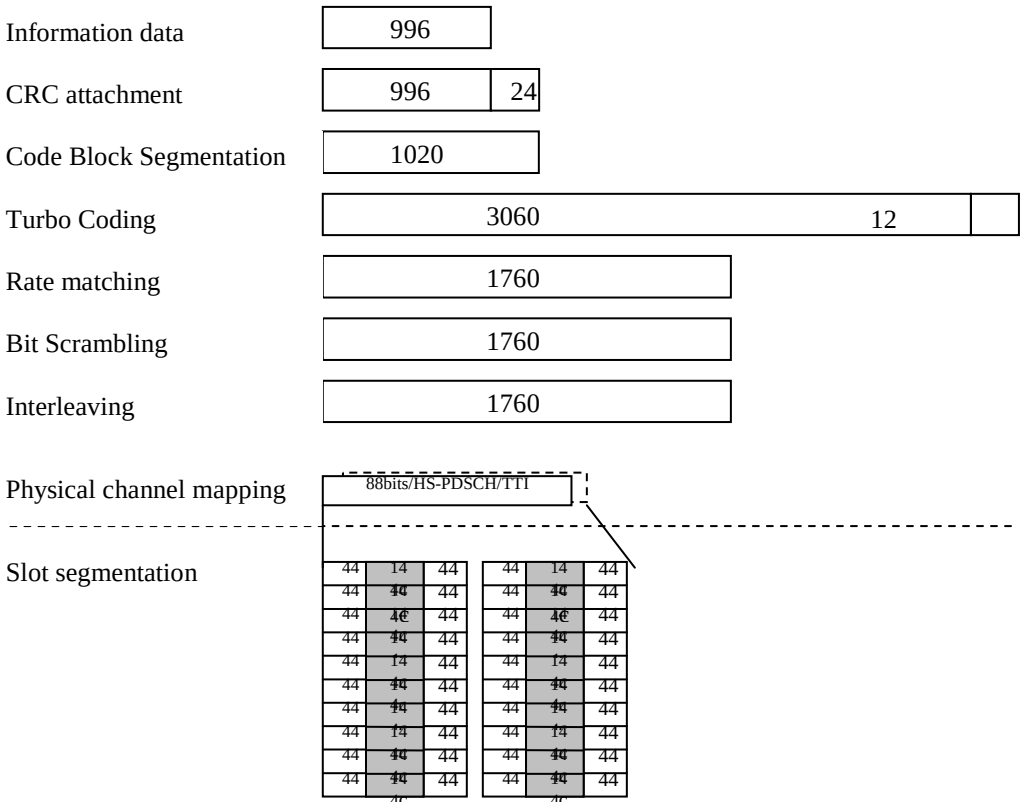

**Figure C.4.1.4: Coding for HS-PDSCH fixed reference channel with 16-QAM modulation for the VA120 Channel – Category 8**

## C.4.2 HSDPA reference measurement channels for 1.28 Mcps TDD option

### C.4.2.1 Reference measurement channels for 0.5Mbps UE class

#### C.4.2.1.1 QPSK modulation scheme

| Parameter | Value |
|---|---|
| Maximum information data rate | 199.2 kbps |
| RU's allocated | 2TS（10*SF16）=20 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 43% |




## C.4.2.2 Reference measurement channels for 1.1Mbps UE class

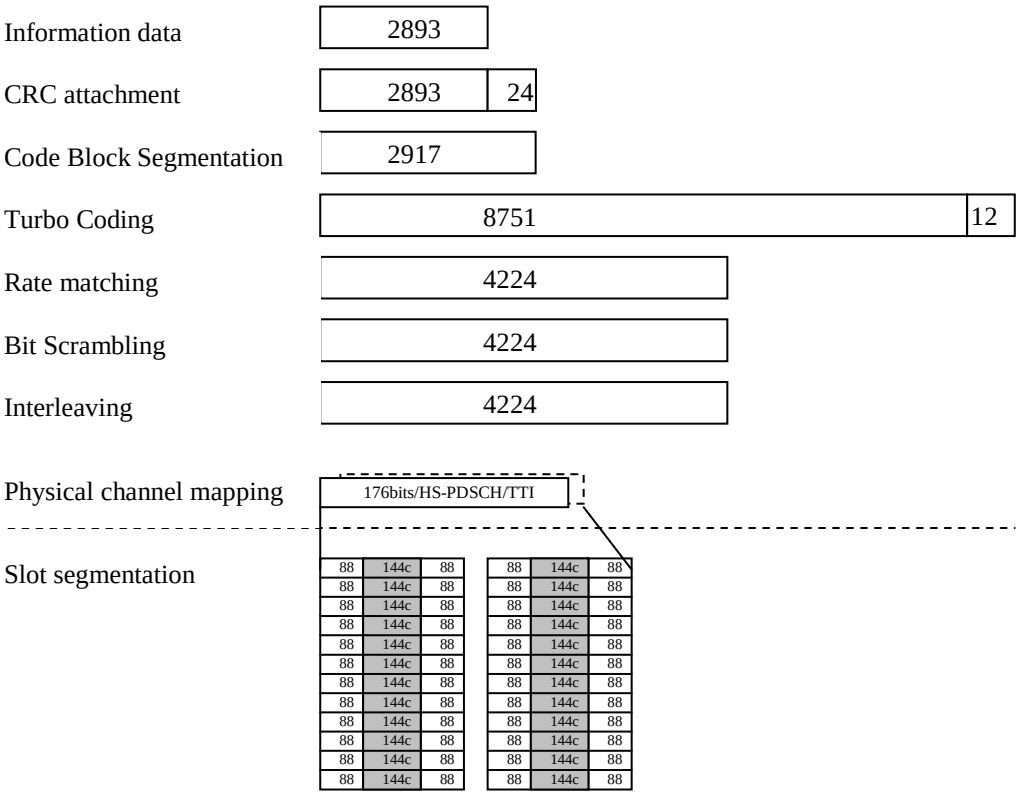
### C.4.2.2.1 QPSK modulation scheme

Reference channel in C.4.2.1.1 applies.

### C.4.2.2.2 16QAM modulation scheme

**Table A.13-2**

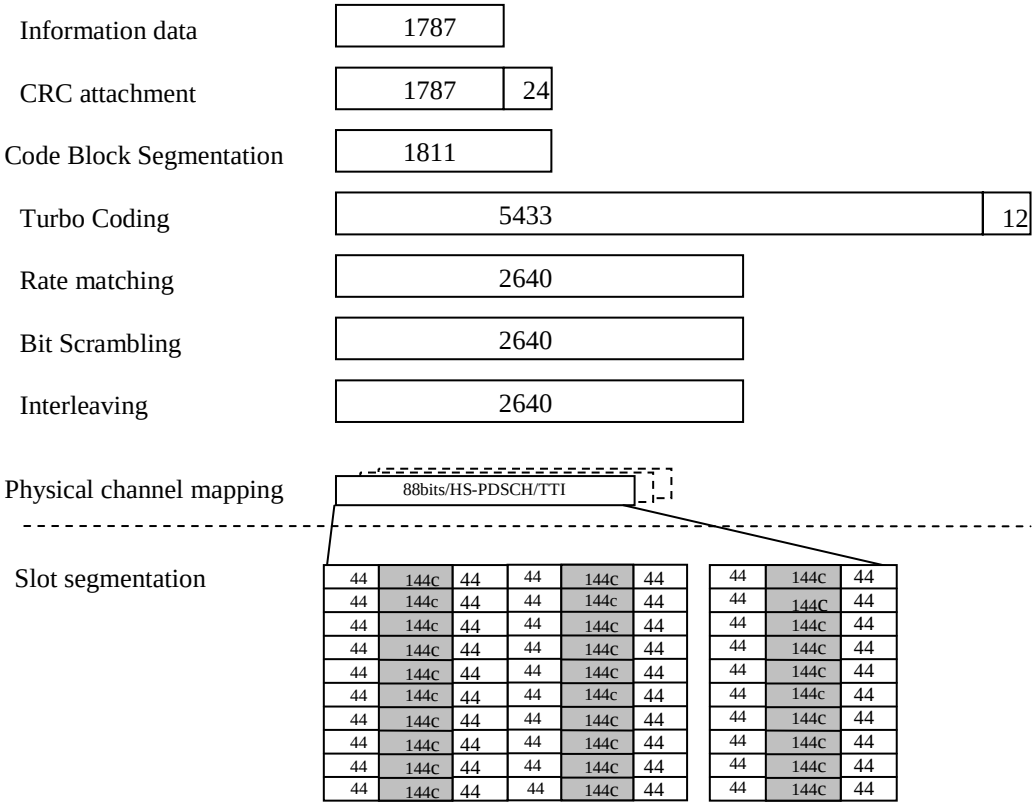
| Parameter | Value |
|---|---|
| Maximum information data rate | 578.6 kbps |
| RU's allocated | 2TS（12*SF16）=24 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 52% |




## C.4.2.3 Reference measurement channels for 1.6Mbps UE class

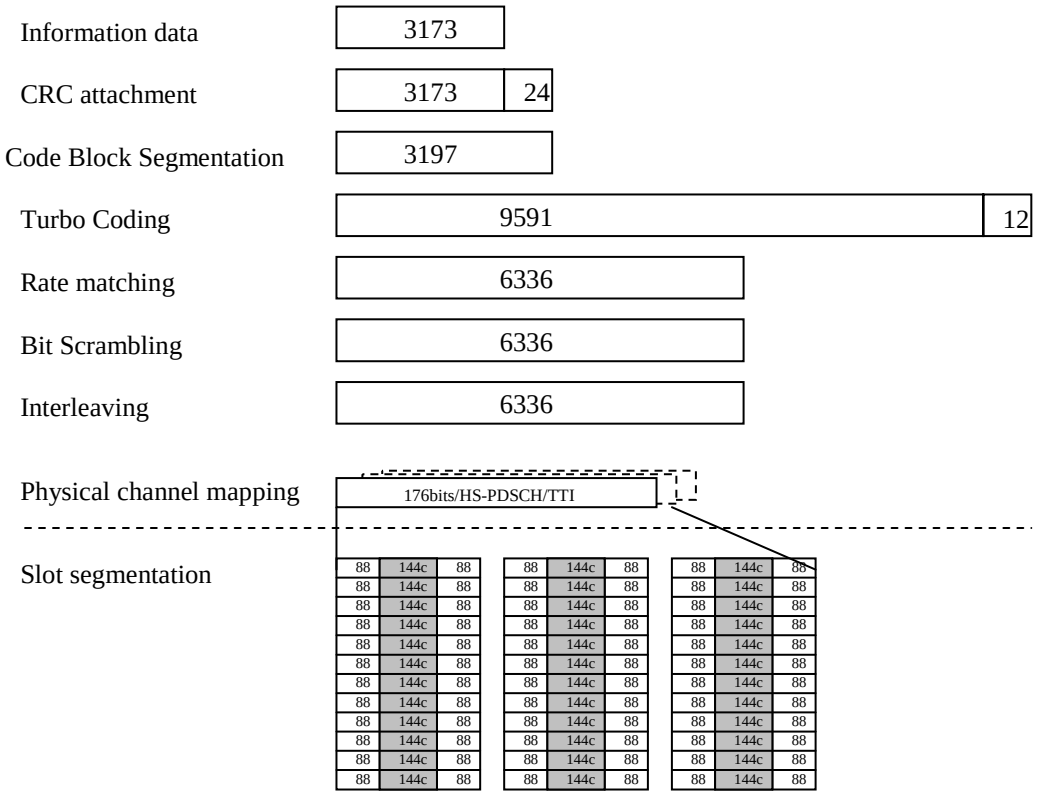
### C.4.2.3.1 QPSK modulation scheme

| Parameter | Value |
|---|---|
| Maximum information data rate | 357.4 kbps |
| RU's allocated | 3TS（10*SF16）=30 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 52% |




## C.4.2.3.2 16QAM modulation scheme

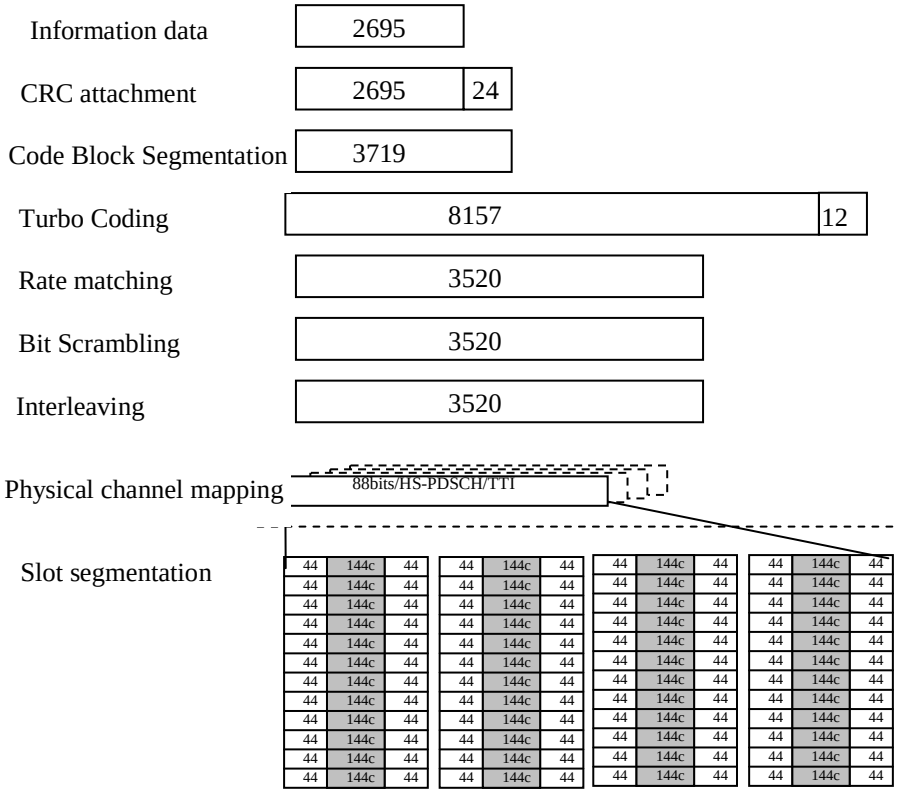
| Parameter | Value |
|---|---|
| Maximum information data rate | 634.6 kbps |
| RU's allocated | 3TS（12*SF16）=36 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 34% |




## C.4.2.4 Reference measurement channels for 2.2Mbps UE class

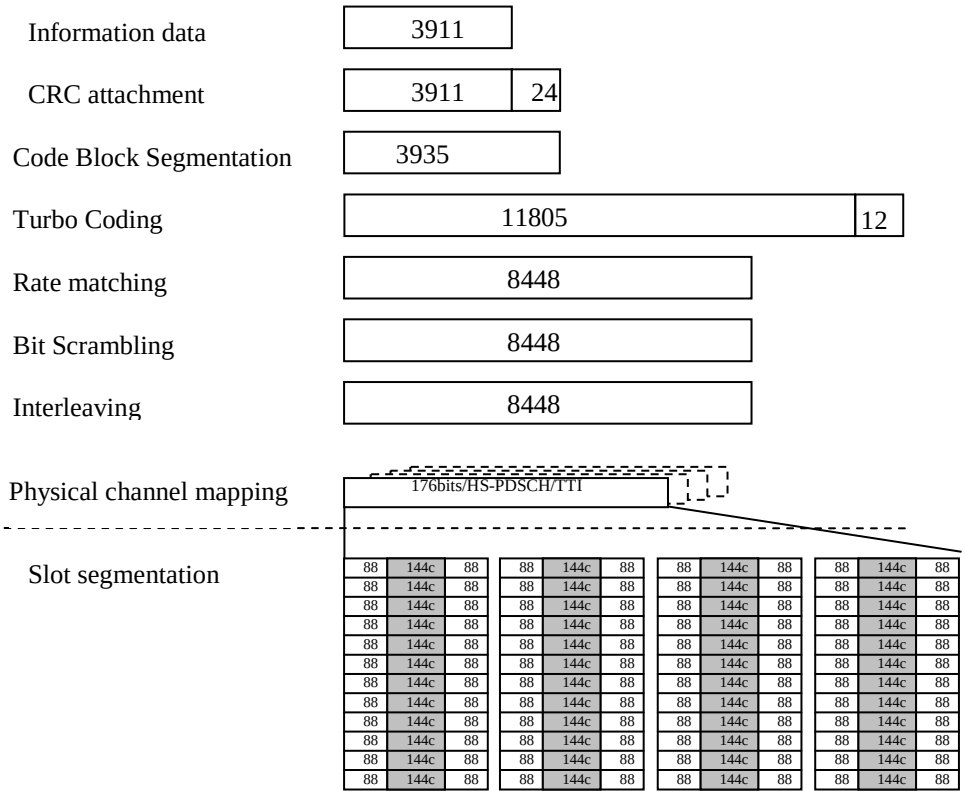
### C.4.2.4.1 QPSK modulation scheme

| Parameter | Value |
|---|---|
| Maximum information data rate | 539 kbps |
| RU's allocated | 4TS（10*SF16）=40 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 57% |




## C.4.2.4.2 16QAM modulation scheme

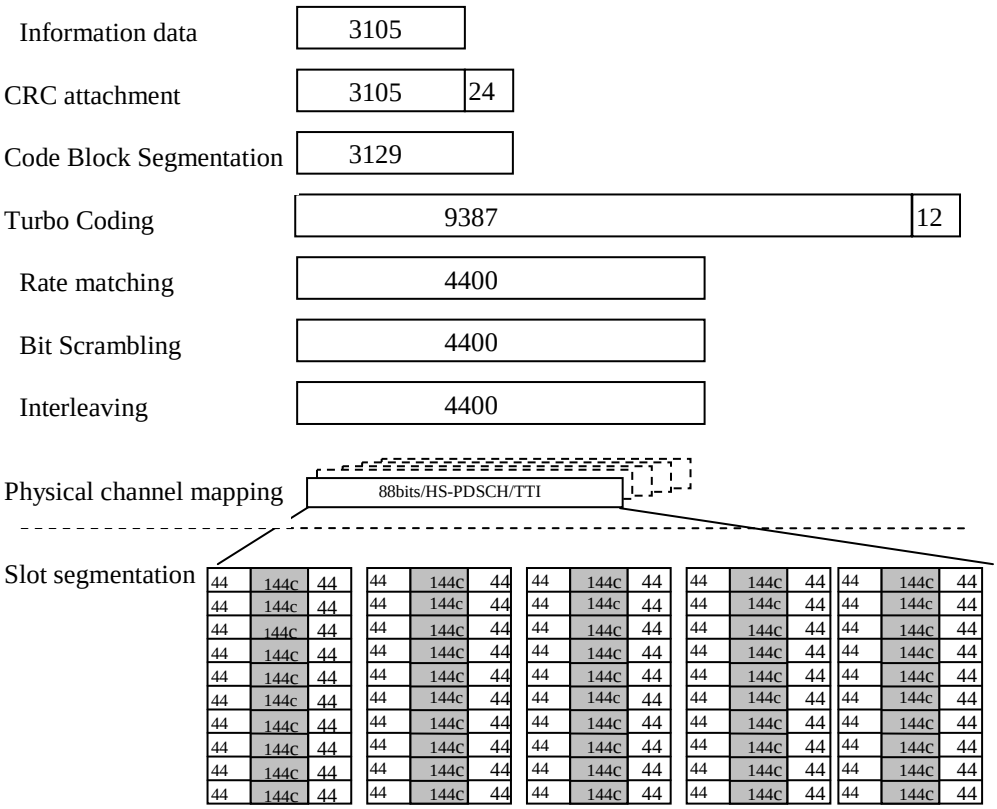
| Parameter | Value |
|---|---|
| Maximum information data rate | 782.2 kbps |
| RU's allocated | 4TS（12*SF16）=48 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 29% |




## C.4.2.5 Reference measurement channels for 2.8Mbps UE class

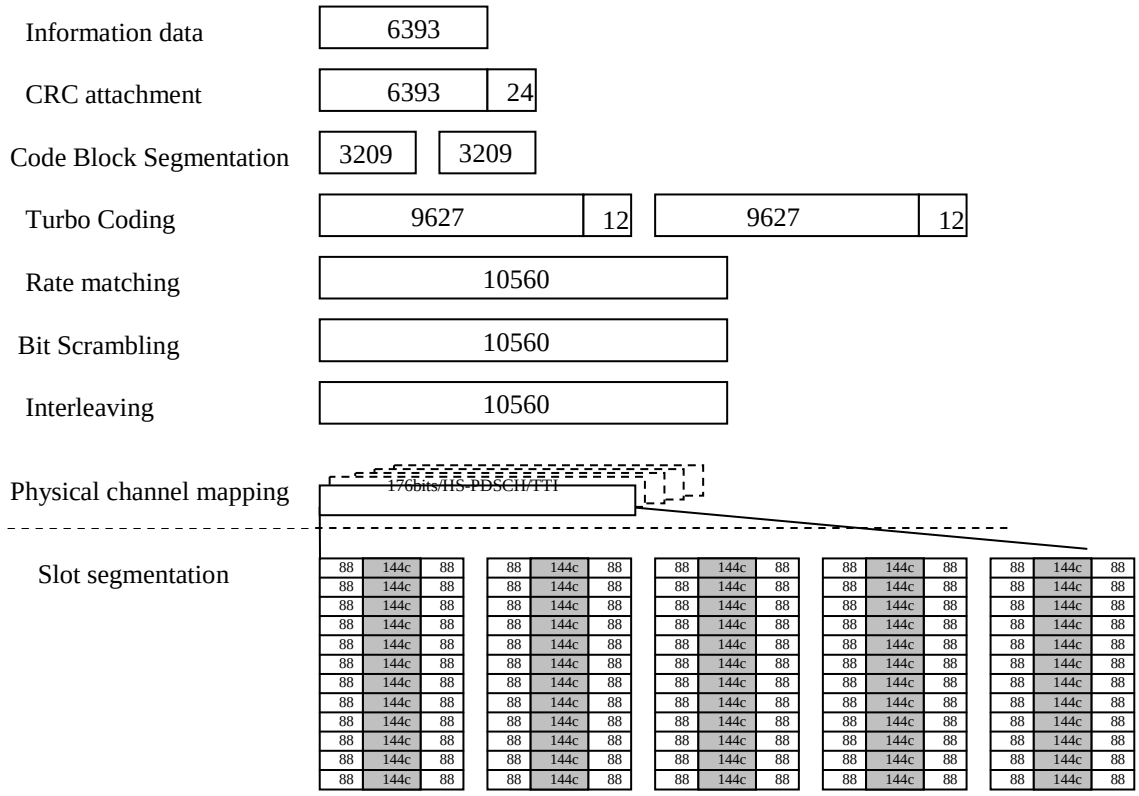
### C.4.2.5.1 QPSK modulation scheme

| Parameter | Value |
|---|---|
| Maximum information data rate | 621 kbps |
| RU's allocated | 5TS（10*SF16）=50 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 53% |




## C.4.2.5.2 16QAM modulation scheme

| Parameter | Value |
|---|---|
| Maximum information data rate | 1278.6 kbps |
| RU's allocated | 5TS（12*SF16）=60 RU/5ms |
| Midamble | 144chips |
| Puncturing level at code rate 1/3 | 45% |




# C.4.2A HSDPA reference measurement channels for 7,68 Mcps TDD option

## C.4.2A.1 Reference measurement channels for 5,3 Mbps - Category 8 - UE

### C.4.2A.1.1 QPSK modulation scheme for test 1, 2, 3 & 4

**Table C.4.2A.1a: HS-PDSCH fixed reference channel for the PA3, PB3, VA30 and VA120 Channel models - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | Mbps | 1.7612 |
| Number of HARQ Processes | Processes | 3 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 17612 |
| Number Code Blocks | Blocks | 4 |
| Total Available of Soft Channel bits in UE | Bits | 211968 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 70656 |
| Number of coded bits per TTI | Bits | 35328 |
| Coding Rate | | 1/2 |
| Number of HS-PDSCH Timeslots | Slots | 4 |
| Number of HS-PDSCH codes per TS | Codes | 32 |
| Spreading factor | SF | 32 |

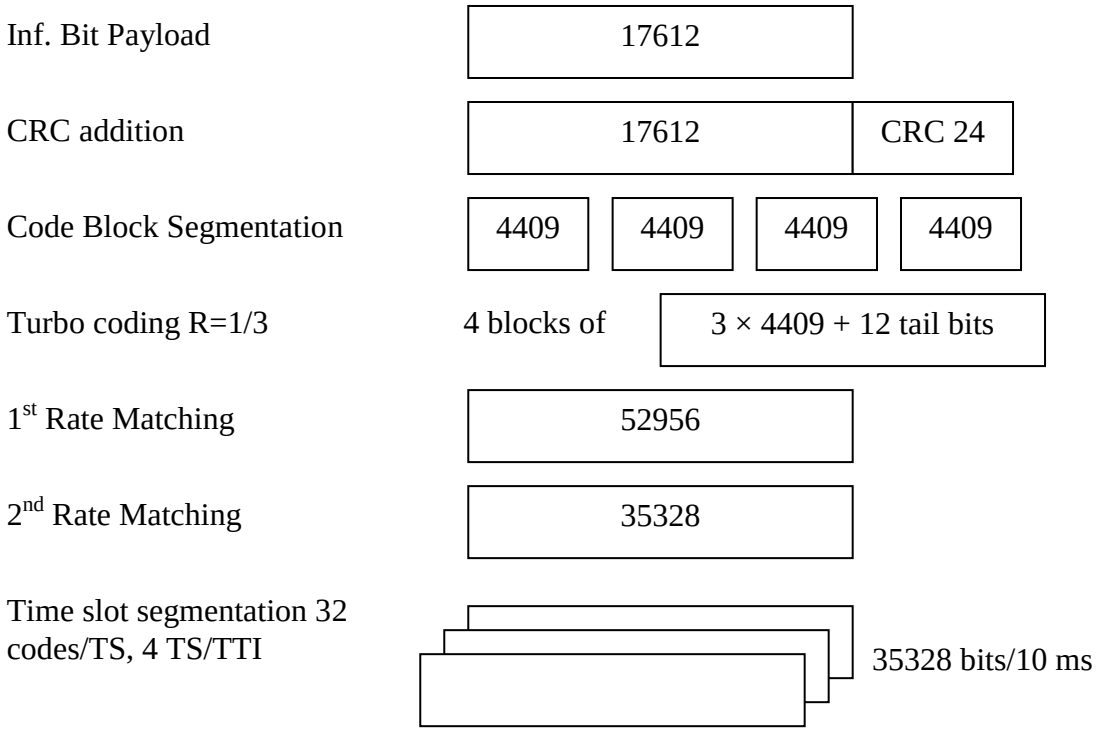



**Figure C.4.2A.1b: Coding for HS-PDSCH fixed reference channel with QPSK modulation for the PA3, PB3, VA30 and VA120 Channels - Category 8**

### C.4.2A.1.2 16QAM modulation scheme for test 1, 2, 3 & 4

**Table C.4.2A.1c: HS-PDSCH fixed reference channel for the PA3, PB3, VA30 and VA120 Channel models - Category 8**

| Parameter | Unit | Value |
|---|---|---|
| Modulation | | **16-QAM** |
| Maximum information bit throughput | Mbps | 3.5066 |
| Number of HARQ Processes | Processes | 3 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 35066 |
| Number Code Blocks | Blocks | 7 |
| Total Available of Soft Channel bits in UE | Bits | 211968 |
| Number of Soft Channel bit per HARQ Proc. | Bits | 70656 |
| Number of coded bits per TTI | Bits | 70656 |
| Coding Rate | | ½ |
| Number of HS-PDSCH Timeslots | Slots | 4 |
| Number of HS-PDSCH codes per TS | Codes | 32 |
| Spreading factor | SF | 32 |

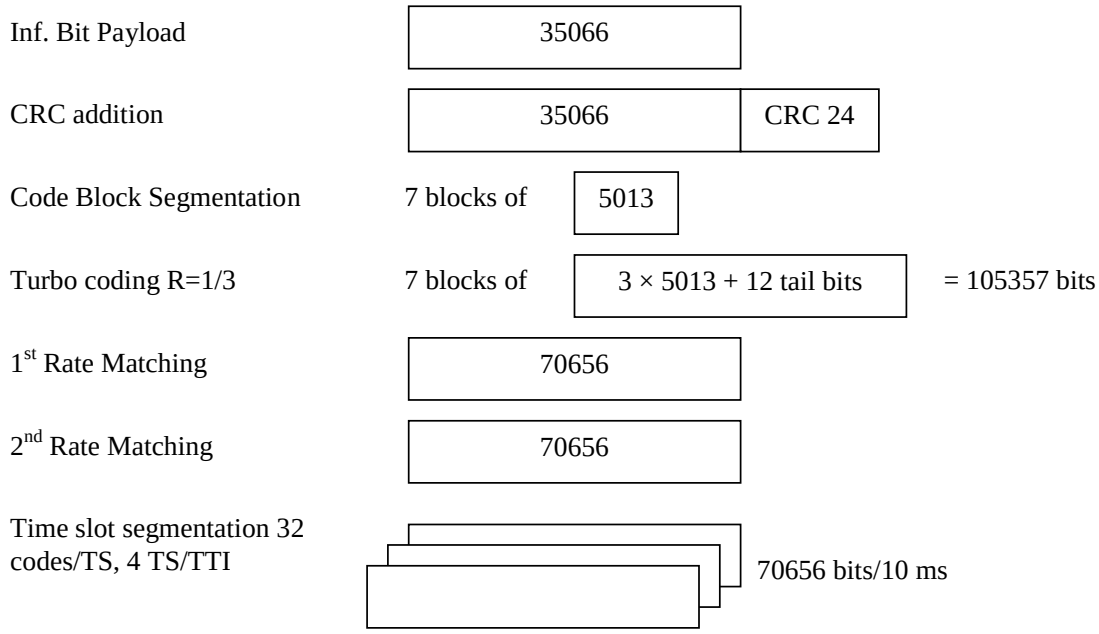



**Figure C.4.2A.1d: Coding for HS-PDSCH fixed reference channel with 16-QAM modulation for the PA3 PB3, VA30 and VA120 Channels - Category 8**

## C.4.3 Variable Reference Channel definition for 3,84 Mcps and 1,28 Mcps TDD options

The variable reference measurement channels are defined by:

a) The maximum information bit payload that is determined by the UE capability class under test and the allocated resource units (and hence implicitly by the CQI table applicable to the UE under test as derived from TS25.321).

b) The most recently received UE CQI report.

# C.5 Downlink reference parameters for MBMS tests

## C.5.1 MCCH

### C.5.1.1 3.84 Mcps TDD Option

The parameters for the MCCH demodulation tests are specified in Table C.5.1.1a and Table C.5.1.1b.

**Table C.5.1.1a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | kbps | 22.8 |
| Channel symbol rate | ksps | 11.4 |
| Slot Format #i | - | 3 |
| TFCI | - | ON |

**Table C.5.1.1b: Transport channel parameters for S-CCPCH**

| Parameter | MCCH |
|---|---|
| User Data Rate | 7.2 kbps |
| Number Transport Channel | 1 |
| Transport Block Size | 581 |
| Transport Block Set Size | 581 |
| RLC SDU block size | 4088 |
| Transmission Time Interval | 80 ms |
| Repetition period | 640 ms |
| Modification period | 1280 ms |
| Type of Error Protection | Turbo |
| Coding Rate | 1/3 |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

## C.5.1.2 1.28 Mcps TDD Option

The parameters for the MCCH demodulation tests are specified in Table C5.1.2a and Table C5.1.2b.

**Table C5.1.2a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | kbps | 17.6 |
| Channel symbol rate | ksps | 8.8 |
| Slot Format | | No TPC&SS |
| SF | - | 16 |
| TFCI | - | ON |

**Table C5.1.2b: Transport channel parameters for S-CCPCH**

| Parameter | MCCH |
|---|---|
| User Data Rate | 7.6 kbps |
| Number Transport Channel | 1 |
| Transport Block Size | 72 |
| Transport Block Set Size | 72 |
| RLC SDU block size | 4088 |
| Transmission Time Interval | 10 ms |
| Repetition period | 640 ms |
| Modification period | 1280 ms |
| Type of Error Protection | Convolutional code 1/3 |
| Coding Rate | 1/3 |
| Rate Matching attribute | 160 |
| Size of CRC | 16 |
| TFCI | ON |

## C.5.1.3 7.68 Mcps TDD Option

The parameters for the MCCH demodulation tests are specified in Table C.5.1.3a and Table C.5.1.3b.

**Table C.5.1.3a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level |
|---|---|---|
| Channel bit rate | kbps | 22.8 |
| Channel symbol rate | ksps | 11.4 |
| Slot Format #i | - | 3 |
| TFCI | - | ON |

**Table C.5.1.3b: Transport channel parameters for S-CCPCH**

| Parameter | MCCH |
|---|---|
| User Data Rate | 7.2 kbps |
| Number Transport Channel | 1 |
| Transport Block Size | 581 |
| Transport Block Set Size | 581 |
| RLC SDU block size | 4088 |
| Transmission Time Interval | 80 ms |
| Repetition period | 640 ms |
| Modification period | 1280 ms |
| Type of Error Protection | Turbo |
| Coding Rate | 1/3 |
| Rate Matching attribute | 256 |
| Size of CRC | 16 |

# C.5.2 MTCH

## C.5.2.1 3.84 Mcps TDD Option

The parameters for the MTCH demodulation tests are specified in Table C.5.2.1a and Table C.5.2.1b.

**Table C.5.2.1a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level | Level |
|---|---|---|---|
| User Data Rate | kpbs | 256 | 128 |
| Channel bit rate | kbps | 388.8 | 388.8 |
| Channel symbol rate | ksps | 194.4 | 194.4 |
| Slot Format #i | - | 3 and 0 | 3 and 0 |
| TFCI | - | ON | ON |

**Table C.5.2.1b: Transport channel parameters for S-CCPCH**

| Parameter | MTCH | |
|---|---|---|
| User Data Rate | 256 kbps | 128 kbps |
| Number of Transport Channel | 1 | 1 |
| Transport Block Size | 2561 | 2561 |
| Transport Block Set Size | 10244 | 5122 |
| Nr of transport blocks/TTI | 4 | 2 |
| RLC SDU block size | 10160 | 5072 |
| Transmission Time Interval | 40 ms | 40 ms |
| Type of Error Protection | Turbo | Turbo |
| Coding Rate | 1/3 | 1/3 |
| Rate Matching attribute | 256 | 256 |
| Size of CRC | 16 | 16 |

## C.5.2.2 1.28 Mcps TDD Option

The parameters for the MTCH demodulation tests are specified in Table C5.2.2a and Table C5.2.2b.

**Table C5.2.2a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level | Level | level | level |
|---|---|---|---|---|---|
| User Data Rate | kpbs | 128 | 64 | 384 | 192 |
| Channel bit rate | kbps | 246.4 | 140.8 | 563.2 | 281.6 |
| Channel symbol rate | ksps | 123.2 | 70.4 | 140.8 | 140.8 |
| Slot Format | - | No TPC&SS | No TPC&SS | No TPC SS | No TPC SS |
| TFCI | - | ON | ON | ON | ON |

**Table C5.2.2b: Transport channel parameters for S-CCPCH**

| Parameter | MTCH | | | |
|---|---|---|---|---|
| User Data Rate | 128 kbps | 64 kbps | 384kbps | 192kbps |
| Number of Transport Channel | 1 | 1 | 1 | 1 |
| Transport Block Size | 2561 | 1281 | 2561 | 2561 |
| Transport Block Set Size | 5122 | 2562 | 15366 | 7683 |
| Nr of transport blocks/TTI | 2 | 2 | 6 | 3 |
| RLC SDU block size | 5072 | 2512 | 15248 | 7616 |
| Transmission Time Interval | 40 ms | 40 ms | 40ms | 40ms |
| Type of Error Protection | Turbo | Turbo | Turbo | Turbo |
| Coding Rate | 1/3 | 1/3 | 1/3 | 1/3 |
| Rate Matching attribute | 2448 | 1392 | 256 | 256 |
| Size of CRC | 16 | 16 | 16 | 16 |

Parameters for combined MTCH demodulation and cell identification requirements are defined in Table C5.2.2c.

**Table C5.2.2c: Cell reselection parameters**

| Parameter | Unit | Value |
|---|---|---|
| Serving cell in the initial condition | - | Cell1 |
| Neighbour cells | - | Cell 2 and cell 3 |
| Cell_selection_and_reselection_quality_measure | - | P-CCPCH |
| Qrxlevmin | dBm | -103 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| Treselection | seconds | 4 |
| Sintrasearch | dB | not sent |
| IE "FACH Measurement occasion info" | - | not sent |

## C.5.2.3 7.68 Mcps TDD Option

The parameters for the MTCH demodulation tests are specified in Table C.5.2.3a and Table C.5.2.3b.

**Table C.5.2.3a: Physical channel parameters for S-CCPCH**

| Parameter | Unit | Level | Level |
|---|---|---|---|
| User Data Rate | kpbs | 256 | 128 |
| Channel bit rate | kbps | 388.8 | 388.8 |
| Channel symbol rate | ksps | 194.4 | 194.4 |
| Slot Format #i | - | 3 and 0 | 3 and 0 |
| TFCI | - | ON | ON |

**Table C.5.2.3b: Transport channel parameters for S-CCPCH**

| Parameter | MTCH | |
|---|---|---|
| User Data Rate | 256 kbps | 128 kbps |
| Number of Transport Channel | 1 | 1 |
| Transport Block Size | 2561 | 2561 |
| Transport Block Set Size | 10244 | 5122 |
| Nr of transport blocks/TTI | 4 | 2 |
| RLC SDU block size | 10160 | 5072 |
| Transmission Time Interval | 40 ms | 40 ms |
| Type of Error Protection | Turbo | Turbo |
| Coding Rate | 1/3 | 1/3 |
| Rate Matching attribute | 256 | 256 |
| Size of CRC | 16 | 16 |

# C.6 E-DCH Reference measurement channels

## C.6.1 E-DCH Fixed Reference Channels

### C.6.1.1 3,84 Mcps TDD Option

#### C.6.1.1.1 Fixed Reference Channel 1 (FRC1)

**Table C.6.1.1a: E-DCH Fixed Reference Channel 1 (3.84 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 34.7 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 347 |
| Number Code Blocks | Blocks | 1 |
| Number of coded bits per TTI | Bits | 1200 |
| Coding Rate | | 0.312 |
| Modulation | | QPSK |
| Number of E-DCH Timeslots | Slots | 6 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 16 |
| Number of E-UCCH per TTI | | 4 |

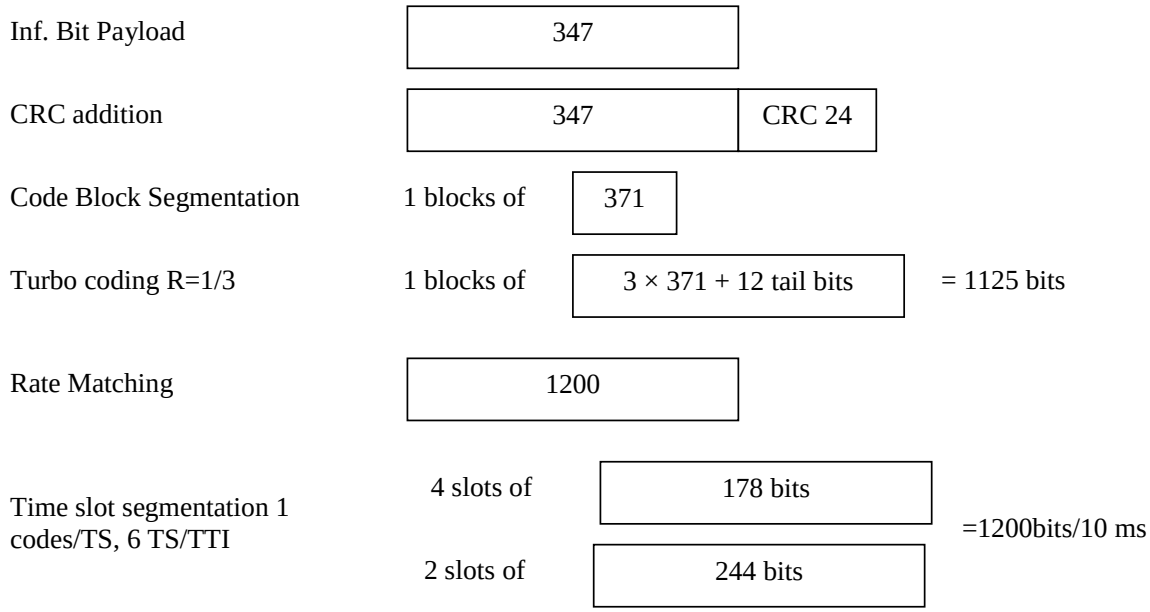



**Figure A.9: Coding for E-DCH FRC1 (3.84 Mcps TDD Option)**

### C.6.1.1.2 Fixed Reference Channel 2 (FRC2)

**Table C.6.1.1b: E-DCH Fixed Reference Channel 2 (3.84 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 1083.1 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 10831 |
| Number Code Blocks | Blocks | 3 |
| Number of coded bits per TTI | Bits | 22272 |
| Coding Rate | | 0.488 |
| Modulation | | 16QAM |
| Number of E-DCH Timeslots | Slots | 6 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 2 |
| Number of E-UCCH per TTI | | 2 |

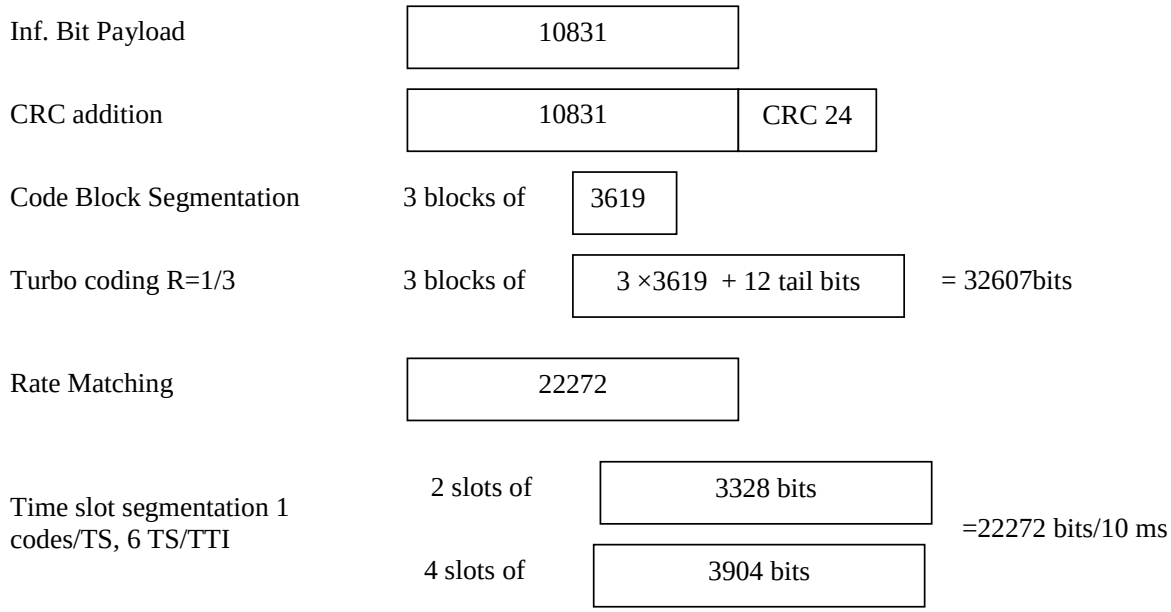



**Figure A.10: Coding for E-DCH FRC2 (3.84 Mcps TDD Option)**

### C.6.1.1.3 Fixed Reference Channel 3 (FRC3)

**Table C.6.1.1c: E-DCH Fixed Reference Channel 3 (3.84 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 2073.7 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 20737 |
| Number Code Blocks | Blocks | 5 |
| Number of coded bits per TTI | Bits | 28992 |
| Coding Rate | | 0.716 |
| Modulation | | 16QAM |
| Number of E-DCH Timeslots | Slots | 4 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 1 |
| Number of E-UCCH per TTI | | 2 |

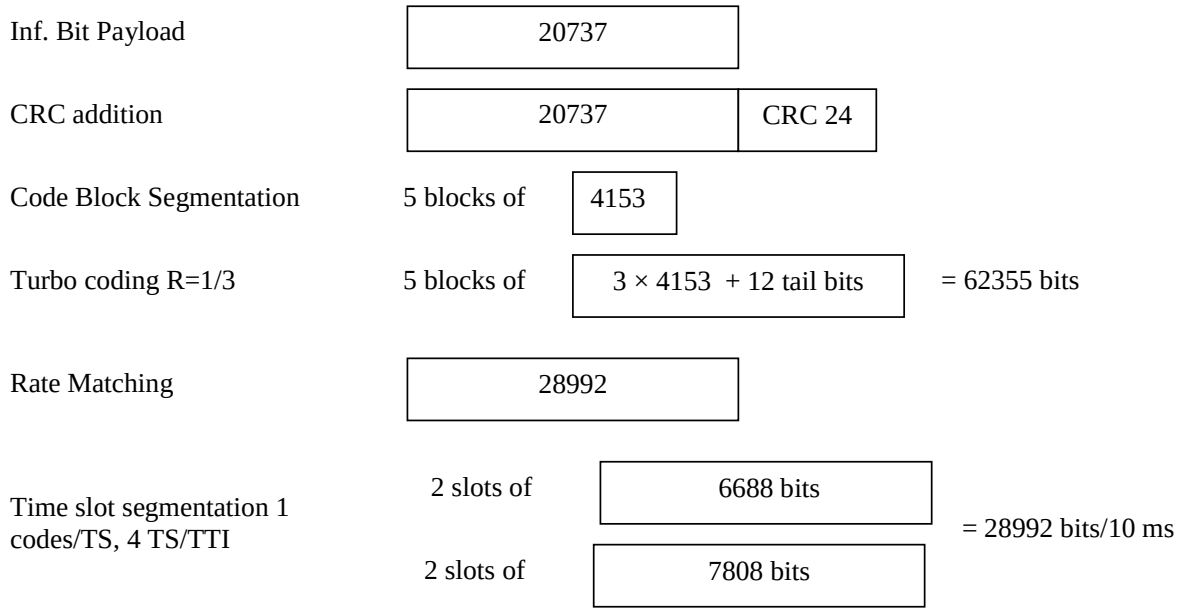



**Figure A.11: Coding for E-DCH FRC3 (3.84 Mcps TDD Option)**

## C.6.1.2 1.28Mcps TDD Option

### C6.1.2.1 E-DCH Fixed reference channel 1(FRC1)

**Table A.9: E-DCH Fixed reference channel 1 (1.28Mcps TDD option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 57.4 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 287 |
| Number Code Blocks | Blocks | 1 |
| Number of coded bits per TTI | Bits | 311 |
| Coding Rate | | 0.54754 |
| Modulation | | QPSK |
| Number of E-DCH Timeslots | Slots | 2 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 4 |
| Number of E-UCCH per TTI | | 4 |

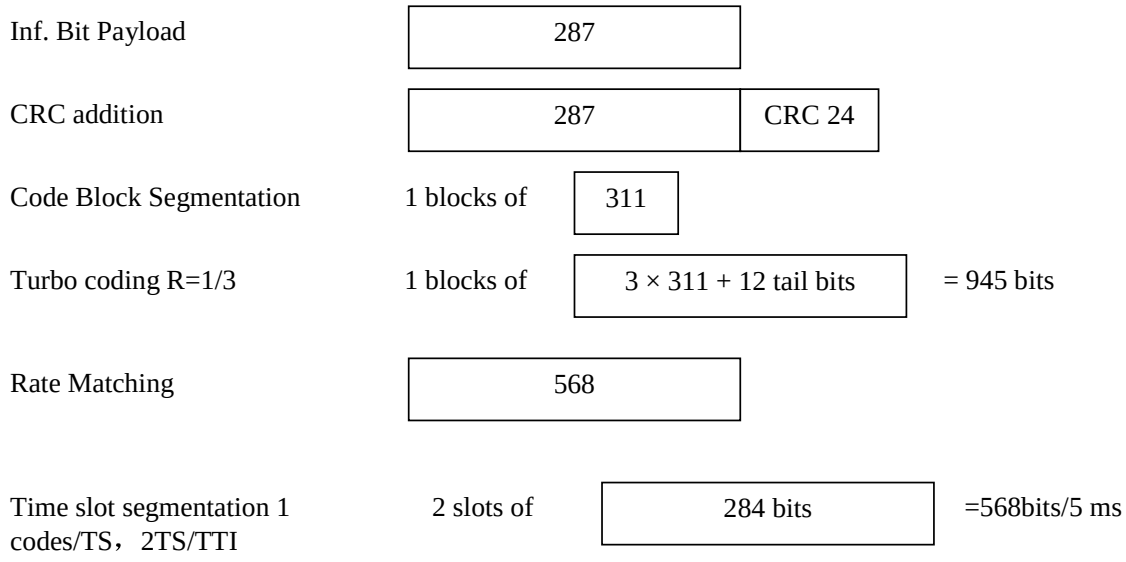



Figure A.12: Coding for E-DCH FRC1 (1.28 Mcps TDD Option)

## C6.1.2.2 Fixed reference channel 2(FRC2)

**Table A.11: E-DCH Fixed reference channel 2 (1.28Mcps TDD option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 514 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 2570 |
| Number Code Blocks | Blocks | 1 |
| Number of coded bits per TTI | Bits | 2594 |
| Coding Rate | | 0.63454 |
| Modulation | | 16QAM |
| Number of E-DCH Timeslots | Slots | 3 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 2 |
| Number of E-UCCH per TTI | | 2 |

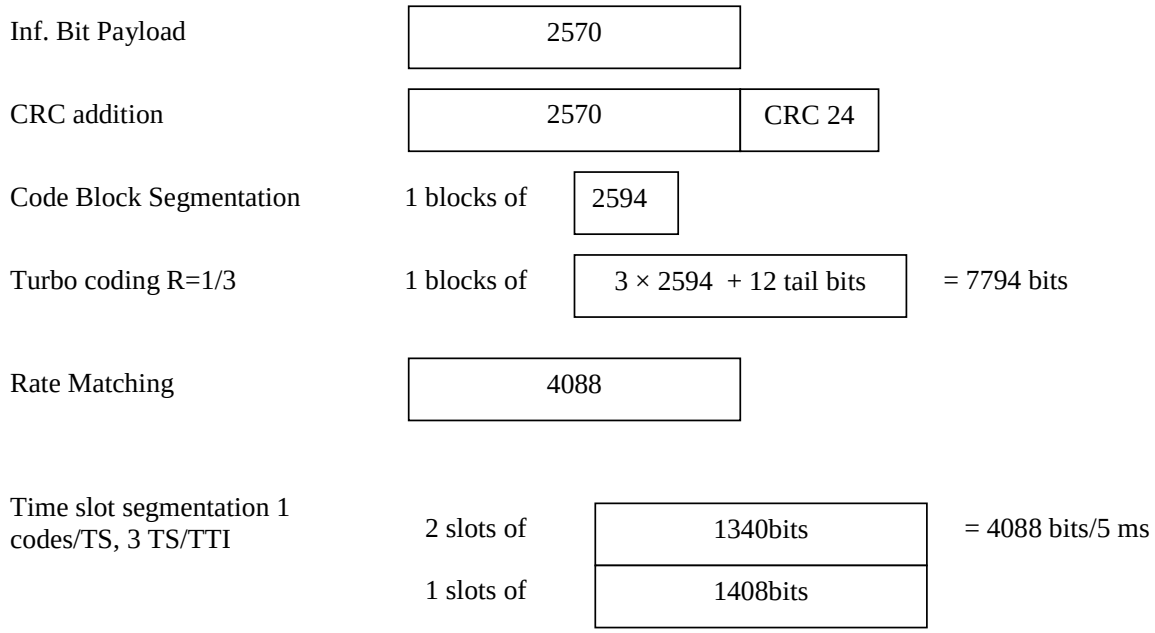



Figure A.13: Coding for E-DCH FRC2 (1.28 Mcps TDD Option)C.6.1.3 7,68 Mcps TDD Option

### C.6.1.3.1 Fixed Reference Channel 1 (FRC1)

**Table C.6.1.3a: E-DCH Fixed Reference Channel 1 (7.68 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 35.9 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 359 |
| Number Code Blocks | Blocks | 1 |
| Number of coded bits per TTI | Bits | 1200 |
| Coding Rate | | 0.323 |
| Modulation | | QPSK |
| Number of E-DCH Timeslots | Slots | 6 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 32 |
| Number of E-UCCH per TTI | | 4 |

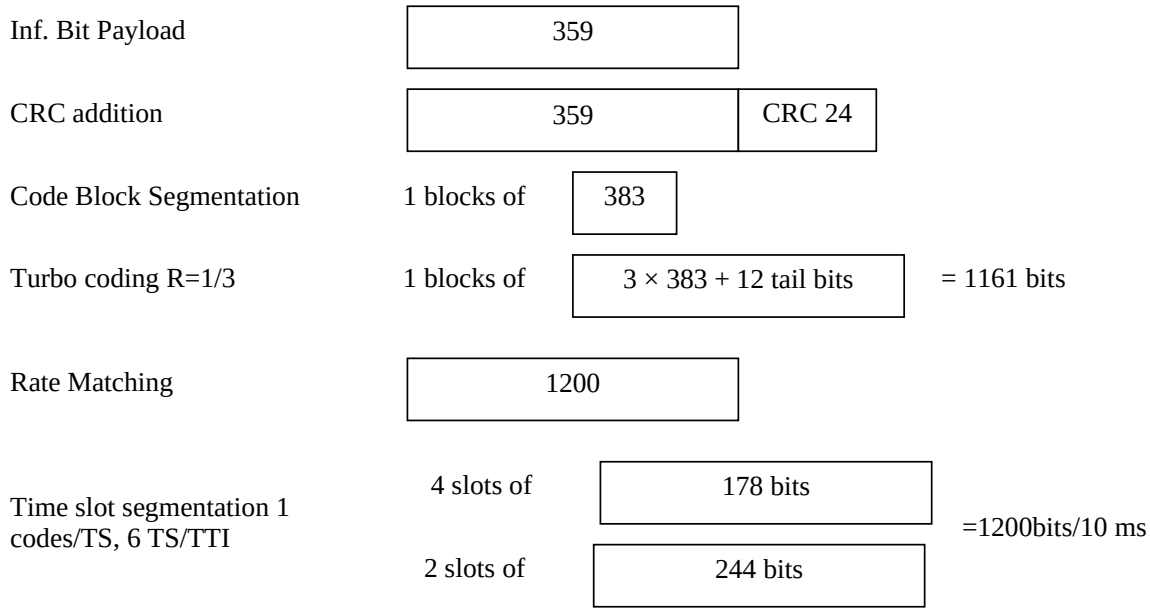



**Figure A.12: Coding for E-DCH FRC1 (7.68 Mcps TDD Option)**

### C.6.1.3.2 Fixed Reference Channel 2 (FRC2)

**Table C.6.1.3b: E-DCH Fixed Reference Channel 2 (7.68 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 1083.1 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 10831 |
| Number Code Blocks | Blocks | 3 |
| Number of coded bits per TTI | Bits | 22272 |
| Coding Rate | | 0.488 |
| Modulation | | 16QAM |
| Number of E-DCH Timeslots | Slots | 6 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 4 |
| Number of E-UCCH per TTI | | 2 |

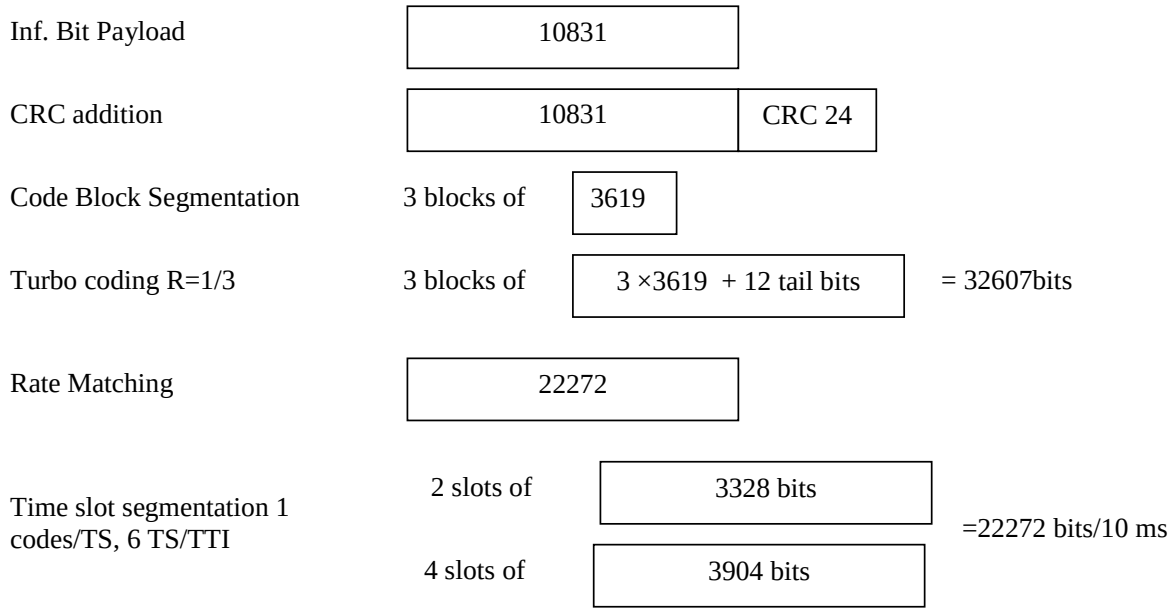



**Figure A.13: Coding for E-DCH FRC2 (7.68 Mcps TDD Option)**

### C.6.1.3.3 Fixed Reference Channel 3 (FRC3)

**Table C.6.1.3c: E-DCH Fixed Reference Channel 3 (7.68 Mcps TDD Option)**

| Parameter | Unit | Value |
|---|---|---|
| Maximum information bit throughput | kbps | 2085.1 |
| Information Bit Payload ( $N_{INF}$ ) | Bits | 20851 |
| Number Code Blocks | Blocks | 5 |
| Number of coded bits per TTI | Bits | 28992 |
| Coding Rate | | 0.720 |
| Modulation | | 16QAM |
| Number of E-DCH Timeslots | Slots | 6 |
| Number of E-DCH codes per TS | Codes | 1 |
| Spreading factor | SF | 2 |
| Number of E-UCCH per TTI | | 2 |

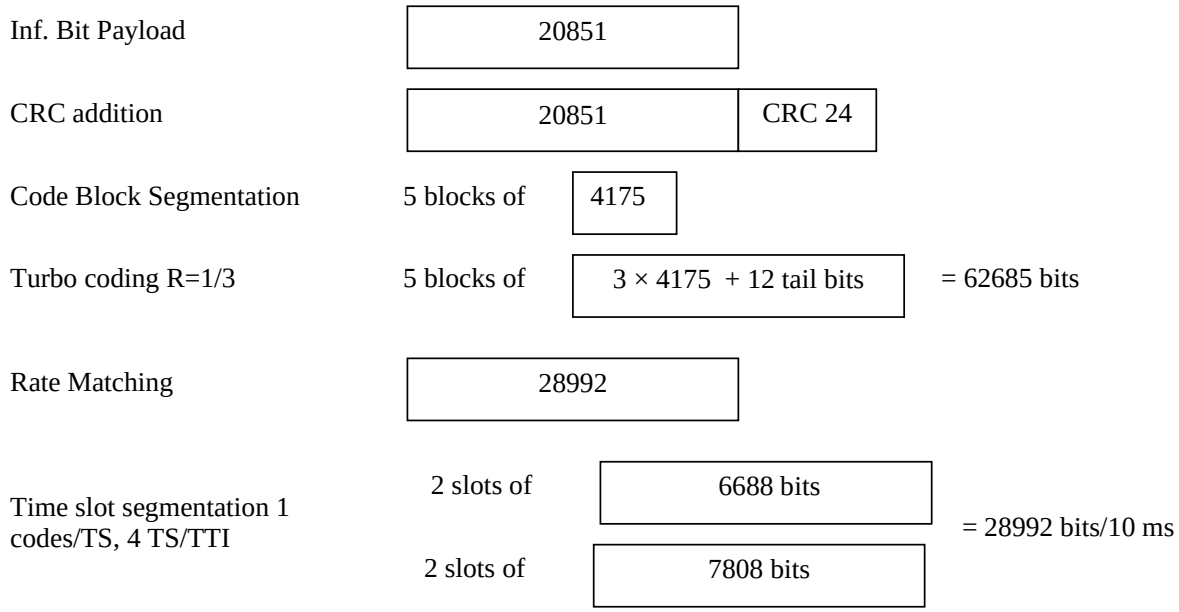



**Figure A.14: Coding for E-DCH FRC3 (7.68 Mcps TDD Option)**

# Annex D (normative): Propagation conditions

## D.1 Test Environments

Table D.1 details the test services, the information data and the propagation conditions.

**Table D.1: Test Environments for UE Performance Specifications**

| Test Services | Information Data Rate | Static | Multipath Case 1 | Multipath Case 2 | Multipath Case 3 |
|---|---|---|---|---|---|
| | | Performance metric | | | |
| Paging Message | | | - | - | - |
| FACH Message | | | - | - | - |
| Circuit Switched Services | 12,2 kbps | BLER < | BLER < | BLER < | BLER < |
| | 64 kbps | BLER < | BLER < | BLER < | BLER < |
| | 144 kbps | BLER < | BLER < | BLER < | BLER < |
| | 384 kbps | BLER < | BLER < | BLER < | BLER < |
| | 2048 kbps | BLER < | - | - | - |
| Packet Switched Data | TBD | TBD | TBD | TBD | TBD |

## D.2 Propagation Conditions

### D.2.1 Static propagation condition

The propagation for the static performance measurement is an Additive White Gaussian Noise (AWGN) environment. No fading and multi-paths exist for this propagation model.

### D.2.2 Multi-path fading propagation conditions

#### D.2.2.1 3,84 Mcps TDD Option

Table D.2.2.1 shows propagation conditions that are used for the performance measurements in multi-path fading environment. All taps have classical Doppler spectrum.

**Table D.2.2.1: 3,84 Mcps Propagation Conditions for Multi path Fading Environments**

| Case 1, speed 3km/h | | Case 2, speed 3 km/h | | Case 3, speed 120 km/h | | Case 4, speed 3 km/h | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 976 | -10 | 976 | 0 | 260 | -3 | 976 | 0 |
| | | 12000 | 0 | 521 | -6 | | |
| | | | | 781 | -9 | | |

**Table D.2.2.1A: Propagation Conditions for Multi-Path Fading Environments for HSDPA Performance Requirements**

| ITU Pedestrian A Speed 3km/h (PA3) | | ITU Pedestrian B Speed 3Km/h (PB3) | | ITU vehicular A Speed 30km/h (VA30) | | ITU vehicular A Speed 120km/h (VA120) | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 110 | -9.7 | 200 | -0.9 | 310 | -1.0 | 310 | -1.0 |
| 190 | -19.2 | 800 | -4.9 | 710 | -9.0 | 710 | -9.0 |
| 410 | -22.8 | 1200 | -8.0 | 1090 | -10.0 | 1090 | -10.0 |
| | | 2300 | -7.8 | 1730 | -15.0 | 1730 | -15.0 |
| | | 3700 | -23.9 | 2510 | -20 | 2510 | -20 |

**Table D.2.2.1B: Propagation Conditions for Multi-Path Fading Environments for MTCH and MCCH Performance Requirements under an extended delay spread environment**

| Extended Delay Spread Speed 3km/h (EDS) | |
|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 |
| 310 | -1 |
| 710 | -9 |
| 1090 | -10 |
| 1730 | -15 |
| 2510 | -20 |
| 12490 | -10 |
| 12800 | -11 |
| 13200 | -19 |
| 13580 | -20 |
| 14220 | -25 |
| 15000 | -30 |
| 27490 | -20 |
| 27800 | -21 |
| 28200 | -29 |
| 28580 | -30 |
| 29220 | -35 |
| 30000 | -40 |

## D.2.2.2 1,28 Mcps TDD Option

Table D2.2.2 shows propagation conditions that are used for the performance measurements in multi-path fading environment. All taps have classical Doppler spectrum.

**Table D.2.2.2.1 : 1,28Mcps Propagation Conditions for Multi-Path Fading Environments**

| Case 1, speed 3km/h | | Case 2, speed 3km/h | | Case 3, speed 120km/h | |
|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 |
| 2928 | -10 | 2928 | 0 | 781 | -3 |
| | | 12000 | 0 | 1563 | -6 |
| | | | | 2344 | -9 |

**Table D.2.2.2.2: 1,28Mcps Propagation Conditions for Multi-Path Fading Environments for HSDPA Performance Requirements**

| ITU Pedestrian A Speed 3km/h (PA3) | | ITU Pedestrian B Speed 3Km/h (PB3) | | ITU vehicular A Speed 30km/h (VA30) | | ITU vehicular A Speed 120km/h (VA120) | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 110 | -9.7 | 200 | -0.9 | 310 | -1.0 | 310 | -1.0 |
| 190 | -19.2 | 800 | -4.9 | 710 | -9.0 | 710 | -9.0 |
| 410 | -22.8 | 1200 | -8.0 | 1090 | -10.0 | 1090 | -10.0 |
| | | 2300 | -7.8 | 1730 | -15.0 | 1730 | -15.0 |
| | | 3700 | -23.9 | 2510 | -20 | 2510 | -20 |

**Table D.2.2.2.3: Propagation Conditions for Multi-Path Fading Environments for MBSFN Demodulation Performance Requirements**

| MBSFN model 1 | | MBSFN channel model 2 | |
|---|---|---|---|
| Speed for Band a, b, c 30 km/h | | Speed for Band a, b, c 30 km/h | |
| Speed for Band d: 23 km/h | | Speed for Band d: 23 km/h | |
| Speed for Band e: 26 km/h | | Speed for Band e: 26km/h | |
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0.0 | 0 | 0.0 |
| 310 | -1.0 | 310 | -1.0 |
| 710 | -9.0 | 710 | -9.0 |
| 1090 | -10.0 | 1090 | -10.0 |
| 1730 | -15.0 | 1730 | -15.0 |
| 2510 | -20.0 | 2510 | -20.0 |
| 2734 | -6.6 | 5859 | -6.8 |
| 3044 | -7.6 | 6169 | -7.8 |
| 3444 | -15.6 | 6569 | -15.8 |
| 3824 | -16.6 | 6949 | -16.8 |
| 4464 | -21.6 | 7589 | -21.8 |
| 5469 | -8.5 | 10938 | -13.3 |
| 5779 | -9.5 | 11248 | -14.3 |
| 6179 | -17.5 | 11648 | -22.3 |
| 6559 | -18.5 | 12028 | -23.3 |
| 8428 | -12.6 | 15459 | -15.0 |
| 8738 | -13.6 | 15769 | -16.0 |
| 9138 | -21.6 | 16169 | -24.0 |

Note: D.2.2.2.3 shows propagation conditions that are used for MBSFN demodulation performance measurements in multi-path fading environment. All taps have classical Doppler spectrum.

In the case of Rx diversity, the fading of the signals and the AWGN signals provided in each receiver antenna port shall be independent.

## D.2.2.3 7,68 Mcps TDD Option

Tables D.2.2.3.1 and D.2.2.3.2 shows propagation conditions that are used for the performance measurements in multi-path fading environment for frequency bands a, b and c in section 4.2. All taps have classical Doppler spectrum.

**Table D.2.2.3.1: Propagation Conditions for Multi path Fading Environmentsfor operations referenced in 4.2 a), 4.2 b) and 4.2 c)**

| Case 1 speed 3km/h | | Case 2 speed 3 km/h | | Case 3 speed 120 km/h | | CASE 4 speed 50 km/h * | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 976 | -10 | 976 | 0 | 260 | -3 | 976 | -10 |
| | | 12000 | 0 | 521 | -6 | | |
| | | | | 781 | -9 | | |

**Table D.2.2.3.2: Propagation Conditions for Multi-Path Fading Environments for HSDPA Performance Requirements for operations referenced in 4.2 a), 4.2 b) and 4.2 c)**

| ITU Pedestrian A Speed 3km/h (PA3) | | ITU Pedestrian B Speed 3Km/h (PB3) | | ITU vehicular A Speed 30km/h (VA30) | | ITU vehicular A Speed 120km/h (VA120) | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 110 | -9.7 | 200 | -0.9 | 310 | -1.0 | 310 | -1.0 |
| 190 | -19.2 | 800 | -4.9 | 710 | -9.0 | 710 | -9.0 |
| 410 | -22.8 | 1200 | -8.0 | 1090 | -10.0 | 1090 | -10.0 |
| | | 2300 | -7.8 | 1730 | -15.0 | 1730 | -15.0 |
| | | 3700 | -23.9 | 2510 | -20 | 2510 | -20 |

Tables D.2.2.3.3 and D.2.2.3.4 shows propagation conditions that are used for the performance measurements in multi-path fading environment for frequency band d in section 4.2.

**Table D.2.2.3.3: Propagation Conditions for Multi path Fading Environments for operations referenced in 4.2 d)**

| Case 1 speed 2.3km/h | | Case 2 speed 2.3 km/h | | Case 3 speed 92 km/h | | Case 4 speed 38 km/h * | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 976 | -10 | 976 | 0 | 260 | -3 | 976 | -10 |
| | | 12000 | 0 | 521 | -6 | | |
| | | | | 781 | -9 | | |

**Table D.2.2.3.4: Propagation Conditions for Multi-Path Fading Environments for HSDPA Performance Requirements for operations referenced in 4.2 d)**

| ITU Pedestrian A Speed 2.3km/h (PA3) | | ITU Pedestrian B Speed 2.3Km/h (PB3) | | ITU vehicular A Speed 23 km/h (VA30) | | ITU vehicular A Speed 92 km/h (VA120) | |
|---|---|---|---|---|---|---|---|
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 110 | -9.7 | 200 | -0.9 | 310 | -1.0 | 310 | -1.0 |
| 190 | -19.2 | 800 | -4.9 | 710 | -9.0 | 710 | -9.0 |
| 410 | -22.8 | 1200 | -8.0 | 1090 | -10.0 | 1090 | -10.0 |
| | | 2300 | -7.8 | 1730 | -15.0 | 1730 | -15.0 |
| | | 3700 | -23.9 | 2510 | -20 | 2510 | -20 |

**Table D.2.2.3.5: Propagation Conditions for Multi-Path Fading Environments for MTCH and MCCH Performance Requirements under an extended delay spread environment**

| Extended Delay Spread | | | |
|---|---|---|---|
| Operations referenced in 4.2 a), 4.2 b) and 4.2 c) Speed 3km/h (EDS) | | Operations referenced in 4.2 d) Speed 2.3km/h (EDS) | |
| Relative Delay [ns] | Relative Mean Power [dB] | Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 | 0 | 0 |
| 310 | -1 | 310 | -1 |
| 710 | -9 | 710 | -9 |
| 1090 | -10 | 1090 | -10 |
| 1730 | -15 | 1730 | -15 |
| 2510 | -20 | 2510 | -20 |
| 12490 | -10 | 12490 | -10 |
| 12800 | -11 | 12800 | -11 |
| 13200 | -19 | 13200 | -19 |
| 13580 | -20 | 13580 | -20 |
| 14220 | -25 | 14220 | -25 |
| 15000 | -30 | 15000 | -30 |
| 27490 | -20 | 27490 | -20 |
| 27800 | -21 | 27800 | -21 |
| 28200 | -29 | 28200 | -29 |
| 28580 | -30 | 28580 | -30 |
| 29220 | -35 | 29220 | -35 |
| 30000 | -40 | 30000 | -40 |

# Annex E (normative): Common RF test conditions

## E.1 General

This normative annex specifies the common RF test conditions that are needed for setting a connection and channels that are needed during a connection.

## E.2 Connection Set-up

Table E.2 describes the downlink Physical Channels that are required for connection set up.

**Table E.2: Downlink Physical Channels required for connection set-up**

| Physical Channel |
|---|
| PSCH |
| PCCPCH |
| DPCH |
| |
| PICH |

## E.3 During connection

The following clauses describes the uplink and downlink Physical Channels that are transmitted during a connection i.e., when measurements are done.

### E.3.1 Measurement of Tx Characteristics

Table E.3.1 is applicable for measurements on the Transmitter Characteristics (Clause 5) with the exception of clause 5.2 (UE maximum output power), 5.3 (Frequency Stability), 5.4.1 (Uplink Power Control). For these cases certain parameters are defined individually.

**Table E.3.1.1: Common downlink test condition RF parameters**

| Physical Channel | Power |
|---|---|
| DPCH | -93 dBm Data content PRBS |
| PSCH | TBD |
| PCCPCH | TBD |
| | |
| PICH | TBD |
| | |
| OCNS | Nothing |

**Table E.3.1.2: Common TX test parameters**

| Parameter | Value/description |
|---|---|
| UL Reference measurement channel | 12,2kbps, according to annex C.2.1 |
| Uplink Power Control | SS level and signalling values such that UE transmits maximum power. |
| Data content | real life (sufficient irregular) |

## E.3.2 Measurement of Rx Characteristics

Table E.3.2 is applicable for measurements on the Receiver Characteristics (Clause 6)

**Table E.3.2: Downlink Physical Channels transmitted during a connection.**

| Physical Channel | Power |
|---|---|
| CPICH | TBD |
| PCCPCH | TBD |
| SCH | TBD |
| PICH | TBD |
| DPCH | TBD |

## E.3.3 Measurement of Performance requirements

Table E.3.3 is applicable for measurements on the Performance requirements (Clause 7)

**Table E.3.3: Downlink Physical Channels transmitted during a connection.**

| Physical Channel | Power | Note |
|---|---|---|
| CPICH | TBD | |
| PCCPCH | TBD | |
| SCH | TBD | |
| PICH | TBD | |
| DPCH | TBD | |
| OCNS | TBD | |

# Annex F (normative): General test conditions and declarations

The requirements of this clause apply to all applicable tests in the present document.

Many of the tests in the present document measure a parameter relative to a value that is not fully specified in the UE specifications. For these tests, the Minimum Requirement is determined relative to a nominal value specified by the manufacturer.

When specified in a test, the manufacturer shall declare the nominal value of a parameter, or whether an option is supported.

.In all the relevant clauses in this clause all Bit Error Ratio (BER), Block Error Ratio (BLER) measurements shall be carried out according to the general rules for statistical testing in annex F.6.

## F.1 Acceptable uncertainty of Test System

The maximum acceptable uncertainty of the Test System is specified below for each test, where appropriate. The Test System shall enable the stimulus signals in the test case to be adjusted to within the specified range, and the equipment under test to be measured with an uncertainty not exceeding the specified values. All tolerances and uncertainties are absolute values, and are valid for a confidence level of 95 %, unless otherwise stated.

A confidence level of 95% is the measurement uncertainty tolerance interval for a specific measurement that contains 95% of the performance of a population of test equipment.

For RF tests it should be noted that the uncertainties in clause F.1 apply to the Test System operating into a nominal 50 ohm load and do not include system effects due to mismatch between the DUT and the Test System.

### F.1.1 Measurement of test environments

The measurement accuracy of the UE test environments defined in Annex G, Test environments shall be.

- Pressure ±5 kPa.
- Temperature ±2 degrees.
- Relative Humidity ±5 %.
- DC Voltage ±1,0 %.
- AC Voltage ±1,5 %.
- Vibration 10 %.
- Vibration frequency 0,1 Hz.

The above values shall apply unless the test environment is otherwise controlled and the specification for the control of the test environment specifies the uncertainty for the parameter.

## F.1.2 Measurement of transmitter

**Table F.1.2: Maximum Test System Uncertainty for transmitter tests**

| Clause | Maximum Test System Uncertainty |
|---|---|
| 5.2 UE Maximum Output Power | ±0.7 dB |
| 5.3 Frequency Stability | ± 10Hz |
| 5.4.1 Uplink power control | Relative<br>±[0.3] dB |
| 5.4.2 Minimum Transmit Power | ±1.0 dB |
| 5.4.3 Transmit OFF Power | ±1.5 dB |
| 5.4.4 Transmit ON/OFF Power | TBD |
| 5.4.5 Out-of-synchronisation handling of output power | ±0,4 dB |
| 5.5.1 Occupied Bandwidth | ±100 kHz |
| 5.5.2.1 Spectrum emission mask | ±1.5 dB |
| 5.5.2.2 ACLR | ± 0,8 dB |
| 5.5.3 Spurious emissions | ± 2.0 dB for UE and coexistenece bands for results > -60 dBm<br><br>± 3.0 dB for results < -60 dBm<br><br>Outside above:<br>f≤2.2GHz : ± 1.5 dB<br>2.2 GHz < f ≤ 4 GHz :<br>± 2.0 dB<br>f > 4 GHz : ±4.0 dB |
| 5.6 Transmit intermodulation: | Will be based on BS, need to work out freq and level ranges. |
| 5.7.1 Transmit modulation: EVM | ±2.5 % |
| 5.7.2 Transmit modulation: peak code domain error | ±1 dB |

## F.1.3 Measurement of receiver

**Table F.1.3 Maximum Test System Uncertainty for receiver tests**

| Clause | Maximum Test System Uncertainty |
|---|---|
| 6.2 Reference Sensitivity Level | ± 0.7 dB |
| 6.3 maximum input level: | TBD |
| 6.4 Adjacent Channel Selectivity (ACS) | Overall system uncertainty<br>± 1.1 dB |
| 6.5 Blocking Characteristics (3,84 Mcps TDD option) | Using ± 0.7 dB for signal and interferer as currently defined, and 68 dB ACLR @ 10 MHz.System error with f <15 MHz offset:<br>± 1.4 dB<br><br>f >= 15 MHz offset and f ≤ 2.2 GHz: ± 1.0 dB<br>2.2 GHz < f ≤ 4 GHz : ±1.7 dB<br>f > 4 GHz: ±3.1 dB |
| 6.5 Blocking Characteristics (1,28 Mcps TDD option) | Using ± 0.7 dB for signal and interferer as currently defined, and 68 dB ACLR @ 3,2 MHz.System error with f <4,8 MHz offset:<br>± 1.4 dB<br><br>f >= 4,8 MHz offset and f ≤ 2.2 GHz: ± 1.0 dB<br>2.2 GHz < f ≤ 4 GHz : ±1.7 dB<br>f > 4 GHz: ±3.1 dB |
| 6.6 Spurious Response | f < 2.2 GHz: ±1.0dB<br>2.2 < f < 4GHz: ±1.7dB<br>f > 4 GHz: ±3.1dB |
| 6.7 Intermodulation Characteristics | ± 1.3 dB<br><br>with Formula =<br>$\sqrt{(2 \cdot CW_level_error)^2 + (\mathrm{mod}_level_error)^2 + (wanted_signal_level_error)^2}$<br>(Using CW interferer ±0.5 dB,<br>modulated interferer ±0.5 dB,<br>wanted signal ±0.7 dB) |
| 6.8 Spurious Emissions | ± 3.0 dB for UE receive band<br>Outside above:<br>f≤2.2GHz : ± 2.0dB<br>2.2 GHz < f ≤ 4 GHz : ± 2.0 dB<br>f > 4 GHz : ±4.0 dB |

## F.1.4 Performance requirement

**Table F.1.4 Maximum Test System Uncertainty for Performance Requirements**

| Clause | Maximum Test System Uncertainty |
|---|---|
| 7.6 Uplink Power Control | 0.3 dB (relative tolerance for 10 dB stepsize) |

## F.1.5 Requirements for support of RRM

TBD

## F.1.6 Performance requirement (HSDPA)

**Table F.1.6: Maximum Test System Uncertainty for Performance Requirements (HSDPA)**

| Clause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 9.3.1 HS-DSCH throughput for Fixed Reference Channels | $\hat{I}_{or}/I_{oc}$ ±0.6 dB<br>$I_{oc}$ ±1.0 dB<br>$\frac{E_c}{I_{or}}$ ±0.1 dB | 0.1 dB uncertainty in Ec/Ior ratio<br>Worst case gain uncertainty due to the fader from the calibrated static profile is ±0.5 dB per output<br>In addition the same ±0.3 dB $\hat{I}_{or}/I_{oc}$ ratio error as 7.2.<br>These are uncorrelated so can be RSS.<br>Overall error in $\hat{I}_{or}/I_{oc}$ is $(0.5^2 + 0.3^2)^{0.5}$ = 0.6 dB |
| 9.3.2 HS-DSCH throughput for Variable Reference Channels | Same as 9.3.1 | Same as 9.3.1 |
| 9.3.3 Reporting of HS-DSCH Channel Quality Indicator | No test system uncertainty applied | |
| 9.3.4 HS-SCCH Detection Performance | Same as 9.3.1 | Same as 9.3.1 |

# F.2 Test Tolerances (This clause is informative)

The Test Tolerances defined in this clause have been used to relax the Minimum Requirements in the present document to derive the Test Requirements.

The Test Tolerances are derived from Test System uncertainties, regulatory requirements and criticality to system performance. As a result, the Test Tolerances may sometimes be set to zero.

The test tolerances should not be modified for any reason e.g. to take account of commonly known test system errors (such as mismatch, cable loss, etc.).

## F.2.1 Transmitter

**Table F.2.1 Test Tolerances for transmitter tests.**

| Clause | Test Tolerance |
|---|---|
| 5.2 Maximum Output Power | 0,7 dB |
| 5.3 UE Frequency Stability | 10 Hz |
| 5.4.1.1 Uplink Power Control, Initial Accuracy | +1,0 dB |
| 5.4.1.2 Uplink Power Control, differential Accuracy | 1dB step: 0,1 dB<br>2 dB step: 0,15 dB<br>3dB step: 0,2 dB<br>10 dB step: 0,5 dB<br>20 dB step: 0,7 dB<br>30 dB step: 0,7 dB<br>>30 dB step: 1,0 dB |
| 5.4.1.3 Open Loop Power Control | +1,0 dB |
| 5.4.2 Minimum Transmit Power | 1,0 dB |
| 5.4.5 Out-of-synchronisation handling of output power: $\frac{\Sigma DPCH_E_c}{I_{or}}$ | 0,4 dB |
| 5.4.5 Out-of-synchronisation handling of output power: transmit ON/OFF time | 0 ms |
| 5.5.1 Transmit OFF power | 1,5 dB |
| 5.5.1 Occupied Bandwidth | 0 kHz |
| 5.5.2.1 Spectrum emission mask | 1,5 dB |
| 5.5.2.2 Adjacent Channel Leakage Power Ratio (ACLR) | 0.8 dB |
| 5.5.3 Spurious Emissions | 0 dB |
| 5.6 Transmit Intermodulation | 0 dB |
| 5.7.1 Error Vector Magnitude | 0% |
| 5.7.2 Peak code domain error | 1,0 dB |

NOTE: Unless explicitly specified test tolerances are set to be equal in the case of 3,84 Mcps TDD Option as well as in the case of 1,28 Mcps TDD Option.

## F.2.2 Receiver

**Table F.2.2 Test Tolerances for receiver tests.**

| Clause | Test Tolerance |
|---|---|
| 6.2 Reference sensitivity level | 0.7 dB |
| 6.4 Adjacent channel selectivity | 0 dB |
| 6.5 Blocking characteristics | 0 dB |
| 6.6 Spurious Response | 0 dB |
| 6.7 Intermodulation Characteristics | 0 dB |
| 6.8 Spurious emissions | 0 dB |

NOTE: Unless explicitly specified test tolerances are set to be equal in the case of 3,84 Mcps TDD Option as well as in the case of 1,28 Mcps TDD Option.

## F.2.3 Performance requirements

**Table F.2.3 Test Tolerances for Performance Requirements.**

| Clause | Test Tolerance |
|---|---|
| 7.2, Demodulation in Static Propagation Condition | 0.3 dB for $\hat{I}_{or}/I_{oc}$ |
| 7.3, Demodulation of DCH in Multiplath Fading conditions | 0.6 dB for $\hat{I}_{or}/I_{oc}$ |
| 7.4, Base Station Transmit diversity modes | 0.8 dB for $\hat{I}_{or}/I_{oc}$ |
| 7.6 Uplink Power Control | 0.5 dB (relative tolerance for 10 dB stepsize) |

## F.2.4 Requirements for support of RRM

### F.2.4.1 TDD/TDD Cell Reselection on intra-frequency cells

#### F.2.4.1.1 3,84 Mcps TDD Option

Void.

#### F.2.4.1.2 1,28 Mcps Option

The test tolerances TT1 and TT2 given in table 8.2.3.1.5.2 shall not exceed [0,2 dB] for the interacting cells 1, 2 and [0,7 dB] for the additional (e.g. noniteracting) cells 3-6, respectively with a DRX cycle length given in table F.2.4.1A.

**Table F.2.4.1A: $T_{measureNTDD}$, $T_{evaluateNTDD}$, $T_{measureTDD}$, $T_{evaluateTDD}$, $T_{measureFDD}$, $T_{evaluateFDD}$ and $T_{measureGSM}$ (1,28 Mcps TDD Option)**

| DRX cycle length [s] | $N_{serv}$ [number of successive measurements] | $T_{measureNTDD}$ [s] (number of DRX cycles) | $T_{evaluateNTDD}$ [s] (number of DRX cycles) | $T_{measureTDD}$ [s] (number of DRX cycles) | $T_{evaluateTDD}$ [s] (number of DRX cycles) |
|---|---|---|---|---|---|
| 0,08 | 4 | 0,64 (8 DRX cycles) | 2,56 (32 DRX cycles) | 0,64 (8 DRX cycles) | 2,56 (32 DRX cycles) |
| 0,16 | 4 | 0,64 (4) | 2,56 (16) | 0,64 (4) | 2,56 (16) |
| 0,32 | 4 | 1,28 (4) | 5,12 (16) | 1,28 (4) | 5,12 (16) |
| 0,64 | 4 | 1,28 (2) | 5,12 (8) | 1,28 (2) | 5,12 (8) |
| 1,28 | 2 | 1,28 (1) | 6,4 (5) | 1,28 (1) | 6,4 (5) |
| 2,56 | 2 | 2,56 (1) | 7,68 (3) | 2,56 (1) | 7,68 (3) |
| 5,12 | 1 | 5,12 (1) | 10,24 (2) | 5,12 (1) | 10,24 (2) |

NOTE: In idle mode, UE shall support DRX cycles lengths 0.64, 1,28, 2.56 and 5.12 s.

#### F.2.4.1.3 7,68 Mcps TDD Option

Void

## F.2.5 Performance requirements (HSDPA)

**Table F.2.5: Test Tolerances for Performance Requirements (HSDPA).**

| Clause | Test Tolerance |
|---|---|
| 9.3.1 HS-DSCH throughput for Fixed Reference Channels | 0.6 dB for $\hat{I}_{or}/I_{oc}$<br>0.1 dB for Ec/Ior |
| 9.3.2 HS-DSCH throughput for Variable Reference Channels | Same as 9.3.1 |
| 9.3.3 Reporting of HS-DSCH Channel Quality Indicator | No test tolerances applied |
| 9.3.4 HS-SCCH Detection Performance | Same as 9.3.1 |

# F.3 Interpretation of measurement results

The measurement results returned by the Test System are compared - without any modification - against the Test Requirements as defined by the shared risk principle.

The Shared Risk principle is defined in ETR 273 Part 1 sub-part 2 clause 6.5.

The actual measurement uncertainty of the Test System for the measurement of each parameter shall be included in the test report.

The recorded value for the Test System uncertainty shall be, for each measurement, equal to or lower than the appropriate figure in clause F.1 of the present document.

If the Test System for a test is known to have a measurement uncertainty greater than that specified in clause F.1, it is still permitted to use this apparatus provided that an adjustment is made value as follows.

Any additional uncertainty in the Test System over and above that specified in clause F.1 shall be used to tighten the Test Requirement – making the test harder to pass. (For some tests e.g. receiver tests, this may require modification of stimulus signals). This procedure will ensure that a Test System not compliant with clause F.1does not increase the chance of passing a device under test where that device would otherwise have failed the test if a Test System compliant with clause F.1 had been used.

# F.4 Derivation of Test Requirements (This clause is informative)

The Test Requirements in the present document have been calculated by relaxing the Minimum Requirements of the core specification using the Test Tolerances defined in clause F.2. When the Test Tolerance is zero, the Test Requirement will be the same as the Minimum Requirement. When the Test Tolerance is non-zero, the Test Requirements will differ from the Minimum Requirements, and the formula used for this relaxation is given in table F.4.

# F.4.1 3.84Mcps TDD Option

**Table F.4.1 Derivation of Test Requirements**

| Test | Minimum Requirement in TS 25.102 | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|
| 5.2 Maximum Output Power | Power single code<br>Power class 1 (30 dBm)<br>Tolerance = +1/-3 dB<br>Power class 2 (24 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (21 dBm)<br>Tolerance = +2/-2 dB<br>Power multi code<br>Power class 2 (21 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (18 dBm)<br>Tolerance = ±2 dB<br>Power class 4 (10 dBm)<br>Tolerance = ± 4 dB | 0.7 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>For power classes 1 and 2 (single and multi):<br>Upper Tolerance limit = +1.7 dB<br>Lower Tolerance limit = -3.7 dB<br>For power class 3 (single and milti):<br>Upper Tolerance limit = +2.7 dB<br>Lower Tolerance limit = -2.7 dB<br>For power class 4 (single and multi)<br>Upper Tolerance limit = +4.7 dB<br>Lower Tolerance limit = -4.7 dB |
| 5.3 UE Frequency Stability | The UE modulated carrier frequency shall be accurate to within ±0.1 ppm compared to the carrier frequency received from the Node B. | 10 Hz | Formula: modulated carrier frequency error + TT<br><br>modulated carrier frequency error = ±(0.1 ppm + 10 Hz). |
| 5.4.1.1 Uplink Power Control, Initial accuracy | ± 9dB normal conditions<br>± 12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>± 10dB (normal conditions)<br>± 13dB (extreme conditions) |
| 5.4.1.2 Uplink Power Control, differential accuracy | SIR Target / Pow Step Tol<br><1 dB / 0.5 dB<br><2dB / 1 dB<br><3 dB / 1.5 dB<br><10 dB / 2 dB<br><20dB / 4 dB<br><30 dB / 6 dB<br>> 30 dB / 9 dB | TT<br>0.1 dB<br>0.15 dB<br>0.2 dB<br>0.5 dB<br>0.7 dB<br>0.7 dB<br>1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT |
| 5.4.1.3 Open Loop Power Control | ± 9dB normal conditions<br>± 12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>± 10dB (normal conditions)<br>± 13dB (extreme conditions) |
| 5.4.2 Minimum Transmit Power | UE minimum transmit power shall be less than –44 dBm | 1.0 dB | Formula:<br>UE minimum transmit power + TT<br>UE minimum transmit power = –43 dBm |

<table>
<tr><td>5.4.5 Out-of-synchronisation handling of output power:</td><td>$\frac{\Sigma DPCH_E_c}{I_{or}}$ levels<br>before A –4.6 dB<br>AB: -10 dB<br>BD: -16 dB<br>DE: -12 dB<br>EF: -6 dB<br>transmit ON/OFF time<br>200ms</td><td>0,4 dB for $\frac{\Sigma DPCH_E_c}{I_{or}}$<br>0 ms for timing measurement</td><td>Formulas:<br>Ratio between A and B + TT<br>Ratio between B and D - TT<br>Ratio between D and E - TT<br>Ratio between E and F + TT<br>transmit ON/OFF time + TT timing<br><br>$\frac{\Sigma DPCH_E_c}{I_{or}}$ levels:<br>Before A: -4.6<br>AB: -10 + 0,4 dB<br>BD: -16 – 0,4 dB<br>DE: -12 - 0,4 dB<br>EF: -6 + 0,4 dB<br>Uncertainty of OFF power measurement is handled by Transmit OFF power test and uncertainty of ON power measurement is handled by Minimum output power test.transmit ON/OFF time<br>200ms</td></tr>
<tr><td>5.5.1 Transmit OFF power</td><td>Transmit OFF power shall be less than –65 dBm</td><td>1.5 dB</td><td>Formula: Transmit OFF power + TT<br>Transmit OFF power = –63.5 dBm</td></tr>
<tr><td>5.5.1 Occupied Bandwidth</td><td>The occupied channel bandwidth shall be less than 5 MHz based on a chip rate of 3,84 Mcps.</td><td>0 kHz</td><td>Formula: occupied channel bandwitdh: + TT<br><br>occupied channel bandwidth = 5.0 MHz</td></tr>
<tr><td>5.5.2.1 Spectrum emission mask</td><td>Minimum requirement defined in TS25.101 table 6.10.<br>The lower limit shall be –50 dBm / 3,84 MHz or which ever is higher.</td><td>1.5 dB</td><td>Formula: Minimum requirement + TT<br>Lower limit + TT<br>Add 1.5 to Minimum requirement entries in TS25.101 table 6.10<br>The lower limit shall be –48.5 dBm / 3,84 MHz or which ever is higher.</td></tr>
<tr><td>5.5.2.2 Adjacent Channel Leakage Power Ratio (ACLR)</td><td>Power Classes 2 and 3:<br>UE channel +5 MHz or -5 MHz, ACLR limit: 33 dB<br>UE channel +10 MHz or -10 MHz, ACLR limit: 43 dB</td><td>0.8 dB</td><td>Formula: ACLR limit - TT<br>Power Classes 2 and 3:<br>UE channel +5 MHz or -5 MHz, ACLR limit: 32.2 dB<br>UE channel +10 MHz or -10 MHz, ACLR limit: 42.2 dB</td></tr>
<tr><td rowspan="7">5.5.3 Spurious Emissions</td><td colspan="2"></td><td></td><td colspan="2">Formula: Minimum Requirement+ TT<br>Add zero to all the values of Minimum Requirements in table 5.5.3</td></tr>
<tr><td>Frequency Band</td><td>Minimum Requirement</td><td></td><td>Frequency Band</td><td>Minimum Requirement</td></tr>
<tr><td>9 kHz ≤ f < 150 kHz</td><td>–36dBm /1kHz</td><td>0 dB</td><td>9kHz ≤ f < 1GHz</td><td>–36dBm /1kHz</td></tr>
<tr><td>150 kHz ≤ f < 30 MHz</td><td>–36dBm /10kHz</td><td>0 dB</td><td>150 kHz ≤ f < 30 MHz</td><td>–36dBm /10kHz</td></tr>
<tr><td>30 MHz ≤ f < 1000 MHz</td><td>–36dBm /100kHz</td><td>0 dB</td><td>30 MHz ≤ f < 1000 MHz</td><td>–36dBm /100kHz</td></tr>
<tr><td rowspan="3">1 GHz ≤ f < 12.75 GHz</td><td rowspan="3">–30dBm /1MHz</td><td>0 dB</td><td>1 GHz ≤ f < 2.2 GHz</td><td>–30dBm /1MHz</td></tr>
<tr><td>0 dB</td><td>2.2 GHz ≤ f < 4 GHz</td><td>–30dBm /1MHz</td></tr>
<tr><td>0 dB</td><td>4 GHz ≤ f < 12.75 GHz</td><td>–30dBm /1MHz</td></tr>
</table>

| | | | | | |
|---|---|---|---|---|---|
| | 925 MHz < f < 935 MHz | –67dBm /100kHz | 0 dB | 925 MHz < f < 935 MHz | –67dBm /100kHz |
| | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz | 0 dB | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz |
| | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz | 0 dB | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz |
| | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz | 0 dB | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz |
| 5.6 Transmit Intermodulation | Intermodulation Product<br>5MHz -31 dBc<br>10MHz -41 dBc | | 0 dB | Formula: Intermodulation Product + TT<br>Intermodulation Product<br>5MHz -31 dBc<br>10MHz -41 dBc | |
| 5.7.1 Error Vector Magnitude | The Error Vector Magnitude shall not exceed 17.5 % | | 0% | Formula: EVM limit + TT<br>EVM limit = 17.5 % | |
| 5.7.2 Peak code domain error | The peak code domain error shall not exceed -21dB | | ±1.0 dB | Formula: Peak code domain error + TT<br>Peak code domain error = -20 dB | |
| 6.2 Reference sensitivity level | $\hat{I}_{or}$ = -105 dBm / 3,84 MHz<br>BER limit = 0.001 | | 0.7 dB | Formula: $\hat{I}_{or}$+ T<br>BER limit unchanged<br><br>$\hat{I}_{or}$ = -104.3 dBm / 3,84 MHz | |
| 6.4 Adjacent Channel Selectivity | $\hat{I}_{or}$ = -91 dBm / 3,84 MHz<br>Ioac (modulated) = -52 dBm/3,84 MHz<br>BER limit = 0.001 | | 0 dB | Formula: $\hat{I}_{or}$ unchanged<br>Ioac - TT<br>BER limit unchanged<br><br>Ioac = -52 dBm/3,84 MHz | |
| 6.5 Blocking Characteristics | See table 6.5.2a and 6.5.2b. in TS34.122<br>BER limit = 0.001 | | 0 dB | Formula:<br>$I_{blocking}$ (modulated) - TT (dBm/3,84MHz)<br>$I_{blocking}$ (CW) - TT (dBm)<br>BER limit unchanged | |
| 6.6 Spurious Response | Iblocking(CW) –44 dBm<br>Fuw:<br>Spurious response frequencies<br>BER limit = 0.001 | | 0 dB | Formula: $I_{blocking}$ (CW) - TT (dBm)<br>Fuw unchanged<br>BER limit unchanged<br><br>Iblocking(CW) -44 dBm | |
| 6.7 Intermodulation Characteristics | Iouw1 (CW) -46 dBm<br>Iouw2 (modulated) –46 dBm / 3,84 MHz<br>Fuw1 (offset) 10 MHz<br>Fuw2 (offset) 20 MHz<br><br>BER limit = 0.001 | | 0 dB | Formula: TBD<br>BER limit unchanged. | |
| 6.8 Spurious Emissions | | | | Formula: Maximum level+ TT<br>Add zero to all the values of Maximum Level in table 6.8.1. | |
| | Frequency Band | Maximum level | | Frequency Band | Maximum level |
| | 9kHz ≤ f < 1GHz | -57dBm /100kHz | 0 dB | 9kHz ≤ f < 1GHz | -57dBm /100kHz |
| | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz | -60 dBm / 3,84MHz | 0 dB | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz | -60 dBm / 3,84MHz |
| | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz | -47 dBm/1MHz | 0 dB | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz | -47 dBm/1MHz |

<table>
<tr><td rowspan="3"></td><td rowspan="3">1GHz ≤ f ≤ 12.75GHz</td><td rowspan="3">-47dBm /1MHz</td><td>0 dB</td><td>1GHz ≤ f ≤ 2.2GHz</td><td>-47dBm /1MHz</td></tr>
<tr><td>0 dB</td><td>2.2GHz < f ≤ 4GHz</td><td>-47dBm /1MHz</td></tr>
<tr><td>0 dB</td><td>4GHz < f ≤ 12.75GHz</td><td>-47dBm /1MHz</td></tr>
<tr><td rowspan="3">7.6 Downlink Power Control</td><td></td><td>TS #1,9</td><td>TS #7,14</td><td rowspan="3">0.5 dB for 10 dB change in output power, 0 otherwise.</td><td colspan="2" rowspan="3">Formula for 10 dB change in transmit power: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT</td></tr>
<tr><td>1st frame</td><td>15 ±4.0 dBm</td><td>5 ±0.5 dBm</td></tr>
<tr><td>2nd frame</td><td>15 ±4.0 dBm</td><td>15 ±4.0 dBm</td></tr>
<tr><td>9.3.1 HS-DSCH throughput for Fixed Reference Channels</td><td colspan="3">$\frac{E_c}{I_{or}}$ -10,-9.5 dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 10 dB</td><td>0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$</td><td colspan="2">Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged</td></tr>
<tr><td>9.3.2 HS-DSCH throughput for Variable Reference Channels</td><td colspan="3">$\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 8 and 10 dB</td><td>0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$</td><td colspan="2">Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged</td></tr>
<tr><td>9.3.3 Reporting of HS-DSCH Channel Quality Indicator</td><td colspan="3"></td><td>No test tolerances applied</td><td colspan="2"></td></tr>
<tr><td>9.3.4 HS-SCCH Detection Performance</td><td colspan="3">$\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 12 and 16 dB</td><td>0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$</td><td colspan="2">Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged</td></tr>
<tr><td colspan="7">* Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a).</td></tr>
</table>

# F.4.2 1.28Mcps TDD Option

## F.4.2.1 Transmitter

**Table F.4.2.1 Derivation of Transmitter Test Requirements**

| Test | Minimum Requirement in TS 25.102 | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|
| 5.2 Maximum Output Power | Power single code<br>Power class 1 (33 dBm)<br>Tolerance = +1/-3 dB<br>Power class 2 (24 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (21 dBm)<br>Tolerance = +2/-2 dB<br>Power multi code<br>Power class 2 (21 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (18 dBm)<br>Tolerance = ±2 dB<br>Power class 4 (27 dBm)<br>Tolerance = +1/-3 dB | 0.7 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>For power classes 1 and 2 (single and multi):<br>Upper Tolerance limit = +1.7 dB<br>Lower Tolerance limit = -3.7 dB<br>For power class 3 (single and milti):<br>Upper Tolerance limit = +2.7 dB<br>Lower Tolerance limit = -2.7 dB<br>For power class 4 (single and multi)<br>Upper Tolerance limit = +1.7 dB<br>Lower Tolerance limit = -3.7 dB |
| 5.3 UE Frequency Stability | The UE modulated carrier frequency shall be accurate to within □0.1 ppm compared to the carrier frequency received from the Node B. | 10 Hz | Formula: modulated carrier frequency error + TT<br><br>modulated carrier frequency error = □(0.1 ppm + 10 Hz). |
| 5.4.1.1 Uplink Power Control, Initial accuracy | □9dB normal conditions<br>□12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>□10dB (normal conditions)<br>□13dB (extreme conditions) |
| 5.4.1.2 Uplink Power Control, differential accuracy | SIR Target / Pow Step Tol / TT<br><1 dB / 0.5 dB / 0.1 dB<br><2dB / 1 dB / 0.15 dB<br><3 dB / 1.5 dB / 0.2 dB<br><10 dB / 2 dB / 0.5 dB<br><20dB / 4 dB / 0.7 dB<br><30 dB / 6 dB / 0.7 dB<br>> 30 dB / 9 dB / 1.0 dB | | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT |
| 5.4.1.3 Open Loop Power Control | □9dB normal conditions<br>□12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>□10dB (normal conditions)<br>□13dB (extreme conditions) |
| 5.4.2 Minimum Transmit Power | UE minimum transmit power shall be less than –49 dBm | 1.0 dB | Formula:<br>UE minimum transmit power + TT<br>UE minimum transmit power = –48 dBm |

| | | | | | |
|---|---|---|---|---|---|
| 5.4.5 Out-of-synchronisation handling of output power: | $\frac{\Sigma DPCH_E_c}{I_{or}}$ levels<br>before A –2.4 dB<br>AB: -10 dB<br>BD: -16 dB<br>DE: -12 dB<br>EF: -6 dB<br>transmit ON/OFF time 200ms | | 0,4 dB for $\frac{\Sigma DPCH_E_c}{I_{or}}$<br>0 ms for timing measurement | Formulas:<br>Ratio between A and B + TT<br>Ratio between B and D - TT<br>Ratio between D and E - TT<br>Ratio between E and F + TT<br>transmit ON/OFF time + TT timing<br><br>$\frac{\Sigma DPCH_E_c}{I_{or}}$ levels:<br>Before A: -4.6<br>AB: -10 + 0,4 dB<br>BD: -16 – 0,4 dB<br>DE: -12 - 0,4 dB<br>EF: -6 + 0,4 dB<br>Uncertainty of OFF power measurement is handled by Transmit OFF power test and uncertainty of ON power measurement is handled by Minimum output power test.transmit ON/OFF time<br>200ms | |
| 5.5.1 Transmit OFF power | Transmit OFF power shall be less than –65 dBm | | 1.5 dB | Formula: Transmit OFF power + TT<br>Transmit OFF power = –63.5 dBm | |
| 5.5.1 Occupied Bandwidth | The occupied channel bandwidth shall be less than 1.6 MHz based on a chip rate of 1.28 Mcps. | | 0 kHz | Formula: occupied channel bandwitdh: + TT<br><br>occupied channel bandwidth = 1.6 MHz | |
| 5.5.2.1 Spectrum emission mask | Minimum requirement defined in TS25.102 table 6.5A.<br>The lower limit shall be –55 dBm / 1.28 MHz or which ever is higher. | | 1.5 dB | Formula: Minimum requirement + TT<br>Lower limit + TT<br>Add 1.5 to Minimum requirement entries in TS25.102 table 6.5A<br>The lower limit shall be –53.5 dBm / 1.28 MHz or which ever is higher. | |
| 5.5.2.2 Adjacent Channel Leakage Power Ratio (ACLR) | Power Classes 2 and 3:<br>UE channel +1.6 MHz or -1.6MHz, ACLR limit: 33 dB<br>UE channel +3.2 MHz or -3.2 MHz, ACLR limit: 43 dB | | 0.8 dB | Formula: ACLR limit - TT<br>Power Classes 2 and 3:<br>UE channel +1.6 MHz or -1.6 MHz, ACLR limit: 32.2 dB<br>UE channel +3.2 MHz or -3.2MHz, ACLR limit: 42.2 dB | |
| 5.5.3 Spurious Emissions | | | | Formula: Minimum Requirement+ TT<br>Add zero to all the values of Minimum Requirements in table 5.5.3 | |
| | Frequency Band | Minimum Requirement | | Frequency Band | Minimum Requirement |
| | 9 kHz ≤ f < 150 kHz | –36dBm /1kHz | 0 dB | 9kHz ≤ f < 1GHz | –36dBm /1kHz |
| | 150 kHz ≤ f < 30 MHz | –36dBm /10kHz | 0 dB | 150 kHz ≤ f < 30 MHz | –36dBm /10kHz |
| | 30 MHz ≤ f < 1000 MHz | –36dBm /100kHz | 0 dB | 30 MHz ≤ f < 1000 MHz | –36dBm /100kHz |
| | 1 GHz ≤ f < 12.75 GHz | –30dBm /1MHz | 0 dB | 1 GHz ≤ f < 2.2 GHz | –30dBm /1MHz |
| | | | 0 dB | 2.2 GHz ≤ f < 4 GHz | –30dBm /1MHz |
| | | | 0 dB | 4 GHz ≤ f < 12.75 GHz | –30dBm /1MHz |

| | | | | | |
|---|---|---|---|---|---|
| | 925 MHz < f < 935 MHz | –67dBm /100kHz | 0 dB | 925 MHz < f < 935 MHz | –67dBm /100kHz |
| | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz | 0 dB | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz |
| | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz | 0 dB | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz |
| | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz | 0 dB | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz |
| 5.6 Transmit Intermodulation | Intermodulation Product<br>1.6MHz -31 dBc<br>3.2MHz -41 dBc | | 0 dB | Formula: Intermodulation Product + TT<br>Intermodulation Product<br>1.6MHz -31 dBc<br>3.2MHz -41 dBc | |
| 5.7.1 Error Vector Magnitude | The Error Vector Magnitude shall not exceed 17.5 % | | 0% | Formula: EVM limit + TT<br>EVM limit = 17.5 % | |
| 5.7.2 Peak code domain error | The peak code domain error shall not exceed -21dB | | □1.0 dB | Formula: Peak code domain error + TT<br>Peak code domain error = -20 dB | |

## F.4.2.2 Receiver

**Table F.4.2.2 Derivation of Receiver Test Requirements**

| Test | Minimum Requirement in TS 25.102 | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|
| 6.2 Reference sensitivity level | Îor = -108 dBm / 1.28 MHz<br>BER limit = 0.001 | 0.7 dB | Formula: Îor+ T<br>BER limit unchanged<br><br>Îor = -107.3 dBm / 1.28MHz |
| 6.4 Adjacent Channel Selectivity | Îor = -91 dBm / 1.28 MHz<br>Ioac (modulated) = -54 dBm/1.28 MHz<br>BER limit = 0.001 | 0 dB | Formula: Îor unchanged<br>Ioac - TT<br>BER limit unchanged<br><br>Ioac = -54 dBm/1.28MHz |
| 6.5 Blocking Characteristics | See table 6.5.2a and 6.5.2b. in TS34.122<br>BER limit = 0.001 | 0 dB | Formula:<br>$I_{blocking}$ (modulated) - TT (dBm/1.28MHz)<br>$I_{blocking}$ (CW) - TT (dBm)<br>BER limit unchanged |
| 6.6 Spurious Response | Iblocking(CW) –44 dBm<br>Fuw:<br>Spurious response frequencies<br>BER limit = 0.001 | 0 dB | Formula: $I_{blocking}$ (CW) - TT (dBm)<br>Fuw unchanged<br>BER limit unchanged<br><br>Iblocking(CW) -44 dBm |
| 6.7 Intermodulation Characteristics | Iouw1 (CW) -46 dBm<br>Iouw2 (modulated) –46 dBm / 1.28 MHz<br>Fuw1 (offset) 3.2MHz<br>Fuw2 (offset) 6.4 MHz<br><br>BER limit = 0.001 | 0 dB | Formula: TBD<br>BER limit unchanged. |
| 6.8 Spurious Emissions | | | Formula: Maximum level+ TT<br>Add zero to all the values of Maximum Level in table 6.8.1. |
| | Frequency Band / Maximum level | | Frequency Band / Maximum level |
| | 9kHz ≤ f < 1GHz / -57dBm /100kHz | 0 dB | 9kHz ≤ f < 1GHz / -57dBm /100kHz |
| | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz / -60 dBm / 3,84MHz | 0 dB | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz / -60 dBm / 3,84MHz |
| | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz / -47 dBm/1MHz | 0 dB | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz / -47 dBm/1MHz |
| | 1GHz ≤ f ≤ 12.75GHz / -47dBm /1MHz | 0 dB | 1GHz ≤ f ≤ 2.2GHz / -47dBm /1MHz |
| | | 0 dB | 2.2GHz < f ≤ 4GHz / -47dBm /1MHz |
| | | 0 dB | 4GHz < f ≤ 12.75GHz / -47dBm /1MHz |
| 7.6 Downlink Power Control | TS #1,9 / TS #7,14 | 0.5 dB for 10 dB change in output power, 0 otherwise. | Formula for 10 dB change in transmit power: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT |
| | 1st frame: 15 ±4.0 dBm / 5 ±0.5 dBm | | |
| | 2nd frame: 15 ±4.0 dBm / 15 ±4.0 dBm | | |

## F.4.2.3 Performance requirement

**Table F.4.2.3 Derivation of Performance Test Requirements**

| Test | Minimum Requirement in TS 25.102 | | | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|---|---|
| 7.2, Demodulation in Static Propagation Condition | $\hat{I}_{or}/I_{oc}$ =2.4 to 3.6dB<br>$I_{oc}$ = -60 dBm/1.28MHz | | | 0.3 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\hat{I}_{or}/I_{oc}$ = Minimum Requirement + TT<br>$I_{oc}$ unchanged<br>$\hat{I}_{or}/I_{oc}$ =2.7 to 3.9 dB |
| 7.3.1, Demodulation of DCH in Multiplath Fading case 1 | $\hat{I}_{or}/I_{oc}$ =15.8 to 23.9dB<br>$I_{oc}$ = -60 dBm/1.28MHz | | | 0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\hat{I}_{or}/I_{oc}$ = Minimum Requirement + TT<br>$I_{oc}$ unchanged<br>$\hat{I}_{or}/I_{oc}$ =16.4 to 24.5 dB |
| 7.3.2, Demodulation of DCH in Multiplath Fading case 2 | $\hat{I}_{or}/I_{oc}$ =9.8 to 14.4dB<br>$I_{oc}$ = -60 dBm/1.28MHz | | | 0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\hat{I}_{or}/I_{oc}$ = Minimum Requirement + TT<br>$I_{oc}$ unchanged<br>$\hat{I}_{or}/I_{oc}$ =10.4 to 15 dB |
| 7.3.3, Demodulation of DCH in Multiplath Fading case 3 | $\hat{I}_{or}/I_{oc}$ =9.0 to 14.3dB<br>$I_{oc}$ = -60 dBm/1.28MHz | | | 0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\hat{I}_{or}/I_{oc}$ = Minimum Requirement + TT<br>$I_{oc}$ unchanged<br>$\hat{I}_{or}/I_{oc}$ =9.6 to 14.9 dB |
| 7.6 Downlink Power Control | | TS #1,9 | TS #7,14 | 0.5 dB for 10 dB change in output power, 0 otherwise. | Formula for 10 dB change in transmit power: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT |
| | 1st frame | 15 ±4.0 dBm | 5 ±0.5 dBm | | |
| | 2nd frame | 15 ±4.0 dBm | 15 ±4.0 dBm | | |

## F.4.2.4 Performance of HSDPA

**Table F.4.2.4 Derivation of Performance of HSDPA Test Requirements**

| Test | Minimum Requirement in TS 25.102 | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|
| 9.3.1 HS-DSCH throughput for Fixed Reference Channels | $\frac{E_c}{I_{or}}$ -10,-9.5 dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 10 dB | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged |
| 9.3.2 HS-DSCH throughput for Variable Reference Channels | $\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 8 and 10 dB | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged |
| 9.3.3 Reporting of HS-DSCH Channel Quality Indicator | | No test tolerances applied | |
| 9.3.4 HS-SCCH Detection Performance | $\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 12 and 16 dB | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged |

# F.4.3 7.68Mcps TDD Option

**Table F.4.3 Derivation of Test Requirements**

| Test | Minimum Requirement in TS 25.102 | Test Tolerance (TT) | Test Requirement in TS 34.122 |
|---|---|---|---|
| 5.2 Maximum Output Power | Power single code<br>Power class 1 (30 dBm)<br>Tolerance = +1/-3 dB<br>Power class 2 (24 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (21 dBm)<br>Tolerance = +2/-2 dB<br>Power multi code<br>Power class 2 (21 dBm)<br>Tolerance = +1/-3 dB<br>Power class 3 (18 dBm)<br>Tolerance = ±2 dB<br>Power class 4 (10 dBm)<br>Tolerance = ± 4 dB | 0.7 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>For power classes 1 and 2 (single and multi):<br>Upper Tolerance limit = +1.7 dB<br>Lower Tolerance limit = -3.7 dB<br>For power class 3 (single and milti):<br>Upper Tolerance limit = +2.7 dB<br>Lower Tolerance limit = -2.7 dB<br>For power class 4 (single and multi)<br>Upper Tolerance limit = +4.7 dB<br>Lower Tolerance limit = -4.7 dB |
| 5.3 UE Frequency Stability | The UE modulated carrier frequency shall be accurate to within □0.1 ppm compared to the carrier frequency received from the Node B. | 10 Hz | Formula: modulated carrier frequency error + TT<br><br>modulated carrier frequency error = □(0.1 ppm + 10 Hz). |
| 5.4.1.1 Uplink Power Control, Initial accuracy | □9dB normal conditions<br>□12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>□10dB (normal conditions)<br>□13dB (extreme conditions) |
| 5.4.1.2 Uplink Power Control, differential accuracy | SIR Target / Pow Step Tol<br><1 dB / 0.5 dB<br><2dB / 1 dB<br><3 dB / 1.5 dB<br><10 dB / 2 dB<br><20dB / 4 dB<br><30 dB / 6 dB<br>> 30 dB / 9 dB | TT<br>0.1 dB<br>0.15 dB<br>0.2 dB<br>0.5 dB<br>0.7 dB<br>0.7 dB<br>1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT |
| 5.4.1.3 Open Loop Power Control | □9dB normal conditions<br>□12dB extreme conditions | 1.0 dB | Formula: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT<br>Nominal expected TX power<br>□10dB (normal conditions)<br>□13dB (extreme conditions) |
| 5.4.2 Minimum Transmit Power | UE minimum transmit power shall be less than –44 dBm | 1.0 dB | Formula:<br>UE minimum transmit power + TT<br>UE minimum transmit power = –43 dBm |

| | | | | | |
|---|---|---|---|---|---|
| 5.4.5 Out-of-synchronisation handling of output power: | $\frac{\Sigma DPCH_E_c}{I_{or}}$ levels<br>before A –4.6 dB<br>AB: -10 dB<br>BD: -16 dB<br>DE: -12 dB<br>EF: -6 dB<br>transmit ON/OFF time 200ms | | 0,4 dB for $\frac{\Sigma DPCH_E_c}{I_{or}}$<br>0 ms for timing measurement | Formulas:<br>Ratio between A and B + TT<br>Ratio between B and D - TT<br>Ratio between D and E - TT<br>Ratio between E and F + TT<br>transmit ON/OFF time + TT timing<br><br>$\frac{\Sigma DPCH_E_c}{I_{or}}$ levels:<br>Before A: -4.6<br>AB: -10 + 0,4 dB<br>BD: -16 – 0,4 dB<br>DE: -12 - 0,4 dB<br>EF: -6 + 0,4 dB<br>Uncertainty of OFF power measurement is handled by Transmit OFF power test and uncertainty of ON power measurement is handled by Minimum output power test.transmit ON/OFF time<br>200ms | |
| 5.5.1 Transmit OFF power | Transmit OFF power shall be less than –65 dBm | | 1.5 dB | Formula: Transmit OFF power + TT<br>Transmit OFF power = –63.5 dBm | |
| 5.5.1 Occupied Bandwidth | The occupied channel bandwidth shall be less than 5 MHz based on a chip rate of 3,84 Mcps. | | 0 kHz | Formula: occupied channel bandwitdh: + TT<br><br>occupied channel bandwidth = 5.0 MHz | |
| 5.5.2.1 Spectrum emission mask | Minimum requirement defined in TS25.101 table 6.10.<br>The lower limit shall be –50 dBm / 3,84 MHz or which ever is higher. | | 1.5 dB | Formula: Minimum requirement + TT<br>Lower limit + TT<br>Add 1.5 to Minimum requirement entries in TS25.101 table 6.10<br>The lower limit shall be –48.5 dBm / 3,84 MHz or which ever is higher. | |
| 5.5.2.2 Adjacent Channel Leakage Power Ratio (ACLR) | Power Classes 2 and 3:<br>UE channel +5 MHz or -5 MHz, ACLR limit: 33 dB<br>UE channel +10 MHz or -10 MHz, ACLR limit: 43 dB | | 0.8 dB | Formula: ACLR limit - TT<br>Power Classes 2 and 3:<br>UE channel +5 MHz or -5 MHz, ACLR limit: 32.2 dB<br>UE channel +10 MHz or -10 MHz, ACLR limit: 42.2 dB | |
| 5.5.3 Spurious Emissions | | | | Formula: Minimum Requirement+ TT<br>Add zero to all the values of Minimum Requirements in table 5.5.3 | |
| | Frequency Band | Minimum Requirement | | Frequency Band | Minimum Requirement |
| | 9 kHz ≤ f < 150 kHz | –36dBm /1kHz | 0 dB | 9kHz ≤ f < 1GHz | –36dBm /1kHz |
| | 150 kHz ≤ f < 30 MHz | –36dBm /10kHz | 0 dB | 150 kHz ≤ f < 30 MHz | –36dBm /10kHz |
| | 30 MHz ≤ f < 1000 MHz | –36dBm /100kHz | 0 dB | 30 MHz ≤ f < 1000 MHz | –36dBm /100kHz |
| | 1 GHz ≤ f < 12.75 GHz | –30dBm /1MHz | 0 dB | 1 GHz ≤ f < 2.2 GHz | –30dBm /1MHz |
| | | | 0 dB | 2.2 GHz ≤ f < 4 GHz | –30dBm /1MHz |
| | | | 0 dB | 4 GHz ≤ f < 12.75 GHz | –30dBm /1MHz |

| | | | | | |
|---|---|---|---|---|---|
| | 925 MHz < f < 935 MHz | –67dBm /100kHz | 0 dB | 925 MHz < f < 935 MHz | –67dBm /100kHz |
| | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz | 0 dB | 935 MHz ≤ f ≤ 960 MHz | –79dBm /100kHz |
| | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz | 0 dB | 1805 MHz < f ≤ 1880 MHz | –71dBm /100kHz |
| | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz | 0 dB | 1884.5 MHz ≤ f ≤ 1919.6 MHz | –41dBm* /300kHz |
| 5.6 Transmit Intermodulation | Intermodulation Product<br>5MHz -31 dBc<br>10MHz -41 dBc | | 0 dB | Formula: Intermodulation Product + TT<br>Intermodulation Product<br>5MHz -31 dBc<br>10MHz -41 dBc | |
| 5.7.1 Error Vector Magnitude | The Error Vector Magnitude shall not exceed 17.5 % | | 0% | Formula: EVM limit + TT<br>EVM limit = 17.5 % | |
| 5.7.2 Peak code domain error | The peak code domain error shall not exceed -21dB | | □1.0 dB | Formula: Peak code domain error + TT<br>Peak code domain error = -20 dB | |
| 6.2 Reference sensitivity level | Îor = -105 dBm / 3,84 MHz<br>BER limit = 0.001 | | 0.7 dB | Formula: Îor+ T<br>BER limit unchanged<br><br>Îor = -104.3 dBm / 3,84 MHz | |
| 6.4 Adjacent Channel Selectivity | Îor = -91 dBm / 3,84 MHz<br>Ioac (modulated) = -52 dBm/3,84 MHz<br>BER limit = 0.001 | | 0 dB | Formula: Îor unchanged<br>Ioac - TT<br>BER limit unchanged<br><br>Ioac = -52 dBm/3,84 MHz | |
| 6.5 Blocking Characteristics | See table 6.5.2a and 6.5.2b. in TS34.122<br>BER limit = 0.001 | | 0 dB | Formula:<br>$I_{blocking}$ (modulated) - TT (dBm/3,84MHz)<br>$I_{blocking}$ (CW) - TT (dBm)<br>BER limit unchanged | |
| 6.6 Spurious Response | Iblocking(CW) –44 dBm<br>Fuw:<br>Spurious response frequencies<br>BER limit = 0.001 | | 0 dB | Formula: $I_{blocking}$ (CW) - TT (dBm)<br>Fuw unchanged<br>BER limit unchanged<br><br>Iblocking(CW) -44 dBm | |
| 6.7 Intermodulation Characteristics | Iouw1 (CW) -46 dBm<br>Iouw2 (modulated) –46 dBm / 3,84 MHz<br>Fuw1 (offset) 10 MHz<br>Fuw2 (offset) 20 MHz<br><br>BER limit = 0.001 | | 0 dB | Formula: TBD<br>BER limit unchanged. | |
| 6.8 Spurious Emissions | | | | Formula: Maximum level+ TT<br>Add zero to all the values of Maximum Level in table 6.8.1. | |
| | Frequency Band | Maximum level | | Frequency Band | Maximum level |
| | 9kHz ≤ f < 1GHz | -57dBm /100kHz | 0 dB | 9kHz ≤ f < 1GHz | -57dBm /100kHz |
| | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz | -60 dBm / 3,84MHz | 0 dB | 1.9-1.92 GHz<br>2.01-2.025GHz<br>2.11-2.170GHz | -60 dBm / 3,84MHz |
| | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz | -47 dBm/1MHz | 0 dB | 1 –1.9GHz,<br>1.92–2.01 GHz<br>2.025–2.11GHz | -47 dBm/1MHz |
| | 1GHz ≤ f ≤ | -47dBm | 0 dB | 1GHz ≤ f ≤ 2.2GHz | -47dBm |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 12.75GHz | | /1MHz | | | /1MHz |
| | | | | 0 dB | 2.2GHz < f ≤ 4GHz | -47dBm /1MHz |
| | | | | 0 dB | 4GHz < f ≤ 12.75GHz | -47dBm /1MHz |
| 7.6 Downlink Power Control | | TS #1,9 | TS #7,14 | 0.5 dB for 10 dB change in output power, 0 otherwise. | Formula for 10 dB change in transmit power: Upper Tolerance limit + TT<br>Lower Tolerance limit – TT | |
| | 1st frame | 15 ±4.0 dBm | 5 ±0.5 dBm | | | |
| | 2nd frame | 15 ±4.0 dBm | 15 ±4.0 dBm | | | |
| 9.3.1 HS-DSCH throughput for Fixed Reference Channels | $\frac{E_c}{I_{or}}$ -10,-9.5 dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 10 dB | | | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged | |
| 9.3.2 HS-DSCH throughput for Variable Reference Channels | $\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 8 and 10 dB | | | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged | |
| 9.3.3 Reporting of HS-DSCH Channel Quality Indicator | | | | No test tolerances applied | | |
| 9.3.4 HS-SCCH Detection Performance | $\frac{E_c}{I_{or}}$ -10dB<br>$I_{oc}$ = -60 dBm<br>$\hat{I}_{or}/I_{oc}$ = 12 and 16 dB | | | 0.1 dB for $\frac{E_c}{I_{or}}$<br>0.6 dB for $\hat{I}_{or}/I_{oc}$ | Formulas:<br>$\frac{E_c}{I_{or}}$ = ratio + TT<br>$\hat{I}_{or}/I_{oc}$ = ratio + TT<br>$I_{oc}$ unchanged | |
| * Applicable for transmission in 2010-2025 MHz as defined in subclause 4.2 (a). | | | | | | |

# F.5 Acceptable uncertainty of Test Equipment (This clause is informative)

This informative clause specifies the critical parameters of the components of an overall Test System (e.g. Signal generators, Signal Analyzers etc.) which are necessary when assembling a Test System that complies with clause F.1 Acceptable Uncertainty of Test System. These Test Equipment parameters are fundamental to the accuracy of the overall Test System and are unlikely to be improved upon through System Calibration.

## F.5.1 Transmitter measurements

**Table F.5.1 Equipment accuracy for transmitter measurements**

| Test | Equipment accuracy | Test conditions |
|---|---|---|
| 5.2 UE Maximum Output Power | Not applicable | |
| 5.3 Frequency Stability | ± 10Hz | |
| 5.4.1 Uplink power control | | |
| 5.4.2 Minimum Transmit Power | | |
| 5.4.3 Transmit OFF Power | | |
| 5.4.4 Transmit ON/OFF Power | | |
| 5.4.5 Out-of-synchronisation handling of output power $\frac{DPCCH_E_c}{I_{or}}$ | | |
| 5.4.5 Out-of-synchronisation handling of output power: transmit ON/OFF time | | |
| 5.5.1 Occupied Bandwidth | 100 kHz | |
| 5.5.2.1 Spectrum emission mask | Not applicable | |
| 5.5.2.2 ACLR | | |
| 5.5.3 Spurious emissions | | |
| 5.5.3 Spurious emissions: additional | | |
| 5.6 Transmit intermodulation: | Not applicable | |
| 5.7.1 Transmit modulation: EVM | 2.5 % | |
| 5.7.2 Transmit modulation: peak code domain error | ±[1 dB] | |

## F.5.2 Receiver measurements

**Table F.5.2 Equipment accuracy for receiver measurements**

| Clause | Equipment accuracy | Test conditions |
|---|---|---|
| 6.2 Reference Sensitivity Level | Not applicable | |
| 6.3 maximum input level: | Not applicable | |
| 6.4 Adjacent Channel Selectivity (ACS) | Not applicable | |
| 6.5 Blocking Characteristics | Not applicable | |
| 6.6 Spurious Responce | Not applicable | |
| 6.7 Intermodulation Characteristics | Not applicable | |
| 6.8 Spurious Emissions | Not applicable | |

## F.5.3 Performance measurements

**Table G.3 Equipment accuracy for performance measurements**

| Clause | Equipment accuracy | Test conditions |
|---|---|---|
| TBD | TBD | |

## F.5.5 Performance measurements (HSDPA)

**Table F.5.5: Equipment accuracy for performance measurements (HSDPA)**

| Clause | Equipment accuracy | Test conditions |
|---|---|---|
| 9.3.1 HS-DSCH throughput for Fixed Reference Channels | $\frac{E_c}{I_{or}}$ ±0.1 dB | -10 and -9.5dB |
| 9.3.2 HS-DSCH throughput for Variable Reference Channels | Same as 9.3.1 | -10dB |
| 9.3.3 Reporting of HS-DSCH Channel Quality Indicator | Same as 9.3.1 | -10dB |
| 9.3.4 HS-SCCH Detection Performance | Same as 9.3.1 | -10dB |

# F.6 General rules for statistical testing

## F.6.1 Statistical testing of receiver BER/BLER performance

### F.6.1.1 Error Definition

1) Bit Error Ratio (BER)

   The Bit Error Ratio is defined as the ratio of the bits wrongly received to all data bits sent. The bits are the information bits above the convolutional/turbo decoder

2) Block Error Ratio (BLER)

   A Block Error Ratio is defined as the ratio of the number of erroneous blocks received to the total number of blocks sent. An erroneous block is defined as a Transport Block, the cyclic redundancy check (CRC) of which is wrong.

### F.6.1.2 Test Method

Each test is performed in the following manner:

a) Setup the required test conditions.

b) Record the number of samples tested and the number of occurred events (bit error or block error)

c) Stop the test at a stop criterion which is minimum test time or an early pass or an early fail event.

d) Once the test is stopped decide according to the pass fail decision rules ( subclause F.6.1.7)

### F.6.1.3 Test Criteria

The test shall fulfil the following requirements:

a) good pass fail decision

   1) to keep reasonably low the probability (risk) of passing a bad unit for each individual test;

   2) to have high probability of passing a good unit for each individual test;

b) good balance between test time and statistical significance

   3) to perform measurements with a high degree of statistical significance;

   4) to keep the test time as low as possible.

## F.6.1.4 Calculation assumptions

### F.6.1.4.1 Statistical independence

It is assumed, that error events are rare (lim BER BLER → 0) independent statistical events. However the memory of the convolutional /turbo coder is terminated after one TTI. Samples and errors are summed up every TTI. So the assumption of independent error events is justified.

In the BLER test with fading there is the memory of the multipath fading channel which interferes the statistical independence. A minimum test time is introduced to average fluctuations of the multipath fading channel. So the assumption of independent error events is justified approximately.

### F.6.1.4.2 Applied formulas

The formulas, applied to describe the BER BLER test, are primarily based on the following experiment:

(1) After having observed a certain number of errors (**ne**) the number of samples are counted to calculate BER BLER.

Provisions are made (note 1) such that the complementary experiment is valid as well:

(2) After a certain number of samples (**ns**) the number of errors, occurred, are counted to calculate BER BLER.

Experiment (1) stipulates to use the following Chi Square Distribution with degree of freedom ne:
2*dchisq(2*NE,2*ne) for all calculations.

Experiment (2) stipulates to use the Poisson Distribution: dpois(ne,NE)

(NE: average of the distribution)

To determine the early stop conditions, the following inverse cumulative operation is applied:

0.5 * qchisq(D,2*ne). This is applicable for experiment (1) and (2).

D: wrong decision risk per test step

Note: other inverse cumulative operations are available, however only this is suited for experiment (1) and (2).

### F.6.1.4.3 Test procedure

The test procedure is as follows:

During a running measurement for a UE ns (number of samples) and ne (number of errors) are accumulated and from this the preliminary BER BLER is calculated. Then new samples up to the next error are taken. The entire past and the new samples are basis for the next preliminary BER BLER. Depending on the result at every step, the UE can pass, can fail or must continue the test.

As early pass- and early fail-UEs leave the statistical totality under consideration, the experimental conditions are changed every step resulting in a distribution that is truncated more and more towards the end of the entire test. Such a distribution can not any more be handled analytically. The unchanged distribution is used as an approximation to calculate the early fail and early pass bounds.

## F.6.1.5 Definition of good pass fail decision

This is defined by the probability of wrong decision F at the end of the test. The probability of a correct decision is 1-F.

The probability (risk) to fail a good DUT shall be ≤ F according to the following definition: A DUT is failed, accepting a probability of ≤ Fthat the DUT is still better than the specified error ratio (Test requirement).

The probability to pass a bad DUT shall be ≤ Faccording to the following definition: A DUT is passed, accepting a probability of ≤ Fthat the DUT is still worse than M times the specified error ratio. (M>1 is the bad DUT factor).

This definitions lead to an early pass and an early fail limit:

Early fail: ber≥ $berlim_{fail}$

$$ber\lim_{fail}(D,ne) = \frac{2*ne}{qchisq(D,2*ne)} \quad (1)$$

For ne≥[5]

Early pass: ber≤berlimbad$_{pass}$

$$ber\lim bad_{pass}(D,ne) = \frac{2*ne*M}{qchisq(1-D,2*ne)} \quad (2)$$

For ne ≥1

With

ber (normalized BER,BLER): BER,BLER according to F.6.1.1 divided by Test requirement

D: wrong decision probability for a test step. This is a numerically evaluated fraction of F, the wrong decision probability at the end of the test. see table F.6.1.1

ne: Number of error events

M: bad DUT factor see table F.6.1.1

qchisq: inverse cumulative chi squared distribution

## F.6.1.6 Good balance between testtime and statistical significance

There independent test parameters are introduced into the test and shown in Table F.6.1.6.1. These are the obvious basis of test time and statistical significance. From the first two of them four dependent test parameters are derived. The third independent test parameter is justified separately.

**Table F.6.1.6.1 independent and dependent test parameters**

| Independent test parameters | | | Dependent test parameters | | |
|---|---|---|---|---|---|
| Test Parameter | Value | Reference | Test parameter | Value | Reference |
| Bad DUT factor M | 1.5 | Table F.6.1.8 | Early pass/fail condition | Curves | Subclause F.6.1.5 Figure 6.1.9 |
| Final probability of wrong pass/fail decision F | 0.2% 0.02%, note 2 | Subclause F.6.1.5 | Target number of error events | 345 | Table 6.1.8 |
| | | | Probability of wrong pass/fail decision per test step D | 0.0085% 0.0008% and 0.008%, note 2 | |
| | | | Test limit factor TL | 1.234 | Table 6.1.8 |
| Minimum test time | | Table F.6.1.6.2 | | | |

The minimum test time is derived from the following justification:

1) For no propagation conditions and static propagation condition

   No early fail calculated from fractional number of errors <1 see note 1

2) For multipath fading condition

   No stop of the test until 999wavelengths are crossed during relevant UE reception timeslots, relevant for BER BLER testing, with the speed given in the fading profile.

**Table F.6.1.6.2 : minimum Test time**

| Fading profile | Minimum test time |
|---|---|
| Multipath propagation 3 km/h | [164 s*TSPF/TSRX*)] |
| Multipath propagation 50 km/h | [9.8 s* TSPF/TSRX] |
| Multipath propagation 120 km/h | [4.1 s * TSPF/TSRX] |
| Multipath propagation 250 km/h | [2s* TSPF/TSRX] |
| *) TSPF = Time slots per frame, TSRX = relevant UE reception timeslots per frame, relevant for the BER BLER test | |

TSPF and TSRX form the prolongation factor and depend on the user data rate and the TDD Option (3,84 Mchip/s, 1,28 Mchip/s or 7.68Mcps)

**Table F.6.1.6.3 : Prolongation factor for minimum Test time**

| User Data rate | TSPF/TSRX for TDD 3,84 Mchip/s and 7,68Mchip/s | TSPF/TSRX for TDD 1,28 Mchip/s |
|---|---|---|
| 12.2 kbit/s | 15/1 | 7/1 |
| 64 kbit/s | 15/1 | 7/1 |
| 144 kbit/s | 15/1 | 7/2 |
| 384 kbit/s | 15/3 | 7/4 |

In table F.6.1.8 the minimum test time is converted in minimum number of samples

## F.6.1.7 Pass fail decision rules

No decision is allowed before the minimum test time is elapsed

1) If minimum Test time < time for target number of error events then the following applies: The required confidence level 1-F (= correct decision probability) shall be achieved. This is fulfilled at an early pass or early fail event.

   For BER: For every TTI (Transmit Time Interval) sum up the number of bits (ns) and the number if errors (ne) from the beginning of the test and calculate

   $BER_1$ (including the artificial error at the beginning of the test (Note 1))and

   $BER_0$ (excluding the artificial error at the beginning of the test (Note 1)).

   If $BER_0$ is above the early fail limit, fail the DUT.

   If $BER_1$ is below the early pass limit, pass the DUT.

   Otherwise continue the test

   For BLER: For every block sum up the number of blocks (ns) and the number if erroneous blocks (ne) from the beginning of the test and calculate,

   $BLER_1$ (including the artificial error at the beginning of the test (Note 1))and

   $BLER_0$ (excluding the artificial error at the beginning of the test (Note 1)).

   If $BLER_1$ is below the early pass limit, pass the DUT.

   If $BLER_0$ is above the early fail limit, fail the DUT.

   Otherwise continue the test

2) If the minimum test time >= time for target error events, then the test runs for the minimum test time and the decision is done by comparing the result with the test limit.

   For BER:

For every TTI (Transmit Time Interval) sum up the number of bits (ns) and the number if errors (ne) from the beginning of the test and calculate $BER_0$

For BLER:

For every block sum up the number of blocks (ns) and the number of erroneous blocks (ne) from the beginning of the test and calculate $BLER_0$

If $BER_0$/$BLER_0$ is above the test limit, fail the DUT.

If $BER_0$/$BLER_0$ is on or below the test limit, pass the DUT.

## F.6.1.8 Test conditions for BER,BLER tests

**Table F.6.1.8: Test conditions for a single BER/BLER tests**

| Type of test (BER) | Test requirement (BER/BLER) | Test limit (BER/BLER)= Test requirement (BER/BLER)x TL TL | Target number of error events (time) | Minimum number of samples | Prob that good unit will fail = Prob that bad unit will pass [%] | Bad unit BER/BLER factor M |
|---|---|---|---|---|---|---|
| Reference Sensitivity Level | 0.001 | 1.234 | 345 (22.9s) | Note 1 | 0.2 | 1.5 |
| Maximum Input Level | 0.001 | 1.234 | 345 (22.9s) | Note 1 | 0.2 | 1.5 |
| Adjacent Channel Selectivity | 0.001 | 1.234 | 345 (22.9s) | Note 1 | 0.2 | 1.5 |
| Blocking Characteristics Pass condition Note 2 | 0.001 | 1.251 | 403 (26.4s) | Note 1 | 0.2 | 1.5 |
| Blocking Characteristics Fail condition Note 2 | 0.001 | 1.251 | 403 (26.4s) | Note 1 | 0.02 | 1.5 |
| Spurious Response | 0.001 | 1.234 | 345 (22.9s) | Note 1 | 0.2 | 1.5 |
| Intermodulation Characteristics | 0.001 | 1.234 | 345 (22.9s) | Note 1 | 0.2 | 1.5 |
| HS-SCCH Detection Performance | 0.01 | FFS | FSS | Note 1 | 0.2 | 1.5 |

**Table F.6.1.8-2: Test conditions for BLER tests**

| Type of test (BLER) | Information Bit rate | Test requirement (BER/BLER) | Test limit (BER/BLER)= Test requirement (BER/BLER)x TL<br><br>TL | Target number of error events (time) | Minimum number of samples<br><br>TDD 3,84 Mchip/s | Minimum number of samples<br><br>TDD 1,28 Mchip/s | Prob that bad unit will pass = Prob that good unit will fail [%] | Bad unit BER/BLER factor M |
|---|---|---|---|---|---|---|---|---|
| Demodulation in Static Propagation conditions | 12.2<br>64<br><br>144<br><br>384 | 0.01<br>0.1<br>0.01<br>0.1<br>0.01<br>0.1<br>0.01 | 1.234 | 345<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(27.96s)<br>(279.58s) | Note1 | Note1 | 0.2 | 1.5 |
| Demodulation of DCH in Multi-path Fading conditions | | | | | | | | |
| 3km/h (Case 1, Case 2) | 12.2<br>64<br><br>144<br><br>384 | 0.01<br>0.1<br>0.01<br>0.1<br>0.01<br>0.1<br>0.01 | 1.234 | 345<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(27.96s)<br>(279.58s) | [1350]<br>[1350]<br>[1350]<br>[1350]<br>[1350]<br>[450]<br>[450] | 57400<br>57400<br>57400<br><br>28700<br>28700<br>14350<br>14350 | 0.2 | 1.5 |
| 120 km/h (Case3) | 12.2<br>64<br><br>144<br><br>384 | 0.01<br>0.1<br>0.01<br>0.1<br>0.01<br>0.1<br>0.01 | 1.234 | 345<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(55.92s)<br>(559.16s)<br>(27.96s)<br>(279.58s) | [34]<br>[34]<br>[34]<br>[34]<br>[34]<br>[12]<br>[12] | 1435<br>1435<br>1435<br>718<br>718<br>359<br>359 | 0.2 | 1.5 |
| 250 km/h | | | | | | | | |
| Power control in the downlink | | | | Not applicable | | | | |

## F.6.1.9 Practical Use (informative)

See figure F.6.1.9:

The early fail limit represents formula (1) in F.6.1.5 The range of validity is ne>7, >8 in case of blocking test] to ne =345

The early pass limit represents the formula (2) in F.6.1.5 The range of validity is ne=1 to ne =345. See note 1

The intersection co-ordinates of both curves are: Target number of errors ne = 345 and test limit TL = 1.234

The range of validity for TL is ne>345

A typical BER BLER test, calculated form the number of samples and errors (F.6.1.2.(b)) using experimental method (1) or (2) (see F.6.1.4. calculation assumptions) runs along the yellow trajectory. With an errorless sample the trajectory goes down vertically. With an erroneous sample it goes up right. The tester checks if the BER BLER test intersects the early fail or early pass limits. The real time processing can be reduced by the following actions:

BLER is calculated only in case of an error event.

BER is calculated only in case of an error event within a TTI.

So the early fail limit cannot be missed by errorless samples.

The check against the early pass limit may be done by transforming formula (2) in F.6.1.5 such that the tester checks against a Limit-Number-of-samples (NL(ne)) depending on the current number of errors.

Early pass if

$$NL(ne) \geq \frac{qchisq(1-D, 2*ne)}{2*TR*M}$$

TR: test requirement (0.001)

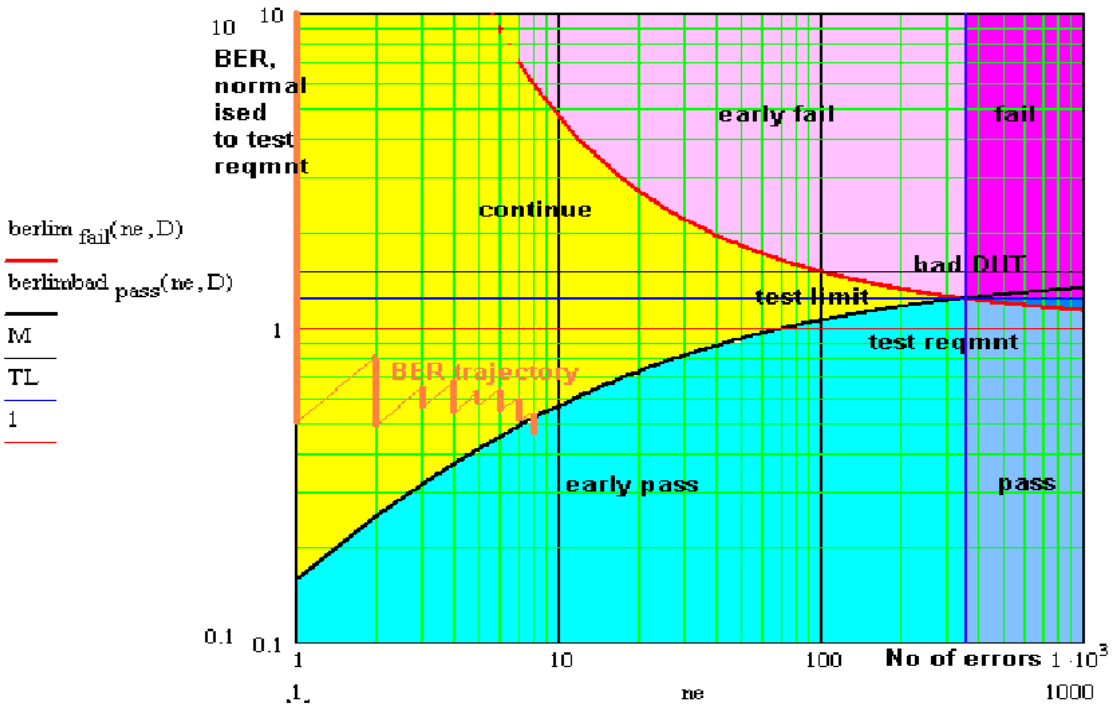



**Figure F.6.1.9**

Note 1: At the beginning of the test, an artificial error is introduced. This ensures that an ideal DUT meets the valid range of the early pass limit. In addition this ensures that the complementary experiment (F.6.1.4. bullet point (2)) is applicable as well.

Due to the nature of the test, namely discrete error events, the early fail condition shall not be valid, when fractional errors <1 are used to calculate the early fail limit: Any early fail decision is postponed until number of errors ne >7. In the blocking test any early fail decision is postponed until number of errors ne >8

Note2: F=0.2% is intended to be used for a test containing a few BER/BLER tests (e.g. receiver sensitivity is repeated 12 times). For a test containing many BER/BLER tests (e.g. blocking test) this value is not appropriate for a single BER/BLER test.

The blocking test contains approx. 12750 single BER tests. A DUT on the limit will fail approx. 25 to 26 times due to statistical reasons (wrong decision probability [0.2]%). 24 fails are allowed in the blocking test but they are reserved for spurious responses. This shall be solved by the following rule:

All passes (based on F=0.2%) are accepted, including the wrong decisions due to statistical reasons.

An early fail limit based on F=0.02%instead of 0.2% is established, that ensures that wrong decisions due to statistical reasons are reduced to 2 to 3.

These asymmetric test conditions ensure that a DUT on the test limit consumes hardly more test time for a blocking test than in the symmetric case and on the other hand discriminates sufficiently between statistical fails and spurious response cases.

# F.6.2 Statistical testing of RRM delay performance

## F.6.2.1 Test Method

Each test is performed in the following manner:

a) Setup the required test conditions.

b) Measure the delay repeated times. Start each repetition after sufficient time, such that each delay test is independent from the previous one. The delay-times, measured, are simplified to:

a good delay, if the measured delay is ≤ limit.

a bad delay, if the measured delay is > limit

c) Record the number of delays (ns), tested, and the number of bad delays (ne)

d) Stop the test at an early pass or an early fail event.

e) Once the test is stopped, decide according to the pass fail decision rules ( subclause F.6.2.7)

## F.6.2.2 Bad Delay Ratio (ER)

The Bad Delay Ratio (ER) is defined as the ratio of bad delays (ne) to all delays (ns).

(1-ER is the success ratio)

## F.6.2.3 Test Criteria

The test shall fulfil the following requirements:

a) good pass fail decision

1) to keep reasonably low the probability (risk) of passing a bad unit for each individual test;

2) to have high probability of passing a good unit for each individual test;

b) good balance between test-time and statistical significance

3) to perform measurements with a high degree of statistical significance;

4) to keep the test time as low as possible.

## F.6.2.4 Calculation assumptions

### F.6.2.4.1 Statistical independence

It is arranged by test conditions, that bad delays are independent statistical events.

### F.6.2.4.2 Applied formulas

The specified ER is 10% in most of the cases. This stipulates to use the binomial distribution to describe the RRM delay statistics. With the binomial distribution optimal results can be achieved. However the inverse cumulative operation for the binomial distribution is not supported by standard mathematical tools. The use of the Poisson or Chi Square Distribution requires ER→0. Using one of this distributions instead of the binomial distribution gives sub-optimal results in the conservative sense: a pass fail decision is done later than optimal and with a lower wrong decision risk than predefined.

The formulas, applied to describe the RRM delay statistics test, are based on the following experiment:

(1) After having observed a certain number of bad delays (**ne**) the number of all delays (**ns**) are counted to calculate ER. Provisions are made (note 1) such that the complementary experiment is valid as well:

(2) After a certain number of delays (**ns**) the number of bad delays (**ne**), occurred, are counted to calculate ER.

Experiment (1) stipulates to use the Chi Square Distribution with degree of freedom ne: 2*dchisq(2*NE,2*ne).

Experiment (2) stipulates to use the Poisson Distribution: dpois(ne,NE)

(NE: mean value of the distribution)
To determine the early stop conditions, the following inverse cumulative operation is applied:
0.5 * qchisq(D,2*ne) for experiment (1) and (2)
D: wrong decision risk per test step
Note: other inverse cumulative operations are available, however only this is suited for experiment (1) and (2).

### F.6.2.4.3 Approximation of the distribution

The test procedure is as follows:

During a running measurement for a UE ns (Number of Delays) and ne (Number of bad delays) are accumulated and from this the preliminary ER is calculated. Then new samples up to the next bad delay are taken. The entire past and the new samples are basis for the next preliminary ER. Depending on the result at every step, the UE can pass, can fail or must continue the test.

As early pass- and early fail-UEs leave the statistical totality under consideration, the experimental conditions are changed every step resulting in a distribution that is truncated more and more towards the end of the entire test. Such a distribution can not any more be handled analytically. The unchanged distribution is used as an approximation to calculate the early fail and early pass bounds.

## F.6.2.5 Definition of good pass fail decision.

This is defined by the probability of wrong decision F at the end of the test. The probability of a correct decision is 1- F.

The probability (risk) to fail a good DUT shall be ≤ F according to the following definition: A DUT is failed, accepting a probability of ≤ F that the DUT is still better than the specified bad delay ratio (Test requirement).

The probability (risk) to pass a bad DUT shall be ≤ F according to the following definition: A DUT is passed, accepting a probability of ≤ F that the DUT is still worse than M times the specified bad delay ratio. (M>=1 is the bad DUT factor).

This definitions lead to an early pass and an early fail limit:

Early fail: er≥ er**lim**$_{fail}$

$$er\lim_{fail}(D,ne) = \frac{2*ne}{qchisq(D,2*ne)} \quad (1)$$

For ne ≥ [5],

Early pass: er ≤ er**lim**bad$_{pass}$

$$er\lim bad_{pass}(D,ne) = \frac{2*ne*M}{qchisq(1-D,2*ne)} \quad (2)$$

For ne ≥1,

With,

er (normalized ER): ER according to F.6.2.2 divided by specified ER

D: wrong decision probability for a test step . This is a numerically evaluated fraction of F, the wrong decision probability at the end of the test. see table F.6.2.6.1

ne: Number of bad delays

M: bad DUT factor see table F.6.2.6.1

qchisq: inverse cumulative chi squared distribution

## F.6.2.6 Good balance between test-time and statistical significance

Two independent test parameters are introduced into the test and shown in Table F.6.2.6.1. These are the obvious basis of test time and statistical significance. From them four dependent test parameters are derived.

**Table F.6.2.6: Independent and dependent test parameters**

| Independent test parameters | | | Dependent test parameters | | |
|---|---|---|---|---|---|
| **Test Parameter** | **Value** | **Reference** | **Test parameter** | **Value** | **Reference** |
| Bad DUT factor M | 1.5 | Table F.6.1.8 | Early pass/fail condition | Curves | Subclause F.6.2.5 Figure 6.2.9 |
| Final probability of wrong pass/fail decision F | 5% | Table F.6.2.8 | Target number of bad delays | 154 | Table 6.2.8 |
| | | | Probability of wrong pass/fail decision per test step D | 0.6 % | |
| | | | Test limit factor TL | 1.236 | Table 6.2.8 |

## F.6.2.7 Pass fail decision rules

The required confidence level 1-F (= correct decision probability) shall be achieved. This is fulfilled at an early pass or early fail event. Sum up the number of all delays (ns) and the number of bad delays from the beginning of the test and calculate:

$ER_1$ (including the artificial error at the beginning of the test (Note 1))and

$ER_0$ (excluding the artificial error at the beginning of the test (Note 1)).

If $ER_0$ is on or above the early fail limit, fail the DUT.

If $ER_1$ is on or below the early pass limit, pass the DUT.

Otherwise continue the test.

## F.6.2.8 Test conditions for RRM delay tests

**Table F.6.2.8: Test conditions for a single RRM delay tests(3.84 Mcps TDD option)**

| Type of test | Test requirement Delay (s) | Test requirement (ER) | Testlimit(ER)= Test requirement (ER)x TL TL | Target number of bad delays | Prob that good unit will fail = Prob that bad unit will pass [%] | Bad unit factor M |
|---|---|---|---|---|---|---|
| 8.2.2.1 TDD/TDD Cell re-selection in Idle mode (single carrier) | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.2.2.2 TDD/TDD Cell re-selection in Idle Mode (mulit-carrier) | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.2.2.3 TDD/FDD Cell re-selection in Idle Mode | 8 | | | | | |
| 8.2.2.4 UTRAN to GSM cell re-selection in Idle Mode | 27.9 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.1 TDD/TDD handover | 40 ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.2 TDD/FDD handover | 100 ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.3 TDD/GSM handover | 40ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.4 TDD/TDD Cell Re-selection in CELL_FACH | 2.5 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.5 TDD/TDD Cell Re-selection in CELL_PCH | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.6 TDD/TDD Cell Re-selection in URA_PCH | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |

**Table F.6.2.8A: Test conditions for a single RRM delay tests(1.28Mcps TDD option)**

| Type of test | Test requirement Delay (s) | Test requirement (ER) | Testlimit(ER)= Test requirement (ER)x TL TL | Target number of bad delays | Prob that good unit will fail = Prob that bad unit will pass [%] | Bad unit factor M |
|---|---|---|---|---|---|---|
| 8.2.2.1 TDD/TDD Cell re-selection in Idle mode (single carrier) | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.2.2.2 TDD/TDD Cell re-selection in Idle Mode (multi-carrier) | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.2.2.3 TDD/FDD Cell re-selection in Idle Mode | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.2.2.4 UTRAN to GSM cell re-selection Scenario 4 | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.2.2.5 UTRAN to GSM cell re-selection in Idle Mode Scenario 4A | 27.9 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.1A TDD/TDD handover | 160 ms | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.2A TDD/FDD handover | 100 ms | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.3 TDD/GSM handover | 90ms | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.4.1 TDD/TDD Cell Re-selection in CELL_FACH (single carrier) | 1.6 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.4.2 TDD/TDD Cell Re-selection in CELL_FACH (multi-carrier) | 2 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.5 TDD/TDD Cell Re-selection in CELL_PCH | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |
| 8.3.6 TDD/TDD Cell Re-selection in URA_PCH | 8 | 0.1 | 1.236 | 154 | 5 | 1.5 |

**Table F.6.2.8B: Test conditions for a single RRM delay tests(7.68 Mcps TDD option)**

| Type of test | Test requirement Delay (s) | Test requirement (ER) | Testlimit(ER)= Test requirement (ER)x TL TL | Target number of bad delays | Prob that good unit will fail = Prob that bad unit will pass [%] | Bad unit factor M |
|---|---|---|---|---|---|---|
| 8.2.2.1 TDD/TDD Cell re-selection in Idle mode (single carrier) | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.2.2.2 TDD/TDD Cell re-selection in Idle Mode (mulit-carrier) | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.2.2.3 TDD/FDD Cell re-selection in Idle Mode | 8 | | | | | |
| 8.2.2.4 UTRAN to GSM cell re-selection in Idle Mode | 27.9 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.1B TDD/TDD handover | 40 ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.2B TDD/FDD handover | 100 ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.3 TDD/GSM handover | 40ms | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.4 TDD/TDD Cell Re-selection in CELL_FACH | 2.5 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.5 TDD/TDD Cell Re-selection in CELL_PCH | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |
| 8.3.6 TDD/TDD Cell Re-selection in URA_PCH | 8 | 0.1 | [1.236] | [154] | [5] | [1.5] |

## F.6.2.9 Practical Use (informative)

See figure F.6.2.9:

The early fail limit represents formula (1) in F.6.2.5. The range of validity is ne≥ 5 to ne = 154

The early pass limit represents the formula (2) in F.6.2.5. The range of validity is ne=1 to ne =154. See note 1. The intersection co-ordinates of both curves are : target number of bad delays ne = 154 and test limit TL = 1.236 .

A typical delay test, calculated form the number of samples and errors (F.6.2.2) using experimental method (1) or (2) (see F.6.2.4.2. calculation assumptions) runs along the yellow trajectory. With an good delay the trajectory goes down vertically. With a bad delay it jumps up right. The tester checks if the ER test intersects the early fail or early pass limits.

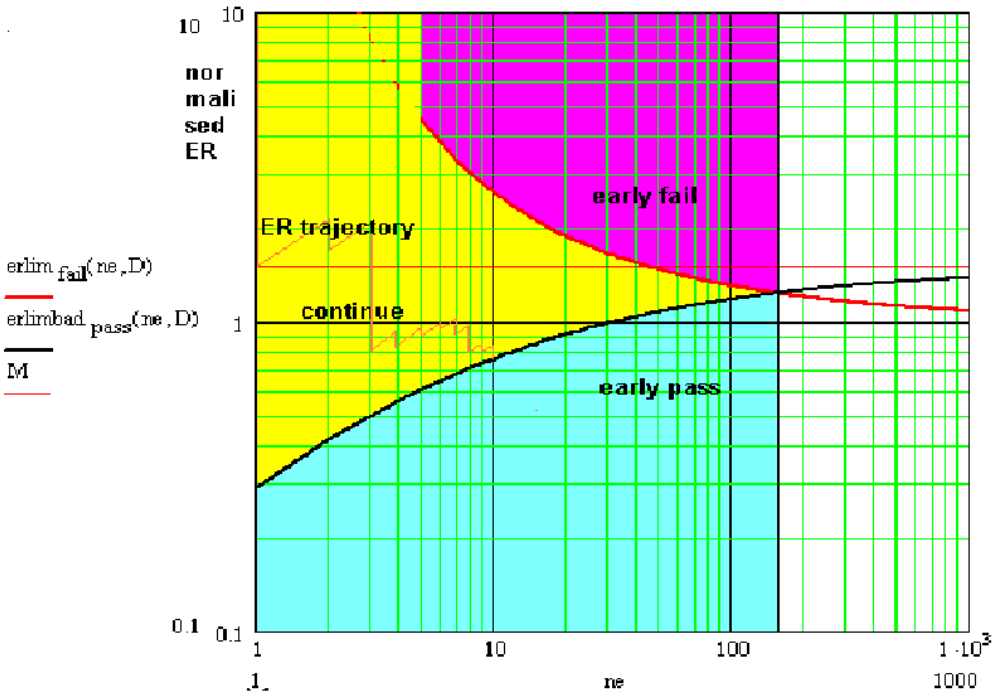



**Figure F.6.2.9**

Note 1: At the beginning of the test, an artificial bad delay is introduced. This ensures that an ideal DUT meets the valid range of the early pass limit. In addition this ensures that the complementary experiment (F.6.2.4.2. bullet point (2)) is applicable as well. For the check against the early fail limit the artificial bad delay sample, introduced at the beginning of the test , is disregarded.

Due to the nature of the test, namely discrete bad delay events, the early fail condition shall not be valid, when fractional bad delays <1 are used to calculate the early fail limit: Any early fail decision is postponed until number of errors ne ≥ [5 ].

# F.6.3 Statistical Testing of HSDPA Receiver Performance

## F.6.3.1 Definition

Information Bit Throughput R:

The measured information bit throughput R is defined as the sum (in kilobits) of the information bit payloads (excluding the 24-bit HS-DSCH CRC) successfully received during the test interval, divided by the duration of the test interval (in seconds).

## F.6.3.2 Mapping throughput to block error ratio

a) In measurement practice the UE indicates successfully received information bit payload by signalling an ACK to the SS.
If payload is received, but damaged and cannot be decoded, the UE signals a NACK.

b) Only the ACK and NACK signals, not the data bits received, are accessible to the SS.
The number of bits is known in the SS from knowledge of what payload was sent.

c) For fixed reference channel the number of bits in a TTI is fixed during one test.

d) The time in the measurement interval is composed of successful TTIs (ACK), unsuccessful TTIs (NACK) and DTX-TTIs.

e) DTX-TTIs occur regularly according to the test. (regDTX).
In real live this is the time when other UEs are served.
regDTX vary from test to test but are fixed within the test.

f) Additional DTX-TTIs occur statistically when the UE is not responding ACK or NACK where it should. (statDTX)
This may happen when the UE was not expecting data or decided that the data were not intended for it.

The pass / fail decision is done by observing the:

- number of NACKs
- number of ACKs and
- number of statDTXs (regDTX is implicitly known to the SS)

The ratio `(NACK + statDTX)/(NACK+ statDTX +ACK)` is the Block Error Ratio BLER. For fixed referece channel ,taking into account the time consumed by the ACK-, NACK-, and DTX-TTIs (regular and statistical), BLER can be mapped unambiguously to throughput for any single test.

## F.6.3.3 Bad DUT factor

Note: Data throughput in a communication system is of statistical nature and must be measured and decided pass or fail. The specified limit of throughput related to the ideal throughput in different throughput tests is in the range of a few % to near 100%. To make it comparable with BER, we define the complement of the relative throughput: BLER as defined above. Complementary this is in the range of near 100% down to a few % For e.g. BLER = 1%, the currently in BER BLER used Bad DUT factor M=1.5 is highly meaningful. For e.g. BLER = 99%, the currently used M=1.5 obviously meaningless.

An appropriate definition of the bad DUT factor is illustrated in figure F.6.3.3: constant and variable Bad DUT factor.

It illustrates how to find the Bad BLER when the nominal BLER is given.

1) In the range 0%< nominal BLER>10% the Bad DUT factor is constant 1.5
2) In the range 90%< bad BLER>100% it decreases to 1. (symmetrical to (1))
3) The range in between is interpolated by an arc section.

The example shows: nominal BLER=35,6% → bad BLER=47.67.5% → M=1.34

(blue mapping)

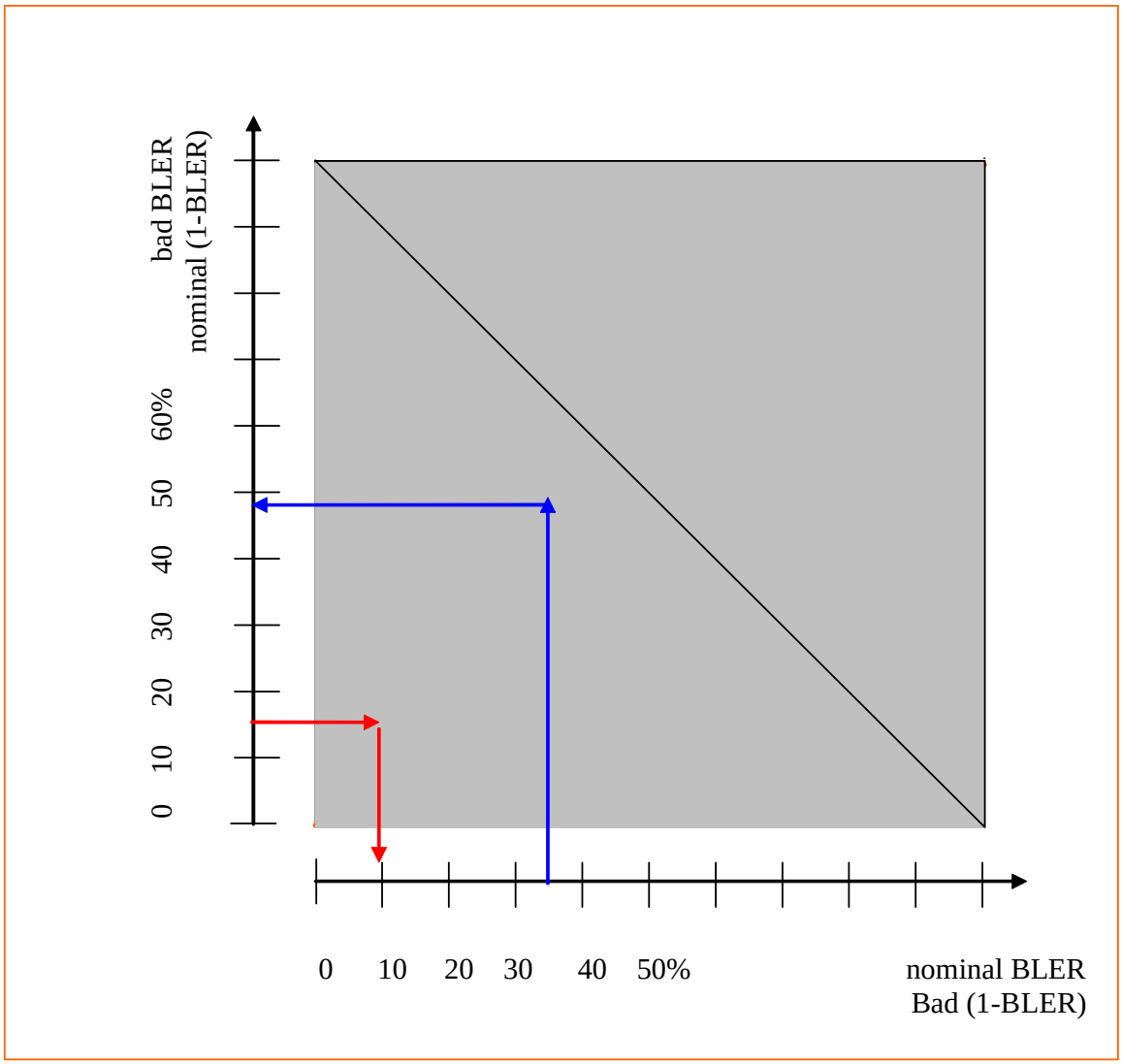



**Figure F.6.3.3: constant and variable Bad DUT factor**

Formula:

For 0 < BLER<= 0.1 M = 1.5

For 0.1 <BLER <0.9

$$M(BLER) := \frac{\sqrt{r^2 - (BLER - 2.35)^2}}{BLER} - \frac{1.35}{BLER}$$

For 0.9 <= BLER < 1 M(BLER)= 2/3BLER + 1/3

With BLER: nominal Block Error Ratio (0<BLER<1)

With r = 2.70415 (Radius of the arc)

### F.6.3.3.1 Bad DUT factor, range of applicability

Inaccuracy is one practical reason to avoid the grey shaded area of figure F.6.3.3: constant and variable Bad DUT factor. For BLER near 1 the Bad DUT factor M is near 1. For M=1,exactly, the pass and fail criteria do not intersect. The test never is finalised.

For M near 1 the pass and fail criteria exhibit a very smooth intersection. In addition the binomial distribution and its inverse are of discrete nature. Therefore the test limit and the number of samples is calculable only very ambiguous.

It is proposed to apply the bad DUT factor only in the not shaded area of figure F.6.3.3.

This is done by the following:

BLER mode:

Use BLER as defined above in the range of 0 to 50%, use M >1 as defined above.

The Test Limit will be > the Test Requirement in the table F.6.3.5. below.

Relative Throughput mode:

If BLER is in the range 50 to 100%, use 1-BLER instead. Use m<1 instead of M.

1-BLER is the relative throughput with respect to the ideal throughput.

As a consequence, the Test Limit < Test Requirement

Formula for m:

For 0 < (1-BLER) <= 0.15 m = 1/1.5

For 0.15 <(1-BLER) <.85 $$m := \frac{2.35 - \sqrt{r^2 - [(1 - BLER) + 1.35]^2}}{(1 - BLER)}$$

In the figure F.6.3.3: this is represented by the red mapping.

The tables F.6.3.5. below distinguishe between m and M.

## F.6.3.4 Minimum Test time

Same as with BER BLER there is a minimum test time is necessary for multipath fading profiles with the same justification:

| profile | Minimum Test time |
|---|---|
| PA3, PB3 | 164s |
| VA30 | 16.4s |
| VA 120 | 4.1s |

## F.6.3.5 Pass fail decision rules

The purpose of tables F.6.3.5.1 to F.6.3.5.2 is to decide throughput pass or fail.

(the Ior/Ioc levels are only for reference)

Meaning of a decision:

- A passed DUT is not worse than a Bad DUT with 95%confidence level.
- A failed DUT is not better than a Limit DUT with 95%confidence level.

The minimum Test Time is

1) the minimum test time due to statistical reasons

   ( To ensure the confidence level, the test must be continued until a certain number of samples (NACK+ statDTX +ACK) is reached.)

2) the minimum test time due to multipath fading.

The longer test time applies. It is marked in table F.6.3.5. which one applies.

Statistical independence:

If a process works within an incremental redundancy sequence, the samples are not independent. The incremental redundancy sequence for every process must be finalised, successfully or unsuccessfully, on or beyond the minimum test time.

Then the BLER (or 1-BLER) is compared with the Test Limit to decide pass or fail.

Note: It is FFS, if correlation within groups of retransmissions may influence the confidence level of the test.

Formula:

The theory, to derive the minimum number of samples and the Test Limit, takes into consideration that BLER is in the range of near 0% to near 100%. Hence it is based on the binomial distribution and its inverse cumulative function: qbinom:

For the BLER test mode:

$ne_{low}$=qbinom(D,ns,M*$BLER_{limit}$) (1)

$ne_{high}$=qbinom(1-D,ns,$BLER_{limit}$) (2)

given: 1-D: confidence level= 95%

$BLER_{limit}$=Block error ratio at the limit

M: Bad DUT factor >1

Input: ns: number of samples (NACK+ statDTX + ACK)

Output ne: number of events (NACK+ statDTX)

The intersection of (1) and (2) is the Test Limit with the coordinates: ns and ne

For the Relative Throughput test mode:

$ne_{low}$=qbinom(D,ns,1-$BLER_{limit}$) (3)

$ne_{high}$=qbinom(1-D,ns,m*(1-$BLER_{limit}$)) (4)

given: 1-D: confidence level= 95%

1-$BLER_{limit}$= Relative Throughtput at the limit

m: Bad DUT factor <1

Input: ns: number of samples (NACK+ statDTX + ACK)

Output ne: number of events (ACK)

The intersection of (3) and (4) is the Test Limit with the coordinates: ns and ne

Note: In contrast to BER BLER test, this approach does not contain any test time optimisation.

(early pass, early fail)

Nomenclature used in the tables F.6.3.5 below:

- NACK+ statDTX + ACK is summarised as No of samples
- NACK+ statDTX is summarised as No of errors
- ACK is summarised as No of successes
- In the BLER test mode the ratio: No of errors/ No of samples is recorded. In this mode a pass is below the test limit
- In the Relative Throughput test mode (1-BLER) the ratio: No of successes/ No of samples is recorded. In this mode a pass is above the test limit

- The test mode, used, is indicated in the rightmost column with BL or RT
- The transition from the BL to the RT test mode can also be seen in the column relative test requirement: BLER%→ (1-BLER%)
- The generic term for No of errors (BLER mode) or No of successes (Relative Throughput mode) is No of events. This is used in the table column Test Limit.

**Table F.6.3.5.1: Test case 9.3.1.3 for fixed reference measurement channel requirements for 0.5 Mbps UE class QPSK of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=10dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=199.2kbps)<br>No of events / No of samples in % | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 QPSK | PA3 | 375160 | 19.68% | 54/226 (M =1.4527) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 170 | 14.66% | 52/289 (M=1.4845) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 161 | 19.18% | 55/236 (M=1.4562) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 153 | 23.20% | 57/204 (M=1.4277) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.2: Test case 9.3.1.5 for fixed reference measurement channel requirements for 1.1 Mbps UE class 16QAMof 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=15dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=578.6kbps)<br>No of events / No of samples in %<br>No of events/No of samples in %<br>BL → (RT) | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 388 | 32.94% | 69/179 (M =1.3573) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 347 | 40.03% | 78/169 (M=1.3085) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 316 | 45.39% | 84/163 (M=1.2735) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 274 | 52.74%→ (47.36%) | 70/170 (m=0.7459) | N.A. | 4.1s(fading) | RT |

**Table F.6.3.5.3: Test case 9.3.1.7 for fixed reference measurement channel requirements for 1.6 Mbps UE class QPSK of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=10dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=357.4kbps) No of events / No of samples in % | Test limit expressed as No of events / min No of samples ( Bad DUT factor) | Min No of samples ( number of events to pass) Mandatory, if applicable | Test time in s Mandatory if fading, Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 270 | 24.45% | 59/201 (M =1.4186) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 278 | 22.22% | 57/212 (M=1.4348) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 259 | 27.53% | 60/183 (M=1.3962) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 242 | 32.29% | 65/171 (M=1.3620) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.4: Test case 9.3.1.7 for fixed reference measurement channel requirements for 1.6 Mbps UE class 16QAM of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=15dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=634.6kbps) No of events / No of samples in % | Test limit expressed as No of events / min No of samples ( Bad DUT factor) | Min No of samples ( number of events to pass) Mandatory, if applicable | Test time in s Mandatory if fading, Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 488 | 23.10% | 57/205 (M =1.4284) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 471 | 25.78% | 58/188 (M=1.4089) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 431 | 32.18% | 65/172 (M=1.3634) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 377 | 40.59% | 78/167 (M=1.3047) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.5: Test case 9.3.1.9 for fixed reference measurement channel requirements for 2.2 Mbps UE class QPSK of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=10dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=539kbps)<br>No of events / No of samples in % | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 360 | 33.21% | 67/172 (M =1.3554) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 343 | 36.36% | 71/168 (M=1.3334) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 320 | 40.63% | 79/169 (M=1.3045) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 275 | 48.98% | 92/167 (M=1.2509) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.6: Test case 9.3.1.9 for fixed reference measurement channel requirements for 2.2 Mbps UE class 16QAM of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=15dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=782.2kbps)<br>No of events / No of samples in % | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 615 | 21.38% | 57/220 (M =1.4408) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 606 | 22.53% | 56/206 (M=1.4325) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 554 | 29.17% | 63/182 (M=1.3843) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 493 | 36.97% | 71/165 (M=1.3292) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.7: Test case 9.3.1.11 for fixed reference measurement channel requirements for 2.8 Mbps UE class QPSK of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=10dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=621kbps)<br>No of events / No of samples in % | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 461 | 25.76% | 60/195 (M =1.4090) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 470 | 24.32% | 59/202 (M=1.4196) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 438 | 29.47% | 63/180 (M=1.3822) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 409 | 34.14% | 69/173 (M=1.3489) | N.A. | 4.1s(fading) | BL |

**Table F.6.3.5.8: Test case 9.3.1.11 for fixed reference measurement channel requirements for 2.8 Mbps UE class 16QAM of 1.28Mcps HS-DSCH**

| Single link performance<br>Ior/Ioc=15dB | Absolute Test requirement (kbps) | | Relative Test requirement (normalized to ideal=1278.6kbps)<br>No of events / No of samples in % | Test limit expressed as No of events / min No of samples<br>( Bad DUT factor) | Min No of samples<br>( number of events to pass)<br>Mandatory, if applicable | Test time in s<br>Mandatory if fading,<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| Test1 | PA3 | 890 | 30.39% | 64/178 (M =1.3755) | N.A. | 164s (fading) | BL |
| Test2 | PB3 | 810 | 36.65% | 72/169 (M=1.3315) | N.A | 164s (fading) | BL |
| Test3 | VA30 | 730 | 42.91% | 80/163 (M=1.2895) | N.A. | 16.4s(fading) | BL |
| Test4 | VA120 | 630 | 50.73%→ ( 49.27% ) | 71/165 (m=0.7528) | N.A. | 4.1s(fading) | RT |

**Table F.6.3.5.9 Test case 9.2.1 demodulation of 3.84Mcps HS-DSCH(QPSK)**

| Single link Performance<br>QPSK<br><br>Test number | Absolute Test requirement (kbps) | | Relative test requirement (normalized to: ideal for tests 1,2,3=2649.6 kbps ideal for test 4=2317.6 kbps)<br><br>No of events/No of samples in %<br>BL → (RT) | Test limit expressed as No of events/min No of samples<br><br>(Bad DUT factor) | Min No of samples<br><br>(number of events to pass)<br><br>Mandatory if applicable | Test time in s<br><br>Mandatory if fading<br><br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| 1<br>( $\hat{I}_{or} / I_{oc}$ = 8,5 dB) | PA3 | 1300 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A. | 164s (fading) | RT |
| 2<br>( $\hat{I}_{or} / I_{oc}$ = 9,0 dB) | PB3 | 1300 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A | 164s (fading) | RT |
| 3<br>( $\hat{I}_{or} / I_{oc}$ = 9,75 dB) | VA30 | 1300 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A | 164s (fading) | RT |
| 4<br>( $\hat{I}_{or} / I_{oc}$ = 11,5 dB) | VA120 | 1400 | 39.59% | 80/175<br>(M=1.312) | N.A | 164s (fading) | BL |

**Table F.6.3.5.10 Test case 9.2.1 demodulation of 3.84Mcps HS-DSCH(16QAM)**

| Single link Performance<br>QPSK<br><br>Test number | Absolute Test requirement (kbps) | | Relative test requirement (normalized to: ideal for tests 1,2,3=5299.6 kbps ideal for tests 4=3477.3 kbps)<br><br>No of events/No of samples in %<br>BL → (RT) | Test limit expressed as No of events/min No of samples<br><br>(Bad DUT factor) | Min No of samples<br><br>(number of events to pass)<br><br>Mandatory if applicable | Test time in s<br><br>Mandatory if fading<br><br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| 1<br>( $\hat{I}_{or} / I_{oc}$ = 16.0 dB) | PA3 | 2600 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A. | 164s (fading) | RT |
| 2<br>( $\hat{I}_{or} / I_{oc}$ = 17.5 dB) | PB3 | 2600 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A | 164s (fading) | RT |
| 3<br>( $\hat{I}_{or} / I_{oc}$ = 18.5 dB) | VA30 | 2600 | 50.94%→<br>(49.06%) | 77/179<br>(m = 0.752) | N.A | 164s (fading) | RT |
| 4<br>( $\hat{I}_{or} / I_{oc}$ = 14.5 dB) | VA120 | 1600 | 53.99%%→<br>(46.01%) | 72/180<br>(M=0.741) | N.A | 164s (fading) | BL |

**Table F.6.3.5.11 Test case 9.4.1 demodulation of 7.68Mcps HS-DSCH(QPSK)**

| Single link Performance<br>QPSK<br>Test number | Absolute Test requirement (kbps) | | Relative test requirement (normalized to: ideal =2119.68 kbps)<br>No of events/No of samples in %<br>BL → (RT) | Test limit expressed as No of events/min No of samples<br>(Bad DUT factor) | Min No of samples<br>(number of events to pass)<br>Mandatory if applicable | Test time in s<br>Mandatory if fading<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| 1 ( $\hat{I}_{or}/I_{oc}$ = 5,2 dB) | PA3 | 880 | 58.48%→ (41.52%) | 68/190 (m = 0.7261) | N.A. | 164s (fading) | RT |
| 2 ( $\hat{I}_{or}/I_{oc}$ = 5,5 dB) | PB3 | 880 | 58.48%→ (41.52%) | 68/190 (m = 0.7261) | N.A | 164s (fading) | RT |
| 3 ( $\hat{I}_{or}/I_{oc}$ = 6,2 dB) | VA30 | 880 | 58.48%→ (41.52%) | 68/190 (m = 0.7261) | N.A | 164s (fading) | RT |
| 4 ( $\hat{I}_{or}/I_{oc}$ = 6,2 dB) | VA120 | 880 | 58.48%→ (41.52%) | 68/190 (m = 0.7261) | N.A | 164s (fading) | BL |

**Table F.6.3.5.12 Test case 9.4.1 demodulation of 7.68Mcps HS-DSCH(16QAM)**

| Single link Performance<br>QPSK<br>Test number | Absolute Test requirement (kbps) | | Relative test requirement (normalized to: ideal =4239.36 kbps)<br>No of events/No of samples in %<br>BL → (RT) | Test limit expressed as No of events/min No of samples<br>(Bad DUT factor) | Min No of samples<br>(number of events to pass)<br>Mandatory if applicable | Test time in s<br>Mandatory if fading<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|---|
| 1 ( $\hat{I}_{or}/I_{oc}$ = 11.1 dB) | PA3 | 1765 | 58.37%→ (41.63%) | 77/179 (m = 0.7264) | N.A. | 164s (fading) | RT |
| 2 ( $\hat{I}_{or}/I_{oc}$ = 13.2 dB) | PB3 | 1765 | 58.37%→ (41.63%) | 77/179 (m = 0.752) | N.A | 164s (fading) | RT |
| 3 ( $\hat{I}_{or}/I_{oc}$ = 13.7 dB) | VA30 | 1765 | 58.37%→ (41.63%) | 77/179 (m = 0.752) | N.A | 164s (fading) | RT |
| 4 ( $\hat{I}_{or}/I_{oc}$ = 13.6 dB) | VA120 | 1765 | 58.37%→ (41.63%) | 72/180 (M=0.741) | N.A | 164s (fading) | BL |

**Table F.6.3.5.13 Maximum Input Level for HS-PDSCH Reception (16QAM)**

| Maximum Input Level for HS-PDSCH Reception (16QAM)<br>16 QAM | Absolute Test requirement (kbps) | Relative test requirement (normalized to ideal=578.6 kbps)<br>No of events/No of samples in % | Test limit expressed as No of events/min No of samples<br>(Bad DUT factor) | Min No of samples<br>(number of events to pass)<br>Mandatory if applicable | Test time in s<br>Mandatory if fading<br>Informative and approx. if statistical | BL / RT |
|---|---|---|---|---|---|---|
| | 500 | 13.58% | 53/317<br>(M=1.4899) | N.A | 2.8s (stat) | BL |

# Annex G (normative): Environmental conditions

## G.1 General

This normative annex specifies the environmental requirements of the UE. Within these limits the requirements of the present documents shall be fulfilled.

## G.2 Environmental requirements

The requirements in this clause apply to all types of UE(s)

### G.2.1 Temperature

The UE shall fulfil all the requirements in the full temperature range of:

| | |
|---|---|
| +15°C to +35°C | for normal conditions (with relative humidity of 25 % to 75 %) |
| -10°C to +55°C | for extreme conditions (see IEC publications 68-2-1 and 68-2-2) |

Outside this temperature range the UE, if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in TS 25.102 [1] for extreme operation.

Some tests in the present document are performed also in extreme temperature conditions. These test conditions are denoted as TL (temperature low, –10°C) and TH (temperature high, +55°C).

### G.2.2 Voltage

The UE shall fulfil all the requirements in the full voltage range, i.e. the voltage range between the extreme voltages.

The manufacturer shall declare the lower and higher extreme voltages and the approximate shutdown voltage. For the equipment that can be operated from one or more of the power sources listed below, the lower extreme voltage shall not be higher, and the higher extreme voltage shall not be lower than that specified below.

| Power source | Lower extreme voltage | Higher extreme voltage | Normal conditions voltage |
|---|---|---|---|
| AC mains | 0,9 * nominal | 1,1 * nominal | nominal |
| Regulated lead acid battery | 0,9 * nominal | 1,3 * nominal | 1,1 * nominal |
| Non regulated batteries:<br>Leclanché / lithium<br>Mercury/nickel & cadmium | <br>0,85 * nominal<br>0,90 * nominal | <br>Nominal<br>Nominal | <br>Nominal<br>Nominal |

Outside this voltage range the UE if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in [1] TS 25.102 for extreme operation. In particular, the UE shall inhibit all RF transmissions when the power supply voltage is below the manufacturer declared shutdown voltage.

Some tests in the present document are performed also in extreme voltage conditions. These test conditions are denoted as VL (lower extreme voltage) and VH (higher extreme voltage).

## G.2.3 Vibration

The UE shall fulfil all the requirements when vibrated at the following frequency/amplitudes:

| Frequency | ASD (Acceleration Spectral Density) random vibration |
|---|---|
| 5 Hz to 20 Hz | 0,96 $m^2/s^3$ |
| 20 Hz to 500 Hz | 0,96 $m^2/s^3$ at 20 Hz, thereafter –3 dB/Octave |

Outside the specified frequency range the UE, if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in TS 25.102 [1] for extreme operation.

## G.2.4 Specified frequency range

The manufacturer shall declare, which of the frequency bands defined in clause 4.2 is supported by the UE.

Some tests in the present document are performed also in low, mid and high range of the operating frequency band of the UE. The UARFCN's to be used for low, mid and high range are defined in TS 34.108 [3] clause 5.1.2.

# Annex H (normative): Terminal Baseline and Service Implementation Capabilities (TDD)

## H.1 Baseline Implementation Capabilities of 3.84 Mcps TDD

**Table H.1: Baseline implementation capabilities of 3.84 Mcps TDD**

| Capability TDD | Clause | UE* | Comments |
|---|---|---|---|
| Chip rate 3,84 Mcps | | M | |
| Frequency bands: (uplink and downlink) | | | |
| 1900-1920 MHz | | M | |
| 2010-2025 MHz | | M | |
| 1850-1910 MHz | | M | |
| 1930-1990 MHz | | M | |
| 1910-1930 MHz | | M | |
| Other spectrum | | O | As Declared |
| Carrier raster 200 kHz | | M | |
| UE maximum output power | 6.2.1 | M | 2, 3 |

(* M = mandatory, O = optional)

- The special conformance testing functions and the logical test interface as specified in TS 34.109 [3]. This issue is currently under investigation.
- Uplink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.1.
- Downlink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.2.

## H.1A Baseline Implementation Capabilities of 1.28 Mcps TDD

**Table H.1A: Baseline implementation capabilities of 1.28 Mcps TDD**

| Capability TDD | Clause | Value range |
|---|---|---|
| Chip rate 1.28 Mcps | | Yes |
| Frequency bands: (uplink and downlink) | 25.102 clause 5.2[1] | a), b), c), a+b), a+c), b+c), a+b+c) |
| UE maximum output power | 25.102 clause 6.2.1[1] | 2, 3 |

- The special conformance testing functions and the logical test interface as specified in TS 34.109 [3]. This issue is currently under investigation.
- Uplink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.1.
- Downlink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.2.

# H.1B Baseline Implementation Capabilities of 7.68 Mcps TDD

**Table H.1: Baseline implementation capabilities of 7.68 Mcps TDD**

| Capability TDD | Clause | UE* | Comments |
|---|---|---|---|
| Chip rate 7,68 Mcps | | M | |
| Frequency bands: (uplink and downlink) | | | |
| 1900-1920 MHz | | M | |
| 2010-2025 MHz | | M | |
| 1850-1910 MHz | | M | |
| 1930-1990 MHz | | M | |
| 1910-1930 MHz | | M | |
| 2570-2620 MHz | | M | |
| Other spectrum | | O | As Declared |
| Carrier raster 200 kHz | | M | |
| UE maximum output power | 6.2.1 | M | 2, 3 |

(* M = mandatory, O = optional)

- The special conformance testing functions and the logical test interface as specified in TS 34.109 [3]. This issue is currently under investigation.
- Uplink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.1.3.
- Downlink reference measurement channel 12,2 kbps (TDD), TS 25.102 [1] clause A.2.2.3.

# H.2 Service Implementation Capabilities:

- Downlink reference measurement channel 64 kbps (TDD), TS 25.102 clause A.2.3.
- Downlink reference measurement channel 144 kbps (TDD), TS 25.102 clause A.2.4.
- Down-link reference measurement channel 384 kbps (TDD), TS 25.102 clause A.2.5.
- BCH Reference Measurement Channel(TDD),TS 25.102 clause A.2.6.

# Annex I (normative): Default Message Contents

This Annex contains the default values of common messages, other than those described in TS 34.108. The messages are primarily concerning the RRM test cases in clause 8 and unless indicated otherwise in specific test cases, shall be transmitted and checked by the system simulator. The necessary messages are listed in alphabetical order.

In this Annex, decimal values are normally used. However, sometimes, a hexadecimal value, indicated by an "H", or a binary value, indicated by a "B" is used.

Contents of MEASUREMENT REPORT message for Intra frequency 3,84 Mcps option TDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Intra-frequency measured results list | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - SFN-SFN observed time difference | Checked that this IE is present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Intra frequency 1,28 Mcps option TDD test cases

| Information Element/Group name | Semantics description |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Intra-frequency measured results list | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - COUNT-C-SFN frame difference | Checked that this IE is present |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is absent |
| - Timeslot ISCP | Checked that this IE is absent |
| Measured Results on RACH | Checked that this IE is absent |
| Additional Measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Intra frequency 7,68 Mcps option TDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Intra-frequency measured results list | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - COUNT-C-SFN frame difference | Checked that this IE is present |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Inter frequency 3,84 Mcps option TDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Inter-frequency measured results list | |
| - UTRA Carrier RSSI | Checked that this IE is present |
| - Inter-frequency cell measurement results | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - SFN-SFN observed time difference | Checked that this IE is present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Inter frequency 1,28 Mcps option TDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Inter-frequency measured results list | |
| - UTRA Carrier RSSI | Checked that this IE is present |
| - Inter-frequency cell measurement results | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - SFN-SFN observed time difference | Checked that this IE is present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Inter frequency 7,68 Mcps option TDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Inter-frequency measured results list | |
| - UTRA Carrier RSSI | Checked that this IE is present |
| - Inter-frequency cell measurement results | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - SFN-SFN observed time difference | Checked that this IE is present |
| - Cell synchronisation information | |
| - CHOICE mode | TDD |
| - COUNT-C-SFN frame difference | Checked that this IE is present |
| - OFF | Checked that this IE is present |
| - CHOICE mode | TDD |
| - Cell Parameters ID | 4 |
| - Primary CCCPCH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Timeslot ISCP | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for Inter frequency FDD test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Inter-frequency measured results list | |
| - UTRA Carrier RSSI | Checked that this IE is present |
| - Inter-frequency cell measurement results | |
| - Cell measured results | |
| - Cell Identity | Not present |
| - SFN-SFN observed time difference | Checked that this IE is present |
| - Cell synchronisation information | |
| -Tm | Checked that this IE is present |
| - OFF | Checked that this IE is present |
| - CHOICE mode | FDD |
| - Primary CPICH info | Checked that this IE is present |
| - Primary scrambling code | 150 |
| - CPICH Ec/N0 | Checked that this IE is present |
| - CPICH RSCP | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |

Contents of MEASUREMENT REPORT message for inter – RAT test cases

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | The presence of this IE is dependent on IXIT statements in TS 34.123-2. If integrity protection is indicated to be active, this IE shall be present with the values of the sub IEs as stated below. Else, this IE and the sub-IEs shall be absent. |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Inter-RAT measured results list | |
| - CHOICE system | GSM |
| - Measured GSM cells | Checked that this IE is present |
| - GSM carrier RSSI | Checked that this IE is present |
| - Pathloss | Checked that this IE is present |
| - Observed time difference to GSM cell | Checked that this IE is present |
| Measured results on RACH | Checked that this IE is absent |
| Additional measured results | Checked that this IE is absent |
| Event results | Checked that this IE is absent |
| Inter-RAT cell info indication | Checked that this IE is absent |

# Annex J (informative): Change history

| Meeting -1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version - Current | Version -New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-22 | TP-030281 | 181 | 1 | Addition of LCR GSM neighbour reporting | F | 4.9.0 | 5.0.0 | T1-031615 |
| TP-22 | TP-030281 | 182 | 1 | Addition of LCR GSM handover test | F | 4.9.0 | 5.0.0 | T1-031616 |
| TP-22 | TP-030281 | 183 | 1 | Update to LCR GSM RSSI measurement | F | 4.9.0 | 5.0.0 | T1-031617 |
| TP-22 | TP-030281 | 184 | 1 | Update to inter frequency measurements | F | 4.9.0 | 5.0.0 | T1-031618 |
| TP-22 | TP-030281 | 185 | 1 | Correction of LCR ISCP test case | F | 4.9.0 | 5.0.0 | T1-031619 |
| TP-22 | TP-030281 | 186 | 1 | Addition of TDD HSDPA section & creation Rel 5 | B | 4.9.0 | 5.0.0 | T1-031620 |
| TP-22 | TP-030281 | 187 | 1 | HSDPA HS DSCH throughput (fixed and variable) | B | 4.9.0 | 5.0.0 | T1-031621 |
| TP-22 | TP-030281 | 188 | 1 | Addition of Reporting of HS DSCH CQI | B | 4.9.0 | 5.0.0 | T1-031622 |
| TP-22 | TP-030281 | 189 | 1 | Addition of HS-SCCH Detection Performance | B | 4.9.0 | 5.0.0 | T1-031623 |
| RP-29 | RP-050514 | 190 | - | CR to 34.122 Rel-5: Correction to Cell Re-selection in CELL_FACH for 1.28Mcps TDD | F | 5.0.0 | 5.1.0 | R5-051217 |
| RP-29 | RP-050514 | 191 | - | CR to 34.122 Rel-5: Correction to Cell Re-selection in idle for 1.28Mcps TDD | F | 5.0.0 | 5.1.0 | R5-051218 |
| RP-29 | RP-050514 | 192 | - | CR to 34.122 Rel-5:Correction to UE Measurements Procedures for 1.28Mcps TDD | F | 5.0.0 | 5.1.0 | R5-051219 |
| RP-29 | RP-050514 | 193 | - | CR to 34.122 Rel-5:Correction to UE Measurements Performance Requirements for 1.28Mcps TDD | F | 5.0.0 | 5.1.0 | R5-051220 |
| RP-30 | RP-050777 | 194 | - | Adding terminal Baseline and Service Implementation Capabilities of 1.28Mcps TDD | F | 5.1.0 | 5.2.0 | R5-051886 |
| RP-30 | RP-050777 | 195 | - | Correction to power control downlink – 1.28 Mcps TDD option | F | 5.1.0 | 5.2.0 | R5-051887 |
| RP-30 | RP-050777 | 196 | - | Correction to blocking exceptions for 1.28 Mcps TDD option | F | 5.1.0 | 5.2.0 | R5-051888 |
| RP-30 | RP-050777 | 197 | - | Name correction of logical and transport channels | F | 5.1.0 | 5.2.0 | R5-051889 |
| RP-30 | RP-050777 | 198 | - | Correction to power definitions | F | 5.1.0 | 5.2.0 | R5-051890 |
| RP-30 | RP-050777 | 199 | - | Completing 1.28Mcps TDD/FDD cell reselection in idle mode | F | 5.1.0 | 5.2.0 | R5-051891 |
| RP-30 | RP-050777 | 200 | - | Adding OCNS_Ec/Ior and correcting Ioc | F | 5.1.0 | 5.2.0 | R5-051892 |
| RP-30 | RP-050777 | 201 | - | Correction of Cell Re-selection in CELL_FACH | F | 5.1.0 | 5.2.0 | R5-051893 |
| RP-31 | RP-060163 | 202 | - | Correcting cell reselection test procedure | F | 5.2.0 | 5.3.0 | R5-060453 |
| RP-31 | RP-060146 | 203 | - | Adding statistical Testing of HSDPA Receiver Performance | F | 5.2.0 | 5.3.0 | R5-060034 |
| RP-31 | RP-060146 | 204 | - | Completing the test of reporting of HS-DSCH Channel Quality Indicator | F | 5.2.0 | 5.3.0 | R5-060033 |
| RP-31 | RP-060146 | 205 | - | Completing HS-SCCH Detection Performance test | F | 5.2.0 | 5.3.0 | R5-060032 |
| RP-31 | RP-060146 | 206 | - | Completing HS-DSCH throughput for variable reference channels | F | 5.2.0 | 5.3.0 | R5-060031 |
| RP-31 | RP-060146 | 207 | - | Completing HS-DSCH throughput for fixed reference channels | F | 5.2.0 | 5.3.0 | R5-060030 |
| RP-31 | RP-060146 | 208 | - | Adding general test conditions and declarations for HSDPA | F | 5.2.0 | 5.3.0 | R5-060029 |
| RP-32 | RP-060329 | 209 | - | Update of HCR reference channels | F | 5.3.0 | 5.4.0 | R5-061150 |
| RP-32 | RP-060328 | 210 | - | Correcting the handover delay for 1.28Mcps TDD option | F | 5.3.0 | 5.4.0 | R5-061426 |
| RP-32 | RP-060328 | 211 | - | Correcting inter RAT cell re-selection for 1.28Mcps TDD option | F | 5.3.0 | 5.4.0 | R5-061427 |
| RP-32 | RP-060336 | 212 | - | Addition of HCR HS-DSCH throughput for fixed reference channels | F | 5.3.0 | 5.4.0 | R5-061142 |
| RP-32 | RP-060336 | 213 | - | Addition of HCR HS-DSCH throughput for variable reference channels | F | 5.3.0 | 5.4.0 | R5-061143 |
| RP-32 | RP-060336 | 214 | - | Addition of HCR reporting of HS-DSCH channel quality indicator | F | 5.3.0 | 5.4.0 | R5-061144 |
| RP-32 | RP-060336 | 215 | - | Addition of HCR HSDPA reference channels | F | 5.3.0 | 5.4.0 | R5-061145 |
| RP-32 | RP-060336 | 216 | - | Addition of HCR multi-path fading environments for HSDPA performance requirements | F | 5.3.0 | 5.4.0 | R5-061146 |
| RP-32 | RP-060336 | 217 | - | Addition of statistical testing of HSDPA receiver performance for HCR to Annex F | F | 5.3.0 | 5.4.0 | R5-061147 |
| RP-33 | RP-060552 | 218 | - | Correction to Handover to GSM for 1.28Mcps TDD option | F | 5.4.0 | 5.5.0 | R5-062124 |
| RP-33 | RP-060552 | 219 | - | Correcting the RRC re-establishment delay for 1.28Mcps TDD option | F | 5.4.0 | 5.5.0 | R5-062125 |
| RP-33 | RP-060552 | 220 | - | Correcting the statistical testing of receiver BER/BLER performance | F | 5.4.0 | 5.5.0 | R5-062126 |
| RP-33 | RP-060552 | 221 | - | Correcting the statistical testing of RRM delay performance | F | 5.4.0 | 5.5.0 | R5-062128 |

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| RP-33 | RP-060549 | 222 | - | Clarification of Tx spurious emission level from 3.84 Mcps TDD UE into PHS band. | F | 5.4.0 | 5.5.0 | R5-062267_r1 |
| RP-34 | RP-060732 | 223 | - | Adding test case of GSM measurement of without BSIC verification required | F | 5.5.0 | 5.6.0 | R5-063106 |
| RP-34 | RP-060732 | 224 | - | Correction to GSM measurement for 1.28Mcps TDD option | F | 5.5.0 | 5.6.0 | R5-063154 |
| RP-34 | RP-060747 | 225 | - | HS-SCCH detection test for HCR TDD | F | 5.5.0 | 5.6.0 | R5-063218 |
| RP-35 | RP-070104 | 228 | | Correcting the inter frequency measurement for 1.28Mcps TDD option | F | 5.6.0 | 5.7.0 | R5-070119 |
| RP-35 | RP-070104 | 229 | | Correcting the intra frequency measurement test case for 1.28Mcps TDD option | F | 5.6.0 | 5.7.0 | R5-070120 |
| RP-35 | RP-070104 | 230 | | Correcting the test case of open loop power control for 1.28Mcps TDD option | F | 5.6.0 | 5.7.0 | R5-070121 |
| RP-35 | RP-070104 | 231 | | Adding the test criteria in RRM test cases for 1.28Mcps TDD option | F | 5.6.0 | 5.7.0 | R5-070122 |
| RP-35 | RP-070104 | 232 | | Adding test case of Timing Advance for 1.28Mcps TDD option | F | 5.6.0 | 5.7.0 | R5-070123 |
| RP-35 | RP-070105 | 227 | | 34.122 v5.7.0 pointer to Release 6 version | F | 5.6.0 | 5.7.0 | R5-070076 |
| RP-35 | RP-070111 | 226 | | 34.122 v6.0.0 pointer to Release 7 version | F | 5.6.0 | 6.0.0 | R5-070075 |
| RP-35 | RP-070091 | 233 | | Addition of 7.68Mcps to Frequency bands and channel arrangement (section 4 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070175 |
| RP-35 | RP-070091 | 234 | | Addition of 7.68Mcps tests to transmitter characteristics (section 5 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070538 |
| RP-35 | RP-070091 | 235 | | Addition of 7.68Mcps tests to receiver characteristics (section 6 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070539 |
| RP-35 | RP-070091 | 236 | | Addition of 7.68Mcps tests to performance requirements (section 7 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070178 |
| RP-35 | RP-070091 | 237 | | Addition of 7.68Mcps tests to connected mode mobility requirements for support of RRM (section 8.3 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070540 |
| RP-35 | RP-070091 | 238 | | Addition of 7.68Mcps tests to RRC connection control for support of RRM (section 8.4 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070180 |
| RP-35 | RP-070091 | 239 | | Addition of 7.68Mcps tests to idle mode requirements for support of RRM (section 8.2 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070181 |
| RP-35 | RP-070091 | 240 | | Addition of 7.68Mcps tests to UE measurement procedures (section 8.6 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070182 |
| RP-35 | RP-070091 | 241 | | Addition of 7.68Mcps tests to measurements performance requirements (section 8.7 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070183 |
| RP-35 | RP-070091 | 242 | | Addition of 7.68Mcps tests to timing charcteristics for support of RRM (section 8.5 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070184 |
| RP-35 | RP-070091 | 243 | | Addition of 7.68Mcps tests to Performance requirements for HSDPA (section 9 of 34.122) | F | 5.6.0 | 7.0.0 | R5-070541 |
| RP-35 | RP-070091 | 244 | | Addition of 7.68Mcps reference measurement channels to Annex C of 34.122 | F | 5.6.0 | 7.0.0 | R5-070542 |
| RP-35 | RP-070091 | 245 | | Addition of 7.68Mcps propagation conditions to Annex D of 34.122 | F | 5.6.0 | 7.0.0 | R5-070543 |
| RP-35 | RP-070091 | 246 | | Addition of 7.68Mcps cases to Annex F of 34.122 | F | 5.6.0 | 7.0.0 | R5-070544 |
| RP-35 | RP-070091 | 247 | | Addition of 7.68Mcps to Annex H of 34.122 | F | 5.6.0 | 7.0.0 | R5-070545 |
| RP-35 | RP-070091 | 248 | | Addition of 7.68Mcps messages to Annex I of 34.122 | F | 5.6.0 | 7.0.0 | R5-070190 |
| RP-35 | RP-070092 | 249 | | Addition of measurement parameters for MBMS tests to Annex C of 34.122 | F | 5.6.0 | 7.0.0 | R5-070192 |
| RP-35 | RP-070093 | 250 | | Addition of E-DCH reference measurement channels for E-DCH tests to Annex C of 34.122 | F | 5.6.0 | 7.0.0 | R5-070191 |
| RP-36 | RP-070366 | 251 | | Addition of E-AGCH demodulation performance tests for 3.84 and 7.68Mcps TDD | F | 7.0.0 | 7.1.0 | R5-071281 |
| RP-36 | RP-070364 | 252 | | Addition of Scenario 2C: 3,84 Mcps TDD cell re-selection for 7,68 Mcps TDD UE (section 8 of 34.122) | F | 7.0.0 | 7.1.0 | R5-071285 |
| RP-36 | RP-070365 | 253 | | MTCH demodulation tests for 3.84 and 7.68Mcps TDD | F | 7.0.0 | 7.1.0 | R5-071288 |
| RP-36 | RP-070364 | 254 | | Addition of Scenario 2D: 3,84 Mcps TDD cell re-selection for 7,68 Mcps TDD UE (section 8 of 34.122) | F | 7.0.0 | 7.1.0 | R5-071313 |
| RP-36 | RP-070344 | 255 | | Correcting the test requirement for 1.28Mcps TDD option | F | 7.0.0 | 7.1.0 | R5-071425 |
| RP-36 | RP-070344 | 256 | | Correcting the message content for 1.28Mcps TDD option | F | 7.0.0 | 7.1.0 | R5-071426 |
| RP-36 | RP-070344 | 257 | | Correcting the Spectrum emission mask for 1.28Mcps TDD option | F | 7.0.0 | 7.1.0 | R5-071427 |
| RP-37 | RP-070609 | 258 | - | Addition of MCCH demodulation tests for 3.84Mcps and 7.68Mcps | F | 7.1.0 | 7.2.0 | R5-072294 |
| RP-37 | RP-070610 | 259 | - | Addition of Detection of E-DCH HARQ ACK Indicator Channel (E-HICH) for 3.84Mcps and 7.68Mcps TDD | F | 7.1.0 | 7.2.0 | R5-072293 |
| RP-37 | RP-070616 | 260 | - | Adding the test case of MTCH demodulation | F | 7.1.0 | 7.2.0 | R5-072382 |

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| | | | | performance for 1.28Mcps TDD option | | | | |
| RP-37 | RP-070616 | 261 | - | Adding the test case of MTCH demodulation and cell identification for 1.28Mcps TDD option | F | 7.1.0 | 7.2.0 | R5-072383 |
| RP-38 | RP-070876 | 262 | | Introduction of multi-frequency operation for LCR TDD and corrections | F | 7.2.0 | 7.3.0 | R5-073331 |
| RP-38 | RP-070876 | 263 | | Correcting the test case of maximum output power for 1.28Mcps TDD option | F | 7.2.0 | 7.3.0 | R5-073293 |
| RP-38 | RP-070876 | 264 | | Updating of HSDPA demodulation performance requirements for 1.28Mcps TDD option | F | 7.2.0 | 7.3.0 | R5-073294 |
| RP-38 | RP-070878 | 265 | | Deleting test case of MCCH demodulation requirement for 1.28Mcps TDD option | F | 7.2.0 | 7.3.0 | R5-073295 |
| RP-38 | RP-070878 | 266 | | Correcting test case of MTCH demodulation requirement for 1.28Mcps TDD option | F | 7.2.0 | 7.3.0 | R5-073296 |
| RP-39 | RP-080102 | 267 | | Correcting the pass fail decision rule in annex F6.3.5 for 1.28Mcps TDD option | F | 7.3.0 | 7.4.0 | R5-080389 |
| RP-39 | RP-080102 | 268 | | Adding the test tolerance of performance requiremet for 1.28Mcps TDD option | F | 7.3.0 | 7.4.0 | R5-080095 |
| RP-39 | RP-080102 | 269 | | Adding the test parameter of midamble allocation for performance requiremet for LCR TDD | F | 7.3.0 | 7.4.0 | R5-080096 |
| RP-39 | RP-080102 | 270 | | Adding the test case of maximum input level for HS-PDSCH reception for 1.28Mcps TDD option | F | 7.3.0 | 7.4.0 | R5-080097 |
| RP-39 | RP-080115 | 271 | | Adding the test case of E-HICH detection parameter for 1.28Mcps TDD Option | F | 7.3.0 | 7.4.0 | R5-080191 |
| RP-39 | RP-080115 | 272 | | Adding the test case of E-AGCH detection parameter for 1.28Mcps TDD Option | F | 7.3.0 | 7.4.0 | R5-080192 |
| RP-40 | RP-080377 | 0273 | | Adding the test case of E-AGCH detection parameters for 1.28Mcps TDD Option | F | 7.4.0 | 7.5.0 | R5-081103 |
| RP-40 | RP-080377 | 0274 | | Adding the test case of E-HICH detection parameter for 1.28Mcps TDD Option | F | 7.4.0 | 7.5.0 | R5-081104 |
| RP-40 | RP-080377 | 0275 | | Adding the test case of EVM with E-DCH16QAM for 1.28Mcps TDD Option | F | 7.4.0 | 7.5.0 | R5-081105 |
| RP-40 | RP-080365 | 0276 | | Modifiying the test parameter of HS-SCCH for 1.28Mcps TDD Options | F | 7.4.0 | 7.5.0 | R5-081106 |
| RP-41 | RP-080557 | 0277 | | Correction to test case RRC re-establishment delay to an known target cell test parameters | F | 7.5.0 | 7.6.0 | R5-083057 |
| RP-41 | RP-080557 | 0278 | | Correction to test case RRC re-establishment delay to an unknown target cell test parameters | F | 7.5.0 | 7.6.0 | R5-083058 |
| RP-41 | RP-080557 | 0279 | | Correcting the test case of spurious emissions | F | 7.5.0 | 7.6.0 | R5-083303 |
| RP-41 | RP-080557 | 0280 | | Adding the test tolerance of performance requirement for HSDPA | F | 7.5.0 | 7.6.0 | R5-083304 |
| RP-41 | RP-080569 | 0281 | | Adding the test case of MBSFN capable UE for LCR TDD | F | 7.5.0 | 7.6.0 | R5-083305 |
| RP-41 | RP-080568 | 0282 | | Addition of RF tests for MBSFN for 3.84Mcps and 7.68Mcps TDD | F | 7.6.0 | 8.0.0 | R5-083320 |
| RP-41 | RP-080568 | 0283 | | Update of annex with new test configuration and multi path channel definition required for MBSFN testing for 3.84Mcps and 7.68Mcps TDD | F | 7.6.0 | 8.0.0 | R5-083823 |
| RP-41 | RP-080565 | 0284 | | Adding the test case of UE Transmission Power Headroom for 1.28Mcps TDD Option | F | 7.5.0 | 7.6.0 | R5-083824 |
| RP-41 | RP-080561 | 0285 | | Update of Detection of E-DCH HARQ ACK Indicator Channel (E-HICH) and Demodulation of E-DCH Absolute Grant Channel (E-AGCH) tests for 3.84Mcps and 7.68Mcps TDD | F | 7.6.0 | 8.0.0 | R5-083826 |

# History

| Document history | | |
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